

INTEGRATED EDITION • 2026

Economies, Accounts and Money

*Mathematics, Statistics, Microeconomics, Accounting, Finance,
Macroeconomics and Monetary Economics as One Connected System*

Seven Parts • Fifty-Two Chapters • Two Mathematical Appendices

A text for upper-undergraduate study, with Advanced Analysis sections
bridging to first-year graduate work.

*The ability to explain one economic event at every level—
numbers, evidence, incentives, accounts, value, aggregates and money—
without contradiction.*

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How to Use This Book

The idea behind the book

Most students meet mathematics, statistics, microeconomics, accounting, finance, macroeconomics and monetary economics as separate courses taught by separate departments, with separate notations and separate vocabularies. The result is predictable. A student who can derive a demand curve cannot say what happens to a bank's balance sheet when a household borrows to buy a car; a student who can prepare a cash-flow statement cannot value the company whose cash flows it records; a student who can run a regression cannot say whether its coefficient measures a cause; a student who can recite the Taylor rule cannot explain what a central bank physically does on the morning it raises its policy rate.

This book treats these subjects as one connected body of knowledge about one economy:

- **Mathematics** supplies the language: quantities, functions, growth, optimization, matrices and present values.
- **Statistics and econometrics** supply the discipline of evidence: probability, distributions, inference, regression, causal identification and forecasting.
- **Microeconomics** explains how households and firms make decisions and how markets coordinate those decisions through prices.
- **Accounting** explains how economic activity is recorded and measured, firm by firm, in balance sheets, income statements and cash-flow statements.
- **Finance** explains how risk and time are priced, how investments are chosen and valued, and how firms are financed.
- **Macroeconomics** adds up decisions and accounts into economy-wide output, income, employment, saving, investment and growth.
- **Monetary economics** explains how money, banks, credit, interest rates, central banks and expectations interact with the real economy.

The test of whether you have understood the book is whether you can take a single event—a central-bank rate increase, a bank failure, an oil-price shock, a takeover, a war—and explain it at every level without contradicting yourself.

Who the book is for

The primary reader is an upper-undergraduate student who wants the depth of a strong university programme, or a professional who needs the whole toolkit of economic and financial analysis in one place. The core text assumes no prior economics, statistics, accounting or finance, and only school algebra; it moves quickly and uses calculus freely once Part I has developed it. Material that goes beyond the undergraduate core is placed in clearly marked *Advanced Analysis* boxes. You can skip every one of those boxes on a first reading and lose nothing needed later in the core text; conversely, a reader preparing for graduate study should work through them.

About this edition

This integrated edition combines two works: the economics, accounting and money text *Economics, Accounts and Money*, and a course in mathematics, statistics and finance. Their overlapping material has been merged rather than repeated. Parts I and II develop mathematics and statistics in full, with proofs where they illuminate; Part V develops corporate finance, investments and risk management on the foundations laid by the accounting chapters; and the economics chapters draw on both, with cross-references in place of repetition.

Prerequisites

You should be comfortable with arithmetic, basic algebra and graphs. Part I develops everything else that the book uses: percentages and index numbers, functions, exponentials and logarithms, derivatives and optimization with and without constraints (including the Lagrangian method), integrals, matrices and present values. Part II develops probability, statistics and econometrics from first principles. No prior accounting is assumed; Chapters 10 and 27 build double-entry bookkeeping from first principles.

The order of ideas

A deliberate rule governs the sequence: *no concept is used before it has been explained*. If a formula combines x and z , both x and z have been defined earlier. This rule produced several departures from the conventional ordering of textbooks:

1. The mathematics comes first (Part I): constrained optimization and the Lagrangian (Chapter 6) precede consumer theory, and present value (Chapter 9) precedes accounting for bonds, capital budgeting and the yield curve.
2. Probability and statistics (Part II) precede economics and finance: expected value and variance precede the economics of risk (Chapter 25); covariance and diversification (Chapter 14) precede the capital asset pricing model (Chapter 34); and regression (Chapter 16) precedes the estimation of betas and demand elasticities.
3. Game theory (Chapter 23) precedes oligopoly (Chapter 24), because Cournot and Bertrand competition are applications of Nash equilibrium.
4. Accounting (Part IV) precedes finance (Part V), because valuation, credit analysis and financial modelling work from the financial statements; and both precede macroeconomics (Part VI), because national accounts, sector balances and bank balance sheets are much easier to understand once you can read a firm's accounts.
5. Finance precedes money and banking (Part VII): the term structure of interest rates (Chapter 33) and covered interest parity (Chapter 42) are derived in finance and used in the chapters on monetary transmission and open-economy policy.
6. Risk and asymmetric information (Chapter 25) precede banking, because deposit insurance, collateral and capital requirements are responses to information problems.

Structure

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43	National Accounts and the Open-Economy Accounting Framework
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45	Consumption, Saving, Investment and Labour Markets
46	Short-Run Fluctuations and Stabilization
47	Modern Macroeconomic Frameworks
Part VII	Money, Banking and Integration
48	Money, Commercial Banking and Payments
49	Central-Bank Operations and Monetary Transmission

50	Inflation, Expectations, Credibility and Policy Rules
51	Unconventional Policy and Financial Stability
52	Open-Economy Monetary Economics, Crises and the Integrated Capstone
	Mathematical appendices: difference and differential equations; dynamic optimization

Reading paths

The integrated programme (three to four semesters). All seven Parts in order.

Quantitative foundations. Chapters 1–19: the mathematics and statistics needed for economics, finance and data analysis.

Intermediate economics. Chapters 1, 10, 20–26 and 43–47, drawing on Parts I and II as needed and reading Chapter 27 as an accounting bridge.

Corporate finance and valuation. Chapters 9, 13–16, 27–31 and 32–42.

Money, banking and macroeconomics. Chapters 10, 25, 27, 30, 33, 39 and 43–52.

Professional economic and financial analysis. Chapters 1, 10, 22, 24, 27–32, 37, 39, 43, 46 and 48–52, emphasizing cases rather than Advanced Analysis boxes.

How each chapter is built

Chapters follow a recurring rhythm: a motivating question; the measurement problem it raises; the model; its derivation; worked examples; an application to real institutions or data; a case; the limitations of the model; and exercises. Each chapter ends with a summary, a list of key terms, and exercises with *full worked solutions*. The solutions are part of the teaching: many of them develop a point further than the chapter does. Try each exercise seriously before reading its solution.

The boxes

Definition 0.1. Definition

A precise statement of what a term means. When a word is printed in **bold** in the text, it is being defined and appears in the index.

Proposition 0.2. Proposition

A result that follows logically from stated assumptions. Propositions are proved or derived in the text unless the proof is beyond the book's level, in which case a reference is given.

Worked Example 0.3. Worked Example

A problem solved step by step with numbers you can check.

► Common error

A mistake that students (and sometimes professionals) commonly make, with the correction.

Check your reasoning

Short questions to test whether you have understood the preceding section before moving on. Answers follow immediately after in small print or are left for reflection.

Advanced Analysis: Optional

Material at the level of first-year graduate study.

Bridge. A connection to another part of the book, pointing backward to where a concept was built or forward to where it will be used.

Case: Case study

A real episode—a company, a market, a policy or a crisis—analysed with the chapter's tools. Facts are drawn from published sources; where figures are uncertain or disputed, the text says so.

Notation and conventions

A book spanning seven subjects needs a single notation. The following conventions hold throughout, except where a section explicitly says otherwise.

Symbol	Meaning
$t = 0, 1, 2, \dots$	Discrete time periods (years unless stated).
x_t	Value of variable x in period t .
$\dot{x}$	Time derivative dx/dt in continuous-time models.
g_x, g	Growth rate of x ; g alone is a growth rate of output, dividends or cash flows.
i	Nominal interest rate (per period, as a decimal: 5% is 0.05).
r	Real interest rate in macroeconomics; a discount rate or required return in finance.
r_f, r_E, r_D	Safe rate; cost of equity; cost of debt.
π	Inflation rate. (Profit is written Π , capital pi, to avoid a clash.)
π^e	Expected inflation.
P	Price level (an index) in macroeconomics; the price of a single good in microeconomics, where p is also used; the price of a bond or share in finance. Context always makes this clear.
Y	Real output (real GDP).
C, I, G, X, M	Consumption, investment, government purchases, exports, imports.
S	Saving; T taxes net of transfers. In finance, S is also a spot price.
K, L, A	Capital, labour, total factor productivity.
D, E, V	In finance: debt, equity and firm value $V = D + E$.
τ	Tax rate.
$u(\cdot), U(\cdot)$	Utility function.
β	Subjective discount factor ($0 < \beta < 1$) in economics; a regression coefficient in statistics; an asset's market beta in finance.
δ	Depreciation rate.
ε	Elasticity (with subscript identifying which), or a random error term in statistics.
$\mathbb{E}[\cdot], \text{Var}(\cdot), \text{Cov}(\cdot, \cdot)$	Expectation, variance and covariance.
μ, σ, ρ	Mean, standard deviation and correlation of a random variable.
$\hat{\theta}$	An estimate of the parameter θ computed from data.

Φ, ϕ	Standard normal distribution function and density.
$\ln$	Natural logarithm.
$\approx$	Approximately equal (the approximation is always stated).
$\equiv$	Identity: true by definition, for all values of the variables.

Identities versus equations. The sign $\equiv$ is reserved for accounting identities, which hold by construction, such as $\text{Assets} \equiv \text{Liabilities} + \text{Equity}$. The sign $=$ is used for behavioural relationships and equilibrium conditions, which may or may not hold in the world. Distinguishing the two is one of the central skills taught in this book.

Money amounts and units. Money amounts carry a currency symbol or are stated in “currency units” (cu) when the currency does not matter. Parentheses around a number in a financial statement denote a negative amount or an outflow, as in accounting practice: (250) means -250 .

Debits and credits. Journal entries are written with debits first and credits indented beneath them. “Dr” means debit (left side) and “Cr” means credit (right side).

Accounting standards. Two systems of financial reporting are treated in parallel: International Financial Reporting Standards (IFRS), issued by the International Accounting Standards Board (IASB), and United States Generally Accepted Accounting Principles (U.S. GAAP), issued by the Financial Accounting Standards Board (FASB). Where they differ in a way that matters, a comparison is given.

Data. Where the book discusses data, it names the producing institution (for example the U.S. Bureau of Economic Analysis, the Bureau of Labor Statistics, the Bank for International Settlements, the European Central Bank, the Deutsche Bundesbank, the People’s Bank of China or Bank Negara Malaysia). Official statistics are revised; figures quoted in the text should be understood as the published values at the time of writing, and you should consult the source for current values.

A note on accuracy

Every derivation in this book has been worked through, every numerical example recalculated, and every set of financial statements checked to balance. Institutional descriptions (accounting standards, central-bank operating procedures, regulation, market practice) describe arrangements as they stood in 2026; such arrangements change, and where a detail matters for a decision you should confirm it against the issuing body’s current documents. Market data quoted in cases are rounded, and the sources are listed at the end of the book.

Part I

Foundations: Reasoning and Mathematics

Economic Reasoning, Institutions and Scarcity

After studying this chapter you should be able to

1. Define scarcity and explain why it forces choice, and distinguish the three questions every economic system must answer.
2. Calculate opportunity costs, including implicit costs, and explain why sunk costs should not affect forward-looking decisions.
3. Construct a production possibilities frontier, explain its shape, and use it to discuss efficiency and growth.
4. Distinguish absolute from comparative advantage and demonstrate numerically that specialization and trade can make both parties better off.
5. Apply the marginal decision rule and explain why optimal choices equate marginal benefit and marginal cost.
6. Explain how institutions—property rights, contracts, courts and money—shape incentives.
7. Define economic equilibrium and distinguish it from optimality.
8. Distinguish positive from normative statements, and efficiency from equity.
9. Distinguish accounting identities from behavioural relationships, and stocks from flows.

A motivating question

In the autumn of 2022, households and firms across Europe, and in Germany above all, faced natural-gas prices several times higher than a year earlier. Governments debated capping prices, subsidizing bills, rationing supply or simply letting prices rise. Each of those options has consequences that can be traced at four levels. At the level of the individual household, a higher price changes what is sensible to consume. At the level of the firm, it changes costs, profits, and the numbers reported in financial statements. At the level of the whole economy, it changes measured inflation, output and the trade balance. At the level of the monetary system, it changes what the central bank must decide about interest rates.

This chapter develops the ideas that every one of those analyses relies on: scarcity, opportunity cost, marginal reasoning, incentives, institutions, equilibrium, and the difference between what *is* and what *ought to be*. It also introduces a distinction that is surprisingly often forgotten, even by professionals: the difference between an *accounting identity*, which is true by definition, and a *behavioural relationship*, which describes how people act and may be false.

1.1 Scarcity and the Economic Problem

1.1.1 Scarcity

Definition 1.1. Scarcity

A good or resource is **scarce** when the quantity people would want at a zero price exceeds the quantity available. Scarcity is a relationship between wants and availability, not a statement that something is rare.

Sand in a desert is abundant in the everyday sense, but beach-quality sand delivered to a construction site is scarce: people would use more of it if it were free, and producing and transporting it uses resources that could have done something else. Air is, in most places, not scarce: at a zero price nobody wants more air than is available. Clean air in a polluted city, however, is scarce, and much of Chapter 26 concerns what happens when a scarce resource has no price.

The resources used to produce goods and services are traditionally grouped into four **factors of production**:

- **Labour**: human time and effort, including skill (often called *human capital*).
- **Capital**: produced goods used to produce other goods—machines, buildings, software, vehicles. In economics, “capital” in this sense means physical (or intangible) productive assets, not money. Money used to buy capital goods is *financial capital*; the distinction matters throughout this book.
- **Land** and natural resources: the surface of the earth, minerals, water, the radio spectrum.
- **Entrepreneurship** or organization: the coordination of the other factors and the bearing of the risk that a venture will fail.

1.1.2 Three questions every economy must answer

Because resources are scarce, every society, whatever its political system, must answer three questions:

1. **What** goods and services will be produced, and in what quantities?
2. **How** will they be produced—with which combinations of labour, capital and land, and with which technologies?
3. **For whom** will they be produced—how will output be distributed among people?

There are only a few broad mechanisms for answering these questions: *tradition* (do what was done before), *command* (a central authority decides), and *markets* (voluntary exchange at prices that adjust). Every real economy mixes all three. Households typically allocate resources internally by tradition and negotiation; firms internally allocate by command (a manager assigns tasks rather than auctioning them among employees); and firms and households transact with each other mainly through markets. Much of economics is about understanding when each mechanism works well and when it fails. A recurring theme of Part III is that markets coordinate enormous amounts of information cheaply, through prices, but that they fail in identifiable circumstances—market power, externalities, public goods and asymmetric information.

Key idea. Economics is the study of how people make choices under scarcity, and of how those individual choices interact to determine what is produced, how, and for whom.

1.2 Opportunity Cost

1.2.1 Definition

Definition 1.2. Opportunity cost

The **opportunity cost** of a choice is the value of the best alternative forgone by making that choice.

Three features of the definition deserve emphasis. First, it refers to the *best* alternative, not the sum of all alternatives: if you have one free evening and three possible uses for it, choosing one gives up only the one you would otherwise have chosen, not all three. Second, it is measured in terms of *value to the decision maker*, which may or may not be a money amount. Third, it concerns *alternatives that are actually available*; a forgone option that was never feasible is not a cost.

1.2.2 Explicit and implicit costs

Definition 1.3. Explicit and implicit costs

An **explicit cost** requires an outlay of money (wages paid, materials bought). An **implicit cost** is the value of a resource the decision maker already owns and uses in the activity, measured by what that resource could earn in its best alternative use.

Worked Example 1.4. The economic cost of running a small business

Olena leaves a salaried job paying 48,000 per year to run her own bakery. In its first year, the bakery has revenue of 150,000, pays 70,000 for ingredients and rent, and pays 30,000 to an employee. Olena has put 100,000 of her own savings into ovens and fittings; had she left the money in a deposit account, it would have earned 4% interest per year. Ignore depreciation and taxes.

Explicit costs are the money outlays: $70,000 + 30,000 = 100,000$.

Accounting profit (in the simple sense used here) is revenue minus explicit costs: $150,000 - 100,000 = 50,000$.

Implicit costs are the forgone salary, 48,000, and the forgone interest on her savings, $0.04 \times 100,000 = 4,000$. Total implicit costs: 52,000.

Economic profit is revenue minus *all* opportunity costs:

$$150,000 - 100,000 - 52,000 = -2,000.$$

The bakery reports a healthy accounting profit, yet Olena is 2,000 worse off than if she had kept her job and her deposit. Whether she should continue depends on expected future profits and on non-monetary considerations (she may value being her own employer); but the economic calculation shows that the 50,000 accounting figure does not measure her gain from running the business.

Bridge to Part IV. The distinction between accounting profit and economic profit reappears in Chapter 21 (where firms maximize economic profit), in Chapter 27 (where you will see exactly what accounting profit measures and why it omits implicit costs), and in Chapter 31 (where measures such as economic value added attempt to charge a cost for equity capital, partially closing the gap).

1.2.3 Sunk costs

Definition 1.5. Sunk cost

A **sunk cost** is a cost that has already been incurred and cannot be recovered whatever decision is now made.

Because a sunk cost is the same under every option, it cannot affect which option is best. Only costs and benefits that *differ* between the available options—sometimes called **relevant costs**—should enter a forward-looking decision.

Worked Example 1.6. The half-built factory

A firm has spent 40 million building a factory. Finishing it will cost a further 25 million. Once completed, the factory will generate future cash flows with a present value (a concept defined in Chapter 9; for now, treat it as the value today of the future cash) of 30 million. If the firm abandons the project, it can sell the site and partial structure for 3 million.

Options now:

- *Finish*: pay 25, receive 30. Net: +5 million.
- *Abandon*: receive 3. Net: +3 million.

Finishing is better by 2 million. The 40 million already spent appears in neither calculation, because it has been spent whichever option is chosen.

Now suppose the completion cost is 28 million instead. Finishing yields $30 - 28 = 2$; abandoning yields 3. The firm should abandon—even though “we have already spent 40 million” may feel like a reason to continue. The instinct to continue a failing project because of past spending is called the **sunk cost fallacy**.

► Common error: “We can’t stop now, we’ve invested too much”

Past spending matters only through its effect on the options now available. A half-built factory is worth more than an empty field if it lowers the cost of completion; that effect is captured in the completion cost. The amount already spent adds nothing further to the decision. Accounting records sunk costs (as assets, and later as depreciation or impairment—Chapter 29), but the *decision* about whether to continue should look only forward.

Check your reasoning

1. You paid 80 for a non-refundable concert ticket. On the evening, a friend offers you free tickets to a match you would enjoy more. What is the opportunity cost of going to the concert?
2. A university charges no tuition. Is attending free?

Answers. (1) The value to you of the match you would forgo; the 80 is sunk and is irrelevant. Since you prefer the match, you should go to the match. (2) No. The largest cost of a degree for most students is the income forgone by not working full-time, plus living costs that exceed what one would otherwise incur.

1.3 The Production Possibilities Frontier

1.3.1 Construction

Consider an economy that can produce only two goods, which we call food and machines, using a fixed quantity of resources and a given technology. The **production possibilities frontier** (PPF) shows the maximum quantity of one good that can be produced for each quantity of the other.

Definition 1.7. Production possibilities frontier

Given fixed resources and technology, the **production possibilities frontier** is the set of output combinations (x_1, x_2) such that it is impossible to produce more of one good without producing less of the other.

Suppose the economy has 100 workers, and output of each good depends on how many workers are assigned to it:

$$F = 10\sqrt{L_F}, \quad M = 5\sqrt{L_M}, \quad L_F + L_M = 100,$$

where F is food, M is machines, and L_F, L_M are workers in each sector. The square root captures *diminishing returns*: each additional worker adds less output than the previous one, because the other resources in the sector (land, tools) are fixed. Table 1.1 lists several points.

Table 1.1: Points on the production possibilities frontier

L_F	L_M	Food $F = 10\sqrt{L_F}$	Machines $M = 5\sqrt{L_M}$	Label
100	0	100.0	0.0	A
84	16	91.7	20.0	B
64	36	80.0	30.0	C
36	64	60.0	40.0	D
0	100	0.0	50.0	E

Eliminating L_F and L_M gives the frontier directly. From the production functions, $L_F = (F/10)^2$ and $L_M = (M/5)^2$; substituting into $L_F + L_M = 100$:

$$\frac{F^2}{100} + \frac{M^2}{25} = 100.$$

This is a quarter-ellipse in the positive quadrant, bowed outward from the origin (Figure 1.1).

1.3.2 What the PPF shows

Efficiency. A point on the frontier is **productively efficient**: there is no way to produce more of one good without giving up some of the other. A point inside the frontier, such as U in Figure 1.1, means some resources are idle or misallocated—unemployment during a recession is the most important real-world example, and Chapter 46 studies it.

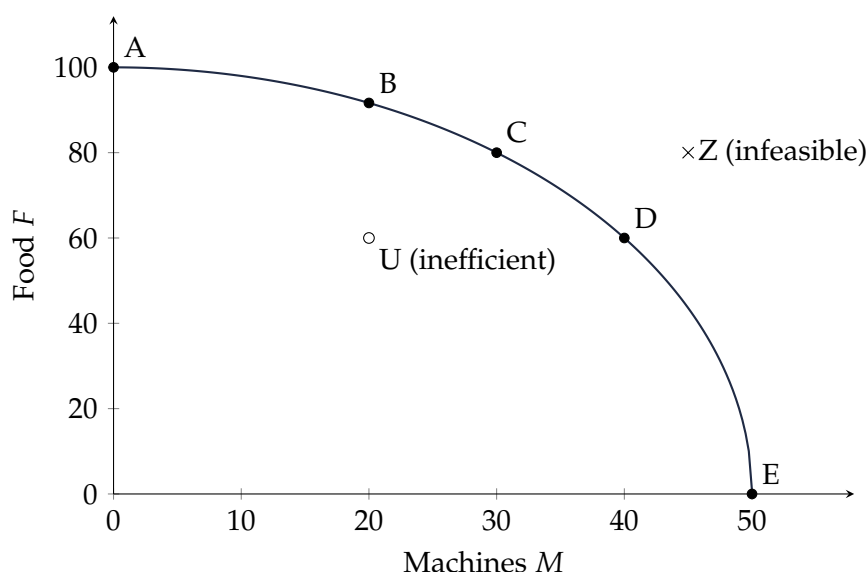


Figure 1.1: A production possibilities frontier with diminishing returns in both sectors. Points on the curve are efficient; points inside, such as U, waste resources; points outside, such as Z, are unattainable with current resources and technology.

Feasibility. A point outside the frontier, such as Z, cannot be produced with current resources and technology. Scarcity is precisely the fact that the frontier exists: wants extend beyond it.

Trade-offs and opportunity cost. Moving along the frontier, more of one good requires less of the other. The slope measures the opportunity cost: the amount of food given up per additional machine. Between C and D, machines rise by 10 and food falls by 20, so each machine costs, on average over that range, 2 units of food. Between A and B, machines rise by 20 and food falls by only 8.3, an average cost of about 0.42 food per machine.

Increasing opportunity cost. The opportunity cost of machines rises as more machines are produced. That is why the frontier bows outward (is *concave* to the origin). The reason here is diminishing returns in each sector: as workers move from food to machines, each additional worker adds less to machine output and subtracts more from food output. More generally, frontiers bow outward whenever resources are *specialized*—when some resources are better suited to one good than the other. The first resources moved into machine production are those least useful in farming and most useful in manufacturing; later, the economy must transfer excellent farmland and skilled farmers whose loss hurts food output a great deal.

We can compute the opportunity cost at a point exactly once derivatives are available (Chapter 5). Differentiating $F^2/100 + M^2/25 = 100$ implicitly with respect to M :

$$\frac{2F}{100} \frac{dF}{dM} + \frac{2M}{25} = 0 \implies \frac{dF}{dM} = -\frac{4M}{F}.$$

At C, $M = 30$, $F = 80$: the slope is $-120/80 = -1.5$. The **marginal rate of transformation**, the absolute value of this slope, is 1.5 food per machine at C, and rises from zero at A (where $M = 0$) toward infinity at E (where $F = 0$).

Growth. The frontier shifts outward when resources increase (more workers, more capital) or technology improves. A technological improvement specific to one good shifts the frontier out mainly along that good's axis. The choice of a point on today's frontier affects tomorrow's

frontier: an economy that produces more machines (capital goods) and less food (consumption) today has more capital tomorrow and a larger frontier. This intertemporal trade-off between consumption now and capacity later is the core of growth theory in Chapter 44.

► **Common error: “A straight-line PPF means no trade-off”**

A linear frontier has a *constant* opportunity cost, not a zero one. The trade-off still exists; it simply does not change as the mix changes. Linear frontiers arise when resources are equally productive in both uses relative to one another, as in the comparative-advantage model below.

1.4 Specialization, Comparative Advantage and Trade

1.4.1 Absolute and comparative advantage

Definition 1.8. Absolute and comparative advantage

A producer has an **absolute advantage** in a good if it can produce the good using fewer resources than another producer. A producer has a **comparative advantage** in a good if its *opportunity cost* of producing that good is lower than another producer's.

The insight, due to David Ricardo (1817), is that gains from trade depend on comparative, not absolute, advantage. Even a producer that is worse at producing everything has a comparative advantage in something.

Worked Example 1.9. Ricardian gains from trade

Two countries, Home and Foreign, each have 100 hours of labour. Labour is the only input, and the hours needed per unit are constant:

	Hours per unit of cloth	Hours per unit of wine
Home	1	2
Foreign	4	4

Home has an absolute advantage in both goods: it needs fewer hours for each.

Opportunity costs. At Home, one unit of wine uses 2 hours that could make 2 cloth, so wine costs 2 cloth (and cloth costs 1/2 wine). In Foreign, one wine uses 4 hours that could make 1 cloth, so wine costs 1 cloth (and cloth costs 1 wine). Foreign has the lower opportunity cost of wine (1 cloth < 2 cloth); Home has the lower opportunity cost of cloth (1/2 wine < 1 wine). So Foreign has a comparative advantage in wine and Home in cloth.

Production possibilities. Home: $C_H + 2W_H = 100$, so maximum cloth 100 or maximum wine 50. Foreign: $4C_F + 4W_F = 100$, maximum cloth 25 or maximum wine 25.

Autarky (no trade). Suppose each country spends half its hours on each good. Home produces and consumes 50 cloth and 25 wine; Foreign produces and consumes 12.5 cloth and 12.5 wine.

Specialization and trade. Home produces only cloth (100 units); Foreign produces only wine (25 units). They trade at a price of 1.5 cloth per wine, which lies between the two countries' autarky opportunity costs (1 and 2). Home exports 30 cloth and imports 20 wine.

Consumption after trade:

- Home: $100 - 30 = 70$ cloth and 20 wine.
- Foreign: 30 cloth and $25 - 20 = 5$ wine.

Are both better off? Compare each country's post-trade consumption with what it could have produced alone. If Home wanted 20 wine without trade, it would need 40 hours, leaving 60 hours for 60 cloth. With trade it has 70 cloth and 20 wine: 10 more cloth. If Foreign wanted 5 wine alone, it would use 20 hours, leaving 80 hours for 20 cloth. With trade it has 30 cloth and 5 wine: 10 more cloth. Both countries consume a bundle that lies *outside* their own production possibilities frontier.

Proposition 1.10. Gains from trade under comparative advantage

If two producers have different opportunity costs of producing two goods, and they trade at any relative price strictly between those opportunity costs, each can consume a bundle outside its own production possibilities frontier by specializing in the good in which it has a comparative advantage.

Argument. Let Home's opportunity cost of wine be c_H cloth and Foreign's be c_F , with $c_F < c_H$. Let the trade price be p cloth per wine, with $c_F < p < c_H$. Home specializes in cloth. For each unit of wine Home obtains, it gives up p cloth through trade rather than c_H cloth through its own production; because $p < c_H$, trade is the cheaper way to get wine. Foreign specializes in wine. For each unit of wine it exports, it receives p cloth through trade, whereas giving up one unit of wine in its own production would yield only c_F cloth; because $p > c_F$, trade is the cheaper way for Foreign to obtain cloth. So each country's consumption possibilities through trade lie strictly outside its own frontier. $\square$

What comparative advantage does and does not say. It says that specialization according to opportunity cost *can* make both parties better off in aggregate. It does not say that everyone within each country gains: in the example, if Home's wine producers must move into cloth, they may lose specific skills or income. How the gains are distributed within countries, and whether losers are compensated, is a distributional question (Section 1.9). It also does not say which goods a country will in fact export; that also depends on demand, transport costs, policy and the other factors of production, which the simple model omits.

Check your reasoning

A lawyer types twice as fast as her assistant. Should she do her own typing?

Answer. Usually not. Her opportunity cost of an hour of typing is an hour of legal work, which may be worth many times what the assistant earns. The assistant, though slower, has the comparative advantage in typing.

1.5 Marginal Reasoning

1.5.1 Thinking at the margin

Most economic choices are not all-or-nothing. A firm chooses *how many* workers to hire, a household *how much* to save, a central bank *how far* to raise interest rates. Such choices are made by comparing the extra benefit of a little more with the extra cost of a little more.

Definition 1.11. Marginal benefit and marginal cost

The **marginal benefit** (MB) of an activity is the additional benefit from one more unit of it. The **marginal cost** (MC) is the additional cost of one more unit.

Proposition 1.12. The marginal decision rule

If total net benefit is to be as large as possible, the activity should be expanded as long as marginal benefit exceeds marginal cost, and contracted when marginal cost exceeds marginal benefit. For activities that can be varied continuously, an interior optimum requires $MB = MC$.

The logic is simple. If $MB > MC$ for the next unit, doing it adds more to benefit than to cost, raising net benefit; so you are not yet at the optimum. If $MB < MC$ for the last unit, dropping it saves more cost than it loses in benefit; so you have gone too far.

Worked Example 1.13. How long should a café stay open?

A café is deciding how many hours to open in the evening beyond its normal closing time. Each extra hour costs 60 in wages, heating and electricity. The expected extra revenue (net of the cost of the food and drink sold) falls hour by hour as fewer customers arrive:

Extra hour	MB	MC	Net of this hour	Total net benefit
1st	150	60	+90	90
2nd	110	60	+50	140
3rd	75	60	+15	155
4th	50	60	−10	145
5th	30	60	−30	115

Total net benefit is maximized at three extra hours (155). The third hour is worth doing because its marginal benefit (75) exceeds its marginal cost (60); the fourth is not, because $50 < 60$. Notice that the average net benefit per hour is highest after one hour (90 per hour), but maximizing the *average* is not the goal; maximizing the *total* is. Stopping at one hour would give up 65 of attainable net benefit.

► Common error: maximizing averages instead of totals

A firm that hires workers until output *per worker* is highest, or a student who studies until marks *per hour* are highest, will usually stop too early. The objective is almost always a total (profit, utility, welfare), and the relevant comparison is marginal.

1.5.2 Marginal, average and total

For any quantity measured in total (cost, revenue, output), the **average** is the total divided by the number of units, and the **marginal** is the change in the total from one more unit. Two relationships between them hold generally and will be used many times:

- If the marginal is *above* the average, the average is *rising*; if the marginal is *below* the average, the average is *falling*.
- Therefore, when the average reaches a minimum or maximum, the marginal equals the average.

A student's grade-point average illustrates the first rule: if the next course grade (the marginal) is above the current average, the average rises. We prove the second rule formally for cost curves in Chapter 21.

1.6 Incentives and Institutions

1.6.1 Incentives

An **incentive** is anything that changes the costs or benefits of an action and therefore tends to change behaviour. Prices are the most obvious incentives, but taxes, subsidies, fines, reputations, social norms and contract terms are all incentives too. A central working assumption of economics is that people respond predictably to incentives: when the cost of an action rises, other things equal, people tend to do less of it.

Incentives sometimes produce outcomes their designers did not intend. A frequently cited example concerns cobra bounties in colonial Delhi, which reportedly led people to breed cobras for the reward; whether or not every detail of that story is accurate, the mechanism—a reward for a measured output encourages manufacturing the measured output rather than achieving the underlying goal—is real and pervasive. The accounting frauds studied in Chapter 29 are an example: compensation tied to reported earnings created incentives to manipulate reported earnings.

1.6.2 Institutions

Definition 1.14. Institutions

Institutions are the formal and informal rules that structure economic, political and social interaction: laws, property rights, contract enforcement, regulations, and customs and norms. In Douglass North's phrase, they are "the rules of the game."

Institutions matter because they determine the incentives people face. Several institutions are foundational for everything in this book:

Property rights. A **property right** is the right to use a resource, to exclude others from it, to earn income from it, and to transfer it. Without secure property rights, investment is discouraged: why build a factory if it can be seized? Why improve land you may lose? When nobody owns a resource, it tends to be overused, a problem known as the *tragedy of the commons* and analysed in Chapter 26.

Contract enforcement. Most economic exchange is not simultaneous: goods are delivered now and paid for later, loans are made now and repaid later. Such exchange requires confidence that promises will be kept. Courts, reputation and collateral provide it. Chapter 25 shows how the difficulty of enforcing and monitoring contracts shapes lending, insurance and employment.

Money. A generally accepted medium of exchange reduces the cost of trade by eliminating the need for a *double coincidence of wants*—finding someone who has what you want and wants what you have. Part VII studies what money is, who creates it and how its value is managed.

Accounting and disclosure rules. Investors and lenders can finance firms they do not manage only if they can trust the numbers those firms report. Accounting standards and auditing are

institutions that make such trust possible (Part IV).

Central banks and the monetary framework. Rules governing the central bank—its mandate, its independence, its operating procedures—shape expectations about inflation and interest rates (Chapters 49–50).

Bridge. When a later chapter says “the model assumes well-defined property rights” or “perfect contract enforcement,” it is assuming an institutional background. Many of the most interesting problems in economics arise when that background is weak or missing.

1.7 Equilibrium

1.7.1 The idea

Economists analyse the interaction of many decision makers through the concept of equilibrium.

Definition 1.15. Equilibrium

An **equilibrium** is a situation in which no participant, given the actions of all the others and the constraints they face, has an incentive to change their behaviour. Equivalently, it is a state that, once reached, persists until something outside the system changes.

The most familiar example is a competitive market: at the equilibrium price, the quantity buyers want to buy equals the quantity sellers want to sell. If the price were higher, sellers would want to sell more than buyers would buy; unsold goods would push the price down. If the price were lower, buyers would want more than sellers offer; shortages would push the price up. Chapter 22 develops this model in detail. Chapter 23 introduces a more general concept, Nash equilibrium, in which each participant’s strategy is the best response to the others’ strategies.

1.7.2 Equilibrium is not the same as optimality

An equilibrium is a situation in which nobody wants to change *individually*. That does not mean the outcome is good for the group. In a crowded stadium, if everybody stands up to see better, standing is an equilibrium (sitting down when others stand makes you worse off), yet everyone would prefer that all sit. Pollution, overfishing, bank runs and arms races all have this structure. Whether an equilibrium is efficient is a separate question that must be answered by analysis, and much of Part III is devoted to it.

1.7.3 Comparative statics

Much of economic analysis consists of **comparative statics**: comparing an equilibrium before and after a change in some external condition. How does the equilibrium price of oil change when a producer cuts output? How does equilibrium employment change when the minimum wage rises? The method is always the same: (1) describe the initial equilibrium; (2) identify which relationship the change directly shifts; (3) find the new equilibrium; (4) compare. Comparative statics ignores the path between equilibria. Where the path matters—as it often does in macroeconomics—we use dynamic analysis (Chapters 44–47).

1.8 Models, Assumptions and Evidence

1.8.1 What a model is

Definition 1.16. Economic model

An **economic model** is a simplified representation of economic behaviour, usually expressed in words, diagrams or equations, that isolates particular mechanisms in order to explain or predict outcomes.

A model is like a map. A map at a scale of 1:1 would be useless; a map is valuable because it leaves things out. A subway map distorts distances but is excellent for choosing a route; it would be useless for planning a walk. Likewise, every economic model is “wrong” in the sense of omitting features of reality. The relevant question is whether its omissions matter for the question being asked.

Endogenous and exogenous variables. In a model, **endogenous variables** are determined within the model (for example, price and quantity in a market model). **Exogenous variables** are taken as given from outside (for example, the weather’s effect on a harvest, or the tax rate set by government). Comparative statics traces how endogenous variables respond to changes in exogenous ones.

Ceteris paribus. Economic statements such as “a higher price reduces quantity demanded” are made *ceteris paribus*—other things equal. In reality, many things change at once. Separating the effect of one change from others is the central difficulty of empirical economics, taken up in Chapter 17.

Assumptions. Every model rests on assumptions. Some are simplifications made for convenience (two goods rather than millions); others are substantive (people are rational; prices adjust instantly). A good practice when reading any model is to ask: which assumptions drive the result? If a conclusion survives the relaxation of an assumption, the assumption was a convenience; if it disappears, the assumption was essential.

1.8.2 Rationality

Most models in this book assume that decision makers are **rational** in a specific, limited sense: they have consistent preferences over outcomes and choose the best available option given those preferences and the information they have. This does not assume people are selfish (preferences may include concern for others), all-knowing (information may be limited) or calculating (people may behave *as if* optimizing through habit or experience). Behavioural economics documents systematic departures from this benchmark—present bias, loss aversion, limited attention—and several chapters discuss where those departures matter. The benchmark remains useful because it generates precise predictions and because departures from it are defined relative to it.

1.9 Positive and Normative Analysis

Definition 1.17. Positive and normative statements

A **positive statement** is a claim about how the world is or would be, which can in principle be tested against evidence. A **normative statement** is a claim about how the world ought to be, which depends on values.

“A 10% increase in the price of cigarettes reduces cigarette consumption by about 4%” is positive: it may be true or false, and data can test it. “The government should raise cigarette taxes” is normative: it depends on how one weighs health, individual liberty, the burden on poorer smokers, and government revenue. Positive analysis can inform normative debate—by telling us what a tax would actually do—but it cannot settle it alone.

1.9.1 Efficiency and equity

Two criteria recur in normative economic analysis.

Definition 1.18. Pareto efficiency

An allocation is **Pareto efficient** if there is no feasible alternative allocation that makes at least one person better off without making anyone worse off. A change that makes at least one person better off and nobody worse off is a **Pareto improvement**.

Pareto efficiency is a weak criterion in one sense: a very unequal allocation—one person owns everything—can be Pareto efficient, since any change would make that person worse off. It is also a demanding criterion in another sense: almost every real policy creates some losers, so strict Pareto improvements are rare. Economists therefore often use the **Kaldor–Hicks criterion**: a change is an improvement if the winners gain enough that they *could* compensate the losers and still be better off, whether or not compensation is actually paid. The gains-from-trade argument above passes this test. Whether compensation *should* be paid is a question of equity.

Equity concerns the fairness of the distribution of outcomes or opportunities. There is no single accepted measure; different ethical frameworks (utilitarian, egalitarian, libertarian, Rawlsian) give different answers. Economics can clarify the trade-offs—for example, how much efficiency a given redistribution costs—but does not settle the value judgement.

► Common error: “efficient” means “good”

An efficient allocation has no waste. It may still be one that many people find unjust. Conversely, an inefficient policy may be justified on equity grounds. Keep the two criteria separate and state which you are using.

1.10 Accounting Identities, Behavioural Relationships, Stocks and Flows

This section introduces two distinctions that the rest of the book relies on constantly. Confusing them is one of the most common sources of error in economic argument.

1.10.1 Identities versus behavioural relationships

Definition 1.19. Accounting identity

An **accounting identity** is an equation that holds by definition, because of the way its terms are defined and measured. It is true for all values of its variables and cannot be falsified by any data. We write identities with the symbol $\equiv$.

Definition 1.20. Behavioural relationship

A **behavioural relationship** describes how decision makers respond to their circumstances—for example, how consumption depends on income, or how investment depends on interest rates. It is a claim about behaviour and can be false.

Definition 1.21. Equilibrium condition

An **equilibrium condition** states that some balance must hold in equilibrium—for example, that quantity supplied equals quantity demanded. Unlike an identity, it need not hold out of equilibrium.

Three identities appear repeatedly:

1. **The accounting equation** for any entity: $\text{Assets} \equiv \text{Liabilities} + \text{Equity}$. Equity is *defined* as assets minus liabilities, so the equation cannot fail.
2. **The national income identity**: $Y \equiv C + I + G + (X - M)$. Measured output is *defined* as the sum of spending on final goods, with unsold output counted as investment in inventories, so the identity holds by construction (Chapter 43).
3. **The quantity equation**: $MV \equiv PY$, where M is the money stock, P the price level, Y real output, and V the velocity of money. Because V is *defined* as PY/M , the equation is an identity (Chapter 48).

An identity says that some quantities must move together, but it says nothing on its own about *why* they move or which causes which. Consider the quantity equation. Written as an identity, $MV \equiv PY$ is true whatever happens. It becomes the *quantity theory of money*—a behavioural claim—only when one adds assumptions such as “ V is stable” and “ Y is determined by real factors independent of M .” Under those additional assumptions, an increase in M raises P proportionally. Whether those assumptions hold is an empirical question, and Chapter 48 shows that they often fail in the short run.

Worked Example 1.22. Reading an identity correctly

In a closed economy without government, the identity $Y \equiv C + I$ together with the definition of saving $S \equiv Y - C$ implies $S \equiv I$: saving equals investment, always. Does it follow that “if households decide to save more, investment must rise”? Not from the identity alone. If households try to save more by consuming less, firms may sell less, accumulate unsold inventories (which count as investment), and then cut production and income. Measured saving and investment will still be equal at every moment, but the *outcome* might be lower income and lower desired investment. Which happens depends on behaviour—how firms respond to lower sales, how interest rates respond, what the central bank does. The identity constrains the outcome; it does not determine it. Chapter 46 develops this as the “paradox of thrift.”

► **Common error: treating an identity as a theory**

“Government deficits must crowd out private investment because $S = I + (G - T)$ ” is not a valid argument. The identity (derived in Chapter 43) shows that *some* combination of higher private saving, lower investment, or a larger external deficit must accompany a larger government deficit. Which combination occurs depends on behaviour and on monetary policy.

1.10.2 Stocks and flows

Definition 1.23. Stock and flow

A **stock** is a quantity measured at a point in time (a date). A **flow** is a quantity measured over a period of time (per unit of time).

The water in a bathtub at 8:00 p.m. is a stock; the rate at which water pours from the tap, in litres per minute, is a flow. The change in the stock over a period equals inflows minus outflows over that period:

$$\text{Stock}_t \equiv \text{Stock}_{t-1} + \text{Inflows}_t - \text{Outflows}_t.$$

This **stock-flow identity** is everywhere in economics and accounting:

Stock (at a date)	Related flow (over a period)	Where
Capital stock K_t	Investment I_t minus depreciation δK_{t-1}	Ch. 17
Government debt B_t	Deficit (spending minus revenue, plus interest)	Ch. 20
A firm's cash balance	Net cash flow in the cash-flow statement	Ch. 15
Retained earnings	Net income minus dividends	Ch. 11
Net foreign assets	Current-account balance (plus valuation changes)	Ch. 16
Bank deposits	New lending, repayments, payments in and out	Ch. 21
The number of unemployed	Flows into and out of unemployment	Ch. 18

Every flow must carry a time unit. A wage of “3,000” is meaningless until we know whether it is per week, per month or per year; a GDP of “25 trillion dollars” is annual unless stated otherwise. A stock carries a date: “government debt was 30 trillion *on 31 December*.”

► **Common error: comparing stocks with flows**

Statements such as “the country's debt is 100% of GDP” compare a stock (debt at a date) with a flow (output over a year). The ratio is meaningful—it tells us how many years of total output the debt represents—but it is not a fraction of anything in the ordinary sense, and it would change if GDP were measured per quarter rather than per year. Always check that the time units are consistent.

1.10.3 Balance sheets and income statements: a first look

The two distinctions come together in accounting. A **balance sheet** records stocks at a date: what an entity owns (assets) and owes (liabilities), with equity as the difference. An **income statement** records flows over a period: revenue and expenses. The two are linked: profit (a flow) for a period, less dividends paid, adds to retained earnings (part of the equity stock). Chapter 10 introduces these statements; Chapter 27 builds them from individual transactions.

1.11 One Event, Four Lenses

To see how the rest of the book fits together, consider an increase in the central bank's policy interest rate from 3% to 4%. Here is a preview of how each part of the book would analyse it; none of the terms used is expected to be clear yet, and each is defined in the chapter indicated.

Microeconomics. Higher interest rates raise the price of spending today relative to spending in the future. Households with debts face higher payments; savers earn more. Firms find that projects must earn more to be worth undertaking. How much prices of goods respond depends on market structure: a firm with market power may absorb higher financing costs differently from a competitive firm (Chapters 20–24, 45).

Accounting. A firm with floating-rate debt reports higher interest expense and lower net income. The market value of fixed-rate bonds held by banks falls; depending on how those bonds are classified, the loss may or may not appear in reported profit or equity. Banks' net interest margins change (Chapters 28–30, 48).

Macroeconomics. Lower consumption and investment reduce aggregate demand. Output and employment tend to fall relative to what they would otherwise have been; inflation tends to fall with a lag. Government interest costs rise, affecting debt dynamics (Chapters 43–47).

Monetary economics. The central bank must *implement* the higher rate: by raising the interest it pays on banks' reserve balances and the rates on its lending and borrowing facilities, so that short-term market rates move up. Longer-term rates respond according to expectations of future policy. The exchange rate may appreciate. The ultimate effects are uncertain and arrive with lags (Chapters 48–52).

Each lens alone is incomplete. The micro lens does not tell you how large the aggregate effect will be; the macro lens does not tell you which firms will fail; the accounting lens does not tell you why the central bank acted; the monetary lens does not tell you how households will respond. The rest of the book builds each lens and then, in Chapter 52, uses them together.

Case: Price controls and queues: U.S. gasoline in the 1970s

In 1973–74 and again in 1979, disruptions to oil supply sharply reduced the quantity of crude oil available at previously prevailing prices. In the United States, federal price controls prevented gasoline prices from rising to the level at which quantity demanded would have fallen to match available supply. Long queues at filling stations resulted, along with rationing schemes such as allowing purchases only on odd or even days depending on the last digit of the licence plate.

The tools of this chapter illuminate the episode. *Scarcity* increased: at the controlled price, people wanted more gasoline than was available. With the price not allowed to rise, the good had to be allocated by another mechanism—waiting in line (a form of *rationing by time*) and administrative rules (a form of *command*). The *opportunity cost* of gasoline to a buyer was not only the controlled price paid at the pump but also the value of the time spent queuing, which was very different for a retiree and a surgeon. Because time spent queuing is a cost to the buyer that is not a receipt to anyone else, it is a real resource cost to society, unlike a higher price, which is a transfer from buyer to seller. *Equity* arguments were made for the controls (they protected poorer drivers from higher prices); *efficiency* arguments were made against them (they wasted time and misdirected

gasoline away from its most valuable uses). Chapter 22 analyses price ceilings formally and measures these losses.

1.12 Limitations of the Framework

Three limitations of this chapter's framework deserve mention before we move on.

First, opportunity cost is conceptually clear but often hard to measure. The value of a forgone alternative may depend on information not available at the time of the decision.

Second, the rational-choice benchmark describes behaviour reasonably well in many repeated, high-stakes market settings, but less well in others—decisions involving very long horizons, very small probabilities, or strong emotions. Where this matters, later chapters say so.

Third, equilibrium analysis tells us where a system tends to settle, not how long it takes to get there, whether it gets there at all, or what happens along the way. Macroeconomic fluctuations and financial crises are, in large part, about what happens along the way.

Summary

- Scarcity—wants exceeding available resources at a zero price—forces every society to decide what to produce, how, and for whom.
- The opportunity cost of a choice is the value of the best alternative forgone. Economic costs include implicit costs; sunk costs are irrelevant to forward-looking decisions.
- The production possibilities frontier shows efficient output combinations; its slope is the opportunity cost, which typically rises as specialized resources are reallocated.
- Comparative advantage, not absolute advantage, determines the gains from specialization; both parties can gain from trade at a price between their opportunity costs.
- Optimal choices expand an activity until marginal benefit equals marginal cost. Averages are not the right guide.
- Institutions—property rights, contract enforcement, money, disclosure rules, central banks—determine incentives.
- An equilibrium is a state where nobody wishes to change behaviour unilaterally; it need not be efficient.
- Positive statements can be tested; normative statements depend on values. Pareto efficiency and equity are distinct criteria.
- Identities hold by definition and cannot be falsified; behavioural relationships can. Stocks are measured at a date and flows over a period, linked by the stock-flow identity.

Key terms

scarcity; factors of production; opportunity cost; explicit cost; implicit cost; economic profit; sunk cost; sunk cost fallacy; production possibilities frontier; productive efficiency; marginal rate of transformation; absolute advantage; comparative advantage; marginal benefit; marginal cost; marginal decision rule; incentive; institutions; property rights; equilibrium; comparative statics; model; endogenous variable;

exogenous variable; *ceteris paribus*; rationality; positive statement; normative statement; Pareto efficiency; Kaldor–Hicks criterion; equity; accounting identity; behavioural relationship; equilibrium condition; stock; flow; stock-flow identity.

Exercises

Exercise 1.1 (Opportunity cost). A student can work 20 hours per week in a shop at 15 per hour, or spend those hours on an unpaid internship that she expects will raise her starting salary after graduation. Tuition is 9,000 per year and is paid whichever she chooses. What is the weekly opportunity cost of the internship? Is tuition relevant to this choice?

Exercise 1.2 (Economic profit). A software developer quits a job paying 90,000 per year to start a firm. In year one the firm earns revenue of 260,000 and pays 150,000 in salaries, rent and cloud services. The founder has invested 200,000 of her own savings, which could have earned 5% per year elsewhere. Compute accounting profit (revenue minus explicit costs) and economic profit.

Exercise 1.3 (Sunk costs). A pharmaceutical firm has spent 300 million developing a drug. Completing trials will cost 120 million more. If the drug is approved (probability 0.6), the value of its future cash flows at the time of approval will be 250 million; if not, 0. Alternatively, the firm can sell the project now to another company for 20 million. Treat all amounts as values today and assume the firm is risk-neutral (it maximizes expected value; probability and expectation are formalized in Chapters 12 and 13). Should it complete the trials or sell? What role does the 300 million play?

Exercise 1.4 (PPF). An economy's frontier is given by $F^2/100 + M^2/25 = 100$ as in the text.

- (a) What is the maximum food output if 40 machines are produced?
- (b) Compute the marginal rate of transformation at $M = 40$.
- (c) Suppose a technological improvement doubles machine output from any number of workers, so that $M = 10\sqrt{L_M}$. Derive the new frontier and state the new maximum machine output.

Exercise 1.5 (Comparative advantage). In one day, Anna can write 4 reports or code 2 programs; Ben can write 3 reports or code 3 programs. Each works one day, and output is proportional to time spent on each task.

- (a) Who has an absolute advantage in each task?
- (b) Compute each person's opportunity cost of a program.
- (c) Who should specialize in what? Within what range must the price of a program (in reports) lie for both to gain from trade?

Exercise 1.6 (Gains from trade). Using Example 1.9, suppose the trade price is 1.2 cloth per wine and Home still imports 20 wine. How much cloth does Home export? Compute each country's consumption and compare with the text's price of 1.5. Which country gains more at the lower price, and why?

Exercise 1.7 (Marginal reasoning). A farmer applies fertilizer. Each 10 kg bag costs 30. The extra revenue from each successive bag is 120, 90, 65, 45, 28, 15. How many bags should she apply? What is her maximum net gain from fertilizer? If the price of fertilizer rises to 50 per bag, how many bags should she apply?

Exercise 1.8 (Average and marginal). A student's grade average after four courses is 3.2. She takes a fifth course and receives 3.7. What is her new average? Explain in terms of marginal and average why it rose.

Exercise 1.9 (Positive and normative). Classify each statement as positive or normative, and for positive statements say what evidence could test it:

- (a) Raising the minimum wage reduces employment among teenagers.
- (b) The minimum wage should be raised because full-time workers deserve a living income.
- (c) Inflation in the euro area was higher in 2022 than in 2019.
- (d) Central banks should target 2% inflation.
- (e) A 1 percentage-point increase in the policy rate reduces inflation by about 0.3 percentage points after two years.

Exercise 1.10 (Identities). For each equation, state whether it is an accounting identity, a behavioural relationship, or an equilibrium condition, and explain.

- (a) $\text{Assets} = \text{Liabilities} + \text{Equity}$.
- (b) $C = 200 + 0.8(Y - T)$.
- (c) $\text{Quantity supplied} = \text{quantity demanded}$.
- (d) $MV = PY$.
- (e) $\text{End-of-year cash} = \text{start-of-year cash} + \text{cash inflows} - \text{cash outflows}$.

Exercise 1.11 (Stocks and flows). Classify each as a stock or a flow and give an appropriate unit: (a) a firm's inventory; (b) a country's exports; (c) household wealth; (d) a bank's deposits; (e) depreciation; (f) the unemployment rate; (g) interest paid on a loan.

Exercise 1.12 (Stock-flow accounting). A country's capital stock at the end of 2025 is 400 (billions of constant-price currency units). During 2026 gross investment is 60 and depreciation is 5% of the end-2025 stock. What is the end-2026 capital stock? What gross investment would be needed in 2026 just to keep the stock constant?

Exercise 1.13 (Equilibrium and efficiency). Two fishing firms share a lake. Each can send 1 or 2 boats. Explain, in words, why an equilibrium in which each sends 2 boats might exist even if both firms would earn more if each sent 1. What institutional arrangement might change the outcome? (Chapter 23 formalizes this reasoning.)

Exercise 1.14 (Integrating the lenses). A large rise in the world price of natural gas affects a country that imports all its gas. Give one consequence at each of the four levels (micro, accounting, macro, monetary). You do not yet need the formal models; use the concepts of this chapter.

Solutions

Solution 1.1. The weekly opportunity cost of the internship is the forgone wage, $20 \times 15 = 300$ per week, since working in the shop is the best alternative use of the hours. (If the internship also has non-monetary costs or benefits—it may be more or less pleasant—those should be valued too.) Tuition is paid under either option, so it does not differ between them and is irrelevant to this particular choice. It would be relevant to the separate choice of whether to be a student at all.

Solution 1.2. Accounting profit = $260,000 - 150,000 = 110,000$. Implicit costs: forgone salary 90,000 plus forgone interest $0.05 \times 200,000 = 10,000$, total 100,000. Economic profit = $110,000 - 100,000 = 10,000$. The founder is 10,000 better off than in her best alternative (ignoring non-monetary factors and risk).

Solution 1.3. Expected value of completing: $0.6 \times 250 - 120 = 150 - 120 = 30$ million. Selling yields 20 million. Completing is better by 10 million in expected value. The 300 million already spent is sunk: it is the same whichever option is chosen and plays no role. Note that the firm will, overall, not recover its 300 million even in expectation; but that is irrelevant now. Had the expected value of completing been below 20, the firm should sell despite the 300 million spent.

Solution 1.4. (a) With $M = 40$: $F^2/100 = 100 - 1600/25 = 100 - 64 = 36$, so $F^2 = 3600$, $F = 60$. (b) $dF/dM = -4M/F = -160/60 \approx -2.67$. The MRT is about 2.67 food per machine. (c) Now $L_M = (M/10)^2$ and $L_F = (F/10)^2$, so $F^2/100 + M^2/100 = 100$, i.e. $F^2 + M^2 = 10,000$: a quarter-circle of radius 100. Maximum machine output is 100 (all workers in machines: $10\sqrt{100} = 100$). Maximum food is unchanged at 100. The frontier pivots outward along the machine axis.

Solution 1.5. (a) Anna writes more reports ($4 > 3$), so she has the absolute advantage in reports. Ben codes more programs ($3 > 2$), so he has the absolute advantage in programs. (b) Anna: a program uses half a day, which could produce 2 reports, so a program costs 2 reports. Ben: a program uses one-third of a day, which could produce 1 report, so a program costs 1 report. (c) Ben has the lower opportunity cost of programs and should specialize in programs; Anna, whose opportunity cost of a report is $1/2$ program (versus Ben's 1 program), should specialize in reports. Both gain if the price of a program lies strictly between 1 and 2 reports.

Solution 1.6. Home exports $1.2 \times 20 = 24$ cloth. Home consumes $100 - 24 = 76$ cloth and 20 wine; Foreign consumes 24 cloth and 5 wine. Compared with the no-trade benchmark in the text (60 cloth for Home with 20 wine; 20 cloth for Foreign with 5 wine), Home now gains 16 cloth and Foreign 4 cloth, whereas at 1.5 each gained 10. At the lower price, wine is cheaper, so the wine importer (Home) captures more of the gains. The *terms of trade*—the relative price at which trade occurs—determine the division of the gains; the total gain (20 extra cloth here) is the same in both cases because production is unchanged.

Solution 1.7. With $MC = 30$: bags 1–4 have MB of 120, 90, 65, 45, all above 30; the fifth has $MB\ 28 < 30$. Apply 4 bags. Net gain = $(120 + 90 + 65 + 45) - 4 \times 30 = 320 - 120 = 200$. With $MC = 50$: bags 1–3 have MB above 50; the fourth has $45 < 50$. Apply 3 bags; net gain = $275 - 150 = 125$.

Solution 1.8. Total grade points after four courses: $4 \times 3.2 = 12.8$. After five: $12.8 + 3.7 = 16.5$. New average: $16.5/5 = 3.3$. The marginal grade (3.7) exceeded the previous average (3.2), so the average rose.

Solution 1.9. (a) Positive: test with data comparing employment before and after minimum-wage changes, ideally using comparison regions (Chapter 17 discusses such designs). (b) Normative: rests on a value judgement about deserving. (c) Positive: check official inflation statistics (Eurostat's HICP). (d) Normative, although it can be informed by positive evidence about the effects of different targets. (e) Positive: estimate with time-series or quasi-experimental methods; the figure is a claim about magnitudes.

Solution 1.10. (a) Identity: equity is defined as assets minus liabilities. (b) Behavioural relationship: a consumption function, which may or may not describe how households behave. (c) Equilibrium condition: holds only at the market-clearing price. (d) Identity if V is defined as PY/M ; it becomes a theory only with additional assumptions. (e) Identity: the stock-flow identity applied to cash.

Solution 1.11. (a) Stock (units or currency value at a date). (b) Flow (currency per year or quarter). (c) Stock (currency at a date). (d) Stock (currency at a date). (e) Flow (currency per year). (f) Neither a pure stock nor a flow: it is a ratio of two stocks (unemployed persons to labour force) at a date, so it is a stock ratio measured at a point in time (usually as a monthly average). (g) Flow (currency per period).

Solution 1.12. Depreciation = $0.05 \times 400 = 20$. End-2026 stock = $400 + 60 - 20 = 440$. To keep the stock constant, gross investment must equal depreciation: 20.

Solution 1.13. If the other firm sends 2 boats, sending 1 boat leaves the lake almost equally depleted but gives a firm a smaller share of the catch, so each firm may prefer 2 whatever the other does. Both

then send 2 and overfish, earning less than if both had sent 1. This is an equilibrium (neither wants to deviate alone) that is not efficient. Institutional remedies include assigning property rights to the lake to one owner, enforceable quotas, a tax per boat, or an agreement enforced by repeated interaction and reputation.

Solution 1.14. Many answers are possible; one per level: *Micro*: households face a higher opportunity cost of heating and reduce consumption; firms using gas intensively face higher marginal costs and may reduce output or raise prices. *Accounting*: gas-using firms report higher cost of sales and lower profit; gas distributors with fixed-price contracts may report losses on those contracts. *Macro*: the import bill rises, worsening the trade balance; measured inflation rises; real incomes fall. *Monetary*: the central bank must decide whether to raise interest rates to prevent the price increase from raising inflation expectations, trading this off against weaker output.

Numbers, Ratios and Index Numbers

After studying this chapter you should be able to

1. Attach units, periods and populations to every quantity, and use unit analysis to check a calculation.
2. Use summation notation and its rules.
3. Distinguish percentage changes from percentage-point changes, combine successive changes by multiplying growth factors, and explain why percentage changes with different bases do not cancel.
4. Propagate rounding and measurement error through sums, products and ratios.
5. Choose correctly among arithmetic, weighted, geometric and harmonic means.
6. Decompose a change in a value into price and volume components, and a change in a total into contributions from its parts.
7. Construct, rebase and splice index numbers, and express a weighted index as a weighted mean of price relatives.

A motivating question

A company's annual report states that "revenue grew 5.6 per cent, driven by 10 per cent volume growth." A newspaper reports that a country's policy rate "rose by 0.5 per cent," when it went from 2.0 to 2.5 per cent. A fund advertises that its returns "averaged 5 per cent a year" over two years in which it gained 20 per cent and then lost 10 per cent. Each statement uses familiar arithmetic, and each can mislead. The company's price fell; the policy rate rose by half a percentage point but by a quarter in relative terms; and the fund's investors actually earned less than 4 per cent a year.

Before any economics, accounting or finance can be done, numbers must be read correctly: what is being measured, in what unit, over what period, relative to what base. This chapter develops that discipline. None of the arithmetic is difficult. What is difficult—and what distinguishes careful analysts from careless ones—is asking of every number: *per what, over what period, and compared with what?*

2.1 Quantities, Units and Notation

2.1.1 Every number has a unit

A **quantity** is a number together with a unit: 120 million euros, 40 hours, 3.5 per cent per year. Chapter 1 distinguished *stocks*, measured at a point in time (cash of 120 million on 31 December), from *flows*, measured per unit of time (revenue of 120 million per year). Mixing the two is one of the commonest errors in economics and finance: dividing a stock by a flow gives a number of *periods* (debt of 300 divided by annual earnings of 100 is "3 years of earnings"),

not a percentage.

Key idea. Carry units through every calculation as if they were algebraic symbols. If the units of the answer are not the units of the question, the calculation is wrong, however plausible the number looks.

Worked Example 2.1. Unit analysis

A factory produces 4,800 units a month using 30 workers who each work 160 hours a month. Labour productivity is

$$\frac{4,800 \text{ units/month}}{30 \text{ workers} \times 160 \text{ hours/(worker} \cdot \text{month)}} = \frac{4,800 \text{ units}}{4,800 \text{ hour}} = 1 \text{ unit per hour.}$$

The months and the workers cancel, leaving units per hour. If the wage is 20 euros per hour, labour cost per unit is 20 euros/hour $\div$ 1 unit/hour = 20 euros per unit. A figure of “revenue per employee” (euros per employee per year) is *not* productivity in this sense: it mixes output with prices and says nothing about profit.

2.1.2 Summation notation

Economics constantly adds up many similar terms: expenditures across goods, cash flows across years, observations in a sample. We write

$$\sum_{i=1}^n x_i = x_1 + x_2 + \cdots + x_n,$$

read “the sum of x_i from $i = 1$ to n .” The index i is a placeholder; any letter will do. Subscripts *label*: x_2 is the second observation, not x squared. Three rules follow directly from the properties of addition:

$$\sum_{i=1}^n a x_i = a \sum_{i=1}^n x_i, \quad \sum_{i=1}^n (x_i + y_i) = \sum_{i=1}^n x_i + \sum_{i=1}^n y_i, \quad \sum_{i=1}^n c = nc.$$

The **arithmetic mean** is $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$. A consequence used throughout the statistics chapters is that deviations from the mean sum to zero:

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - n\bar{x} = n\bar{x} - n\bar{x} = 0.$$

► Summation does not pass through products or ratios

In general $\sum x_i y_i \neq (\sum x_i)(\sum y_i)$ and $\sum (x_i / y_i) \neq (\sum x_i) / (\sum y_i)$. The average of firms’ profit margins is not the margin of their combined profits and revenues. Which one is relevant depends on the question: a typical firm’s margin, or the margin of the industry as a whole.

2.2 Ratios, Percentages and Changes

2.2.1 Ratios and rates

A **ratio** compares two quantities by division. Ratios need compatible scopes. A *profit margin* divides a flow (profit) by a flow (revenue) over the same period; a *debt-to-equity ratio* divides a

stock by a stock at the same date; a *return on assets* divides a flow by a stock and must say which stock—opening, closing or average (Chapter 31). A ratio whose numerator and denominator come from different dates, currencies or accounting conventions is not wrong because of the arithmetic but because it measures nothing well defined.

A **rate** is a ratio with a time dimension or a population in the denominator: 3.5 per cent per year, 12 deaths per 1,000 people, 5.2 per cent of the labour force. Always name the denominator. “Unemployment fell” could mean fewer unemployed people or a lower unemployment rate; the two can move in opposite directions if the labour force shrinks (Chapter 43).

2.2.2 Percentage change

The **percentage change** from an initial value x_0 to a final value x_1 is

$$\% \Delta x = \frac{x_1 - x_0}{x_0} \times 100.$$

The *base*—the denominator—is the initial value. It follows that a rise and an equal fall in percentage terms do not cancel. From 80 to 100 is a rise of 25 per cent; from 100 back to 80 is a fall of 20 per cent, because the base is now 100.

2.2.3 Growth factors

It is often clearer to work with the **growth factor** $1 + g$, where $g = (x_1 - x_0)/x_0$ is the growth rate as a decimal: $x_1 = (1 + g)x_0$. Successive changes *multiply*:

$$x_2 = (1 + g_2)(1 + g_1)x_0.$$

A rise of 25 per cent followed by a fall of 20 per cent gives $1.25 \times 0.80 = 1$: back to the start. A rise of 10 per cent followed by a rise of 5 per cent gives $1.10 \times 1.05 = 1.155$, a total rise of 15.5 per cent, not 15. The difference, $0.10 \times 0.05 = 0.005$, is the *cross term*; it is negligible for small changes and large for large ones. Adding growth rates is an approximation; multiplying growth factors is exact. Chapter 4 shows that logarithms convert the multiplication into addition exactly.

Worked Example 2.2. Price, volume and revenue

A firm sells 1,000 units at 50 euros, then 1,100 units at 48 euros. Revenue rises from 50,000 to 52,800, a rise of 5.6 per cent. Volume rose 10 per cent and price fell 4 per cent. The exact combination is $1.10 \times 0.96 = 1.056$. Adding $10 - 4 = 6$ per cent would overstate revenue growth. Note what the calculation does *not* show: whether profit rose, which depends on costs (Chapter 3).

2.2.4 Percentage points

When the quantity being changed is itself a percentage, two different comparisons are possible. If an interest rate rises from 2.0 to 2.5 per cent, it has risen by 0.5 **percentage points** (the difference of the two percentages) and by 25 per cent (the relative change, $0.5/2.0$). Central banks announce changes in percentage points, often in *basis points* (one basis point is 0.01 percentage point, so 0.5 points is 50 basis points). Saying the rate “rose 0.5 per cent” is ambiguous at best.

► “The share rose 3 per cent”

If women’s share of board seats rises from 20 to 23 per cent, the share rose by 3 percentage points and by 15 per cent. Reports that confuse the two can exaggerate or understate a change by a large factor. When the base is small, the gap is especially large: a default rate rising from 1 to 2 per cent is a rise of 1 percentage point but a doubling.

2.3 Precision, Rounding and Measurement Error

Economic data are measured with error, and reported figures are rounded. A GDP growth rate of 2.3 per cent means something between 2.25 and 2.35 at best, and first estimates are revised, often by more than that (Chapter 10).

2.3.1 How errors combine

Suppose a quantity x is measured with a small relative error e_x (for instance, $e_x = 0.01$ means up to 1 per cent) and y with relative error e_y . Then:

- For a *product* xy or a *ratio* x/y , relative errors approximately *add* in the worst case: $(1 + e_x)(1 + e_y) \approx 1 + e_x + e_y$.
- For a *sum* $x + y$, *absolute* errors add in the worst case.
- For a *difference* $x - y$ of two similar numbers, the absolute error is the sum of the absolute errors, but the relative error can be huge, because the difference itself is small.

Worked Example 2.3. Error in a small difference

Revenue is 1,000 and costs are 950, each known to within 1 per cent. Profit is 50. The absolute error in revenue is up to 10 and in costs up to 9.5, so profit lies between about 30.5 and 69.5—an uncertainty of almost 40 per cent in profit from 1 per cent errors in its components. The same arithmetic explains why a country’s saving rate, measured as the difference between two large aggregates (income and consumption), is revised so heavily, and why net exports and inventory changes contribute so much to revisions of GDP growth.

Rounding. Round only at the end of a calculation; rounding intermediate results compounds errors. Report as many digits as the data justify and no more: a valuation of “1,234,567 euros” built on a growth forecast known only to the nearest percentage point conveys false precision.

2.4 Averages

The word “average” covers several different calculations. Choosing the wrong one is a frequent and consequential error in finance.

2.4.1 Weighted means

A **weighted mean** of values $x_1, \dots, x_n$ with non-negative weights $w_1, \dots, w_n$ is

$$\bar{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}.$$

If the weights are normalized to sum to one, $\bar{x}_w = \sum_i w_i x_i$. A weighted mean with non-negative weights always lies between the smallest and largest of the x_i . The weights must be the ones the question implies. If two projects require 20 and 80 of capital and earn 10 and 5 per cent on it over the same period, the combined return is $(20 \times 0.10 + 80 \times 0.05)/100 = 6$ per cent, not the unweighted average of 7.5 per cent. The cost of capital (Chapter 35), the return on a portfolio (Chapter 14) and every price index are weighted means.

2.4.2 Geometric means

For growth over successive periods, the relevant average is the constant rate that would produce the same cumulative change: the **geometric mean** of the growth factors. If growth factors are $1 + g_1, \dots, 1 + g_n$,

$$1 + \bar{g}_G = [(1 + g_1)(1 + g_2) \cdots (1 + g_n)]^{1/n}.$$

Worked Example 2.4. The fund that averaged 5 per cent

A fund gains 20 per cent and then loses 10 per cent. The arithmetic mean of its returns is $(20 - 10)/2 = 5$ per cent. But 100 invested became $100 \times 1.20 \times 0.90 = 108$: a cumulative gain of 8 per cent over two years. The constant annual rate that gives the same result is $\sqrt{1.08} - 1 = 3.92$ per cent: the geometric mean. An investor who reads “averaged 5 per cent” as her annual compound return overestimates it by more than a point.

The geometric mean is never larger than the arithmetic mean (the *AM–GM inequality*), and the gap grows with volatility: approximately, the geometric mean return is the arithmetic mean minus half the variance of returns (a result proved in Chapter 13). The arithmetic mean is the right estimate of the *expected* return in one future period; the geometric mean describes *realized* compound growth. Both are correct answers to different questions.

2.4.3 Harmonic means

When equal amounts of *money* are spent at different prices, the average price paid is the **harmonic mean** of the prices:

$$\bar{p}_H = \frac{n}{\sum_{i=1}^n 1/p_i}.$$

Worked Example 2.5. Regular investing

An investor puts 1,200 euros into a fund each month for three months, at unit prices of 10, 12 and 8. She buys $120 + 100 + 150 = 370$ units for 3,600 euros: an average cost of 9.73 per unit. The arithmetic mean of the prices is 10; the harmonic mean is $3/(1/10 + 1/12 + 1/8) = 9.73$. Buying a fixed money amount automatically buys more units when prices are low. Averaging price–earnings ratios across firms raises the same issue: the harmonic mean of P/E ratios corresponds to the ratio of total price to total earnings when equal amounts are invested in each firm.

2.5 Decomposing Changes

2.5.1 Price and volume

Many economic values are products of a price and a quantity: revenue = pq , nominal GDP = price level $\times$ real GDP, wage bill = wage $\times$ hours. Because the growth factor of a product is

the product of the growth factors,

$$\frac{p_1 q_1}{p_0 q_0} = \frac{p_1}{p_0} \times \frac{q_1}{q_0},$$

the change in value splits exactly into a price factor and a volume factor. In growth rates the split is exact only with the cross term: $g_{pq} = g_p + g_q + g_p g_q$. Distinguishing price from volume is the central task of national accounting (Chapter 10) and of business analysis: revenue growth that comes from raising prices has very different implications from growth that comes from selling more.

2.5.2 Contributions to growth

If a total is a sum of parts, $X = \sum_j X_j$, the growth of the total is a weighted mean of the growth rates of the parts, weighted by their initial shares:

$$g_X = \frac{\sum_j (X_{j,1} - X_{j,0})}{X_0} = \sum_j \frac{X_{j,0}}{X_0} g_j.$$

Each term $s_j g_j$, with $s_j = X_{j,0}/X_0$, is the **contribution** of part j to the growth of the total, measured in percentage points. Contributions add up exactly to total growth. Chapter 43 uses this to find which components of spending drove GDP growth.

Worked Example 2.6. Contributions

A company has two divisions with revenues of 600 and 400. Next year the first grows 5 per cent and the second 20 per cent. Total revenue grows from 1,000 to $630 + 480 = 1,110$: 11 per cent. Contributions: $0.6 \times 5 = 3$ points and $0.4 \times 20 = 8$ points, summing to 11. The smaller division, growing fast, accounts for almost three-quarters of the company's growth.

2.6 Index Numbers

2.6.1 Simple indexes

An **index number** expresses a series relative to its value in a *base period*, conventionally set to 100:

$$I_t = 100 \times \frac{x_t}{x_{\text{base}}}.$$

Indexes make series with different units comparable in growth terms and make percentage changes easy to read: an index moving from 100 to 112 has risen 12 per cent. A move from 112 to 120 is 8 *index points* but $8/112 = 7.1$ per cent. Index points, like percentage points, must not be confused with percentages.

2.6.2 Rebasing and splicing

To *rebase* an index to a new base period, divide the whole series by its value in the new base period and multiply by 100. Rebasing changes no growth rate.

Statistical agencies periodically revise the composition of their indexes and start a new series. *Splicing* links an old and a new series at an overlap period so that growth rates are preserved: if the old series is 125 in 2020 and the new series (2020 = 100) is 103 in 2021, the spliced value for 2021 on the old base is $125 \times 1.03 = 128.75$.

Worked Example 2.7. Rebasing

A price index is 100 in 2015, 108 in 2020 and 117 in 2024. Rebased to 2020 = 100, it is $100 \times 100/108 = 92.59$ in 2015, 100 in 2020 and $100 \times 117/108 = 108.33$ in 2024. Prices rose 8.33 per cent from 2020 to 2024 on either base: $117/108 = 108.33/100$.

2.6.3 Weighted indexes as means of relatives

Most economic indexes combine many items. Let $p_{i,0}$ and $p_{i,1}$ be the prices of item i in the base and current periods, and $q_{i,0}$ the base-period quantities. The cost of the base-period basket rises by the factor

$$\frac{\sum_i p_{i,1} q_{i,0}}{\sum_i p_{i,0} q_{i,0}} = \sum_i s_{i,0} \frac{p_{i,1}}{p_{i,0}}, \quad s_{i,0} = \frac{p_{i,0} q_{i,0}}{\sum_j p_{j,0} q_{j,0}}.$$

The equality follows by writing $p_{i,1} q_{i,0} = (p_{i,1}/p_{i,0}) p_{i,0} q_{i,0}$ and dividing by the base-period total. So a fixed-basket price index is a weighted mean of *price relatives* $p_{i,1}/p_{i,0}$, weighted by base-period expenditure shares. If food takes 60 per cent of spending and its price rises 5 per cent, and energy takes 40 per cent and rises 20 per cent, the index rises by $0.6 \times 5 + 0.4 \times 20 = 11$ per cent. Chapter 10 names this construction (the Laspeyres index), compares it with alternatives that use current-period quantities, and shows how the choice affects measured inflation.

Bridge. Weighted means, growth factors and index numbers are the arithmetic of the whole book: portfolio returns and the cost of capital (Part V) are weighted means; compound interest and economic growth (Chapters 4 and 44) are products of growth factors; price indexes (Chapter 10) convert nominal values into real ones; and contributions to growth organize the national accounts (Chapter 43).

Case: The Mars Climate Orbiter: a unit error

On 23 September 1999, NASA's Mars Climate Orbiter was lost as it approached Mars. It passed far closer to the planet than planned and was destroyed or skipped away into space. The investigation board found that one piece of ground software supplied the thrust impulse of the spacecraft's small course-correcting burns in pound-force seconds, an English unit, while the navigation software that used those figures expected newton seconds, the metric unit specified in the interface agreement. Each figure was therefore too small by a factor of about 4.45. Individually tiny, the errors accumulated over months of flight into a large error in the spacecraft's trajectory.

Why the case belongs in an economics book. Every step of the calculation was arithmetically correct; the failure lay in what the numbers *meant*. The same kind of error occurs constantly in economic and financial work: a monthly interest rate treated as annual, a figure in thousands read as millions, a nominal series compared with a real one, local currency mixed with dollars, a stock divided by a flow and called a percentage. The defences are the same as the board recommended: state units explicitly at every interface between people, models and data sets; check that the units of every result are those expected; and test the output against an independent rough estimate.

Summary

- Every quantity has a unit, a period and a population. Unit analysis catches many errors; stocks divided by flows give periods, not percentages.

- Summation notation: constants factor out, sums split, $\sum c = nc$; deviations from the mean sum to zero. Sums do not pass through products or ratios.
- Percentage changes depend on their base and do not cancel; successive changes multiply as growth factors. Percentage-point changes differ from percentage changes.
- Relative errors add for products and ratios; absolute errors add for sums; differences of similar numbers are imprecise. Round at the end.
- Weighted means need the right weights; geometric means describe compound growth; harmonic means give average prices for equal money amounts.
- Values split exactly into price and volume factors; growth of a total is the share-weighted sum of contributions of its parts.
- Indexes can be rebased and spliced without changing growth rates. A fixed-basket index is a weighted mean of price relatives with base-period expenditure weights.

Key terms

quantity; unit analysis; summation notation; arithmetic mean; ratio; rate; percentage change; base; growth factor; cross term; percentage point; basis point; relative error; absolute error; weighted mean; geometric mean; harmonic mean; AM–GM inequality; price and volume decomposition; contribution to growth; index number; base period; index point; rebasing; splicing; price relative; expenditure share.

Exercises

Exercise 2.1 (Units). A bank's loans are 800 million euros and its annual net interest income is 24 million euros. (a) What is net interest income as a percentage of loans, and what are its units? (b) An analyst says the bank's "loans are 33 times its income." What are the units of 33, and how should it be interpreted?

Exercise 2.2 (Summation). With $x = (3, 5, 10)$ and $y = (2, 4, 1)$, compute $\sum x_i y_i$, $(\sum x_i)(\sum y_i)$, $\sum (x_i - \bar{x})$ and $\sum (x_i - \bar{x})^2$.

Exercise 2.3 (Successive changes). A price rises 25 per cent and then falls 20 per cent. Starting from 80, where does it end? A second price falls 20 per cent and then rises 25 per cent. Where does it end? Explain.

Exercise 2.4 (Percentage points). The unemployment rate rises from 4 to 5 per cent. By how many percentage points and by what percentage has it risen? A bank's non-performing loan ratio falls from 2.5 to 2.0 per cent. Express the change in basis points and in per cent.

Exercise 2.5 (Error propagation). A firm's revenue is estimated as price times volume. Price is known to within 1 per cent and volume to within 3 per cent. What is the worst-case relative error in revenue? If costs are 90 per cent of revenue and are known exactly, what is the worst-case relative error in profit?

Exercise 2.6 (Weighted and geometric means). A portfolio holds 30 per cent in asset A, which returns 12 per cent, and 70 per cent in asset B, which returns -2 per cent, over the same year. (a) What is the portfolio return? (b) An investment returns 30, -20 and 10 per cent in three successive years. Find the arithmetic mean return, the cumulative return and the geometric mean return.

Exercise 2.7 (Harmonic mean). A firm buys 50,000 euros of fuel in each of four quarters at prices of 1.60, 2.00, 1.80 and 1.60 euros per litre. What average price per litre does it pay? Compare with the arithmetic mean of the prices.

Exercise 2.8 (Contributions). A country's exports consist of electronics (60 per cent of exports), palm oil (15 per cent) and other goods (25 per cent). In a year, electronics exports grow 8 per cent, palm oil falls 12 per cent and other goods grow 2 per cent. Find total export growth and each contribution.

Exercise 2.9 (Indexes). An index is 250 in 2019 (base 2005 = 100). A new series starts with 2019 = 100 and reads 104 in 2020 and 110 in 2021. (a) Splice the new series onto the old base. (b) Rebase the spliced series to 2020 = 100. (c) What was the percentage change from 2020 to 2021?

Exercise 2.10 (Weighted index). A household spends 50 per cent of its budget on food, 30 per cent on housing and 20 per cent on transport in the base period. Over the year, food prices rise 4 per cent, housing costs 2 per cent and transport prices fall 5 per cent. By how much does the cost of the base basket rise?

Exercise 2.11 (Diagnose the error). "Our return on equity rose from 10 per cent to 12 per cent, a 2 per cent improvement, and because we averaged returns of 15 per cent and -5 per cent in the last two years, our investors earned 5 per cent a year." Identify two errors and correct them.

Solutions

Solution 2.1. (a) $24/800 = 3$ per cent per year: a flow divided by a stock, so the unit is "per year." (b) $800/24 = 33.3$ years: a stock divided by an annual flow is a number of years—it would take 33 years of net interest income to equal the loan book. It is not a percentage or a multiple in any economically meaningful sense without that interpretation.

Solution 2.2. $\sum x_i y_i = 6 + 20 + 10 = 36$; $(\sum x_i)(\sum y_i) = 18 \times 7 = 126$. $\bar{x} = 6$, so deviations are -3, -1, 4: their sum is 0, and $\sum (x_i - \bar{x})^2 = 9 + 1 + 16 = 26$.

Solution 2.3. $80 \times 1.25 \times 0.80 = 80$; $80 \times 0.80 \times 1.25 = 80$. Multiplication is commutative, so the order of the changes does not matter; the rise and fall cancel because $1.25 \times 0.8 = 1$, not because 25 and 20 are "equal."

Solution 2.4. 1 percentage point; a rise of $1/4 = 25$ per cent. The NPL ratio falls by 50 basis points, or by $0.5/2.5 = 20$ per cent.

Solution 2.5. Revenue: up to $1 + 3 = 4$ per cent (more precisely, $1.01 \times 1.03 - 1 = 4.03$ per cent). Profit is 10 per cent of revenue; an error of 4 per cent of revenue is 40 per cent of profit.

Solution 2.6. (a) $0.3 \times 12 + 0.7 \times (-2) = 3.6 - 1.4 = 2.2$ per cent. (b) Arithmetic mean $(30 - 20 + 10)/3 = 6.67$ per cent. Cumulative factor $1.3 \times 0.8 \times 1.1 = 1.144$: a 14.4 per cent gain. Geometric mean $1.144^{1/3} - 1 = 4.59$ per cent.

Solution 2.7. Litres: $31,250 + 25,000 + 27,778 + 31,250 = 115,278$ for 200,000 euros: 1.735 euros per litre (the harmonic mean). The arithmetic mean price is 1.75; the firm pays less on average because it buys more litres when prices are low.

Solution 2.8. Contributions: electronics $0.60 \times 8 = 4.8$ points; palm oil $0.15 \times (-12) = -1.8$; other $0.25 \times 2 = 0.5$. Total growth = 3.5 per cent.

Solution 2.9. (a) 2020: $250 \times 1.04 = 260$; 2021: $250 \times 1.10 = 275$. (b) Divide by 260: 2019: 96.15; 2020: 100; 2021: 105.77. (c) $275/260 - 1 = 5.77$ per cent (equivalently $110/104 - 1$).

Solution 2.10. $0.5 \times 4 + 0.3 \times 2 + 0.2 \times (-5) = 2 + 0.6 - 1 = 1.6$ per cent.

Solution 2.11. ROE rose by 2 percentage points, or by 20 per cent, not “2 per cent.” Investors’ compound return was $\sqrt{1.15 \times 0.95} - 1 = \sqrt{1.0925} - 1 = 4.52$ per cent a year, not 5 per cent: the arithmetic mean of returns overstates compound growth.

Algebra, Functions and Equations

After studying this chapter you should be able to

1. Translate verbal economic statements into algebraic expressions and equations, and rearrange formulas to isolate any variable.
2. Define a function, its domain and range, and work with composite, inverse and piecewise functions.
3. Interpret the slope and intercept of a linear function, and derive inverse demand from demand.
4. Build cost, revenue and profit functions, and find break-even points subject to capacity.
5. Solve quadratic equations, find the vertex of a parabola, and use it to maximize revenue without calculus.
6. Solve systems of linear equations by substitution and elimination, and classify systems with one, none or infinitely many solutions.
7. Solve linear inequalities and absolute-value inequalities, and describe feasible sets such as budget sets.
8. Distinguish marginal from average rates in a piecewise-linear tax schedule.

A motivating question

A small manufacturer sells a product for 30 euros; each unit costs 18 euros in materials and labour, and the factory costs 24,000 euros a period to run whatever it produces. How many units must it sell to cover its costs? The answer, 2,000, takes one line of algebra. But the factory can make only 1,500 units a period. The equation has a solution that the firm cannot reach, and a manager who solved the equation without writing down the constraint would conclude that the business can break even when it cannot.

Algebra is the language in which economic models are written. It lets us state relationships precisely—how cost depends on output, how quantity demanded depends on price—and solve for the values that satisfy them. This chapter develops that language. Its recurring lesson, which the manufacturer illustrates, is that an equation is only half of a model: the other half is the set of values the variables are allowed to take.

3.1 Expressions, Equations and Formulas

3.1.1 Variables and expressions

A **variable** is a symbol that stands for a quantity: q for output, p for price, r for an interest rate. An *expression* combines variables and numbers with operations: $pq - (F + cq)$. Evaluating an expression means substituting values: with $p = 30$, $c = 18$, $F = 24,000$ and $q = 1,500$, the

expression $pq - (F + cq)$ equals $45,000 - 51,000 = -6,000$.

The usual conventions govern the order of operations: brackets first, then powers, then multiplication and division, then addition and subtraction. The *distributive law* $a(b + c) = ab + ac$ is used constantly to expand and factor expressions: $pq - cq - F = (p - c)q - F$.

3.1.2 Solving linear equations

An **equation** asserts that two expressions are equal. To *solve* it for a variable is to find the values of that variable that make it true. A linear equation in one unknown is solved by performing the same operation on both sides until the unknown is isolated.

Worked Example 3.1. A linear equation

Solve $4(3x - 2) + 5 = 33$. Expanding, $12x - 8 + 5 = 33$, so $12x - 3 = 33$, $12x = 36$ and $x = 3$. Check: $4(9 - 2) + 5 = 28 + 5 = 33$. Checking by substitution is cheap and catches most errors.

An equation may have no solution: $7x + 4 = 7(x - 2)$ simplifies to $4 = -14$, which is false for every x . It may hold for every value: $2(x + 1) = 2x + 2$ is an *identity*. Chapter 1 made the same distinction in economics between identities, true by definition, and behavioural equations, true only for particular values or in particular circumstances.

3.1.3 Rearranging formulas

A *formula* expresses one quantity in terms of others. Rearranging it to isolate a different variable uses the same rules as solving an equation.

Worked Example 3.2. Rearranging a simple-interest formula

A loan of principal P at simple interest rate r per year for t years is repaid with $A = P(1 + rt)$. To find the rate implied by a repayment, divide by P , subtract 1 and divide by t :

$$r = \frac{A/P - 1}{t}.$$

If 10,000 is repaid with 11,200 after two years, $r = (1.12 - 1)/2 = 6$ per cent a year. The domain matters: the formula requires $P > 0$ and $t > 0$. Chapter 4 replaces simple interest by compound interest, and the rearrangement then requires roots or logarithms.

3.2 Functions

3.2.1 Definition

Definition 3.3. Function

A **function** f assigns to each element x of a set called the **domain** exactly one value $f(x)$. We write $y = f(x)$, call x the *argument* or *independent variable* and y the *dependent variable*. The set of values f takes is its **range**.

Economic examples are everywhere. A *demand function* $Q = D(P)$ gives the quantity buyers wish to purchase at each price. A *cost function* $C(q)$ gives the cost of producing q units. A *production function* $Y = F(K, L)$ gives output from capital and labour; it is a function of two

variables, studied in Chapter 5. The requirement that each argument have exactly one value matters: a demand function says that at a given price there is one quantity demanded, not several.

Domains. Write down the domain before solving anything. Prices and quantities are usually non-negative; a denominator cannot be zero; a logarithm requires a positive argument (Chapter 4); a capacity limit bounds output from above. Many errors in applied work are not errors of algebra but solutions outside the domain.

3.2.2 Linear functions

A **linear function** has the form $y = a + bx$. The constant a is the **intercept** (the value of y when $x = 0$) and b is the **slope**:

$$b = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

for any two points on the line. The slope is the change in y per unit change in x , and its units are units of y per unit of x . For a linear function it is the same everywhere.

Worked Example 3.4. A linear demand function

Suppose the demand for bus rides in a town is $Q = 1,200 - 400P$, where Q is rides per day and P is the fare in currency units. The intercept 1,200 is the number of rides demanded at a zero fare. The slope -400 says that each 1-unit increase in the fare reduces daily rides by 400. At $P = 1.50$, $Q = 1,200 - 600 = 600$. At $P = 3$, $Q = 0$: no rides are demanded at a fare of 3 or more (the *choke price*). The economically meaningful domain is $0 \leq P \leq 3$.

3.2.3 Inverse functions

If $y = f(x)$ and each y in the range corresponds to exactly one x , we can solve for x in terms of y , obtaining the **inverse function** $x = f^{-1}(y)$. A function has an inverse on an interval if it is strictly increasing or strictly decreasing there. Solving $Q = 1,200 - 400P$ for P gives the **inverse demand function**

$$P = 3 - \frac{Q}{400} = 3 - 0.0025Q.$$

Inverse demand tells us the highest price at which a given quantity can be sold—equivalently, the value the marginal buyer places on the good. Economists conventionally draw demand curves with price on the vertical axis, so the curve you see in a diagram is the graph of the inverse demand function. This convention, inherited from Alfred Marshall, confuses many students: the “slope” of a demand curve in a diagram is $\Delta P / \Delta Q$, the reciprocal of $\Delta Q / \Delta P$.

3.2.4 Composite functions

If $y = g(x)$ and $z = f(y)$, then $z = f(g(x))$, the **composite function** $f \circ g$. Revenue as a function of price is a composite: quantity depends on price through demand, and revenue is price times quantity, so $R(P) = P \cdot D(P)$. For the bus company, $R(P) = P(1,200 - 400P) = 1,200P - 400P^2$, a quadratic function of the fare (Section 3.4).

3.2.5 Piecewise functions

A **piecewise function** uses different formulas on different parts of its domain. Income-tax schedules, tiered electricity tariffs, bank fees and loan covenants (Chapter 39) are all piecewise. The case at the end of the chapter works through a real tax schedule.

3.3 Cost, Revenue and Break-Even

3.3.1 The linear cost model

Suppose a firm's cost of producing q units in a period is

$$C(q) = F + cq, \quad q \geq 0,$$

where F is **fixed cost**, incurred whatever the output, and c is **variable cost** per unit. In this model c is also the extra cost of one more unit. If the firm sells every unit at a price p , revenue is $R(q) = pq$ and profit is

$$\Pi(q) = R(q) - C(q) = (p - c)q - F.$$

The term $p - c$ is the **contribution margin** per unit: the amount each unit contributes toward covering fixed cost. The **break-even quantity** solves $\Pi(q) = 0$:

$$q_{BE} = \frac{F}{p - c}, \quad p > c.$$

If $p \leq c$, no quantity breaks even: each unit loses money before any fixed cost is covered.

Worked Example 3.5. Break-even with a capacity constraint

With $p = 30$, $c = 18$ and $F = 24,000$, profit is $\Pi(q) = 12q - 24,000$ and $q_{BE} = 24,000/12 = 2,000$. If capacity is 1,500 units, the feasible domain is $0 \leq q \leq 1,500$, and on it profit never reaches zero: the best the firm can do is $\Pi(1,500) = 18,000 - 24,000 = -6,000$. To break even it must raise its price, cut its costs, or expand capacity. Solving the equation without the inequality would have hidden the problem.

3.3.2 Average cost

Average cost per unit is

$$AC(q) = \frac{C(q)}{q} = \frac{F}{q} + c, \quad q > 0.$$

It is a *rational function* (a ratio of polynomials) that falls toward c as output rises, because fixed cost is spread over more units; its graph approaches the horizontal line $AC = c$, an *asymptote*, and rises without limit as q approaches zero. The distinction between average and marginal cost—here $F/q + c$ against c —is central to Chapter 21. Chapter 31 develops break-even and contribution-margin analysis as tools of managerial accounting, where costs are rarely as simple as this model.

3.4 Quadratic Functions

3.4.1 The parabola and its vertex

A **quadratic function** has the form $f(x) = ax^2 + bx + c$ with $a \neq 0$. Its graph is a parabola, opening upward if $a > 0$ and downward if $a < 0$. *Completing the square* rewrites it as

$$f(x) = a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a}.$$

Because the squared term is never negative, the function reaches its minimum (if $a > 0$) or maximum (if $a < 0$) where the squared term is zero: at the **vertex** $x^* = -b/(2a)$, where $f(x^*) = c - b^2/(4a)$.

Worked Example 3.6. Maximizing revenue without calculus

A firm faces inverse demand $p = 100 - 2q$. Revenue is $R(q) = (100 - 2q)q = -2q^2 + 100q$. Here $a = -2$ and $b = 100$, so revenue is maximized at $q^* = -100/(2 \times (-2)) = 25$, where $p = 50$ and $R = 1,250$. The bus company of Example 3.4 has $R(P) = -400P^2 + 1,200P$, maximized at $P^* = 1,200/800 = 1.50$ with revenue 900. Chapter 5 obtains the same answers by setting a derivative to zero, a method that works for functions far more general than quadratics.

3.4.2 Roots and the quadratic formula

The **roots** of $ax^2 + bx + c = 0$ are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The *discriminant* $b^2 - 4ac$ determines how many real roots there are: two if it is positive, one (a repeated root at the vertex) if it is zero, none if it is negative. The formula follows from setting the completed-square form equal to zero.

Worked Example 3.7. The break-even range

Continue Example 3.6 with cost $C(q) = 800$. Profit is $\Pi(q) = -2q^2 + 100q - 800$. Break-even requires $q^2 - 50q + 400 = 0$ (dividing by -2), so

$$q = \frac{50 \pm \sqrt{2,500 - 1,600}}{2} = \frac{50 \pm 30}{2} = 10 \text{ or } 40.$$

The firm makes a profit for outputs between 10 and 40, with maximum profit $1,250 - 800 = 450$ at $q = 25$. With a downward-sloping demand curve, selling more is not always better: beyond 40 units the price has fallen so far that revenue no longer covers cost.

3.5 Systems of Linear Equations

3.5.1 Two equations in two unknowns

Many economic questions require several conditions to hold at once: supply equals demand; assets equal liabilities plus equity; each market in an economy clears. A **system of equations** is solved by finding values of the unknowns that satisfy every equation simultaneously. With two linear equations in two unknowns, two methods suffice.

Substitution: solve one equation for one unknown and substitute into the other. *Elimination*: add a multiple of one equation to the other so that one unknown disappears.

Worked Example 3.8. Market equilibrium

Buyers' demand is $Q_d = 100 - 2P$ and sellers' supply is $Q_s = -20 + 4P$. Equilibrium requires $Q_d = Q_s$: $100 - 2P = -20 + 4P$, so $6P = 120$, $P^* = 20$ and $Q^* = 60$. Graphically, the solution is the intersection of the two lines (Figure 3.1). Chapter 22 develops the economics behind both curves and asks what happens to the solution when the curves shift.

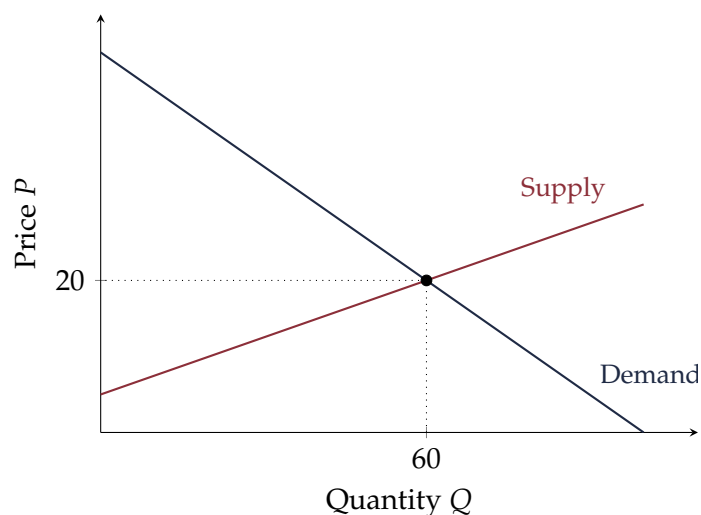


Figure 3.1: The solution of a system of two linear equations is the intersection of their graphs. Here the demand line $P = 50 - Q/2$ and the supply line $P = 5 + Q/4$ (the inverse functions of Example 3.8) intersect at $Q = 60$, $P = 20$.

Worked Example 3.9. Financing weights

A firm has assets of 100, financed by debt D and equity E , and its policy is that debt be 40 per cent of total financing. Then $D + E = 100$ and $D = 0.4(D + E)$. Substituting the first into the second gives $D = 40$, so $E = 60$. The algebra is trivial; the economics is not: whether the 40 per cent refers to book values or market values changes the answer greatly, as Chapter 35 shows.

3.5.2 How many solutions?

Two linear equations in two unknowns describe two straight lines. There are three possibilities:

1. The lines cross once: a unique solution.
2. The lines are parallel (same slope, different intercepts): no solution. The system is *inconsistent*—for instance, a demand and a supply curve with the same slope that never meet.
3. The lines coincide: infinitely many solutions. One equation is a multiple of the other and adds no information.

With more unknowns, the same three cases arise. As a rule, a unique solution requires as many independent equations as unknowns. Chapter 8 develops systematic methods for large

systems and a test—the determinant—that tells us which case we are in.

3.6 Inequalities and Feasible Sets

3.6.1 Linear inequalities

An **inequality** states that one expression is less than (or no greater than) another. Inequalities are solved like equations with one exception: multiplying or dividing both sides by a negative number reverses the direction of the inequality. For example, $-3x + 5 \leq 11$ gives $-3x \leq 6$, and dividing by -3 gives $x \geq -2$.

3.6.2 Absolute value

The **absolute value** $|x|$ is the distance of x from zero. The inequality $|x - a| \leq d$ says that x lies within d of a : $a - d \leq x \leq a + d$. A central bank that aims to keep an overnight interest rate within a quarter of a point of a 3 per cent target requires $|i - 3| \leq 0.25$, i.e. $2.75 \leq i \leq 3.25$ (Chapter 49); an inflation target of “2 per cent plus or minus 1 point” is $|\pi - 2| \leq 1$.

3.6.3 Feasible sets

Constraints on several variables define a **feasible set**: the combinations of values that satisfy all of them. A household with income 100 that can buy good x at price 2 and good y at price 4 faces the **budget constraint** $2x + 4y \leq 100$, with $x \geq 0$ and $y \geq 0$. The feasible set is a triangle whose outer edge, the *budget line* $2x + 4y = 100$, has slope $-2/4 = -0.5$: the price of x in units of y . Adding a second constraint—say a limit of $x \leq 30$ imposed by rationing—cuts a corner off the triangle.

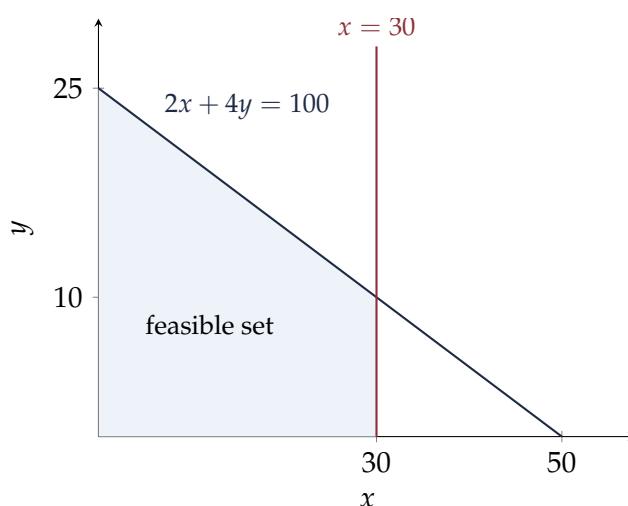


Figure 3.2: A feasible set defined by linear inequalities: the budget constraint $2x + 4y \leq 100$, non-negativity, and a rationing limit $x \leq 30$. Its corners are $(0, 0)$, $(30, 0)$, $(30, 10)$ and $(0, 25)$.

Optimization problems in economics choose the best point in a feasible set. When the objective is itself linear—maximize profit $6x + 5y$ from two products subject to capacity limits, say—the best point is always at a corner of the feasible set, a result that underlies *linear programming* (Chapter 6). When the objective is non-linear, as utility usually is, the best point may lie on an edge, and calculus is needed to find it (Chapters 5 and 6).

Case: Marginal and average tax rates: the U.S. income tax

Income taxes are the most familiar piecewise-linear functions. For the 2024 tax year, the U.S. federal income tax for a single filer applied rates of 10 per cent on the first 11,600 dollars of taxable income, 12 per cent on income from 11,600 to 47,150, 22 per cent from 47,150 to 100,525, 24 per cent from 100,525 to 191,950, and rising rates of 32, 35 and 37 per cent above that.

Computing the function. For taxable income of 60,000 dollars, tax is

$$\begin{aligned} &0.10 \times 11,600 + 0.12 \times (47,150 - 11,600) + 0.22 \times (60,000 - 47,150) \\ &= 1,160 + 4,266 + 2,827 = 8,253. \end{aligned}$$

The *average* tax rate is $8,253/60,000 = 13.8$ per cent. The *marginal* tax rate—the tax on one more dollar, the slope of the tax function at 60,000—is 22 per cent.

Why the distinction matters. A common belief is that moving into a higher bracket can reduce after-tax income. It cannot, in a schedule of this kind: the higher rate applies only to income *above* the threshold, so the tax function is continuous and its slope never reaches 100 per cent. Decisions about working more, saving more or realizing a capital gain depend on the marginal rate; the share of income paid in tax is measured by the average rate. The same distinction between marginal and average quantities—marginal and average cost, marginal and average product—runs through all of economics (Chapter 1).

Where the slope can exceed the statutory rate. When benefits are withdrawn as income rises, the combined schedule of taxes and benefits can have marginal rates far above any tax bracket—sometimes above 70 per cent for low-income families in several countries. These effective marginal rates, which are the slopes of a combined piecewise function, matter for labour supply (Chapter 20).

Summary

- Equations are solved by doing the same thing to both sides; always check by substitution. Some equations have no solution; identities hold for all values.
- A function assigns one value to each point of its domain. Linear functions have constant slope; inverse demand is the inverse of the demand function. Composite and piecewise functions are common in economics.
- In the linear cost model, profit is $(p - c)q - F$, and break-even output is $F/(p - c)$ —if it is feasible. Average cost $F/q + c$ falls toward marginal cost.
- A quadratic reaches its maximum or minimum at the vertex $-b/(2a)$; its roots follow from the quadratic formula.
- Systems of linear equations have one, none or infinitely many solutions; a unique solution requires as many independent equations as unknowns.
- Multiplying an inequality by a negative number reverses it. Constraints define feasible sets; linear objectives are optimized at corners.
- Marginal rates are slopes; average rates are ratios. They differ whenever a function is not proportional.

Key terms

variable; expression; distributive law; equation; identity; formula; function; domain; range; linear function; intercept; slope; inverse function; inverse demand function; composite function; piecewise function; fixed cost; variable cost; contribution margin; break-even quantity; average cost; asymptote; quadratic function; vertex; completing the square; roots; discriminant; system of equations; substitution; elimination; inconsistent system; inequality; absolute value; feasible set; budget constraint; budget line; marginal tax rate; average tax rate.

Exercises

Exercise 3.1 (Equations). Solve: (a) $5(x - 4) = 2x + 7$; (b) $\frac{x}{3} + \frac{x}{6} = 9$; (c) $3(2x + 1) = 6x + 5$.

Exercise 3.2 (Rearranging). The price of a bond that pays F in one year is $P = F/(1 + y)$. Solve for the yield y . Find y when $F = 1,000$ and $P = 960$.

Exercise 3.3 (Inverse demand). Demand is $Q = 500 - 25P$. (a) Find the inverse demand function and the choke price. (b) What is the domain of the demand function in economically meaningful terms? (c) Write revenue as a function of P .

Exercise 3.4 (Break-even). A product sells for 24, costs 15 per unit to make, and has fixed cost 18,000. (a) Find the break-even output. (b) If capacity is 2,500 units, what is the maximum profit? (c) What price would the firm need to break even at an output of 1,500?

Exercise 3.5 (Quadratics). Inverse demand is $p = 60 - 0.5q$ and cost is $C(q) = 1,000$. Find the revenue-maximizing output and price, the break-even outputs, and the maximum profit.

Exercise 3.6 (Equilibrium). Demand is $Q_d = 240 - 3P$ and supply $Q_s = 60 + 3P$. Find the equilibrium. A tax causes supply to become $Q_s = 30 + 3P$ (in terms of the price buyers pay). Find the new equilibrium.

Exercise 3.7 (Systems). Classify each system and solve where possible: (a) $x + 2y = 10$, $3x - y = 2$; (b) $2x + 4y = 8$, $x + 2y = 5$; (c) $2x + 4y = 8$, $x + 2y = 4$.

Exercise 3.8 (Inequalities). Solve $7 - 2x > 3x - 8$. Solve $|q - 1,000| \leq 50$ and interpret it as a production tolerance.

Exercise 3.9 (Feasible sets). A firm makes two products using a machine available for 120 hours and a workforce available for 150 hours. Each unit of product A needs 2 machine hours and 1 labour hour; each unit of B needs 1 machine hour and 3 labour hours. Write the constraints, find the corners of the feasible set, and determine which corner maximizes profit $40A + 50B$.

Exercise 3.10 (Tax rates). Using the 2024 U.S. schedule in the case, find the tax, average rate and marginal rate for taxable incomes of 40,000 and 150,000 dollars.

Exercise 3.11 (Price, volume and a constraint). A business sells 1,000 units at 40 each. Variable cost is 24 per unit and fixed cost for the period is 8,000. It considers cutting its price to 39, which it forecasts will raise sales to 1,100 units. Capacity is 1,050 units unless it spends 1,000 on expansion this period. Compute operating profit for each feasible option, and state what evidence you would want before acting.

Solutions

Solution 3.1. (a) $5x - 20 = 2x + 7$, $3x = 27$, $x = 9$. (b) Multiply by 6: $2x + x = 54$, $x = 18$. (c) $6x + 3 = 6x + 5$ gives $3 = 5$: no solution.

Solution 3.2. $1 + y = F/P$, so $y = F/P - 1$. With the numbers, $y = 1,000/960 - 1 = 4.17$ per cent.

Solution 3.3. (a) $P = 20 - Q/25 = 20 - 0.04Q$; the choke price is 20. (b) $0 \leq P \leq 20$ (so that $0 \leq Q \leq 500$). (c) $R(P) = P(500 - 25P) = 500P - 25P^2$.

Solution 3.4. (a) $18,000/(24 - 15) = 2,000$. (b) At capacity, $9 \times 2,500 - 18,000 = 4,500$. (c) Need $(p - 15) \times 1,500 = 18,000$, so $p - 15 = 12$, $p = 27$.

Solution 3.5. $R = 60q - 0.5q^2$, maximized at $q = 60/(2 \times 0.5) = 60$, $p = 30$, $R = 1,800$. Break-even: $-0.5q^2 + 60q - 1,000 = 0$, i.e. $q^2 - 120q + 2,000 = 0$, $q = (120 \pm \sqrt{14,400 - 8,000})/2 = (120 \pm 80)/2 = 20$ or 100 . Maximum profit $= 1,800 - 1,000 = 800$ at $q = 60$.

Solution 3.6. $240 - 3P = 60 + 3P$: $P = 30$, $Q = 150$. With the tax: $240 - 3P = 30 + 3P$: $P = 35$, $Q = 135$. Buyers pay 5 more; quantity falls by 15.

Solution 3.7. (a) From the second, $y = 3x - 2$; substituting, $x + 6x - 4 = 10$, so $x = 2$, $y = 4$: unique. (b) The first equation divided by 2 is $x + 2y = 4$, which contradicts $x + 2y = 5$: parallel lines, no solution. (c) The equations are the same line: infinitely many solutions $(x, 2 - x/2)$.

Solution 3.8. $15 > 5x$, so $x < 3$. $950 \leq q \leq 1,050$: output must be within 50 units of the 1,000 target.

Solution 3.9. Machine: $2A + B \leq 120$; labour: $A + 3B \leq 150$; $A, B \geq 0$. Corners: $(0, 0)$, $(60, 0)$, $(0, 50)$ and the intersection of the two constraints: from $B = 120 - 2A$, $A + 360 - 6A = 150$, so $A = 42$, $B = 36$. Profits: 0; 2,400; 2,500; $40 \times 42 + 50 \times 36 = 1,680 + 1,800 = 3,480$. The best corner is $(42, 36)$.

Solution 3.10. At 40,000: $1,160 + 0.12 \times 28,400 = 1,160 + 3,408 = 4,568$; average 11.4 per cent; marginal 12 per cent. At 150,000: $1,160 + 4,266 + 0.22 \times 53,375 + 0.24 \times 49,475 = 1,160 + 4,266 + 11,742.50 + 11,874 = 29,042.50$; average 19.4 per cent; marginal 24 per cent.

Solution 3.11. Current: revenue 40,000, profit $(40 - 24) \times 1,000 - 8,000 = 8,000$. Price 39 without expansion: output is capped at 1,050, revenue 40,950, profit $15 \times 1,050 - 8,000 = 7,750$. Price 39 with expansion: revenue 42,900, profit $15 \times 1,100 - 8,000 - 1,000 = 7,500$. Both new options raise revenue but lower profit. Before acting, ask whether the 1,100 forecast is a credible causal estimate of the response to the price cut (Chapter 17), whether the expansion would bring benefits in later periods, and whether variable cost stays at 24 near capacity.

Exponentials, Logarithms, Growth and Sequences

After studying this chapter you should be able to

1. Apply the laws of exponents, including negative and fractional exponents.
2. Model compound growth with exponential functions, convert between compounding frequencies, and use continuous growth with the number e .
3. Use logarithms to solve for growth rates and times, and derive the rule of 70.
4. Approximate growth rates by log differences, and apply the rules for the growth of products, ratios and powers.
5. Compute compound annual growth rates and annualize growth over shorter periods.
6. Sum arithmetic and geometric sequences, finite and infinite, and state when an infinite series converges.
7. Solve first-order linear difference equations, find their steady states, and determine stability and half-lives.

A motivating question

In 1978 China's output per person was among the lowest in the world. Over the following three decades it grew by close to 10 per cent a year. How much richer did that make the average Chinese citizen? Adding 10 per cent a year for 28 years suggests incomes rose by 280 per cent, a little under four-fold. The true answer is more than fourteen-fold. The difference arises because growth compounds: each year's 10 per cent is applied to a larger base than the year before.

Compounding governs interest on savings and debts, the growth of economies and populations, inflation, the returns on investments and the spread of new technologies. Its mathematics—powers, exponential functions and their inverses, logarithms—is the subject of this chapter, together with two tools built from the same ideas: sums of geometric sequences, which underlie present values and the spending multiplier, and difference equations, which describe how an economic variable evolves from one period to the next.

4.1 Powers and Roots

For a positive number a and a positive integer n , the power a^n is a multiplied by itself n times. The **laws of exponents** follow from counting factors:

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn}, \quad (ab)^n = a^n b^n.$$

Requiring these laws to hold for all exponents fixes the meaning of other powers: $a^0 = 1$, $a^{-n} = 1/a^n$, and $a^{1/n} = \sqrt[n]{a}$, the positive number whose n th power is a . Hence $a^{m/n} = (\sqrt[n]{a})^m$.

For example, $8^{2/3} = (\sqrt[3]{8})^2 = 4$ and $1.08^{1/2} = 1.0392$.

Parentheses determine what an exponent acts on: $(2x)^2 = 4x^2$ but $2x^2$ is twice x^2 . *Scientific notation* writes large and small numbers as $m \times 10^k$ with $1 \leq m < 10$: world GDP of roughly 110 trillion dollars is about 1.1×10^{14} dollars.

4.2 Exponential Growth

4.2.1 Compound growth

A quantity growing at a constant rate g per period is multiplied by the growth factor $1 + g$ each period (Chapter 2). After t periods,

$$x_t = x_0(1 + g)^t.$$

As a function of t , this is an **exponential function**: the variable appears in the exponent. Its defining property is that it changes by a constant *proportion* in each period, whereas a linear function changes by a constant *amount*. Over short horizons the two look similar; over long horizons exponential growth overwhelms linear growth. At 10 per cent a year, $1.1^{28} = 14.4$: the fourteen-fold rise of the motivating question.

4.2.2 Compounding frequency

Interest rates are quoted per year but often compounded more frequently. A *nominal* annual rate j compounded m times a year means that the rate j/m is applied m times, so one unit grows in a year to $(1 + j/m)^m$. The **effective annual rate** is

$$r_{\text{eff}} = \left(1 + \frac{j}{m}\right)^m - 1.$$

A credit card charging 12 per cent a year compounded monthly has an effective annual rate of $1.01^{12} - 1 = 12.68$ per cent. Rate and period must always match: applying an annual rate to a monthly cash flow, or a monthly rate for a year without compounding, is the unit error of Chapter 2 in another form.

4.2.3 Continuous growth and the number e

As the number of compounding periods rises, $(1 + j/m)^m$ increases but approaches a limit. That limit is e^j , where

$$e = \lim_{m \rightarrow \infty} \left(1 + \frac{1}{m}\right)^m \approx 2.71828.$$

Growth at a *continuously compounded* rate g therefore gives

$$x(t) = x_0 e^{gt}.$$

At 5 per cent for 10 years, continuous compounding turns 1,000 into $1,000e^{0.5} = 1,648.72$, compared with $1,000 \times 1.05^{10} = 1,628.89$ with annual compounding. Continuous time makes growth algebra exact rather than approximate, which is why the growth models of Chapter 44 and many finance models (Chapter 41) are written in it. The function e^x is called the *natural exponential function*, often written $\exp(x)$.

4.3 Logarithms

4.3.1 Definition and laws

The **natural logarithm** $\ln x$, defined for $x > 0$, is the inverse of the exponential function: $y = \ln x$ exactly when $e^y = x$. So $\ln(e^x) = x$, $e^{\ln x} = x$, $\ln 1 = 0$ and $\ln e = 1$. Because exponents add when numbers multiply, logarithms turn multiplication into addition. For $x, y > 0$ and any real a :

$$\ln(xy) = \ln x + \ln y, \quad \ln(x/y) = \ln x - \ln y, \quad \ln(x^a) = a \ln x.$$

A logarithm to another base b satisfies $\log_b x = \ln x / \ln b$. In this book “log” means the natural logarithm unless stated otherwise.

4.3.2 Solving for times and rates

Logarithms solve equations in which the unknown is an exponent. If $x_t = x_0(1 + g)^t$, taking logs gives $\ln(x_t/x_0) = t \ln(1 + g)$, so

$$t = \frac{\ln(x_t/x_0)}{\ln(1 + g)}, \quad g = \left(\frac{x_t}{x_0}\right)^{1/t} - 1.$$

The second formula needs a root rather than a logarithm. At 6 per cent a year, 1,000 reaches 2,500 after $\ln 2.5 / \ln 1.06 = 15.7$ years.

4.3.3 Doubling times and the rule of 70

A quantity doubles when $(1 + g)^t = 2$, i.e.

$$t = \frac{\ln 2}{\ln(1 + g)} \approx \frac{0.693}{g} \approx \frac{70}{100g}.$$

This is the **rule of 70**: divide 70 by the growth rate in per cent to get the approximate doubling time. At 7 per cent growth, income doubles in about 10 years (exactly 10.24); at 2 per cent, in about 35 years. China’s growth of close to 10 per cent a year for three decades implied a doubling roughly every seven years (Chapter 44 discusses why such rates are rare and why they slow). The approximation $\ln(1 + g) \approx g$ is explained next.

4.4 Growth Rates and Logarithms

4.4.1 The log approximation

For small g , $\ln(1 + g) \approx g$. (Chapter 5 shows that the error is roughly $g^2/2$; at $g = 0.03$ the approximation gives 0.03 against the true $\ln 1.03 = 0.02956$.) Therefore

$$\ln x_t - \ln x_{t-1} = \ln \frac{x_t}{x_{t-1}} = \ln(1 + g_x) \approx g_x.$$

The change in the log of a variable approximates its growth rate. This is why economists plot variables such as GDP on a *log scale*: equal vertical distances represent equal *percentage* changes, and a variable growing at a constant rate appears as a straight line. On an ordinary scale, the same series curves upward ever more steeply and makes early growth look negligible.

4.4.2 Growth rates of products and ratios

Taking logs turns products into sums, which gives rules used constantly in macroeconomics and finance:

Proposition 4.1. Growth of products and ratios

If $z = xy$, then $g_z \approx g_x + g_y$. If $z = x/y$, then $g_z \approx g_x - g_y$. If $z = x^a$, then $g_z \approx a g_x$. In continuous time these hold exactly.

Proof. $\ln z = \ln x + \ln y$, so $\Delta \ln z = \Delta \ln x + \Delta \ln y$; each $\Delta \ln$ approximates the corresponding growth rate. The other cases are identical with $\ln z = \ln x - \ln y$ and $\ln z = a \ln x$. In continuous time, the growth rate of x is $\dot{x}/x = d \ln x / dt$ exactly (Chapter 5), and differentiating the log relationships with respect to t gives the rules exactly. $\square$

Worked Example 4.2. Real GDP and GDP per person

Nominal GDP equals the price level times real GDP: $Y^n = PY$. If nominal GDP grows by 6% and the price level by 4%, real GDP grows by about $6 - 4 = 2\%$. (Exactly: $1.06/1.04 - 1 = 1.923\%$.) If population grows by 0.5%, real GDP per person grows by about $2 - 0.5 = 1.5\%$.

4.4.3 Compound annual growth and annualization

The **compound annual growth rate** (CAGR) over n years is the constant rate that turns the starting value into the final value: $\text{CAGR} = (x_n/x_0)^{1/n} - 1$. It is the geometric mean of the annual growth factors minus one (Chapter 2). If a firm's revenue grows from 200 to 350 over five years, its CAGR is $1.75^{0.2} - 1 = 11.8$ per cent a year.

Growth over a period shorter than a year is *annualized* by compounding: a quarterly growth rate g_q corresponds to an annual rate $(1 + g_q)^4 - 1$. A quarterly rise in GDP of 0.5 per cent is reported in the United States as $1.005^4 - 1 = 2.0$ per cent at an annual rate, and a quarterly fall of 9 per cent as $0.91^4 - 1 = -31.4$ per cent (Chapter 43). Annualizing a single volatile quarter can exaggerate its significance; year-on-year comparisons are smoother.

4.5 Sequences and Series

4.5.1 Arithmetic and geometric sequences

A **sequence** is an ordered list of numbers $a_1, a_2, a_3, \dots$. In an *arithmetic sequence* each term adds a constant d : $a_t = a_1 + (t - 1)d$. In a *geometric sequence* each term multiplies by a constant ratio q : $a_t = a_1 q^{t-1}$. Linear growth produces arithmetic sequences; compound growth produces geometric ones. A *series* is the sum of the terms of a sequence.

Arithmetic series. The sum of the first n terms of an arithmetic sequence is n times the average of the first and last terms: $S_n = n(a_1 + a_n)/2$. (Write the sum forwards and backwards and add: each of the n pairs sums to $a_1 + a_n$.) The sum $1 + 2 + \dots + 100 = 100 \times 101/2 = 5,050$.

4.5.2 Geometric series

Geometric series appear whenever a quantity is discounted, compounded or passed on in fixed proportion.

Proposition 4.3. Geometric series

For $q \neq 1$,

$$\sum_{t=0}^{n-1} q^t = 1 + q + q^2 + \cdots + q^{n-1} = \frac{1 - q^n}{1 - q}.$$

If $|q| < 1$, the infinite series converges:

$$\sum_{t=0}^{\infty} q^t = \frac{1}{1 - q}.$$

Proof. Let $S = 1 + q + \cdots + q^{n-1}$. Then $qS = q + q^2 + \cdots + q^n$, so $S - qS = 1 - q^n$ and $S = (1 - q^n)/(1 - q)$. If $|q| < 1$, $q^n \rightarrow 0$ as $n \rightarrow \infty$, giving $1/(1 - q)$. If $|q| \geq 1$, the terms do not shrink and the infinite sum has no finite value. $\square$

Three uses appear later in this book. The *spending multiplier*: an extra 100 of spending, of which a fraction 0.75 is re-spent at each round, raises total spending by $100(1 + 0.75 + 0.75^2 + \cdots) = 100/(1 - 0.75) = 400$ (Chapter 46). *Present values*: the value today of a constant payment every period is a geometric series with ratio $1/(1 + i)$ (Chapter 9). *Expected durations*: if a price survives each period with probability θ , its expected life is a geometric sum (Chapter 47).

4.6 Difference Equations

Many economic variables evolve according to a rule linking this period's value to last period's: capital accumulates (Chapter 44), unemployment adjusts through flows into and out of jobs (Chapter 45), debt rolls over (Chapter 47), inflation depends on expected inflation (Chapter 50). The simplest such rule is the first-order linear **difference equation**

$$x_{t+1} = a x_t + b,$$

with constants a and b , and a given starting value x_0 .

Steady state. A **steady state** (or fixed point) $\bar{x}$ is a value that reproduces itself: $\bar{x} = a\bar{x} + b$. If $a \neq 1$,

$$\bar{x} = \frac{b}{1 - a}.$$

Solution. Subtract the steady-state equation from the difference equation: $x_{t+1} - \bar{x} = a(x_t - \bar{x})$. The deviation from steady state is multiplied by a each period—a geometric sequence—so

$$x_t = \bar{x} + a^t(x_0 - \bar{x}).$$

Proposition 4.4. Stability

If $|a| < 1$, then $x_t \rightarrow \bar{x}$ from any starting point: the steady state is **stable**. If $|a| > 1$, deviations grow without bound unless $x_0 = \bar{x}$: the steady state is **unstable**. If $0 < a < 1$, convergence is monotonic; if $-1 < a < 0$, x_t oscillates around $\bar{x}$ with shrinking amplitude.

Worked Example 4.5. Convergence

Let $x_{t+1} = 0.5x_t + 10$ with $x_0 = 0$. The steady state is $10/(1 - 0.5) = 20$. The path is $x_1 = 10$, $x_2 = 15$, $x_3 = 17.5$, $x_4 = 18.75$, approaching 20. The formula agrees: $x_4 = 20 + 0.5^4(0 - 20) = 20 - 1.25 = 18.75$. Each period, half the remaining gap is closed.

Half-life. The number of periods needed to close half the gap solves $|a|^t = 1/2$, i.e. $t = \ln 0.5 / \ln |a|$. In Chapter 44 this gives the speed at which poor economies converge toward their steady states. Appendix A extends the analysis to second-order and stochastic difference equations and to their continuous-time counterparts.

Case: Moore's law: exponential growth on a log scale

In 1965 Gordon Moore, later a co-founder of Intel, observed that the number of components that could be placed economically on an integrated circuit had been doubling every year, and predicted that the trend would continue. In 1975 he revised the doubling time to about two years. For roughly half a century the prediction held remarkably well: the number of transistors on leading microprocessors rose from a few thousand in the early 1970s to tens of billions by the 2020s.

The arithmetic. A doubling every two years is a growth factor of $2^{1/2} = 1.414$ a year, or 41 per cent; in continuously compounded terms the rate is $\ln 2/2 = 0.35$ a year. Over 50 years it implies a factor of 2^{25} , about 34 million. No linear trend could produce such numbers, and a chart of transistor counts on an ordinary scale shows nothing for decades and then an almost vertical line. On a log scale, it is close to a straight line whose slope is the growth rate.

Why it matters for economics. The falling price of computing power drove much of the measured growth in productivity from the mid-1990s (Chapter 44), and it creates measurement problems: price indexes must account for the fact that each year's computer is far more powerful than last year's (Chapter 10). The slowing of Moore's law since the 2010s, as physical limits approached, is one reason economists debate whether technological progress will be as fast in the future. The lesson for data analysis is general: plot growing quantities on a log scale, where constant growth rates appear as straight lines and changes in growth rates as changes in slope.

Summary

- Exponent laws: $a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, $a^{-n} = 1/a^n$, $a^{1/n} = \sqrt[n]{a}$.
- Compound growth gives $x_t = x_0(1 + g)^t$; a nominal rate j compounded m times a year has effective rate $(1 + j/m)^m - 1$; continuous growth gives $x_0 e^{gt}$.
- Logarithms invert exponentials and turn products into sums; they solve for times and rates. Doubling time is about $70/(100g)$.
- Log differences approximate growth rates. Growth rates of products add, of ratios subtract, of powers scale.
- CAGR is the geometric-mean growth rate; short-period growth is annualized by compounding.
- Geometric series sum to $(1 - q^n)/(1 - q)$, and to $1/(1 - q)$ when infinite with $|q| < 1$.

- $x_{t+1} = ax_t + b$ has steady state $b/(1-a)$, solution $\bar{x} + a^t(x_0 - \bar{x})$, and is stable if $|a| < 1$; the half-life is $\ln 0.5 / \ln |a|$.

Key terms

laws of exponents; scientific notation; exponential function; effective annual rate; continuous compounding; the number e ; natural logarithm; rule of 70; log approximation; log scale; growth rules; compound annual growth rate; annualization; sequence; series; arithmetic sequence; geometric sequence; geometric series; convergence; difference equation; steady state; stability; half-life.

Exercises

Exercise 4.1 (Exponents). Simplify: (a) x^3x^{-5} ; (b) $(a^2b)^3/(ab^2)$; (c) $27^{2/3}$; (d) $K^{0.3}L^{0.7}$ when K and L are both multiplied by 4.

Exercise 4.2 (Compounding). Find the effective annual rate of 8 per cent compounded (a) quarterly, (b) daily (365 days), (c) continuously.

Exercise 4.3 (Logarithms). Solve for t : (a) $1.05^t = 3$; (b) $e^{0.04t} = 2$. (c) What constant annual growth rate turns 80 into 200 in 12 years?

Exercise 4.4 (Rule of 70). Using the rule of 70 and the exact formula, find the doubling times at growth rates of 1, 3 and 10 per cent. When is the rule least accurate, and why?

Exercise 4.5 (Growth rules). Labour productivity Y/L grows at 1.5 per cent and hours worked at 0.8 per cent. The GDP deflator rises 2.5 per cent. Approximately how fast do real GDP and nominal GDP grow?

Exercise 4.6 (CAGR). A country's GDP per person was 4,000 dollars in 2000 and 11,000 dollars in 2024. Find the compound annual growth rate. How many years would it take to double again at that rate?

Exercise 4.7 (Annualizing). A country's real GDP falls 2.1 per cent in one quarter and rises 1.4 per cent in the next. Annualize each rate and find the change over the half-year.

Exercise 4.8 (Series). Compute: (a) $\sum_{t=1}^{20} (3 + 2t)$; (b) $\sum_{t=0}^9 0.9^t$; (c) $\sum_{t=1}^{\infty} 0.8^t$.

Exercise 4.9 (Multiplier). Households spend 60 per cent of any extra income. Using a geometric series, find the total rise in spending that follows an initial rise of 50. How much of it occurs in the first five rounds (including the initial one)?

Exercise 4.10 (Difference equation). $x_{t+1} = 0.8x_t + 4$, $x_0 = 40$. Find the steady state, x_5 , and the half-life of deviations.

Exercise 4.11 (Stability). For each equation, find the steady state and state whether it is stable, and whether the path is monotonic or oscillating: (a) $x_{t+1} = 1.1x_t - 5$; (b) $x_{t+1} = -0.5x_t + 6$.

Solutions

Solution 4.1. (a) $x^{-2} = 1/x^2$. (b) $a^6b^3/(ab^2) = a^5b$. (c) $(\sqrt[3]{27})^2 = 9$. (d) $(4K)^{0.3}(4L)^{0.7} = 4^{0.3+0.7}K^{0.3}L^{0.7} = 4K^{0.3}L^{0.7}$: output quadruples.

Solution 4.2. (a) $1.02^4 - 1 = 8.24$ per cent. (b) $(1 + 0.08/365)^{365} - 1 = 8.328$ per cent. (c) $e^{0.08} - 1 = 8.329$ per cent.

Solution 4.3. (a) $t = \ln 3 / \ln 1.05 = 22.5$. (b) $t = \ln 2 / 0.04 = 17.3$. (c) $g = 2.5^{1/12} - 1 = 7.93$ per cent.

Solution 4.4. Rule: 70, 23.3 and 7 years. Exact: $\ln 2 / \ln 1.01 = 69.7$, $\ln 2 / \ln 1.03 = 23.4$, $\ln 2 / \ln 1.10 = 7.27$. The rule is least accurate at high growth rates, because $\ln(1 + g) \approx g$ deteriorates as g grows.

Solution 4.5. Real GDP = $(Y/L) \times L$: about $1.5 + 0.8 = 2.3$ per cent. Nominal GDP = deflator $\times$ real GDP: about $2.3 + 2.5 = 4.8$ per cent.

Solution 4.6. $(11,000/4,000)^{1/24} - 1 = 2.75^{1/24} - 1 = 4.30$ per cent a year. Doubling time $\ln 2 / \ln 1.0430 = 16.4$ years.

Solution 4.7. $0.979^4 - 1 = -8.14$ per cent; $1.014^4 - 1 = 5.72$ per cent. Over the half-year, $0.979 \times 1.014 - 1 = -0.73$ per cent.

Solution 4.8. (a) Terms run from 5 to 43: $20 \times (5 + 43) / 2 = 480$. (b) $(1 - 0.9^{10}) / 0.1 = (1 - 0.3487) / 0.1 = 6.513$. (c) $0.8 / (1 - 0.8) = 4$ (the series starts at $t = 1$).

Solution 4.9. Total = $50 / (1 - 0.6) = 125$. First five rounds: $50(1 - 0.6^5) / (1 - 0.6) = 50 \times 0.92224 / 0.4 = 115.3$, about 92 per cent of the total.

Solution 4.10. Steady state $4 / (1 - 0.8) = 20$. $x_t = 20 + 0.8^t(40 - 20)$, so $x_5 = 20 + 20 \times 0.32768 = 26.55$. Half-life = $\ln 0.5 / \ln 0.8 \approx 3.1$ periods.

Solution 4.11. (a) $\bar{x} = -5 / (1 - 1.1) = 50$; $|a| = 1.1 > 1$: unstable—any deviation from 50 grows. (b) $\bar{x} = 6 / 1.5 = 4$; $|a| = 0.5 < 1$: stable, oscillating because $a < 0$.

Derivatives and Unconstrained Optimization

After studying this chapter you should be able to

1. Explain limits and continuity, and define the derivative as the limit of an average rate of change.
2. Differentiate using the standard rules, including the chain rule and the rules for exponentials and logarithms.
3. Interpret first derivatives as marginal quantities and second derivatives as curvature; distinguish concave and convex functions.
4. Use linear and quadratic (Taylor) approximations, and find roots numerically by Newton's method.
5. Compute and interpret elasticities.
6. Solve unconstrained optimization problems in one variable, including boundary solutions.
7. Compute partial derivatives, total differentials and the slopes of level curves; apply homogeneity and Euler's theorem.
8. Solve unconstrained optimization problems in several variables and check second-order conditions.

A motivating question

A firm is producing 100 units a week. Should it produce 101? The answer depends on whether the extra revenue from the 101st unit exceeds its extra cost. Chapter 1 called this *thinking at the margin*, and argued that most economic decisions—how much to produce, to work, to save, to invest—are best analysed this way. Chapter 3 found the best output for a firm with a quadratic revenue function by locating the vertex of a parabola. That trick fails for almost every other function.

Calculus makes marginal reasoning precise and general. The derivative of a function measures the rate at which it changes—the *marginal* cost, revenue, utility or product—and an optimum is where a marginal benefit equals a marginal cost, that is, where a derivative is zero. This chapter develops the derivative and its uses for functions of one and several variables. Chapter 6 adds constraints.

5.1 Limits and Continuity

The **limit** of $f(x)$ as x approaches a , written $\lim_{x \rightarrow a} f(x) = L$, is the value L that $f(x)$ gets arbitrarily close to as x gets close to a (without necessarily equalling it). For example, $(x^2 -$

$9)/(x - 3)$ is undefined at $x = 3$, but for every other x it equals $x + 3$, so its limit as $x \rightarrow 3$ is 6. Limits also describe behaviour at infinity: $\lim_{q \rightarrow \infty} (F/q + c) = c$, which says that average cost in the linear cost model approaches marginal cost as output grows (Chapter 3).

A function is **continuous** at a if $\lim_{x \rightarrow a} f(x) = f(a)$: its graph has no break there. Tax schedules (Chapter 3) are continuous but have kinks; a fee that is charged only above a threshold produces a jump—a discontinuity—which creates strong incentives to stay just below the threshold. Much of calculus requires continuity, and economists look for bunching of behaviour at discontinuities as evidence of how people respond to incentives (Chapter 17).

5.2 Derivatives

5.2.1 Definition

For a nonlinear function, the slope changes along the curve. The slope of the straight line joining two points on the graph is the *average rate of change* over the interval. As the two points move closer, that line approaches the *tangent* to the curve, and its slope approaches the slope of the curve at the point.

Definition 5.1. Derivative

The **derivative** of f at x is

$$f'(x) = \frac{df}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h},$$

provided the limit exists. It is the slope of the tangent to the graph of f at x , and the instantaneous rate of change of f with respect to x . Its units are units of f per unit of x .

Worked Example 5.2. Derivative from the definition

Let $f(x) = x^2$. Then

$$\frac{f(x+h) - f(x)}{h} = \frac{x^2 + 2xh + h^2 - x^2}{h} = 2x + h.$$

As $h \rightarrow 0$, this tends to $2x$. So $f'(x) = 2x$: at $x = 3$ the slope of x^2 is 6.

A function is *differentiable* at x if the limit exists. Differentiable functions are continuous, but continuous functions need not be differentiable: at a kink, such as the point where a tax rate changes, the slopes from the left and right differ and the derivative does not exist.

In economics, the derivative is the precise version of the *marginal* concept of Chapter 1. If $C(q)$ is total cost, then $C'(q)$ is **marginal cost**: the rate at which cost rises with output. If $R(q)$ is revenue, $R'(q)$ is **marginal revenue**. If $u(c)$ is utility from consumption, $u'(c)$ is marginal utility. When output is measured in discrete units, the derivative approximates the extra cost of one more unit: $C(q+1) - C(q) \approx C'(q)$.

5.2.2 Rules of differentiation

One rarely uses the definition directly. The following rules, each provable from the definition, cover every function in this book. Let a and n be constants.

Rule	Statement	Example
Constant	$\frac{d}{dx}a = 0$	$\frac{d}{dx}7 = 0$
Power	$\frac{d}{dx}x^n = nx^{n-1}$ (any real n)	$\frac{d}{dx}\sqrt{x} = \frac{1}{2}x^{-1/2}$
Constant multiple	$(af)' = af'$	$(5x^3)' = 15x^2$
Sum	$(f + g)' = f' + g'$	$(x^2 + 3x)' = 2x + 3$
Product	$(fg)' = f'g + fg'$	$(xe^x)' = e^x + xe^x$
Quotient	$(f/g)' = (f'g - fg')/g^2$	$\left(\frac{x}{1+x}\right)' = \frac{1}{(1+x)^2}$
Chain	$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$	$\frac{d}{dx}(1+2x)^3 = 6(1+2x)^2$
Exponential	$\frac{d}{dx}e^x = e^x$	$\frac{d}{dx}e^{0.03x} = 0.03e^{0.03x}$
Logarithm	$\frac{d}{dx}\ln x = 1/x$ ($x > 0$)	$\frac{d}{dx}\ln(5x) = 1/x$
General base	$\frac{d}{dx}a^x = a^x \ln a$	$\frac{d}{dt}1.05^t = 1.05^t \ln 1.05$

The last row follows from the chain rule, since $a^x = e^{x \ln a}$. The logarithm example uses $\ln(5x) = \ln 5 + \ln x$, whose derivative is $0 + 1/x$.

Worked Example 5.3. Marginal cost

A firm's total cost of producing q units is $C(q) = 500 + 20q + 0.1q^2$. Marginal cost is

$$C'(q) = 20 + 0.2q.$$

At $q = 100$, marginal cost is 40. Check with discrete units: $C(101) - C(100) = (500 + 2020 + 1020.1) - (500 + 2000 + 1000) = 40.1$, close to 40. The difference, 0.1, arises because marginal cost is rising over the unit.

Growth rates as derivatives. If a variable $x(t)$ changes continuously over time, its rate of change is written $\dot{x} = dx/dt$, and its growth rate is $\dot{x}/x$. By the chain rule, $d \ln x / dt = (1/x)\dot{x}$: the growth rate is exactly the time derivative of the log. This is the fact behind Theorem 4.1 of Chapter 4.

5.2.3 Second derivatives, concavity and convexity

The derivative $f'(x)$ is itself a function of x ; its derivative is the **second derivative**, $f''(x)$. It measures how the slope changes.

Definition 5.4. Concave and convex functions

A twice-differentiable function is **concave** on an interval if $f''(x) \leq 0$ there (its slope is non-increasing; the graph bends downward) and **convex** if $f''(x) \geq 0$ (its slope is non-decreasing; the graph bends upward). It is *strictly* concave or convex if the inequality is strict. Equivalently, f is concave if the straight line joining any two points of its graph lies on or below the graph.

Economic interpretations:

- A utility function $u(c)$ with $u' > 0$ and $u'' < 0$ exhibits *diminishing marginal utility*: each extra unit of consumption adds less utility than the previous one. This concavity underlies risk aversion (Chapter 25) and consumption smoothing (Chapter 45).
- A production function $f(L)$ with $f'' < 0$ exhibits *diminishing marginal product*.
- A cost function with $C'' > 0$ has rising marginal cost.

- A bond's price is a convex function of its yield, a property called convexity that matters for interest-rate risk (Chapter 33).

5.3 Approximation

5.3.1 Linear and quadratic approximation

Near a point a , a differentiable function is close to its tangent line:

$$f(a + h) \approx f(a) + f'(a)h.$$

Adding a second-order term improves the approximation:

$$f(a + h) \approx f(a) + f'(a)h + \frac{1}{2}f''(a)h^2.$$

These are the first terms of the **Taylor expansion**; for many functions, adding the terms $f^{(k)}(a)h^k/k!$ for higher derivatives $f^{(k)}$ gives an infinite series that equals $f(a + h)$ exactly. For the logarithm around $a = 1$, with $f'(1) = 1$, $f''(1) = -1$ and $f'''(1) = 2$,

$$\ln(1 + g) = g - \frac{g^2}{2} + \frac{g^3}{3} - \dots$$

This justifies the approximation $\ln(1 + g) \approx g$ used in Chapter 4 and shows that its error is about $g^2/2$: at $g = 0.10$, $\ln 1.1 = 0.0953$, while $g - g^2/2 = 0.0950$.

In finance, the first-order term of a bond's price as a function of its yield gives *duration*, and the second-order term gives *convexity* (Chapter 33); in statistics, the approximation of a nonlinear function of an estimate by its tangent gives the *delta method*.

5.3.2 Newton's method

Many equations in economics and finance cannot be solved by algebra—the internal rate of return of a project (Chapter 9) and the yield of a bond (Chapter 33) are examples. **Newton's method** finds a root of $f(x) = 0$ by repeatedly replacing the function by its tangent line and solving the linear equation: from a guess x_k ,

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}.$$

Worked Example 5.5. Finding a rate numerically

Find the rate r at which $f(r) = -1,000 + 400/(1 + r) + 400/(1 + r)^2 + 400/(1 + r)^3 = 0$. The derivative is $f'(r) = -400/(1 + r)^2 - 800/(1 + r)^3 - 1,200/(1 + r)^4$. Starting from $r_0 = 0.05$: $r_1 = 0.0937$, $r_2 = 0.0970$, $r_3 = 0.09701$, after which the digits no longer change. The equation says that 1,000 invested now buys 400 at the end of each of three years; the root, 9.70 per cent, is the investment's rate of return (Chapter 9 calls it the internal rate of return). Spreadsheet functions such as IRR and RATE use methods of this kind.

Newton's method converges very fast near a root, but it can fail if the starting guess is poor or the derivative is close to zero; in practice it is combined with checks that the answer makes sense.

5.4 Elasticity

Slopes depend on units: the slope of a demand curve changes if we measure quantity in kilograms rather than tonnes. Economists therefore often measure responsiveness in percentage terms.

Definition 5.6. Elasticity

The **elasticity** of y with respect to x is the percentage change in y associated with a 1% change in x :

$$\varepsilon_{y,x} = \frac{dy/y}{dx/x} = \frac{dy}{dx} \cdot \frac{x}{y} = \frac{d \ln y}{d \ln x}.$$

The last form follows from the chain rule: $d \ln y = dy/y$ and $d \ln x = dx/x$. It shows that an elasticity is the slope of a relationship drawn in logs. Elasticities are unit-free. Because logs turn products into sums, the elasticity of a product is the sum of the elasticities: the elasticity of revenue PQ with respect to price is $1 + \varepsilon_{Q,P}$.

Worked Example 5.7. Elasticity of linear demand

For $Q = 1,200 - 400P$, $dQ/dP = -400$. The price elasticity of demand is

$$\varepsilon = -400 \cdot \frac{P}{1,200 - 400P}.$$

At $P = 1.50$, $Q = 600$: $\varepsilon = -400 \times 1.5/600 = -1$. At $P = 1$, $Q = 800$: $\varepsilon = -0.5$. At $P = 2$, $Q = 400$: $\varepsilon = -2$. Although the slope of a linear demand curve is constant, its elasticity is not: demand becomes more elastic (larger in absolute value) at higher prices. Revenue is maximized where $\varepsilon = -1$, at $P = 1.50$, because the elasticity of revenue with respect to price, $1 + \varepsilon$, is zero there—the same answer the vertex gave in Chapter 3.

Worked Example 5.8. Constant-elasticity demand

If $Q = AP^{-b}$ with $A, b > 0$, then $\ln Q = \ln A - b \ln P$, so $\varepsilon = d \ln Q / d \ln P = -b$ at every price. For example, $Q = 500P^{-1.5}$ has elasticity -1.5 everywhere: a 1% price rise reduces quantity by 1.5%. Such functions are often used in empirical work because the coefficient in a regression of $\ln Q$ on $\ln P$ is the elasticity (Chapter 16).

An elasticity is a local measure. For a large change in price, evaluate the whole function rather than multiplying a local elasticity by a large percentage change. Chapter 20 uses elasticities to classify demand and Chapter 24 uses them to derive a monopolist's markup.

5.5 Optimization in One Variable

5.5.1 First-order and second-order conditions

Economic agents maximize or minimize: consumers maximize utility, firms maximize profit and minimize cost.

Proposition 5.9. Conditions for a maximum

Let f be twice differentiable. If x^* is an interior maximum of f , then $f'(x^*) = 0$ (the **first-order condition**). If $f'(x^*) = 0$ and $f''(x^*) < 0$, then x^* is a strict local maximum (the **second-order condition**). If f is concave everywhere, any x^* with $f'(x^*) = 0$ is a global maximum. For a minimum, reverse the inequalities: $f''(x^*) > 0$, and convexity guarantees a global minimum.

The intuition: at the top of a smooth hill, the ground is flat ($f' = 0$); it is a top rather than a valley or a ledge if the slope is falling as you pass through it ($f'' < 0$). A point where $f'(x) = 0$ is a *stationary* or *critical* point; it may be a maximum, a minimum or neither (as for x^3 at zero).

Worked Example 5.10. Profit maximization

A firm faces inverse demand $P = 100 - 2q$ and has cost $C(q) = 50$ (a fixed cost of 50 and no variable cost; unrealistic, but it keeps the arithmetic transparent). Profit is revenue minus cost:

$$\Pi(q) = (100 - 2q)q - 50 = 100q - 2q^2 - 50.$$

First-order condition: $\Pi'(q) = 100 - 4q = 0$, so $q^* = 25$. Second-order condition: $\Pi''(q) = -4 < 0$, so this is a maximum (in fact global, since Π is concave). Price $P = 100 - 50 = 50$; profit $= 50 \times 25 - 50 = 1,200$.

The first-order condition has an economic reading: $100 - 4q$ is marginal revenue (the derivative of revenue $100q - 2q^2$) and marginal cost is zero, so the firm sets marginal revenue equal to marginal cost, as the marginal decision rule of Chapter 1 requires.

5.5.2 Boundaries and corners

The first-order condition applies to *interior* optima. If the variable is restricted to an interval— $q \geq 0$, or q no greater than capacity—the optimum may be at an end point, where f' need not be zero. A complete search compares the value of the function at every interior stationary point with its values at the end points.

Worked Example 5.11. A capacity limit

Revenue is $R(q) = 40q - 0.01q^2$ and cost $C(q) = 10q + 1,000$. Profit $\Pi(q) = 30q - 0.01q^2 - 1,000$ has $\Pi'(q) = 30 - 0.02q$, zero at $q = 1,500$, with $\Pi'' = -0.02 < 0$. If capacity is 1,200 units, $\Pi'(q) > 0$ throughout the feasible interval $[0, 1,200]$: profit rises all the way to the boundary, and the best feasible output is 1,200, where marginal revenue (16) still exceeds marginal cost (10). The rule “marginal revenue equals marginal cost” is useless if the output it prescribes is not feasible; the gap between them at the boundary measures the value of extra capacity (Chapter 6).

5.6 Functions of Several Variables**5.6.1 Partial derivatives**

Most economic functions have several arguments. For $z = f(x, y)$, the **partial derivative** with respect to x , written $\partial f / \partial x$ or f_x , is the derivative with respect to x treating y as a constant. It measures the effect of changing x *holding y fixed*—the mathematical form of *ceteris paribus*.

Worked Example 5.12. Marginal products

Let output be $Y = F(K, L) = K^{0.3}L^{0.7}$. The marginal product of labour is

$$F_L = \frac{\partial Y}{\partial L} = 0.7K^{0.3}L^{-0.3} = 0.7\frac{Y}{L},$$

and the marginal product of capital is $F_K = 0.3K^{-0.7}L^{0.7} = 0.3Y/K$. With $K = 100$ and $L = 100$, $Y = 100$, so $F_L = 0.7$ and $F_K = 0.3$. The second partial derivative $F_{LL} = 0.7(-0.3)K^{0.3}L^{-1.3} < 0$: the marginal product of labour diminishes as more labour is used with a fixed capital stock. The cross-partial $F_{LK} = 0.21K^{-0.7}L^{-0.3} > 0$: more capital raises the marginal product of labour. (For well-behaved functions, $F_{LK} = F_{KL}$: the order of differentiation does not matter. This is *Young's theorem*.)

The vector of partial derivatives $\nabla f = (f_x, f_y)$ is the **gradient**. It points in the direction in which f increases fastest.

5.6.2 The total differential and the chain rule

If x and y both change by small amounts, the change in $z = f(x, y)$ is approximately

$$dz = f_x dx + f_y dy.$$

This **total differential** adds up the separate effects. If x and y both depend on time, dividing by dt gives the multivariable chain rule $\dot{z} = f_x \dot{x} + f_y \dot{y}$. Applied to $\ln Y = \ln A + \alpha \ln K + (1 - \alpha) \ln L$, it gives the growth-accounting equation of Chapter 44: $g_Y = g_A + \alpha g_K + (1 - \alpha)g_L$.

5.6.3 Level curves and implicit differentiation

A **level curve** of f is the set of (x, y) along which f is constant: $f(x, y) = \bar{z}$. An indifference curve (Chapter 20) is a level curve of a utility function; an isoquant (Chapter 21) is a level curve of a production function. Along a level curve $dz = 0$, so $f_x dx + f_y dy = 0$, and the slope of the curve is

$$\left. \frac{dy}{dx} \right|_{f=\bar{z}} = -\frac{f_x}{f_y}, \quad f_y \neq 0.$$

This **implicit function rule** is how we will find the slopes of indifference curves and isoquants without solving for y . Chapter 1 used the same idea to find the slope of the production possibilities frontier.

5.6.4 Homogeneous functions**Definition 5.13. Homogeneity**

A function $f(x, y)$ is **homogeneous of degree k** if $f(tx, ty) = t^k f(x, y)$ for all $t > 0$.

The Cobb–Douglas function $x^a y^b$ is homogeneous of degree $a + b$: $(tx)^a (ty)^b = t^{a+b} x^a y^b$. For production functions, the degree of homogeneity describes *returns to scale*: degree 1 means that doubling all inputs doubles output (constant returns, Chapter 21). A demand function that depends only on relative prices and real income is homogeneous of degree 0 in prices and income: doubling all prices and income leaves demand unchanged (Chapter 20).

Proposition 5.14. Euler's theorem for homogeneous functions

If f is homogeneous of degree k and differentiable, then $xf_x + yf_y = kf(x, y)$.

Proof. Differentiate $f(tx, ty) = t^k f(x, y)$ with respect to t : $xf_x(tx, ty) + yf_y(tx, ty) = kt^{k-1}f(x, y)$. Set $t = 1$. $\square$

For a constant-returns production function, Euler's theorem says $KF_K + LF_L = F(K, L)$: if each factor is paid its marginal product, total payments exactly exhaust output. In Example 5.12, $100 \times 0.3 + 100 \times 0.7 = 100 = Y$. This result underlies the factor-share calculations of growth accounting (Chapter 44).

5.7 Optimization in Several Variables

To maximize $f(x, y)$, the first-order conditions are that both partial derivatives vanish:

$$f_x(x^*, y^*) = 0, \quad f_y(x^*, y^*) = 0.$$

At a peak, the surface is flat in every direction.

Worked Example 5.15. A two-product firm

A firm produces two goods in quantities x and y . Because the goods share a production line, profit is

$$\Pi(x, y) = 60x + 40y - x^2 - xy - y^2.$$

First-order conditions:

$$\Pi_x = 60 - 2x - y = 0, \quad \Pi_y = 40 - x - 2y = 0.$$

From the first, $y = 60 - 2x$. Substituting into the second, $40 - x - 120 + 4x = 0$, so $x^* = 80/3 \approx 26.67$ and $y^* = 60 - 160/3 = 20/3 \approx 6.67$. Profit at the optimum is $2800/3 \approx 933.33$.

5.7.1 Second-order conditions

Collect the second partial derivatives in the **Hessian matrix**

$$H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{pmatrix}.$$

A critical point is a strict local maximum if $f_{xx} < 0$ and $\det H = f_{xx}f_{yy} - f_{xy}^2 > 0$; a strict local minimum if $f_{xx} > 0$ and $\det H > 0$; a saddle point—a maximum in one direction and a minimum in another—if $\det H < 0$. If the maximum conditions hold everywhere, f is strictly concave and the critical point is the global maximum. In Example 5.15, $\Pi_{xx} = -2$, $\Pi_{yy} = -2$, $\Pi_{xy} = -1$, so $\det H = 4 - 1 = 3 > 0$ with $\Pi_{xx} < 0$ everywhere: the critical point is the global maximum.

Worked Example 5.16. Checking concavity of a production function

For $f(K, L) = K^{0.3}L^{0.6}$: $f_{KK} = -0.21K^{-1.7}L^{0.6} < 0$, $f_{LL} = -0.24K^{0.3}L^{-1.4} < 0$, $f_{KL} = 0.18K^{-0.7}L^{-0.4}$. Then $f_{KK}f_{LL} - f_{KL}^2 = (0.0504 - 0.0324)K^{-1.4}L^{-0.8} > 0$: the function is strictly concave (it has decreasing returns, $0.3 + 0.6 < 1$). With exponents summing

to one, the determinant is exactly zero: a constant-returns function is concave but not strictly so, which is why the scale of a competitive constant-returns firm is indeterminate (Chapter 21).

Advanced Analysis: Many variables

With n variables, the first-order conditions are $\nabla f(x^*) = 0$, and the second-order condition for a strict local maximum is that the Hessian be *negative definite*: $h'Hh < 0$ for every non-zero vector of changes h . A practical test is that the leading principal minors of H —the determinants of its top-left 1×1 , 2×2 , $\dots$, $n \times n$ submatrices—alternate in sign, beginning negative. Chapter 8 develops the matrix algebra behind these conditions.

Case: How high can a tax rate go? The top of the Laffer curve

A tax rate of zero raises no revenue; so does a rate of 100 per cent if nobody earns taxable income that is fully confiscated. Somewhere in between, revenue is maximized. The shape of revenue as a function of the tax rate is often called the *Laffer curve*, after the economist Arthur Laffer, who popularized it in the 1970s. Calculus locates its peak.

A simple model. Suppose taxable income responds to the net-of-tax rate $1 - \tau$ with constant elasticity e : $z(\tau) = z_0(1 - \tau)^e$. Revenue is $R(\tau) = \tau z_0(1 - \tau)^e$. By the product and chain rules,

$$R'(\tau) = z_0(1 - \tau)^{e-1}[(1 - \tau) - e\tau],$$

which is zero when $1 - \tau = e\tau$, i.e. at $\tau^* = 1/(1 + e)$. With $e = 0.25$, the revenue-maximizing rate is 80 per cent; with $e = 1$, 50 per cent. The whole answer hinges on the elasticity.

The evidence. Economists estimate the *elasticity of taxable income*—how much reported income responds to changes in marginal tax rates, through work effort, tax avoidance and income shifting—using changes in tax schedules as natural experiments (Chapter 17). A widely cited survey by Saez, Slemrod and Giertz (2012) concluded that the best estimates for the United States lie in a range of roughly 0.12 to 0.40. For the top tax rate, only income above the top threshold is taxed at that rate, which adds a term reflecting the distribution of top incomes; Diamond and Saez (2011), using an elasticity of 0.25, calculated a revenue-maximizing top rate of about 73 per cent (the formula becomes $1/(1 + ae)$, with $a \approx 1.5$ describing the thickness of the top tail).

The lessons. First, a first-order condition turns a political slogan into a quantitative question. Second, maximizing revenue is not the same as maximizing welfare: a government that cares about the efficiency costs of taxation (Chapter 22) or about incentives to innovate would choose lower rates, and one that places great weight on redistribution might accept rates near the peak. Third, the elasticity is not a fixed constant of nature: it depends on how easy the tax system makes avoidance, which is itself a policy choice.

Summary

- The derivative is the limit of the average rate of change: the slope of the tangent and the marginal quantity. Differentiable functions are continuous; kinks have no derivative.
- Power, product, quotient, chain, exponential and logarithm rules cover the functions used in this book. The growth rate is the time derivative of the log.
- Second derivatives measure curvature: concave functions have diminishing slopes,

convex functions increasing ones.

- Taylor approximations $f(a+h) \approx f(a) + f'(a)h + \frac{1}{2}f''(a)h^2$ explain the log approximation and underlie duration and convexity. Newton's method finds roots numerically.
- Elasticities are log-log slopes, unit-free and local; revenue is maximized where the price elasticity of demand is -1 .
- Interior optima satisfy first-order conditions; second-order conditions or concavity confirm a maximum; boundaries must be checked.
- Partial derivatives hold other variables constant; level curves have slope $-f_x/f_y$; Euler's theorem gives $xf_x + yf_y = kf$ for homogeneous functions.
- Multivariable optima set all partial derivatives to zero; the Hessian determines whether a critical point is a maximum, minimum or saddle.

Key terms

limit; continuity; derivative; differentiable; marginal cost; marginal revenue; power rule; product rule; quotient rule; chain rule; second derivative; concave; convex; Taylor expansion; linear approximation; Newton's method; elasticity; first-order condition; second-order condition; stationary point; boundary solution; partial derivative; gradient; total differential; level curve; implicit function rule; homogeneous function; returns to scale; Euler's theorem; Hessian matrix; saddle point; negative definite; elasticity of taxable income; Laffer curve.

Exercises

Exercise 5.1 (Derivatives). Differentiate: (a) $3x^4 - 2x + 7$; (b) $\ln(x^2 + 1)$; (c) xe^{-2x} ; (d) $(2x + 1)/(x - 3)$; (e) 100×1.04^t with respect to t .

Exercise 5.2 (Marginal and discrete). $C(q) = 100 + 3q + 0.5q^2$. Find marginal cost at $q = 10$ and the exact cost of the eleventh unit. Explain the difference.

Exercise 5.3 (Concavity). Classify each function as concave, convex or neither on $x > 0$: (a) $\sqrt{x}$; (b) $x^2 - 4x$; (c) $\ln x$; (d) $x^3 - 3x$.

Exercise 5.4 (Taylor approximation). Use a first- and a second-order approximation to estimate $\sqrt{1.1}$ from $f(x) = \sqrt{x}$ at $x = 1$. Compare with the true value.

Exercise 5.5 (Elasticity). Demand is $Q = 200 - 4P$. Compute the price elasticity at $P = 20$. At what price is demand unit-elastic, and what is revenue there?

Exercise 5.6 (Profit maximization). A firm sells at a fixed market price of 40 and has cost $C(q) = 100 + 4q + 0.5q^2$. Find the profit-maximizing output and the resulting profit. Verify the second-order condition. What if capacity is 30?

Exercise 5.7 (Newton's method). Apply two iterations of Newton's method to $f(x) = x^2 - 2$, starting from $x_0 = 1$. What number are you approximating?

Exercise 5.8 (Partial derivatives). For $U(x, y) = x^{0.4}y^{0.6}$, find U_x , U_y and the slope of the indifference curve through $(10, 20)$.

Exercise 5.9 (Homogeneity). Show that $F(K, L) = 2K^{0.5}L^{0.5}$ is homogeneous of degree one and verify Euler's theorem at $K = 16$, $L = 4$.

Exercise 5.10 (Two variables). Find and classify the critical point of $f(x, y) = 10x + 8y - x^2 + xy - y^2$.

Exercise 5.11 (Laffer). Using the simple model of the case, find the revenue-maximizing tax rate if $e = 0.5$. If the actual rate is 40 per cent, would a small rise in the rate raise or lower revenue?

Solutions

Solution 5.1. (a) $12x^3 - 2$. (b) Chain rule: $2x/(x^2 + 1)$. (c) Product rule: $e^{-2x} - 2xe^{-2x} = e^{-2x}(1 - 2x)$. (d) Quotient rule: $[2(x - 3) - (2x + 1)]/(x - 3)^2 = -7/(x - 3)^2$. (e) $100 \times 1.04^t \ln 1.04$.

Solution 5.2. $C'(10) = 3 + 10 = 13$. $C(11) - C(10) = (100 + 33 + 60.5) - (100 + 30 + 50) = 13.5$. The derivative is a local slope; because C is convex, marginal cost rises over the unit, so the discrete cost exceeds $C'(10)$ by $\frac{1}{2}C'' \times 1^2 = 0.5$.

Solution 5.3. (a) $f'' = -\frac{1}{4}x^{-3/2} < 0$: concave. (b) $f'' = 2$: convex. (c) $f'' = -1/x^2$: concave. (d) $f'' = 6x > 0$ for $x > 0$: convex on $x > 0$ (but not on the whole real line).

Solution 5.4. $f(1) = 1$, $f'(1) = 1/2$, $f''(1) = -1/4$. First order: $1 + 0.05 = 1.05$. Second order: $1.05 - \frac{1}{8}(0.01) = 1.04875$. True: 1.048809.

Solution 5.5. At $P = 20$, $Q = 120$ and $\varepsilon = -4 \times 20/120 = -0.667$. Unit elasticity requires $4P/(200 - 4P) = 1$, so $P = 25$, $Q = 100$, revenue 2,500—the maximum revenue. For a linear demand curve, unit elasticity is always at the midpoint of the curve.

Solution 5.6. $\Pi = 40q - 100 - 4q - 0.5q^2$. FOC: $36 - q = 0$, so $q^* = 36$ (price equals marginal cost $4 + q$). $\Pi = 1,440 - 100 - 144 - 648 = 548$. $\Pi'' = -1 < 0$: a maximum. With capacity 30, $\Pi' > 0$ on $[0, 30]$, so $q = 30$ and $\Pi = 1,200 - 100 - 120 - 450 = 530$.

Solution 5.7. $x_1 = 1 - (1 - 2)/2 = 1.5$; $x_2 = 1.5 - (2.25 - 2)/3 = 1.41667$. The root is $\sqrt{2} = 1.41421$.

Solution 5.8. $U_x = 0.4x^{-0.6}y^{0.6} = 0.4U/x$, $U_y = 0.6U/y$. Slope $= -U_x/U_y = -(0.4/0.6)(y/x) = -(2/3)(20/10) = -4/3$.

Solution 5.9. $F(tK, tL) = 2t^{0.5}K^{0.5}t^{0.5}L^{0.5} = tF(K, L)$. At $(16, 4)$: $F = 2 \times 4 \times 2 = 16$; $F_K = K^{-0.5}L^{0.5} = 2/4 = 0.5$; $F_L = K^{0.5}L^{-0.5} = 4/2 = 2$. $KF_K + LF_L = 8 + 8 = 16 = F$.

Solution 5.10. $f_x = 10 - 2x + y = 0$, $f_y = 8 + x - 2y = 0$. From the first, $y = 2x - 10$; then $8 + x - 4x + 20 = 0$, $x = 28/3$, $y = 26/3$. $f_{xx} = -2$, $f_{yy} = -2$, $f_{xy} = 1$: $\det H = 4 - 1 = 3 > 0$ with $f_{xx} < 0$: a global maximum.

Solution 5.11. $\tau^* = 1/1.5 = 66.7$ per cent. At 40 per cent, $R'(\tau)$ has the sign of $(1 - \tau) - e\tau = 0.6 - 0.2 = 0.4 > 0$: a small rise would raise revenue.

Constrained Optimization

After studying this chapter you should be able to

1. Solve equality-constrained problems by substitution and by the Lagrangian method, and interpret the tangency condition.
2. Interpret the Lagrange multiplier as a shadow price, and state and apply the envelope theorem.
3. Solve a cost-minimization problem and identify its multiplier with marginal cost.
4. Handle inequality and non-negativity constraints with the Kuhn–Tucker conditions and complementary slackness.
5. Formulate and solve linear programs graphically, explain why optima occur at corners, and compute shadow prices through duality.

A motivating question

A household cannot spend more than its income; a firm must deliver a contracted quantity at least cost; a factory's machines run at most 120 hours a week; a pension fund may not borrow. Almost every economic decision is a choice *subject to constraints*, and the constraints often matter more than the objective. Chapter 5 showed that an unconstrained optimum is where a derivative is zero. With a constraint, the derivative of the objective need not be zero at the best point: a consumer would always like more of everything, but income runs out.

This chapter develops the tools for constrained problems. The central idea, due to Joseph-Louis Lagrange, is that a constraint can be given a *price*. At the optimum, that price—the Lagrange multiplier—tells us exactly how much the objective would rise if the constraint were relaxed by one unit: what an extra unit of income is worth to a consumer, what an extra hour of machine time is worth to a factory, what the capital requirement costs a bank. These shadow prices are among the most useful numbers in economics.

6.1 The Problem and Two Methods

The typical problem is

$$\max_{x,y} f(x,y) \quad \text{subject to} \quad g(x,y) = c.$$

The function f is the *objective*; $g(x,y) = c$ is the *constraint*.

6.1.1 Method 1: substitution

If the constraint can be solved for one variable, substitute it into the objective and solve an unconstrained problem.

Worked Example 6.1. Utility maximization by substitution

A consumer has utility $U(x, y) = x^{0.5}y^{0.5}$, income 100, and faces prices $p_x = 2$, $p_y = 4$. The budget constraint is $2x + 4y = 100$, so $y = 25 - 0.5x$. Maximizing U is equivalent to maximizing $\ln U = 0.5 \ln x + 0.5 \ln y$ (since $\ln$ is increasing, it preserves the location of the maximum and simplifies the algebra):

$$V(x) = 0.5 \ln x + 0.5 \ln(25 - 0.5x).$$

$$V'(x) = \frac{0.5}{x} - \frac{0.25}{25 - 0.5x} = 0 \Rightarrow 0.5(25 - 0.5x) = 0.25x \Rightarrow 12.5 = 0.5x \Rightarrow x^* = 25.$$

Then $y^* = 25 - 12.5 = 12.5$. The consumer spends $2 \times 25 = 50$ on x and $4 \times 12.5 = 50$ on y : half of income on each. Utility $= \sqrt{25 \times 12.5} = \sqrt{312.5} \approx 17.68$.

6.1.2 Method 2: the Lagrangian

Substitution becomes unwieldy with many variables or constraints that cannot be solved explicitly. The Lagrangian method handles such cases and yields an extra piece of economic information.

Form the **Lagrangian**

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda [c - g(x, y)],$$

where λ is a new variable called the **Lagrange multiplier**. Then set all three partial derivatives to zero:

$$\begin{aligned}\mathcal{L}_x &= f_x - \lambda g_x = 0, \\ \mathcal{L}_y &= f_y - \lambda g_y = 0, \\ \mathcal{L}_\lambda &= c - g(x, y) = 0.\end{aligned}$$

The third condition simply restates the constraint. Dividing the first by the second gives

$$\frac{f_x}{f_y} = \frac{g_x}{g_y},$$

which says that at the optimum *the level curve of the objective is tangent to the constraint*: their slopes $-f_x/f_y$ and $-g_x/g_y$ are equal (the implicit function rule of Chapter 5).

Proposition 6.2. Lagrange's method

Suppose f and g are continuously differentiable, (x^*, y^*) solves the problem above, and $(g_x, g_y) \neq (0, 0)$ at (x^*, y^*) . Then there exists a number λ^* such that (x^*, y^*, λ^*) satisfies the three first-order conditions.

Why does this work? Consider moving along the constraint. At the optimum, no small movement along the constraint can raise f . If the level curve of f crossed the constraint rather than touching it, one could move along the constraint into a region of higher f . So at the optimum the two curves must be tangent, which is exactly the ratio condition. The first-order conditions are necessary, not sufficient; if f is concave and g is linear (or, more generally, the constraint set is convex), as in most problems in this book, they are also sufficient.

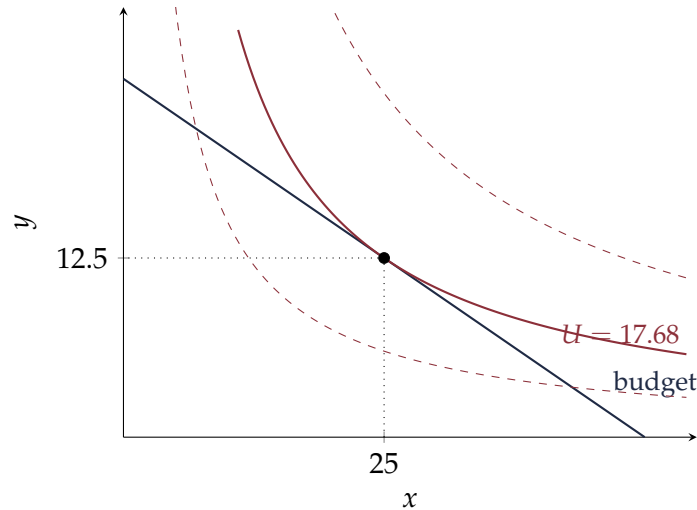


Figure 6.1: Constrained maximum in Example 6.1: the highest level curve of utility that touches the budget line is tangent to it at $(25, 12.5)$. Dashed curves are lower (inside the budget) and unattainable (outside) levels of utility.

Worked Example 6.3. The same consumer, by Lagrangian

$$\mathcal{L} = x^{0.5}y^{0.5} + \lambda(100 - 2x - 4y).$$

$$\mathcal{L}_x = 0.5x^{-0.5}y^{0.5} - 2\lambda = 0,$$

$$\mathcal{L}_y = 0.5x^{0.5}y^{-0.5} - 4\lambda = 0,$$

$$\mathcal{L}_\lambda = 100 - 2x - 4y = 0.$$

Dividing the first by the second: $y/x = 2/4$, so $x = 2y$. The constraint gives $2(2y) + 4y = 100$, so $y^* = 12.5$, $x^* = 25$, as before. From the first condition,

$$\lambda^* = \frac{0.5 (12.5/25)^{0.5}}{2} = \frac{0.5 \times 0.7071}{2} \approx 0.1768.$$

6.2 Shadow Prices and the Envelope Theorem

6.2.1 What the multiplier means

Proposition 6.4. Interpretation of the Lagrange multiplier

Let $V(c)$ be the maximum value of f subject to $g(x, y) = c$. Then, under standard regularity conditions,

$$V'(c) = \lambda^*.$$

The multiplier measures how much the optimal value of the objective rises when the constraint is relaxed by one unit.

In Example 6.3, income is the constraint constant, and $\lambda^* \approx 0.1768$ is the **marginal utility of income**: an extra unit of income would raise maximum utility by about 0.1768. We can check: with income m , the consumer spends half on each good, so $x = m/4$, $y = m/8$, and $V(m) = \sqrt{m^2/32} = m/\sqrt{32}$. Then $V'(m) = 1/\sqrt{32} \approx 0.1768$, exactly as the proposition says.

Lagrange multipliers are therefore **shadow prices**: they put a value on a constraint. In a firm's cost-minimization problem (below and Chapter 21), the multiplier on the output constraint is marginal cost. In the household's intertemporal problem (Chapter 45), the multiplier on the budget constraint is the marginal value of wealth. In an optimal policy problem, the multiplier tells us what a constraint costs society. A shadow price is a local, marginal valuation inside a model: it is not a market price that anyone will necessarily pay.

6.2.2 The envelope theorem

Theorem 6.4 is a special case of a general result about how optimal values respond to parameters.

Proposition 6.5. The envelope theorem

Let $V(a) = \max_x f(x, a)$ subject to $g(x, a) = 0$, with Lagrangian $\mathcal{L} = f(x, a) - \lambda g(x, a)$. Then

$$V'(a) = \left. \frac{\partial \mathcal{L}}{\partial a} \right|_{x=x^*(a), \lambda=\lambda^*(a)}.$$

For an unconstrained problem, $V'(a) = \partial f / \partial a$ evaluated at the optimum.

The indirect effects of a through the optimal choice $x^*(a)$ vanish because, at the optimum, the first-order conditions make the objective insensitive to small changes in x . Only the *direct* effect of the parameter remains. For the income constraint written as $g = p_x x + p_y y - m$, $\partial \mathcal{L} / \partial m = \lambda$, which gives Theorem 6.4. For a price, $\partial \mathcal{L} / \partial p_x = -\lambda x$: a rise in the price of x lowers maximum utility by the multiplier times the quantity bought, as if the consumer's income had fallen by x per unit rise in price. The envelope theorem underlies Roy's identity and Shephard's lemma in Chapters 20 and 21, and the menu-cost argument of Chapter 47: near an optimum, small mistakes in a choice variable cost very little.

6.2.3 Cost minimization

The same tools solve the problem of producing a given output at least cost, which is the mirror image of maximizing output for a given cost.

Worked Example 6.6. Least-cost production

A firm produces output $q = K^{0.5}L^{0.5}$ from capital K , rented at $r = 1$ per unit, and labour L , paid $w = 4$. To produce $q = 10$ at least cost, minimize $wL + rK$ subject to $K^{0.5}L^{0.5} = 10$. The Lagrangian is $\mathcal{L} = 4L + K + \lambda(10 - K^{0.5}L^{0.5})$. The first-order conditions $4 = \lambda 0.5K^{0.5}L^{-0.5}$ and $1 = \lambda 0.5K^{-0.5}L^{0.5}$ imply, on division, $K/L = 4$: the ratio of marginal products equals the ratio of input prices. With $K = 4L$, the constraint gives $2L = 10$, so $L = 5$, $K = 20$, and cost is $20 + 20 = 40$. The multiplier is $\lambda = 4 / (0.5 \times \sqrt{20/5}) = 4$. Producing one more unit would cost about 4: the multiplier is marginal cost. Indeed, for this technology the minimum cost of producing q is $C(q) = 2q\sqrt{wr} = 4q$, whose derivative is 4.

Advanced Analysis: Second-order conditions with constraints

For a constrained maximum, the objective need only curve downward *along the constraint*, not in every direction. With one constraint and two variables, the condition is that the

bordered Hessian

$$\bar{H} = \begin{pmatrix} 0 & g_x & g_y \\ g_x & \mathcal{L}_{xx} & \mathcal{L}_{xy} \\ g_y & \mathcal{L}_{xy} & \mathcal{L}_{yy} \end{pmatrix}$$

have a positive determinant at the critical point. Quasi-concave objectives—those whose upper level sets are convex, such as utility functions with convex indifference curves (Chapter 20)—with linear constraints always satisfy it.

6.3 Inequality Constraints

6.3.1 Binding and slack constraints

Budgets are really inequalities—spending must not *exceed* income—and quantities must be non-negative. When preferences are such that more is always better, the budget constraint binds and the equality form is correct. Other constraints may or may not bind: a borrowing limit matters only for a household that wants to borrow; a capacity limit matters only if the unconstrained optimum exceeds it (Example 5.11 of Chapter 5). A constraint that holds with equality at the optimum is *binding*; one that holds with strict inequality is *slack*.

Worked Example 6.7. A corner solution

Let $U = \ln(1+x) + y$, with $p_x = 1$, $p_y = 1$, income m . Substituting $y = m - x$, maximize $\ln(1+x) + m - x$. The derivative is $1/(1+x) - 1$, which is negative for every $x > 0$ and zero only at $x = 0$. So the consumer buys $x^* = 0$ and spends all income on y . Now let $p_x = 0.5$. The objective becomes $\ln(1+x) + (m - 0.5x)$ and its derivative is $1/(1+x) - 0.5$, zero at $x = 1$. If $m \geq 0.5$, the consumer buys $x^* = 1$ and $y^* = m - 0.5$. At a high enough price, the good is not bought at all. Such **corner solutions** are common: most consumers buy none of most goods.

6.3.2 The Kuhn–Tucker conditions

For $\max f(x, y)$ subject to $g(x, y) \leq c$, $x \geq 0$, $y \geq 0$, form $\mathcal{L} = f + \lambda(c - g)$. The **Kuhn–Tucker conditions** are

$$\begin{aligned} \mathcal{L}_x &\leq 0, \quad x \geq 0, \quad x\mathcal{L}_x = 0; & \mathcal{L}_y &\leq 0, \quad y \geq 0, \quad y\mathcal{L}_y = 0; \\ c - g &\geq 0, \quad \lambda \geq 0, \quad \lambda(c - g) = 0. \end{aligned}$$

The **complementary slackness** conditions say that either a variable is positive and its first-order condition holds with equality, or it is zero and the first-order condition may hold as a strict inequality; and either the constraint binds or its shadow price is zero. A constraint that does not bind is worth nothing at the margin. In Example 6.7 with $p_x = 1$, $\mathcal{L}_x = 1/(1+x) - \lambda$ and $\mathcal{L}_y = 1 - \lambda$. Since $y > 0$, $\lambda = 1$, and at $x = 0$, $\mathcal{L}_x = 1 - 1 = 0 \leq 0$: the conditions hold at $x = 0$.

Worked Example 6.8. A binding borrowing constraint

A household maximizes $\ln c_1 + \beta \ln c_2$ subject to its lifetime budget $c_1 + c_2/(1+r) = W$, where $W = y_1 + y_2/(1+r)$, and the constraint $c_1 \leq y_1$: it cannot borrow. (Chapter 9 explains why future amounts are divided by $1+r$.) Without the borrowing constraint, the Lagrangian conditions give $c_1 = W/(1+\beta)$. If $W/(1+\beta) \leq y_1$, the constraint is slack and its multiplier is zero. If $W/(1+\beta) > y_1$, it binds: $c_1 = y_1$, $c_2 = (1+r)(W - y_1) = y_2$, and its multiplier is $\mu = 1/y_1 - \beta(1+r)/c_2 > 0$ (from the first-order conditions $1/c_1 =$

$\lambda + \mu$ and $\beta/c_2 = \lambda/(1+r)$)—the utility value of being allowed to borrow one more unit. Liquidity constraints of this kind explain why many households spend transfers immediately (Chapter 45).

6.4 Linear Programming

6.4.1 The problem

When both the objective and the constraints are linear, the problem is a **linear program**. Chapter 3 met one: a factory makes products A and B , each unit of A needing 2 machine hours and 1 labour hour and each unit of B needing 1 machine hour and 3 labour hours, with 120 machine hours and 150 labour hours available. With profit contributions of 40 and 50 per unit, the problem is

$$\max 40A + 50B \quad \text{s.t.} \quad 2A + B \leq 120, \quad A + 3B \leq 150, \quad A, B \geq 0.$$

Proposition 6.9. Optimal corners

If a linear program has an optimal solution, then an optimal solution occurs at a corner (vertex) of the feasible set.

Sketch. The feasible set is a convex polygon. The level curves of a linear objective are parallel straight lines. Moving a level line in the direction of increasing value, the last point at which it touches the polygon is a corner, or an entire edge containing two corners if the objective is parallel to that edge. $\square$

Evaluating the objective at the four corners $(0,0)$, $(60,0)$, $(0,50)$ and $(42,36)$ gives 0, 2,400, 2,500 and 3,480. The optimum produces 42 units of A and 36 of B , using all machine and labour hours. For large problems with thousands of variables, the *simplex method*, devised by George Dantzig in 1947, moves from corner to corner, improving the objective at each step, and modern solvers handle millions of variables.

6.4.2 Shadow prices and duality

How much would one more machine hour be worth? Relaxing the machine constraint to 121 moves the optimal corner to the intersection of $2A + B = 121$ and $A + 3B = 150$: $A = 42.6$, $B = 35.8$, with profit 3,494. One more machine hour is worth 14.

Every linear program has a **dual** program that prices its constraints directly. Let y_1 and y_2 be the values of a machine hour and a labour hour. The dual asks for the cheapest valuation of the factory's resources such that no product is worth more than the resources it uses:

$$\min 120y_1 + 150y_2 \quad \text{s.t.} \quad 2y_1 + y_2 \geq 40, \quad y_1 + 3y_2 \geq 50, \quad y_1, y_2 \geq 0.$$

Since both products are made, both dual constraints bind: $2y_1 + y_2 = 40$ and $y_1 + 3y_2 = 50$, giving $y_1 = 14$ and $y_2 = 12$. The dual's minimum value is $120 \times 14 + 150 \times 12 = 3,480$ —exactly the primal's maximum. This is the *duality theorem*: at the optimum, the value of the output equals the shadow value of the resources used, and each resource's shadow price is its Lagrange multiplier. A resource that is not fully used has a shadow price of zero (complementary slackness again). A manager deciding whether to rent more machine time should pay up to 14 an hour for it; a product whose resources are worth more at shadow prices than its profit contribution should not be made.

Bridge. Linear programming is used to allocate capital among projects when funds are rationed (Chapter 32), to plan production and logistics, to construct portfolios subject to limits (Chapter 34), and in the input–output analysis of whole economies (Chapter 8). The general lesson—that constrained optima come with prices for the constraints—reappears in Chapter 22, where market prices in competitive equilibrium play exactly the role of shadow prices.

Case: The diet problem: from Stigler to the simplex method

In 1945 the economist George Stigler posed a question that sounds trivial: what is the cheapest diet that meets an adult’s nutritional needs? He took 77 foods, their 1939 prices and their content of nine nutrients, and the recommended daily allowances of each nutrient. The problem is a linear program: choose quantities of each food, with non-negative amounts, to minimize cost subject to nine inequality constraints (at least the recommended amount of each nutrient).

Heuristics versus optimization. With no algorithm for such problems, Stigler searched by clever trial and error and found a diet costing 39.93 dollars a year at 1939 prices, based mainly on wheat flour, evaporated milk, cabbage, spinach and dried navy beans. When Dantzig’s simplex method became available, the problem was solved exactly in 1947 by a team using desk calculators—a computation reported to have taken many person-days—and the true optimum was 39.69 dollars a year. Stigler’s heuristic had come within a quarter of a dollar, but only the algorithm could prove that nothing cheaper existed.

What the solution teaches. The optimum is a corner: only a handful of the 77 foods appear, one for each binding constraint, just as Theorem 6.9 predicts. The dual solution gives the shadow price of each nutrient—how much the minimum cost would rise if the requirement were raised slightly—and nutrients whose constraint is slack (the diet provides more than required) have a shadow price of zero. Stigler himself noted that no one would want to eat the diet: the model omitted taste and variety, which a person values but the objective function ignored. That is a lesson for every optimization in economics and finance: the solution is only as good as the objective and constraints written down.

Recognition. Leonid Kantorovich, who had developed linear programming for planning production in the Soviet Union in 1939, and Tjalling Koopmans, who applied it to transport and resource allocation, shared the 1975 Nobel prize “for their contributions to the theory of optimum allocation of resources.”

Summary

- Constrained optima equate the slope of the objective’s level curve with the slope of the constraint; the Lagrangian $f + \lambda(c - g)$ delivers this tangency together with the constraint.
- The multiplier is the shadow price of the constraint: $V'(c) = \lambda^*$. The envelope theorem gives the effect of any parameter on the optimal value as its direct effect at the optimum.
- In cost minimization, the ratio of marginal products equals the ratio of input prices and the multiplier is marginal cost.
- Inequality constraints are handled by the Kuhn–Tucker conditions: complementary slackness means that a slack constraint has a zero shadow price and a variable at zero may have a negative marginal value.

- Linear programs attain their optima at corners of the feasible set. The dual program prices the constraints; at the optimum, primal and dual values coincide.

Key terms

objective; constraint; substitution; Lagrangian; Lagrange multiplier; tangency condition; shadow price; marginal utility of income; envelope theorem; cost minimization; bordered Hessian; quasi-concavity; binding constraint; slack constraint; corner solution; Kuhn–Tucker conditions; complementary slackness; linear program; feasible set; corner theorem; simplex method; dual program; duality theorem.

Exercises

Exercise 6.1 (Lagrangian). Maximize $U = x^{1/3}y^{2/3}$ subject to $x + 2y = 90$. Find x^* , y^* and λ^* , and verify that λ^* equals the derivative of maximum utility with respect to income.

Exercise 6.2 (Substitution). A farmer has 400 metres of fencing to enclose a rectangular field along a straight river; no fence is needed on the river side. What dimensions maximize the area, and what is the shadow value of an extra metre of fencing?

Exercise 6.3 (Envelope theorem). In Example 6.3, use the envelope theorem to find the effect on maximum utility of a small rise in p_x . Check by computing V as a function of p_x .

Exercise 6.4 (Cost minimization). A firm with $q = K^{0.5}L^{0.5}$ faces $w = 9$ and $r = 4$. Find the cost-minimizing inputs to produce $q = 12$, the minimum cost and marginal cost.

Exercise 6.5 (Corner solution). A consumer has $U = 2\sqrt{x} + y$, income 10, $p_y = 1$. Find the optimal bundle when $p_x = 0.5$ and when $p_x = 0.05$. Explain what happens if income were only 5 with $p_x = 0.05$.

Exercise 6.6 (Kuhn–Tucker). A household with $\ln c_1 + \ln c_2$ has $y_1 = 40$, $y_2 = 176$, $r = 10$ per cent, and cannot borrow. Find its consumption and the multiplier on the borrowing constraint. What would it consume if it could borrow freely?

Exercise 6.7 (Linear programming). Maximize $3x + 2y$ subject to $x + y \leq 8$, $x + 3y \leq 12$, $x \leq 5$, $x, y \geq 0$. Find the optimal corner and the shadow prices of the three constraints.

Exercise 6.8 (Duality). In the factory example of the text, suppose the profit contribution of B rises to 90. Find the new optimal plan and the shadow prices. Which resource has become more valuable, and why?

Exercise 6.9 (Interpretation). A bank's regulator imposes a minimum ratio of capital to assets. In a model of the bank's profit maximization, the multiplier on this constraint is 0.04 per unit of capital. Explain what this number means and why it is not the bank's cost of issuing shares.

Solutions

Solution 6.1. Tangency: $\frac{(1/3)x^{-2/3}y^{2/3}}{(2/3)x^{1/3}y^{-1/3}} = \frac{y}{2x} = \frac{1}{2}$, so $y = x$. Then $3x = 90$, $x^* = y^* = 30$: one-third of income (30) is spent on x and two-thirds (60) on y , matching the exponents. $\lambda^* = MU_x/p_x = (1/3)(y/x)^{2/3} = 1/3$. With income m , $x = m/3$, $y = m/3$, so $V(m) = (m/3)^{1/3}(m/3)^{2/3} = m/3$ and $V'(m) = 1/3 = \lambda^*$.

Solution 6.2. With sides x (two of them, perpendicular to the river) and y (parallel), maximize xy subject to $2x + y = 400$. Substituting, $A = x(400 - 2x)$, maximized at $x = 100$, $y = 200$, area 20,000. With fencing F , $x = F/4$, $y = F/2$, $A = F^2/8$, so $A'(F) = F/4 = 100$ square metres per metre. The Lagrangian $xy + \lambda(400 - 2x - y)$ gives $y = 2\lambda$ and $x = \lambda$, so $\lambda = 100$: the multiplier is the shadow value of fencing, as Theorem 6.4 states.

Solution 6.3. $\partial \mathcal{L} / \partial p_x = -\lambda x = -0.1768 \times 25 = -4.42$. Directly: $x = m/(2p_x)$, $y = m/(2p_y)$, $V = m/(2\sqrt{p_x p_y})$; $\partial V / \partial p_x = -m/(4p_x^{3/2} p_y^{1/2}) = -100/(4 \times 2^{1.5} \times 2) = -4.42$.

Solution 6.4. Tangency: $K/L = w/r = 9/4$, so $K = 2.25L$; $q = 1.5L = 12$ gives $L = 8$, $K = 18$. Cost $= 72 + 72 = 144$. $C(q) = 2q\sqrt{wr} = 12q$, so marginal cost is 12 (and average cost is also 12, since returns to scale are constant).

Solution 6.5. Interior condition: $MU_x/p_x = MU_y/p_y$, i.e. $x^{-1/2}/p_x = 1$, so $x = 1/p_x^2$. At $p_x = 0.5$: $x = 4$, spending 2, $y = 8$. At $p_x = 0.05$: $x = 400$, costing 20, more than income; so $y = 0$ (a corner) and $x = 10/0.05 = 200$. With income 5 the corner is $x = 100$, $y = 0$: the non-negativity constraint on y binds.

Solution 6.6. $W = 40 + 176/1.1 = 200$; unconstrained $c_1 = W/2 = 100 > 40$, so the constraint binds: $c_1 = 40$, $c_2 = 176$. $\lambda = \beta/c_2 \times (1 + r) = 1.1/176 = 0.00625$; $\mu = 1/40 - 0.00625 = 0.01875 > 0$. Free borrowing: $c_1 = 100$, $c_2 = 110$.

Solution 6.7. At $x = 5$, the constraint $x + 3y \leq 12$ allows $y \leq 7/3$, tighter than $x + y \leq 8$ (which allows 3). Corners: $(0, 0)$, $(5, 0)$, $(5, 7/3)$, $(0, 4)$; the constraint $x + y \leq 8$ never binds. Values: 0, 15, $15 + 14/3 = 19.67$, 8. Optimum $(5, 7/3)$, value $59/3$. Shadow prices: $x + y \leq 8$ is slack, so its price is 0. The dual conditions for the two binding constraints, $y_2 + y_3 = 3$ (for x) and $3y_2 = 2$ (for y), give $y_2 = 2/3$ and $y_3 = 7/3$. Check: $12 \times \frac{2}{3} + 5 \times \frac{7}{3} = 8 + \frac{35}{3} = \frac{59}{3}$.

Solution 6.8. Corner values: $(60, 0)$: 2,400; $(0, 50)$: 4,500; $(42, 36)$: $1,680 + 3,240 = 4,920$. Still $(42, 36)$. Dual: $2y_1 + y_2 = 40$, $y_1 + 3y_2 = 90$: $y_1 = 6$, $y_2 = 28$ (value $720 + 4,200 = 4,920$). Labour, used intensively by the now more profitable product B , has become more valuable; machine time less so.

Solution 6.9. It is the rate at which the bank's maximum profit would rise if the requirement were relaxed so that one less unit of capital were needed—the marginal cost of the regulation to the bank's objective, inside the model. It is not the cost of issuing shares, which is a market transaction cost; the shadow price reflects everything in the model that makes capital constraining (the forgone profitable lending), and it is local: a large relaxation would not be worth 0.04 per unit throughout.

Integration and Continuous Models

After studying this chapter you should be able to

1. Find antiderivatives using the basic rules, substitution and integration by parts.
2. Interpret a definite integral as accumulated change and as an area, and state the fundamental theorem of calculus.
3. Recover totals from marginal quantities: variable cost from marginal cost, revenue from marginal revenue, stocks from flows.
4. Measure consumer and producer surplus as areas.
5. Value continuous streams of cash flows, including improper integrals over an infinite horizon.
6. Use integrals to compute probabilities from densities and inequality measures from Lorenz curves.
7. Solve simple differential equations for exponential, convergent and logistic growth, and approximate solutions numerically.

A motivating question

A firm knows its marginal cost at every level of output: each extra unit costs a little more than the last. How much does it cost, in total, to raise output from 1,000 to 1,500 units? A central bank knows the rate at which it has been buying bonds every day for a year. How large is its holding now? A government wants to know how much consumers would lose from a tax that raises a price. Each question gives a *rate of change* and asks for a *total*.

Differentiation turns totals into rates; integration turns rates back into totals. This is the second half of calculus. Its economic uses are as wide as those of the derivative: stocks accumulate from flows (Chapter 1), costs accumulate from marginal costs, welfare is measured by areas under demand curves, the value of a stream of income is an integral of discounted payments, and probabilities for continuous quantities are areas under a density. The chapter ends with differential equations, which describe how a quantity evolves when its rate of change depends on its current level.

7.1 Antiderivatives

A function F is an **antiderivative** of f if $F'(x) = f(x)$. Because the derivative of a constant is zero, antiderivatives are determined only up to an added constant: if F is one, so is $F + C$ for any constant C . The *indefinite integral* collects them:

$$\int f(x) \, dx = F(x) + C.$$

Reading the table of derivatives of Chapter 5 backwards gives the basic rules:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1), \quad \int \frac{1}{x} dx = \ln |x| + C, \quad \int e^{ax} dx = \frac{e^{ax}}{a} + C \quad (a \neq 0),$$

together with linearity: $\int [af(x) + bg(x)] dx = a \int f dx + b \int g dx$.

Substitution. The chain rule in reverse: if $u = g(x)$, then $\int f(g(x))g'(x) dx = \int f(u) du$. For example, $\int 2x(1+x^2)^3 dx = (1+x^2)^4/4 + C$, with $u = 1+x^2$.

Integration by parts. The product rule in reverse: $\int u dv = uv - \int v du$. For example, $\int xe^x dx = xe^x - \int e^x dx = (x-1)e^x + C$.

7.2 The Definite Integral

7.2.1 Accumulation and area

Suppose a quantity changes at rate $f(t)$ per unit of time. Over a short interval Δt , it changes by about $f(t)\Delta t$. Dividing an interval $[a, b]$ into many short pieces and adding the changes gives a *Riemann sum* $\sum f(t_k)\Delta t$. As the pieces shrink, the sum approaches a limit, the **definite integral**

$$\int_a^b f(t) dt.$$

Geometrically, it is the area between the graph of f and the horizontal axis from a to b , with area below the axis counted as negative. Economically, it is the *total change* accumulated from a rate: the integral of a rate in euros per year over years is euros.

Proposition 7.1. The fundamental theorem of calculus

If f is continuous on $[a, b]$ and F is any antiderivative of f , then

$$\int_a^b f(t) dt = F(b) - F(a).$$

Conversely, the function $A(x) = \int_a^x f(t) dt$ has derivative $A'(x) = f(x)$.

The theorem connects the two halves of calculus: to accumulate a rate, find a function whose rate of change it is, and take the difference of its values at the end points. The second statement says that the rate at which an accumulated total grows is the current rate—the stock–flow relation of Chapter 1 in continuous form. If a stock K changes at the rate of net investment $I(t)$, then $K(T) = K(0) + \int_0^T I(t) dt$.

7.2.2 Totals from marginals

Worked Example 7.2. Variable cost from marginal cost

Marginal cost is $MC(q) = 20 + 0.2q$ (Example 5.3 of Chapter 5). The cost of raising output from 100 to 150 units is

$$\int_{100}^{150} (20 + 0.2q) dq = [20q + 0.1q^2]_{100}^{150} = (3,000 + 2,250) - (2,000 + 1,000) = 2,250.$$

The total variable cost of the first 100 units is $\int_0^{100} MC \, dq = 2,000 + 1,000 = 3,000$. The integral recovers cost only up to a constant: it cannot tell us the fixed cost of 500, which does not vary with output and so leaves no trace in marginal cost.

Similarly, total revenue is the integral of marginal revenue from zero, and the total product of a variable input is the integral of its marginal product. Whenever a derivative has an economic meaning, so does the integral of it.

7.3 Surplus as Area

A downward-sloping inverse demand curve $P = D(Q)$ gives, for each unit, the maximum price some buyer would pay for it (Chapter 3). If the market price is P^* and Q^* units are bought, the buyer of each unit gains the difference between what she would have paid and what she pays. Adding over all units bought gives **consumer surplus**:

$$CS = \int_0^{Q^*} D(Q) \, dQ - P^* Q^*,$$

the area below the demand curve and above the price. **Producer surplus** is, symmetrically, the area above the supply (marginal cost) curve and below the price. Chapter 22 uses these areas to measure the gains from trade and the losses from taxes and price controls.

Worked Example 7.3. Consumer surplus

With inverse demand $P = 100 - 2Q$ and price 40, buyers purchase $Q^* = 30$. Then

$$CS = \int_0^{30} (100 - 2Q) \, dQ - 40 \times 30 = [100Q - Q^2]_0^{30} - 1,200 = 2,100 - 1,200 = 900,$$

the area of a triangle with base 30 and height $100 - 40 = 60$. For a curved demand function the triangle formula no longer applies but the integral does: with $P = 20e^{-0.1Q}$ and price 5, $Q^* = 10 \ln 4 = 13.86$ and $CS = 200(1 - e^{-0.1Q^*}) - 5Q^* = 150 - 69.3 = 80.7$.

7.4 Continuous Discounting

Chapter 4 showed that with continuous compounding at rate ρ , an amount grows by the factor $e^{\rho t}$ in t years. Conversely, a payment received at time t is worth $e^{-\rho t}$ times its amount today. If income arrives continuously at rate $c(t)$ per year, its present value is

$$PV = \int_0^T c(t) e^{-\rho t} \, dt.$$

For a constant stream c ,

$$PV = c \int_0^T e^{-\rho t} \, dt = c \left[\frac{-e^{-\rho t}}{\rho} \right]_0^T = \frac{c}{\rho} (1 - e^{-\rho T}).$$

Worked Example 7.4. Continuous versus year-end payments

A machine produces value at a constant rate of 100 euros a year for two years, and the continuously compounded discount rate is 5 per cent. Its present value is $100(1 -$

$e^{-0.1})/0.05 = 190.33$. If instead 100 were received at the end of each year, the value would be $100e^{-0.05} + 100e^{-0.10} = 185.61$. The difference reflects timing, not error: money received evenly through the year arrives, on average, earlier than money received at the year's end. Always use the timing that matches the cash flows (Chapter 9).

7.4.1 Improper integrals

As $T \rightarrow \infty$, $e^{-\rho T} \rightarrow 0$, so a constant perpetual stream is worth

$$\int_0^{\infty} c e^{-\rho t} dt = \frac{c}{\rho}.$$

An integral over an infinite interval is *improper*: it is defined as the limit of integrals over finite intervals, and exists only if that limit is finite. A stream growing at rate g , $c(t) = c_0 e^{gt}$, has present value $\int_0^{\infty} c_0 e^{-(\rho-g)t} dt = c_0/(\rho - g)$ if $g < \rho$, and no finite value if $g \geq \rho$. This is the continuous-time version of the growing perpetuity used in equity valuation (Chapters 9 and 37), and the condition $g < \rho$ is the reason valuations explode when an analyst assumes permanent growth close to the discount rate.

7.5 Probabilities and Inequality as Areas

7.5.1 Densities

For a quantity that can take any value in a range—a return, a waiting time, an income—probabilities attach to intervals rather than to single points. A **probability density** $f(x) \geq 0$ gives probabilities as areas: the probability that the quantity lies between a and b is $\int_a^b f(x) dx$, and the total area is $\int_{-\infty}^{\infty} f(x) dx = 1$. The probability of any exact value is zero. For example, if a delivery time is equally likely to be anywhere between 2 and 6 days, its density is $1/4$ on $[2, 6]$ and zero elsewhere, and the probability of a delivery within 3 days is $\int_2^3 \frac{1}{4} dx = 0.25$. Chapter 13 develops densities, expectations and the normal distribution.

7.5.2 The Lorenz curve and the Gini coefficient

Rank a population from poorest to richest. The **Lorenz curve** $L(p)$ gives the share of total income received by the poorest fraction p of the population. If everyone had the same income, $L(p) = p$, the 45-degree line of equality. With inequality, the curve sags below it. The **Gini coefficient** is twice the area between the line of equality and the Lorenz curve:

$$G = 2 \int_0^1 [p - L(p)] dp = 1 - 2 \int_0^1 L(p) dp.$$

It ranges from 0 (perfect equality) to 1 (one person has everything). For the Lorenz curve $L(p) = p^2$, $\int_0^1 p^2 dp = 1/3$ and $G = 1/3$; for $L(p) = p^3$, $G = 1 - 2/4 = 0.5$.

7.6 Differential Equations

A **differential equation** relates a function to its own derivatives. Many dynamic economic models take the form $\dot{x} = h(x)$: the rate of change of a quantity depends on its current level.

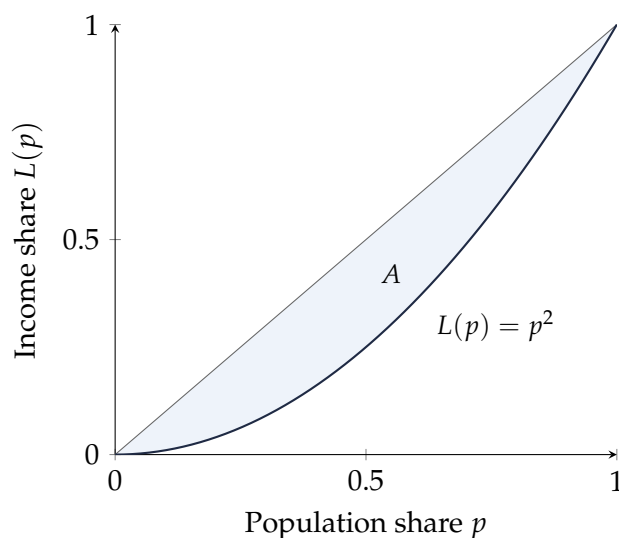


Figure 7.1: A Lorenz curve. The Gini coefficient is twice the shaded area A between the line of equality and the curve; here $G = 1/3$.

7.6.1 Exponential growth and convergence

Constant growth. If $\dot{x} = gx$, then $\dot{x}/x = g$, so $d \ln x / dt = g$; integrating, $\ln x(t) = \ln x_0 + gt$, and

$$x(t) = x_0 e^{gt}.$$

This is the continuous compound growth of Chapter 4.

Convergence to a steady state. If $\dot{x} = -\lambda(x - x^*)$ with $\lambda > 0$, the gap $x - x^*$ grows at the constant rate $-\lambda$, so

$$x(t) = x^* + (x_0 - x^*)e^{-\lambda t}.$$

The gap shrinks by the proportion λ per unit of time, and its half-life is $\ln 2 / \lambda$. This is the continuous-time counterpart of the difference equation $x_{t+1} = ax_t + b$ (Chapter 4); Chapter 44 uses it to find the speed at which economies converge to their steady states.

7.6.2 Logistic growth

New technologies, products and financial innovations typically spread slowly, then rapidly, then slowly again as the market saturates. The **logistic equation**

$$\dot{x} = kx \left(1 - \frac{x}{\bar{x}}\right)$$

captures this: growth is proportional to the current level (more users bring more imitators) and to the remaining room (fewer potential adopters remain). Its solution, found by separating variables, is

$$x(t) = \frac{\bar{x}}{1 + \frac{\bar{x} - x_0}{x_0} e^{-kt}},$$

an S-shaped curve. If adoption starts at 1 per cent of a market that saturates at 100 per cent, with $k = 0.5$ a year, adoption reaches half the market when $e^{-kt} = 1/99$, after $\ln 99 / 0.5 = 9.2$ years.

7.6.3 Numerical solution

Most differential equations used in modern economics cannot be solved with formulas and are solved numerically. The simplest method, due to Euler, steps forward using the current rate: $x(t + \Delta t) \approx x(t) + h(x(t))\Delta t$. For $\dot{x} = 0.05x$ from $x(0) = 100$ with steps of one year, it gives 105 and then 110.25 after two years, against the exact $100e^{0.1} = 110.52$: it is annual compounding in disguise. Smaller steps improve accuracy; refined methods improve it much faster. Appendix A discusses systems of differential equations and phase diagrams.

Case: Measuring inequality with the Gini coefficient

The Gini coefficient, proposed by the Italian statistician Corrado Gini in 1912, is the most widely reported measure of inequality. It is computed from the Lorenz curve of survey data on household income or consumption, approximating the integral by a sum over income groups.

Orders of magnitude. Estimates published by the World Bank place income or consumption Ginis at around 0.25–0.30 in the Nordic countries and several central European countries, somewhat above 0.3 in Germany, around 0.4 in the United States and Malaysia, and above 0.6 in South Africa, among the highest in the world; China's has been estimated at a little under 0.4. These figures should be compared with care, because they differ in what they measure.

What the integral hides. First, the *concept* matters: income before or after taxes and transfers, or consumption, which is typically more equal than income because households smooth consumption (Chapter 45). Redistribution through taxes and transfers lowers Ginis in many European countries by 0.15 or more. Second, the *unit*: household or per person, and how household size is adjusted for. Third, *coverage*: surveys under-represent the very rich, so Ginis computed from tax records are often higher. Fourth, a single number cannot distinguish *where* inequality lies: two Lorenz curves can cross—one society more unequal at the bottom, another at the top—and have the same Gini, because only the total area enters. Analysts therefore also report income shares of the top 1 or 10 per cent and ratios of percentiles (Chapter 11).

The lesson. An integral compresses a whole function into one number. That is its power—a single comparable measure—and its danger: the choice of what to integrate, and how, determines what the number means.

Summary

- Antiderivatives reverse differentiation; substitution and integration by parts reverse the chain and product rules.
- The definite integral accumulates a rate into a total and measures an area. By the fundamental theorem, $\int_a^b f = F(b) - F(a)$ for any antiderivative F .
- Totals follow from marginals: variable cost from marginal cost, stocks from flows. Fixed costs and initial stocks are not recovered by integration.
- Consumer surplus is the area under inverse demand and above the price; producer surplus the area above supply and below the price.
- A continuous stream c for T years discounted at ρ is worth $(c/\rho)(1 - e^{-\rho T})$; a perpetual growing stream is worth $c_0/(\rho - g)$ if $g < \rho$.
- Probabilities of continuous quantities are areas under a density; the Gini coefficient is

$$1 - 2 \int_0^1 L(p) \, dp.$$

- $\dot{x} = gx$ gives exponential growth; $\dot{x} = -\lambda(x - x^*)$ gives convergence with half-life $\ln 2/\lambda$; the logistic equation gives S-shaped diffusion. Euler's method steps forward numerically.

Key terms

antiderivative; indefinite integral; substitution; integration by parts; Riemann sum; definite integral; fundamental theorem of calculus; accumulation; consumer surplus; producer surplus; continuous discounting; improper integral; probability density; Lorenz curve; Gini coefficient; differential equation; exponential growth; convergence; half-life; logistic equation; diffusion; Euler's method.

Exercises

Exercise 7.1 (Antiderivatives). Find: (a) $\int (6x^2 - 4x + 3) \, dx$; (b) $\int e^{-0.05t} \, dt$; (c) $\int \frac{2}{x} \, dx$; (d) $\int x e^{2x} \, dx$.

Exercise 7.2 (Definite integrals). Evaluate: (a) $\int_0^3 2q \, dq$; (b) $\int_1^4 x^{-1/2} \, dx$; (c) $\int_0^{10} e^{-0.1t} \, dt$.

Exercise 7.3 (Cost). Marginal cost is $MC(q) = 5 + 0.04q$ and fixed cost is 2,000. Find total cost at $q = 500$ and the cost of increasing output from 500 to 600.

Exercise 7.4 (Surplus). Inverse demand is $P = 60 - 0.5Q$ and inverse supply is $P = 10 + 0.5Q$. Find the equilibrium, consumer surplus and producer surplus.

Exercise 7.5 (Stocks and flows). A central bank buys bonds at a rate that rises steadily from 0 to 12 billion a month over a year: $b(t) = t$ billion per month for t between 0 and 12 months. By how much does its bond holding rise over the year?

Exercise 7.6 (Continuous discounting). A licence produces income at 50,000 euros a year continuously for 10 years. Find its present value at a continuously compounded rate of 6 per cent. What would a perpetual licence be worth? And one whose income grows at 2 per cent a year forever?

Exercise 7.7 (Density). A density is $f(x) = 2x$ on $[0, 1]$ and zero elsewhere. Check that it integrates to one, and find $P(X \leq 0.5)$ and $P(X > 0.9)$.

Exercise 7.8 (Gini). Find the Gini coefficient for the Lorenz curve $L(p) = 0.4p + 0.6p^2$. Is this society more or less equal than one with $L(p) = p^2$?

Exercise 7.9 (Differential equations). (a) Solve $\dot{x} = -0.2(x - 50)$ with $x(0) = 10$. What is $x(5)$, and what is the half-life? (b) In the logistic model with $k = 0.8$ and $\bar{x} = 100$, starting from $x_0 = 2$, when does adoption reach 50? 90?

Exercise 7.10 (Euler's method). Use Euler's method with steps of 0.5 to approximate $x(1)$ for $\dot{x} = 0.1x$, $x(0) = 1,000$. Compare with the exact value and with steps of 1.

Solutions

Solution 7.1. (a) $2x^3 - 2x^2 + 3x + C$. (b) $-20e^{-0.05t} + C$. (c) $2 \ln |x| + C$. (d) By parts with $u = x$, $dv = e^{2x} \, dx$: $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$.

Solution 7.2. (a) $[q^2]_0^3 = 9$. (b) $[2x^{1/2}]_1^4 = 4 - 2 = 2$. (c) $10(1 - e^{-1}) = 6.32$.

Solution 7.3. $C(q) = 2,000 + 5q + 0.02q^2$. $C(500) = 2,000 + 2,500 + 5,000 = 9,500$. $C(600) - C(500) = (3,000 + 7,200) - (2,500 + 5,000) = 2,700$.

Solution 7.4. $60 - 0.5Q = 10 + 0.5Q$: $Q = 50$, $P = 35$. $CS = \frac{1}{2} \times 50 \times (60 - 35) = 625$. $PS = \frac{1}{2} \times 50 \times (35 - 10) = 625$.

Solution 7.5. $\int_0^{12} t \, dt = 72$ billion.

Solution 7.6. $50,000(1 - e^{-0.6})/0.06 = 50,000 \times 0.45119/0.06 = 375,990$. Perpetual: $50,000/0.06 = 833,333$. Growing: $50,000/0.04 = 1,250,000$.

Solution 7.7. $\int_0^1 2x \, dx = 1$. $P(X \leq 0.5) = 0.25$. $P(X > 0.9) = 1 - 0.81 = 0.19$.

Solution 7.8. $\int_0^1 (0.4p + 0.6p^2) \, dp = 0.2 + 0.2 = 0.4$, so $G = 1 - 0.8 = 0.2$: more equal than $L = p^2$ ($G = 1/3$).

Solution 7.9. (a) $x(t) = 50 - 40e^{-0.2t}$; $x(5) = 50 - 40e^{-1} = 35.3$; half-life $\ln 2/0.2 = 3.47$. (b) $x(t) = 100/(1 + 49e^{-0.8t})$. Reaches 50 when $49e^{-0.8t} = 1$: $t = \ln 49/0.8 = 4.86$. Reaches 90 when $49e^{-0.8t} = 1/9$: $t = \ln 441/0.8 = 7.61$.

Solution 7.10. Steps of 0.5: $1,000 \times 1.05^2 = 1,102.50$. Exact: $1,000e^{0.1} = 1,105.17$. One step of 1: 1,100. Halving the step roughly halves the error (5.17 against 2.67).

Vectors, Matrices and Linear Systems

After studying this chapter you should be able to

1. Represent bundles, portfolios and data as vectors, and compute dot products.
2. Add, multiply and transpose matrices, and state the rules of matrix algebra.
3. Solve linear systems by Gaussian elimination, determine when a unique solution exists, and compute inverses and determinants.
4. Apply Cramer's rule and the matrix form of the implicit function theorem to comparative statics.
5. Build and solve a Leontief input–output model.
6. Classify quadratic forms as definite or semidefinite, and connect definiteness to second-order conditions.
7. Compute eigenvalues, use them to determine the stability of linear dynamic systems, and find the steady state of a Markov chain.

A motivating question

When the price of steel rises, the cost of cars rises; but steel is made with machines that contain steel, and electricity generated in plants built with steel, and transported by trucks that contain steel. To find how much the price of cars must ultimately rise, one must follow every one of these interdependencies at once. Wassily Leontief showed in the 1930s that such problems, for an economy with dozens or hundreds of industries, reduce to a single equation—not in numbers but in *matrices*—and can be solved in one step.

Matrices are tables of numbers treated as single objects. They organize systems of many equations (Chapter 3), the second-order conditions of optimization with many variables (Chapter 5), the risks of portfolios of many assets (Chapter 14), regressions with many explanatory variables (Chapter 16) and the dynamics of economies with many interacting variables (Chapter 47). This chapter develops the algebra and its most important economic applications.

8.1 Vectors

A **vector** is an ordered list of numbers, written as a column:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

A consumption bundle, a set of prices, the weights of a portfolio and a column of observations in a data set are all vectors. Vectors of the same length are added element by element and multiplied by a number (a *scalar*) element by element.

The **dot product** (or inner product) of two vectors of the same length is

$$\mathbf{x} \cdot \mathbf{y} = \mathbf{x}'\mathbf{y} = \sum_{i=1}^n x_i y_i,$$

where $\mathbf{x}'$ (“ $\mathbf{x}$ transpose”) is $\mathbf{x}$ written as a row. Dot products are everywhere in economics: the cost of a bundle $\mathbf{q}$ at prices $\mathbf{p}$ is $\mathbf{p}'\mathbf{q}$; the return on a portfolio with weights $\mathbf{w}$ and asset returns $\mathbf{r}$ is $\mathbf{w}'\mathbf{r}$; a weighted mean (Chapter 2) is a dot product of weights and values. With weights 0.6 and 0.4 on assets returning 10 and -5 per cent, the portfolio returns $0.6 \times 0.10 + 0.4 \times (-0.05) = 4$ per cent.

The *length* of a vector is $\|\mathbf{x}\| = \sqrt{\mathbf{x}'\mathbf{x}}$. Two vectors are *orthogonal* if their dot product is zero. Orthogonality reappears in regression, where the residuals are orthogonal to every explanatory variable (Chapter 16).

8.2 Matrices

8.2.1 Operations

A **matrix** is a rectangular array of numbers; an $m \times n$ matrix A has m rows and n columns, with typical element a_{ij} in row i and column j . Matrices of the same size are added element by element. The product AB of an $m \times n$ matrix A and an $n \times p$ matrix B is the $m \times p$ matrix whose (i, j) element is the dot product of row i of A and column j of B :

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

The product is defined only when the number of columns of A equals the number of rows of B .

Property	Statement
Associative	$(AB)C = A(BC)$
Distributive	$A(B + C) = AB + AC$
Not commutative	in general $AB \neq BA$, even when both are defined
Transpose	A' swaps rows and columns; $(AB)' = B'A'$
Identity	I has ones on the diagonal and zeros elsewhere; $AI = IA = A$
Symmetric	$A = A'$, as for Hessians and covariance matrices

A system of m linear equations in n unknowns can be written compactly as $A\mathbf{x} = \mathbf{b}$, where A holds the coefficients, $\mathbf{x}$ the unknowns and $\mathbf{b}$ the constants.

8.3 Solving Linear Systems

8.3.1 Gaussian elimination

The systematic version of the elimination method of Chapter 3 is **Gaussian elimination**: use each equation in turn to eliminate one unknown from the equations below it, until the last equation contains one unknown; then substitute back upward. Three operations are allowed, none of which changes the solution: swapping two equations, multiplying an equation by a non-zero number, and adding a multiple of one equation to another.

Worked Example 8.1. Three equations

Solve $x + y + z = 6$, $2x - y + z = 3$, $x + 2y - z = 2$. Subtract twice the first equation from the second, and the first from the third:

$$-3y - z = -9, \quad y - 2z = -4.$$

Add one-third of the first of these to the second: $-\frac{7}{3}z = -7$, so $z = 3$. Then $-3y - 3 = -9$ gives $y = 2$, and $x = 6 - 2 - 3 = 1$. Check in the original third equation: $1 + 4 - 3 = 2$.

8.3.2 Rank, determinants and inverses

Elimination reveals how many equations are genuinely independent. The **rank** of a matrix is the number of linearly independent rows (equivalently, columns). For a square system $A\mathbf{x} = \mathbf{b}$ with n equations and n unknowns, there is a unique solution for every $\mathbf{b}$ exactly when A has full rank n . Then A has an **inverse** A^{-1} , with $A^{-1}A = AA^{-1} = I$, and

$$\mathbf{x} = A^{-1}\mathbf{b}.$$

A square matrix without an inverse is *singular*: its equations are redundant or contradictory, the parallel and coincident lines of Chapter 3.

The **determinant** $\det A$ is a single number that is non-zero exactly when A is invertible. For a 2×2 matrix,

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \det A = ad - bc, \quad A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

For larger matrices the determinant can be computed by expanding along a row (a sum of each element times the determinant of the smaller matrix left after deleting its row and column, with alternating signs) or, more efficiently, as the product of the diagonal elements after elimination. The coefficient matrix of Example 8.1 has determinant 7.

8.3.3 Cramer's rule and comparative statics

Cramer's rule gives each unknown as a ratio of determinants: $x_i = \det A_i / \det A$, where A_i is A with its i th column replaced by $\mathbf{b}$. It is inefficient for large systems but ideal for the small systems of comparative statics, because it shows directly how each unknown depends on each constant.

Worked Example 8.2. Two related markets

Two goods are substitutes. Demand and supply in the two markets are

$$Q_1 = 75 - 2p_1 + p_2, \quad Q_1 = -10 + 3p_1; \quad Q_2 = 60 + p_1 - 3p_2, \quad Q_2 = 5 + 2p_2.$$

Equating demand and supply in each market gives the system

$$\begin{pmatrix} 5 & -1 \\ -1 & 5 \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix} = \begin{pmatrix} 85 \\ 55 \end{pmatrix}.$$

The determinant is $25 - 1 = 24$. By Cramer's rule, $p_1 = (85 \times 5 + 55)/24 = 20$ and $p_2 = (5 \times 55 + 85)/24 = 15$, with $Q_1 = 50$ and $Q_2 = 35$. Now let demand for good 1 rise, raising the first constant from 85 to $85 + \Delta$. Cramer's rule gives $\partial p_1 / \partial \Delta = 5/24$ and $\partial p_2 / \partial \Delta = 1/24$: a rise of 12 in demand for good 1 raises its price by 2.5 and, because

buyers switch to the substitute, raises the price of good 2 by 0.5. Analysing either market alone would miss the second effect and understate the first.

Advanced Analysis: The implicit function theorem

Suppose a model consists of m equations $\mathbf{F}(\mathbf{x}, \mathbf{z}) = \mathbf{0}$ in m endogenous variables $\mathbf{x}$ and some exogenous variables $\mathbf{z}$. If the Jacobian matrix $\partial \mathbf{F} / \partial \mathbf{x}$ (the matrix of partial derivatives of each equation with respect to each endogenous variable) is non-singular at a solution, then near that solution $\mathbf{x}$ is a differentiable function of $\mathbf{z}$, with

$$\frac{\partial \mathbf{x}}{\partial \mathbf{z}} = - \left(\frac{\partial \mathbf{F}}{\partial \mathbf{x}} \right)^{-1} \frac{\partial \mathbf{F}}{\partial \mathbf{z}}.$$

This is the general basis of comparative statics: it generalizes the slope rule $dy/dx = -f_x/f_y$ of Chapter 5 to many equations. Applied to the goods- and money-market equations of the IS–LM model (Chapter 46), it shows that a rise in government purchases raises output by less than the simple spending multiplier, because it also raises the interest rate.

8.4 The Input–Output Model

Leontief's model divides an economy into n industries. Let a_{ij} be the amount of industry i 's output needed to produce one unit of industry j 's output, collected in the *technical coefficients matrix* A . If $\mathbf{x}$ is the vector of gross outputs and $\mathbf{d}$ the vector of final demands (consumption, investment, government purchases and exports), each industry's output must cover both the intermediate inputs other industries need and final demand:

$$\mathbf{x} = A\mathbf{x} + \mathbf{d} \implies \mathbf{x} = (I - A)^{-1}\mathbf{d}.$$

The matrix $(I - A)^{-1}$, the **Leontief inverse**, gives the total—direct and indirect—output of each industry needed to deliver one unit of final demand for each product.

Worked Example 8.3. A two-industry economy

Producing one unit of manufactures uses 0.2 units of manufactures and 0.4 units of services; producing one unit of services uses 0.3 units of manufactures and 0.1 units of services. Final demand is 100 for manufactures and 50 for services. Then

$$I - A = \begin{pmatrix} 0.8 & -0.3 \\ -0.4 & 0.9 \end{pmatrix}, \quad \det = 0.72 - 0.12 = 0.60, \quad (I - A)^{-1} = \begin{pmatrix} 1.5 & 0.5 \\ 0.667 & 1.333 \end{pmatrix},$$

and gross outputs are $x_1 = 1.5 \times 100 + 0.5 \times 50 = 175$ and $x_2 = 0.667 \times 100 + 1.333 \times 50 = 133.3$. Check manufactures: intermediate use $0.2 \times 175 + 0.3 \times 133.3 = 75$ plus final demand 100 equals 175. The first column of the inverse says that delivering one extra unit of manufactures to final users requires 1.5 units of manufacturing output and 0.667 units of services in total.

The same matrix answers the price question of the motivating example: if each industry's price covers its input costs plus value added per unit $\mathbf{v}$, prices satisfy $\mathbf{p}' = \mathbf{p}'A + \mathbf{v}'$, so $\mathbf{p}' = \mathbf{v}'(I - A)^{-1}$, and a cost shock in one industry spreads through all the others. Chapter 10 shows that value added, not gross output, is what GDP measures; the input–output model makes precise why summing gross outputs would double-count.

8.5 Quadratic Forms and Definiteness

A **quadratic form** is an expression $\mathbf{h}'A\mathbf{h} = \sum_i \sum_j a_{ij}h_ih_j$ for a symmetric matrix A . The matrix is

- **positive definite** if $\mathbf{h}'A\mathbf{h} > 0$ for every non-zero $\mathbf{h}$, and *positive semidefinite* if $\mathbf{h}'A\mathbf{h} \geq 0$;
- **negative definite** if $\mathbf{h}'A\mathbf{h} < 0$ for every non-zero $\mathbf{h}$, and *negative semidefinite* if ≤ 0 .

For a 2×2 symmetric matrix, positive definiteness is equivalent to $a_{11} > 0$ and $\det A > 0$, and negative definiteness to $a_{11} < 0$ and $\det A > 0$.

Definiteness organizes much of the book. A function is strictly concave where its Hessian is negative definite, which is the second-order condition for a maximum (Chapter 5). A covariance matrix Σ is always positive semidefinite, because the variance of any portfolio, $\mathbf{w}'\Sigma\mathbf{w}$, cannot be negative (Chapter 14). In regression, $X'X$ is positive definite exactly when the explanatory variables are not perfectly collinear, which is when the least-squares estimate is unique (Chapter 16).

8.6 Eigenvalues and Dynamics

8.6.1 Eigenvalues

A number λ is an **eigenvalue** of a square matrix A if $A\mathbf{v} = \lambda\mathbf{v}$ for some non-zero vector $\mathbf{v}$, an *eigenvector*. Multiplying an eigenvector by A merely rescales it. Eigenvalues solve the *characteristic equation* $\det(A - \lambda I) = 0$. For a 2×2 matrix this is the quadratic

$$\lambda^2 - (\text{tr } A)\lambda + \det A = 0,$$

where $\text{tr } A = a_{11} + a_{22}$ is the trace. The product of the eigenvalues equals the determinant and their sum equals the trace. A symmetric matrix is positive definite exactly when all its eigenvalues are positive.

8.6.2 Stability of linear systems

Eigenvalues govern the dynamics of systems $\mathbf{x}_{t+1} = A\mathbf{x}_t$, the many-variable version of the difference equations of Chapter 4. Each solution is a combination of terms $\lambda^t\mathbf{v}$, so the system converges to zero from every starting point if and only if every eigenvalue has modulus less than one. For example, $A = \begin{pmatrix} 0.5 & 0.2 \\ 0.1 & 0.4 \end{pmatrix}$ has trace 0.9 and determinant 0.18, so $\lambda = (0.9 \pm \sqrt{0.81 - 0.72})/2 = 0.6$ or 0.3 : the system is stable, and deviations die out at a rate governed mainly by the larger eigenvalue. Complex eigenvalues produce oscillations—cycles—whose damping depends on their modulus (Appendix A).

8.6.3 Markov chains

A **Markov chain** describes an individual moving between states with fixed probabilities. Chapter 45 models workers who move between employment and unemployment: each month, an employed worker loses her job with probability $s = 0.02$, and an unemployed worker finds one with probability $f = 0.30$. The *transition matrix*, whose rows give the probabilities of moving from each state, is

$$P = \begin{pmatrix} 0.98 & 0.02 \\ 0.30 & 0.70 \end{pmatrix}.$$

If π'_t is the row vector of the shares of the labour force employed and unemployed, then $\pi'_{t+1} = \pi'_t P$. A *stationary distribution* satisfies $\pi' = \pi' P$; here it gives an unemployment rate of $s/(s+f) = 0.02/0.32 = 6.25$ per cent. The eigenvalues of P are 1 (always, for a transition matrix) and $1 - s - f = 0.68$; the second tells us that deviations from the stationary distribution shrink by 32 per cent a month. Markov chains are also used for credit-rating migrations (Chapter 39) and for regime-switching models of the business cycle.

Advanced Analysis: Determinacy in rational-expectations models

In the New Keynesian model of Chapter 47, the output gap and inflation can be written $E_t \mathbf{z}_{t+1} = A \mathbf{z}_t$, where both variables are *jump* (non-predetermined) variables that adjust freely today in response to news. Blanchard and Kahn (1980) showed that a model of this kind has a unique stable solution if and only if the number of eigenvalues of A outside the unit circle equals the number of jump variables. With two jump variables, both eigenvalues must lie outside the unit circle; if one lies inside, there is a continuum of stable solutions and expectations can be self-fulfilling (indeterminacy). Working out this condition for the three-equation model gives the Taylor principle.

Case: Input–output tables: from Leontief to global supply chains

Wassily Leontief constructed the first input–output table of the U.S. economy in the 1930s, with data for 1919, and published a larger table for 1939 during the Second World War, when planners used it to estimate the steel, aluminium and rubber that military production would require. He received the Nobel prize in 1973 for developing the method.

In official statistics. Input–output analysis is now built into national accounting. Statistical agencies publish *supply and use tables* showing, industry by industry, the goods and services each industry buys from every other and what it sells to final users; the U.S. Bureau of Economic Analysis, Eurostat, the Department of Statistics Malaysia and others compile them as part of the System of National Accounts (Chapter 43). International organizations link national tables into inter-country tables, such as the OECD's Inter-Country Input–Output tables, which trace how a smartphone assembled in one country embodies components, design and services from many others.

Uses. Governments use the Leontief inverse to estimate the total employment and output supported by a change in final demand—the *input–output multiplier* of a new factory or a fall in exports. Trade economists use inter-country tables to measure *trade in value added*: a country's exports may contain a large share of imported inputs, so gross export figures overstate the domestic value it adds. During the COVID-19 pandemic and the semiconductor shortage of 2021, analysts used the tables to trace how shortages in one industry propagated to others.

Limits. The model assumes fixed technical coefficients—no substitution between inputs when relative prices change—and no capacity constraints. For small changes over short horizons, that is a reasonable approximation; for large shocks, such as a war or a sudden embargo, firms substitute and prices adjust, and general-equilibrium models (Chapter 22) are needed. The fixed-coefficient assumption is the Leontief production function of Chapter 21.

Summary

- Vectors hold bundles, prices, weights and data; dot products give costs, portfolio returns and weighted means.
- Matrix multiplication combines rows with columns; it is associative and distributive but not commutative. $(AB)' = B'A'$.
- Gaussian elimination solves linear systems. A square system has a unique solution when the matrix has full rank, equivalently a non-zero determinant; then $\mathbf{x} = A^{-1}\mathbf{b}$.
- Cramer's rule and the implicit function theorem give comparative statics in interdependent systems.
- In the input–output model, gross outputs are $(I - A)^{-1}\mathbf{d}$; the Leontief inverse captures direct and indirect requirements.
- Definiteness of quadratic forms gives second-order conditions, the non-negativity of portfolio variance and the uniqueness of least-squares estimates.
- Eigenvalues determine stability: $\mathbf{x}_{t+1} = A\mathbf{x}_t$ is stable if all eigenvalues have modulus below one. Markov chains have stationary distributions and converge at a rate set by the second eigenvalue.

Key terms

vector; scalar; dot product; transpose; orthogonal; matrix; matrix product; identity matrix; symmetric matrix; Gaussian elimination; rank; inverse; singular matrix; determinant; Cramer's rule; Jacobian; implicit function theorem; input–output model; technical coefficients; Leontief inverse; value added; quadratic form; positive and negative definite; eigenvalue; eigenvector; characteristic equation; trace; stability; Markov chain; transition matrix; stationary distribution; Blanchard–Kahn condition; supply and use tables; trade in value added.

Exercises

Exercise 8.1 (Dot products). A household buys the bundle $\mathbf{q} = (10, 4, 2)'$ at prices $\mathbf{p} = (3, 12, 25)'$. What does it spend? Next year prices are $(3.3, 12, 24)'$. What does the same bundle cost, and what is the percentage change?

Exercise 8.2 (Matrix products). Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & 0 \\ 1 & 2 \end{pmatrix}$. Compute AB , BA and $(AB)'$, and verify that $(AB)' = B'A'$.

Exercise 8.3 (Elimination). Solve $2x + y - z = 3$, $x - y + 2z = 7$, $3x + 2y + z = 10$.

Exercise 8.4 (Inverse). Find the inverse of $\begin{pmatrix} 4 & 3 \\ 2 & 2 \end{pmatrix}$ and use it to solve $4x + 3y = 18$, $2x + 2y = 10$.

Exercise 8.5 (Singularity). For which value of k is $\begin{pmatrix} 2 & k \\ 3 & 6 \end{pmatrix}$ singular? For that k , does $2x + ky = 4$, $3x + 6y = 6$ have no solution or infinitely many?

Exercise 8.6 (Comparative statics). In Example 8.2, supply of good 2 falls: its supply curve becomes $Q_2 = -7 + 2p_2$. Rewrite the system, find the new prices using Cramer's rule, and explain why the price of good 1 changes.

Exercise 8.7 (Input–output). In Example 8.3, final demand for services rises by 30. By how much must each industry's gross output rise? Why does manufacturing output rise even though final demand for manufactures is unchanged?

Exercise 8.8 (Definiteness). Classify $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, $\begin{pmatrix} -4 & 2 \\ 2 & -1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}$ as positive definite, negative (semi)definite or indefinite.

Exercise 8.9 (Eigenvalues). Find the eigenvalues of $\begin{pmatrix} 0.9 & 0.3 \\ 0.2 & 0.4 \end{pmatrix}$. Is the system $\mathbf{x}_{t+1} = A\mathbf{x}_t$ stable?

Exercise 8.10 (Markov chain). Each year, 90 per cent of firms rated investment grade keep that rating and 10 per cent are downgraded to speculative grade; 20 per cent of speculative-grade firms are upgraded and 80 per cent stay. Find the stationary distribution. If all firms start investment grade, what share is speculative grade after two years?

Solutions

Solution 8.1. $\mathbf{p}'\mathbf{q} = 30 + 48 + 50 = 128$. New cost $33 + 48 + 48 = 129$: a rise of 0.78 per cent—a fixed-basket index (Chapter 2).

Solution 8.2. $AB = \begin{pmatrix} 6 & 4 \\ 3 & 6 \end{pmatrix}$, $BA = \begin{pmatrix} 4 & 8 \\ 1 & 8 \end{pmatrix} \neq AB$. $(AB)' = \begin{pmatrix} 6 & 3 \\ 4 & 6 \end{pmatrix}$; $B'A' = \begin{pmatrix} 4 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 6 & 3 \\ 4 & 6 \end{pmatrix}$.

Solution 8.3. From the second, $x = 7 + y - 2z$. Substituting: $2(7 + y - 2z) + y - z = 3$ gives $3y - 5z = -11$; $3(7 + y - 2z) + 2y + z = 10$ gives $5y - 5z = -11$. Subtracting, $2y = 0$, so $y = 0$, $z = 11/5 = 2.2$, $x = 7 - 4.4 = 2.6$. Check: $5.2 + 0 - 2.2 = 3$.

Solution 8.4. $\det = 8 - 6 = 2$; inverse $= \frac{1}{2} \begin{pmatrix} 2 & -3 \\ -2 & 4 \end{pmatrix}$. $x = (2 \times 18 - 3 \times 10)/2 = 3$, $y = (-2 \times 18 + 4 \times 10)/2 = 2$.

Solution 8.5. $\det = 12 - 3k = 0$ at $k = 4$. Then the first equation is $2x + 4y = 4$, i.e. $x + 2y = 2$, and the second $3x + 6y = 6$, i.e. $x + 2y = 2$: the same line, infinitely many solutions.

Solution 8.6. The second equation becomes $60 + p_1 - 3p_2 = -7 + 2p_2$, i.e. $-p_1 + 5p_2 = 67$ (the constant is the demand intercept minus the supply intercept, so a fall in supply raises it from 55 to 67). Then $p_1 = (85 \times 5 + 67)/24 = 492/24 = 20.5$ and $p_2 = (5 \times 67 + 85)/24 = 420/24 = 17.5$. Good 2's price rises by 2.5; buyers switch toward the substitute, good 1, whose price rises by 0.5. Tracing each constant back to the economics fixes the sign of every comparative-static effect.

Solution 8.7. $\Delta\mathbf{x} = (I - A)^{-1}(0, 30)' = (0.5 \times 30, 1.333 \times 30)' = (15, 40)'$. Services use manufactured inputs (0.3 per unit), and those manufactures use services and manufactures in turn; the indirect requirements raise manufacturing output.

Solution 8.8. First: $a_{11} = 2 > 0$, $\det = 5 > 0$: positive definite. Second: $a_{11} = -4 < 0$, $\det = 4 - 4 = 0$: negative semidefinite (not definite). Third: $\det = 1 - 9 = -8 < 0$: indefinite.

Solution 8.9. Trace 1.3, determinant $0.36 - 0.06 = 0.30$: $\lambda = (1.3 \pm \sqrt{1.69 - 1.2})/2 = (1.3 \pm 0.7)/2 = 1.0$ or 0.3 . One eigenvalue equals 1, so the system is not stable in the strict sense: deviations along that eigenvector persist forever (a unit root, Chapter 18).

Solution 8.10. Stationary share speculative $= 0.1/(0.1 + 0.2) = 1/3$, investment grade $2/3$. After one year: $(0.9, 0.1)$; after two: speculative $= 0.9 \times 0.1 + 0.1 \times 0.8 = 0.17$.

The Time Value of Money

After studying this chapter you should be able to

1. Compute future and present values, and match discount rates to the timing, currency, inflation basis and risk of cash flows.
2. Value ordinary annuities, annuities due, deferred annuities, perpetuities and growing perpetuities.
3. Compute loan payments and build an amortization schedule that splits payments into interest and principal.
4. Compute net present value and internal rate of return, and state the NPV rule.
5. Price a bond from its cash flows and explain why bond prices and yields move in opposite directions.
6. Convert between nominal and real interest rates with the Fisher relation.
7. Solve a two-period intertemporal choice problem and interpret the Euler equation.

A motivating question

In November 2022 a single ticket won the largest lottery jackpot in history: an advertised 2.04 billion dollars. The winner could instead take a “cash option” of 997.6 million dollars. Was the jackpot really worth twice the cash option? No—the 2.04 billion was the sum of thirty payments spread over twenty-nine years, and a dollar received in 2051 is not worth a dollar received in 2022. Which offer was better depends on what the winner could earn on money in the meantime.

A unit of money today is worth more than a unit of money in the future, because money available today can be invested to earn interest, and because people generally prefer to consume sooner. This is the **time value of money**. Every decision that involves money at different dates—saving and borrowing, valuing bonds and shares, investing in machines and factories, pricing leases and pensions, choosing between a lump sum and an annuity—rests on it. This chapter builds the tools, which later chapters use in consumer theory (Chapter 20), accounting for bonds, leases and pensions (Chapters 29 and 30), corporate finance (Part V) and macroeconomics (Chapter 45).

9.1 Future and Present Value

9.1.1 Future value

If an amount PV is invested at interest rate i per period, compounded each period, then after n periods it grows to

$$FV = PV(1 + i)^n.$$

Compound interest means that interest earns interest (Chapter 4). With $PV = 1,000$, $i = 0.05$ and $n = 10$: $FV = 1,000 \times 1.05^{10} = 1,628.89$. Simple interest (interest only on the original amount) would give $1,000 + 10 \times 50 = 1,500$.

9.1.2 Present value

Reversing the calculation, the value today of an amount FV received in n periods is its **present value**:

$$PV = \frac{FV}{(1+i)^n}.$$

The rate i is called the **discount rate** and $1/(1+i)^n$ the **discount factor**. The present value of 1,000 received in 5 years at 6% is $1,000/1.06^5 = 747.26$: someone who can borrow and lend at 6% is indifferent between 747.26 today and 1,000 in five years, because 747.26 invested at 6% grows to exactly 1,000.

For a stream of cash flows $CF_1, CF_2, \dots, CF_n$ received at the ends of periods $1, \dots, n$:

$$PV = \sum_{t=1}^n \frac{CF_t}{(1+i)^t}.$$

9.1.3 Matching the rate to the cash flows

A discount rate is not a free-floating number. It must match the cash flows in five respects:

1. **Period.** A monthly cash flow needs a monthly rate; an annual effective rate r corresponds to a monthly rate of $(1+r)^{1/12} - 1$, not $r/12$ (Chapter 4).
2. **Timing.** Payments at the start of each period are discounted one period less than payments at the end.
3. **Currency.** Euro cash flows need a euro rate; ringgit cash flows a ringgit rate.
4. **Inflation.** Nominal cash flows need a nominal rate; cash flows in constant prices need a real rate (Section 9.6).
5. **Risk.** A risky cash flow is worth less than a safe one with the same expected value; Chapters 34 and 35 show how risk enters the discount rate.

A mismatch in any of these produces a precise-looking but meaningless number.

9.2 Annuities and Perpetuities

9.2.1 Ordinary annuities

An **annuity** pays a constant amount A at the end of each of n periods. Its present value is a geometric series (Theorem 4.3 of Chapter 4), which we can evaluate in closed form.

Proposition 9.1. Present value of an annuity

$$PV = \sum_{t=1}^n \frac{A}{(1+i)^t} = \frac{A}{i} \left[1 - \frac{1}{(1+i)^n} \right].$$

Proof. Let $v = 1/(1+i)$ and $S = v + v^2 + \dots + v^n$. Then $vS = v^2 + \dots + v^{n+1}$, so $S - vS = v - v^{n+1}$, giving $S = v(1 - v^{n+1})/(1 - v)$. Since $1 - v = i/(1+i)$, we have $v/(1 - v) = 1/i$, so $S = (1 - v^{n+1})/i$. Multiply by A . $\square$

Annuities due. If the payments arrive at the *start* of each period—rent, insurance premiums, many leases—each is received one period earlier, so the present value is the ordinary-annuity value multiplied by $(1 + i)$.

Deferred annuities. A stream that begins later is valued in two steps: value it as an annuity at the date one period before its first payment, then discount that value back to today. An annuity of 1,000 a year paid at the ends of years 6 to 10 is worth $1,000 \times (1 - 1.05^{-5})/0.05 = 4,329.48$ at the end of year 5 and $4,329.48/1.05^5 = 3,392.26$ today at 5 per cent.

9.2.2 Perpetuities

A **perpetuity** pays A forever. Letting $n \rightarrow \infty$, $v^n \rightarrow 0$ (because $v < 1$ when $i > 0$), so

$$PV_{\text{perpetuity}} = \frac{A}{i}.$$

A **growing perpetuity** pays A at the end of period 1, growing at rate $g < i$ thereafter. Its present value is

$$PV = \frac{A}{i - g},$$

obtained by the same geometric-sum argument with ratio $(1 + g)/(1 + i) < 1$. If $g \geq i$, the sum has no finite value. This formula is the basis of the dividend-discount model (Chapter 31) and of terminal values in company valuation (Chapter 37).

Worked Example 9.2. Annuity, perpetuity, growing perpetuity

At $i = 5\%$: an annuity of 100 per year for 10 years is worth $100 \times (1 - 1.05^{-10})/0.05 = 772.17$; paid at the start of each year it is worth $772.17 \times 1.05 = 810.78$; a perpetuity of 100 per year is worth $100/0.05 = 2,000$; a perpetuity starting at 100 and growing at 2% is worth $100/(0.05 - 0.02) = 3,333.33$. Notice how much of the perpetuity's value lies beyond year 10: $2,000 - 772.17 = 1,227.83$.

9.3 Loans and Amortization

The annuity formula also gives the constant payment A that repays a loan of L over n periods at rate i : set $L = PV$ and solve,

$$A = \frac{iL}{1 - (1 + i)^{-n}}.$$

A 200,000 mortgage over 25 years at 6% per year with annual payments requires $A = 0.06 \times 200,000 / (1 - 1.06^{-25}) = 15,645.34$ per year.

Each payment covers the interest due on the outstanding balance, and the rest repays principal. An **amortization schedule** tracks the split.

Worked Example 9.3. An amortization schedule

A loan of 10,000 at 6 per cent is repaid in three equal annual payments of $A = 600 / (1 - 1.06^{-3}) = 3,741.10$.

Year	Opening balance	Interest (6%)	Principal repaid	Closing balance
1	10,000.00	600.00	3,141.10	6,858.90
2	6,858.90	411.53	3,329.56	3,529.34
3	3,529.34	211.76	3,529.34	0.00

Interest falls and principal repayment rises over time, because interest is charged on a shrinking balance. The closing balance at any date equals the present value of the remaining payments at the loan rate—for example, $3,741.10/1.06 = 3,529.34$ at the end of year 2. The borrower's income statement records the interest column as an expense; the principal column reduces the liability on its balance sheet (Chapter 30).

9.4 Net Present Value and the Internal Rate of Return

A project that requires an outlay CF_0 today (negative) and generates cash flows $CF_1, \dots, CF_n$ has **net present value**

$$NPV = CF_0 + \sum_{t=1}^n \frac{CF_t}{(1+i)^t}.$$

Proposition 9.4. The NPV rule

If the firm can borrow and lend at rate i , undertaking a project with $NPV > 0$ raises the firm's value by NPV , and undertaking one with $NPV < 0$ lowers it.

The reasoning is that the owners could replicate a project's cash flows by borrowing or lending at i ; a project with positive NPV delivers them more cheaply. The **internal rate of return** (IRR) is the discount rate at which $NPV = 0$.

Worked Example 9.5. NPV and IRR

A machine costs 1,000 and yields 400 at the end of each of the next three years. At $i = 8\%$:

$$NPV = -1,000 + 400 \times \frac{1 - 1.08^{-3}}{0.08} = -1,000 + 1,030.84 = 30.84.$$

The project is worth undertaking. Its IRR solves $400(1 - (1+r)^{-3})/r = 1,000$; by Newton's method (Example 5.5 of Chapter 5), $r = 9.70\%$. Since the IRR exceeds the 8% cost of funds, the IRR rule agrees with the NPV rule here.

The two rules can disagree when comparing mutually exclusive projects of different scale or timing, and a project may have several IRRs if its cash flows change sign more than once. For these reasons the NPV rule is the reliable one. Chapter 32 develops capital budgeting in depth: which cash flows to include, how to compare projects, and how to value flexibility.

9.5 Bond Prices and Yields

A standard **bond** pays a fixed **coupon** c each period and its **face value** F at maturity n . If the market discount rate for such a bond is y , its price is

$$P_B = \sum_{t=1}^n \frac{c}{(1+y)^t} + \frac{F}{(1+y)^n} = \frac{c}{y} \left[1 - \frac{1}{(1+y)^n} \right] + \frac{F}{(1+y)^n} :$$

an annuity plus a single payment. The rate y that equates this sum to the observed market price is the bond's **yield to maturity**.

Worked Example 9.6. Bond price and yield

A three-year bond with face value 1,000 pays an annual coupon of 50 (a 5% coupon rate). If the market yield is 6%:

$$P_B = \frac{50}{1.06} + \frac{50}{1.06^2} + \frac{1,050}{1.06^3} = 47.17 + 44.50 + 881.60 = 973.27.$$

If the market yield falls to 4%, the price rises to 1,027.75. If the yield equals the coupon rate, 5%, the price equals face value, 1,000.

Key idea. Bond prices and yields move in opposite directions. When market interest rates rise, the fixed payments of an existing bond are discounted more heavily and its price falls. This single fact links monetary policy to bank balance sheets: when a central bank raises rates, the market value of banks' holdings of fixed-rate bonds falls (Chapters 30, 48 and 51). Chapter 33 measures how *much* a price moves when yields change.

9.6 Real and Nominal Interest Rates

The interest rates above are *nominal*: they measure the growth of money. What savers care about is the growth of *purchasing power*. If the nominal rate is i and inflation is π , one unit of money buys $1/P_0$ goods today and grows to $(1+i)$ units, which buy $(1+i)/P_1 = (1+i)/[P_0(1+\pi)]$ goods next period. The **real interest rate** r therefore satisfies

$$1+r = \frac{1+i}{1+\pi}, \quad \text{so} \quad r \approx i - \pi$$

when the rates are small. This is the **Fisher relation**. With $i = 5\%$ and $\pi = 3\%$, $r = 1.05/1.03 - 1 = 1.94\% \approx 2\%$. Chapter 50 develops the Fisher relation with expected rather than actual inflation.

Consistency matters in valuation: nominal cash flows (which rise with inflation) are discounted at nominal rates; real cash flows (in constant prices) at real rates. Done consistently, both give the same present value. Mixing them—discounting cash flows in constant prices at a nominal rate, say—understates value, and the error compounds with the horizon.

9.7 Intertemporal Choice: A First Look

Present value lets us write down a consumer's choice across time. This is the first appearance of a model that returns in Chapters 20, 45 and 47.

9.7.1 The two-period problem

A household lives for two periods. It receives income y_1 and y_2 , and can borrow or lend at real interest rate r . It chooses consumption c_1, c_2 . Whatever it saves in period 1, $s = y_1 - c_1$ (negative if it borrows), becomes $(1+r)s$ in period 2, which it consumes along with y_2 :

$$c_2 = y_2 + (1+r)(y_1 - c_1).$$

Dividing by $(1 + r)$ and rearranging gives the **intertemporal budget constraint**:

$$c_1 + \frac{c_2}{1+r} = y_1 + \frac{y_2}{1+r} \equiv W.$$

The present value of consumption must equal the present value of income, which we call lifetime wealth W .

Preferences are represented by lifetime utility $u(c_1) + \beta u(c_2)$, where u is increasing and concave and $\beta \in (0, 1)$ is the **discount factor**: the household values a unit of utility next period at β units of utility today.

9.7.2 The Euler equation

The Lagrangian (Chapter 6) is

$$\mathcal{L} = u(c_1) + \beta u(c_2) + \lambda \left[W - c_1 - \frac{c_2}{1+r} \right].$$

First-order conditions: $u'(c_1) = \lambda$ and $\beta u'(c_2) = \lambda/(1+r)$. Eliminating λ :

$$u'(c_1) = \beta(1+r) u'(c_2).$$

This is the **Euler equation**. It is a marginal condition. Giving up one unit of consumption today costs $u'(c_1)$ in utility; saving it yields $1+r$ units tomorrow, each worth $\beta u'(c_2)$ in today's utility. At the optimum, the two must be equal.

Worked Example 9.7. Two-period consumption with log utility

Let $u(c) = \ln c$, so $u'(c) = 1/c$. The Euler equation becomes $c_2 = \beta(1+r)c_1$. Substituting into the budget constraint: $c_1 + \beta c_1 = W$, so

$$c_1 = \frac{W}{1+\beta}, \quad c_2 = \frac{\beta(1+r)W}{1+\beta}.$$

With $y_1 = 100$, $y_2 = 110$, $r = 0.10$, $\beta = 0.9$: $W = 100 + 110/1.1 = 200$. Then $c_1 = 200/1.9 = 105.26$, and $c_2 = 0.9 \times 1.1 \times 105.26 = 104.21$. Check: $105.26 + 104.21/1.1 = 105.26 + 94.74 = 200$. The household borrows 5.26 in period 1, so that consumption in the two periods is almost equal even though income is not.

Bridge. The Euler equation is the microeconomic foundation of the modern theory of aggregate consumption (Chapter 45) and, through it, of the “IS curve” in the New Keynesian model (Chapter 47). A higher real interest rate raises $\beta(1+r)$, tilting consumption toward the future: this is the *intertemporal substitution* channel through which monetary policy affects spending.

Case: The 2.04-billion-dollar jackpot: lump sum or annuity?

The Powerball jackpot of November 2022 was advertised at 2.04 billion dollars, payable as an annuity: one payment immediately and 29 further annual payments, each 5 per cent larger than the one before. The alternative cash option was 997.6 million dollars.

The annuity's structure. If the first payment is P and payments grow at 5 per cent, the thirty payments sum to $P(1.05^{30} - 1)/0.05 = 66.44P$, a geometric series. Setting this equal to 2,040 million gives $P = 30.7$ million dollars, rising to about 126 million in the

final year.

The implied discount rate. The two offers are equally valuable at the discount rate r that makes the present value of the annuity equal to the cash option:

$$\sum_{k=0}^{29} \frac{30.7 \times 1.05^k}{(1+r)^k} = 997.6.$$

Solving numerically gives $r \approx 4.4$ per cent. At a discount rate of 4 per cent, the annuity is worth about 1,062 million—more than the cash option; at 5 per cent, well under it. In November 2022, yields on long-term U.S. Treasury bonds were close to 4 per cent, so the lottery was, in effect, offering the annuity on terms similar to buying government bonds. (The lottery funds the annuity by buying such bonds with the cash amount.)

What the calculation leaves out. Taxes on lottery winnings, which differ between a lump sum and an annuity; the winner's own opportunities to borrow and invest, which set her personal discount rate; the risk of dying before the payments end (the payments then go to the estate); and self-control—some winners prefer an annuity precisely because it prevents them from spending everything at once. Most winners choose the lump sum. The headline jackpot, a simple sum of payments at different dates, is the one number that means nothing economically.

Summary

- $FV = PV(1+i)^n$ and $PV = FV/(1+i)^n$. The discount rate must match the period, timing, currency, inflation basis and risk of the cash flows.
- Annuity: $PV = (A/i)[1 - (1+i)^{-n}]$; annuity due: multiply by $1+i$; perpetuity: A/i ; growing perpetuity: $A/(i-g)$ for $g < i$.
- Loan payments invert the annuity formula; amortization schedules split each payment into interest on the outstanding balance and principal.
- NPV sums discounted cash flows; positive-NPV projects create value. The IRR sets NPV to zero; NPV is the more reliable rule.
- A bond is an annuity plus a final payment; its price falls when its yield rises.
- The Fisher relation: $1+r = (1+i)/(1+\pi)$, so $r \approx i - \pi$. Discount nominal flows at nominal rates and real flows at real rates.
- The intertemporal budget constraint equates the present values of consumption and income; the Euler equation $u'(c_1) = \beta(1+r)u'(c_2)$ characterizes optimal saving.

Key terms

time value of money; compound interest; future value; present value; discount rate; discount factor; annuity; annuity due; deferred annuity; perpetuity; growing perpetuity; loan payment; amortization schedule; net present value; NPV rule; internal rate of return; bond; coupon; face value; yield to maturity; nominal interest rate; real interest rate; Fisher relation; intertemporal budget constraint; lifetime wealth; discount factor (utility); Euler equation.

Exercises

Exercise 9.1 (Present value). A lottery winner may take 1,000,000 now or 60,000 at the end of each year for 25 years. Which is worth more if the interest rate is (a) 4%, (b) 3%?

Exercise 9.2 (Timing). A lease requires payments of 12,000 a year for five years. Find its present value at 7 per cent if payments are made (a) at the end of each year, (b) at the start of each year.

Exercise 9.3 (Deferred annuity). A pension pays 20,000 a year at the end of each year from year 21 to year 40. What is it worth today at 5 per cent?

Exercise 9.4 (Loan schedule). A car loan of 24,000 at 9 per cent a year is repaid in four equal annual payments. Find the payment and build the amortization schedule.

Exercise 9.5 (Monthly payments). A mortgage of 300,000 is repaid monthly over 30 years. The quoted rate is 6 per cent a year, compounded monthly. Find the monthly payment and the effective annual rate.

Exercise 9.6 (NPV and IRR). A project costs 5,000 and returns 1,500 at the end of each of four years. Find its NPV at 8 and 10 per cent, and bracket its IRR.

Exercise 9.7 (Bond pricing). A two-year bond with face value 1,000 pays an 8% annual coupon. Find its price if the yield is 10%. Explain without calculation why the price is below 1,000.

Exercise 9.8 (Real and nominal). A project produces 1,000 a year in today's prices for three years; prices rise 3 per cent a year. The nominal discount rate is 8.15 per cent. Value the project (a) by discounting nominal cash flows at the nominal rate and (b) by discounting real cash flows at the real rate. Why do the answers agree?

Exercise 9.9 (Two-period choice). A household with $u(c) = \ln c$ and $\beta = 1$ receives $y_1 = 50$ and $y_2 = 165$; $r = 10\%$. Find c_1 , c_2 and saving in period 1. What would happen if the household could not borrow?

Exercise 9.10 (Growing annuity). Show that the present value of n payments, the first of A at the end of period 1 and each growing at rate g , discounted at $i \neq g$, is $\frac{A}{i-g} \left[1 - \left(\frac{1+g}{1+i} \right)^n \right]$. What happens as $n \rightarrow \infty$?

Solutions

Solution 9.1. (a) PV of the annuity $= 60,000(1 - 1.04^{-25})/0.04 = 937,325$, below 1,000,000: take the lump sum. (b) At 3%, $PV = 1,044,789$: take the annuity. The ranking depends on the interest rate because the annuity's payments are spread far into the future.

Solution 9.2. (a) $12,000(1 - 1.07^{-5})/0.07 = 49,202.37$. (b) $49,202.37 \times 1.07 = 52,646.54$.

Solution 9.3. Value at the end of year 20: $20,000(1 - 1.05^{-20})/0.05 = 249,244.21$. Value today: $249,244.21/1.05^{20} = 93,937.52$.

Solution 9.4. $A = 0.09 \times 24,000/(1 - 1.09^{-4}) = 7,408.05$. Year 1: interest 2,160.00, principal 5,248.05, balance 18,751.95. Year 2: interest 1,687.68, principal 5,720.37, balance 13,031.58. Year 3: interest 1,172.84, principal 6,235.21, balance 6,796.37. Year 4: interest 611.67, principal 6,796.37, balance 0.

Solution 9.5. Monthly rate 0.005, $n = 360$: $A = 0.005 \times 300,000/(1 - 1.005^{-360}) = 1,798.65$. Effective annual rate $1.005^{12} - 1 = 6.17$ per cent.

Solution 9.6. At 8 per cent: $-5,000 + 1,500 \times 3.31213 = -31.81$. At 10 per cent: $-5,000 + 1,500 \times 3.16987 = -245.20$. NPV is negative at both, so the IRR is below 8 per cent; at 7 per cent, $\text{NPV} = -5,000 + 1,500 \times 3.38721 = 80.82 > 0$, so the IRR lies between 7 and 8 per cent (it is about 7.71 per cent).

Solution 9.7. $P = 80/1.1 + 1,080/1.1^2 = 72.73 + 892.56 = 965.29$. The coupon rate (8%) is below the market yield (10%), so investors will pay less than face value; the gain from buying below face value compensates for the low coupon.

Solution 9.8. Real rate $1.0815/1.03 - 1 = 5.00$ per cent. (a) Nominal flows 1,030, 1,060.90, 1,092.73 discounted at 8.15 per cent: $952.38 + 907.03 + 863.84 = 2,723.25$. (b) $1,000 \times (1 - 1.05^{-3})/0.05 = 2,723.25$. They agree because each nominal factor $1.03^t/1.0815^t$ equals the real factor $1/1.05^t$.

Solution 9.9. $W = 50 + 165/1.1 = 200$. With $\beta = 1$, $c_1 = W/2 = 100$ and $c_2 = (1.1)(100) = 110$. Saving $= 50 - 100 = -50$: the household borrows 50 and repays 55 in period 2, leaving $165 - 55 = 110$. Without borrowing, it would consume $c_1 = 50$ and $c_2 = 165$: lower lifetime utility ($\ln 50 + \ln 165 \approx 9.02$ versus $\ln 100 + \ln 110 \approx 9.31$). Borrowing constraints are important in explaining why consumption tracks current income more closely than the model predicts (Chapter 45).

Solution 9.10. The payments are $A(1+g)^{t-1}$, so $PV = \frac{A}{1+i} \sum_{t=0}^{n-1} q^t$ with $q = (1+g)/(1+i)$. By Theorem 4.3, this is $\frac{A}{1+i} \cdot \frac{1-q^n}{1-q}$, and $(1+i)(1-q) = i-g$, giving the result. If $g < i$, $q^n \rightarrow 0$ and the value tends to the growing perpetuity $A/(i-g)$; if $g \geq i$, it grows without limit.

Measuring Firms and Economies

After studying this chapter you should be able to

1. State the accounting equation and trace the effect of business transactions on assets, liabilities and equity.
2. Construct a simple balance sheet, income statement and cash-flow statement from a list of transactions, and show how they articulate.
3. Distinguish accrual accounting from cash accounting and explain why profit and cash flow differ.
4. Define value added and explain why GDP counts only final goods and services.
5. Measure GDP by the production, expenditure and income approaches and show why the three must agree.
6. Distinguish GDP, GNI and net domestic product, and explain what GDP omits.
7. Construct Laspeyres, Paasche and Fisher price indexes; distinguish the CPI from the GDP deflator; and explain chain-weighting.
8. Derive the sector financial balances identity and interpret it.
9. Explain why official data are revised and why that matters.

A motivating question

In 2015, Ireland's official statistics office reported that the country's real GDP had grown by about 26 per cent in a single year. No Irish household experienced anything like a quarter more income; Irish factories did not suddenly produce a quarter more goods. The explanation lay in accounting: a handful of multinational firms had moved ownership of very valuable intellectual property—patents and software rights—into Irish subsidiaries, and aircraft-leasing companies had brought large fleets onto Irish balance sheets. Under international statistical rules, the output generated by those assets was now recorded as Irish production.

The episode illustrates the central theme of this chapter: the numbers economists and investors use are *constructed* according to rules, and understanding the rules is part of understanding the numbers. The chapter introduces two measurement systems that the rest of the book uses constantly—business accounting, which measures an individual firm, and national accounting, which measures a whole economy—and shows that they are built on the same logic.

10.1 Why Measurement Comes First

Economic theory makes predictions about quantities: output, profit, investment, prices, saving, debt. Before testing a theory or making a decision, we need to know how those quantities are measured. Consider the following statements:

- “The firm made a profit of 5 million last year.”
- “The economy grew by 2.5 per cent.”
- “Inflation was 3 per cent.”
- “The country is running a current-account deficit of 4 per cent of GDP.”

Each rests on definitions and conventions. A profit figure depends on when revenue is recognized and how assets are depreciated. A growth figure depends on which prices are used to value output. An inflation figure depends on which basket of goods is priced. A current-account figure depends on where a multinational’s profits are booked. Two people who disagree about the economy may, on inspection, be disagreeing about measurement.

Business accounting and national accounting share a common structure. Both distinguish stocks (balance sheets) from flows (income and expenditure). Both rely on double entry: every transaction is recorded twice, so the records balance. Both are subject to identities that hold by construction. This chapter introduces each system at the level needed for Parts III and VI; Part IV develops business accounting fully and Chapter 43 develops national accounts fully.

10.2 The Accounting Equation

10.2.1 Assets, liabilities and equity

Definition 10.1. Assets, liabilities and equity

An **asset** is a resource controlled by an entity as a result of past events, from which economic benefits are expected to flow. A **liability** is a present obligation of the entity, arising from past events, that is expected to require an outflow of economic resources. **Equity** is the residual interest in the assets after deducting all liabilities.

These definitions follow the conceptual framework of the International Accounting Standards Board (IASB); the U.S. Financial Accounting Standards Board (FASB) uses closely similar concepts. Chapter 27 examines them carefully. For now, three observations suffice. Assets include cash, amounts owed to the firm by customers (*accounts receivable*), goods held for sale (*inventory*) and equipment. Liabilities include loans and amounts the firm owes to suppliers (*accounts payable*). Equity is not a separate pile of resources; it is a claim—the owners’ claim on whatever is left after creditors are paid.

Because equity is *defined* as assets minus liabilities, the following holds by construction:

The accounting equation.

$$\text{Assets} \equiv \text{Liabilities} + \text{Equity}.$$

Every asset an entity controls is financed either by creditors (liabilities) or by owners (equity).

10.2.2 How transactions affect the equation

Every transaction changes at least two items in the equation, in a way that keeps it in balance. This is the principle of **double entry**. There are only four possibilities:

1. An asset increases and another asset decreases (buying equipment for cash).

2. An asset and a liability (or equity) both increase (borrowing cash from a bank).
3. An asset and a liability (or equity) both decrease (repaying a supplier).
4. A liability or equity item increases and another decreases (converting a loan into shares).

Income and expenses are changes in equity. Earning revenue increases equity (the owners are better off); incurring an expense decreases it. Profit for a period—revenue minus expenses—is therefore the change in equity arising from the firm's operations, as distinct from changes arising from owners putting money in or taking it out.

Worked Example 10.2. The first month of Harbour Coffee Ltd

Harbour Coffee Ltd starts business on 1 January. During January:

- (1) The owners buy shares for 50,000 cash.
- (2) The company borrows 20,000 from a bank.
- (3) It buys espresso machines and furniture for 30,000 cash.
- (4) It buys coffee, milk and cups for 8,000 on credit from a supplier.
- (5) It sells drinks for 15,000 cash; the supplies used cost 5,000.
- (6) It supplies an office on credit for 3,000, using supplies that cost 1,000.
- (7) It pays wages of 6,000.
- (8) It pays the supplier 4,000 of the amount owed.
- (9) The equipment is expected to last five years; one month's *depreciation* (the allocation of its cost to the period in which it is used) is 500.
- (10) One month's interest on the loan, 100, has accrued but will be paid in March.

Table 10.1 records each transaction's effect. Revenues and expenses are shown in the retained-earnings column because they change equity.

Table 10.1: Harbour Coffee: effects of January transactions on the accounting equation

	Assets				Liabilities			Equity	
	Cash	Receiv.	Invent.	Equip.	Loan	Payable	Int. pay.	Share cap.	Ret. earn.
(1)	+50,000							+50,000	
(2)	+20,000				+20,000				
(3)	-30,000			+30,000					
(4)			+8,000			+8,000			
(5)	+15,000		-5,000						+10,000
(6)		+3,000	-1,000						+2,000
(7)	-6,000								-6,000
(8)	-4,000					-4,000			
(9)				-500					-500
(10)							+100		-100
End	45,000	3,000	2,000	29,500	20,000	4,000	100	50,000	5,400

Total assets at 31 January are $45,000 + 3,000 + 2,000 + 29,500 = 79,500$. Liabilities plus equity are $20,000 + 4,000 + 100 + 50,000 + 5,400 = 79,500$. The equation balances, as it must: each row of the table balanced individually.

10.3 The Financial Statements

From the transaction record we can prepare the three principal financial statements. Chapter 27 treats their format and the rules behind them in detail.

10.3.1 The balance sheet

The **balance sheet** (or *statement of financial position*) lists assets, liabilities and equity at a date. It is a statement of stocks.

Harbour Coffee Ltd — Balance sheet at 31 January			
Assets		Liabilities and equity	
Cash	45,000	Accounts payable	4,000
Accounts receivable	3,000	Interest payable	100
Inventory	2,000	Bank loan	20,000
Equipment (net)	29,500	<i>Total liabilities</i>	<i>24,100</i>
		Share capital	50,000
		Retained earnings	5,400
		<i>Total equity</i>	<i>55,400</i>
Total assets	79,500	Total liabilities and equity	79,500

10.3.2 The income statement

The **income statement** (or *statement of profit or loss*) reports revenue and expenses over a period. It is a statement of flows.

Harbour Coffee Ltd — Income statement for January	
Revenue	18,000
Cost of goods sold	(6,000)
Gross profit	12,000
Wages	(6,000)
Depreciation	(500)
Operating profit	5,500
Interest expense	(100)
Net income (ignoring tax)	5,400

Revenue of 18,000 includes the 3,000 credit sale to the office, even though that customer has not yet paid. Cost of goods sold includes only the supplies *used* (6,000), not all the supplies bought (8,000). Depreciation of 500 is an expense even though no cash was paid in January. Interest of 100 is an expense even though it will be paid in March. These are the hallmarks of accrual accounting, discussed below.

10.3.3 The cash-flow statement

The **cash-flow statement** reports cash receipts and payments over the period, classified into three activities:

- **Operating activities:** cash flows from the firm's main business—customers, suppliers, employees.
- **Investing activities:** cash spent on or received from long-lived assets and investments.

- **Financing activities:** cash received from or paid to owners and lenders.

Harbour Coffee Ltd — Cash-flow statement for January	
Cash received from customers	15,000
Cash paid to suppliers	(4,000)
Cash paid to employees	(6,000)
Net cash from operating activities	5,000
Purchase of equipment	(30,000)
Net cash used in investing activities	(30,000)
Issue of shares	50,000
Bank loan	20,000
Net cash from financing activities	70,000
Net increase in cash	45,000
Cash at 1 January	0
Cash at 31 January	45,000

(Under IFRS, interest paid may be classified as operating or financing; under U.S. GAAP it is operating. No interest was paid in January, so the choice does not arise here.)

10.3.4 Articulation

The statements are not independent reports; they **articulate**, meaning they are linked by identities:

1. Net income for the period, less dividends, equals the change in retained earnings on the balance sheet: $0 + 5,400 - 0 = 5,400$.
2. The net change in cash in the cash-flow statement equals the change in cash on the balance sheet: $45,000 - 0$.
3. Net income and operating cash flow differ by precisely the accrual items. Starting from net income:

Net income	5,400
Add: depreciation (an expense that used no cash)	500
Less: increase in accounts receivable (revenue not yet received)	(3,000)
Less: increase in inventory (cash tied up in unsold stock)	(2,000)
Add: increase in accounts payable (purchases not yet paid for)	4,000
Add: increase in interest payable (expense not yet paid)	100
Net cash from operating activities	5,000

This reconciliation is called the **indirect method** of presenting operating cash flow; the first presentation, listing actual receipts and payments, is the **direct method**. Both give the same total. Chapter 31 uses this reconciliation to assess the *quality* of reported earnings: when net income persistently runs far ahead of operating cash flow, analysts ask why.

10.3.5 Accrual versus cash accounting

Definition 10.3. Accrual accounting

Under **accrual accounting**, revenue is recognized when the entity satisfies its obligation to the customer (when it delivers goods or services), and expenses are recognized when the resources are used up in generating revenue—regardless of when cash is received or paid. Under **cash accounting**, revenue and expenses are recorded only when cash changes hands.

Why accrual? Because cash timing can badly misrepresent performance. Harbour Coffee spent 30,000 on equipment in January; under cash accounting January would show a large loss, followed by years of apparently high profits as the equipment is used without further payment. Accrual accounting spreads the cost over the equipment's useful life through depreciation, matching it to the revenue the equipment helps generate. Cash, however, remains essential: a profitable firm that runs out of cash fails. The cash-flow statement exists because accrual profit, however informative, is not cash.

► Common error: “profit is cash”

Profit measures whether operations created value during the period, on accrual principles. Cash flow measures whether the firm's bank balance rose. A rapidly growing firm is often profitable but short of cash, because it must pay for inventory and extend credit to customers before it is paid. Many firms that fail are profitable on paper.

Check your reasoning

1. Harbour Coffee receives 2,000 in February from a customer for a catering job to be done in April. Is this February revenue?
2. Depreciation is sometimes called a “non-cash expense.” When was the cash associated with it paid?

Answers. (1) No. The service has not been performed; the firm has an obligation to the customer, recorded as a liability (“deferred revenue” or a “contract liability”). It becomes revenue in April. (2) When the equipment was bought (January, in the investing section). Depreciation allocates that earlier payment to the periods that benefit.

10.4 From Firms to the Nation: Value Added

10.4.1 The double-counting problem

To measure an economy's output, it might seem natural to add up the revenue of every firm. That would badly overstate output. Suppose a farmer grows wheat and sells it to a miller for 100; the miller grinds it into flour and sells it to a baker for 250; the baker bakes bread and sells it to households for 400. Total revenue is $100 + 250 + 400 = 750$, but the economy has produced only 400 of bread. The wheat is counted three times (once in each firm's sales) and the flour twice.

Definition 10.4. Intermediate and final goods

An **intermediate good** is used up in producing another good within the period (the wheat, the flour). A **final good** is purchased for final use: consumption, investment, government use or export (the bread).

Definition 10.5. Value added

A firm's **value added** is the value of its output minus the value of the intermediate goods and services it purchased from other firms:

$$\text{Value added} = \text{Output} - \text{Intermediate consumption.}$$

In the example the farmer's value added is 100 (assuming no purchased inputs), the miller's is $250 - 100 = 150$, and the baker's is $400 - 250 = 150$. Their sum, 400, equals the value of the final good. This is no coincidence: summing value added cancels every intermediate purchase, because each intermediate sale is revenue to one firm and an intermediate cost to another.

Definition 10.6. Gross domestic product

Gross domestic product (GDP) is the market value of all final goods and services produced within a country's borders in a given period. Equivalently, it is the sum of value added by all producers resident in the country, plus taxes on products less subsidies on products.

(The final clause adjusts for the fact that value added is usually measured at producers' prices, which exclude taxes on products such as sales taxes and VAT, while final expenditure is measured at purchasers' prices, which include them. We ignore product taxes in our examples.)

10.5 Three Ways to Measure GDP

Every unit of final spending is someone's revenue, and every unit of value added is paid out as someone's income—wages to workers, profits to owners, interest to lenders, rent to landlords, or taxes to government. GDP can therefore be measured three ways, and the three must agree.

1. **Production approach:** sum of value added across all producers.
2. **Expenditure approach:** sum of spending on final goods and services:

$$Y \equiv C + I + G + (X - M),$$

where C is household consumption, I is investment (business spending on equipment and structures, residential construction, and changes in inventories), G is government purchases of goods and services, X is exports and M is imports. Imports are subtracted because C , I , G and X include spending on goods produced abroad, which are not domestic production.

3. **Income approach:** sum of incomes generated in production: compensation of employees, plus gross operating surplus (profits, interest, rent and depreciation of firms), plus mixed income of self-employed people, plus taxes on production less subsidies.

Worked Example 10.7. Three measures of GDP in a small economy

Extend the bread chain. The farm pays wages of 60 and keeps a profit of 40; the mill pays wages of 100 and keeps a profit of 50; the bakery pays wages of 110 and keeps a profit of 40. In addition, a machine-maker builds an oven for 50, sold to the bakery as investment; it pays wages of 30 and keeps a profit of 20, and uses no purchased inputs. Households buy all the bread.

Producer	Output	Intermediate inputs	Value added	Wages	Profit
Farm	100	0	100	60	40
Mill	250	100	150	100	50
Bakery	400	250	150	110	40
Machine-maker	50	0	50	30	20
Total	800	350	450	300	150

Production: total value added = 450. *Expenditure:* $C = 400$ (bread), $I = 50$ (oven); $Y = 450$. *Income:* wages 300 plus profits 150 = 450.

The oven is a final good—it is investment—even though the bakery uses it in production, because it is not *used up* within the period. Only its depreciation is. This is the distinction between a capital good and an intermediate good.

10.5.1 Inventories and the identity

Why must the expenditure measure equal the production measure even if firms produce goods that nobody buys? Because unsold output is counted as **inventory investment**, part of I . If the bakery produced bread worth 420 but sold only 400, the extra 20 would be recorded as an increase in inventories: GDP would be 470 by production, and $C + I = 400 + (50 + 20) = 470$ by expenditure. The expenditure identity holds by construction because inventory change is defined as the difference between production and sales. This construction is exactly what makes $Y \equiv C + I + G + X - M$ an identity rather than an equilibrium condition, as Chapter 1 emphasized.

10.5.2 The circular flow

The equality of the three measures can be pictured as a **circular flow** (Figure 10.1). Households supply labour and capital to firms and receive incomes; they spend on goods and services, which becomes firms' revenue. Some income is saved, taxed or spent on imports; these *leakages* from the flow are matched, in the national accounts, by *injections*—investment, government purchases and exports. Financial markets channel saving into investment. Part VI builds on this picture.

10.6 GDP, GNI and What GDP Leaves Out**10.6.1 Domestic versus national**

GDP measures production *within a country's borders*, regardless of who owns the factors of production. **Gross national income** (GNI) measures income received by a country's *residents*, wherever it is earned:

$$\text{GNI} = \text{GDP} + \text{primary income received from abroad} - \text{primary income paid abroad}.$$

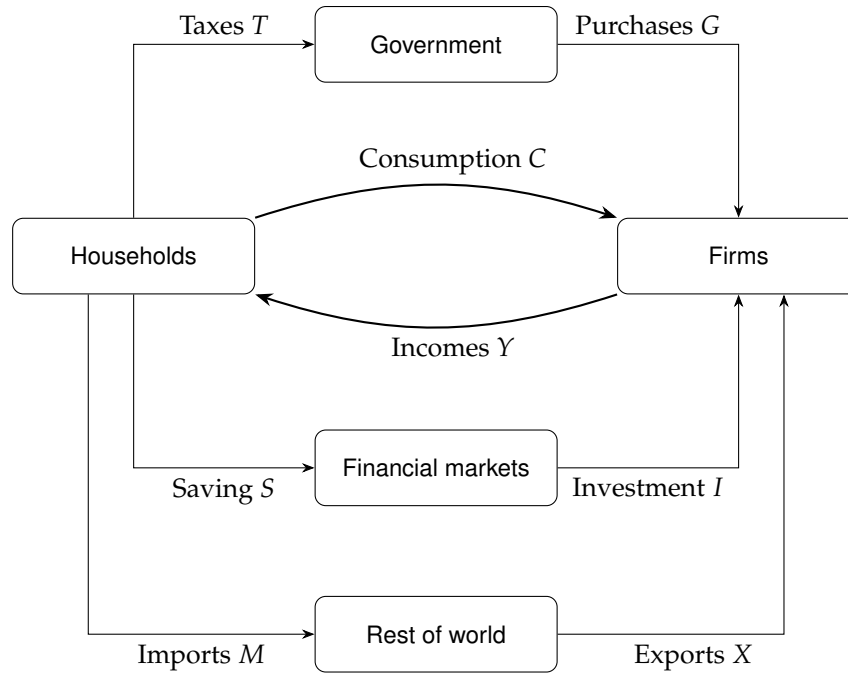


Figure 10.1: The circular flow of income and expenditure. Leakages (saving, taxes, imports) are matched by injections (investment, government purchases, exports).

Primary income consists mainly of compensation of employees and investment income (profits, dividends, interest). For most large economies GDP and GNI are close. For countries hosting large foreign-owned firms, GDP can greatly exceed GNI, because much of the profit generated domestically belongs to foreigners. Ireland is the extreme case: its statistics office developed a modified measure, GNI*, precisely because GDP had ceased to be a reliable guide to Irish living standards.

10.6.2 Gross versus net

“Gross” in GDP means that depreciation (called *consumption of fixed capital* in national accounts) has not been deducted. **Net domestic product** (NDP) is GDP minus depreciation. Net measures are conceptually better guides to sustainable income, because depreciation is output needed merely to keep the capital stock intact; gross measures are used more often because depreciation is hard to measure.

10.6.3 What GDP omits

GDP measures market production and some government production. It is not a measure of welfare, and was never designed to be one. It omits or mismeasures:

- **Household production:** cooking, cleaning, childcare done within the household. (Paid childcare is counted; the same care by a parent is not.)
- **Leisure:** two economies with equal GDP, one working 1,400 hours per person per year and the other 2,000, are not equally well off.
- **Environmental depletion and damage:** extracting oil counts as output, but the depletion of the resource is not deducted, and pollution is not subtracted.
- **Distribution:** average income can rise while most households’ income falls.

- **Quality change and new goods:** statisticians adjust for quality, imperfectly. A smart-phone replaces many earlier devices and services at a small fraction of their combined cost.
- **The informal and illegal economy:** large in many developing countries and not negligible in rich ones; statistical agencies estimate some of it.

It includes one notable imputation: the rent that owner-occupiers would pay to themselves if they rented their homes (*imputed rent*). Without it, GDP would fall whenever tenants bought their homes, though nothing real had changed.

10.7 Nominal and Real: Price Indexes

10.7.1 The problem

Nominal GDP can rise because more is produced or because prices are higher. To separate the two, we value output at constant prices, producing **real GDP**, and we measure the general level of prices with a **price index**.

10.7.2 Fixed-basket indexes

Suppose a representative household buys a basket of goods. Let $p_{k,t}$ be the price of good k in period t and $q_{k,t}$ the quantity.

Definition 10.8. Laspeyres, Paasche and Fisher indexes

The **Laspeyres price index** uses base-period quantities:

$$L = \frac{\sum_k p_{k,1} q_{k,0}}{\sum_k p_{k,0} q_{k,0}}.$$

The **Paasche price index** uses current-period quantities:

$$P = \frac{\sum_k p_{k,1} q_{k,1}}{\sum_k p_{k,0} q_{k,1}}.$$

The **Fisher ideal index** is their geometric mean, $F = \sqrt{L \cdot P}$.

Worked Example 10.9. Three price indexes

A household's annual purchases:

	p_0	q_0	p_1	q_1
Bread (loaves)	2.00	100	2.20	95
Phone	500	1	450	1.2
Rent (months)	800	12	840	12

Base-period cost of the base basket: $200 + 500 + 9,600 = 10,300$. Current cost of the base basket: $220 + 450 + 10,080 = 10,750$. Laspeyres: $10,750/10,300 = 1.0437$, inflation 4.37%. Current basket at current prices: $209 + 540 + 10,080 = 10,829$; at base prices: $190 + 600 + 9,600 = 10,390$. Paasche: $10,829/10,390 = 1.0423$, inflation 4.23%. Fisher: $\sqrt{1.0437 \times 1.0423} = 1.0430$, inflation 4.30%.

10.7.3 Substitution bias

The Laspeyres index exceeds the Paasche index in the example, and this is typical. When a good becomes relatively cheaper, households buy more of it. The Laspeyres index ignores that substitution, since it prices the *old* basket; it therefore overstates the increase in the cost of maintaining a given standard of living. The Paasche index uses the new basket and tends to understate it. This **substitution bias** was one of several sources of upward bias in the U.S. Consumer Price Index identified by the 1996 Boskin Commission, which estimated the total upward bias at about 1.1 percentage points per year at the time; statistical agencies have since adopted methods to reduce it.

10.7.4 The CPI and the GDP deflator

Two price indexes are widely used:

- The **consumer price index** (CPI) measures the price of a basket of goods and services bought by households, including imported goods. In the United States it is compiled by the Bureau of Labor Statistics; in the euro area the harmonized index (HICP) is compiled by Eurostat. The basket is updated periodically, so the CPI is approximately a Laspeyres index with regularly refreshed weights.
- The **GDP deflator** is nominal GDP divided by real GDP. It covers all domestically produced final goods—including investment goods and exports—but excludes imports.

The two can diverge. A rise in the price of imported oil raises the CPI directly but affects the GDP deflator only indirectly. A rise in the price of domestically produced machinery raises the deflator but not the CPI. Central banks often focus on consumer prices; the U.S. Federal Reserve's preferred measure is a third index, the price index for personal consumption expenditures (PCE), produced by the Bureau of Economic Analysis.

10.7.5 Real GDP and chain-weighting

Real GDP values each year's output at the prices of some base year. But the choice of base year matters, especially when relative prices change a lot.

Worked Example 10.10. Why the base year matters

An economy produces apples and computers.

	Year 1		Year 2	
	Quantity	Price	Quantity	Price
Apples	100	1	110	1.2
Computers	10	100	15	80

Nominal GDP: year 1, $100 + 1,000 = 1,100$; year 2, $132 + 1,200 = 1,332$.

At year-1 prices, real GDP in year 2 is $110 + 1,500 = 1,610$: growth of 46.4%. At year-2 prices, real GDP in year 1 is $120 + 800 = 920$, and year 2 is 1,332: growth of 44.8%. Using early-year prices gives more weight to computers, whose output grew fast and whose price fell; the growth estimate is higher.

Chain-weighting resolves the arbitrariness by computing growth between each pair of adjacent years with a Fisher index of quantities: $\sqrt{1.4636 \times 1.4478} = 1.4557$, growth of 45.6%. Chaining these year-to-year growth rates produces a real GDP series whose growth does not depend on a distant base year. The U.S. Bureau of Economic Analysis has used chain-weighted (Fisher) measures since 1996; most other statistical agencies,

including Germany's, use annually chained indexes.

A drawback of chain-weighting is that real components (such as real consumption and real investment) no longer add up exactly to real GDP. This is a price worth paying for growth rates that do not depend on the base year, but it means that "shares of real GDP" should be computed from nominal data.

10.8 Sector Financial Balances

10.8.1 Derivation

Combining the expenditure identity with the way income is used gives one of the most useful identities in macroeconomics. Households receive income Y , pay taxes net of transfers T , consume C and save the rest; so private saving is $S \equiv Y - T - C$. (For now we treat all private income as passing through households; Chapter 43 separates firms and households.) Substituting $Y \equiv C + I + G + X - M$:

$$S \equiv C + I + G + X - M - T - C,$$

$$(S - I) + (T - G) + (M - X) \equiv 0.$$

The sector balances identity.

$$\underbrace{(S - I)}_{\text{private surplus}} + \underbrace{(T - G)}_{\text{government surplus}} + \underbrace{(M - X)}_{\text{rest-of-world surplus}} \equiv 0.$$

The financial surpluses of the three sectors must sum to zero. If one sector spends more than its income, another must spend less than its income.

The logic is the same as in business accounting: every financial asset is someone else's liability. A sector that runs a surplus acquires financial claims on other sectors; a sector in deficit issues claims. Globally, the claims must net to zero.

Worked Example 10.11. Sector balances

An economy has $Y = 1,000$, $C = 580$, $I = 200$, $G = 240$, $T = 200$, $X = 150$, $M = 170$. Check the expenditure identity: $580 + 200 + 240 + 150 - 170 = 1,000$. Private saving: $1,000 - 200 - 580 = 220$.

Private surplus: $S - I = 20$. Government surplus: $T - G = -40$ (a deficit of 40). Rest-of-world surplus: $M - X = 20$: foreigners earn 20 more from selling to this country than they spend on its exports, and so accumulate 20 of claims on it. Sum: $20 - 40 + 20 = 0$. The government's deficit of 40 is financed half by domestic private surplus and half by foreigners.

10.8.2 Interpretation

The **current-account balance** is (approximately—Chapter 43 adds income flows) $X - M$. The identity can be rearranged as

$$X - M \equiv (S - I) + (T - G) :$$

a country's current-account surplus equals its national saving minus its investment. Countries that save more than they invest domestically lend the difference abroad.

Germany and China illustrate this. Germany's current-account surplus reached around 8 per cent of GDP in the mid-2010s, reflecting high saving by households and firms, a government that ran budget surpluses in several of those years, and subdued domestic investment. China's surplus peaked close to 10 per cent of GDP in 2007, reflecting extraordinarily high national saving, and then fell substantially as domestic investment and consumption rose relative to saving. The United States has run current-account deficits almost every year since the early 1980s: it has invested more than it has saved, financing the gap by selling assets to, and borrowing from, the rest of the world.

► **Common error: the identity does not say which way causation runs**

The identity holds whatever the cause. A larger U.S. government deficit might be accompanied by more private saving, less investment, or a larger current-account deficit. Whether the "twin deficits" (fiscal and external) move together is a behavioural question that depends on interest rates, exchange rates and monetary policy (Chapters 46 and 52).

10.9 Data Quality and Revisions

Official statistics are estimates, built from surveys, tax records, administrative data and models. They are published quickly and revised as better information arrives. In the United States, the Bureau of Economic Analysis publishes an *advance* estimate of quarterly GDP about a month after the quarter ends, followed by *second* and *third* estimates in the next two months, annual updates, and occasional comprehensive revisions that can change the level and growth of GDP for many years back. Germany's Federal Statistical Office (Destatis), China's National Bureau of Statistics and Malaysia's Department of Statistics (DOSM) follow similar cycles.

Revisions can be large relative to the quantities that policy-makers care about. An initial estimate of 2 per cent growth may later become 1 per cent or 3 per cent. This has two consequences for later chapters. First, central banks make decisions using data they know to be imprecise, which affects how aggressively they should respond (Chapter 50). Second, judging a past policy decision requires using the data available at the time, not today's revised data.

10.10 Bridging the Two Systems

The business and national accounts are designed to connect, but their concepts differ in ways that matter:

Firm accounts	National accounts	Relationship
Revenue	Output	A firm's revenue is not its contribution to GDP; only value added is.
Cost of goods sold, purchased services	Intermediate consumption	Deducted to obtain value added.
Wages expense	Compensation of employees	Part of value added paid to labour.
Operating profit plus depreciation (roughly)	Gross operating surplus	The part of value added paid to capital.
Purchase of equipment (investing cash flow)	Gross fixed capital formation (<i>I</i>)	Counted in GDP when bought; depreciated thereafter in both systems.
Increase in inventory	Change in inventories (part of <i>I</i>)	Unsold output is investment.
Balance sheet	Sector balance sheets and financial accounts	Every financial asset is another sector's liability.

► **Common error: “GDP rises whenever firms report higher revenue”**

If a firm's revenue rises because it buys more inputs from other firms and passes them through, its value added may not change at all. If it rises because prices rose, real GDP may not change. And if a firm books revenue in one country that was generated in another—as multinationals routinely do for tax reasons—the location of recorded GDP can move without any change in real activity. Ireland in 2015 is the example.

Case: Ireland's 26 per cent growth

Ireland's Central Statistics Office initially reported real GDP growth of about 26 per cent for 2015. The largest contributors were the relocation to Ireland of intellectual property assets by multinational firms and the expansion of aircraft leasing. Once a company resident in Ireland owns a patent, the royalties and product sales generated by that patent count as Irish output; and the purchase of the patent counts as Irish investment in intellectual property products, an asset category included in *I* under the international System of National Accounts. Net exports rose because the goods manufactured abroad under contract for Irish-resident firms (“contract manufacturing”) were recorded as Irish exports.

Nothing in the statistics was wrong in the sense of violating the rules; the rules had simply been designed for a world in which production was tied to physical location. The episode led the statistics office to publish modified GNI (GNI*), which excludes the depreciation of foreign-owned intellectual property and leased aircraft and the retained earnings of firms that have moved their legal residence to Ireland. The lessons for this book: measured GDP reflects accounting conventions about *where* and *to whom* value added is attributed; GDP and GNI can diverge sharply; and a firm's accounting decisions—where to locate assets and book profits—can move national aggregates.

Summary

- $\text{Assets} \equiv \text{Liabilities} + \text{Equity}$. Every transaction changes at least two items and keeps the equation in balance.
- The balance sheet reports stocks at a date; the income statement and cash-flow statement report flows over a period. They articulate: net income less dividends changes retained

earnings, and the change in cash equals net cash flow.

- Accrual accounting recognizes revenue when earned and expenses when resources are used; profit and operating cash flow differ by the accruals.
- GDP is the value of final goods and services produced domestically, equal to total value added. It can be measured by production, expenditure ($C + I + G + X - M$) or income, and the three agree by construction.
- GNI adds net primary income from abroad; NDP subtracts depreciation. GDP omits household production, leisure, environmental damage and distribution.
- Laspeyres indexes tend to overstate and Paasche indexes to understate cost-of-living changes; Fisher and chain-weighted indexes reduce the bias. The CPI includes imports; the GDP deflator does not.
- The sector balances identity $(S - I) + (T - G) + (M - X) \equiv 0$ says that sectoral surpluses and deficits must offset. It constrains outcomes but does not determine causation.
- Official data are revised; analyses should record which vintage they use.

Key terms

asset; liability; equity; accounting equation; double entry; depreciation; balance sheet; income statement; cash-flow statement; operating, investing and financing activities; articulation; indirect method; direct method; accrual accounting; cash accounting; intermediate good; final good; value added; GDP; production, expenditure and income approaches; inventory investment; circular flow; GNI; net domestic product; imputed rent; real GDP; price index; Laspeyres, Paasche and Fisher indexes; substitution bias; CPI; GDP deflator; PCE price index; chain-weighting; sector financial balances; current-account balance; data revision.

Exercises

Exercise 10.1 (Transactions). For each transaction, state its effect on assets, liabilities and equity: (a) a firm buys inventory for 5,000 cash; (b) it buys inventory for 5,000 on credit; (c) it pays a supplier 2,000; (d) it sells goods costing 3,000 for 4,500 on credit; (e) it pays a dividend of 1,000; (f) it repays 10,000 of a bank loan.

Exercise 10.2 (Harbour Coffee's February). In February Harbour Coffee: sells drinks for 20,000 cash, using supplies that cost 7,000; buys supplies for 9,000 on credit; collects the 3,000 owed by the office; pays the supplier 10,000; pays wages of 7,000; records depreciation of 500; accrues a further 100 of interest (none paid). Starting from the 31 January balance sheet, prepare the 28 February balance sheet, the February income statement, and February operating cash flow by both the direct and indirect methods.

Exercise 10.3 (Accrual versus cash). A software firm signs a contract on 1 December to provide a year of service for 12,000, paid in full on signing. Its financial year ends on 31 December. How much revenue is recognized in the year ending 31 December? What liability appears on the balance sheet? What is the effect on operating cash flow?

Exercise 10.4 (Value added). A mine produces iron ore worth 200 and sells it to a steelmaker. The steelmaker also buys energy worth 100 and produces steel worth 600, of which 450 is sold to a carmaker and 150 is exported. The carmaker buys parts worth 300 from other domestic firms (which used no purchased inputs) and produces cars worth 1,200, all sold to households.

Compute each firm's value added and GDP. Check your answer with the expenditure approach. (Assume the energy producer used no purchased inputs.)

Exercise 10.5 (What counts in GDP?). State whether each is included in this year's GDP of the country concerned, and if so, in which expenditure component: (a) a household buys a newly built house; (b) a household buys a ten-year-old house; (c) a firm buys a new delivery van; (d) a government pays unemployment benefits; (e) a government pays teachers' salaries; (f) a German firm buys machinery produced in China; (g) a Malaysian factory produces chips that are not sold by the end of the year; (h) you buy shares in a company.

Exercise 10.6 (GDP and GNI). A country's GDP is 500. Foreign-owned firms operating in the country earn profits of 60, which they send home. Residents of the country earn 15 in interest on foreign bonds and 5 in wages from working across the border. Compute GNI. Which measure better indicates residents' incomes?

Exercise 10.7 (Price indexes). Using Example 10.9, suppose the price of phones in period 1 had been 350 instead of 450, with quantities unchanged. Recompute the Laspeyres, Paasche and Fisher indexes. Why does the gap between Laspeyres and Paasche change?

Exercise 10.8 (CPI versus deflator). A country imports all its oil. The world price of oil doubles, and nothing else changes immediately. Explain the likely effect on (a) the CPI, (b) the GDP deflator.

Exercise 10.9 (Real growth). Using the data in Example 10.10, compute the GDP deflator for year 2 using year 1 as the base, and state the inflation rate implied. Then explain why a chain-weighted deflator would give a different figure.

Exercise 10.10 (Sector balances). An economy has $Y = 2,000$, $C = 1,300$, $I = 400$, $G = 450$, $T = 380$, $M = 500$. Compute exports, private saving, and the three sectoral balances. Which sector is financing which?

Exercise 10.11 (Interpreting the identity). "A country that cuts its government deficit will automatically improve its current account." Use the sector balances identity to evaluate this claim.

Exercise 10.12 (Revisions). A central bank raises interest rates after an advance estimate shows 4 per cent annual growth. Two years later the figure is revised to 2 per cent. Was the decision a mistake? What else would you need to know?

Exercise 10.13 (Integration). Harbour Coffee (Example 10.2) is part of a national economy. What was its contribution to January GDP by the production approach? (Assume the supplies it used were bought from other domestic firms, and that value added is output minus intermediate inputs used; output equals revenue plus the change in its own-produced inventory, which is zero here.) Why does the purchase of equipment not reduce its value added?

Solutions

Solution 10.1. (a) Inventory +5,000, cash -5,000: assets unchanged in total. (b) Inventory +5,000, accounts payable +5,000. (c) Cash -2,000, accounts payable -2,000. (d) Receivables +4,500, inventory -3,000: assets +1,500; equity (retained earnings) +1,500 (revenue 4,500 less cost of goods sold 3,000). (e) Cash -1,000, retained earnings -1,000: a dividend is a distribution to owners, not an expense, so it does not appear in the income statement. (f) Cash -10,000, loan -10,000.

Solution 10.2. *Balance sheet at 28 February.* Cash: $45,000 + 20,000 + 3,000 - 10,000 - 7,000 = 51,000$. Receivables: $3,000 - 3,000 = 0$. Inventory: $2,000 + 9,000 - 7,000 = 4,000$. Equipment: $29,500 - 500 =$

29,000. Total assets: 84,000. Accounts payable: $4,000 + 9,000 - 10,000 = 3,000$. Interest payable: 200. Loan: 20,000. Share capital: 50,000. Retained earnings: $5,400 + 5,400 = 10,800$ (see below). Total: $3,000 + 200 + 20,000 + 50,000 + 10,800 = 84,000$. It balances.

Income statement. Revenue 20,000; cost of goods sold 7,000; wages 7,000; depreciation 500; interest 100. Net income 5,400.

Operating cash flow, direct. Receipts from customers $20,000 + 3,000 = 23,000$; paid to suppliers 10,000; paid to employees 7,000. Net 6,000.

Indirect. Net income 5,400 + depreciation 500 + decrease in receivables 3,000 – increase in inventory 2,000 – decrease in payables 1,000 + increase in interest payable 100 = 6,000. Check: change in cash = 6,000, with no investing or financing flows.

Solution 10.3. One month of service has been provided, so revenue of $12,000/12 = 1,000$ is recognized. The remaining 11,000 is a liability (deferred revenue or contract liability): the firm owes 11 months of service. Operating cash flow includes the full 12,000 received. This is why subscription businesses often report operating cash flow well above net income while they are growing.

Solution 10.4. Mine: 200. Energy producer: 100. Steelmaker: $600 - 200 - 100 = 300$. Parts makers: 300. Carmaker: $1,200 - 450 - 300 = 450$. GDP = $200 + 100 + 300 + 300 + 450 = 1,350$. Expenditure: consumption (cars) 1,200 plus exports (steel) 150 = 1,350.

Solution 10.5. (a) Yes, residential investment (*I*). (b) No: the house was produced in an earlier year. Any estate agent's commission is a current service and is counted. (c) Yes, business investment (*I*). (d) No: a transfer payment, not a purchase of output. It appears in *T* as a negative (net taxes). (e) Yes, government purchases (*G*): teachers' services are output valued at cost. (f) In Germany: counted in German investment *I* but subtracted in imports *M*, so no net effect on German GDP. In China: included in exports, hence in Chinese GDP. (g) Yes, in Malaysia's GDP as inventory investment. (h) No: a financial transaction, an exchange of assets, not production.

Solution 10.6. GNI = $500 - 60 + 15 + 5 = 460$. GNI is the better guide to residents' incomes, because the 60 of profits belongs to foreigners.

Solution 10.7. Base basket at period-1 prices: $220 + 350 + 10,080 = 10,650$; Laspeyres = $10,650/10,300 = 1.0340$. Current basket at current prices: $209 + 420 + 10,080 = 10,709$; at base prices 10,390; Paasche = 1.0307. Fisher = $\sqrt{1.0340 \times 1.0307} \approx 1.0323$. The gap between Laspeyres and Paasche widens (0.33 points versus 0.14) because the relative price change is larger, and the Paasche index gives more weight to the good whose price fell (households bought 1.2 phones rather than 1).

Solution 10.8. (a) The CPI rises, because oil and its products (petrol, heating) are in the consumer basket. (b) The GDP deflator is not directly affected, because imported oil is not domestic production. If anything, measured real GDP is unchanged and the deflator unchanged at first; later, domestic firms may raise prices to pass on higher energy costs, which does raise the deflator.

Solution 10.9. Deflator (year 1 base) = nominal/real = $1,332/1,610 = 0.827$, implying -17.3% inflation: the fall in computer prices dominates when year-2 output is valued with year-1's high computer prices, which inflate real GDP. A chain-weighted deflator is the ratio of nominal growth to chain-weighted real growth: $(1,332/1,100)/1.4557 = 1.2109/1.4557 = 0.832$, i.e. about -16.8% . The base-year choice affects measured real growth, and hence measured inflation.

Solution 10.10. $X = Y - C - I - G + M = 2,000 - 1,300 - 400 - 450 + 500 = 350$. Private saving $S = Y - T - C = 2,000 - 380 - 1,300 = 320$. Private balance $S - I = -80$; government $T - G = -70$; rest of world $M - X = 150$. Sum: $-80 - 70 + 150 = 0$. Both the private sector and the government are in deficit, financed by foreigners, who run a surplus of 150 vis-à-vis this country (the country's current-account deficit).

Solution 10.11. The identity $X - M \equiv (S - I) + (T - G)$ shows that if $T - G$ rises, the current account improves *unless* the private balance $S - I$ falls by an offsetting amount. It might: fiscal tightening may reduce incomes and hence private saving, or lower interest rates may raise investment. Nothing is automatic; the outcome depends on behaviour. The claim confuses an identity with a theory.

Solution 10.12. Not necessarily. A decision should be judged against the information available when it was made, together with the costs of waiting. One would want to know what other indicators showed

at the time (employment, inflation, surveys), how uncertain the estimate was known to be, and what the costs of acting and of not acting were. If the central bank reasonably believed growth was strong and inflation rising, raising rates may have been sensible even though the growth estimate was later revised down.

Solution 10.13. Output 18,000 (revenue), intermediate inputs used 6,000 (supplies). Value added 12,000. This is paid out as wages (6,000), interest (100), depreciation (500) and profit (5,400): the sum is 12,000, the income approach at the level of one firm. The equipment is a capital good, not an intermediate input; its purchase is investment in GDP. Only its depreciation, which is part of gross value added (as a component of gross operating surplus), reflects its use. That is why the measure is called *gross* value added.

Part II

Evidence: Probability, Statistics and Econometrics

Data, Measurement and Description

After studying this chapter you should be able to

1. Identify the units, variables, population and estimand of an empirical question, and distinguish cross-sectional, time-series and panel data.
2. Recognize the main sources of bias in how data are collected: selection, non-response, survivorship and measurement error.
3. Explain why surveys use stratification, clustering and weights, and compute a weighted estimate.
4. Summarize a distribution by its centre, spread and shape, choosing between mean and median, and compute variances, standard deviations, quantiles and z-scores.
5. Describe inequality with percentile ratios and top income shares, complementing the Gini coefficient.
6. Choose honest graphical displays: histograms, empirical distribution functions, box plots, time plots and scatter plots.
7. Classify missing data and describe the practices that make an empirical analysis reproducible.

A motivating question

In 1936 the *Literary Digest*, an American magazine, predicted that the Republican challenger Alf Landon would defeat President Franklin Roosevelt. Its prediction rested on about 2.4 million returned ballots—one of the largest opinion surveys ever conducted. Roosevelt won by a landslide, carrying 46 of the 48 states. A young pollster named George Gallup, using a sample of a few tens of thousands, predicted the result correctly. How can a sample of millions be so wrong and a far smaller one right?

The answer is that the size of a data set matters much less than *how* it was produced. Every statistic in this book—GDP, inflation, unemployment, a company's earnings, the estimated effect of a policy—is the product of decisions about what to measure, whom to ask, and how to record and summarize the answers. This chapter begins Part II with those decisions. It develops the vocabulary of data, the ways in which data can mislead, and the numerical and graphical summaries used to describe what data contain. The following chapters build the probability theory needed to reason about uncertainty, and the methods of inference, regression and causal analysis that turn data into evidence.

11.1 The Anatomy of a Data Set

11.1.1 Units, variables and observations

A data set records the values of **variables** for a set of **units**. The units may be people, households, firms, banks, countries or months; each row of a table is an **observation** on one unit (or one unit at one date). Variables are

- *quantitative*—measured on a numerical scale: income in euros, hours worked, a firm’s revenue. Quantitative variables may be *discrete* (number of children) or *continuous* (a price);
- *categorical*—recording membership of a group: employment status (employed, unemployed, inactive), industry, country. Categories may be ordered (credit ratings AAA, AA, A, ...) or unordered.

As Chapter 2 insisted, every quantitative variable needs a unit, a period and a definition. “Income” could be before or after tax, individual or household, monthly or annual.

11.1.2 Three shapes of data

- **Cross-sectional data** observe many units at one time: a survey of 10,000 households in 2024; the financial statements of all listed firms for one year.
- **Time-series data** observe one unit at many dates: quarterly GDP of Malaysia since 1991; the daily closing value of a stock index.
- **Panel data** observe many units at many dates: the same households interviewed every year; the annual accounts of the same firms over twenty years.

Panel data are especially valuable because they allow a unit to be compared with itself over time, which controls for characteristics that do not change (Chapter 18). Time series raise their own problems, because observations close together in time are rarely independent.

11.1.3 Populations, samples and estimands

The **population** is the whole set of units about which we want to learn: all households in Germany, all loans a bank might make, all months in which a policy could be in force. A **sample** is the subset we actually observe. The **estimand** is the precise quantity we want to know about the population—for example, the average after-tax income of households in Germany in 2024, or the proportion of small-business loans that default within two years.

Key idea. Before computing anything, state the population and the estimand. A sample statistic is useful only as an estimate of some population quantity, and whether it is a good estimate depends on how the sample relates to the population—a question that no amount of computation on the sample can answer by itself.

11.2 How Data Can Mislead

11.2.1 Selection

A sample is *representative* if it resembles the population in the respects that matter. The most reliable way to obtain one is **random sampling**: every unit in the population has a known, non-zero chance of selection, determined by chance rather than by the researcher or the respondent.

Random samples can still differ from the population by luck, but the luck is quantifiable (Chapter 15). Non-random samples can be biased in ways that no formula measures.

Selection bias arises when the way units enter the sample is related to the variable being studied. Surveying customers who visit a bank's website tells you about online customers, not all customers. Studying the returns of firms that are listed today ignores those that went bankrupt or were delisted.

Survivorship bias is a common form. The average past return of mutual funds that exist today overstates the average return of all funds that existed, because poorly performing funds tend to be closed or merged. Databases that drop dead funds, failed banks or delisted firms produce flattering averages.

Non-response bias arises when those who decline to answer differ from those who respond. Response rates to many official household surveys have fallen substantially over recent decades, and statistical agencies adjust for non-response with weights, which work only as far as the characteristics used for weighting capture the differences.

11.2.2 Measurement error

Even for units in the sample, recorded values may differ from true values. People misremember their income, round their hours, or under-report income from informal work. Firms' reported earnings reflect accounting choices (Part IV). Official statistics are revised as more complete information arrives (Chapter 10). Measurement error that is random adds noise; error that is systematic—under-reporting that rises with income, say—adds bias. Chapter 17 shows that even random error in an explanatory variable biases regression estimates toward zero.

11.2.3 Designing surveys

Large official surveys rarely use simple random samples. Instead they combine:

- *Stratification*: dividing the population into groups (regions, firm-size classes) and sampling from each, guaranteeing coverage and often improving precision;
- *Clustering*: sampling groups (villages, schools, city blocks) and then units within them, which is cheaper but reduces precision because units in the same cluster tend to be similar;
- *Weights*: each observation carries a weight equal to the number of population units it represents, correcting for unequal selection probabilities and non-response.

Worked Example 11.1. Why weights matter

A country's population is 70 per cent urban and 30 per cent rural. A survey samples 100 urban and 100 rural households, finding mean incomes of 52 and 31 (thousand). The unweighted mean of all 200 observations is $(52 + 31)/2 = 41.5$, but rural households are over-represented. Each urban household represents $0.7/0.5 = 1.4$ times its share of the sample and each rural household $0.3/0.5 = 0.6$; the weighted mean is $0.7 \times 52 + 0.3 \times 31 = 45.7$. Ignoring the design understates national average income by more than 4,000.

11.3 Describing a Distribution: Centre

The **distribution** of a variable describes what values it takes and how often. Summaries condense a distribution into a few numbers. Consider the annual incomes, in thousands, of nine households:

18, 22, 25, 27, 30, 34, 41, 55, 148.

11.3.1 Mean, median and mode

The **mean** is $\bar{x} = \frac{1}{n} \sum x_i = 400/9 = 44.4$. The **median** is the middle value when the data are ordered—here the fifth, 30—or the average of the two middle values when n is even. The **mode** is the most frequent value, useful mainly for categorical data.

The mean and median differ here because one household has a very high income. The mean is sensitive to extreme values: without the 148, it falls to 31.5, while the median barely moves (to 28.5). The distribution is **skewed** to the right: a long tail of high values pulls the mean above the median. Income and wealth distributions are almost always right-skewed, which is why statistical agencies report median household income as the measure of the “typical” household. A *trimmed mean*, which drops a fixed share of the highest and lowest values before averaging, is a compromise; several central banks track trimmed-mean inflation, which excludes the largest price changes in either direction, as a measure of underlying inflation (Chapter 50).

► “The average household earns 44,000”

In the example, seven of the nine households earn less than the mean. When a distribution is skewed, the mean describes no typical unit. Report the median, or both, and say which. Conversely, when the *total* matters—total tax revenue, total spending—the mean is the right summary, because the total equals the mean times the number of units.

11.3.2 Quantiles

The p th **quantile** (or 100 p th percentile) is the value below which a fraction p of the observations lie. The median is the 0.5 quantile; the *quartiles* Q_1 and Q_3 are the 0.25 and 0.75 quantiles. Several conventions exist for computing quantiles in small samples; one simple rule takes the median of the lower and upper halves. For the incomes above, Q_1 is the median of 18, 22, 25, 27, which is 23.5, and Q_3 the median of 34, 41, 55, 148, which is 48.

11.4 Describing a Distribution: Spread and Shape

11.4.1 Range and interquartile range

The *range* (maximum minus minimum, here 130) depends entirely on the two most extreme observations. The **interquartile range** $IQR = Q_3 - Q_1 = 24.5$ measures the spread of the middle half of the data and is insensitive to outliers. A common rule flags as possible *outliers* values more than $1.5 \times IQR$ above Q_3 or below Q_1 : here anything above $48 + 36.75 = 84.75$, so 148 is flagged. An outlier is not necessarily an error—high incomes are real—but it deserves a check.

11.4.2 Variance and standard deviation

The sample **variance** is the average squared deviation from the mean, with divisor $n - 1$:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

(The divisor $n - 1$ rather than n corrects for the fact that deviations are measured from the sample mean, which is itself fitted to the data; Chapter 15 explains why.) The **standard deviation** $s = \sqrt{s^2}$ is in the same units as the data. For the nine incomes, $s^2 = 1,631.3$ and $s = 40.4$ thousand. Squaring gives large deviations great weight: the single household at 148 contributes more than 80 per cent of the sum of squared deviations.

The *coefficient of variation* $s/\bar{x}$ measures spread relative to the mean and is unit-free: here $40.4/44.4 = 0.91$.

11.4.3 Standardizing: z-scores

The z-score of an observation measures how many standard deviations it lies from the mean:

$$z_i = \frac{x_i - \bar{x}}{s}.$$

The household earning 148 has $z = (148 - 44.4)/40.4 = 2.56$. z-scores make variables with different units comparable: a firm whose leverage is two standard deviations above its industry mean and whose profitability is one standard deviation below is unusual in a precise sense. Chapter 13 shows that for normally distributed data, about 95 per cent of observations have $|z| < 1.96$.

11.4.4 Shape

Beyond centre and spread, distributions differ in *skewness* (asymmetry) and *kurtosis* (the heaviness of the tails relative to a normal distribution). Financial returns typically have heavy tails: extreme days are far more common than a normal distribution would predict, a fact with large consequences for risk management (Chapter 34).

11.5 Describing Inequality

Chapter 7 defined the Lorenz curve and the Gini coefficient. Because the Gini compresses a whole distribution into one number, analysts also use measures that show *where* inequality lies:

- **Percentile ratios:** the ratio of the 90th to the 10th percentile ($P90/P10$) compares the top and bottom of the distribution while ignoring extremes; $P90/P50$ and $P50/P10$ split it into upper and lower halves.
- **Top income shares:** the share of total income received by the top 10 or top 1 per cent. Because surveys under-represent the very rich, research led by Thomas Piketty and Emmanuel Saez estimated top shares from income-tax records, finding that the top 1 per cent's share of pre-tax income in the United States roughly doubled between the late 1970s and the 2010s.
- **Concentration ratios** for firms: the market share of the largest four firms in an industry, or the Herfindahl–Hirschman index (the sum of squared market shares), used in competition policy (Chapter 24).

11.6 Displaying Data

Good graphs reveal patterns that summaries hide; bad graphs create patterns that are not there.

- A **histogram** groups a quantitative variable into intervals and draws bars whose *areas* are proportional to frequencies. Its appearance depends on the choice of intervals, so try several.
- The *empirical distribution function* plots, for each value x , the fraction of observations at or below x . It needs no choice of intervals and makes quantiles easy to read.
- A **box plot** draws a box from Q_1 to Q_3 with a line at the median, and “whiskers” to the most extreme values within $1.5 \times \text{IQR}$, plotting outliers separately (Figure 11.1). Side-by-side box plots compare distributions across groups.
- A *time plot* shows a series against time. Growing series belong on a log scale (Chapter 4).
- A **scatter plot** shows two quantitative variables, one point per unit, revealing association, nonlinearity and outliers (Chapter 14).

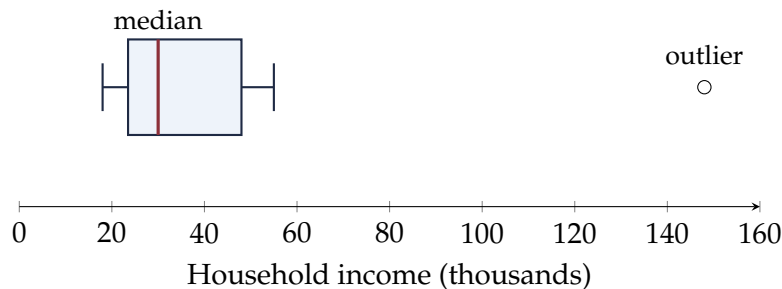


Figure 11.1: Box plot of the nine household incomes. The box spans the interquartile range (23.5 to 48); the line inside it marks the median (30); the whiskers reach the most extreme values within 1.5 IQRs of the box; the household at 148 is plotted separately.

► Graphs that mislead

Truncated vertical axes exaggerate small changes; axes with different scales on the left and right can make any two series appear to move together; three-dimensional effects distort areas; cumulative totals always rise and hide slowdowns; and plotting a growing series on an ordinary scale makes early growth invisible. An honest graph states its units, starts bar charts at zero, labels sources and dates, and uses a log scale for growth.

11.7 Missing Data and Reproducibility

11.7.1 Missing data

Real data sets have gaps: a household skips the income question; a firm stops reporting; a country's statistics lapse during a war. What to do depends on *why* the data are missing:

- *Missing completely at random* (MCAR): missingness is unrelated to anything, observed or not. Dropping incomplete observations loses precision but causes no bias.
- *Missing at random* (MAR): missingness depends only on observed variables (older respondents skip questions more often, and age is recorded). Weighting or imputation using the

observed variables can correct the bias.

- *Missing not at random* (MNAR): missingness depends on the missing value itself (high earners decline to report income). No method based only on the data corrects this without assumptions; sensitivity analysis is essential.

Imputation fills gaps with predicted values; *multiple imputation* does so several times with random variation, so that the final uncertainty reflects the uncertainty about the missing values.

11.7.2 Reproducibility

A good empirical analysis records, for every series, the producing institution, its exact definition, the frequency, any transformation applied (logs, growth rates, seasonal adjustment, deflation), the sample period and the date the data were retrieved—and, where licences allow, keeps a frozen copy of the data together with the code that turns it into every table and figure. Without this, neither you nor anyone else can check the result or update it when the data are revised. When evaluating a past decision—a central bank’s rate choice, a firm’s investment—use the data available at the time (the *real-time vintage*), not today’s revised figures.

Case: The *Literary Digest* poll of 1936

The design. The *Literary Digest* had correctly called several previous presidential elections. In 1936 it mailed about 10 million mock ballots to people drawn from its own subscriber list, telephone directories and lists of automobile owners. About 2.4 million were returned. The results pointed to a clear victory for Alf Landon.

The outcome. Roosevelt won about 61 per cent of the popular vote and 46 of 48 states. Gallup, with a far smaller but more carefully constructed sample, predicted a Roosevelt victory.

What went wrong. Two biases compounded. First, *selection*: in the depths of the Depression, households with telephones, cars and magazine subscriptions were richer than average, and richer voters leaned Republican. The sampling frame—the list from which the sample was drawn—did not represent the population of voters. Second, *non-response*: fewer than a quarter of those mailed replied, and people who felt strongly against the incumbent may have been more likely to return a ballot. Later analysis of the episode suggests that non-response contributed at least as much as the flawed frame.

Why it still matters. The episode is the classic warning that a large biased sample is worse than a small representative one: more data make a biased estimate more *precise*, not more *accurate*. The warning applies directly to “big data” in economics and finance: web-scraped prices, card-transaction records, social-media sentiment and app-usage data cover millions of observations but only those people and firms who use a particular platform. Such data can be extremely valuable, especially for tracking changes quickly (Chapter 18), but their relation to the population of interest must be established, not assumed.

Summary

- Data record variables for units; data are cross-sectional, time series or panels. State the population and the estimand before computing.
- Random sampling allows quantifiable uncertainty; selection, survivorship and non-response bias do not. Measurement error adds noise or bias.

- Surveys use stratification, clustering and weights; weighted estimates correct for unequal selection.
- The mean is sensitive to extremes; the median is robust. Skewed distributions have mean and median far apart.
- Spread is measured by the IQR, the variance $s^2 = \sum (x_i - \bar{x})^2 / (n - 1)$ and the standard deviation; z-scores standardize.
- Percentile ratios and top shares show where inequality lies; the Gini summarizes it.
- Honest graphs show units and sources, start bars at zero and use log scales for growth.
- Missing data may be MCAR, MAR or MNAR; the right treatment depends on which. Reproducible work records sources, definitions, transformations, vintages and code.

Key terms

unit; variable; observation; quantitative and categorical variables; cross-sectional data; time-series data; panel data; population; sample; estimand; random sampling; sampling frame; selection bias; survivorship bias; non-response bias; measurement error; stratification; clustering; survey weights; distribution; mean; median; mode; skewness; trimmed mean; quantile; percentile; quartile; interquartile range; outlier; variance; standard deviation; coefficient of variation; z-score; kurtosis; percentile ratio; top income share; Herfindahl–Hirschman index; histogram; empirical distribution function; box plot; scatter plot; missing completely at random; missing at random; missing not at random; imputation; real-time vintage.

Exercises

Exercise 11.1 (Data shapes). Classify each data set as cross-sectional, time series or panel, and name its units: (a) the monthly unemployment rate of Germany since 1991; (b) the 2023 annual reports of all firms in the FTSE Bursa Malaysia KLCI index; (c) the credit scores of the same 5,000 borrowers measured every quarter for three years.

Exercise 11.2 (Estimands). A bank wants to know how likely it is that loans to small restaurants will default. It has records of all restaurant loans it made in the past five years. State a precise estimand, and give two reasons why the bank's records might not represent the loans it will make next year.

Exercise 11.3 (Survivorship). A database contains the ten-year returns of the 200 equity funds that exist today. Their average annual return is 7.5 per cent. Fifty other funds were closed during the period, with average returns (while open) of 2 per cent. Explain why 7.5 per cent overstates the performance an investor could have expected from choosing a fund ten years ago.

Exercise 11.4 (Weights). A survey samples 400 small firms and 100 large firms. Large firms are 5 per cent of all firms. Mean employment growth is 1 per cent among small firms and 6 per cent among large firms. Compute the unweighted and properly weighted mean growth rate across firms. What would you weight by if you wanted employment growth in the economy as a whole?

Exercise 11.5 (Centre and spread). For the data 4, 7, 7, 9, 12, 15, 30: find the mean, median, mode, quartiles (by the halves rule, excluding the median), IQR and sample standard deviation. Is any value an outlier by the $1.5 \times \text{IQR}$ rule?

Exercise 11.6 (z-scores). Firm A has a return on assets of 9 per cent in an industry with mean 5 per cent and standard deviation 2 per cent. Firm B has ROA of 14 per cent in an industry with mean 10 per cent and standard deviation 4 per cent. Which is more exceptional relative to its industry?

Exercise 11.7 (Mean or median). For each question, say whether the mean or the median is the better summary, and why: (a) the typical house price in a city; (b) total household spending in a country; (c) the typical time to find a job after a layoff.

Exercise 11.8 (Inequality). In a population of ten people, incomes are 5, 8, 10, 12, 15, 18, 20, 25, 37 and 50. Find the share of total income received by the top 10 per cent and the top 20 per cent, and the ratio of the 90th to the 10th percentile (using the ninth and first values).

Exercise 11.9 (Missing data). In a survey of household finances, 30 per cent of households decline to report their wealth. Give an example of a pattern consistent with each of MCAR, MAR and MNAR, and say how each would affect an estimate of mean wealth computed from respondents only.

Exercise 11.10 (Diagnose the graph). A chart shows a bank's share price rising from 98 to 102 over a month, with the vertical axis running from 97 to 103. A second chart shows the bank's cumulative number of customers since its founding. What impression might each chart give, and how would you redraw them honestly?

Solutions

Solution 11.1. (a) Time series; the unit is Germany (observed monthly). (b) Cross-section; units are firms. (c) Panel; units are borrowers observed quarterly.

Solution 11.2. Estimand: the probability that a loan of the kind the bank will make to small restaurants next year defaults within, say, two years. The records may not represent next year's loans because (i) the bank's lending standards or the applicants may change (the population shifts); (ii) the past five years had particular economic conditions (for example, a pandemic or a boom) that may not recur; (iii) recent loans have not had time to default, so default rates computed from them are too low.

Solution 11.3. An investor choosing ten years ago could have picked any of the 250 funds. The average across all 250, weighting the closed funds' returns for the time they were open, is well below 7.5 per cent; including them roughly $(0.8 \times 7.5 + 0.2 \times 2 = 6.4$ per cent, as a crude equal-weight approximation) shows the size of the bias.

Solution 11.4. Unweighted: $(400 \times 1 + 100 \times 6)/500 = 2.0$ per cent. Weighted by population shares: $0.95 \times 1 + 0.05 \times 6 = 1.25$ per cent. For aggregate employment growth, weight each firm by its initial employment: large firms, though few, employ many workers, so employment-weighted growth would be much closer to 6 per cent.

Solution 11.5. Mean = $84/7 = 12$; median 9; mode 7. $Q_1 = \text{median of } 4, 7, 7 = 7$; $Q_3 = \text{median of } 12, 15, 30 = 15$; IQR = 8. Deviations: $-8, -5, -5, -3, 0, 3, 18$; squares sum to $64 + 25 + 25 + 9 + 0 + 9 + 324 = 456$; $s^2 = 76$, $s = 8.72$. Upper fence $15 + 12 = 27$: 30 is an outlier.

Solution 11.6. $z_A = (9 - 5)/2 = 2$; $z_B = (14 - 10)/4 = 1$. Firm A is more exceptional relative to its industry, although its ROA is lower.

Solution 11.7. (a) Median: house prices are right-skewed. (b) Mean (or the total itself): the total equals the mean times the number of households. (c) Median: durations are right-skewed, with a few very long spells.

Solution 11.8. Total income 200. Top 10 per cent (one person): $50/200 = 25$ per cent. Top 20 per cent: $87/200 = 43.5$ per cent. $P_{90}/P_{10} = 37/5 = 7.4$.

Solution 11.9. MCAR: a random subset of interview forms was lost—no bias, less precision. MAR: older households decline more often, and age is observed; the respondents-only mean is biased if wealth varies with age, but weighting by age corrects it. MNAR: the wealthiest decline most often because of what their wealth is; the respondents-only mean understates mean wealth, and correcting it requires assumptions or outside data (such as tax records).

Solution 11.10. The truncated axis makes a 4 per cent rise look like a surge; redraw with an axis that includes a meaningful baseline, or show percentage changes. The cumulative chart must always rise, hiding any slowdown or loss of customers; plot customers at each date (a stock) or new customers per period (a flow) instead.

Probability and Bayes' Rule

After studying this chapter you should be able to

1. Define sample spaces, events and probabilities, and apply the complement and addition rules.
2. Count outcomes with the multiplication principle, permutations and combinations.
3. Define independence, and compute probabilities for repeated independent trials.
4. Compute conditional probabilities from two-way tables and probability trees.
5. Apply the law of total probability and Bayes' rule, including in odds form with likelihood ratios.
6. Recognize the base-rate fallacy, the prosecutor's fallacy and the error of assuming independence.

A motivating question

A bank uses a credit-scoring model that flags 80 per cent of applicants who would go on to default, and wrongly flags only 20 per cent of those who would repay. An applicant is flagged. How likely is it that she would default? Most people answer “about 80 per cent.” If one applicant in ten would default, the correct answer is under 31 per cent. The error—ignoring how rare defaulters are to begin with—is so common that it has a name, and it has led to bad lending decisions, wrongful medical diagnoses and, as the case at the end of this chapter shows, a wrongful conviction for murder.

Uncertainty pervades economics and finance: whether a borrower repays, whether a recession starts, whether a drug trial succeeds, what a share will be worth next year. Probability is the mathematics of uncertainty. This chapter develops its rules, with particular attention to *conditional* probability: how the probability of one event changes when we learn that another has occurred. That idea underlies learning from evidence, the information economics of Chapter 25, statistical inference (Chapter 15) and Bayesian methods (Chapter 19).

12.1 Events and Probabilities

12.1.1 Sample spaces and events

An **experiment** is any process with an uncertain outcome: lending to a borrower, observing next quarter's GDP, drawing a household at random from a population. The set of all possible outcomes is the **sample space** Ω . An **event** is a set of outcomes—for example, “the borrower defaults within two years,” or “GDP growth is negative.” Events combine by the operations of set theory: $A \cup B$ (“A or B,” at least one occurs), $A \cap B$ (“A and B”) and A^c (“not A”). Events are *mutually exclusive* if they cannot both occur.

12.1.2 The axioms of probability

A **probability** $P(A)$ assigns to each event a number such that

1. $0 \leq P(A) \leq 1$;
2. $P(\Omega) = 1$;
3. if A and B are mutually exclusive, $P(A \cup B) = P(A) + P(B)$.

These three axioms, set out by Andrey Kolmogorov in 1933, imply everything else. In particular,

$$P(A^c) = 1 - P(A), \quad P(A \cup B) = P(A) + P(B) - P(A \cap B),$$

where the subtraction corrects for counting outcomes in both A and B twice.

Interpretations. A probability may be read as a long-run frequency (the proportion of times an event occurs in many repetitions), as a proportion of a population (the share of households with a given characteristic, when one is drawn at random), or as a degree of belief (an investor's probability that a merger will be approved). The rules are the same under each interpretation; the *Bayesian* view of probability as a degree of belief becomes important in Chapter 19.

12.2 Counting

When all outcomes are equally likely, the probability of an event is the number of outcomes in it divided by the number in the sample space. Counting large sets requires three tools.

The multiplication principle. If one choice can be made in m ways and a second, for each of those, in n ways, the pair can be made in mn ways. A four-digit PIN has $10^4 = 10,000$ possibilities.

Permutations. The number of ordered arrangements of k objects chosen from n distinct objects is

$$P(n, k) = n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!},$$

where $n! = n(n-1) \cdots 2 \cdot 1$ and $0! = 1$. A board of 8 can choose a chair and a deputy in $P(8, 2) = 56$ ways.

Combinations. The number of unordered selections of k objects from n is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!},$$

because each unordered group of k can be ordered in $k!$ ways. An investor can choose 3 stocks from a list of 10 in $\binom{10}{3} = 120$ ways. Combinations reappear in the binomial distribution (Chapter 13).

12.3 Conditional Probability

12.3.1 Definition

Definition 12.1. Conditional probability

The probability of A given that B has occurred, for $P(B) > 0$, is

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

Learning that B occurred shrinks the relevant sample space to B ; the conditional probability is the share of B 's probability that also belongs to A . Rearranging gives the *multiplication rule* $P(A \cap B) = P(A | B)P(B) = P(B | A)P(A)$.

12.3.2 Two-way tables

Conditional probabilities are easiest to see in a table of counts.

Worked Example 12.2. Unemployment and education

A survey of 1,000 adults in the labour force records education and employment status:

	Employed	Unemployed	Total
Degree	380	20	400
No degree	540	60	600
Total	920	80	1,000

The unemployment rate is $P(U) = 80/1,000 = 8$ per cent. Among graduates, $P(U | D) = 20/400 = 5$ per cent; among non-graduates, $P(U | D^c) = 60/600 = 10$ per cent. But $P(D | U) = 20/80 = 25$ per cent: a quarter of the unemployed are graduates. $P(U | D)$ and $P(D | U)$ answer different questions and are very different numbers. Confusing them—the “inverse fallacy”—is the error at the heart of this chapter's case.

12.3.3 Independence

Events A and B are **independent** if learning that one occurred does not change the probability of the other: $P(A | B) = P(A)$, equivalently

$$P(A \cap B) = P(A)P(B).$$

In Example 12.2, education and unemployment are not independent. Independence is a strong assumption, and assuming it when it is false is one of the most consequential errors in finance (Chapter 51).

Repeated trials. If n events are independent, each with probability p , the probability that none occurs is $(1 - p)^n$, so the probability that at least one occurs is $1 - (1 - p)^n$. A portfolio of 20 independent loans, each with a 5 per cent default probability, has a probability $1 - 0.95^{20} = 64$ per cent of at least one default. If defaults are positively correlated—because they share exposure to one local economy—the probability of *no* defaults is higher, but so is the probability of *many*: correlation fattens the tails, which is exactly what matters for a bank's solvency.

12.4 Total Probability and Bayes' Rule

12.4.1 Trees and the law of total probability

If events $B_1, \dots, B_k$ are mutually exclusive and together cover the sample space—a *partition*—then any event A can be split across them:

$$P(A) = \sum_{j=1}^k P(A \mid B_j) P(B_j).$$

This is the **law of total probability**. A probability tree draws it: the first branches give the probabilities of the B_j , the second branches the conditional probabilities of A given each, and the probability of A is the sum over the paths that end in A . In Example 12.2, $P(U) = 0.05 \times 0.4 + 0.10 \times 0.6 = 0.08$.

12.4.2 Bayes' rule

Combining the multiplication rule with the law of total probability lets us reverse a conditional probability:

$$P(B \mid A) = \frac{P(A \mid B) P(B)}{P(A \mid B) P(B) + P(A \mid B^c) P(B^c)}.$$

This is **Bayes' rule**, named after the Reverend Thomas Bayes, whose essay on it was published posthumously in 1763. It tells us how to update a **prior** probability $P(B)$ into a **posterior** probability $P(B \mid A)$ after observing evidence A . It is the logic of learning from signals, used in Chapter 25 (screening and signalling) and Chapter 51 (bank runs).

Worked Example 12.3. Screening loan applicants

Ten per cent of a bank's applicants would default if granted a loan. A credit-scoring model raises a warning flag for 80% of those who would default, but also for 20% of those who would not. If an applicant is flagged, what is the probability of default?

$$P(D \mid F) = \frac{0.8 \times 0.1}{0.8 \times 0.1 + 0.2 \times 0.9} = \frac{0.08}{0.26} \approx 0.308.$$

In counts: out of 1,000 applicants, 100 would default and 80 of them are flagged; 900 would repay and 180 of them are flagged. Of the 260 flagged, 80—about 31 per cent—would default. Although the model catches most defaulters, fewer than a third of flagged applicants would default, because non-defaulters are so much more numerous.

► The base-rate fallacy

Ignoring the prior probability (the *base rate*, here 10 per cent) and treating $P(F \mid D) = 0.8$ as if it were $P(D \mid F)$ is the **base-rate fallacy**. It inflates the apparent reliability of every screening test applied to a rare condition: fraud detection, medical tests, early-warning indicators of financial crises. Before interpreting a signal, ask how common the thing it signals is.

12.4.3 Odds and likelihood ratios

Bayes' rule is simplest in terms of *odds*, the ratio of the probability of an event to the probability of its complement:

$$\underbrace{\frac{P(B | A)}{P(B^c | A)}}_{\text{posterior odds}} = \underbrace{\frac{P(A | B)}{P(A | B^c)}}_{\text{likelihood ratio}} \times \underbrace{\frac{P(B)}{P(B^c)}}_{\text{prior odds}}.$$

Evidence multiplies the odds by its **likelihood ratio**: how much more likely the evidence is if B is true than if it is false. In Example 12.3, prior odds of default are $0.1/0.9 = 1/9$, the likelihood ratio of a flag is $0.8/0.2 = 4$, and posterior odds are $4/9$, a probability of $4/13 = 30.8$ per cent. If a second, independent test also flags the applicant, the odds are multiplied by its likelihood ratio again: with another ratio of 4, the posterior odds become $16/9$, a probability of 64 per cent. Evidence accumulates multiplicatively—which is also why a series of *non*-independent signals that merely repeat one another must not be counted as if each were new evidence.

Case: The prosecutor's fallacy: the Sally Clark case

In 1999 Sally Clark, an English solicitor, was convicted of murdering her two baby sons, who had died suddenly in 1996 and 1998. A central piece of evidence was statistical. A paediatrician testified that the probability of a sudden infant death in a family like hers was about 1 in 8,543, and that the probability of two such deaths was therefore about $1/8,543^2$, or 1 in 73 million.

The first error: independence. Squaring the probability assumes that the two deaths were independent events. But genetic and environmental factors that raise the risk of sudden infant death are shared within a family; given one such death, a second is considerably more likely than the base rate. The figure of 1 in 73 million grossly understated the probability of two natural deaths.

The second error: inverting the conditional. Even if the probability of two natural deaths were very small, that is $P(\text{evidence} | \text{innocence})$. The jury needed $P(\text{innocence} | \text{evidence})$. By Bayes' rule, that depends also on how likely the evidence is under the alternative—and a double infant murder by a mother is itself extremely rare. The relevant comparison is between two rare explanations, not between one rare explanation and certainty. Treating a small $P(\text{evidence} | \text{innocence})$ as a small probability of innocence is known as the **prosecutor's fallacy**.

The outcome. In 2001 the Royal Statistical Society issued a public statement that the figure had no statistical basis. Clark's conviction was overturned on a second appeal in 2003, after it emerged that pathology evidence suggesting a natural cause for one death had not been disclosed. She had spent more than three years in prison, and died in 2007. Several other convictions that relied on similar testimony were reviewed.

The lessons for economics and finance. The same two errors appear in risk management: multiplying small probabilities of default as if defaults were independent (the error that made many mortgage securities appear almost riskless before 2007, Chapter 51), and reading a model's low probability of observing some data under a hypothesis as a low probability of the hypothesis itself (the misinterpretation of p -values discussed in Chapter 15).

Summary

- Probabilities satisfy Kolmogorov's axioms; $P(A^c) = 1 - P(A)$ and $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

- Counting uses the multiplication principle, permutations $n!/(n-k)!$ and combinations $\binom{n}{k}$.
- $P(A | B) = P(A \cap B)/P(B)$. $P(A | B)$ and $P(B | A)$ are different.
- Independence means $P(A \cap B) = P(A)P(B)$. For independent trials, $P(\text{at least one}) = 1 - (1 - p)^n$; correlation fattens tails.
- The law of total probability sums over a partition; Bayes' rule reverses conditional probabilities and updates priors into posteriors.
- In odds form, posterior odds equal the likelihood ratio times the prior odds; independent evidence multiplies.
- Ignoring base rates, assuming independence and inverting conditionals are the three classic errors.

Key terms

experiment; sample space; event; mutually exclusive; probability; Kolmogorov's axioms; complement rule; addition rule; multiplication principle; permutation; factorial; combination; conditional probability; multiplication rule; two-way table; independence; repeated trials; partition; law of total probability; probability tree; Bayes' rule; prior; posterior; base-rate fallacy; odds; likelihood ratio; prosecutor's fallacy.

Exercises

Exercise 12.1 (Addition rule). In a region, 30 per cent of firms export, 20 per cent import and 12 per cent do both. What proportion export or import? What proportion do neither?

Exercise 12.2 (Counting). (a) How many ways can an analyst rank 5 of 12 funds in order? (b) How many different committees of 4 can be chosen from 9 directors? (c) What is the probability that a randomly chosen committee of 4 includes both of two particular directors?

Exercise 12.3 (Two-way table). Of 500 small-business loans, 200 were to firms with collateral; 10 of these defaulted. Of the 300 without collateral, 45 defaulted. Find $P(\text{default})$, $P(\text{default} | \text{collateral})$, $P(\text{collateral} | \text{default})$. Are default and collateral independent?

Exercise 12.4 (Repeated trials). A payment system has 5 independent servers, each available 99 per cent of the time. The system works if at least one server is available. What is the probability that the system is down? What if the servers are in the same building and fail together with probability 0.5 per cent whenever any one fails?

Exercise 12.5 (Total probability). In a given year, an economy is in recession with probability 0.15. A firm's sales fall with probability 0.7 in a recession and 0.2 otherwise. What is the probability that its sales fall?

Exercise 12.6 (Bayes' rule). A leading indicator signals "recession" in 70% of the years that precede a recession and in 10% of the years that do not. Recessions follow 20% of years. Given a signal, what is the probability of a recession?

Exercise 12.7 (Fraud detection). One card transaction in 1,000 is fraudulent. A detection system flags 95 per cent of fraudulent transactions and 2 per cent of legitimate ones. What proportion of flagged transactions are fraudulent? What would the bank need to change to raise that proportion to one half?

Exercise 12.8 (Likelihood ratios). An analyst believes there is a 25 per cent chance that a company will cut its dividend. The company then announces a delay in its results, which happens with probability 0.6 before dividend cuts and 0.15 otherwise. Update the probability using odds. A second, independent warning sign with likelihood ratio 2 appears. What is the probability now?

Exercise 12.9 (Diagnose the error). “Only 1 per cent of honest firms have revenue patterns like this one’s, so there is a 99 per cent chance this firm is committing fraud.” Explain the error, and state what additional information is needed.

Exercise 12.10 (Price of independence). A structured bond pays off unless both of two loans default. Each loan defaults with probability 0.05. Find the probability of loss if defaults are independent. If instead the probability that the second defaults, given that the first does, is 0.5, what is the probability of loss? How many times riskier is the bond?

Solutions

Solution 12.1. $P(E \cup I) = 0.30 + 0.20 - 0.12 = 0.38$; neither: 0.62.

Solution 12.2. (a) $P(12, 5) = 12 \times 11 \times 10 \times 9 \times 8 = 95,040$. (b) $\binom{9}{4} = 126$. (c) Committees containing both: choose the other 2 from 7, $\binom{7}{2} = 21$; probability $21/126 = 1/6$.

Solution 12.3. $P(D) = 55/500 = 0.11$; $P(D | C) = 10/200 = 0.05$; $P(C | D) = 10/55 = 0.18$. Not independent: $P(D | C) = 0.05 \neq 0.11$.

Solution 12.4. Independent: $0.01^5 = 10^{-10}$. With a common failure mode, the probability of simultaneous failure is at least 0.005 times the probability that one fails, which is about $0.01 \times 0.005 = 5 \times 10^{-5}$ —hundreds of thousands of times larger. Redundancy protects only against independent failures.

Solution 12.5. $0.7 \times 0.15 + 0.2 \times 0.85 = 0.105 + 0.17 = 0.275$.

Solution 12.6. $P(R | S) = 0.7 \times 0.2 / (0.7 \times 0.2 + 0.1 \times 0.8) = 0.14 / 0.22 \approx 0.636$.

Solution 12.7. $0.95 \times 0.001 / (0.95 \times 0.001 + 0.02 \times 0.999) = 0.00095 / 0.02093 = 4.5$ per cent. To reach one half, the false-positive rate must fall to the point where 0.02×0.999 is replaced by about 0.00095: a false-positive rate of about 0.095 per cent, roughly twenty times lower—or the system must be applied only to transactions with a much higher base rate of fraud.

Solution 12.8. Prior odds 1/3; likelihood ratio $0.6/0.15 = 4$; posterior odds 4/3: probability $4/7 = 57$ per cent. Second sign: odds 8/3, probability $8/11 = 73$ per cent.

Solution 12.9. It confuses $P(\text{pattern} | \text{honest}) = 0.01$ with $P(\text{honest} | \text{pattern})$. We need the base rate of fraud and the probability of the pattern among fraudulent firms. If 1 firm in 500 commits fraud and 50 per cent of frauds show the pattern, $P(\text{fraud} | \text{pattern}) = 0.001 / (0.001 + 0.01 \times 0.998) = 9$ per cent.

Solution 12.10. Independent: $0.05^2 = 0.0025$. Correlated: $0.05 \times 0.5 = 0.025$: ten times riskier. Ratings that assumed independence would have understated the risk tenfold.

Random Variables, Distributions and Expectation

After studying this chapter you should be able to

1. Define discrete and continuous random variables, and describe them by probability mass functions, densities and distribution functions.
2. Compute expected values, apply linearity of expectation, and use Jensen's inequality.
3. Compute variances and standard deviations, and apply the rules for linear transformations.
4. Define conditional expectation and state the law of iterated expectations.
5. Use the Bernoulli, binomial, Poisson, uniform, exponential, normal and lognormal distributions.
6. Explain why the geometric mean return falls short of the arithmetic mean by about half the variance.
7. Compute probabilities and quantiles of the normal distribution, including a simple value at risk.

A motivating question

On 19 October 1987 the U.S. stock market fell by about 20 per cent in a single day. If daily returns followed a normal distribution with the volatility typical of the preceding years, a fall that large would be a more than twenty-standard-deviation event, with a probability so small that it should not be expected to happen once in many times the age of the universe. Yet it happened, and falls of 7 or 10 per cent in a day, which the same model says should be vanishingly rare, have occurred repeatedly since.

To reason about uncertain quantities—returns, incomes, losses, the number of defaults in a loan portfolio—we need a language for describing *all* the values a quantity might take and how likely each is. That language is the random variable and its distribution. This chapter develops it, together with the expected value and variance, which summarize a distribution's centre and spread. The distributions introduced here are the building blocks of statistical inference (Chapter 15), of the economics of risk (Chapter 25) and of asset pricing (Chapter 34). The motivating question is a warning that the most convenient distribution is not always the right one.

13.1 Random Variables

A **random variable** assigns a number to each outcome of an experiment: the return on a share next year, the number of loans in a portfolio that default, the income of a household drawn at random. We write random variables with capital letters (X) and particular values with

lower-case letters (x).

13.1.1 Discrete random variables

A *discrete* random variable takes a finite or countable set of values $x_1, x_2, \dots$, with probabilities $p_k = P(X = x_k)$ that are non-negative and sum to one. The list of probabilities is the **probability mass function**.

13.1.2 Continuous random variables

A *continuous* random variable can take any value in an interval. As Chapter 7 explained, it is described by a **density** $f(x) \geq 0$ whose area over an interval gives the probability of that interval:

$$P(a \leq X \leq b) = \int_a^b f(x) dx, \quad \int_{-\infty}^{\infty} f(x) dx = 1.$$

The probability of any single exact value is zero; a density is not a probability, and can exceed one.

13.1.3 The distribution function

For any random variable, the **cumulative distribution function** (CDF) is $F(x) = P(X \leq x)$. It rises from 0 to 1. For a continuous variable, F is the integral of the density and the density is the derivative of F (the fundamental theorem of calculus, Chapter 7). The p th *quantile* is the value x_p with $F(x_p) = p$ —the population counterpart of the sample quantiles of Chapter 11.

13.2 Expectation

13.2.1 Definition and linearity

Definition 13.1. Expected value

The **expected value** of a discrete random variable is the probability-weighted average of its values:

$$\mathbb{E}[X] = \sum_k p_k x_k.$$

For a continuous variable, $\mathbb{E}[X] = \int x f(x) dx$. It is often written μ .

The expected value is the long-run average if the experiment were repeated many times (Chapter 15 makes this precise), and the centre of gravity of the distribution. It need not be a value the variable can take: the expected number of children per family may be 1.7.

Expectation is *linear*: for constants a, b and any random variables X, Y ,

$$\mathbb{E}[a + bX] = a + b \mathbb{E}[X], \quad \mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

The second rule holds whether or not X and Y are independent. The expected return on a portfolio is therefore the weighted average of the expected returns of its assets (Chapter 14).

13.2.2 Functions of random variables and Jensen's inequality

For a function g , $\mathbb{E}[g(X)] = \sum_k p_k g(x_k)$ (or $\int g(x) f(x) dx$). In general

$$\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$$

for a nonlinear g . If g is concave (Chapter 5), **Jensen's inequality** states that

$$\mathbb{E}[g(X)] \leq g(\mathbb{E}[X]),$$

with the inequality reversed for convex g . Geometrically, a concave function lies above its chords, so averaging the function's values gives less than evaluating it at the average.

Worked Example 13.2. Utility of a gamble

A person with utility $u(w) = \sqrt{w}$ faces a gamble paying 0 or 100 with equal probability. Its expected value is 50, but its expected utility is $0.5\sqrt{0} + 0.5\sqrt{100} = 5$, less than $u(50) = 7.07$. She would accept any sure amount above 25 (since $\sqrt{25} = 5$) in place of the gamble. This inequality is the source of risk aversion, developed in Chapter 25.

13.3 Variance

Definition 13.3. Variance

The **variance** of X is the expected squared deviation from the mean:

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mu^2.$$

The **standard deviation** is $\sigma_X = \sqrt{\text{Var}(X)}$, measured in the same units as X .

The second formula follows by expanding the square and using linearity: $\mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu^2 + \mu^2$. Rules: $\text{Var}(a + bX) = b^2 \text{Var}(X)$. Adding a constant does not change dispersion; scaling by b scales the standard deviation by $|b|$. In finance, the standard deviation of returns is called **volatility**.

Worked Example 13.4. Return on an investment

An investment returns -10% with probability 0.25, 5% with probability 0.5 and 20% with probability 0.25. Then

$$\mathbb{E}[R] = 0.25(-10) + 0.5(5) + 0.25(20) = 5\%,$$

$$\text{Var}(R) = 0.25(-15)^2 + 0.5(0)^2 + 0.25(15)^2 = 112.5 (\%^2), \quad \sigma_R \approx 10.61\%.$$

13.3.1 The arithmetic and geometric mean return

Chapter 2 showed that the compound (geometric mean) return on an investment is below the arithmetic mean of its returns, and promised a formula for the gap. A second-order Taylor approximation (Chapter 5) of $\ln(1 + R)$ around zero gives $\ln(1 + R) \approx R - R^2/2$. Taking expectations,

$$\mathbb{E}[\ln(1 + R)] \approx \mu - \frac{1}{2}(\sigma^2 + \mu^2) \approx \mu - \frac{1}{2}\sigma^2$$

when the mean return μ is small. Since $\mathbb{E}[\ln(1 + R)]$ is the long-run growth rate of wealth (in log terms), the compound return is approximately the arithmetic mean minus half the variance. For the investment in Example 13.4, $0.05 - \frac{1}{2}(0.1061)^2 = 4.44$ per cent, close to the exact $\mathbb{E}[\ln(1 + R)] = 4.36$ per cent in logs (4.46 per cent as a compound rate). *Volatility drag* explains why two investments with the same expected return can have very different long-run growth, and why leverage that raises volatility can lower long-run wealth.

13.4 Conditional Expectation

If X and Y are random variables, the **conditional expectation** $\mathbb{E}[X \mid Y = y]$ is the expected value of X computed from the conditional distribution of X given $Y = y$ (Chapter 12). As a function of the random variable Y , $\mathbb{E}[X \mid Y]$ is itself random. The **law of iterated expectations** states that averaging the conditional expectation over Y recovers the unconditional expectation:

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X].$$

If a firm's expected profit is 10 in a boom and 2 in a recession, and a boom has probability 0.6, its expected profit is $0.6 \times 10 + 0.4 \times 2 = 6.8$: the law of total probability applied to expectations.

Conditional expectations are the best forecasts in a precise sense: $\mathbb{E}[X \mid Y]$ minimizes the expected squared forecast error among all functions of Y . This is why the theory of *rational expectations* (Chapter 47) models people's forecasts as conditional expectations given their information, written $\mathbb{E}_t[\cdot]$, and why regression (Chapter 16) can be understood as estimating a conditional expectation.

13.5 Discrete Distributions

13.5.1 Bernoulli and binomial

A **Bernoulli** random variable equals 1 ("success") with probability p and 0 otherwise. Its mean is p and its variance $p(1 - p)$. The default of a single loan, the purchase decision of a single customer and the outcome of a single trial are Bernoulli.

The number of successes in n independent Bernoulli trials with the same p has the **binomial distribution**:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n,$$

with mean np and variance $np(1 - p)$. The combination $\binom{n}{k}$ counts the orderings in which k successes can occur (Chapter 12).

Worked Example 13.5. Defaults in a loan portfolio

A portfolio holds 20 independent loans, each with a 5 per cent probability of default. The number of defaults is binomial with $n = 20$, $p = 0.05$: mean 1, variance 0.95. The probability of no defaults is $0.95^{20} = 0.358$; of exactly one, $20 \times 0.05 \times 0.95^{19} = 0.377$; of two or more, $1 - 0.358 - 0.377 = 0.264$.

13.5.2 Poisson

When events occur independently at a constant average rate λ per period—bank failures per year, insurance claims per month, customer arrivals per hour—the number that occur in a period often follows the **Poisson distribution**:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

Its mean and variance are both λ . It is the limit of the binomial as n grows large and p small with $np = \lambda$ fixed. If a banking system averages two failures a year, the probability of a year with none is $e^{-2} = 0.135$, and of four or more $1 - e^{-2}(1 + 2 + 2 + 4/3) = 0.143$. Observing variance far above the mean in count data—"overdispersion"—signals that events cluster, as bank failures do in crises.

13.6 Continuous Distributions

13.6.1 Uniform and exponential

A **uniform** random variable on $[a, b]$ has density $1/(b - a)$ on that interval, mean $(a + b)/2$ and variance $(b - a)^2/12$. It represents complete ignorance within known bounds and appears in auction models (Chapter 26).

The **exponential distribution** with rate λ has density $\lambda e^{-\lambda x}$ for $x \geq 0$, mean $1/\lambda$ and $P(X > x) = e^{-\lambda x}$. It describes waiting times between Poisson events. If unemployed workers find jobs at a constant rate with a mean spell of 5 months, the probability that a spell exceeds 10 months is $e^{-10/5} = 0.135$. The exponential distribution is *memoryless*: the probability of waiting another month does not depend on how long one has already waited. Real unemployment spells are not memoryless—job-finding rates fall with the length of unemployment—which is one reason economists study duration dependence.

13.6.2 The normal distribution

The **normal distribution** $N(\mu, \sigma^2)$ has the familiar bell-shaped density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right],$$

symmetric about its mean μ . About 68 per cent of its probability lies within one standard deviation of the mean, 95 per cent within 1.96 standard deviations, and 99.7 per cent within three. Any normal variable can be *standardized*: $Z = (X - \mu)/\sigma$ has the *standard normal* distribution $N(0, 1)$, whose CDF is written Φ . Probabilities for any normal variable come from Φ , tabulated or computed by software: $P(X \leq x) = \Phi((x - \mu)/\sigma)$. Sums of independent normal variables are normal, and the central limit theorem (Chapter 15) explains why the normal distribution approximates so many averages.

z	0	0.5	1	1.645	1.96	2.576
$\Phi(z)$	0.500	0.691	0.841	0.950	0.975	0.995

Worked Example 13.6. The probability of a loss

Suppose the annual return on a diversified equity portfolio is normal with mean 7 per cent and standard deviation 16 per cent. The probability of a loss in a year is $\Phi((0 - 7)/16) = \Phi(-0.44) = 0.33$: about one year in three. The probability of losing more than 25 per cent is $\Phi(-2) = 0.023$, about one year in 44—if the normal model were right about the tails, which, as the case below shows, it is not.

Value at risk. Risk managers summarize the tail of a loss distribution by its quantiles. The 5 per cent **value at risk** (VaR) over a period is the loss that will be exceeded with probability 5 per cent. If a portfolio's monthly return is normal with mean 0.5 per cent and standard deviation 4 per cent, its 5 per cent quantile is $0.5 - 1.645 \times 4 = -6.08$ per cent: the monthly 5 per cent VaR is a loss of 6.08 per cent of the portfolio. VaR says nothing about how large losses are *beyond* that threshold; Chapter 34 discusses expected shortfall and the limits of normal-based risk measures.

13.6.3 The lognormal distribution

If $\ln X \sim N(\mu, \sigma^2)$, then X is **lognormal**. Lognormal variables are always positive and right-skewed, which makes them natural models for asset prices, incomes and firm sizes. Their mean is

$$\mathbb{E}[X] = e^{\mu + \sigma^2/2}.$$

The term $\sigma^2/2$ is a Jensen's-inequality correction: because the exponential is convex, the mean of X exceeds the exponential of the mean of $\ln X$, which is the median of X . The same term is the volatility drag of the previous section seen from the other side.

13.6.4 Distributions for inference

Three further distributions arise from sums of squared normal variables and are used to construct confidence intervals and tests: the *chi-square* distribution (for variances), Student's *t* distribution (for means when the variance is estimated; it has heavier tails than the normal, converging to it as the sample grows) and the *F* distribution (for comparing variances and testing several coefficients at once). Chapters 15 and 16 introduce each where it is needed.

Case: Black Monday and the tails of the distribution

The event. On Monday 19 October 1987 the Dow Jones Industrial Average fell 22.6 per cent and the S&P 500 index about 20.5 per cent, the largest one-day percentage falls in their histories. Markets around the world fell sharply the same day or the next.

What the normal model says. Daily returns on the S&P 500 in the years before 1987 had a standard deviation of roughly 1 per cent. Under a normal distribution with that volatility, a fall of 20 per cent is a twenty-standard-deviation event. The probability $\Phi(-20)$ is of the order of 10^{-89} : not a once-in-a-century event but one that should never be observed in the lifetime of the universe.

What the data say. Daily returns on stock indexes, exchange rates and commodities have *fat tails*: large moves are far more frequent than the normal distribution predicts. Moves of five standard deviations or more, which a normal model says should occur about once in several thousand years of trading days, occur in most years somewhere in world markets. Returns also show *volatility clustering*: calm periods and turbulent periods, rather than a constant standard deviation (Chapter 18 models this).

Why it matters. Risk measures built on the normal distribution—including many value-at-risk models used by banks before 2008—systematically understate the probability of large losses. Pricing models for options that assume lognormal prices (Chapter 41) misprice deep out-of-the-money options, which is why after 1987 the prices of options protecting against crashes rose permanently relative to the model (the “volatility smile”). Regulators responded after 2008 by requiring banks to use stressed scenarios and expected shortfall, which looks beyond the VaR threshold (Chapter 51). The general lesson: a distribution is a model, chosen for convenience; its tails, where the damage is done, are exactly where convenience and reality most often diverge.

Summary

- Random variables are described by probability mass functions (discrete), densities (continuous) and cumulative distribution functions.
- $\mathbb{E}[X]$ is the probability-weighted average; expectation is linear. Jensen's inequality: $\mathbb{E}[g(X)] \leq g(\mathbb{E}[X])$ for concave g .

- $\text{Var}(X) = \mathbb{E}[X^2] - \mu^2$; $\text{Var}(a + bX) = b^2 \text{Var}(X)$. The compound return is about the arithmetic mean minus half the variance.
- The law of iterated expectations: $\mathbb{E}[\mathbb{E}[X | Y]] = \mathbb{E}[X]$. Conditional expectations are optimal forecasts.
- Binomial: $\binom{n}{k} p^k (1-p)^{n-k}$, mean np ; Poisson: mean and variance λ ; exponential: memoryless waiting times.
- The normal distribution is standardized by $z = (x - \mu)/\sigma$; 95 per cent of its mass lies within 1.96 standard deviations. VaR is a quantile of the loss distribution.
- Lognormal variables have mean $e^{\mu + \sigma^2/2}$. Financial returns have fatter tails than the normal.

Key terms

random variable; probability mass function; density; cumulative distribution function; quantile; expected value; linearity of expectation; Jensen's inequality; variance; standard deviation; volatility; volatility drag; conditional expectation; law of iterated expectations; Bernoulli distribution; binomial distribution; Poisson distribution; overdispersion; uniform distribution; exponential distribution; memoryless; normal distribution; standard normal; value at risk; lognormal distribution; chi-square distribution; t distribution; F distribution; fat tails; volatility clustering.

Exercises

Exercise 13.1 (Expectation and variance). X is 0 with probability 0.5, 100 with probability 0.3 and 200 with probability 0.2. Find $\mathbb{E}[X]$, $\text{Var}(X)$ and σ_X .

Exercise 13.2 (Linear transformation). A firm's profit is $\Pi = 3Q - 50$, where Q is random with mean 40 and standard deviation 10. Find the mean and standard deviation of profit.

Exercise 13.3 (Jensen). An investor with utility $u(w) = \ln w$ and wealth 100 is offered a bet that raises wealth to 150 or lowers it to 60 with equal probability. Find the expected wealth, expected utility and certainty equivalent. Should she accept?

Exercise 13.4 (Volatility drag). Two investments both have an expected annual return of 8 per cent; their volatilities are 10 and 30 per cent. Approximate the long-run compound return of each. What does this imply for comparing funds by their average returns?

Exercise 13.5 (Iterated expectations). A bank's loan losses are 1 per cent of its loans in a normal year and 6 per cent in a recession year. The probability of a recession is 0.15. Find the expected loss rate.

Exercise 13.6 (Binomial). A salesperson makes 10 independent calls, each ending in a sale with probability 0.3. Find the probabilities of no sales, exactly 3 sales and at least one sale, and the mean and variance of sales.

Exercise 13.7 (Poisson). Claims on an insurer's small policy book arrive at an average of 3 per month. Find the probability of no claims in a month and of more than 5 claims.

Exercise 13.8 (Normal). Monthly household spending is normal with mean 3,000 and standard deviation 600. What proportion of households spend more than 4,000? Below what amount do 90 per cent of households spend? (Use $\Phi(1.667) = 0.952$ and $\Phi(1.282) = 0.90$.)

Exercise 13.9 (Value at risk). A trading portfolio worth 50 million has daily returns that are normal with mean 0 and standard deviation 1.2 per cent. Find the 1 per cent daily VaR in currency. Why might the true 1 per cent loss be larger?

Exercise 13.10 (Lognormal). The log of a share's price in a year is normal with mean $\ln 100 + 0.05$ and standard deviation 0.3. Find the median and the mean of the price in a year.

Solutions

Solution 13.1. $\mathbb{E}[X] = 30 + 40 = 70$. $\mathbb{E}[X^2] = 0.3 \times 10,000 + 0.2 \times 40,000 = 11,000$. $\text{Var}(X) = 11,000 - 4,900 = 6,100$; $\sigma_X \approx 78.1$.

Solution 13.2. $\mathbb{E}[\Pi] = 3 \times 40 - 50 = 70$; $\sigma_\Pi = 3 \times 10 = 30$.

Solution 13.3. Expected wealth 105. Expected utility $0.5 \ln 150 + 0.5 \ln 60 = 0.5(5.0106 + 4.0943) = 4.5525$. Certainty equivalent $e^{4.5525} = 94.87$, below current wealth 100, so she declines, although the bet has a positive expected value.

Solution 13.4. $0.08 - 0.01/2 = 7.5$ per cent; $0.08 - 0.09/2 = 3.5$ per cent. Average (arithmetic) returns overstate long-run growth, and by more for volatile funds; compare compound returns.

Solution 13.5. $0.85 \times 1 + 0.15 \times 6 = 1.75$ per cent.

Solution 13.6. $P(0) = 0.7^{10} = 0.0282$; $P(3) = \binom{10}{3} 0.3^3 0.7^7 = 120 \times 0.027 \times 0.08235 = 0.2668$; $P(\geq 1) = 0.9718$. Mean 3, variance 2.1.

Solution 13.7. $P(0) = e^{-3} = 0.0498$. $P(\leq 5) = e^{-3}(1 + 3 + 4.5 + 4.5 + 3.375 + 2.025) = e^{-3} \times 18.4 = 0.916$; $P(> 5) = 0.084$.

Solution 13.8. $z = (4,000 - 3,000)/600 = 1.667$; $P = 1 - 0.952 = 4.8$ per cent. 90th percentile: $3,000 + 1.282 \times 600 = 3,769$.

Solution 13.9. The 1 per cent quantile of a standard normal is -2.326 : $\text{VaR} = 2.326 \times 1.2\% = 2.79$ per cent of 50 million = 1.40 million. Fat tails and volatility clustering make extreme losses more likely than the normal model predicts, especially in turbulent periods.

Solution 13.10. Median $= e^{\ln 100 + 0.05} = 100e^{0.05} = 105.13$. Mean $= 100e^{0.05 + 0.045} = 100e^{0.095} = 109.97$.

Covariance, Correlation and Diversification

After studying this chapter you should be able to

1. Describe the joint distribution of two random variables and define independence of random variables.
2. Compute and interpret covariance and correlation, in populations and in samples, and recognize their limitations.
3. Derive the variance of a sum and use it to explain diversification.
4. Compute the expected return and variance of a portfolio, using matrix notation for many assets.
5. Distinguish systematic from idiosyncratic risk, and show why diversification cannot remove average covariance.
6. Find the minimum-variance combination of two assets and the optimal mix of a risky and a safe asset under a mean–variance objective.

A motivating question

An investor holds shares in a single company whose annual return has a standard deviation of 30 per cent. She could instead split the same money equally among ten similar companies, each just as risky. How much risk would she remove? If the ten returns were unrelated, the portfolio's standard deviation would fall to under 10 per cent. In practice, company returns tend to move together, and the answer is about 18 per cent. However many companies she adds, she can never get below about 16 per cent.

The reason lies in how random variables move *together*—their covariance. Covariance determines how much risk can be diversified away, how banks' losses on many loans accumulate, how a firm's revenue in different markets combines, and, in Chapter 34, which risks investors are rewarded for bearing. This chapter develops covariance and correlation, the mathematics of portfolios, and the limits of diversification.

14.1 Joint Distributions

Two random variables X and Y are described together by their **joint distribution**: for discrete variables, the probabilities $P(X = x, Y = y)$ of each pair of values. The distribution of X alone—its *marginal* distribution—is obtained by summing over y . X and Y are **independent random variables** if the joint probabilities factor, $P(X = x, Y = y) = P(X = x)P(Y = y)$ for every pair, so that learning the value of one tells us nothing about the other (Chapter 12).

Returns on assets are naturally described jointly across *states of the world*. Suppose next year's economy may be in a recession (probability 0.25), normal (0.5) or a boom (0.25), and two assets return:

State	Probability	Asset 1	Asset 2
Recession	0.25	−10%	10%
Normal	0.50	5%	5%
Boom	0.25	20%	0%

Asset 1 is the investment of Example 13.4 of Chapter 13, with mean 5 per cent and variance 112.5 (per cent squared). Asset 2 also has mean 5 per cent, with variance $0.25(5)^2 + 0.5(0)^2 + 0.25(-5)^2 = 12.5$. The two clearly move in opposite directions.

14.2 Covariance and Correlation

Definition 14.1. Covariance and correlation

The **covariance** of X and Y is $\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mu_X\mu_Y$. Their **correlation** is

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X\sigma_Y}, \quad -1 \leq \rho_{XY} \leq 1.$$

Covariance is positive when X and Y tend to be above their means together, and negative when one tends to be above its mean when the other is below. In the table, the covariance of the two assets is $0.25(-15)(5) + 0.5(0)(0) + 0.25(15)(-5) = -37.5$, and their correlation is $-37.5/(10.61 \times 3.54) = -1$: here asset 2's return is an exact linear function of asset 1's.

Correlation rescales covariance to be unit-free. $\rho = \pm 1$ means an exact linear relationship; $\rho = 0$ means no *linear* relationship. Properties used constantly:

$$\text{Cov}(X, X) = \text{Var}(X), \quad \text{Cov}(a + bX, c + dY) = bd \text{Cov}(X, Y).$$

14.2.1 Sample covariance and correlation

From n paired observations (x_i, y_i) , the *sample covariance* is $s_{xy} = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$ and the *sample correlation* $r = s_{xy}/(s_x s_y)$. For the data $x = (2, 4, 6, 8, 10)$ and $y = (1, 3, 2, 5, 4)$, $\bar{x} = 6$, $\bar{y} = 3$, $s_{xy} = 16/4 = 4$, $s_x^2 = 10$, $s_y^2 = 2.5$, and $r = 4/\sqrt{25} = 0.8$. A scatter plot (Chapter 11) should always accompany a correlation.

► What correlation does not measure

(i) *Nonlinearity*: $Y = X^2$ with X symmetric about zero has zero correlation although Y is completely determined by X ; independence implies zero covariance, but zero covariance does not imply independence. (ii) *Outliers*: one extreme point can create or destroy a correlation. (iii) *Causation*: correlation between X and Y may reflect a common cause, reverse causation or selection (Chapter 17). (iv) *Trends*: two unrelated series that both grow over time are highly correlated (Chapter 18). (v) *Instability*: correlations estimated in calm periods often rise sharply in crises, as this chapter's case shows.

14.3 The Variance of a Sum

Proposition 14.2. Variance of a sum

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y).$$

Proof. $\text{Var}(aX + bY) = \mathbb{E}[(a(X - \mu_X) + b(Y - \mu_Y))^2]$. Expanding the square and using linearity of expectation gives $a^2 \mathbb{E}[(X - \mu_X)^2] + b^2 \mathbb{E}[(Y - \mu_Y)^2] + 2ab \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$. $\square$

If X and Y are independent (or merely uncorrelated), the covariance term vanishes and variances simply add. Otherwise the covariance term can raise or lower the total. This formula is the mathematics of *diversification*. A portfolio half in each of the two assets above has variance

$$0.25(112.5) + 0.25(12.5) + 2(0.25)(-37.5) = 28.125 + 3.125 - 18.75 = 12.5,$$

far below the first asset's 112.5, with the same expected return of 5%. Assets whose returns move in opposite directions insure each other. The same logic explains why banks try to diversify their loans across regions and industries, and why concentrated exposures—to one housing market, for example—are dangerous (Chapter 51).

14.4 Portfolios

14.4.1 Two assets

A **portfolio** places fraction w of wealth in asset 1 and $1 - w$ in asset 2 (weights sum to one; a negative weight is a *short* position). Its return is $R_p = wR_1 + (1 - w)R_2$, so by linearity and Theorem 14.2,

$$\mathbb{E}[R_p] = w\mu_1 + (1 - w)\mu_2, \quad \sigma_p^2 = w^2\sigma_1^2 + (1 - w)^2\sigma_2^2 + 2w(1 - w)\rho\sigma_1\sigma_2.$$

The expected return is a weighted average; the standard deviation is *not*, unless $\rho = 1$. For $\rho < 1$, σ_p is below the weighted average of σ_1 and σ_2 : diversification lowers risk without lowering expected return.

Worked Example 14.3. The minimum-variance portfolio

Asset 1 has $\sigma_1 = 20$ per cent, asset 2 has $\sigma_2 = 10$ per cent, and their correlation is 0.2, so $\sigma_{12} = 0.2 \times 0.2 \times 0.1 = 0.004$. Setting the derivative of σ_p^2 with respect to w to zero (Chapter 5) gives the weight that minimizes variance:

$$w^* = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} = \frac{0.01 - 0.004}{0.05 - 0.008} = 0.143.$$

With 14.3 per cent in the riskier asset, the portfolio's variance is 0.00914 and its standard deviation 9.56 per cent—*lower* than that of the safer asset alone. Adding a little of a risky, imperfectly correlated asset reduces total risk.

14.4.2 Many assets

With n assets, collect the weights in a vector $\mathbf{w}$, the expected returns in $\boldsymbol{\mu}$, and the variances and covariances in the **covariance matrix** Σ , whose (i, j) element is $\sigma_{ij} = \text{Cov}(R_i, R_j)$ and whose

diagonal holds the variances. Using the matrix algebra of Chapter 8,

$$\mathbb{E}[R_p] = \mathbf{w}'\boldsymbol{\mu}, \quad \sigma_p^2 = \mathbf{w}'\Sigma\mathbf{w} = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij}.$$

Because the variance of any portfolio cannot be negative, Σ is positive semidefinite. A covariance matrix for n assets has n variances and $n(n-1)/2$ distinct covariances: for 500 shares, nearly 125,000 covariances. Estimating so many from limited data is error-prone, and optimizing portfolios with noisy estimates can produce extreme, unstable weights—one reason practitioners use factor models (Chapter 34), shrink estimates toward simple benchmarks, and impose position limits.

14.4.3 The limits of diversification

Suppose an investor holds n assets in equal weights $1/n$, each with variance $\bar{\sigma}^2$, and every pair has covariance $\bar{\sigma}_{ij}$. Of the n^2 terms in the double sum, n are variances and $n(n-1)$ are covariances, so

$$\sigma_p^2 = \frac{1}{n}\bar{\sigma}^2 + \left(1 - \frac{1}{n}\right)\bar{\sigma}_{ij} \longrightarrow \bar{\sigma}_{ij} \quad \text{as } n \rightarrow \infty.$$

Proposition 14.4. Systematic and idiosyncratic risk

In a large, equally weighted portfolio, the contribution of each asset's own variance vanishes, and the portfolio's variance approaches the average covariance among the assets. Risk specific to individual assets (**idiosyncratic risk**) can be diversified away; risk common to all of them (**systematic risk**) cannot.

Worked Example 14.5. How much does diversification help?

Each share has a standard deviation of 30 per cent and every pair has correlation 0.3, so $\bar{\sigma}^2 = 0.09$ and $\bar{\sigma}_{ij} = 0.027$. The portfolio standard deviation is 30 per cent with one share, 24.2 per cent with two, 18.2 per cent with ten, 16.8 per cent with fifty, and approaches $\sqrt{0.027} = 16.4$ per cent. Most of the benefit comes from the first ten or twenty holdings; beyond that, what remains is the market-wide risk that all shares share.

Theorem 14.4 is the foundation of modern asset pricing. If idiosyncratic risk can be eliminated costlessly by diversifying, investors should not expect to be paid for bearing it: only systematic risk should command a risk premium. Chapter 34 turns this insight into the capital asset pricing model.

14.5 Choosing Risk: Mean and Variance

How much risk should an investor bear? One tractable model, due to Harry Markowitz (1952) and James Tobin (1958), assumes that investors care only about the expected return and variance of their portfolio, trading them off according to a *mean–variance objective*

$$U = \mathbb{E}[R_p] - \frac{\lambda}{2} \sigma_p^2,$$

where $\lambda > 0$ measures risk aversion. Suppose the investor divides her wealth between a safe asset returning r_f and a risky portfolio with expected return μ and variance σ^2 , placing a

fraction w in the risky portfolio. Then $\mathbb{E}[R_p] = r_f + w(\mu - r_f)$ and $\sigma_p^2 = w^2\sigma^2$, and maximizing U gives

$$w^* = \frac{\mu - r_f}{\lambda\sigma^2}.$$

The investor holds more of the risky asset the higher its expected excess return, the lower its variance and the lower her risk aversion.

Worked Example 14.6. The risky share

A safe asset returns 4 per cent; a risky fund has expected return 10 per cent and variance 0.04 (standard deviation 20 per cent). With $\lambda = 3$, $w^* = 0.06/(3 \times 0.04) = 0.5$: half in the fund. The portfolio's expected return is 7 per cent and its standard deviation 10 per cent. With $\lambda = 1$, the formula gives $w^* = 1.5$: borrowing to invest 150 per cent of wealth in the fund. If borrowing is not allowed, the constraint $w \leq 1$ binds and the optimum is the corner $w = 1$ (Chapter 6). An optimum computed from estimated inputs is only as good as the estimates: an error of a few points in μ changes w^* greatly.

Case: Long-Term Capital Management, 1998: when correlations go to one

The fund. Long-Term Capital Management (LTCM), founded in 1994 by the former Salomon Brothers bond trader John Meriwether, counted among its partners Myron Scholes and Robert Merton, who shared the 1997 Nobel prize for their work on option pricing (Chapter 41). Its strategy was to buy securities that it judged cheap and sell similar securities it judged expensive—for example, buying less liquid government bonds and selling more liquid ones of similar maturity—betting that price differences would narrow. Each trade offered a small expected profit with, according to historical data, low volatility, and the fund held hundreds of such positions across many markets, which its models treated as largely uncorrelated. To earn attractive returns from small price differences, it used heavy leverage: balance-sheet assets of the order of 25 times its capital of about 4.7 billion dollars at the start of 1998, and derivatives with notional values above a trillion dollars.

What happened. After Russia defaulted on its domestic debt and devalued in August 1998, investors worldwide fled to the safest and most liquid securities. Almost all of LTCM's positions—in different countries and instruments, but all bets that illiquid assets would converge toward liquid ones—lost money at the same time. Correlations that had been low in the historical data rose toward one. Within weeks the fund had lost most of its capital, about 4.6 billion dollars in total. In September 1998, fearing that a disorderly liquidation would disrupt markets, the Federal Reserve Bank of New York brought together fourteen banks and brokers, which injected 3.6 billion dollars in exchange for control of the fund, which was later wound down.

In terms of this chapter. The diversification in LTCM's models depended on estimated covariances. Theorem 14.4 says that diversification removes idiosyncratic risk but not what the positions have in common; LTCM's positions shared a hidden common factor—exposure to a flight to liquidity—that was invisible in calm-period data. With leverage of 25, a loss of 4 per cent on assets wipes out the capital. The episode anticipated the crisis of 2007–09 (Chapter 51), in which assets that had appeared diversified, such as mortgage securities from many regions, turned out to share exposure to a nationwide fall in house prices.

Summary

- Joint distributions describe how random variables move together; independence means the joint probabilities factor.
- $\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$; correlation $\rho = \text{Cov} / (\sigma_X \sigma_Y)$ lies between -1 and 1 and measures linear association only.
- $\text{Var}(aX + bY) = a^2 \text{Var } X + b^2 \text{Var } Y + 2ab \text{Cov}(X, Y)$: negative or low covariance reduces the risk of a combination.
- Portfolio expected return is $\mathbf{w}'\boldsymbol{\mu}$; variance is $\mathbf{w}'\Sigma\mathbf{w}$. With $\rho < 1$, portfolio volatility is below the weighted average of volatilities.
- In large portfolios, idiosyncratic risk vanishes and variance approaches the average covariance: systematic risk remains.
- A mean–variance investor holds $w^* = (\mu - r_f) / (\lambda \sigma^2)$ in a risky portfolio, subject to constraints.
- Estimated covariances are unstable; correlations tend to rise in crises.

Key terms

joint distribution; marginal distribution; independent random variables; states of the world; covariance; correlation; sample covariance; sample correlation; variance of a sum; diversification; portfolio; portfolio weight; short position; minimum-variance portfolio; covariance matrix; idiosyncratic risk; systematic risk; mean–variance objective; risk aversion coefficient; leverage; flight to liquidity.

Exercises

Exercise 14.1 (Covariance from states). Two assets return $(-5, 8, 15)$ and $(2, 4, 12)$ per cent in three equally likely states. Find each mean and variance, their covariance and correlation.

Exercise 14.2 (Sample correlation). Compute the sample correlation between $x = (1, 2, 3, 4, 5)$ and $y = (2, 4, 5, 4, 5)$.

Exercise 14.3 (Portfolio variance). Two assets have standard deviations 20% and 10% and correlation 0.5. Find the standard deviation of an equal-weighted portfolio. What would it be if the correlation were -1 ?

Exercise 14.4 (Minimum variance). Assets with volatilities 25 and 15 per cent have correlation -0.2 . Find the minimum-variance weight in the first asset and the resulting volatility.

Exercise 14.5 (Diversification). Shares each have volatility 40 per cent and pairwise correlation 0.25. Find the volatility of equally weighted portfolios of 1, 5, 25 and infinitely many shares.

Exercise 14.6 (Loans). A bank holds 100 loans of equal size. Each loan's loss rate has standard deviation 10 per cent. Find the standard deviation of the portfolio loss rate if losses are (a) independent, (b) correlated with pairwise correlation 0.2. What does this imply for how much capital the bank needs?

Exercise 14.7 (Matrix form). With weights $\mathbf{w} = (0.5, 0.3, 0.2)'$, expected returns $(8, 6, 4)$ per cent, volatilities $(20, 15, 5)$ per cent, and correlations 0.5 between assets 1 and 2 and zero otherwise, compute the portfolio's expected return and volatility.

Exercise 14.8 (Mean–variance). A safe asset returns 3 per cent; an equity fund has expected return 8 per cent and volatility 18 per cent. Find the optimal fraction in equities for $\lambda = 2$ and $\lambda = 6$. If the investor cannot borrow, what does she hold when $\lambda = 1$?

Exercise 14.9 (Leverage). A fund has assets of 100 and capital of 4. Its assets fall in value by 3 per cent. What happens to its capital and leverage? Explain why leverage makes the accuracy of covariance estimates especially important.

Solutions

Solution 14.1. Means 6 and 6. Deviations $(-11, 2, 9)$ and $(-4, -2, 6)$. Variances $(121 + 4 + 81)/3 = 68.67$ and $(16 + 4 + 36)/3 = 18.67$. Covariance $(44 - 4 + 54)/3 = 31.33$. Correlation $31.33/\sqrt{68.67 \times 18.67} = 31.33/35.80 = 0.875$.

Solution 14.2. $\bar{x} = 3, \bar{y} = 4$; cross-products sum to 6; $\sum(x - \bar{x})^2 = 10, \sum(y - \bar{y})^2 = 6$. $r = 6/\sqrt{60} = 0.775$.

Solution 14.3. $\text{Cov} = 0.5 \times 20 \times 10 = 100$. Variance $= 0.25(400) + 0.25(100) + 2(0.25)(100) = 175$; standard deviation $\approx 13.2\%$. With correlation -1 , covariance is -200 and variance $100 + 25 - 100 = 25$, standard deviation 5% .

Solution 14.4. $\sigma_{12} = -0.2 \times 0.25 \times 0.15 = -0.0075$. $w^* = (0.0225 + 0.0075)/(0.0625 + 0.0225 + 0.015) = 0.03/0.1 = 0.30$. Variance $= 0.09 \times 0.0625 + 0.49 \times 0.0225 + 2 \times 0.3 \times 0.7 \times (-0.0075) = 0.005625 + 0.011025 - 0.00315 = 0.0135$; volatility 11.6 per cent, below both assets.

Solution 14.5. $\bar{\sigma}^2 = 0.16, \bar{\sigma}_{ij} = 0.04$. $n = 1$: 40 per cent; $n = 5$: $\sqrt{0.032 + 0.032} = 25.3$ per cent; $n = 25$: $\sqrt{0.0064 + 0.0384} = 21.2$ per cent; limit $\sqrt{0.04} = 20$ per cent.

Solution 14.6. (a) $10/\sqrt{100} = 1$ per cent. (b) $\sqrt{0.01/100 + 0.99 \times 0.2 \times 0.01} = \sqrt{0.0001 + 0.00198} = 4.56$ per cent. Correlation raises the portfolio's risk more than fourfold; capital set on the assumption of independence would be far too small.

Solution 14.7. Expected return $0.5 \times 8 + 0.3 \times 6 + 0.2 \times 4 = 6.6$ per cent. Variance $= 0.25 \times 0.04 + 0.09 \times 0.0225 + 0.04 \times 0.0025 + 2 \times 0.5 \times 0.3 \times 0.5 \times 0.2 \times 0.15 = 0.01 + 0.002025 + 0.0001 + 0.0045 = 0.016625$; volatility 12.9 per cent.

Solution 14.8. $\sigma^2 = 0.0324$. $\lambda = 2$: $w^* = 0.05/0.0648 = 0.77$. $\lambda = 6$: 0.26. $\lambda = 1$: unconstrained $w^* = 1.54$; without borrowing, $w = 1$.

Solution 14.9. Assets fall to 97; debt is still 96, so capital falls to 1—a 75 per cent loss—and leverage rises from 25 to 97. With high leverage, small losses destroy capital, so errors in estimated risk—especially underestimated correlations that make many positions lose together—are fatal rather than merely costly.

Sampling, Estimation and Hypothesis Testing

After studying this chapter you should be able to

1. Explain what a sampling distribution is, and derive the mean and standard error of the sample mean.
2. State the law of large numbers and the central limit theorem, and explain why they matter.
3. Evaluate estimators by bias, variance, mean squared error, consistency and efficiency.
4. Construct and correctly interpret confidence intervals for means, proportions and differences, using the normal and t distributions and the bootstrap.
5. Propagate uncertainty through a nonlinear function with the delta method.
6. Carry out hypothesis tests, interpret p -values, and distinguish Type I and Type II errors and power.
7. Recognize the dangers of multiple testing, specification searching and publication bias, and distinguish statistical from economic significance.

A motivating question

A survey of 1,000 households finds that 43 per cent expect prices to rise faster next year. A central bank wants to know the proportion among *all* households. The true figure is almost certainly not exactly 43 per cent. Could it be 40? 48? A second survey next month finds 46 per cent. Have expectations really risen, or is the difference the luck of the draw?

Statistical inference answers such questions. It treats a sample statistic as the realization of a random variable whose distribution—the *sampling distribution*—describes how the statistic would vary across repeated samples. From that distribution come standard errors, confidence intervals and hypothesis tests: ways of stating what the data can and cannot tell us about a population, with a measure of how uncertain we are. This chapter develops the logic and the main tools, and warns about the ways they are commonly misused.

15.1 Sampling Distributions

15.1.1 The sample mean

Suppose $X_1, \dots, X_n$ are independent draws from a population with mean μ and standard deviation σ (Chapter 11 explained why random sampling matters). The **sample mean** $\bar{X} = \frac{1}{n} \sum X_i$ is itself a random variable: a different sample would give a different value. Its distribution across all possible samples is its **sampling distribution**. By linearity of expectation and the

variance of a sum of independent terms (Chapter 14),

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad \text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

The standard deviation of an estimator's sampling distribution is its **standard error**. In practice σ is unknown and is replaced by the sample standard deviation s . Precision improves only with the square root of the sample size: to halve the standard error, quadruple the sample.

Proportions. A proportion is the mean of a Bernoulli variable (Chapter 13), so the sample proportion $\hat{p}$ has mean p and standard error $\sqrt{p(1-p)/n}$, estimated by $\sqrt{\hat{p}(1-\hat{p})/n}$. In the motivating survey, $\text{SE} = \sqrt{0.43 \times 0.57/1,000} = 0.0157$, or 1.6 percentage points.

15.1.2 The law of large numbers and the central limit theorem

Proposition 15.1. Two limit theorems

Law of large numbers: as $n \rightarrow \infty$, $\bar{X}$ converges to μ . *Central limit theorem:* for large n , $\bar{X}$ is approximately normally distributed with mean μ and standard deviation $\sigma/\sqrt{n}$, whatever the distribution of the individual X_i (provided its variance is finite).

The law of large numbers is why insurance works: an insurer with many independent policies can predict its average claim closely, even though each claim is unpredictable (Chapter 25). It is also why it fails when risks are *not* independent: when house prices fall nationwide, mortgage defaults are correlated and pooling provides much less protection (Chapter 51). The central limit theorem is why the normal distribution is so useful: whatever the shape of the population—skewed incomes, lumpy losses—averages of many independent observations are approximately normal, and so are many estimators that behave like averages, including regression coefficients (Chapter 16). “Large” depends on the population: for mildly skewed data, 30 observations may suffice; for heavy-tailed data such as losses, many more are needed.

15.2 Estimators and Their Properties

An **estimator** is a rule for computing an estimate of a population quantity θ from a sample. Good estimators have these properties:

- **Unbiasedness:** $\mathbb{E}[\hat{\theta}] = \theta$. The *bias* is $\mathbb{E}[\hat{\theta}] - \theta$.
- **Low variance:** estimates vary little from sample to sample.
- **Low mean squared error:** $\text{MSE} = \mathbb{E}[(\hat{\theta} - \theta)^2] = \text{bias}^2 + \text{Var}(\hat{\theta})$. A slightly biased estimator with a much smaller variance can be more accurate overall—the logic of shrinkage methods in Chapter 19.
- **Consistency:** $\hat{\theta}$ converges to θ as the sample grows.
- **Efficiency:** among unbiased estimators, the smallest variance.

Why divide by $n - 1$? The sample variance $s^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ is unbiased for σ^2 , whereas dividing by n is biased downward. The reason is that the deviations are measured from $\bar{X}$, which is fitted to the sample and therefore closer to the observations than the unknown μ is. Formally, $\mathbb{E}[\sum (X_i - \bar{X})^2] = n\sigma^2 - n \text{Var}(\bar{X}) = (n-1)\sigma^2$. Fitting the mean “uses up” one *degree of freedom*.

15.3 Confidence Intervals

15.3.1 Construction

By the central limit theorem, $(\bar{X} - \mu)/SE$ is approximately standard normal, so with probability 0.95 it lies between -1.96 and 1.96 . Rearranging,

$$\bar{X} \pm 1.96 SE$$

contains μ in about 95 per cent of samples. It is a 95 per cent **confidence interval**. For the survey, $43 \pm 1.96 \times 1.57$, or 39.9 to 46.1 per cent: values near 40 or 46 are consistent with the data; 48 is not.

► What a confidence interval means

It is *not* correct to say that the true proportion lies in 39.9–46.1 per cent “with probability 0.95.” The proportion is a fixed number; the interval either contains it or not. The 95 per cent describes the *procedure*: across many samples, 95 per cent of intervals built this way would contain the truth. (The Bayesian intervals of Chapter 19 do support probability statements about the parameter, at the cost of specifying prior beliefs.)

15.3.2 Small samples and the t distribution

When σ is estimated by s in a small sample, the extra uncertainty is accounted for by using Student’s t distribution with $n - 1$ degrees of freedom instead of the normal. Its critical values are larger: 2.131 for a 95 per cent interval with 15 degrees of freedom, against 1.96. If 16 firms’ price increases have mean 2.4 per cent and standard deviation 1.2, the standard error is $1.2/4 = 0.3$ and the 95 per cent interval is $2.4 \pm 2.131 \times 0.3 = [1.76, 3.04]$. As n grows, the t distribution converges to the normal. Strictly, the t interval is exact only for normal populations; with larger samples, the central limit theorem makes it approximately right for others.

15.3.3 Differences

To compare two independent groups—customers shown two versions of a website, regions with and without a policy—estimate the difference in means with standard error

$$SE(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}},$$

since variances of independent estimates add.

15.3.4 The bootstrap

For complicated estimators—a median, a ratio, a Gini coefficient—a formula for the standard error may be unavailable. The **bootstrap**, introduced by Bradley Efron in 1979, estimates the sampling distribution by resampling: draw many samples of size n *with replacement* from the observed data, compute the estimator in each, and use the spread of these bootstrap estimates as the standard error (or their 2.5th and 97.5th percentiles as a 95 per cent interval). The sample stands in for the population. The bootstrap requires computing power rather than algebra, and fails when observations are not independent (unless resampling respects the dependence, for example by resampling blocks of a time series) or for extreme quantiles.

15.3.5 The delta method

Often we need the uncertainty of a nonlinear function $g(\hat{\theta})$ of an estimate. A first-order Taylor approximation (Chapter 5) gives $g(\hat{\theta}) \approx g(\theta) + g'(\theta)(\hat{\theta} - \theta)$, so

$$\text{SE}(g(\hat{\theta})) \approx |g'(\hat{\theta})| \text{SE}(\hat{\theta}).$$

This is the **delta method**. If average productivity growth is estimated at 2 per cent a year with a standard error of 0.5 points, the implied doubling time $T = 70/g = 35$ years has $|dT/dg| = 70/g^2 = 17.5$ years per point, so its standard error is about $17.5 \times 0.5 = 8.75$ years. Small uncertainty about a growth rate is large uncertainty about long-horizon outcomes.

15.4 Hypothesis Tests

15.4.1 The logic

A **hypothesis test** asks whether the data are consistent with a specific claim about the population, the **null hypothesis** H_0 —for example, that the mean equals μ_0 , or that two groups have the same mean. The *alternative* H_1 is what we would conclude instead. We compute a *test statistic*, such as

$$t = \frac{\bar{X} - \mu_0}{\text{SE}},$$

which measures how far the estimate lies from the null value in standard errors, and ask how surprising it would be if the null were true.

The **p-value** is the probability, computed assuming the null is true, of a test statistic at least as extreme as the one observed. If it is below a chosen *significance level* α , conventionally 0.05, we *reject* the null. For a two-sided test at the 5 per cent level with a large sample, this means $|t| > 1.96$.

Worked Example 15.2. Testing a mean

A survey of 100 firms finds mean expected price increases of 5.2% with sample standard deviation 10 percentage points. Is this consistent with an average of 3%? The standard error is $10/\sqrt{100} = 1$; $t = (5.2 - 3)/1 = 2.2$; since $2.2 > 1.96$, we reject 3% at the 5% level (the two-sided p -value is 0.028). The 95% confidence interval is $5.2 \pm 1.96 = [3.24, 7.16]$. A test at the 5 per cent level rejects exactly those null values that lie outside the 95 per cent confidence interval.

Worked Example 15.3. An online experiment

A bank shows 2,000 randomly chosen visitors a new application page and 2,000 the old one. Of the first group, 12 per cent apply; of the second, 10 per cent. The standard error of the difference is $\sqrt{0.12 \times 0.88/2,000 + 0.10 \times 0.90/2,000} = 0.0099$, so $t = 0.02/0.0099 = 2.02$ and $p = 0.043$. The difference is statistically significant at the 5 per cent level—barely. Because visitors were randomly assigned, it can be read as causal (Chapter 17).

15.4.2 Two kinds of error

	H_0 true	H_0 false
Reject H_0	Type I error (probability α)	Correct
Do not reject	Correct	Type II error (probability β)

The **power** of a test, $1 - \beta$, is the probability of rejecting a false null—of detecting an effect that is really there. Power rises with the size of the true effect, the sample size and the significance level, and falls with the noise in the data. Before running an experiment, researchers compute the sample size needed for adequate power. To detect a rise in application rates from 10 to 12 per cent with 80 per cent power at the 5 per cent level, each group needs about

$$n \approx \frac{(1.96 + 0.84)^2 [0.10 \times 0.90 + 0.12 \times 0.88]}{0.02^2} \approx 3,840$$

visitors, nearly twice as many as in Example 15.3: that experiment had only about a 50 per cent chance of detecting a true effect of this size.

► Statistical versus economic significance

“Statistically significant” means that an estimate is unlikely to be the product of sampling noise alone; it does not mean the effect is large or important. With enough data, trivially small effects become significant; with little data, important effects may not be. Always report and interpret the size of an estimate and its confidence interval, not just whether it is significant. Conversely, failing to reject a null hypothesis is not evidence that the null is true: an underpowered study that finds “no significant effect” has learned little.

► What a *p*-value is not

A *p*-value of 0.03 is *not* the probability that the null hypothesis is true, nor the probability that the result is a fluke. It is the probability of data this extreme *if* the null were true— $P(\text{data} \mid H_0)$, not $P(H_0 \mid \text{data})$. Confusing the two is the prosecutor’s fallacy of Chapter 12. Whether a rejected null is likely to be false depends also on how plausible the tested effect was beforehand.

15.5 Multiple Testing and the Credibility of Findings

15.5.1 The problem

If a researcher tests 20 hypotheses that are all in fact false leads, each at the 5 per cent level, the probability of at least one “significant” result is $1 - 0.95^{20} = 64$ per cent (Chapter 12). Test enough variables, subgroups, time periods or model specifications, and something will appear significant by chance. Reporting only the successes is sometimes called *p-hacking* or specification searching; when journals publish significant results more readily than null ones, **publication bias** makes the published literature overstate effects.

15.5.2 Remedies

- **Adjust the threshold.** The *Bonferroni* correction tests each of m hypotheses at level α / m (0.0025 for 20 tests), controlling the probability of any false rejection. Less conservative methods control the *false discovery rate*—the expected share of rejections that are false.
- **Pre-register.** Specify the hypotheses, sample and analysis before seeing the data, as randomized trials in medicine and increasingly in economics are required to do.
- **Replicate out of sample.** Test a finding on new data—a later period, another country—that played no part in discovering it.
- **Report everything.** Show how results change across reasonable specifications (robustness analysis), and report effect sizes with confidence intervals.

Case: The factor zoo: multiple testing in finance

The setting. Chapter 34 explains that some characteristics of shares—their exposure to the market, their size, their value relative to accounting book value—have been associated with higher average returns. From the 1970s onward, academics and practitioners searched historical data for more such “factors,” and published hundreds of them: characteristics claimed to predict returns with t -statistics above 2.

The problem. Campbell Harvey, Yan Liu and Heqing Zhu (2016) catalogued over 300 factors proposed in published research and asked what standard of evidence was appropriate given that so many had been tried—and many more tried and not reported. Using methods that account for multiple testing, they argued that a new factor should clear a t -statistic of about 3.0 rather than 2.0, and that many published factors were likely false discoveries. Later studies found that the average return associated with many published anomalies fell substantially after publication, partly because investors traded on them and partly because some had been products of data mining.

The general lesson. Every empirical field faces the same problem when researchers, fund managers or analysts search many possibilities in the same data set. A backtest of an investment strategy that was chosen after trying hundreds of variants on the same history will look far better than the strategy’s future performance is likely to be (Chapter 19). The remedies are the same: fewer, theory-guided hypotheses; higher thresholds; out-of-sample and out-of-period tests; and scepticism toward any single significant result.

Summary

- The sample mean is unbiased with standard error $\sigma / \sqrt{n}$; a proportion has standard error $\sqrt{p(1-p)/n}$.
- The law of large numbers makes averages converge; the central limit theorem makes them approximately normal.
- Estimators are judged by bias, variance, MSE ($\text{bias}^2 + \text{variance}$), consistency and efficiency; s^2 divides by $n - 1$.
- A 95 per cent confidence interval $\hat{\theta} \pm 1.96 \text{SE}$ (or a t critical value in small samples) describes a procedure that covers the truth in 95 per cent of samples. The bootstrap estimates standard errors by resampling; the delta method propagates them through functions.
- Tests compare an estimate with a null value in standard-error units; the p -value is $P(\text{data this extreme} \mid H_0)$. Type I and II errors trade off; power requires adequate samples.
- Multiple testing produces false discoveries; adjust thresholds, pre-register, replicate out of sample and report effect sizes.

Key terms

sample mean; sampling distribution; standard error; law of large numbers; central limit theorem; estimator; bias; mean squared error; consistency; efficiency; degrees of freedom; confidence interval; t distribution; bootstrap; delta method; hypothesis test; null hypothesis; alternative hypothesis; test statistic; p -value; significance level; Type I error; Type II error; power; sample size calculation; mul-

multiple testing; Bonferroni correction; false discovery rate; p -hacking; publication bias; pre-registration; statistical and economic significance.

Exercises

Exercise 15.1 (Standard errors). Household monthly spending has standard deviation 800. What is the standard error of the sample mean with samples of 100, 400 and 1,600? How large a sample is needed for a standard error of 10?

Exercise 15.2 (Confidence interval and test). A sample of 400 households has mean monthly saving rate 3.2% with standard deviation 2 percentage points. Construct a 95% confidence interval and test whether the population mean is 3%.

Exercise 15.3 (Proportions). In a poll of 1,200 voters, 52 per cent support a policy. Construct a 95 per cent confidence interval. Can you reject the hypothesis that support is 50 per cent?

Exercise 15.4 (Small sample). Nine bank branches cut waiting times after a reorganization by an average of 4 minutes, with standard deviation 3 minutes. The t critical value with 8 degrees of freedom is 2.306. Construct a 95 per cent confidence interval for the mean reduction. Is it significantly different from zero?

Exercise 15.5 (Difference in means). In one region, 500 randomly sampled firms report mean investment growth of 4.0 per cent (s.d. 6); in another, 400 firms report 3.1 per cent (s.d. 5). Test whether mean growth differs between the regions.

Exercise 15.6 (MSE). Estimator A is unbiased with standard error 4. Estimator B has bias 1 and standard error 2. Which has the lower mean squared error?

Exercise 15.7 (Delta method). A demand elasticity is estimated at -0.5 with standard error 0.1. An analyst needs $g(\epsilon) = 1/|\epsilon|$, the percentage price rise that reduces quantity by 1 per cent. Find g and its approximate standard error.

Exercise 15.8 (Power). Using the approximation in the text, how many observations per group are needed to detect a rise in a conversion rate from 20 to 22 per cent with 80 per cent power at the 5 per cent level?

Exercise 15.9 (Multiple testing). An analyst backtests 50 trading rules, none of which truly predicts returns, each at the 5 per cent level. How many would she expect to appear significant? What is the probability that at least one does? What threshold would a Bonferroni correction require?

Exercise 15.10 (Interpretation). A study finds that a training programme raises earnings by 1.5 per cent, with a 95 per cent confidence interval from -0.5 to 3.5 per cent, and reports “no significant effect.” What can and cannot be concluded?

Solutions

Solution 15.1. $800/10 = 80$; $800/20 = 40$; $800/40 = 20$. For SE 10, $\sqrt{n} = 80$, so $n = 6,400$.

Solution 15.2. $SE = 2/\sqrt{400} = 0.1$. CI: $3.2 \pm 0.196 = [3.004, 3.396]$. $t = (3.2 - 3)/0.1 = 2 > 1.96$: reject 3% at the 5% level—narrowly, as the interval’s lower bound shows.

Solution 15.3. $SE = \sqrt{0.52 \times 0.48/1,200} = 0.0144$; CI $52 \pm 2.83 = [49.2, 54.8]$ per cent. Since 50 lies inside the interval ($t = 0.02/0.0144 = 1.39$), we cannot reject 50 per cent at the 5 per cent level.

Solution 15.4. $SE = 3/3 = 1$. $CI\ 4 \pm 2.306 = [1.69, 6.31]$ minutes. Zero lies outside: significant at the 5 per cent level ($t = 4$).

Solution 15.5. $SE = \sqrt{36/500 + 25/400} = \sqrt{0.072 + 0.0625} = 0.367$. $t = 0.9/0.367 = 2.45$: reject equality at the 5 per cent level ($p \approx 0.014$).

Solution 15.6. A: $MSE = 0 + 16 = 16$. B: $1 + 4 = 5$. B is more accurate on average despite its bias.

Solution 15.7. $g(-0.5) = 2$. For $\varepsilon < 0$, $g(\varepsilon) = -1/\varepsilon$, so $|g'(\varepsilon)| = 1/\varepsilon^2 = 4$; $SE \approx 4 \times 0.1 = 0.4$. A standard error of 0.1 on the elasticity becomes 0.4 on its reciprocal, because the function is steep near -0.5 .

Solution 15.8. $n \approx 7.84 \times (0.2 \times 0.8 + 0.22 \times 0.78)/0.0004 = 7.84 \times 0.3316/0.0004 = 6,499$ per group.

Solution 15.9. Expected false positives $50 \times 0.05 = 2.5$. $P(\geq 1) = 1 - 0.95^{50} = 0.92$. Bonferroni: $0.05/50 = 0.001$, roughly $|t| > 3.3$.

Solution 15.10. The data are consistent with effects from a small loss to a gain of 3.5 per cent: the study cannot rule out a zero effect, but neither can it rule out an economically meaningful gain. “No significant effect” does not mean “no effect”; the study is imprecise. A larger sample, or pooling with other studies, is needed.

Regression

After studying this chapter you should be able to

1. Interpret a regression as an estimate of a conditional expectation, and derive the ordinary least squares estimator for one and several explanatory variables.
2. Interpret coefficients in multiple regression as partial effects, and state the Frisch–Waugh–Lovell theorem.
3. Write OLS in matrix form and explain its geometry.
4. State the assumptions under which OLS is unbiased and efficient, and compute standard errors, t tests and F tests.
5. Use robust and clustered standard errors, and diagnose outliers, leverage and multicollinearity.
6. Interpret models with logarithms, indicator variables, interactions and quadratic terms.
7. Distinguish prediction from explanation, and evaluate predictions out of sample.

A motivating question

Workers with more schooling earn more. How much more does each extra year bring, on average? Workers with more experience also earn more, and in many data sets older workers have less schooling than younger ones. A simple comparison of the earnings of workers with different schooling mixes the effect of schooling with that of experience. What we want is the association between earnings and schooling *holding experience fixed*.

Regression is the workhorse tool for such questions. It summarizes how the average of one variable changes with others, can hold measured variables constant, and comes with standard errors that quantify its precision. It is used to estimate demand elasticities (Chapter 20), the returns to education, the effects of monetary policy (Chapter 49), the betas of shares (Chapter 34), and in almost every empirical claim in economics and finance. This chapter develops regression as a statistical tool: how it is computed, what its coefficients mean, how precise they are, and how its predictions are checked. Chapter 17 asks the harder question of when a regression coefficient can be read as a *causal* effect.

16.1 Simple Regression

16.1.1 Regression as a conditional mean

The **population regression function** $\mathbb{E}[Y \mid X = x]$ gives the average of Y among units with $X = x$. Chapter 13 showed that the conditional expectation is the best predictor of Y given X .

The simplest model assumes it is linear:

$$Y = \beta_0 + \beta_1 X + u, \quad \mathbb{E}[u \mid X] = 0,$$

where u is the *error*: everything other than X that affects Y . The assumption $\mathbb{E}[u \mid X] = 0$ says that the error averages zero at every value of X —that nothing omitted is systematically related to X . It is the assumption on which everything depends.

16.1.2 The least squares line

Given observations (x_i, y_i) , $i = 1, \dots, n$, the **ordinary least squares** (OLS) line $\hat{y} = b_0 + b_1 x$ chooses b_0 and b_1 to minimize the sum of squared **residuals** $e_i = y_i - b_0 - b_1 x_i$:

$$\min_{b_0, b_1} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2.$$

Proposition 16.1. The OLS estimator

The solution is

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{s_{xy}}{s_x^2}, \quad b_0 = \bar{y} - b_1 \bar{x}.$$

Proof. Setting the partial derivative with respect to b_0 to zero gives $-2 \sum (y_i - b_0 - b_1 x_i) = 0$, so $\bar{y} = b_0 + b_1 \bar{x}$: the line passes through the point of means. Substituting $b_0 = \bar{y} - b_1 \bar{x}$, the objective becomes $\sum [(y_i - \bar{y}) - b_1(x_i - \bar{x})]^2$; its derivative with respect to b_1 is $-2 \sum (x_i - \bar{x})[(y_i - \bar{y}) - b_1(x_i - \bar{x})]$, which is zero at the stated b_1 . The objective is a convex quadratic, so this is the minimum (Chapter 5). $\square$

The slope is the sample covariance divided by the sample variance of x (Chapter 14); equivalently $b_1 = r s_y / s_x$, where r is the correlation. The first-order conditions imply that the residuals sum to zero and are uncorrelated with x .

Worked Example 16.2. A regression by hand

Data: $x = (1, 2, 3, 4, 5)$, $y = (2, 4, 5, 4, 5)$. Means: $\bar{x} = 3$, $\bar{y} = 4$. Deviations of x : $(-2, -1, 0, 1, 2)$; of y : $(-2, 0, 1, 0, 1)$. Cross products sum to $4 + 0 + 0 + 0 + 2 = 6$; squared x -deviations sum to 10. So $b_1 = 0.6$ and $b_0 = 4 - 0.6 \times 3 = 2.2$. Fitted values: 2.8, 3.4, 4.0, 4.6, 5.2; residuals $-0.8, 0.6, 1.0, -0.6, -0.2$, which sum to zero, as they always do when an intercept is included.

16.1.3 Goodness of fit

The R^2 is the share of the variation in y accounted for by the regression:

$$R^2 = 1 - \frac{\sum e_i^2}{\sum (y_i - \bar{y})^2}.$$

In Example 16.2, $\sum e_i^2 = 2.4$ and $\sum (y_i - \bar{y})^2 = 6$, so $R^2 = 0.6$. In simple regression, $R^2 = r^2$. A high R^2 does not mean the relationship is causal, and a low one does not mean x is unimportant: the effect of schooling on earnings is large and precisely estimated in data sets where schooling explains only a small share of the enormous variation in individual earnings.

16.2 Multiple Regression

16.2.1 Partial effects

With several explanatory variables,

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + u,$$

OLS again minimizes the sum of squared residuals. The coefficient β_1 now measures the change in the expected value of Y associated with a one-unit change in X_1 *holding the other included variables fixed*. That is multiple regression's great advantage: it lets us control for measured confounding factors. In a regression of log wages on schooling and experience, the schooling coefficient compares workers with the same experience.

Proposition 16.3. Frisch–Waugh–Lovell

The OLS coefficient on X_1 in a multiple regression equals the slope from a simple regression of Y on the residuals from a regression of X_1 on all the other explanatory variables.

The theorem makes “holding the others fixed” concrete: multiple regression uses only the part of X_1 that is not linearly predictable from the other variables. If schooling were perfectly predictable from experience, there would be no independent variation left to estimate its effect—the problem of *perfect collinearity*, in which OLS has no unique solution. When X_1 is highly (but not perfectly) correlated with the others, little independent variation remains, and its coefficient is imprecisely estimated: **multicollinearity**.

16.2.2 Matrix form

Stack the n observations of the dependent variable in a vector $\mathbf{y}$ and the explanatory variables, with a first column of ones for the intercept, in the $n \times (k+1)$ matrix X . The residuals are $\mathbf{e} = \mathbf{y} - X\mathbf{b}$, and OLS minimizes $\mathbf{e}'\mathbf{e}$. Setting the derivative with respect to $\mathbf{b}$ to zero gives the *normal equations* $X'X\mathbf{b} = X'\mathbf{y}$, so

$$\mathbf{b} = (X'X)^{-1}X'\mathbf{y},$$

provided $X'X$ is invertible—which is exactly the absence of perfect collinearity (Chapter 8).

Worked Example 16.4. OLS by matrices

For the data of Example 16.2,

$$X'X = \begin{pmatrix} 5 & 15 \\ 15 & 55 \end{pmatrix}, \quad X'\mathbf{y} = \begin{pmatrix} 20 \\ 66 \end{pmatrix}, \quad (X'X)^{-1} = \frac{1}{50} \begin{pmatrix} 55 & -15 \\ -15 & 5 \end{pmatrix},$$

so $\mathbf{b} = \frac{1}{50}(55 \times 20 - 15 \times 66, -15 \times 20 + 5 \times 66)' = (2.2, 0.6)'$, the same intercept and slope as before.

Geometry. The normal equations say $X'\mathbf{e} = \mathbf{0}$: the residual vector is orthogonal to every column of X . OLS projects $\mathbf{y}$ onto the space spanned by the explanatory variables; the fitted values are the projection and the residuals the part of $\mathbf{y}$ perpendicular to it. This is why residuals are uncorrelated with every included regressor by construction—and why that fact is no evidence that the error u is unrelated to them.

16.3 Inference

16.3.1 Assumptions

Under the following assumptions, OLS estimates are unbiased and, with a large sample, approximately normal:

1. *Linearity*: the model is linear in its parameters (variables may be transformed).
2. *Random sampling*: observations are independent draws from the population.
3. *No perfect collinearity*.
4. *Zero conditional mean*: $\mathbb{E}[u \mid X_1, \dots, X_k] = 0$.

If in addition the error has constant variance σ^2 whatever the values of the regressors (*homoskedasticity*), the **Gauss–Markov theorem** states that OLS has the smallest variance among all linear unbiased estimators: it is *BLUE*. Assumption 4 fails when relevant variables correlated with the regressors are omitted, when causation runs both ways, or when regressors are measured with error; Chapter 17 examines each.

16.3.2 Standard errors and t tests

Under homoskedasticity the variance matrix of $\mathbf{b}$ is $\sigma^2(X'X)^{-1}$, estimated with $s^2 = \sum e_i^2 / (n - k - 1)$, where $n - k - 1$ is the number of observations minus the number of estimated coefficients. In simple regression, $\text{SE}(b_1) = s / \sqrt{\sum (x_i - \bar{x})^2}$: slopes are estimated more precisely when the errors are small and x varies a lot.

Worked Example 16.5. Standard errors and a t test

In Example 16.2, $\sum e_i^2 = 2.4$, so $s^2 = 2.4/3 = 0.8$. The variance of the slope is $0.8/10 = 0.08$ (equivalently $s^2 \times 5/50$ from $(X'X)^{-1}$), so $\text{SE}(b_1) = 0.283$ and $t = 0.6/0.283 = 2.12$. The standard error of the intercept is $\sqrt{0.8 \times 55/50} = 0.94$. With only three degrees of freedom, the relevant critical value from the t distribution is 3.18, so the slope is not significant at the 5 per cent level despite $t > 1.96$ —a reminder that small samples need exact distributions (Chapter 15).

16.3.3 Testing several coefficients: the F test

To test whether several coefficients are jointly zero—say, whether a set of regional indicators matters—compare the sum of squared residuals of the unrestricted model (SSR_u) with that of the model that imposes the q restrictions (SSR_r):

$$F = \frac{(SSR_r - SSR_u)/q}{SSR_u/(n - k - 1)}.$$

Large values reject the restrictions; under the null, F has an F distribution with q and $n - k - 1$ degrees of freedom. Coefficients can be jointly significant even when none is individually significant, if the variables are correlated with one another.

16.3.4 Robust and clustered standard errors

The formula $\sigma^2(X'X)^{-1}$ is wrong when the error variance varies with the regressors (*heteroskedasticity*: the variance of spending differences is larger among rich households) or when

errors are correlated across observations—within firms over time, within regions, within the same month. The coefficients remain unbiased, but the usual standard errors are misleading, often too small. *Heteroskedasticity-robust* standard errors (due to Halbert White, 1980) correct the first problem; *clustered* standard errors, which allow arbitrary correlation within groups, correct the second. They are the default in applied work. With few clusters—say, a policy studied in eight states—even clustered standard errors are unreliable.

16.4 Diagnostics

- **Residual plots.** Plot residuals against fitted values and each regressor. Curvature suggests a missing nonlinear term; a funnel shape suggests heteroskedasticity.
- **Outliers, leverage and influence.** An observation with an unusual value of x has high *leverage*: it can pull the line toward itself. It is *influential* if dropping it changes the estimates substantially. Check influential observations for errors and report how conclusions depend on them.
- **Multicollinearity.** High correlation among regressors inflates standard errors without biasing coefficients. The remedy is more data or more independent variation, not dropping a relevant variable (which causes bias, Chapter 17).

16.5 Functional Form

16.5.1 Logarithms

The form of the variables determines how a coefficient is read:

Model	Interpretation of β_1
$y = \beta_0 + \beta_1 x$	a one-unit rise in x is associated with a β_1 -unit change in y
$\ln y = \beta_0 + \beta_1 x$	a one-unit rise in x is associated with approximately $100\beta_1\%$ change in y
$y = \beta_0 + \beta_1 \ln x$	a 1% rise in x is associated with a $\beta_1/100$ -unit change in y
$\ln y = \beta_0 + \beta_1 \ln x$	β_1 is an elasticity (Chapter 5)

The approximation in the second row is accurate for small β_1 ; the exact percentage change is $100(e^{\beta_1} - 1)$. Logs are used for positive, right-skewed variables such as income and firm size, where percentage changes are more natural than absolute ones. Predicting y itself from a model for $\ln y$ requires care: $\exp(\widehat{\ln y})$ estimates the conditional *median* rather than the mean, and understates the mean by a Jensen's-inequality factor (Chapter 13).

16.5.2 Indicator variables

An **indicator** (dummy) variable equals 1 for units in a category and 0 otherwise. Its coefficient is the difference in the expected outcome between the category and the omitted *base* category, holding the other regressors fixed. In $\ln w = \beta_0 + \beta_1 S - 0.12 \text{ female} + \dots$, women earn about 12 per cent (exactly $1 - e^{-0.12} = 11.3$ per cent) less than men with the same measured characteristics. A categorical variable with m categories needs $m - 1$ indicators; including all m with an intercept creates perfect collinearity (the “dummy variable trap”).

16.5.3 Interactions and quadratics

An *interaction* term $X_1 X_2$ lets the effect of X_1 depend on X_2 : in $\ln w = \beta_0 + \beta_1 S + \beta_2 \text{ female} + \beta_3 (S \times \text{female})$, the return to schooling is β_1 for men and $\beta_1 + \beta_3$ for women. A *quadratic* term

captures diminishing or increasing effects: in $\ln w = \dots + 0.03X - 0.0005X^2$, the marginal effect of experience is $0.03 - 0.001X$, which is 2 per cent a year at 10 years of experience and zero at 30 years. Coefficients on interacted and quadratic terms must be interpreted together, through derivatives (Chapter 5), not one by one.

16.6 Prediction

A regression built to *predict*—a credit score, a sales forecast, a house-price model—is judged differently from one built to *explain*. For prediction, what matters is accuracy on new data, not whether each coefficient has a causal interpretation. The danger is *overfitting*: with enough regressors, a model can fit the sample perfectly while predicting new observations badly, because it has fitted noise. R^2 always rises when a variable is added; the *adjusted* R^2 penalizes additional regressors, but the reliable check is **out-of-sample validation**: estimate the model on part of the data (the training sample) and measure its errors on data it has not seen (the test sample), or use cross-validation (Chapter 19). For time series, the test sample must come *after* the training sample, so that no information from the future leaks into the model (Chapter 18).

Case: The Mincer earnings regression

The equation. In *Schooling, Experience, and Earnings* (1974), Jacob Mincer proposed the regression

$$\ln w_i = \beta_0 + \beta_1 S_i + \beta_2 X_i + \beta_3 X_i^2 + u_i,$$

where w is earnings, S years of schooling and X years of labour-market experience. He derived it from a model of investment in human capital: if a year of schooling costs a year's earnings and raises future earnings by a constant proportion, the log of earnings is linear in schooling, with slope equal to the rate of return. The quadratic in experience captures on-the-job learning that raises earnings rapidly early in a career and more slowly later.

What it found. Estimated on data from many countries and periods, the Mincer regression typically yields a schooling coefficient of roughly 0.06 to 0.10: each extra year of schooling is associated with about 6 to 10 per cent higher earnings. The experience profile is concave, with earnings growth slowing in mid-career. The equation became one of the most widely estimated in economics, used to compare returns to education across countries and over time; in the United States the estimated return rose substantially from 1980, a finding central to debates about technology and inequality (Chapter 45).

What it cannot tell us. The schooling coefficient is an association holding experience fixed. People who stay longer in school may differ in ability, motivation or family background, which also affect earnings; if so, $\mathbb{E}[u \mid S] \neq 0$, and the coefficient mixes the return to schooling with the return to those characteristics. Whether the bias is upward or downward, and how large, is the question of omitted-variable bias taken up in Chapter 17—where researchers' use of natural experiments to estimate the causal return to schooling is a leading example.

Summary

- A regression estimates a conditional expectation. OLS minimizes squared residuals; the simple slope is s_{xy}/s_x^2 and the line passes through the means.
- R^2 measures fit, not causality or importance.

- Multiple-regression coefficients are partial effects, holding other included regressors fixed; by Frisch–Waugh–Lovell, they use only the independent variation in each regressor.
- In matrix form, $\mathbf{b} = (X'X)^{-1}X'y$; residuals are orthogonal to the regressors.
- Unbiasedness requires $\mathbb{E}[u \mid X] = 0$; with homoskedasticity, OLS is BLUE. Standard errors give t tests; F tests test several restrictions.
- Use robust and clustered standard errors; check residuals, influence and multicollinearity.
- Logs give percentage and elasticity interpretations; indicators compare categories; interactions and quadratics make effects depend on other variables.
- Predictive models are judged out of sample; overfitting fits noise.

Key terms

population regression function; error term; zero conditional mean; ordinary least squares; residual; fitted value; R^2 ; multiple regression; partial effect; Frisch–Waugh–Lovell theorem; perfect collinearity; multicollinearity; normal equations; projection; Gauss–Markov theorem; homoskedasticity; heteroskedasticity; standard error of a coefficient; t test; F test; robust standard errors; clustered standard errors; leverage; influence; log-log model; indicator variable; base category; dummy variable trap; interaction; quadratic term; overfitting; adjusted R^2 ; out-of-sample validation; Mincer earnings regression.

Exercises

Exercise 16.1 (OLS by hand). Data: $x = (0, 1, 2, 3)$, $y = (1, 3, 2, 6)$. Compute the OLS intercept and slope, the residuals and R^2 .

Exercise 16.2 (Slope and correlation). The correlation between household income and spending on restaurants is 0.4; the standard deviation of income is 20,000 and of restaurant spending 1,500. Find the OLS slope of spending on income and interpret it.

Exercise 16.3 (Standard error). In Exercise 16.1, compute the standard error of the slope and test whether it differs from zero at the 5 per cent level (the t critical value with 2 degrees of freedom is 4.30).

Exercise 16.4 (Interpreting log regressions). A regression gives $\ln(\text{wage}) = 1.2 + 0.07 \times \text{years of education}$. Interpret the coefficient approximately and exactly. Why might it not measure the causal effect of education?

Exercise 16.5 (Elasticity). A regression of the log of fuel consumption on the log of the fuel price and the log of income gives coefficients -0.25 and 0.9 . Interpret each. By how much is consumption predicted to fall after a 20 per cent price rise, using the elasticity and exactly?

Exercise 16.6 (Indicators and interactions). A regression of monthly sales (thousands) on an indicator for being in the city centre (1 if so), advertising spending (thousands) and their interaction gives: $\text{sales} = 40 + 15\text{centre} + 2.0\text{adv} + 1.5(\text{centre} \times \text{adv})$. What is the effect of an extra thousand of advertising for a city-centre shop and for other shops? What is the predicted difference between a city-centre shop and another shop when both spend 4 on advertising?

Exercise 16.7 (Quadratic). $\ln w = 1.5 + 0.08S + 0.04X - 0.0008X^2$. At what experience level are predicted earnings highest? What is the marginal effect of experience at 5 and at 20 years?

Exercise 16.8 (*F* test). A regression with 4 regressors and an intercept on 105 observations has $SSR = 400$. Dropping two of the regressors raises SSR to 440. Compute the *F* statistic. Are the two regressors jointly significant at the 5 per cent level (critical value about 3.09)?

Exercise 16.9 (Collinearity). A researcher regresses house prices on floor area in square metres and floor area in square feet. What happens, and why?

Exercise 16.10 (Prediction). A model predicting firms' defaults fits the estimation sample with an R^2 of 0.9 using 60 variables and 100 firms, but predicts poorly the next year. Explain what probably happened and how the modeller should have checked.

Solutions

Solution 16.1. $\bar{x} = 1.5$, $\bar{y} = 3$. Cross-products of deviations: $(-1.5)(-2) + (-0.5)(0) + (0.5)(-1) + (1.5)(3) = 3 + 0 - 0.5 + 4.5 = 7$. Squared x -deviations: $2.25 + 0.25 + 0.25 + 2.25 = 5$. $b_1 = 1.4$, $b_0 = 3 - 1.4 \times 1.5 = 0.9$. Fitted: 0.9, 2.3, 3.7, 5.1; residuals 0.1, 0.7, -1.7, 0.9 (sum zero). $\sum e^2 = 0.01 + 0.49 + 2.89 + 0.81 = 4.2$; $\sum (y - \bar{y})^2 = 4 + 0 + 1 + 9 = 14$; $R^2 = 1 - 4.2/14 = 0.7$.

Solution 16.2. $b_1 = 0.4 \times 1,500/20,000 = 0.03$: each extra unit of income is associated with 0.03 units more restaurant spending—3 cents per euro, on average.

Solution 16.3. $s^2 = 4.2/2 = 2.1$; $SE(b_1) = \sqrt{2.1/5} = 0.648$; $t = 1.4/0.648 = 2.16 < 4.30$: not significant at 5 per cent with so few observations.

Solution 16.4. Approximately, each extra year of education is associated with 7% higher wages; exactly, $e^{0.07} - 1 = 7.25\%$. It may not be causal because of omitted variables such as ability or family background, which are correlated with education and affect wages (Chapter 17).

Solution 16.5. A 1 per cent price rise is associated with a 0.25 per cent fall in consumption, holding income fixed (price elasticity -0.25); a 1 per cent rise in income with a 0.9 per cent rise, holding price fixed. After a 20 per cent price rise: approximately $-0.25 \times 20 = -5$ per cent; exactly, $1.2^{-0.25} - 1 = -4.46$ per cent.

Solution 16.6. City centre: $2.0 + 1.5 = 3.5$ thousand per thousand of advertising; elsewhere 2.0. At advertising 4: difference $= 15 + 1.5 \times 4 = 21$ thousand.

Solution 16.7. $\partial \ln w / \partial X = 0.04 - 0.0016X = 0$ at $X = 25$ years. At 5 years: $0.04 - 0.008 = 3.2$ per cent a year; at 20: $0.04 - 0.032 = 0.8$ per cent.

Solution 16.8. $n - k - 1 = 100$. $F = [(440 - 400)/2]/(400/100) = 20/4 = 5 > 3.09$: jointly significant.

Solution 16.9. One variable is an exact multiple of the other ($1 \text{ m}^2 = 10.764 \text{ ft}^2$), so the columns of X are perfectly collinear, $X'X$ is singular and OLS has no unique solution; software drops one variable or reports an error.

Solution 16.10. With 60 variables and 100 observations, the model overfitted: it fitted noise specific to the estimation sample. The in-sample R^2 is meaningless for prediction. The modeller should have held out a test sample (ideally a later period), used cross-validation, and preferred a simpler model or a penalized method (Chapter 19).

Causal Inference and Research Design

After studying this chapter you should be able to

1. Define causal effects with potential outcomes, and decompose a naive comparison into a treatment effect and selection bias.
2. Use causal diagrams to distinguish confounders, mediators and colliders, and to decide which variables to control for.
3. Derive omitted-variable bias and attenuation bias from measurement error.
4. Analyse randomized experiments, including intention-to-treat and non-compliance.
5. Explain selection on observables, matching, propensity scores and inverse probability weighting.
6. Estimate effects with instrumental variables, regression discontinuity, difference-in-differences and synthetic controls, and state the assumption each requires.

A motivating question

Suppose a newspaper reports that people who own more books earn higher incomes, and that a government proposes to raise incomes by giving books to every household. The first claim is a statement about data; the second is a claim about cause and effect. To evaluate the proposal you need to ask what else differs between households with many books and households with few, and whether giving books would change anything that matters for earnings.

Economics and finance are full of such questions. Does a minimum wage reduce employment? Does a training programme raise earnings? Does a rise in interest rates reduce investment? Do firms with independent boards perform better? In each case, the data show associations, and the question is causal. Chapter 16 developed regression as a tool for describing associations, holding measured variables fixed. This chapter asks when, and by what research designs, an association can be read as a causal effect. The answer, developed over the past four decades by economists and statisticians, is that credible causal claims require an argument about why the comparison being made is not contaminated by selection—and that argument usually rests on a particular source of variation in the data, not on the statistical method.

17.1 Potential Outcomes

17.1.1 The fundamental problem

We want to know the *causal effect* of a treatment—a year of schooling, a training programme, an interest-rate rise—on an outcome. For each unit (person, firm, country) i there are two **potential outcomes**: $Y_i(1)$, the outcome with the treatment, and $Y_i(0)$, the outcome without it. The individual causal effect is $Y_i(1) - Y_i(0)$. We only ever observe one of them: a worker either attends the programme or does not. This is the *fundamental problem of causal inference*. We must

therefore infer *average* effects by comparing treated and untreated units: the **average treatment effect** (ATE) over a population, or the *average treatment effect on the treated* (ATT).

17.1.2 Selection bias

Let $D_i = 1$ if unit i is treated. A naive comparison of average outcomes of the treated and the untreated is

$$\mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0] = \underbrace{\mathbb{E}[Y(1) - Y(0) \mid D = 1]}_{\text{ATT: what we want}} + \underbrace{\mathbb{E}[Y(0) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0]}_{\text{selection bias}}.$$

The identity follows by adding and subtracting $\mathbb{E}[Y(0) \mid D = 1]$. The **selection bias** is the difference the treated and untreated would have shown even without treatment.

Worked Example 17.1. Training and earnings

Workers who join a voluntary training programme earn 30,000 a year afterwards; those who do not earn 25,000. Suppose the programme truly raises its participants' earnings by 2,000, so that without it they would have earned 28,000. Then the 5,000 difference consists of a 2,000 treatment effect and a 3,000 selection bias: the more motivated workers chose to train. The data alone cannot separate the two; only a research design can.

People with more books may earn more partly because the kind of household that buys books also invests in education, has more educated parents, or lives in richer areas. Those differences would exist even without the books. Giving books to everyone would not transfer those other characteristics.

17.2 Causal Diagrams

A **causal diagram** (directed acyclic graph) draws variables as nodes and causal effects as arrows. It makes explicit which variables to control for:

- A **confounder** causes both treatment and outcome (ability $\rightarrow$ schooling and ability $\rightarrow$ earnings). Controlling for a confounder removes bias; omitting it causes bias.
- A **mediator** lies on the causal path (schooling $\rightarrow$ occupation $\rightarrow$ earnings). Controlling for it removes part of the effect one wants to measure: the regression then estimates the effect of schooling on earnings *holding occupation fixed*, not the total effect.
- A **collider** is caused by both treatment and outcome (or by their causes). Controlling for a collider—or selecting the sample on it—*creates* bias where there was none. Studying only firms that survived, only employed workers, or only published studies conditions on a collider.

Such “bad controls” show that adding variables to a regression is not automatically conservative: the right controls are those that block confounding paths without opening new ones, and deciding which they are requires a model of how the variables are generated.

17.3 Omitted Variables and Measurement Error

17.3.1 Omitted-variable bias

Regression makes the confounding problem precise. Suppose the true relationship is

$$y = \beta_0 + \beta_1 x + \beta_2 z + u,$$

where u is unrelated to x and z , but we regress y on x alone. Let δ be the slope from a regression of z on x . Then the slope we estimate tends to $\beta_1 + \beta_2 \delta$.

Proposition 17.2. Omitted-variable bias

Omitting a variable z biases the coefficient on x by $\beta_2 \delta$, which is nonzero if (i) z affects y ($\beta_2 \neq 0$) and (ii) z is correlated with x ($\delta \neq 0$).

Proof. By Theorem 16.1 of Chapter 16, the short-regression slope is $\widehat{\text{Cov}}(x, y) / \widehat{\text{Var}}(x)$. Substituting the true model, $\text{Cov}(x, y) = \beta_1 \text{Var}(x) + \beta_2 \text{Cov}(x, z) + \text{Cov}(x, u)$. With $\text{Cov}(x, u) = 0$, dividing by $\text{Var}(x)$ gives $\beta_1 + \beta_2 \text{Cov}(x, z) / \text{Var}(x) = \beta_1 + \beta_2 \delta$. $\square$

Worked Example 17.3. Returns to schooling

Suppose log wages depend on years of schooling S and an unmeasured ability index A : $\ln w = \beta_0 + 0.08S + 0.10A + u$. Suppose also that more able people get more schooling, so that a regression of A on S has slope $\delta = 0.3$. A regression of $\ln w$ on S alone then estimates about $0.08 + 0.10 \times 0.3 = 0.11$: it overstates the return to a year of schooling as 11% rather than 8%, attributing to schooling some of the effect of ability.

The sign of the bias can be reasoned out even when z is unobserved: if the omitted variable raises the outcome and is positively correlated with the treatment, the estimate is biased upward. That reasoning is often the best one can do—and is a reason to seek designs that do not rely on measuring every confounder.

17.3.2 Measurement error

Chapter 11 noted that variables are measured with error. If the explanatory variable is observed as $x = x^* + e$, where e is random noise uncorrelated with the true value x^* and the outcome, the OLS slope tends to

$$\beta_1 \times \frac{\text{Var}(x^*)}{\text{Var}(x^*) + \text{Var}(e)},$$

biased toward zero by the *reliability ratio* of the measure. If the true variable has variance 9 and the noise variance 3, the reliability is $9/12 = 0.75$, and a true effect of 0.08 appears as 0.06. This **attenuation bias** means that noisy measures make effects look smaller than they are—and, with several regressors, can bias other coefficients in either direction. Error in the *dependent* variable, by contrast, adds noise but no bias.

17.4 Randomized Experiments

17.4.1 Why randomization works

If treatment is assigned at random, it is independent of potential outcomes, so $\mathbb{E}[Y(0) \mid D = 1] = \mathbb{E}[Y(0) \mid D = 0]$: selection bias is zero on average, and a simple difference in means estimates the average treatment effect. Randomization balances *all* characteristics of the treatment and control groups in expectation, measured or not—the only method that guarantees this. Randomized controlled trials (RCTs) are increasingly used in development, labour, public and financial economics: field experiments on microcredit, savings commitment devices, tax-compliance letters, job-search assistance and online pricing (Example 15.3 of Chapter 15).

17.4.2 Non-compliance

In many experiments, not everyone offered a treatment takes it up. Comparing those *assigned* to treatment with those assigned to control estimates the **intention-to-treat** (ITT) effect: the effect of the offer. If some assigned units do not comply, the ITT understates the effect of the treatment itself on those who take it. When only those offered can be treated, the effect on the treated is the ITT divided by the difference in take-up rates.

Worked Example 17.4. An experiment with partial take-up

A bank randomly offers half of 10,000 small-business clients a financial-literacy course. Sixty per cent of those offered attend; no one else can. Twelve months later, those offered have average profits 3 (thousand) higher than those not offered: the ITT is 3. Since the offer raised attendance by 60 percentage points, the effect of attending on those who attend is $3/0.6 = 5$. This assumes the offer affects profits only through attendance.

17.4.3 Limits of experiments

Experiments raise problems of their own: *external validity* (does an effect found in one setting transfer to another?), *general-equilibrium effects* (training everyone may lower the return to training), *spillovers* between treated and untreated units, attrition, and cost and ethics. And many important questions—the effect of a central bank’s policy, of a financial crisis, of a country’s institutions—cannot be randomized at all. Economists therefore rely heavily on *natural experiments* and *quasi-experiments*: situations in which something close to random assignment occurred.

17.5 Selection on Observables

If treatment depends only on observed characteristics X —so that, among units with the same X , treatment is as good as random (*conditional independence* or *unconfoundedness*)—causal effects can be estimated by comparing treated and untreated units with the same X .

- **Regression** controls for X linearly.
- **Matching** pairs each treated unit with untreated units that have similar X .
- The **propensity score** $e(X) = P(D = 1 \mid X)$ summarizes X in one number: Paul Rosenbaum and Donald Rubin (1983) showed that if treatment is unconfounded given X , it is also unconfounded given $e(X)$. Units can be matched or grouped on the estimated score.
- **Inverse probability weighting** weights treated units by $1/e(X)$ and untreated units by $1/(1 - e(X))$, creating a pseudo-population in which treatment is unrelated to X . An untreated unit with $e = 0.2$ gets weight $1/0.8 = 1.25$.

These methods require *overlap*: treated and untreated units with similar X must both exist. They cannot fix confounding by variables that are not observed, which is why the assumption must be defended, not assumed.

17.6 Instrumental Variables

An **instrumental variable** Z is a variable that (i) affects the treatment (*relevance*) and (ii) affects the outcome only through the treatment (*exclusion*), and is as good as randomly assigned.

Variation in treatment induced by the instrument is then free of selection bias.

The Wald estimator. With a binary instrument, the causal effect is the ratio of the instrument's effect on the outcome (the *reduced form*) to its effect on the treatment (the *first stage*):

$$\hat{\beta}_{IV} = \frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0]}.$$

If winning a school-voucher lottery raises the probability of attending a private school from 20 to 70 per cent and raises test scores by 5 points, the estimated effect of private schooling is $5/0.5 = 10$ points. Example 17.4 is the same calculation. More generally, $\hat{\beta}_{IV} = \widehat{\text{Cov}}(Z, Y) / \widehat{\text{Cov}}(Z, X)$; with several instruments or controls, *two-stage least squares* first regresses the treatment on the instruments and controls, then regresses the outcome on the fitted treatment.

Examples. Joshua Angrist and Alan Krueger (1991) used quarter of birth, which affects schooling through compulsory-schooling laws, as an instrument for years of education. Chapter 44 describes the use of settler mortality as an instrument for institutions. Studies of monetary policy use surprises in policy announcements (Chapter 49).

Cautions. The exclusion restriction cannot be tested; it must be argued. *Weak instruments*—those with a small first stage—produce estimates that are biased toward OLS and have unreliable standard errors. And when effects differ across people, IV estimates the *local average treatment effect* (LATE): the effect for “compliers” whose treatment is changed by the instrument, who may not resemble the whole population (Imbens and Angrist 1994).

17.7 Regression Discontinuity

When treatment is assigned by whether a *running variable* crosses a threshold—students above a test score receive a scholarship, loans to firms above a credit score are approved, towns above a population size get more councillors—units just above and just below the threshold are similar in every respect except treatment. A jump in the outcome at the threshold estimates the causal effect for units near it: a **regression discontinuity** design. In a *sharp* design treatment switches fully at the threshold; in a *fuzzy* design only its probability jumps, and the effect is the jump in the outcome divided by the jump in treatment (an IV estimate). The design fails if units can precisely manipulate the running variable to fall on the favourable side; a pile-up of observations just above the threshold is a warning sign. Chapter 25 describes a regression-discontinuity study of credit scores.

17.8 Difference-in-Differences

17.8.1 The basic design

Difference-in-differences compares the change in outcomes for a treated group with the change for an untreated group over the same period. If the two groups would have followed *parallel trends* without treatment, the difference in their changes estimates the effect.

Worked Example 17.5. A difference-in-differences calculation

A region introduces a hiring subsidy. Average employment rate among eligible firms rises from 60% to 66% over the year; in a comparable region without the subsidy it rises from 58% to 61%. The naive before-after change in the treated region is 6 points, but some of that would have happened anyway, as the comparison region's 3-point rise shows. The difference-in-differences estimate is $(66 - 60) - (61 - 58) = 3$ percentage points. In regression form, it is the coefficient on the interaction of a treated-group indicator and a post-period indicator.

17.8.2 Checking and extending the design

Parallel trends cannot be verified for the treatment period, but trends *before* treatment can be checked: an *event study* estimates the treated–control difference in each period before and after, and pre-treatment differences near zero support the design. When different units are treated at different dates (*staggered adoption*), the standard two-way fixed-effects regression can be badly biased if effects vary over time, because already-treated units serve as controls for later-treated ones; newer estimators compare each cohort only with not-yet-treated or never-treated units.

17.8.3 Synthetic control

When only one unit is treated—one country, one state—a **synthetic control** builds a comparison as a weighted average of untreated units chosen to match the treated unit's pre-treatment path (Abadie, Diamond and Hainmueller). For example, to estimate the economic effect of German reunification on West Germany, Abadie, Diamond and Hainmueller (2015) built a "synthetic West Germany" from a combination of other OECD economies that tracked its GDP per person closely before 1990, and estimated that reunification reduced West German GDP per person by roughly 8 per cent by 2003 relative to that counterfactual. Inference uses *placebo* tests: applying the same method to untreated units to see whether the estimated effect is unusual.

► Common error: "the data show that x causes y "

Data show associations. A causal claim requires an argument about why the comparison is not contaminated by selection: an experiment, a natural experiment, or a convincing set of controls. When reading any empirical claim—including those about monetary policy—ask: *compared with what, and why is that comparison valid?*

Case: The credibility revolution: minimum wages and the 2021 Nobel prize

The question. Standard competitive theory predicts that a minimum wage above the market wage reduces employment (Chapter 22). Early studies regressed employment on minimum-wage levels across states and years, but states that raise minimum wages differ from those that do not in many ways, and the results were fragile.

The design. In April 1992 New Jersey raised its minimum wage from 4.25 to 5.05 dollars an hour, while neighbouring eastern Pennsylvania did not. David Card and Alan Krueger surveyed about 400 fast-food restaurants in both states before and after the increase. Fast-food restaurants in the two areas faced similar customers and economic conditions, so Pennsylvania's change could stand for what would have happened in New Jersey without the increase—a difference-in-differences design. Card and Krueger (1994) found no fall in fast-food employment in New Jersey relative to Pennsylvania; if anything, a small rise.

The debate. The result provoked a long debate about both the data (a reanalysis using payroll records reached different conclusions, and Card and Krueger responded with further data) and the parallel-trends assumption. It also spurred theoretical work showing that when employers have monopsony power, a moderate minimum wage can raise employment (Chapter 24). Later research with larger samples and improved designs, including comparisons of contiguous counties across state borders, has generally found small employment effects of moderate minimum-wage increases, with more debate about large ones.

The recognition. In 2021 the Nobel prize in economics was awarded to David Card “for his empirical contributions to labour economics” and to Joshua Angrist and Guido Imbens “for their methodological contributions to the analysis of causal relationships”—in particular, the local average treatment effect framework that clarifies what instrumental-variable and natural-experiment estimates measure. The award recognized a shift, often called the *credibility revolution*, in how empirical economics is done: from asking which variables to put into a regression to asking what source of variation identifies the effect, and why that variation is as good as random.

Summary

- Causal effects compare potential outcomes; only one is observed. A naive comparison equals the effect on the treated plus selection bias.
- Causal diagrams distinguish confounders (control for them), mediators (controlling removes part of the effect) and colliders (controlling creates bias).
- Omitting a relevant correlated variable biases a coefficient by $\beta_2\delta$; random measurement error in a regressor attenuates it by the reliability ratio.
- Randomization removes selection bias; with non-compliance, ITT divided by the take-up difference gives the effect on compliers.
- Selection on observables uses regression, matching, propensity scores or inverse weighting, and requires unconfoundedness and overlap.
- Instrumental variables use variation that affects the outcome only through the treatment; the Wald estimator is reduced form over first stage; estimates are local to compliers.
- Regression discontinuity compares units just either side of a threshold; difference-in-differences relies on parallel trends; synthetic control builds a weighted comparison unit.

Key terms

potential outcomes; fundamental problem of causal inference; average treatment effect; effect on the treated; selection bias; causal diagram; confounder; mediator; collider; bad control; omitted-variable bias; attenuation bias; reliability ratio; randomized controlled trial; intention-to-treat; non-compliance; external validity; natural experiment; selection on observables; unconfoundedness; matching; propensity score; inverse probability weighting; overlap; instrumental variable; relevance; exclusion restriction; reduced form; first stage; Wald estimator; two-stage least squares; weak instrument; local average treatment effect; regression discontinuity; running variable; difference-in-differences; parallel trends; event study; staggered adoption; synthetic control; placebo test; credibility revolution.

Exercises

Exercise 17.1 (Selection bias). Hospital patients have worse health on average than people who are not in hospital. Use the potential-outcomes decomposition to explain why this does not show that hospitals harm health.

Exercise 17.2 (Causal diagrams). In a study of the effect of a firm's R&D spending on its profits, classify each potential control as a confounder, mediator or collider, and say whether to include it: (a) the firm's industry; (b) the number of patents the firm obtains; (c) whether the firm is still in existence at the end of the sample.

Exercise 17.3 (Omitted-variable bias). A regression of house prices on distance from the city centre gives a coefficient of -2 (thousand per kilometre). Houses farther from the centre are larger, and size raises prices. Is the true effect of distance, holding size fixed, more or less negative than -2 ? Explain.

Exercise 17.4 (Attenuation). Self-reported years of schooling have a reliability ratio of 0.9. A regression of log wages on reported schooling gives 0.072. If this is the only source of bias, what is the true coefficient?

Exercise 17.5 (Non-compliance). A randomized job-search programme is offered to 5,000 of 10,000 unemployed workers; 40 per cent of those offered participate, and no one else can. The employment rate after six months is 52 per cent among those offered and 48 per cent among those not offered. Find the ITT and the effect on participants.

Exercise 17.6 (Instrumental variables). In a region, a new bridge randomly reduces travel time to a university for some villages. Villages near the bridge have 30 per cent university attendance against 20 per cent elsewhere, and average adult earnings 3 per cent higher. Estimate the effect of university attendance on earnings, and state the assumptions required.

Exercise 17.7 (Difference-in-differences). A country cuts corporate tax for manufacturing firms only. Investment rises from 10 to 13 per cent of assets in manufacturing and from 9 to 10 per cent in services. Estimate the effect. What would make the estimate invalid? How could you check?

Exercise 17.8 (Regression discontinuity). A bank approves loans automatically for applicants with a credit score of at least 650. Default rates are 4 per cent just below 650 (among those approved manually) and 6 per cent just above. What does this comparison estimate, and what could undermine it?

Exercise 17.9 (Causation). Cities with more police officers per resident have higher crime rates. Does this show that police cause crime? Describe a research design that could estimate the causal effect of police on crime.

Exercise 17.10 (Choosing a design). For each question, propose a research design and state its key assumption: (a) the effect of class size on test scores; (b) the effect of a central bank's surprise rate increase on share prices; (c) the effect of a large country's trade tariff on its exports.

Solutions

Solution 17.1. The naive difference in health between patients and non-patients equals the effect of hospital on patients plus the selection bias $\mathbb{E}[Y(0) \mid \text{hospital}] - \mathbb{E}[Y(0) \mid \text{not}]$, which is strongly negative: people go to hospital because they are ill. A negative naive difference is fully consistent with hospitals improving health.

Solution 17.2. (a) Confounder: industry affects both R&D intensity and profitability; include it. (b) Mediator: R&D raises profits partly through patents; controlling for patents removes that channel—exclude it for the total effect. (c) Collider (selection): survival depends on both R&D and profits; restricting to survivors creates bias—avoid conditioning on it, or model attrition.

Solution 17.3. More negative. Distance is positively correlated with size ($\delta > 0$), and size raises prices ($\beta_2 > 0$), so the short regression is biased upward: $-2 = \beta_1 + \beta_2\delta$ with $\beta_2\delta > 0$, so $\beta_1 < -2$.

Solution 17.4. $0.072/0.9 = 0.08$.

Solution 17.5. ITT = 4 percentage points. Effect on participants = $4/0.4 = 10$ percentage points.

Solution 17.6. Wald estimate = $3/0.10 = 30$ per cent higher earnings from attending university. Assumptions: the bridge's location is as good as random with respect to earnings potential; it affects earnings only through attendance (not, say, by improving access to jobs); and the estimate applies to villagers whose attendance was changed by the bridge.

Solution 17.7. $(13 - 10) - (10 - 9) = 2$ percentage points of assets. It is invalid if manufacturing investment would have risen faster anyway—for example because of a manufacturing-specific boom at the same time. Check that the two sectors' investment trends were parallel in the years before the cut (an event study), and whether other policies changed simultaneously.

Solution 17.8. It compares otherwise similar applicants just either side of the threshold, estimating the effect of automatic approval on default: 2 percentage points higher. It is undermined if applicants can manipulate their scores to cross 650 (bunching above the threshold), or if the manual approvals below the threshold involve extra screening that changes who is approved—so the groups differ in more than approval method.

Solution 17.9. No. Causation may run from crime to police: cities with more crime hire more police. There may also be common causes (large cities have both). One design uses an instrument that changes police numbers for reasons unrelated to crime, such as the timing of electoral cycles or of federal grants for hiring officers; another uses sudden redeployments of police, for example after a terrorist alert, as a natural experiment, comparing crime in affected and unaffected areas before and after (a difference-in-differences design).

Solution 17.10. (a) A randomized experiment assigning pupils to small and large classes, or a regression discontinuity exploiting class-size caps that split classes when enrolment crosses a threshold; assumption: enrolment near the cap is not manipulated. (b) High-frequency event study: the change in share prices in a narrow window around the announcement, using the surprise measured from futures prices; assumption: nothing else systematically moves prices in the window. (c) A synthetic control for the imposing country, or difference-in-differences comparing affected and unaffected products; assumption: the comparison tracks what exports would have been without the tariff.

Time Series, Panel Data and Forecasting

After studying this chapter you should be able to

1. Estimate panel models with fixed effects, and explain why they control for unobserved characteristics that do not change over time.
2. Describe trend, seasonality, stationarity and autocorrelation in a time series.
3. Work with autoregressive models: their means, variances, impulse responses and forecasts.
4. Explain random walks and unit roots, the danger of spurious regression, and cointegration.
5. Interpret vector autoregressions, Granger predictability and impulse responses.
6. Evaluate forecasts honestly: loss functions, benchmarks, rolling out-of-sample tests, look-ahead bias, structural breaks and nowcasting.
7. Model time-varying volatility with ARCH and GARCH.

A motivating question

In the 1970s, after the major currencies began to float, economists built models to explain exchange rates using the variables that theory said mattered: money supplies, incomes, interest rates, inflation. The models fitted past data well. In 1983 Richard Meese and Kenneth Rogoff asked a simple question: do they forecast better than assuming that tomorrow's exchange rate will equal today's? The answer, which surprised the profession and has held up remarkably well since, was no.

Much of the data in economics and finance comes as time series: GDP by quarter, prices by month, returns by day. Time series violate the assumption, used in Chapters 15 and 16, that observations are independent: this month's inflation is informative about next month's. That dependence is both a problem—standard errors and regressions can mislead—and an opportunity, because it makes forecasting possible. This chapter develops the tools for data observed over time, starting with panels, which combine many units with many periods, and then turning to single series, forecasting and volatility.

18.1 Panel Data

18.1.1 Unobserved heterogeneity

Panel data observe the same units repeatedly (Chapter 11). Their great advantage is that each unit can be compared with itself. Write

$$y_{it} = \beta x_{it} + \alpha_i + u_{it},$$

where α_i captures everything about unit i that does not change over time—a firm’s management quality, a country’s geography, a worker’s ability. If α_i is correlated with x_{it} , pooled OLS that ignores it suffers omitted-variable bias (Chapter 17).

18.1.2 Fixed effects

The **fixed-effects** estimator removes α_i by subtracting each unit’s own average over time (the *within transformation*), or equivalently by including an indicator for each unit. Only variation *within* units over time is used.

Worked Example 18.1. Pooled versus within

Two firms are observed for three years. Firm A has advertising $x = 1, 2, 3$ and sales $y = 10, 12, 14$; firm B has $x = 4, 5, 6$ and $y = 6, 8, 10$. Pooling all six observations, the OLS slope is -0.57 : more advertising appears to go with lower sales. Within each firm, however, each extra unit of advertising goes with 2 more units of sales; the fixed-effects slope is 2. Firm B advertises more but has lower sales for reasons of its own—a less attractive product, say—so the pooled comparison across firms is confounded by the firm effect.

Time fixed effects (an indicator for each period) similarly remove shocks common to all units in a period, such as a recession. The two together give the *two-way fixed-effects* model, the regression form of difference-in-differences (Chapter 17). Fixed effects cannot estimate the effect of variables that never change within a unit, and they magnify measurement error, because the within variation is often small relative to noise. The *random-effects* estimator treats α_i as uncorrelated with the regressors, which is more efficient if true but biased if not; the Hausman test compares the two.

Errors for the same unit are typically correlated over time, so standard errors should be clustered by unit (Chapter 16). Panels also lose units over time (*attrition*), which biases estimates if those who leave differ from those who stay.

18.2 Describing a Time Series

18.2.1 Components

A time series typically combines a *trend* (long-run movement, such as rising real GDP), *seasonality* (regular patterns within the year, such as higher retail sales in December), a *cycle* (fluctuations over several years) and irregular noise. Statistical agencies publish *seasonally adjusted* series, which remove the estimated seasonal pattern, so that one month can be compared with the previous one. Comparisons with the same month a year earlier avoid seasonal effects without adjustment.

18.2.2 Stationarity and autocorrelation

A series is (weakly) **stationary** if its mean, variance and the covariance between observations a given distance apart do not change over time. Growth rates, interest-rate spreads and returns are often roughly stationary; levels of GDP, prices and exchange rates are not. The **autocorrelation** at lag k is the correlation between y_t and y_{t-k} ; plotted against k , it is the *autocorrelation function*. A series with no autocorrelation and constant mean and variance is *white noise*—the time-series counterpart of independent draws, and a benchmark against which forecasting models are judged.

18.3 Autoregressive Models

18.3.1 The AR(1) model

The first-order **autoregressive** model, AR(1), is

$$y_t = c + \phi y_{t-1} + \varepsilon_t,$$

where ε_t is white noise with variance σ^2 . It is the difference equation of Chapter 4 with a random shock added. If $|\phi| < 1$, the series is stationary, with

$$\mathbb{E}[y_t] = \mu = \frac{c}{1 - \phi}, \quad \text{Var}(y_t) = \frac{\sigma^2}{1 - \phi^2}.$$

A shock ε_t raises y_{t+k} by ϕ^k : the **impulse response** decays geometrically, faster the smaller ϕ . The coefficient ϕ is the lag-1 autocorrelation and is estimated consistently by regressing y_t on y_{t-1} .

18.3.2 Forecasting with an AR(1)

The best forecast of y_{t+h} given information at t is the conditional expectation (Chapter 13):

$$\mathbb{E}_t[y_{t+h}] = \mu + \phi^h(y_t - \mu).$$

Forecasts revert toward the mean, closing the gap by the factor ϕ each period.

Worked Example 18.2. Forecasting inflation

Suppose annual inflation follows $\pi_t = 0.6 + 0.7\pi_{t-1} + \varepsilon_t$, with shock standard deviation 1 point. The mean is $0.6/0.3 = 2$ per cent and the unconditional variance $1/(1 - 0.49) = 1.96$. If inflation is now 5 per cent, forecasts are 4.1 per cent next year, 3.47 in two years, 3.03 in three and 2.17 in eight, approaching 2. The half-life of the deviation is $\ln 0.5 / \ln 0.7 = 1.9$ years. Forecast uncertainty grows with the horizon, toward the unconditional standard deviation of 1.4 points: this is the logic of central banks' "fan charts," which show widening bands around their inflation forecasts.

More general models add further lags (AR(p)), lagged shocks (*moving-average* terms, giving ARMA models) and seasonal terms. Choosing among them uses information criteria that penalize extra parameters and, better, out-of-sample forecast performance.

18.4 Unit Roots and Cointegration

18.4.1 Random walks

If $\phi = 1$ and $c = 0$, the AR(1) becomes a **random walk**: $y_t = y_{t-1} + \varepsilon_t$. Every shock is permanent; the variance of y_t grows without limit with time; and the best forecast of any future value is today's value. Such a series has a **unit root**. Asset prices in efficient markets behave approximately as random walks (Chapter 34), as does consumption under Hall's permanent-income hypothesis (Chapter 45). Many macroeconomic levels, such as GDP and prices, behave more like random walks with drift than like stationary series around a trend. The *first difference* $\Delta y_t = y_t - y_{t-1}$ of a random walk is stationary; such a series is called *integrated of order one*. Tests such as the Dickey–Fuller test ask whether a series has a unit root, but they have low power: over short samples, a highly persistent stationary series and a random walk are hard to tell apart.

18.4.2 Spurious regression

Regressing one non-stationary series on another produces nonsense. Clive Granger and Paul Newbold (1974) showed by simulation that regressions of one independent random walk on another typically yield “significant” t -statistics and high R^2 : two trending series will appear related whether or not they are. The standard remedy is to work with differences or growth rates, unless the series are cointegrated.

18.4.3 Cointegration

Two non-stationary series are **cointegrated** if some linear combination of them is stationary: they wander, but not apart. Consumption and income, short and long interest rates, and the prices of the same good in two markets are examples. Their long-run relationship is then meaningful, and short-run movements can be modelled with an *error-correction model*, in which the change in one variable responds to how far the two have drifted from their long-run relation. Robert Engle and Clive Granger shared the 2003 Nobel prize for methods for analysing time series with time-varying volatility (Engle) and common trends (Granger).

18.5 Systems of Time Series

A **vector autoregression** (VAR), introduced to macroeconomics by Christopher Sims (1980), regresses each of several variables on lags of all of them—for example, output, inflation and the policy interest rate. VARs summarize the dynamic correlations in the data and produce forecasts.

Granger predictability. A variable x is said to *Granger-cause* y if past values of x help forecast y given past values of y . Despite the name, this is predictability, not causation: stock prices Granger-cause GDP because investors anticipate future output, not because share prices produce it.

Impulse responses. To trace the effect of, say, a monetary-policy shock through a VAR, one must identify which movements in the interest rate are *shocks* rather than responses to the other variables—a causal question (Chapter 17). Identification uses assumptions about timing (a central bank sets rates after seeing current inflation, but output responds only with a lag), external instruments such as policy surprises measured in narrow windows around announcements, or restrictions from theory. The impulse responses of Chapter 49’s discussion of monetary transmission come from such exercises.

18.6 Forecasting in Practice

18.6.1 Evaluating forecasts

A forecast is judged by a *loss function*: commonly the root mean squared error (RMSE), which penalizes large errors heavily, or the mean absolute error. Three principles govern honest evaluation:

1. **Compare with a benchmark.** A model is useful only if it beats simple rules: the random walk (“no change”), the historical mean, the same month last year.

2. **Evaluate out of sample.** Estimate the model on data up to a date, forecast the next period, move the date forward and repeat (a *rolling-origin* evaluation). In-sample fit is not forecasting ability (Chapter 16).
3. **Avoid look-ahead bias.** Use only information available at the time of each forecast—including the data vintage available then, since official data are revised (Chapter 11). Using revised data, or choosing a model after seeing the whole sample, leaks future information into past forecasts.

Worked Example 18.3. A forecast horse race

Over four quarters, growth outturns are 2.1, 1.8, 2.4 and 2.0 per cent. A model forecast 2.0, 2.0, 2.2 and 2.3; the random walk, using the previous quarter's outturn (2.3 before the first quarter), forecast 2.3, 2.1, 1.8 and 2.4. The model's RMSE is $\sqrt{(0.01 + 0.04 + 0.04 + 0.09)/4} = 0.21$; the random walk's is $\sqrt{(0.04 + 0.09 + 0.36 + 0.16)/4} = 0.40$. The model wins in this sample—but four quarters are far too few to be confident, and formal tests of forecast accuracy take the sampling uncertainty into account.

18.6.2 Structural breaks and combination

Relationships change: a new monetary regime, a financial crisis or a pandemic can alter the dynamics of inflation or unemployment. A *structural break* makes a model estimated before it unreliable after it. Tests compare parameters before and after a candidate date; forecasters guard against breaks by weighting recent data more heavily and by *combining* forecasts from several models, which often beats any single one.

18.6.3 Nowcasting

Official GDP for a quarter is published weeks after it ends. *Nowcasting* estimates the current quarter from data that arrive sooner and more often: industrial production, retail sales, surveys, card transactions. The Federal Reserve Bank of Atlanta's GDPNow model, publicly available since 2014, updates an estimate of current-quarter U.S. growth as each data release arrives; many central banks run similar models. High-frequency private data proved especially valuable in 2020, when the economy changed faster than monthly statistics could record—subject to the representativeness concerns of Chapter 11.

18.7 Volatility

Chapter 13 noted that financial returns show *volatility clustering*: turbulent days follow turbulent days. Robert Engle's (1982) *autoregressive conditional heteroskedasticity* (ARCH) model lets the variance of today's shock depend on the size of recent shocks. Its most widely used extension, the **GARCH(1,1)** model of Tim Bollerslev (1986), is

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2,$$

where σ_t^2 is today's conditional variance. A large shock yesterday raises today's variance by α times its square, and the effect decays at rate $\alpha + \beta$. If $\alpha + \beta < 1$, volatility reverts to the long-run variance $\omega/(1 - \alpha - \beta)$. With $\omega = 0.00001$, $\alpha = 0.08$ and $\beta = 0.90$, for daily returns, the long-run variance is 0.0005, a daily volatility of 2.2 per cent, and shocks to volatility have a half-life of $\ln 0.5 / \ln 0.98 \approx 34$ trading days. GARCH models underlie many banks' value-at-risk calculations (Chapter 13) and option-pricing adjustments (Chapter 41).

Case: Meese and Rogoff: the random walk that could not be beaten

The contest. Richard Meese and Kenneth Rogoff (1983) compared the out-of-sample forecasts of several structural models of exchange rates—models in which the exchange rate depends on relative money supplies, incomes, interest rates and inflation—with those of a random walk, for the U.S. dollar against the German mark, the Japanese yen and the British pound, and a trade-weighted dollar, in the 1970s. They gave the structural models every advantage: forecasts used the *actual* future values of the explanatory variables, which no real forecaster would know. Even so, at horizons from one to twelve months, the models did not forecast better than the random walk, and usually worse.

Why it mattered. The result showed that good in-sample fit is no evidence of forecasting power, and that the macroeconomic “fundamentals” of the time explained little of month-to-month exchange-rate movements. Its lesson for method is the three principles of this chapter: evaluate out of sample, against a simple benchmark, without look-ahead.

What followed. Four decades of research have produced only partial exceptions. Models do somewhat better at horizons of several years, consistent with purchasing-power parity holding in the long run (Chapter 52); some fundamentals help when estimated with panel data across many currencies; and changes in interest-rate differentials and risk premia explain exchange-rate movements *after the fact*. But beating the random walk consistently at short horizons remains difficult—which is what one would expect if exchange rates, like other asset prices, respond mainly to news that is by definition unpredictable (Chapter 34).

Summary

- Fixed effects use within-unit variation, removing time-invariant unobserved characteristics; two-way fixed effects also remove common shocks. Cluster standard errors by unit.
- Time series combine trend, seasonality, cycle and noise; stationary series have stable means, variances and autocorrelations.
- The AR(1) $y_t = c + \phi y_{t-1} + \varepsilon_t$ has mean $c/(1 - \phi)$, variance $\sigma^2/(1 - \phi^2)$, impulse response ϕ^k and forecasts $\mu + \phi^h(y_t - \mu)$.
- Random walks have permanent shocks; regressions between non-stationary series are spurious unless the series are cointegrated.
- VARs summarize dynamics; Granger predictability is not causality; impulse responses need identified shocks.
- Honest forecast evaluation uses benchmarks, rolling out-of-sample tests and real-time data; breaks and combination matter; nowcasts use timely indicators.
- GARCH models capture volatility clustering with persistence $\alpha + \beta$ and long-run variance $\omega/(1 - \alpha - \beta)$.

Key terms

panel data; unobserved heterogeneity; fixed effects; within transformation; time fixed effects; two-way fixed effects; random effects; attrition; trend; seasonality; seasonal adjustment; stationarity; autocorrelation; white noise; autoregressive model; impulse response; mean reversion; fan chart; ARMA; random walk; unit root; integrated series; spurious regression; cointegration; error-correction model;

vector autoregression; Granger predictability; identification of shocks; loss function; RMSE; benchmark; rolling-origin evaluation; look-ahead bias; structural break; forecast combination; nowcasting; ARCH; GARCH.

Exercises

Exercise 18.1 (Fixed effects). Explain, using an omitted-variable argument, why a regression of countries' growth rates on their levels of democracy using a pooled cross-section could give a different answer from one with country fixed effects. What variation does each use?

Exercise 18.2 (AR(1)). Unemployment follows $u_t = 1.2 + 0.8u_{t-1} + \varepsilon_t$, with shock standard deviation 0.5. Find the mean and unconditional standard deviation. If unemployment is now 8 per cent, forecast it one, two and five years ahead.

Exercise 18.3 (Impulse response). In the model of Exercise 18.2, a shock raises unemployment by 1 point. How much higher is unemployment expected to be after three years? What is the half-life?

Exercise 18.4 (Random walk). A share price follows a random walk with daily shocks of standard deviation 1.5 per cent. What is the standard deviation of the price change over 25 trading days? What is the best forecast of the price in 25 days?

Exercise 18.5 (Spurious regression). An analyst regresses the level of a country's GDP on the level of the number of mobile phones over 25 years and finds $R^2 = 0.97$. What is wrong, and what should she do instead?

Exercise 18.6 (Granger). Weather forecasts Granger-cause umbrella sales but also Granger-cause rain. Does this show that weather forecasts cause rain? What does it show?

Exercise 18.7 (Forecast evaluation). A fund manager shows a trading model that would have earned 25 per cent a year since 2000. The model was designed in 2024 after testing many variants on the full data. List three reasons the reported performance overstates what an investor could expect.

Exercise 18.8 (RMSE). Outturns are 3, 5, 4, 6. Model A forecast 4, 4, 4, 5; model B forecast 3, 4, 5, 5. Compute both RMSEs and mean absolute errors. Which model is better by each measure?

Exercise 18.9 (GARCH). In a GARCH(1,1) with $\omega = 0.000002$, $\alpha = 0.10$ and $\beta = 0.88$, find the long-run daily variance and volatility. Yesterday's variance was 0.0004 and yesterday's return shock was 3 per cent. Find today's variance.

Solutions

Solution 18.1. Countries differ in geography, history, culture and institutions that affect both democracy and growth; in a pooled cross-section these confound the comparison between countries. Country fixed effects remove all time-invariant differences and use only changes in democracy within countries over time—which removes that confounding but may pick up only short-run effects and is sensitive to measurement error in the (slowly changing) democracy index.

Solution 18.2. Mean $1.2/0.2 = 6$ per cent; standard deviation $0.5/\sqrt{1-0.64} = 0.83$. Forecasts: $6 + 2 \times 0.8 = 7.6$; $6 + 2 \times 0.64 = 7.28$; $6 + 2 \times 0.8^5 = 6.66$.

Solution 18.3. $0.8^3 = 0.512$ points. Half-life $\ln 0.5 / \ln 0.8 = 3.1$ years.

Solution 18.4. $1.5\sqrt{25} = 7.5$ per cent. The best forecast is today's price (adjusted for any drift, which over 25 days is small).

Solution 18.5. Both series trend upward, so a regression of levels shows a high R^2 whether or not they are related (spurious regression). She should regress growth rates on growth rates, test whether the series are cointegrated, or—better still for a causal question—find a research design (Chapter 17).

Solution 18.6. No. Forecasts predict rain because forecasters use information that anticipates it; predictability reflects information, not causation. It shows that forecasts contain information about future rain beyond past rain.

Solution 18.7. (i) Multiple testing and selection: the reported model is the best of many tried on the same data (Chapter 15); (ii) look-ahead bias: choosing variables and parameters with knowledge of the whole period; (iii) omitted costs—transaction costs, market impact, taxes—and possibly survivorship bias in the data (Chapter 11). Only genuine out-of-sample performance after 2024 is a fair test.

Solution 18.8. A: errors $-1, 1, 0, 1$: RMSE $\sqrt{3/4} = 0.866$; MAE 0.75. B: errors $0, 1, -1, 1$: RMSE 0.866; MAE 0.75. They tie on both measures here—a reminder that small samples rarely discriminate between models.

Solution 18.9. Long-run variance $0.000002/0.02 = 0.0001$; volatility 1 per cent a day. Today: $0.000002 + 0.10 \times 0.0009 + 0.88 \times 0.0004 = 0.000002 + 0.00009 + 0.000352 = 0.000444$, a volatility of 2.1 per cent.

Decisions, Simulation and Modern Statistical Methods

After studying this chapter you should be able to

1. Structure a decision under uncertainty with states, actions and payoffs, and compute the expected value of perfect and imperfect information.
2. Use Monte Carlo simulation to study the distribution of an uncertain outcome, and distinguish simulation error from model error.
3. Estimate parameters by maximum likelihood.
4. Model binary outcomes with linear probability, logit and probit models, and evaluate classifiers with confusion matrices and costs.
5. Update beliefs with Bayesian inference, and explain shrinkage and partial pooling.
6. Explain the bias–variance trade-off, cross-validation, and regularized and tree-based learning methods, and their limits for causal questions.
7. Recognize the main specialized models for counts, durations and distributions, and the role of meta-analysis.

A motivating question

In 2018 the online property company Zillow began buying houses directly from their owners, using its price-prediction algorithm to make instant cash offers, and reselling them after light renovation. The business depended on predicting, for each house, what it would sell for months later. In 2021 Zillow bought thousands of homes as the market boomed. By November it had announced that it was closing the business, writing down the value of its inventory by hundreds of millions of dollars and cutting about a quarter of its workforce. Its models were sophisticated. What went wrong?

Statistics is ultimately in the service of decisions: whether to lend, invest, launch a product, buy a house, adopt a policy. This final chapter of Part II connects the tools of the preceding chapters to decisions, and introduces the modern methods—simulation, likelihood, classification, Bayesian updating and machine learning—that analysts now use routinely. Its recurring theme is the difference between a model’s precision and its reliability: uncertainty about the world is not reduced by computing more carefully with the wrong model.

19.1 Decisions Under Uncertainty

19.1.1 States, actions and payoffs

A decision problem specifies the *actions* available, the uncertain *states of the world*, their probabilities, and the *payoff* of each action in each state. A risk-neutral decision maker chooses

the action with the highest expected payoff; a risk-averse one maximizes expected utility (Chapter 25).

Worked Example 19.1. Launching a product

A firm can launch a product or not. The launch succeeds with probability 0.4, earning a profit of 10 (million), and fails with probability 0.6, losing 4. The expected value of launching is $0.4 \times 10 - 0.6 \times 4 = 1.6$, so a risk-neutral firm launches.

19.1.2 The value of information

Before deciding, the firm could gather information. With *perfect* information it would launch only when the product would succeed, earning $0.4 \times 10 = 4$ in expectation. The **expected value of perfect information** is $4 - 1.6 = 2.4$: the most the firm should pay for any study.

Real information is imperfect. Suppose market research returns a positive result with probability 0.8 if the product would succeed and 0.3 if it would fail. By the law of total probability, $P(\text{positive}) = 0.8 \times 0.4 + 0.3 \times 0.6 = 0.5$, and by Bayes' rule (Chapter 12), $P(\text{success} \mid \text{positive}) = 0.32/0.5 = 0.64$ and $P(\text{success} \mid \text{negative}) = 0.08/0.5 = 0.16$. After a positive result, launching is worth $0.64 \times 10 - 0.36 \times 4 = 4.96$; after a negative one, $0.16 \times 10 - 0.84 \times 4 = -1.76$, so the firm does not launch. The expected value with research is $0.5 \times 4.96 = 2.48$, and the *expected value of sample information* is $2.48 - 1.6 = 0.88$. Information is valuable only if it can change a decision; research whose result would not alter the choice is worth nothing, however interesting.

19.2 Monte Carlo Simulation

19.2.1 The method

Many uncertain outcomes depend on several random inputs in ways too complicated for formulas: the net present value of a project whose demand, price and cost are uncertain and correlated; the losses of a loan portfolio over several years; the value of a pension fund in 30 years. **Monte Carlo simulation** draws the random inputs from their assumed distributions many times, computes the outcome for each draw, and uses the resulting collection of outcomes to estimate the outcome's distribution: its mean, its quantiles, the probability of a loss. The name, coined by scientists working on nuclear weapons in the 1940s, refers to the casino.

19.2.2 Two kinds of error

The average of N simulated outcomes estimates the expected outcome with standard error $\sigma/\sqrt{N}$, by the results of Chapter 15: if simulated NPVs have a standard deviation of 50, then 10,000 draws estimate the mean NPV to within about ± 1 (twice $50/100 = 0.5$). More draws reduce this *simulation error* as much as desired.

They do nothing about *model error*. If the distributions assumed for demand are wrong, or the correlation between demand and costs is ignored, a million draws give a very precise answer to the wrong question. The most valuable uses of simulation are therefore often not the headline number but the exploration: which inputs matter most, how results change when assumptions change, and what combinations of events produce the worst outcomes (Chapter 36).

19.3 Maximum Likelihood

For many models there is no least-squares formula. The general principle of **maximum likelihood**, developed by Ronald Fisher in the 1920s, chooses the parameter values that make the observed data most probable. The *likelihood* $L(\theta)$ is the probability (or density) of the observed data as a function of the parameters θ ; the maximum-likelihood estimate maximizes it, usually by maximizing its logarithm (Chapter 5).

Worked Example 19.2. Estimating a default rate

Of 500 loans, 30 defaulted. If each defaults independently with probability p , the likelihood is $L(p) = p^{30}(1-p)^{470}$ (times a constant), and the log-likelihood $30 \ln p + 470 \ln(1-p)$. Setting its derivative $30/p - 470/(1-p)$ to zero gives $\hat{p} = 30/500 = 0.06$: the sample proportion. The curvature of the log-likelihood at the maximum measures how precisely p is estimated; here it gives the familiar standard error $\sqrt{0.06 \times 0.94/500} = 0.011$.

Under standard conditions, maximum-likelihood estimators are consistent, approximately normal in large samples, and efficient. OLS is the maximum-likelihood estimator when regression errors are normal. Likelihood also underlies the models that follow.

19.4 Binary Outcomes and Classification

19.4.1 Models for probabilities

Many outcomes are yes-or-no: default or repay, employed or not, buy or not. A *linear probability model* regresses the 0–1 outcome on explanatory variables by OLS; its coefficients are changes in probability (a coefficient of 0.03 means a unit rise in x raises the probability by 3 percentage points), but its predictions can fall outside $[0, 1]$. The **logit** model keeps probabilities in range by modelling the log-odds as linear:

$$\ln \frac{p}{1-p} = \beta_0 + \beta_1 x, \quad p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}};$$

the *probit* model uses the normal CDF instead. Both are estimated by maximum likelihood. A coefficient is not a change in probability: the *marginal effect* of x in the logit is $\beta_1 p(1-p)$, largest when p is near 0.5. With $\beta_0 = -3$, $\beta_1 = 0.5$ and $x = 2$, the log-odds are -2 , so $p = 0.119$, and the marginal effect is $0.5 \times 0.119 \times 0.881 = 0.052$: a unit rise in x raises the probability by about 5 percentage points at that point.

19.4.2 Classification and costs

Turning predicted probabilities into decisions—approve or reject, investigate or not—requires a threshold. The results are summarized in a *confusion matrix*. Suppose a lender’s model, applied to 1,000 past loans of which 100 defaulted, flags 150 as high risk, including 70 of the defaulters:

	Defaulted	Repaid
Flagged	70 (true positives)	80 (false positives)
Not flagged	30 (false negatives)	820 (true negatives)

The model’s *recall* (share of defaulters caught) is 70 per cent, its *precision* (share of flags that are defaulters) 47 per cent, and its false-positive rate $80/900 = 8.9$ per cent. Lowering the

threshold catches more defaulters at the cost of more false alarms. The right threshold depends on the *costs*: if a missed defaulter costs ten times as much as a rejected good borrower, the threshold should be low. A model that ranks borrowers well (often summarized by the area under the ROC curve) may still be poorly *calibrated*—its predicted probabilities may not match observed frequencies—and calibration matters for pricing and provisioning (Chapter 28).

19.5 Bayesian Inference

19.5.1 Priors and posteriors

Bayesian inference treats unknown parameters as uncertain quantities with probability distributions. A *prior* distribution expresses beliefs before seeing the data; Bayes' rule (Chapter 12) combines it with the likelihood to give a *posterior*:

$$\text{posterior} \propto \text{likelihood} \times \text{prior}.$$

The posterior supports direct probability statements about the parameter: a 95 per cent *credible interval* contains the parameter with posterior probability 0.95—the statement that, as Chapter 15 warned, a confidence interval does not support.

Worked Example 19.3. Default rate of a new branch

A bank's branches typically have default rates around 5 per cent. A new branch has made 20 loans, of which 3 have defaulted: 15 per cent. Represent prior beliefs by a *beta* distribution with parameters $(a, b) = (2, 38)$, whose mean $a/(a + b)$ is 5 per cent and which is equivalent to having seen 40 loans with 2 defaults. The beta prior is *conjugate* to the binomial likelihood: the posterior is beta with parameters $(a + \text{defaults}, b + \text{repayments}) = (5, 55)$, with mean $5/60 = 8.3$ per cent. The estimate is pulled from the raw 15 per cent toward the typical 5 per cent, because 20 loans are too few to be sure the branch is unusual. With 2,000 loans and 300 defaults, the posterior mean would be $302/2,040 = 14.8$ per cent: the data would dominate.

19.5.2 Shrinkage and partial pooling

The example illustrates **shrinkage**: estimates for small groups are pulled toward the overall average, by more the smaller the group and the less the groups really differ. Hierarchical (multilevel) models estimate how much groups differ from the data themselves and shrink accordingly—*partial pooling*, between treating each group separately (no pooling) and ignoring differences (complete pooling). Shrinkage deliberately introduces bias to reduce variance, and so lowers mean squared error (Chapter 15); it is used to rank hospitals, schools, fund managers and bank branches, and to estimate covariance matrices for portfolios (Chapter 14). Posterior distributions for complex models are computed by Markov chain Monte Carlo methods, which simulate draws from the posterior.

19.6 Statistical Learning

19.6.1 The bias–variance trade-off

Prediction errors come from bias (a model too simple to capture the true relationship) and variance (a model so flexible that it fits the noise in the particular sample). Adding flexibility lowers bias and raises variance; the best predictive model balances them. Chapter 16 warned

that in-sample fit rewards overfitting. **Cross-validation** estimates out-of-sample error by splitting the data into k parts, fitting the model on $k - 1$ of them and measuring errors on the remaining part, rotating through all parts.

19.6.2 Regularization and trees

Ridge and lasso. Regularized regressions add a penalty for large coefficients to the least-squares objective: $\sum e_i^2 + \lambda \sum_j \beta_j^2$ (ridge) or $\sum e_i^2 + \lambda \sum_j |\beta_j|$ (lasso), with the penalty strength λ chosen by cross-validation. Both shrink coefficients toward zero; the lasso sets many exactly to zero, selecting variables. Because the penalty depends on the scale of each variable, regressors should be standardized first.

Trees and ensembles. A regression tree splits the data repeatedly on the variable and threshold that best separate outcomes, producing a flowchart of rules. Single trees overfit; *random forests* average many trees grown on resampled data, and *gradient boosting* builds trees sequentially, each correcting the errors of the last. These methods capture interactions and nonlinearities automatically and are among the most accurate tools for prediction with many variables.

19.6.3 Prediction is not causation

A model that predicts well need not tell us what would happen if we intervened. A credit model may find that borrowers who shop at certain stores repay more often; forcing borrowers to shop there would not make them more creditworthy. Machine-learning predictions can, however, *assist* causal analysis: *double machine learning* uses flexible prediction to control for many confounders while estimating a treatment effect, and *causal forests* estimate how treatment effects vary across units—both still relying on the identifying assumptions of Chapter 17. Predictions used to allocate credit, insurance or jobs also raise questions of fairness, because they can reproduce past discrimination recorded in the data.

19.7 A Map of Specialized Models

Several outcomes need models of their own:

- **Counts** (patents, bank failures, claims): Poisson and negative-binomial regression (Chapter 13).
- **Durations** (unemployment spells, time to default, time to adoption): survival and hazard models, which handle spells still in progress (*censoring*).
- **Censored and selected samples** (wages observed only for workers; spending with many zeros): Tobit and Heckman selection models, which rely on strong assumptions about why observations are missing (Chapter 11).
- **Distributions** (effects on the low and high ends, not just the mean): *quantile regression*, which estimates how a chosen quantile of the outcome varies with the regressors.

Evidence synthesis. When many studies estimate the same effect, *meta-analysis* combines them, weighting each by its precision. Publication bias (Chapter 15) shows up as a correlation between estimates and their standard errors—small, imprecise studies report larger effects—and meta-analysts test and correct for it.

Case: Zillow Offers: when a prediction model makes decisions

The business. Zillow, known for its “Zestimate” estimates of house values, launched Zillow Offers in 2018 as an “iBuyer”: it bought homes from sellers who wanted a quick, certain sale, using algorithms to set offer prices, and resold them. Profit per house depended on the difference between the resale price months later and the purchase price plus renovation and holding costs—a thin margin relative to the value of the house.

What happened. In 2021, amid rapid house-price growth, Zillow raised its offers and bought homes at a much faster pace. As price growth slowed later in the year, and renovation took longer amid labour shortages, many homes were worth less than Zillow had paid. In November 2021 the company announced that it would wind down Zillow Offers, reported an inventory write-down of about 300 million dollars in the third quarter with further losses expected, and said it would reduce its workforce by about a quarter. Its chief executive said that the unpredictability of house prices far exceeded what the company had anticipated.

In terms of this chapter. A prediction model that is accurate *on average* can still lose money when it is used to make decisions. First, *adverse selection* (Chapter 25): sellers know their houses better than any algorithm, and accept offers that are too high more readily than offers that are too low, so the houses bought are disproportionately those the model overvalued—a “winner’s curse.” Second, *model risk*: a model estimated during a period of steady price growth embeds that environment; when the market turned, its forecast errors became correlated across thousands of houses rather than cancelling out (Chapter 14). Third, *scale*: buying faster increased exposure to exactly the risk the model could not see. More data and more computation reduce estimation noise; they do not remove selection effects or structural change.

Summary

- Decisions under uncertainty compare expected payoffs (or utilities) across actions. Information has value only if it can change a decision; EVPI bounds what any study is worth.
- Monte Carlo simulation estimates outcome distributions; simulation error falls as $1/\sqrt{N}$, model error does not.
- Maximum likelihood chooses parameters that make the data most probable; it gives the sample proportion for a Bernoulli rate and OLS with normal errors.
- Logit and probit keep probabilities in $[0, 1]$; marginal effects depend on the probability. Classification thresholds should reflect the costs of false positives and negatives.
- Bayesian inference combines priors with likelihoods; shrinkage and partial pooling trade bias for lower variance.
- Predictive learning balances bias and variance, using cross-validation, ridge, lasso and tree ensembles; prediction is not causation.
- Specialized models handle counts, durations, censoring and distributions; meta-analysis combines studies and tests for publication bias.

Key terms

decision problem; expected value of perfect information; expected value of sample information; Monte

Carlo simulation; simulation error; model error; likelihood; maximum likelihood; linear probability model; logit; probit; marginal effect; confusion matrix; recall; precision; false-positive rate; calibration; Bayesian inference; prior; posterior; credible interval; conjugate prior; beta distribution; shrinkage; partial pooling; hierarchical model; Markov chain Monte Carlo; bias–variance trade-off; cross-validation; ridge regression; lasso; regression tree; random forest; gradient boosting; double machine learning; survival model; censoring; quantile regression; meta-analysis; winner’s curse; model risk.

Exercises

Exercise 19.1 (Value of information). A bank can lend 1 million to a firm at a profit of 80,000 if it repays, or lose 400,000 if it defaults; the default probability is 0.1. (a) Should the bank lend? (b) What is the expected value of perfect information about default? (c) An audit costs 20,000 and flags defaulters with probability 0.9 and repayers with probability 0.2. Is the audit worth buying?

Exercise 19.2 (Simulation error). A simulation of a project’s NPV with 1,000 draws gives a mean of 12 and a standard deviation of 40. What is the standard error of the estimated mean? How many draws are needed to reduce it to 0.2? Would that make the estimate reliable?

Exercise 19.3 (Maximum likelihood). Waiting times until 8 customers’ first purchases are exponential with rate λ (Chapter 13), with mean observed time 5 days. Write the log-likelihood and find the maximum-likelihood estimate of λ .

Exercise 19.4 (Logit). A default model has log-odds $-4 + 0.8 \times \text{leverage}$ (leverage measured as debt over earnings). Find the default probability and marginal effect at leverage 3 and at leverage 6.

Exercise 19.5 (Classification). In the confusion matrix of the text, suppose each missed defaulter costs 50 and each rejected good borrower costs 5. Compute the total cost of the model’s errors. A second threshold flags 250 loans, catching 90 defaulters. Compute its confusion matrix and cost. Which threshold is better?

Exercise 19.6 (Bayesian updating). A fund manager’s skill is summarized by the probability p of beating the market in a month, with prior Beta(10, 10). She beats the market in 8 of 12 months. Find the posterior mean and compare it with the raw proportion. How would a sceptical prior Beta(50, 50) change the answer?

Exercise 19.7 (Regularization). Explain why a lasso regression with many candidate predictors of GDP growth might select a small number of variables, and why its coefficients should not be interpreted as causal effects.

Exercise 19.8 (Winner’s curse). An algorithm estimates house values without bias, with an error of plus or minus 5 per cent. A company offers the estimated value on every house, and sellers accept only when the offer exceeds the true value. Explain why the company loses money on average, even though its estimates are unbiased.

Solutions

Solution 19.1. (a) $0.9 \times 80,000 - 0.1 \times 400,000 = 32,000 > 0$: lend. (b) With perfect information, lend only to repayers: $0.9 \times 80,000 = 72,000$; EVPI = 40,000. (c) $P(\text{flag}) = 0.9 \times 0.1 + 0.2 \times 0.9 = 0.27$. $P(D \mid \text{flag}) = 0.09/0.27 = 1/3$: lending would earn $\frac{2}{3} \times 80,000 - \frac{1}{3} \times 400,000 < 0$, so reject. $P(D \mid \text{no flag}) = 0.01/0.73 = 0.0137$: lend, earning $0.9863 \times 80,000 - 0.0137 \times 400,000 = 73,425$. Expected

value with audit = $0.73 \times 73,425 = 53,600$; gain over 32,000 is 21,600, which exceeds the 20,000 cost: buy the audit (narrowly).

Solution 19.2. $SE = 40/\sqrt{1,000} = 1.26$. For 0.2: $\sqrt{N} = 200$, $N = 40,000$. This reduces simulation error only; if the input distributions or correlations are wrong, the estimate remains unreliable.

Solution 19.3. Log-likelihood $8 \ln \lambda - \lambda \sum t_i = 8 \ln \lambda - 40\lambda$. Derivative $8/\lambda - 40 = 0$: $\hat{\lambda} = 0.2$ per day, the reciprocal of the mean waiting time.

Solution 19.4. Leverage 3: log-odds -1.6 , $p = 1/(1 + e^{1.6}) = 0.168$, marginal effect $0.8 \times 0.168 \times 0.832 = 0.112$. Leverage 6: log-odds 0.8 , $p = 0.690$, marginal effect $0.8 \times 0.690 \times 0.310 = 0.171$.

Solution 19.5. First threshold: $30 \text{ missed} \times 50 + 80 \text{ false} \times 5 = 1,500 + 400 = 1,900$. Second: flagged 250 with 90 true positives, so 160 false positives, 10 false negatives, 740 true negatives; cost $10 \times 50 + 160 \times 5 = 500 + 800 = 1,300$. The second, lower threshold is better given these costs.

Solution 19.6. Posterior Beta(18, 14): mean $18/32 = 0.5625$, against the raw 0.667. With Beta(50, 50): posterior Beta(58, 54), mean 0.518—twelve months of good results barely move a strong prior that most managers have no skill.

Solution 19.7. The $|\beta|$ penalty makes it optimal to set coefficients of weak predictors exactly to zero, keeping the few that add most predictive power. The coefficients are biased toward zero by design, depend on which correlated variables happen to be selected, and reflect predictive association, not the effect of intervening on a variable (Chapter 17).

Solution 19.8. Sellers accept exactly when the estimate is too high, so among accepted offers the average error is positive—the company systematically overpays for the houses it buys. Unbiasedness across all houses does not imply unbiasedness among the houses selected by informed counterparties (adverse selection, Chapter 25).

Part III

Microeconomics

Consumer Choice and Demand

After studying this chapter you should be able to

1. State the standard assumptions on preferences and explain what each rules out.
2. Represent preferences by indifference curves and utility functions, and compute the marginal rate of substitution.
3. Construct budget constraints and show how they respond to changes in prices and income.
4. Solve the consumer's problem graphically, by substitution and by the Lagrangian, including corner solutions.
5. Derive Marshallian demand functions and classify goods as normal, inferior, substitutes or complements.
6. Decompose the effect of a price change into substitution and income effects.
7. Compute price, income and cross-price elasticities and relate price elasticity to total expenditure.
8. Derive market demand from individual demands.
9. Apply the model to labour supply and to saving, and explain why higher wages or interest rates have ambiguous effects.
10. Measure consumer surplus and use it to evaluate price changes.

A motivating question

When petrol prices rose sharply in the United States in 2008, and again in 2022, Americans did cut back their driving—but by much less than the price increases, at least at first. When the price of a smartphone model falls, sales of that model rise sharply, partly at the expense of rival models. When a government taxes sugary drinks, consumption falls by an amount that varies greatly from place to place. Why do some quantities respond strongly to prices and others hardly at all? Why does a price change affect not only the good whose price changed but other goods too? And how can we measure how much a price change makes consumers better or worse off?

This chapter builds the theory of consumer choice that answers these questions. It is the foundation for everything that follows about markets. It is also the first place where the constrained-optimization tools of Chapter 6 do real economic work.

20.1 Preferences

20.1.1 Bundles and preference relations

A consumer chooses a **consumption bundle**: a list of quantities of goods. With two goods we write a bundle as (x, y) . Two goods is a useful simplification: y can be interpreted as “spending on everything else,” so the two-good model loses less generality than it seems.

We describe tastes by a **preference relation**. We write $A \succsim B$ (“ A is weakly preferred to B ”) if the consumer regards bundle A as at least as good as B . If $A \succsim B$ and $B \succsim A$, the consumer is *indifferent*, $A \sim B$. If $A \succsim B$ but not $B \succsim A$, A is *strictly preferred*, $A \succ B$.

20.1.2 The assumptions

Definition 20.1. Standard assumptions on preferences

1. **Completeness**: for any bundles A and B , either $A \succsim B$ or $B \succsim A$ (or both). The consumer can always compare.
2. **Transitivity**: if $A \succsim B$ and $B \succsim C$, then $A \succsim C$.
3. **Monotonicity** (more is better): if A contains at least as much of every good as B and more of at least one, then $A \succ B$.
4. **Convexity**: averages are at least as good as extremes. If $A \sim B$, then any mixture $tA + (1 - t)B$, $0 < t < 1$, is at least as good as A .

Completeness and transitivity are the core of *rationality* in the sense of Chapter 1. Without transitivity, a consumer could be led around a cycle: preferring A to B , B to C , and C to A , she would pay a little to swap C for B , a little more to swap B for A , and a little more to swap A for C , ending where she started but poorer (a “money pump”). Monotonicity rules out satiation; it is reasonable for broad goods over relevant ranges (and for “bads” like pollution we redefine the good as its absence). Convexity expresses a taste for variety: most people prefer some food and some clothing to all of one or all of the other.

20.1.3 Indifference curves

An **indifference curve** is the set of bundles among which the consumer is indifferent. The assumptions imply four properties:

1. Indifference curves slope downward (by monotonicity: to stay indifferent with more x , the consumer must have less y).
2. Curves further from the origin represent preferred bundles (monotonicity).
3. Indifference curves cannot cross (transitivity: if they crossed, the crossing bundle would be indifferent to two bundles one of which contains more of both goods than the other).
4. Indifference curves are bowed toward the origin (convexity).

20.1.4 The marginal rate of substitution

Definition 20.2. Marginal rate of substitution

The **marginal rate of substitution** of x for y , MRS_{xy} , is the rate at which the consumer is willing to give up y for one more unit of x while remaining indifferent. It is the absolute value of the slope of the indifference curve.

Convexity is equivalent to a *diminishing* MRS: the more x a consumer already has relative to y , the less y she will give up for yet more x . Someone with ten coats and no boots will trade a coat for a pair of boots readily; someone with ten pairs of boots and one coat will not.

20.2 Utility Functions

20.2.1 Utility as a representation

Definition 20.3. Utility function

A **utility function** $U(x, y)$ represents a preference relation if $U(A) \geq U(B)$ exactly when $A \succsim B$.

A utility function is simply a convenient numerical labelling of indifference curves: higher numbers for preferred curves. It is **ordinal**: only the ranking matters, not the size of differences. If U represents preferences, so does any increasing transformation of it, such as $2U + 7$, $\ln U$ or U^3 . Statements such as “this bundle gives twice as much utility” have no meaning in this framework. (Chapter 25 introduces a different, *cardinal* use of utility for choices under risk.)

Under completeness, transitivity and a technical continuity condition, preferences can always be represented by a continuous utility function. From now on we work with utility functions.

20.2.2 Marginal utility and the MRS

The **marginal utility** of x is $MU_x = \partial U / \partial x$. An indifference curve is a level curve of U , so by the implicit function rule of Chapter 5 its slope is $-MU_x / MU_y$, and

$$MRS_{xy} = \frac{MU_x}{MU_y}.$$

The MRS does not depend on which utility function we use to represent preferences: an increasing transformation $f(U)$ multiplies both marginal utilities by $f'(U)$, which cancels in the ratio. This is reassuring, since the MRS is a property of preferences, while any particular utility number is arbitrary.

20.2.3 Four standard utility functions

Cobb–Douglas. $U = x^a y^{1-a}$, with $0 < a < 1$. (Any Cobb–Douglas $x^\alpha y^\beta$ can be normalized to this form by the transformation $U^{1/(\alpha+\beta)}$.) Then

$$MRS = \frac{ax^{a-1}y^{1-a}}{(1-a)x^ay^{-a}} = \frac{a}{1-a} \cdot \frac{y}{x}.$$

The MRS falls as x/y rises: preferences are convex. Cobb–Douglas is the workhorse example of well-behaved preferences.

Perfect substitutes. $U = ax + by$. The MRS is constant at a/b : the consumer will always trade x for y at the same rate. Two brands of bottled water that a consumer regards as identical are perfect substitutes with $a = b$. Indifference curves are straight lines.

Perfect complements. $U = \min(ax, by)$. The goods are consumed in fixed proportions: left shoes and right shoes ($a = b = 1$), or cars and tyres (one car per four tyres: $U = \min(x_{\text{cars}}, y_{\text{tyres}}/4)$). Indifference curves are L-shaped; extra units of one good without the other add nothing.

Quasi-linear. $U = v(x) + y$, with v increasing and concave. The MRS is $v'(x)$, which depends only on x . As we shall see, demand for x then does not depend on income (over a range), which makes quasi-linear preferences convenient for welfare analysis (Chapter 22).

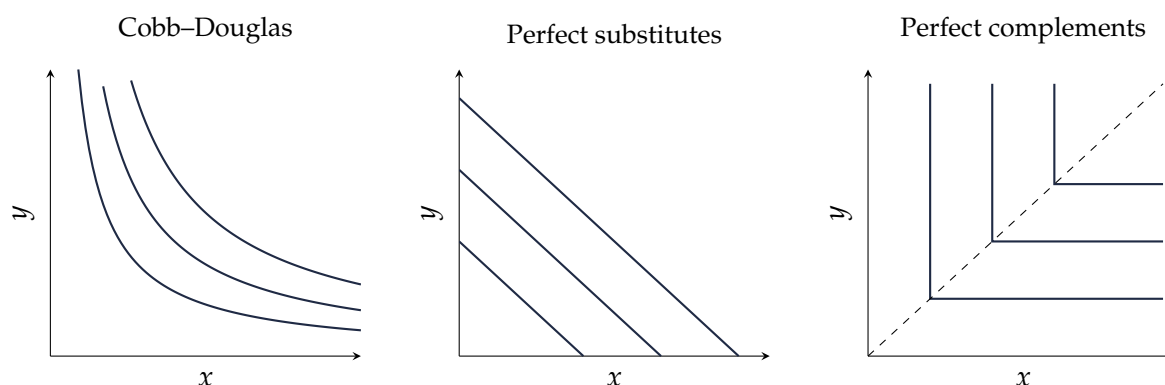


Figure 20.1: Indifference maps for three standard preferences. Utility increases to the north-east in each panel.

20.3 The Budget Constraint

20.3.1 The budget line

A consumer with income m facing prices p_x and p_y can afford any bundle with

$$p_x x + p_y y \leq m.$$

The set of affordable bundles is the **budget set**; its upper boundary, $p_x x + p_y y = m$, is the **budget line**. Solving for y :

$$y = \frac{m}{p_y} - \frac{p_x}{p_y} x.$$

The intercepts m/p_y and m/p_x are the maximum quantities of each good that can be bought. The slope is $-p_x/p_y$: the *market rate* at which y can be exchanged for x . It is the opportunity cost of x measured in units of y .

20.3.2 Shifts and pivots

- An increase in income shifts the budget line outward parallel to itself (slope unchanged).
- An increase in p_x pivots the line inward around the y -intercept (the line becomes steeper).
- Multiplying all prices and income by the same factor leaves the budget line unchanged. Only *relative* prices and *real* income matter. This is why a general inflation in which all prices and incomes rise together should, in principle, leave consumption choices unchanged—a point that becomes contentious in macroeconomics (Chapter 50).

20.3.3 Non-linear budgets

Real budget constraints are often kinked. Quantity discounts, rationing, taxes that apply only above a threshold, and welfare benefits that are withdrawn as earnings rise all produce kinks. The analysis below extends to such cases: the optimum is still the best affordable bundle, but it may lie at a kink.

20.4 Optimal Choice

20.4.1 The tangency condition

The consumer chooses the affordable bundle on the highest indifference curve. With well-behaved preferences and an interior solution, this occurs where an indifference curve is tangent to the budget line (Figure 20.2):

$$MRS_{xy} = \frac{MU_x}{MU_y} = \frac{p_x}{p_y}.$$

The left side is the rate at which the consumer is *willing* to trade y for x ; the right side is the rate at which the market *allows* her to. If the MRS exceeded the price ratio—say she would give up 3 units of y for one more x , but the market charges only 2—she could buy one more x , give up 2 units of y , and be better off. So at an interior optimum the two rates must be equal. Rearranged,

$$\frac{MU_x}{p_x} = \frac{MU_y}{p_y} :$$

the last unit of money spent on each good yields the same marginal utility. This is the marginal decision rule of Chapter 1 applied to the allocation of a budget.

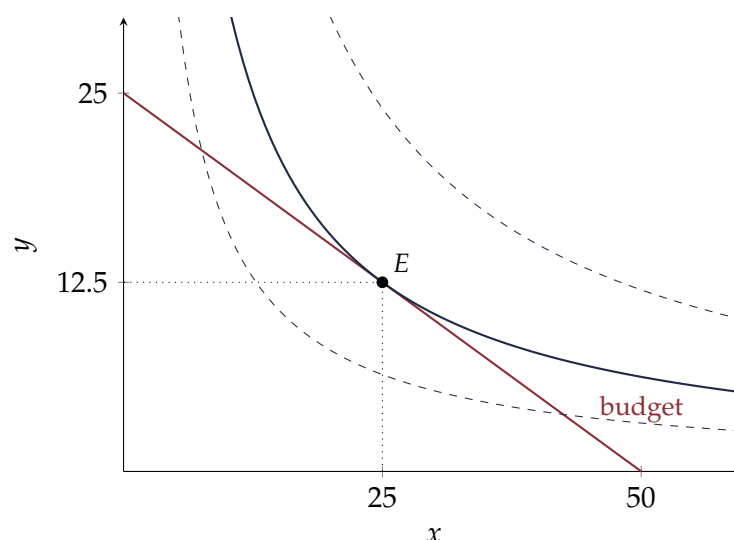


Figure 20.2: The optimal choice E for $U = x^{0.5}y^{0.5}$, $m = 100$, $p_x = 2$, $p_y = 4$ (Example 6.1). The indifference curve through E is tangent to the budget line. Higher curves (dashed, upper) are unaffordable; lower curves are affordable but worse.

20.4.2 Cobb–Douglas demand

Proposition 20.4. Demand with Cobb–Douglas preferences

If $U = x^a y^{1-a}$, the consumer spends a constant share a of income on x and $1 - a$ on y :

$$x^* = \frac{am}{p_x}, \quad y^* = \frac{(1-a)m}{p_y}.$$

Proof. The tangency condition is $\frac{a}{1-a} \frac{y}{x} = \frac{p_x}{p_y}$, so $p_y y = \frac{1-a}{a} p_x x$. Substituting into the budget line: $p_x x + \frac{1-a}{a} p_x x = m$, i.e. $p_x x / a = m$, so $p_x x = am$. Then $p_y y = (1-a)m$. $\square$

Chapter 6 solved the case $a = 0.5$, $m = 100$, $p_x = 2$, $p_y = 4$ and found $x^* = 25$, $y^* = 12.5$, as the formula gives.

20.4.3 Corner solutions

Perfect substitutes. With $U = ax + by$, the consumer compares the utility per unit of money of each good, a/p_x and b/p_y , and spends everything on whichever is higher. If $a/p_x > b/p_y$, $x^* = m/p_x$ and $y^* = 0$. If they are equal, any point on the budget line is optimal. The tangency condition fails because the indifference curve and budget line are both straight lines with, in general, different slopes.

Perfect complements. With $U = \min(x, y)$, the consumer buys equal quantities: $x^* = y^* = m/(p_x + p_y)$. The optimum is at the kink of the indifference curve, where the MRS is not defined. The tangency condition does not apply, but the logic of choosing the best affordable bundle does.

Quasi-linear preferences. With $U = v(x) + y$ and $p_y = 1$, the interior condition is $v'(x) = p_x$, which determines x^* independently of income, as long as income is large enough to buy x^* . The consumer spends every additional unit of income on y . If $v'(0) < p_x$, the consumer buys none of x (Example 6.7).

20.5 Demand Functions

20.5.1 Marshallian demand

Solving the consumer's problem for every combination of prices and income gives the **Marshallian demand functions** (also called ordinary or uncompensated demand):

$$x^* = x(p_x, p_y, m), \quad y^* = y(p_x, p_y, m).$$

Proposition 20.5. Homogeneity of degree zero

Marshallian demands are homogeneous of degree zero in prices and income: $x(tp_x, tp_y, tm) = x(p_x, p_y, m)$ for all $t > 0$.

Proof. Multiplying all prices and income by t leaves the budget set unchanged, so the best affordable bundle is unchanged. $\square$

This property—the absence of *money illusion*—means that demands depend only on relative prices and real income. It is a testable prediction of the theory, and it has important implications for macroeconomics: in the long run, the price level should not affect real choices.

20.5.2 Responses to income

Holding prices fixed and varying income traces out the **income-consumption curve** in the (x, y) plane; plotting x^* against m gives the **Engel curve**, named for Ernst Engel, who studied Prussian and Belgian household budgets in the nineteenth century.

Definition 20.6. Normal and inferior goods

A good is **normal** if demand for it rises with income ($\partial x / \partial m > 0$) and **inferior** if demand falls with income ($\partial x / \partial m < 0$).

Most goods are normal. Inferior goods are typically cheaper substitutes that people abandon as they become richer: intercity bus travel versus air travel, or cheap staple foods. Whether a good is inferior depends on the income range: a good can be normal for the poor and inferior for the rich. Engel's own finding—that the *share* of the budget spent on food falls as income rises, even though spending on food rises—is known as **Engel's law**, and it has held across countries and centuries with remarkable regularity. It helps explain why agriculture's share of employment falls as economies grow, as in China's structural transformation since 1978 (Chapter 44).

20.5.3 Responses to own price

Holding income and the other price fixed and varying p_x traces out the **price-consumption curve**; plotting x^* against p_x gives the individual's **demand curve**. For Cobb–Douglas preferences, $x^* = am/p_x$ is a rectangular hyperbola: spending on x is constant at am whatever the price.

20.5.4 Responses to other prices

Definition 20.7. Gross substitutes and complements

Good x is a **gross substitute** for y if $\partial x / \partial p_y > 0$: when y becomes more expensive, the consumer buys more x . It is a **gross complement** if $\partial x / \partial p_y < 0$.

Coffee and tea are substitutes for many people; coffee and milk are complements. With Cobb–Douglas preferences, $\partial x / \partial p_y = 0$: the two goods are neither gross substitutes nor gross complements, because the budget share of each is fixed. This is a limitation of Cobb–Douglas, not a general property.

20.6 Income and Substitution Effects

20.6.1 Decomposing a price change

When the price of x rises, two things happen at once:

1. x becomes more expensive *relative to* y . Even if the consumer were compensated so as to remain on her original indifference curve, she would substitute toward y . This is the **substitution effect**.

2. The consumer's *purchasing power* falls: with the same income she can no longer afford her original bundle. This is the **income effect**.

The decomposition due to John Hicks proceeds in two steps (Figure 20.3). Starting at the original optimum A , rotate the budget line to the new price ratio but shift it until it is just tangent to the *original* indifference curve, at B . The move $A \rightarrow B$ is the substitution effect: a movement along the original indifference curve, driven purely by the change in relative prices. Then shift the rotated line parallel back to the actual new budget line, reaching the new optimum C . The move $B \rightarrow C$ is the income effect.

Proposition 20.8. Sign of the substitution effect

With convex preferences, the substitution effect of an increase in the price of a good is never positive: the compensated demand for a good never rises when its own price rises.

Argument. Along a convex indifference curve, a steeper price line (higher p_x/p_y) is tangent at a point with a higher MRS, which by diminishing MRS means less x and more y . $\square$

The income effect, by contrast, can have either sign. For a normal good, the fall in purchasing power reduces demand, reinforcing the substitution effect: demand falls unambiguously when price rises. For an inferior good, the income effect works in the opposite direction, partly offsetting the substitution effect.

Definition 20.9. Giffen good

A **Giffen good** is one for which demand *rises* when its own price rises. This requires the good to be inferior and the income effect to be larger than the substitution effect.

Giffen goods are a theoretical possibility that is very rare in practice. The most convincing evidence comes from a field experiment by Jensen and Miller (2008) in Hunan province, China, where very poor households consumed rice as a staple; when the price of rice was subsidized, some of the poorest households bought *less* rice and more meat, implying that, for them, rice was a Giffen good over the relevant range. The conditions are demanding: a staple that takes a very large share of the budget, few substitutes, and households poor enough that a price rise forces them to cut back on the preferred alternatives.

Worked Example 20.10. Computing the decomposition

Let $U = x^{0.5}y^{0.5}$, $m = 100$, $p_y = 1$, and let p_x rise from 1 to 2.

Original choice: $x = 0.5 \times 100/1 = 50$, $y = 50$, $U = 50$ (point A).

New choice: $x = 0.5 \times 100/2 = 25$, $y = 50$, $U = \sqrt{1,250} \approx 35.36$ (point C). Total effect on x : -25 .

Compensated choice (point B): the cheapest bundle giving $U = 50$ at the new prices. Tangency requires $y/x = p_x/p_y = 2$, so $y = 2x$, and $\sqrt{x \cdot 2x} = 50$ gives $x = 50/\sqrt{2} \approx 35.36$, $y \approx 70.71$. Its cost is $2(35.36) + 70.71 = 141.42$: the consumer would need 41.42 more income to be as well off as before.

Substitution effect: $35.36 - 50 = -14.64$. Income effect: $25 - 35.36 = -10.36$. They sum to the total effect, -25 . Both are negative because x is a normal good.

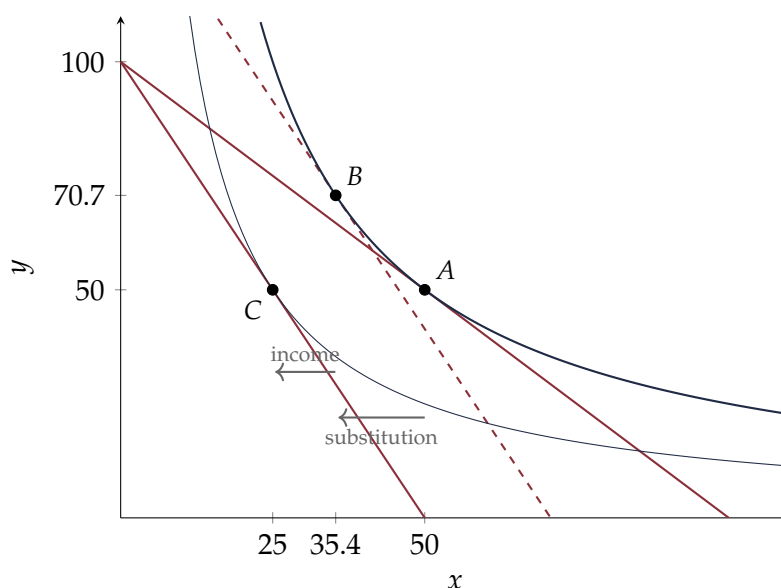


Figure 20.3: Hicks decomposition for $U = x^{0.5}y^{0.5}$, $m = 100$, $p_y = 1$, when p_x rises from 1 to 2. The substitution effect moves the consumer from A to B along the original indifference curve; the income effect moves her from B to C.

Advanced Analysis: Expenditure functions, Hicksian demand and the Slutsky equation

The **expenditure function** $e(p_x, p_y, \bar{U})$ is the minimum expenditure needed to reach utility $\bar{U}$ at prices (p_x, p_y) . The bundle that achieves it gives the **Hicksian** (compensated) demands $h_x(p_x, p_y, \bar{U})$, $h_y(\cdot)$. By the envelope theorem (Chapter 6), differentiating the expenditure function with respect to a price gives the Hicksian demand for that good: $\partial e / \partial p_x = h_x$ (**Shephard's lemma**).

Because Hicksian and Marshallian demands coincide when income equals the expenditure needed to reach the utility actually attained, $h_x(p, \bar{U}) = x(p, e(p, \bar{U}))$ for all p . Differentiating with respect to p_x and using Shephard's lemma:

$$\frac{\partial h_x}{\partial p_x} = \frac{\partial x}{\partial p_x} + \frac{\partial x}{\partial m} h_x \quad \Rightarrow \quad \frac{\partial x}{\partial p_x} = \underbrace{\frac{\partial h_x}{\partial p_x}}_{\text{substitution}} - \underbrace{x \frac{\partial x}{\partial m}}_{\text{income}}.$$

This is the **Slutsky equation**. For the Cobb–Douglas example at $p_x = 1$, $m = 100$: $\partial x / \partial p_x = -m / (2p_x^2) = -50$; $h_x = \bar{U} \sqrt{p_y / p_x}$ so $\partial h_x / \partial p_x = -\frac{1}{2} \bar{U} p_y^{1/2} p_x^{-3/2} = -25$; and $x \partial x / \partial m = 50 \times \frac{1}{2} = 25$. Indeed $-50 = -25 - 25$.

The **indirect utility function** $V(p_x, p_y, m)$ gives maximum utility as a function of prices and income. **Roy's identity**, $x = -(\partial V / \partial p_x) / (\partial V / \partial m)$, recovers Marshallian demand from it. For Cobb–Douglas with $a = \frac{1}{2}$, $V = m / (2\sqrt{p_x p_y})$, and Roy's identity gives $x = m / (2p_x)$.

20.7 Elasticities

20.7.1 Price elasticity of demand

Chapter 5 defined elasticity in general. The **price elasticity of demand** is

$$\varepsilon_{x,p_x} = \frac{\partial x}{\partial p_x} \cdot \frac{p_x}{x}.$$

It is negative for all goods except Giffen goods. Economists often quote its absolute value. Demand is **elastic** if $|\varepsilon| > 1$, **inelastic** if $|\varepsilon| < 1$, and **unit-elastic** if $|\varepsilon| = 1$.

20.7.2 Elasticity and total expenditure

Total expenditure on a good (equivalently, total revenue of its sellers) is $E = p x(p)$. Differentiating:

$$\frac{dE}{dp} = x + p \frac{dx}{dp} = x(1 + \varepsilon).$$

Proposition 20.11. Price elasticity and revenue

A small price increase raises total expenditure if demand is inelastic ($-1 < \varepsilon < 0$), lowers it if demand is elastic ($\varepsilon < -1$), and leaves it unchanged if demand is unit-elastic.

This is why a poor harvest can raise farmers' total income: demand for staple foods is inelastic, so the rise in price more than offsets the fall in quantity. It is also why a firm with market power never chooses to operate where its demand is inelastic (Chapter 24).

20.7.3 Determinants of price elasticity

Demand tends to be more elastic when:

- **Close substitutes are available.** Demand for one brand of petrol is far more elastic than demand for petrol as a whole.
- **The good takes a large share of the budget.** The income effect is then larger.
- **The time horizon is longer.** In the short run, drivers can only drive less; in the long run they can buy more efficient cars, move closer to work, or switch to public transport.
- **The good is a luxury rather than a necessity**, in the sense that it can be done without.

20.7.4 Income and cross-price elasticities

The **income elasticity** $\varepsilon_{x,m} = (\partial x / \partial m)(m/x)$ is positive for normal goods and negative for inferior goods. A normal good with income elasticity greater than 1 is sometimes called a *luxury* (its budget share rises with income); one with elasticity between 0 and 1 a *necessity* (its budget share falls with income; food is the classic case, as Engel's law says). The **cross-price elasticity** $\varepsilon_{x,p_y} = (\partial x / \partial p_y)(p_y/x)$ is positive for gross substitutes and negative for gross complements. Competition authorities use cross-price elasticities to define markets: if a rise in the price of product A sends many customers to product B, A and B are probably in the same market (Chapter 24).

Proposition 20.12. Engel aggregation

Let $s_x = p_x x / m$ and $s_y = p_y y / m$ be budget shares. If the budget constraint binds, the share-weighted average of income elasticities equals one:

$$s_x \epsilon_{x,m} + s_y \epsilon_{y,m} = 1.$$

Proof. Differentiate $p_x x + p_y y = m$ with respect to m : $p_x \partial x / \partial m + p_y \partial y / \partial m = 1$. Multiply and divide each term by the quantity and m : $(p_x x / m)(\partial x / \partial m)(m / x) + (p_y y / m)(\partial y / \partial m)(m / y) = 1$. $\square$

It follows that not all goods can be luxuries or all necessities: if income rises by 1%, total spending rises by 1%.

20.8 Market Demand

The **market demand** for a good is the sum of all individual demands at each price. Graphically, individual demand curves are added *horizontally*: at each price, add the quantities.

Worked Example 20.13. Adding demand curves

Consumer 1 has demand $q_1 = 10 - P$ (for $P \leq 10$, zero above). Consumer 2 has $q_2 = 12 - 3P$ (for $P \leq 4$, zero above). Market demand is

$$Q = \begin{cases} 10 - P, & 4 \leq P \leq 10, \\ 22 - 4P, & 0 \leq P < 4. \end{cases}$$

The market demand curve has a kink at $P = 4$, where consumer 2 enters the market. At $P = 3$, $Q = 7 + 3 = 10$.

Market demand inherits the properties of individual demands: it slopes down (barring Giffen behaviour) and depends on prices and on the distribution of income. In general, it depends on the whole distribution of income, not only on total income, because different consumers spend extra income differently. Only in special cases—when every consumer has the same, linear Engel curves—does market demand depend on aggregate income alone. That special case underlies the “representative consumer” used in much of macroeconomics (Chapter 47), and its fragility is one reason why modern macroeconomics increasingly models heterogeneous households.

20.9 Revealed Preference

Utility functions are not observed; choices are. Paul Samuelson asked what restrictions the theory places on observed choices alone.

Definition 20.14. Revealed preference

If a consumer chooses bundle A when bundle B was also affordable, A is **revealed preferred** to B .

Proposition 20.15. Weak axiom of revealed preference (WARP)

If A is revealed preferred to B (and $A \neq B$), then B must never be revealed preferred to A : whenever the consumer chooses B , A must be unaffordable.

Worked Example 20.16. Checking consistency

At prices $(p_x, p_y) = (1, 2)$, a consumer with income 20 chooses $A = (10, 5)$. At prices $(2, 1)$ with income 20, she chooses $B = (4, 12)$.

Was B affordable when A was chosen? $1 \times 4 + 2 \times 12 = 28 > 20$. No. Was A affordable when B was chosen? $2 \times 10 + 1 \times 5 = 25 > 20$. No. Neither bundle is revealed preferred to the other; the choices are consistent with WARP.

Now suppose that at the second prices, with income 30, she chose $B' = (5, 20)$. Then A cost $25 \leq 30$, so B' is revealed preferred to A . At the first prices, B' cost $5 + 40 = 45 > 20$, so A is not revealed preferred to B' . Still consistent.

Revealed preference shows that the theory is testable: some patterns of choice are inconsistent with *any* utility function. Empirical tests on household and experimental data find that most choices satisfy the axioms, though violations are not rare.

20.10 Labour Supply

The same model explains how much people choose to work. Time is the constrained resource: a person has T hours available, divides them between work h and leisure $\ell = T - h$, earns a wage w per hour and has non-labour income V (from savings, a partner, benefits). Consumption c , with price normalized to 1, satisfies

$$c = wh + V = w(T - \ell) + V \iff c + w\ell = wT + V.$$

The right side is **full income**: the value of all available time plus non-labour income. The wage is the *price of leisure*: each hour of leisure costs w of consumption forgone.

Worked Example 20.17. Labour supply with Cobb–Douglas preferences

Let $U = a \ln c + (1 - a) \ln \ell$. By the Cobb–Douglas rule, the person spends a share $1 - a$ of full income on leisure:

$$\ell^* = \frac{(1 - a)(wT + V)}{w}, \quad h^* = T - \ell^* = aT - \frac{(1 - a)V}{w}.$$

With $a = 0.5$, $T = 16$, $w = 20$, $V = 80$: $\ell^* = 0.5 \times 400/20 = 10$, so $h^* = 6$ hours. If the wage rises to 25: $\ell^* = 0.5 \times 480/25 = 9.6$, $h^* = 6.4$. If $V = 0$, $h^* = aT = 8$ whatever the wage.

A higher wage has two effects on hours worked. The substitution effect raises hours: leisure has become more expensive. The income effect reduces hours if leisure is a normal good: the person is richer and wants more of everything, including leisure. Which dominates is an empirical question. With Cobb–Douglas preferences and no non-labour income, the two effects exactly cancel. At high wages the income effect may dominate, producing a **backward-bending labour supply curve**. The long historical decline in hours worked per person in rich countries, as real wages rose many-fold, suggests that the income effect has been important over long periods; Germany now has some of the lowest average annual hours worked per employed person among OECD countries, and the United States considerably higher hours at

similar income levels, which shows that preferences and institutions—taxes, holidays, labour regulation—matter too.

Bridge. Labour supply is central to macroeconomics. Whether temporary changes in wages induce large changes in hours (the *intertemporal elasticity of labour supply*) is a key question separating real business cycle and New Keynesian views of fluctuations (Chapter 47).

20.11 Intertemporal Choice Revisited

Chapter 9 solved the two-period saving problem with the Euler equation. We can now interpret it with the tools of this chapter. Treat c_1 and c_2 as two goods. The intertemporal budget line

$$c_1 + \frac{c_2}{1+r} = y_1 + \frac{y_2}{1+r}$$

has slope $-(1+r)$: the price of c_1 in terms of c_2 is $1+r$. The Euler equation $u'(c_1) = \beta(1+r)u'(c_2)$ is the tangency condition $MRS = 1+r$.

A rise in the interest rate is a rise in the price of present consumption. Its substitution effect always reduces c_1 , i.e. raises saving. Its income effect depends on whether the household is a lender or a borrower. A lender ($y_1 > c_1$) becomes richer when r rises, which by the income effect raises c_1 ; the net effect on saving is ambiguous. A borrower becomes poorer, which reinforces the substitution effect: saving rises (borrowing falls) unambiguously. In the log-utility example of Chapter 9, $c_1 = W/(1+\beta)$ where $W = y_1 + y_2/(1+r)$: a higher r reduces c_1 only through its effect on the present value of future income; if $y_2 = 0$, c_1 does not respond at all to r , because for log utility the income and substitution effects for a pure lender cancel. This ambiguity is why the response of aggregate saving to interest rates is an empirical question and why monetary policy works partly through channels other than intertemporal substitution (Chapter 49).

20.12 Consumer Surplus

20.12.1 Willingness to pay

The height of an individual's inverse demand curve at a given quantity is the maximum amount she would pay for that unit—her **marginal willingness to pay**. If she pays the market price for every unit, she gains on every unit whose willingness to pay exceeds the price.

Definition 20.18. Consumer surplus

Consumer surplus is the difference between the total amount consumers would be willing to pay for the quantity they buy and the amount they actually pay. It is the area below the demand curve and above the price, up to the quantity purchased.

Worked Example 20.19. Consumer surplus with linear demand

For the bus demand $Q = 1,200 - 400P$ (inverse demand $P = 3 - Q/400$), at a fare of 1.50, 600 rides are demanded. Consumer surplus is the triangle with base 600 and height $3 - 1.5 = 1.5$: $\frac{1}{2} \times 600 \times 1.5 = 450$ per day. If the fare rises to 2, rides fall to 400 and consumer surplus falls to $\frac{1}{2} \times 400 \times 1 = 200$. The loss of 250 has two parts: riders who continue to ride pay 0.50 more on 400 rides (a loss of 200), and 200 rides that no longer

take place lose their surplus, a triangle of $\frac{1}{2} \times 200 \times 0.5 = 50$.

20.12.2 How accurate is consumer surplus?

Consumer surplus measured from the Marshallian demand curve is an exact measure of the consumer's gain only when there are no income effects—for example, with quasi-linear preferences. In general, the exact measures use Hicksian demand: the **compensating variation** (how much income the consumer would need, after a price rise, to be as well off as before) and the **equivalent variation** (how much income she would give up, before the price rise, to avoid it). In Example 20.10, the compensating variation of the price rise is 41.42. The Marshallian consumer-surplus loss, $\int_1^2 50/p \, dp = 50 \ln 2 \approx 34.66$, lies between the equivalent variation and the compensating variation, as it always does for a normal good. When the good is a small share of the budget, income effects are small and the three measures are close. Consumer surplus is therefore a reasonable approximation for most policy analysis, and Chapter 22 uses it to measure the welfare effects of taxes and price controls.

Case: How much do American drivers respond to petrol prices?

Petrol is a textbook example of a good with inelastic short-run demand. Hughes, Knittel and Sperling (2008) estimated that the short-run price elasticity of U.S. gasoline demand was in the range of about -0.03 to -0.08 in 2001–2006, much smaller in absolute value than estimates of roughly -0.2 to -0.3 for 1975–1980. Their explanation draws directly on the determinants of elasticity. By the 2000s, Americans lived in more car-dependent suburbs with fewer practical alternatives to driving; and petrol took a smaller share of household budgets than it had in the late 1970s, so a given price rise had a smaller income effect.

Long-run elasticities are larger, because households can change vehicles, locations and commuting patterns. This gap between short-run and long-run responses has practical consequences. A sudden supply disruption produces a large price spike, because quantity demanded barely adjusts in the short run (Chapter 22). A carbon tax raises little behavioural response at first but more over time. And a central bank facing an oil-price shock must judge how much of the resulting inflation is a one-off adjustment in relative prices and how much will persist (Chapter 50). German households in 2022 faced a similar problem with natural gas, whose short-run demand for heating is also inelastic; industrial users, who could switch fuels or cut production, adjusted faster than households.

20.13 Limitations

The model assumes that consumers know their preferences, understand prices, and optimize. Behavioural research documents systematic departures: people are influenced by how choices are framed, weigh losses more heavily than equivalent gains, procrastinate, and pay limited attention to prices that are not salient (such as sales taxes added at the till rather than displayed on the shelf). None of this makes the model useless—its predictions about the *direction* of responses to prices and incomes are well supported—but it cautions against assuming that observed choices always maximize consumers' own welfare. That caution matters for policy on savings, health and consumer protection.

Summary

- Complete, transitive, monotone and convex preferences are represented by downward-sloping, non-crossing indifference curves bowed toward the origin.
- Utility is ordinal. The MRS equals MU_x/MU_y and is invariant to increasing transformations.
- The budget line has slope $-p_x/p_y$. Only relative prices and real income matter.
- At an interior optimum, $MRS = p_x/p_y$, equivalently $MU_x/p_x = MU_y/p_y$. Corners arise with perfect substitutes and when a good is not worth its price.
- Cobb–Douglas consumers spend constant budget shares. Demands are homogeneous of degree zero.
- A price change has a substitution effect (always non-positive for own price) and an income effect (negative for normal goods). Giffen goods require a dominant income effect for an inferior good.
- Revenue rises with price when demand is inelastic and falls when it is elastic. Elasticity is greater with close substitutes, large budget shares and longer horizons.
- Market demand is the horizontal sum of individual demands.
- Higher wages and interest rates have ambiguous effects on hours worked and saving because income and substitution effects conflict.
- Consumer surplus, the area under demand and above price, measures consumers' gains from trade; it is exact without income effects and a good approximation when income effects are small.

Key terms

consumption bundle; preference relation; completeness; transitivity; monotonicity; convexity; indifference curve; marginal rate of substitution; utility function; ordinal utility; marginal utility; Cobb–Douglas; perfect substitutes; perfect complements; quasi-linear preferences; budget set; budget line; tangency condition; Marshallian demand; homogeneity of degree zero; income-consumption curve; Engel curve; normal good; inferior good; Engel's law; price-consumption curve; gross substitutes; gross complements; substitution effect; income effect; Giffen good; expenditure function; Hicksian demand; Shephard's lemma; Slutsky equation; indirect utility; Roy's identity; price elasticity; elastic; inelastic; income elasticity; cross-price elasticity; Engel aggregation; market demand; revealed preference; WARP; full income; backward-bending labour supply; consumer surplus; compensating variation; equivalent variation.

Exercises

Exercise 20.1 (MRS). Compute the MRS for (a) $U = x^2y$; (b) $U = 2 \ln x + \ln y$; (c) $U = 3x + 2y$; (d) $U = \sqrt{x} + y$. Which pairs of these represent the same preferences?

Exercise 20.2 (Optimal choice). A consumer with $U = x^{0.3}y^{0.7}$ has income 200 and faces $p_x = 3$, $p_y = 7$. Find the optimal bundle and verify the tangency condition.

Exercise 20.3 (Perfect complements). A person always drinks coffee with two sugars: $U = \min(x, y/2)$, where x is cups of coffee and y is sugar cubes. Coffee costs 2 per cup, sugar 0.25 per cube, and income is 50. How many cups does she drink? What is her demand function for coffee? Does a change in the price of sugar affect coffee demand?

Exercise 20.4 (Perfect substitutes). A consumer regards brand A and brand B of rice as perfect substitutes, but values one kilogram of A as equivalent to 1.2 kg of B. A costs 3 per kg and B costs 2.4 per kg. Income for rice is 60. What does she buy? At what price of A would she switch?

Exercise 20.5 (Quasi-linear preferences). $U = 20 \ln x + y$, $p_y = 1$, income m . Find demand for x as a function of p_x and m , stating any condition on m . What happens to demand for x when income rises?

Exercise 20.6 (Income and substitution effects). Using Example 20.10, compute the *Slutsky* version of the decomposition, in which the consumer is compensated so as to be able to afford her original bundle (rather than to reach her original utility). How do the substitution and income effects compare with the Hicks version?

Exercise 20.7 (Inferior goods). Draw indifference curves and budget lines showing a price *fall* for an inferior good that is not a Giffen good. Label the substitution and income effects.

Exercise 20.8 (Elasticity and revenue). The demand for a museum's tickets is $Q = 5,000 - 200P$. The museum currently charges 15. Compute the elasticity. Should the museum raise or lower its price to increase ticket revenue? What price maximizes ticket revenue?

Exercise 20.9 (Income elasticity). A household's food spending rises from 400 to 440 per month when its income rises from 2,000 to 2,400. Prices are unchanged. Compute the income elasticity of food demand (use percentage changes relative to the initial values). Is food a necessity or a luxury for this household? What happens to the food budget share?

Exercise 20.10 (Market demand). There are 100 consumers of type 1, each with demand $q = 20 - 2P$, and 50 of type 2, each with $q = 15 - P$. Find market demand and the quantity demanded at $P = 8$ and $P = 12$.

Exercise 20.11 (Revealed preference). At prices (2, 2) and income 40 a consumer chooses (12, 8). At prices (1, 3) and income 36 she chooses (15, 7). Are these choices consistent with WARP?

Exercise 20.12 (Labour supply). With the preferences of Example 20.17 ($a = 0.5$, $T = 16$), suppose the government introduces a benefit of $V = 120$ that is withdrawn completely if the person works any hours at all. For $w = 20$, compare utility from working the optimal hours (with $V = 0$) with utility from not working and receiving the benefit. What does this imply about the incentive effects of such benefits?

Exercise 20.13 (Interest rates and saving). A person earns $y_1 = 100$, $y_2 = 0$, has $u(c) = \ln c$ and $\beta = 0.95$. How much does she save at $r = 0.02$ and at $r = 0.08$? Explain the result using income and substitution effects.

Exercise 20.14 (Consumer surplus). Market demand for a concert is $P = 120 - 0.02Q$. Tickets are priced at 60. Compute consumer surplus. The promoter raises the price to 80. Compute the change in consumer surplus and decompose it into the loss on tickets still bought and the loss from tickets no longer bought.

Solutions

Solution 20.1. (a) $MU_x = 2xy$, $MU_y = x^2$; $MRS = 2y/x$. (b) $MU_x = 2/x$, $MU_y = 1/y$; $MRS = 2y/x$. (c) $MRS = 3/2$. (d) $MRS = 1/(2\sqrt{x})$. Preferences (a) and (b) are the same: (b) is $\ln$ of (a), an increasing transformation, and the MRS is identical.

Solution 20.2. $x^* = 0.3 \times 200/3 = 20$, $y^* = 0.7 \times 200/7 = 20$. $MRS = (0.3/0.7)(y/x) = 0.3/0.7 = 0.4286$; price ratio $= 3/7 = 0.4286$. The tangency condition holds.

Solution 20.3. She consumes where $y = 2x$. Budget: $2x + 0.25(2x) = 50$, so $2.5x = 50$, $x = 20$ cups and 40 cubes. Demand: $x = m/(p_x + 2p_y)$. A rise in the price of sugar reduces coffee demand: the goods are gross complements. With perfect complements, there is no substitution effect; the whole response is an income effect on the “composite good” coffee-with-sugar.

Solution 20.4. Utility per unit of money: A gives $1.2/3 = 0.4$ (measuring utility in kg-of-B equivalents, 1 kg of A = 1.2); B gives $1/2.4 = 0.4167$. B is better value: she buys $60/2.4 = 25$ kg of B. She would switch to A when $1.2/p_A > 1/2.4$, i.e. $p_A < 2.88$.

Solution 20.5. $MRS = 20/x = p_x$, so $x^* = 20/p_x$, requiring $m \geq p_x x^* = 20$. If $m < 20$, the consumer spends all income on x : $x^* = m/p_x$ and $y^* = 0$. Above $m = 20$, extra income goes entirely to y ; demand for x does not change with income (income elasticity zero).

Solution 20.6. To afford the original bundle (50, 50) at new prices requires $2 \times 50 + 50 = 150$. With income 150 at $p_x = 2$, $x = 0.5 \times 150/2 = 37.5$. Slutsky substitution effect: $37.5 - 50 = -12.5$; income effect: $25 - 37.5 = -12.5$. The Slutsky compensation (50) is larger than the Hicks compensation (41.42), because being able to buy the old bundle leaves the consumer slightly better off than before (she can re-optimize); so the Slutsky substitution effect is smaller in absolute value. For small price changes the two versions converge.

Solution 20.7. When the price of the inferior good falls, the budget line pivots outward. The substitution effect (along the original indifference curve, to the flatter price line) increases quantity. The income effect (a parallel shift outward) *reduces* quantity because the good is inferior. If the income effect is smaller than the substitution effect, the net effect is still an increase: the good is not Giffen. The diagram should show the final point to the right of the original point but to the left of the compensated point.

Solution 20.8. At $P = 15$, $Q = 2,000$; $\epsilon = -200 \times 15/2,000 = -1.5$. Demand is elastic, so lowering the price raises revenue. Revenue $R = P(5,000 - 200P)$ is maximized where $R' = 5,000 - 400P = 0$, $P = 12.50$ (the midpoint, where $\epsilon = -1$), giving $Q = 2,500$ and $R = 31,250$, against 30,000 at $P = 15$.

Solution 20.9. Food spending rises by 10% and income by 20%. With unchanged prices, food quantity rises by 10%, so the income elasticity is $10/20 = 0.5$: food is a necessity. The budget share falls from $400/2,000 = 20\%$ to $440/2,400 = 18.3\%$, consistent with Engel’s law.

Solution 20.10. Type-1 total: $100(20 - 2P) = 2,000 - 200P$ for $P \leq 10$. Type-2 total: $50(15 - P) = 750 - 50P$ for $P \leq 15$. Market demand: $Q = 2,750 - 250P$ for $P \leq 10$; $Q = 750 - 50P$ for $10 < P \leq 15$; 0 above 15. At $P = 8$: $Q = 750$. At $P = 12$: $Q = 150$.

Solution 20.11. Cost of (15, 7) at the first prices: $30 + 14 = 44 > 40$, not affordable; so (12, 8) is not revealed preferred to (15, 7). Cost of (12, 8) at the second prices: $12 + 24 = 36 \leq 36$, affordable; so (15, 7) is revealed preferred to (12, 8). There is no violation: the second revealed preference is not contradicted by the first choice.

Solution 20.12. Working with $V = 0$: $h^* = 8$, $c = 160$, $\ell = 8$; $U = 0.5 \ln 160 + 0.5 \ln 8 = 0.5(5.075) + 0.5(2.079) = 3.577$. Not working with the benefit: $c = 120$, $\ell = 16$; $U = 0.5 \ln 120 + 0.5 \ln 16 = 0.5(4.787) + 0.5(2.773) = 3.780$. The person is better off not working. A benefit withdrawn in full on taking any work imposes an effective marginal tax rate above 100% on the first hours of work, creating a strong disincentive. Tapered withdrawal schemes and in-work credits (such as the U.S. Earned Income Tax Credit) are designed to reduce this problem.

Solution 20.13. With $y_2 = 0$, $W = 100$ whatever r , and $c_1 = W/(1 + \beta) = 100/1.95 = 51.28$: saving is 48.72 at both interest rates. For a pure lender with log utility, the substitution effect of a higher r (save more) is exactly offset by the income effect (she is richer, so she consumes more today). Only c_2 responds: $c_2 = \beta(1 + r)c_1$ rises from 49.69 to 52.62.

Solution 20.14. At $P = 60$, $Q = (120 - 60)/0.02 = 3,000$. $CS = \frac{1}{2} \times 3,000 \times 60 = 90,000$. At $P = 80$, $Q = 2,000$, $CS = \frac{1}{2} \times 2,000 \times 40 = 40,000$. Change: $-50,000$. Loss on tickets still bought: $2,000 \times 20 = 40,000$. Loss on tickets no longer bought: $\frac{1}{2} \times 1,000 \times 20 = 10,000$. Total 50,000.

Technology, Costs and Supply

After studying this chapter you should be able to

1. Describe a firm's technology with a production function and distinguish the short run from the long run.
2. Compute total, average and marginal products and explain diminishing marginal returns.
3. Use isoquants, the marginal rate of technical substitution and returns to scale to describe long-run technology.
4. Solve the cost-minimization problem and derive conditional factor demands and the cost function.
5. Derive short-run cost curves and prove the relationships between marginal and average cost.
6. Explain economies of scale and scope, minimum efficient scale and learning curves.
7. Derive the competitive firm's supply curve, including the shutdown condition.
8. Explain long-run entry and exit and derive long-run industry supply.
9. Measure producer surplus and distinguish economic from accounting costs.

A motivating question

Between 2010 and 2020 the price of solar photovoltaic modules fell by roughly 90 per cent, and by the end of the decade manufacturers in China produced the large majority of the world's modules. Why did costs fall so far? Part of the answer is technological progress; part is that the cost of producing each panel falls as a firm, and an industry, produces more—through larger plants and through accumulated experience. Whether larger firms have lower costs, and why, determines how many firms an industry can support, how competitive it will be, and how prices respond to demand.

This chapter builds the theory of the firm's technology and costs and derives the supply behaviour of a competitive firm and industry. Chapter 22 combines supply with the demand theory of Chapter 20 to analyse markets.

21.1 The Firm and Its Objective

A **firm** is an organization that transforms inputs into outputs. Why do firms exist at all, rather than every task being coordinated through market contracts between individuals? Ronald Coase (1937) answered that using markets is costly: finding trading partners, negotiating, writing and enforcing contracts all take resources. Firms replace many market transactions with internal direction by managers. The boundary of the firm lies where the cost of organizing one more transaction internally equals the cost of using the market. This *transaction-cost* view

recurs in Chapters 25 and 13.

For most of this chapter we assume that a firm aims to maximize *economic* profit: revenue minus all opportunity costs, including the normal return that the owners could earn by investing their capital elsewhere (Chapter 1). The assumption is a simplification—managers may pursue other goals, a problem studied in Chapter 25—but it describes the pressure that competition and owners place on firms, and it generates sharp, testable predictions.

21.2 Production Functions

21.2.1 Technology

Definition 21.1. Production function

A **production function** $q = F(K, L)$ gives the maximum output q obtainable from capital K and labour L with the firm's current technology.

The word “maximum” matters: the production function describes efficient use of inputs. A firm that wastes inputs produces below its production function. Both inputs and output are flows: machine-hours and worker-hours per period, units of output per period.

21.2.2 Short run and long run

Definition 21.2. Short run and long run

The **short run** is a period in which at least one input is fixed. The **long run** is a period long enough for all inputs to be varied.

These are not calendar periods. For a restaurant, the short run might be a few weeks (it cannot quickly add floor space); for a steel mill, several years; for a semiconductor fabrication plant, even longer. Conventionally, capital is taken as fixed in the short run and labour as variable, though in some economies labour is harder to adjust than capital because of employment protection.

21.2.3 Total, average and marginal product

With capital fixed at $\bar{K}$, output depends on labour alone: $q = F(\bar{K}, L) = f(L)$.

Definition 21.3. Average and marginal product

The **average product of labour** is $AP_L = q/L$. The **marginal product of labour** is $MP_L = \partial q / \partial L$: the additional output from one more unit of labour, holding capital fixed.

The law of diminishing marginal returns. As more of a variable input is combined with a fixed input, the marginal product of the variable input eventually declines.

This is an empirical regularity, not a theorem. It holds because the fixed input becomes a bottleneck: adding cooks to a kitchen of fixed size eventually produces crowding rather than meals. Marginal product may rise at first, as specialization becomes possible (one cook chops, another fries), before diminishing returns set in.

Proposition 21.4. Marginal and average product

When $MP_L > AP_L$, average product is rising in L ; when $MP_L < AP_L$, it is falling. Therefore $MP_L = AP_L$ where AP_L reaches its maximum.

Proof. $\frac{d}{dL} \left(\frac{f(L)}{L} \right) = \frac{f'(L)L - f(L)}{L^2} = \frac{1}{L} (MP_L - AP_L)$. The sign of the derivative is the sign of $MP_L - AP_L$; at an interior maximum the derivative is zero. $\square$

Worked Example 21.5. Products of labour

Let $q = 10L^2 - 0.5L^3$ with capital fixed (L in worker-days). Then $MP_L = 20L - 1.5L^2$ and $AP_L = 10L - 0.5L^2$. Marginal product peaks where $20 - 3L = 0$, $L = 6.67$. Average product peaks where $10 - L = 0$, $L = 10$; there $AP_L = 50$ and $MP_L = 200 - 150 = 50$, equal as the proposition requires. Total product peaks where $MP_L = 0$: $L = 13.33$. No profit-maximizing firm would hire beyond that point, since extra workers would reduce output.

21.3 Long-Run Technology

21.3.1 Isoquants

Definition 21.6. Isoquant

An **isoquant** is the set of input combinations (K, L) that produce the same output. It is a level curve of the production function.

Isoquants play the role in production that indifference curves play in consumption, with one difference: output is measurable (cardinal), so the numbers attached to isoquants mean something. Isoquants slope downward (if one input is reduced, the other must increase to maintain output) and are typically convex to the origin.

Definition 21.7. Marginal rate of technical substitution

The **marginal rate of technical substitution**, $MRTS_{LK}$, is the rate at which capital can be reduced when labour is increased by one unit, holding output constant. It is the absolute slope of the isoquant:

$$MRTS_{LK} = \frac{MP_L}{MP_K}.$$

The formula follows from the implicit function rule (Chapter 5) exactly as the MRS did. A *diminishing* MRTS—convex isoquants—means that as a firm substitutes labour for capital, it becomes progressively harder to replace further capital with labour.

21.3.2 Returns to scale

Definition 21.8. Returns to scale

A technology has **constant returns to scale** if $F(tK, tL) = tF(K, L)$ for all $t > 0$; **increasing returns to scale** if $F(tK, tL) > tF(K, L)$ for $t > 1$; and **decreasing returns to scale** if $F(tK, tL) < tF(K, L)$ for $t > 1$.

For the Cobb–Douglas production function $q = AK^\alpha L^\beta$, returns to scale are constant, increasing or decreasing as $\alpha + \beta$ equals, exceeds or falls short of 1 (the homogeneity of Chapter 5).

Returns to scale should not be confused with diminishing marginal returns. Diminishing returns concern increasing *one* input with others fixed; returns to scale concern increasing *all* inputs in proportion. A Cobb–Douglas technology with $\alpha = \beta = 0.5$ has diminishing marginal product of each input and constant returns to scale.

Sources of increasing returns include specialization of labour (Adam Smith’s pin factory), indivisibilities (a railway line or a semiconductor fabrication plant cannot be built at half size), and geometric relationships (a pipeline’s capacity rises with its cross-sectional area, roughly the square of its radius, while its material cost rises with its circumference). Decreasing returns often reflect the difficulty of managing and coordinating very large organizations, or a fixed factor that has been left out of the production function (the entrepreneur’s time, a particular location).

21.3.3 Substitutability and technical change

The ease of substituting between inputs varies. At one extreme are **fixed-proportions** (Leontief) technologies, $q = \min(aK, bL)$, with L-shaped isoquants: one driver per bus. At the other are **perfect-substitute** technologies, $q = aK + bL$, with straight-line isoquants. Cobb–Douglas lies in between. The **elasticity of substitution**—the percentage change in K/L in response to a 1% change in the MRTS—is zero for Leontief, infinite for perfect substitutes, and exactly 1 for Cobb–Douglas.

Technical progress shifts isoquants toward the origin: the same output can be produced with fewer inputs. In $q = AK^\alpha L^{1-\alpha}$, an increase in the parameter A , called **total factor productivity**, raises output from any given inputs. Chapter 44 shows that growth in total factor productivity accounts for much of the long-run growth of living standards.

21.4 Cost Minimization

21.4.1 Isocost lines

A firm that hires labour at wage w and capital at rental rate r (the cost per period of using a unit of capital—explained in Section 21.9) spends

$$C = wL + rK.$$

An **isocost line** is the set of (K, L) with the same cost: $K = C/r - (w/r)L$, with slope $-w/r$.

21.4.2 The tangency condition

To produce a given output q at minimum cost, the firm chooses the point on the q isoquant that lies on the lowest isocost line. At an interior solution, the isocost line is tangent to the isoquant:

$$MRTS_{LK} = \frac{MP_L}{MP_K} = \frac{w}{r}, \quad \text{equivalently} \quad \frac{MP_L}{w} = \frac{MP_K}{r}.$$

The last unit of money spent on each input must yield the same extra output. If an extra unit of money spent on labour yielded more output than one spent on capital, the firm could shift spending from capital to labour and produce the same output more cheaply.

21.4.3 The Lagrangian and the meaning of the multiplier

The problem is

$$\min_{K,L} wL + rK \quad \text{subject to} \quad F(K, L) = q.$$

With $\mathcal{L} = wL + rK + \lambda[q - F(K, L)]$, the first-order conditions are $w = \lambda F_L$, $r = \lambda F_K$ and $F(K, L) = q$. Dividing the first two gives the tangency condition. The multiplier λ is, by Theorem 6.4 of Chapter 6, the increase in minimum cost when the output requirement rises by one unit: it is **marginal cost**.

Definition 21.9. Conditional factor demands and the cost function

The solution gives the **conditional factor demands** $L^c(w, r, q)$ and $K^c(w, r, q)$ —the cost-minimizing input quantities, conditional on producing q . The **cost function** $C(w, r, q) = wL^c + rK^c$ gives the minimum cost of producing q .

Worked Example 21.10. Cost function for Cobb–Douglas technology

Let $q = K^{1/2}L^{1/2}$. Then $MP_L = \frac{1}{2}K^{1/2}L^{-1/2}$, $MP_K = \frac{1}{2}K^{-1/2}L^{1/2}$, and $MRTS = K/L$. Tangency: $K/L = w/r$, so $K = (w/r)L$. Substituting into the output constraint: $q = L\sqrt{w/r}$, so

$$L^c = q\sqrt{\frac{r}{w}}, \quad K^c = q\sqrt{\frac{w}{r}}, \quad C(w, r, q) = 2q\sqrt{wr}.$$

With $w = 16$, $r = 4$ and $q = 100$: $L^c = 100 \times 0.5 = 50$, $K^c = 100 \times 2 = 200$, and $C = 16 \times 50 + 4 \times 200 = 1,600 = 2 \times 100 \times \sqrt{64}$. Marginal cost is $\partial C / \partial q = 2\sqrt{wr} = 16$, constant because returns to scale are constant. From the first-order condition, $\lambda = w / MP_L = 16 / (0.5 \times \sqrt{200/50}) = 16/1 = 16$, confirming that the multiplier is marginal cost.

Because labour is expensive relative to capital ($w/r = 4$), the firm uses four times as much capital as labour. If wages doubled, the cost-minimizing capital-labour ratio would double: firms substitute toward the relatively cheaper input. This substitution is one reason why labour-saving automation is more advanced in high-wage economies such as Germany and why Chinese manufacturers have automated rapidly as Chinese wages have risen.

Advanced Analysis: Properties of the cost function

The cost function is (i) non-decreasing in input prices and output; (ii) homogeneous of degree 1 in input prices (doubling all input prices doubles cost, without changing the input mix); (iii) concave in input prices; and (iv) satisfies **Shephard's lemma** in production: $\partial C / \partial w = L^c(w, r, q)$. The last follows from the envelope theorem, exactly as in consumer theory. In Example 21.10, $\partial C / \partial w = q\sqrt{r/w} = L^c$. Concavity says that when an input price rises, cost rises by less than it would if the firm could not substitute; the curvature measures the value of flexibility.

21.5 Short-Run Costs

21.5.1 Fixed and variable costs

In the short run, some costs do not vary with output.

Definition 21.11. Fixed, variable and total cost

Fixed cost (FC) does not vary with output in the short run (rent on a lease, the cost of the fixed capital). **Variable cost** (VC) varies with output (wages of hourly workers, materials, energy). **Total cost** is $TC(q) = FC + VC(q)$.

From these we define

$$AFC = \frac{FC}{q}, \quad AVC = \frac{VC(q)}{q}, \quad ATC = \frac{TC(q)}{q} = AFC + AVC, \quad MC = \frac{dTC}{dq} = \frac{dVC}{dq}.$$

Marginal cost does not depend on fixed cost, since fixed cost does not change with output.

► Common error: fixed cost versus sunk cost

A fixed cost does not vary with output but may be avoidable by shutting down entirely (a lease that can be terminated). A sunk cost cannot be recovered at all (Chapter 1). In the short-run analysis below, we treat fixed costs as sunk: they must be paid even if output is zero. The distinction matters for entry decisions and for the theory of market power (Chapter 24).

21.5.2 The shapes of cost curves

Short-run marginal cost eventually rises because of diminishing marginal returns: with capital fixed, each extra unit of output requires more and more labour. Formally, if labour is the only variable input, $VC = wL$, so $MC = w dL/dq = w/MP_L$. When MP_L falls, MC rises. Likewise $AVC = wL/q = w/AP_L$: average variable cost is lowest where average product is highest.

Proposition 21.12. Marginal and average cost

Marginal cost cuts average variable cost and average total cost at their minimum points. When $MC < ATC$, ATC is falling; when $MC > ATC$, ATC is rising. The same holds for AVC.

Proof. $\frac{d}{dq} \left(\frac{TC(q)}{q} \right) = \frac{TC'(q)q - TC(q)}{q^2} = \frac{1}{q}(MC - ATC)$. The same argument applies to $VC(q)/q$. □

Worked Example 21.13. A complete set of cost curves

Let $FC = 200$ and $VC(q) = 20q - 2q^2 + 0.1q^3$. Then

$$MC = 20 - 4q + 0.3q^2, \quad AVC = 20 - 2q + 0.1q^2, \quad ATC = \frac{200}{q} + 20 - 2q + 0.1q^2.$$

Minimum AVC: $dAVC/dq = -2 + 0.2q = 0$ at $q = 10$, where $AVC = 20 - 20 + 10 = 10$.
Check: $MC(10) = 20 - 40 + 30 = 10$. *Minimum ATC:* $dATC/dq = -200/q^2 - 2 + 0.2q = 0$, i.e. $q^3 - 10q^2 - 1,000 = 0$, whose root is $q \approx 14.66$, where $ATC \approx 25.81$ and $MC \approx 25.81$.
Minimum MC: $-4 + 0.6q = 0$ at $q = 6.67$, where $MC = 6.67$. Figure 21.1 plots the curves.

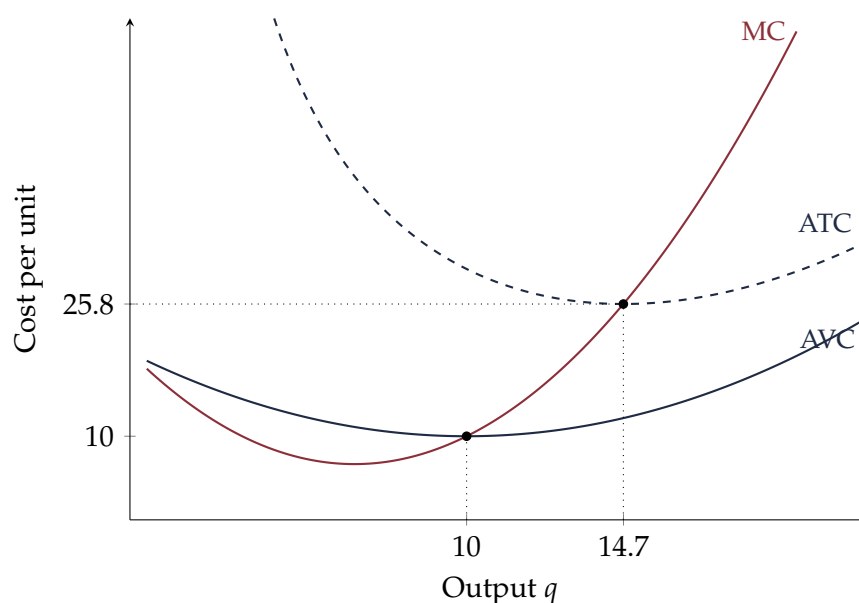


Figure 21.1: Short-run cost curves for Example 21.13. Marginal cost passes through the minima of average variable cost (at $q = 10$) and average total cost (at $q \approx 14.7$).

21.6 Long-Run Costs

21.6.1 The long-run average cost curve

In the long run, the firm can choose its plant size. For each output level, it picks the plant (the level of capital) that minimizes cost. The **long-run average cost** (LRAC) curve therefore lies on or below every short-run average cost curve: it is their lower *envelope*. At the output for which a particular plant is optimal, the short-run and long-run average cost curves touch.

21.6.2 Economies and diseconomies of scale

Definition 21.14. Economies of scale

A firm experiences **economies of scale** over a range of output if long-run average cost falls as output rises, and **diseconomies of scale** if it rises. The output at which LRAC first reaches its minimum is the **minimum efficient scale** (MES).

Economies of scale are closely related to returns to scale: with fixed input prices, increasing returns to scale imply falling LRAC. But the concepts differ when input prices change with the firm's size (a large firm may obtain discounts on materials) or when the firm does not expand all inputs in proportion.

The MES relative to the size of the market largely determines how many firms an industry can support. If the MES of a car plant is several hundred thousand vehicles a year, a country whose market buys a million cars can support only a few efficient plants—which is why car-making tends to be concentrated, and why Malaysia's national car makers, serving a small domestic market, relied on tariff protection and partnerships with foreign manufacturers to reach viable scale. If the MES of a hair salon is a handful of chairs, a city supports thousands of salons.

21.6.3 Economies of scope

Economies of scope exist when producing two products together is cheaper than producing them separately: $C(q_1, q_2) < C(q_1, 0) + C(0, q_2)$. They arise from shared inputs—an airline carrying passengers and cargo on the same flight, a bank using the same branch network for deposits and loans, a technology firm using the same data centres for several services. Economies of scope help explain why firms diversify and why large platforms span many markets (Chapter 24).

21.6.4 Learning curves

Unit costs often fall with *cumulative* output, as workers and managers learn. A common empirical form is

$$c(Q) = c_0 Q^{-b},$$

where Q is cumulative production. Each doubling of cumulative output multiplies unit cost by 2^{-b} ; the percentage fall, $1 - 2^{-b}$, is the **learning rate**. Learning curves were documented in aircraft production in the 1930s and have been measured for semiconductors, chemicals, batteries and solar panels. Learning differs from economies of scale: scale economies depend on the current rate of output, learning on accumulated experience. The distinction matters for strategy—a firm that expects strong learning may price aggressively early on to accumulate experience—and for industrial policy.

21.7 Profit Maximization and Supply in Competitive Markets

21.7.1 Price-taking

A **perfectly competitive** market has many firms selling an identical product, many buyers, full information about prices, and free entry and exit. Each firm is so small relative to the market that it cannot affect the price: it is a **price taker**, facing a horizontal demand curve at the market price P . Chapter 22 discusses how closely real markets fit these assumptions; wheat, many commodity metals and foreign exchange come close.

21.7.2 The short-run supply decision

A price-taking firm's revenue is Pq , so marginal revenue equals P . Profit $\Pi(q) = Pq - TC(q)$ is maximized where

$$P = MC(q), \quad \text{with } MC \text{ rising (the second-order condition).}$$

But the firm also has the option of producing nothing. In the short run, with fixed costs sunk, shutting down gives profit $-FC$. Producing q gives $Pq - VC(q) - FC$. The firm should produce if $Pq - VC(q) \geq 0$, i.e. if price covers average variable cost.

Proposition 21.15. Short-run supply of a competitive firm

A price-taking firm with sunk fixed costs supplies the quantity at which $P = MC$ on the rising part of the MC curve, provided $P \geq \min AVC$. If $P < \min AVC$, it shuts down. Its short-run supply curve is therefore its marginal cost curve above minimum average variable cost.

The price at the minimum of AVC is the **shutdown price**. Between the shutdown price and the minimum of ATC, the firm makes a loss but continues to produce, because revenue covers variable cost and contributes something toward the fixed cost it must pay anyway.

Worked Example 21.16. Supply decisions

Use the cost functions of Example 21.13. The shutdown price is $\min AVC = 10$.

- At $P = 40$: $P = MC$ gives $0.3q^2 - 4q - 20 = 0$, so $q = (4 + \sqrt{16 + 24})/0.6 \approx 17.21$. Revenue ≈ 688.3 ; total cost ≈ 461.5 ; profit ≈ 226.8 .
- At $P = 20$: $0.3q^2 - 4q = 0$, so $q = 13.33$. Revenue 266.67; variable cost 148.15; total cost 348.15; profit -81.48 . The firm makes a loss but produces, because shutting down would lose the full fixed cost of 200.
- At $P = 8$: below the shutdown price of 10. The firm produces nothing and loses 200.

21.7.3 Producer surplus

Definition 21.17. Producer surplus

Producer surplus is revenue minus variable cost: $PS = Pq - VC(q)$. Graphically it is the area above the supply (MC) curve and below the price, up to the quantity supplied.

Producer surplus differs from profit by fixed cost: $\Pi = PS - FC$. In the short run, when fixed costs are sunk, producer surplus measures how much better off the firm is from producing than from shutting down. Chapter 22 uses producer surplus alongside consumer surplus to evaluate market outcomes.

21.7.4 Short-run market supply

With a fixed number of firms, short-run market supply is the horizontal sum of the firms' supply curves, exactly as market demand is the horizontal sum of individual demands.

21.8 The Long Run: Entry and Exit

21.8.1 Long-run equilibrium

In the long run, firms can enter or leave the industry, and every cost is variable. If the market price exceeds the minimum of long-run average cost, firms earn positive economic profit, attracting entry; entry increases supply and lowers the price. If the price is below minimum LRAC, firms make losses and some exit, reducing supply and raising the price.

Proposition 21.18. Long-run competitive equilibrium

With free entry and exit and identical firms, long-run equilibrium requires $P = MC = \min LRAC$: each firm produces at minimum efficient scale and earns zero economic profit.

Zero *economic* profit does not mean zero *accounting* profit. It means that the owners earn exactly the normal return on their capital—what they could earn elsewhere—so that nobody

wants to enter or leave. Positive economic profits are the signal that draws resources into an industry; losses push them out.

21.8.2 Long-run industry supply

The long-run industry supply curve shows how the equilibrium price varies as demand changes, after entry and exit have occurred.

- In a **constant-cost industry**, input prices do not change as the industry expands. New firms enter with the same cost curves, and the long-run supply curve is horizontal at $\min LRAC$. Increases in demand raise output but not the long-run price.
- In an **increasing-cost industry**, expansion bids up the price of some input in limited supply (skilled labour, suitable land, a mineral). Each firm's cost curves shift up as the industry grows, and long-run supply slopes upward.
- In a **decreasing-cost industry**, expansion lowers input prices—for example, because a larger industry supports specialized suppliers, or because industry-wide learning spills over between firms. Long-run supply slopes downward.

Industries with strong external economies of this kind—Silicon Valley, the Stuttgart automotive cluster, the electronics cluster around Penang in Malaysia, or the concentration of solar and battery manufacturing in parts of China—show why production of particular goods often concentrates geographically.

21.9 Economic Costs and Accounting Costs

21.9.1 The user cost of capital

The rental rate r used above is the cost per period of using a unit of capital. When a firm rents equipment, r is simply the rent. When it owns the equipment, r is an *implicit* cost with three parts. Suppose a machine costs P_K , depreciates at rate δ per year, and could otherwise be sold and the money invested at interest rate i . Then the annual cost of using it is approximately

$$r = P_K (i + \delta - g_{P_K}),$$

where g_{P_K} is the rate at which the price of the machine is rising (a capital gain that offsets part of the cost). This is the **user cost of capital**. It reappears in Chapter 45 as a determinant of investment: when the central bank raises interest rates, i rises, the user cost rises, and the optimal capital stock falls.

21.9.2 How accounting measures differ

An accounting income statement (Chapter 10) includes depreciation, which roughly corresponds to $P_K\delta$, and interest paid on debt. It does *not* charge a cost for the equity capital the owners have invested—the forgone return on the owners' money. That is why accounting profit exceeds economic profit, and why an industry in long-run competitive equilibrium, with zero economic profit, reports positive accounting profit equal to the normal return on equity. Accounting depreciation also follows rules (straight-line over an estimated useful life, for instance) that may differ from economic depreciation. Chapter 29 examines these rules, and Chapter 31 shows how analysts adjust reported profits toward an economic measure.

Case: Solar modules: scale, learning and the location of production

Solar photovoltaic modules are a nearly standardized product, sold in a global market where price competition is intense. Their cost fell by roughly 90 per cent over the 2010s. Several mechanisms from this chapter were at work together:

- *Economies of scale*: polysilicon refining, wafer slicing and cell manufacturing involve large indivisible investments; plants producing tens of gigawatts a year have lower average costs than small ones.
- *Learning*: studies of the industry have typically estimated that module costs fell by about 20 per cent for each doubling of cumulative global production over long periods—a learning rate that implies large cost reductions as the industry grew exponentially.
- *Technical progress*: higher cell efficiencies raise output per unit of material, shifting the production function.
- *Input prices and industrial policy*: Chinese producers benefited from low-cost electricity for energy-intensive polysilicon refining, dense supplier networks (external economies), and state support in the form of cheap credit and local-government incentives.

The result was a decreasing-cost industry at the global level, with production concentrated where the scale and cluster advantages were greatest. Falling prices also drove repeated rounds of exit: many European and American manufacturers, and some Chinese ones, left the industry when prices fell below their average variable cost. Solar illustrates that the long-run supply curve of an industry can slope downward, and that the location of an industry reflects not only comparative advantage (Chapter 1) but also scale, learning and policy.

21.10 Limitations

The competitive model treats technology as given and firms as price takers. In many industries, firms are few, products are differentiated and prices are set strategically; Chapters 23 and 24 analyse such markets. The model also treats the firm as a single decision maker with a profit objective. Real firms are organizations whose managers, employees and owners have different information and interests; Chapter 25 examines how contracts and incentives try to align them. Finally, cost curves are rarely observed directly; they must be estimated from data that mix differences in technology, input prices, scale and product quality, and such estimates face the causal-inference problems of Chapter 17.

Summary

- A production function gives maximum output from inputs. In the short run some inputs are fixed; in the long run all are variable.
- Marginal product eventually diminishes when one input is increased with others fixed. Marginal product equals average product at the maximum of average product.
- Isoquants have slope $-MRTS = -MP_L/MP_K$. Returns to scale describe the effect of scaling all inputs.

- Cost minimization equates $MRTS$ with w/r ; the Lagrange multiplier is marginal cost. The cost function summarizes the minimum cost of each output.
- Short-run marginal cost rises because of diminishing returns and cuts AVC and ATC at their minima.
- $LRAC$ is the envelope of short-run average cost curves; economies of scale, scope and learning lower costs as output grows.
- A competitive firm supplies where $P = MC$ if $P \geq \min AVC$, and shuts down otherwise.
- Free entry drives long-run price to $\min LRAC$ and economic profit to zero. Long-run industry supply may be flat, rising or falling.
- Producer surplus is revenue minus variable cost. The user cost of capital includes interest, depreciation and capital gains; accounting profit omits the cost of equity.

Key terms

firm; transaction costs; production function; short run; long run; average product; marginal product; diminishing marginal returns; isoquant; $MRTS$; constant, increasing and decreasing returns to scale; Leontief technology; elasticity of substitution; technical progress; total factor productivity; isocost line; conditional factor demand; cost function; Shephard's lemma; fixed cost; variable cost; AFC ; AVC ; ATC ; marginal cost; $LRAC$; economies of scale; minimum efficient scale; economies of scope; learning curve; perfect competition; price taker; shutdown price; producer surplus; long-run competitive equilibrium; constant-, increasing- and decreasing-cost industries; user cost of capital.

Exercises

Exercise 21.1 (Products). With capital fixed, $q = 6L^2 - 0.2L^3$. Find MP_L and AP_L , the value of L at which AP_L is maximized, and verify that $MP_L = AP_L$ there.

Exercise 21.2 (Returns to scale). Determine the returns to scale of: (a) $q = K^{0.4}L^{0.4}$; (b) $q = 3K + 2L$; (c) $q = \min(K, 2L)$; (d) $q = K^{0.5}L^{0.6}$; (e) $q = KL$.

Exercise 21.3 ($MRTS$). For $q = K^{0.25}L^{0.75}$, compute the $MRTS$ at $K = 16$, $L = 16$ and at $K = 16$, $L = 81$. Interpret the difference.

Exercise 21.4 (Cost minimization). For $q = K^{0.25}L^{0.75}$ with $w = 6$ and $r = 2$, find the cost-minimizing input combination for $q = 100$ and the resulting cost.

Exercise 21.5 (Cost function). Show that for $q = K^{1/2}L^{1/2}$, a 10% rise in both w and r raises the cost of every output by exactly 10% and leaves the cost-minimizing capital-labour ratio unchanged. Which property of cost functions does this illustrate?

Exercise 21.6 (Short-run costs). A firm has $TC = 50 + 4q + q^2$. Find FC , VC , AVC , ATC and MC . Find the output that minimizes ATC and verify that $MC = ATC$ there.

Exercise 21.7 (Supply curve). For the firm in Exercise 21.6, find the short-run supply curve and the shutdown price. How much does it supply at $P = 24$, and what is its profit?

Exercise 21.8 (Long-run equilibrium). Every firm in a competitive industry has the cost function of Exercise 21.6 in the long run (with the fixed cost of 50 avoidable by exit). Market demand is $Q = 1,000 - 20P$. Find the long-run equilibrium price, output per firm, market quantity and number of firms.

Exercise 21.9 (Learning curve). A battery maker's unit cost follows $c = 500Q^{-0.32}$, where Q is cumulative output in thousands. What is the learning rate? By what factor does cost fall between cumulative output of 1 and 64 (thousand)?

Exercise 21.10 (Economies of scope). A firm's cost of producing q_1 and q_2 together is $C = 100 + 2q_1 + 3q_2$. Producing either alone costs $C_1 = 80 + 2q_1$ or $C_2 = 70 + 3q_2$. Are there economies of scope at $q_1 = q_2 = 10$? Explain their source.

Exercise 21.11 (User cost). A machine costs 50,000, depreciates at 12% per year, and its price is expected to be stable. The interest rate is 5%. What is the annual user cost? If the central bank raises rates so that the firm's interest rate becomes 7%, by what percentage does the user cost rise?

Exercise 21.12 (Accounting and economic profit). A competitive industry is in long-run equilibrium. A typical firm has equity of 10 million invested by its owners, who could earn 8% per year elsewhere with similar risk. It has no debt. What accounting profit (before tax) would you expect it to report? Explain.

Solutions

Solution 21.1. $MP_L = 12L - 0.6L^2$; $AP_L = 6L - 0.2L^2$. AP_L is maximized where $6 - 0.4L = 0$, $L = 15$: $AP_L = 90 - 45 = 45$, $MP_L = 180 - 135 = 45$. Equal, as required.

Solution 21.2. (a) Degree 0.8: decreasing. (b) $3tK + 2tL = t(3K + 2L)$: constant. (c) $\min(tK, 2tL) = t \min(K, 2L)$: constant. (d) Degree 1.1: increasing. (e) Degree 2: increasing.

Solution 21.3. $MRTS = MP_L/MP_K = (0.75/0.25)(K/L) = 3K/L$. At (16, 16): 3. At (16, 81): $48/81 \approx 0.59$. With much more labour relative to capital, an extra worker substitutes for much less capital: diminishing MRTS.

Solution 21.4. Tangency: $3K/L = w/r = 3$, so $K = L$. Then $q = L = 100$, $K = 100$. Cost = $6 \times 100 + 2 \times 100 = 800$.

Solution 21.5. $C = 2q\sqrt{wr}$. With $1.1w$ and $1.1r$, $C = 2q\sqrt{1.21wr} = 1.1 \times 2q\sqrt{wr}$. The capital-labour ratio is $K/L = w/r$, unchanged by equal proportional increases. This illustrates homogeneity of degree 1 in input prices: only relative input prices affect the input mix.

Solution 21.6. $FC = 50$; $VC = 4q + q^2$; $AVC = 4 + q$; $ATC = 50/q + 4 + q$; $MC = 4 + 2q$. Min ATC: $-50/q^2 + 1 = 0$, $q = \sqrt{50} \approx 7.07$. There $ATC = 7.07 + 4 + 7.07 = 18.14$ and $MC = 4 + 14.14 = 18.14$.

Solution 21.7. $AVC = 4 + q$ is minimized at $q = 0$, where it equals 4, so the shutdown price is 4. Supply: $P = 4 + 2q$, i.e. $q = (P - 4)/2$ for $P \geq 4$, zero otherwise. At $P = 24$: $q = 10$; revenue 240; cost $50 + 40 + 100 = 190$; profit 50.

Solution 21.8. Long-run price = $\min ATC = 4 + 2\sqrt{50} \approx 18.14$, with each firm producing $\sqrt{50} \approx 7.07$. Market quantity = $1,000 - 20 \times 18.14 = 637.2$. Number of firms = $637.2/7.07 \approx 90.1$. In practice the number of firms must be an integer; with 90 firms the price would be very slightly above min ATC. The approximation is standard in large markets.

Solution 21.9. Each doubling multiplies cost by $2^{-0.32} = 0.80$: a learning rate of 20%. From 1 to 64 is six doublings: cost falls by the factor $0.80^6 = 64^{-0.32} \approx 0.264$, i.e. to about 26% of its initial level (from 500 to about 132).

Solution 21.10. Joint: $100 + 20 + 30 = 150$. Separate: $(80 + 20) + (70 + 30) = 200$. Joint production saves 50: economies of scope. The source is the fixed costs: producing both goods separately requires fixed costs of $80 + 70 = 150$; together only 100, because some fixed inputs (a building, a sales force, a brand) are shared.

Solution 21.11. User cost = $50,000(0.05 + 0.12) = 8,500$ per year. At 7%: $50,000 \times 0.19 = 9,500$: an increase of $1,000/8,500 = 11.8\%$. Interest is a large part of the user cost for long-lived, slowly depreciating assets and a smaller part for rapidly depreciating ones, which is why interest-rate changes affect investment in buildings more than in computers.

Solution 21.12. Zero economic profit means that owners earn exactly their opportunity cost, 8% of 10 million, i.e. 0.8 million. Accounting profit does not deduct this cost of equity, so the firm would report accounting profit (before tax) of about 0.8 million. (Taxes complicate the picture: the owners' alternative return would also be taxed, and the comparison should be made on a consistent basis.)

Competitive Markets, Equilibrium and Welfare

After studying this chapter you should be able to

1. Solve for competitive equilibrium with linear supply and demand, and carry out comparative statics.
2. Explain how elasticities and time horizons determine the size of price and quantity responses.
3. Measure consumer, producer and total surplus, and explain why competitive equilibrium maximizes total surplus.
4. Analyse the incidence and deadweight loss of taxes and subsidies, and derive the incidence formula.
5. Analyse price ceilings, price floors and quotas, including their welfare costs.
6. Analyse tariffs and import quotas in a small open economy.
7. Solve a simple general-equilibrium exchange economy and state the two fundamental welfare theorems and their assumptions.
8. Identify the circumstances in which competitive markets fail to be efficient.

A motivating question

In 2018 the United States imposed tariffs on imports of steel, aluminium and a long list of goods from China. Supporters said the tariffs would be paid by foreign producers; critics said they would be paid by American consumers and firms. Who was right is not a matter of opinion. It depends on the elasticities of supply and demand, and the tools of this chapter can answer it. The same tools tell us who really pays a payroll tax, why rent controls produce housing shortages, why a minimum wage can reduce employment in some markets, and in what precise sense a competitive market is “efficient.”

This chapter brings together the demand theory of Chapter 20 and the supply theory of Chapter 21. It develops the analysis of market equilibrium, uses surplus to evaluate outcomes, and then examines what happens when governments intervene. It ends by stepping back from single markets to the economy as a whole.

22.1 Market Equilibrium

22.1.1 Equilibrium price and quantity

Definition 22.1. Competitive equilibrium

A **competitive equilibrium** in a single market is a price P^* at which the quantity demanded equals the quantity supplied: $Q_d(P^*) = Q_s(P^*)$.

At any other price the market is not in equilibrium. If $P > P^*$, there is **excess supply** (a surplus): sellers cannot sell all they wish, and competition among them pushes the price down. If $P < P^*$, there is **excess demand** (a shortage): buyers cannot buy all they wish, and competition among them pushes the price up. The equilibrium is therefore stable in the sense that deviations set up forces that restore it.

Worked Example 22.2. Solving for equilibrium

Let demand be $Q_d = 100 - 2P$ and supply $Q_s = -20 + 4P$. Setting them equal: $100 - 2P = -20 + 4P$, so $P^* = 20$ and $Q^* = 60$. The inverse functions are $P = 50 - Q/2$ (demand) and $P = 5 + Q/4$ (supply); the demand curve meets the vertical axis at 50 (the choke price) and the supply curve at 5 (the lowest price at which anything is supplied).

22.1.2 Shifts versus movements along curves

A change in the good's own price causes a movement *along* the demand and supply curves: a change in *quantity demanded* or *quantity supplied*. A change in anything else that affects buyers' or sellers' plans *shifts* a curve: a change in *demand* or *supply*.

Shifts demand	Shifts supply
Income (up for normal goods, down for inferior)	Input prices
Prices of substitutes and complements	Technology
Tastes and information	Number of sellers
Number of buyers	Prices of other goods the firms could produce
Expectations of future prices	Expectations of future prices
Taxes and subsidies levied on buyers	Taxes and subsidies levied on sellers; weather, disasters

22.1.3 Comparative statics

With both curves linear, we can derive general formulas. Let demand be $Q_d = a - bP + cZ$ and supply $Q_s = -d + eP$, where Z is a demand shifter (income, say) and $a, b, c, d, e > 0$. Equilibrium requires $a - bP + cZ = -d + eP$, so

$$P^* = \frac{a + d + cZ}{b + e}, \quad Q^* = -d + eP^*.$$

An increase in Z raises the price by $\partial P^* / \partial Z = c / (b + e)$ and quantity by $e c / (b + e)$. The price rises by more, and quantity by less, the less responsive supply is (the smaller e). If supply is perfectly inelastic ($e = 0$), a demand increase raises only the price; if supply is perfectly elastic ($e \rightarrow \infty$), it raises only the quantity.

Key idea. The effect of a shift in one curve is divided between price and quantity according to the slope of the *other* curve. Demand shocks move prices a lot when supply is inelastic; supply shocks move prices a lot when demand is inelastic.

This single principle explains much of the behaviour of commodity markets. Demand for oil is highly inelastic in the short run (Chapter 20), so a small supply disruption causes a large price spike. Supply of housing in cities with strict building restrictions is highly inelastic, so rising demand shows up almost entirely as higher house prices rather than more houses. And because both supply and demand are more elastic in the long run than in the short run, prices typically overshoot after a shock and then partly retrace as quantities adjust.

22.1.4 When both curves shift

If demand and supply both increase, quantity rises, but the effect on price depends on which shift is larger. If demand rises and supply falls, price rises but the effect on quantity is ambiguous. In such cases the direction of one variable can be determined only with quantitative information about the size of the shifts and the slopes.

22.2 Welfare: Measuring the Gains from Trade

22.2.1 Consumer, producer and total surplus

Chapter 20 defined **consumer surplus** as the area below the demand curve and above the price; Chapter 21 defined **producer surplus** as the area above the supply curve and below the price. Their sum is **total surplus**: the total gain to buyers and sellers from trading in the market. Equivalently, total surplus is the area between the demand and supply curves up to the quantity traded, because for each unit traded it is the difference between the buyer's willingness to pay (the height of the demand curve) and the seller's marginal cost (the height of the supply curve).

Worked Example 22.3. Surplus at equilibrium

In Example 22.2, consumer surplus is the triangle between the demand curve and $P = 20$: $\frac{1}{2} \times 60 \times (50 - 20) = 900$. Producer surplus is the triangle between $P = 20$ and the supply curve: $\frac{1}{2} \times 60 \times (20 - 5) = 450$. Total surplus is 1,350.

22.2.2 The efficiency of competitive equilibrium

Proposition 22.4. Competitive equilibrium maximizes total surplus

In a market with no externalities, no market power and full information, total surplus is maximized at the competitive equilibrium quantity.

Proof. Let $D(Q)$ be inverse demand (marginal benefit to buyers) and $S(Q)$ inverse supply (marginal cost to sellers). Total surplus from trading Q units, allocated to the buyers who value them most and produced by the sellers with lowest cost, is $W(Q) = \int_0^Q [D(x) - S(x)] dx$. Then $W'(Q) = D(Q) - S(Q)$, which is zero at the equilibrium quantity, positive below it and negative above it (when D slopes down and S slopes up). So W is maximized at Q^* . $\square$

Three conditions lie behind the result, and each is worth stating:

1. The goods are bought by the consumers who value them most (everyone with willingness to pay above P^* buys; everyone below does not).

2. The goods are produced by the sellers with the lowest costs (every seller with marginal cost below P^* produces).
3. The quantity is right: every unit whose value to a buyer exceeds its cost to a seller is traded, and no unit is traded whose cost exceeds its value.

The market achieves this without anyone knowing the others' valuations or costs. The price aggregates that dispersed information, as Friedrich Hayek emphasized. That is the core of the argument for markets as a coordination mechanism.

► **Common error: "efficient" means "fair"**

The proposition concerns total surplus, not its distribution. A competitive equilibrium may leave some people very poor. The analysis says that if society wants to redistribute, it should prefer methods that do not shrink total surplus more than necessary—not that redistribution is wrong (Chapter 1's distinction between efficiency and equity).

22.3 Taxes and Subsidies

22.3.1 A per-unit tax

Suppose the government levies a tax of t per unit sold. Now the price paid by buyers, P_d , exceeds the price received by sellers, P_s , by the tax:

$$P_d = P_s + t.$$

Equilibrium requires that the quantity buyers want at P_d equal the quantity sellers want at P_s : $Q_d(P_s + t) = Q_s(P_s)$.

Worked Example 22.5. A tax of 6 per unit

With the market of Example 22.2: $100 - 2(P_s + 6) = -20 + 4P_s$, so $88 - 2P_s = -20 + 4P_s$, giving $P_s = 18$, $P_d = 24$ and $Q = 52$.

Incidence: buyers pay 4 more than before (24 versus 20); sellers receive 2 less (18 versus 20). Buyers bear two-thirds of the tax.

Revenue: $6 \times 52 = 312$.

Welfare: consumer surplus falls to $\frac{1}{2} \times 52 \times (50 - 24) = 676$ (from 900); producer surplus falls to $\frac{1}{2} \times 52 \times (18 - 5) = 338$ (from 450). The loss to buyers and sellers is $224 + 112 = 336$, of which 312 goes to the government as revenue. The remaining 24 is lost to everyone: the **deadweight loss**. It is the triangle between the demand and supply curves over the units no longer traded: $\frac{1}{2} \times 6 \times (60 - 52) = 24$.

22.3.2 Who bears the tax?

A striking implication of the analysis is that the incidence of a tax does not depend on who legally pays it. Had the tax of 6 been levied on buyers instead of sellers, the equilibrium would be identical: buyers would pay 18 to sellers plus 6 to the government. The legal incidence (who writes the cheque) and the **economic incidence** (whose real income falls) are different things.

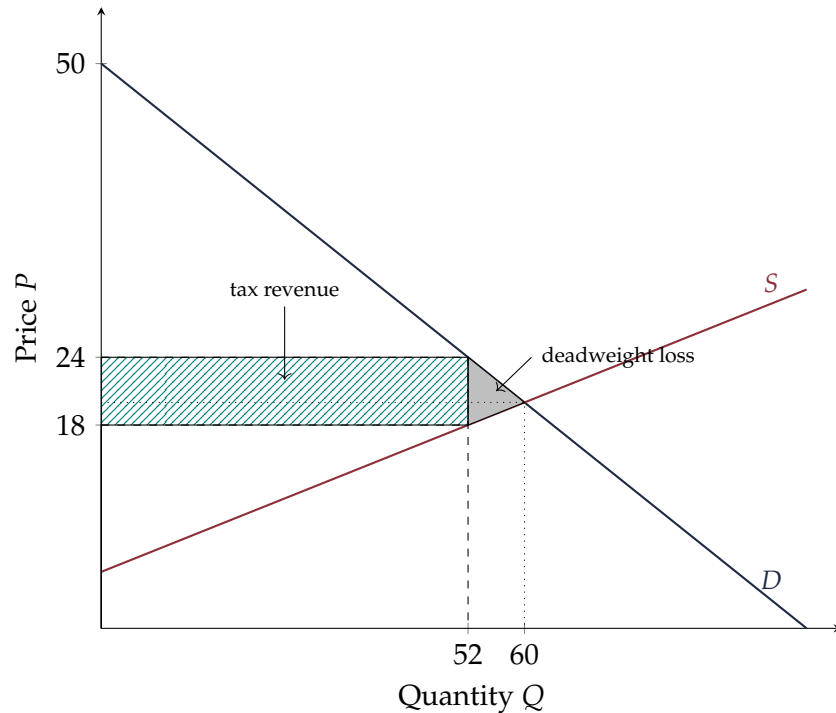


Figure 22.1: A per-unit tax of 6. Buyers pay 24, sellers receive 18, and 52 units are traded. The hatched rectangle is tax revenue; the grey triangle is the deadweight loss.

Proposition 22.6. Tax incidence

For a small per-unit tax, the fraction of the tax borne by buyers is

$$\frac{dP_d}{dt} = \frac{\epsilon_s}{\epsilon_s - \epsilon_d'}$$

and the fraction borne by sellers is $-\epsilon_d/(\epsilon_s - \epsilon_d)$, where $\epsilon_d < 0$ and $\epsilon_s > 0$ are the price elasticities of demand and supply at the initial equilibrium.

Proof. Differentiate $Q_d(P_s + t) = Q_s(P_s)$ with respect to t : $Q_d'(1 + dP_s/dt) = Q_s' dP_s/dt$. So $dP_s/dt = Q_d'/(Q_s' - Q_d')$ and $dP_d/dt = 1 + dP_s/dt = Q_s'/(Q_s' - Q_d')$. Multiply numerator and denominator by P/Q (at the initial equilibrium, where $P_d = P_s = P$) to convert slopes to elasticities. $\square$

Key idea. The more inelastic side of the market bears more of a tax. Whoever cannot easily walk away pays.

In Example 22.5, at the original equilibrium $\epsilon_d = -2 \times 20/60 = -0.67$ and $\epsilon_s = 4 \times 20/60 = 1.33$, so buyers' share is $1.33/(1.33 + 0.67) = 2/3$, as computed. (With linear curves and a non-small tax, the formula with initial elasticities is exact here because the curves are linear.)

Applications are everywhere. A tax on land, whose supply is fixed, falls entirely on landowners. Payroll taxes are legally split between employers and employees in many countries, but because labour supply is fairly inelastic for most workers, most economists believe the bulk of the combined tax is borne by workers in the form of lower wages. Taxes on cigarettes, for which demand is inelastic, are largely passed on to smokers.

22.3.3 Deadweight loss and the size of the tax

For a small tax the deadweight loss is approximately the triangle $\frac{1}{2}t\Delta Q$. Since ΔQ is itself proportional to t , the deadweight loss rises with the *square* of the tax: doubling a tax roughly quadruples its deadweight loss. Two practical lessons follow. It is more efficient to raise a given revenue with many small taxes on broad bases than a few large taxes on narrow bases. And taxes on goods with inelastic supply or demand cause smaller deadweight losses, because they change quantities less.

Tax revenue $tQ(t)$ first rises with t , but because Q falls as t rises, revenue eventually peaks and then falls; at a high enough tax, nothing is traded and revenue is zero. The curve relating revenue to the tax rate is sometimes called a **Laffer curve**. Its existence is not in doubt; where actual tax rates lie on it is an empirical question, and for most broad taxes in most countries rates are well below the revenue-maximizing level.

22.3.4 Subsidies

A per-unit subsidy s is a negative tax: $P_s = P_d + s$. It increases the quantity traded above the efficient level, and the cost to taxpayers exceeds the combined gain to buyers and sellers by a deadweight loss, again a triangle between the curves. The incidence formula applies with the same logic: the more inelastic side captures more of the subsidy. A subsidy for first-time home buyers in a market with inelastic housing supply mainly raises house prices, benefiting sellers.

22.4 Price Controls

22.4.1 Price ceilings

A **price ceiling** is a legal maximum price. If set above the equilibrium price it has no effect. If set below, it is *binding*: quantity supplied falls, quantity demanded rises, and a shortage results.

Worked Example 22.7. A binding ceiling

In the market of Example 22.2, suppose a ceiling of 15. Then $Q_s = -20 + 60 = 40$ and $Q_d = 100 - 30 = 70$: a shortage of 30. Only 40 units are traded. The welfare analysis depends on who gets them.

If the 40 units go to the buyers who value them most (the best case), consumer surplus is the area between the demand curve and the price of 15 over the first 40 units. The demand curve at $Q = 40$ is at $50 - 20 = 30$, so consumer surplus is $\frac{1}{2} \times 40 \times (50 - 30) + 40 \times (30 - 15) = 400 + 600 = 1,000$. Producer surplus is $\frac{1}{2} \times 40 \times (15 - 5) = 200$. Total surplus is 1,200, compared with 1,350 at equilibrium: a deadweight loss of 150, the triangle $\frac{1}{2} \times 20 \times (30 - 15)$.

Consumers as a group gain (1,000 versus 900) at the expense of producers (200 versus 450), but total surplus falls.

The best-case analysis understates the true cost of a ceiling, for three reasons:

1. *Misallocation*: the scarce units are unlikely to go to the buyers who value them most. If they are allocated randomly, by connections, or by who arrives first, some go to buyers with low willingness to pay while some high-value buyers go without.
2. *Rent-seeking and queuing*: buyers spend time and effort competing for the scarce units—waiting in queues, searching, lobbying. These costs dissipate part of the consumer gain (Chapter 1's gasoline queues).

3. *Quality deterioration and black markets*: sellers who cannot raise prices may reduce quality (landlords cut maintenance), or sell illegally at higher prices.

22.4.2 Price floors

A **price floor** is a legal minimum price. If binding (above equilibrium), it creates a surplus: sellers want to sell more than buyers want to buy. Agricultural price supports are a classic case; to maintain the floor, governments have bought and stored the surplus, as the European Union did on a large scale in the 1980s (the “butter mountains” and “wine lakes” of the Common Agricultural Policy).

The most debated price floor is the **minimum wage**. In a competitive labour market, a binding minimum wage reduces employment: firms hire fewer workers at the higher wage, while more people want to work. Workers who keep their jobs gain; those who lose them, or cannot find one, lose; and total surplus falls. Chapter 24 shows that in a labour market where employers have market power (*monopsony*), a moderate minimum wage can *raise* employment. Which model better describes a particular labour market is an empirical question, and the evidence—including the difference-in-differences studies described in Chapter 17—is mixed and depends on the level of the minimum wage relative to typical local wages. Germany’s introduction of a national minimum wage in 2015 was studied closely; most studies found small effects on employment and substantial increases in hourly wages at the bottom of the distribution, with some reduction in hours and in marginal part-time jobs.

22.4.3 Quotas

A **quota** limits the quantity that may be sold. If the quota $\bar{Q}$ is below the equilibrium quantity, the price rises to the level at which buyers want exactly $\bar{Q}$: $P_d = D(\bar{Q})$. Sellers’ marginal cost at $\bar{Q}$ is $S(\bar{Q}) < P_d$. The difference, $D(\bar{Q}) - S(\bar{Q})$, is a **quota rent** per unit, accruing to whoever holds the right to sell. Taxi medallions in cities that limit the number of licences are an example: the medallion’s market price capitalizes the stream of quota rents, which is why medallion values collapsed when ride-hailing platforms effectively expanded supply.

22.5 International Trade

22.5.1 Free trade in a small open economy

A **small open economy** can trade as much as it wishes at a given world price P_w , which it cannot influence. If P_w is below the domestic equilibrium price, the country imports: domestic buyers purchase $Q_d(P_w)$, domestic producers supply $Q_s(P_w)$, and imports fill the gap. Compared with no trade, consumers gain more than producers lose: the net gain is the gain from trade of Chapter 1, now measured in surplus.

22.5.2 Tariffs

A **tariff** is a tax on imports. With a per-unit tariff τ , the domestic price rises to $P_w + \tau$.

Worked Example 22.8. A tariff

Domestic demand is $Q_d = 100 - 2P$ and supply $Q_s = -20 + 4P$ (the market of Example 22.2), and the world price is 10. Under free trade, $Q_d = 80$, $Q_s = 20$, imports = 60.

A tariff of 5 raises the domestic price to 15: $Q_d = 70$, $Q_s = 40$, imports = 30. *Consumer surplus* falls by the area to the left of the demand curve between prices 10 and 15: $5 \times (80 + 70)/2 = 375$. *Producer surplus* rises by the area to the left of the supply curve between 10 and 15: $5 \times (20 + 40)/2 = 150$. *Tariff revenue*: $5 \times 30 = 150$. *Net loss*: $375 - 150 - 150 = 75$. It consists of two triangles: a *production distortion*, $\frac{1}{2} \times 5 \times (40 - 20) = 50$, because domestic firms produce units at a marginal cost above the world price; and a *consumption distortion*, $\frac{1}{2} \times 5 \times (80 - 70) = 25$, because consumers forgo units they value above the world price.

An **import quota** limiting imports to 30 would produce the same domestic price and the same deadweight loss. The difference lies in who receives the rectangle that the tariff would have raised as revenue: with a quota, it goes to whoever holds import licences. If the quota is administered by the exporting country—a “voluntary export restraint,” like the limits Japan agreed to on car exports to the United States in the 1980s—the rents go to foreign firms, and the importing country loses more than under an equivalent tariff.

22.5.3 Large countries and the terms of trade

A large country's tariff reduces world demand for the good enough to lower its world price. Then part of the tariff is borne by foreign exporters, and the importing country's losses are partly offset by a *terms-of-trade gain*. In principle a large country can gain from a small tariff at its trading partners' expense—though retaliation can leave everyone worse off. How much of a tariff foreign exporters bear depends, by the incidence logic of Theorem 22.6, on the elasticity of foreign export supply.

Case: Who paid for the 2018 U.S. tariffs?

In 2018 the United States imposed tariffs on steel and aluminium from most countries and on several hundred billion dollars of imports from China; China retaliated with tariffs on U.S. goods, particularly agricultural products. Several teams of economists, including Amiti, Redding and Weinstein (2019) and Fajgelbaum, Goldberg, Kennedy and Khandelwal (2020), used detailed import-price data to measure incidence. They found close to complete pass-through: the prices paid by U.S. importers, inclusive of the tariff, rose roughly one-for-one with the tariff, while the prices charged by foreign exporters barely fell. In the language of this chapter, foreign export supply was highly elastic—Chinese exporters had other markets, and U.S. buyers could switch to suppliers in Vietnam, Mexico and elsewhere—so the incidence fell almost entirely on U.S. buyers. The deadweight-loss analysis also applies. U.S. firms that used steel as an input faced higher costs, which reduced their competitiveness; U.S. farmers lost export sales to China, and the government paid them compensation. Imports from China fell, but imports from other countries rose to replace them, a pattern called *trade diversion*. The episode illustrates the chapter's central lessons: legal and economic incidence differ; elasticities determine who pays; and the deadweight losses from trade barriers fall partly on the imposing country itself.

22.6 General Equilibrium

22.6.1 Why partial equilibrium is not enough

The analysis so far is **partial equilibrium**: it studies one market, holding prices in other markets and incomes fixed. That is appropriate when the market is small relative to the economy. But a tax on labour affects wages, which affect the demand for every good; a rise in the price of oil affects the cost of producing almost everything. **General equilibrium** analysis studies all markets simultaneously, recognizing that prices and incomes are jointly determined.

22.6.2 An exchange economy

The simplest general-equilibrium model has two consumers, A and B, and two goods, x and y , with no production. Each consumer starts with an **endowment** of the goods and can trade at market prices. The set of possible allocations can be drawn in an **Edgeworth box**, whose width is the total quantity of x and whose height is the total quantity of y ; each point in the box is an allocation, measured from A's origin at the bottom left and B's origin at the top right.

Definition 22.9. Walrasian equilibrium

A **Walrasian (competitive) equilibrium** is a set of prices and an allocation such that (i) each consumer's bundle maximizes her utility subject to her budget constraint at those prices, where her income is the market value of her endowment, and (ii) every market clears: total demand equals total endowment for each good.

Worked Example 22.10. A two-person exchange economy

Consumer A has $U_A = x^{0.5}y^{0.5}$ and endowment $(8, 2)$; consumer B has $U_B = x^{0.5}y^{0.5}$ and endowment $(2, 8)$. Normalize $p_y = 1$ and let p be the price of x . A's income is $8p + 2$ and B's is $2p + 8$. By the Cobb–Douglas rule each spends half on each good:

$$x_A = \frac{8p + 2}{2p}, \quad x_B = \frac{2p + 8}{2p}.$$

Market clearing for x : $x_A + x_B = 10$, i.e. $(10p + 10)/(2p) = 10$, so $p = 1$. Then $x_A = y_A = 5$ and $x_B = y_B = 5$. Each consumer trades 3 units of the good she has in abundance for 3 of the other. Utility rises from $\sqrt{16} = 4$ to 5 for both.

We found p by clearing the market for x only. The market for y then clears automatically: $y_A + y_B = 5 + 5 = 10$. This is no accident.

Proposition 22.11. Walras' law

If every consumer spends her entire income, the total value of excess demand across all markets is zero at any prices: $\sum_k p_k z_k(p) = 0$, where z_k is the excess demand for good k . Hence if all markets but one clear, the last also clears.

Proof. Each consumer's budget constraint says that the value of her purchases equals the value of her endowment, so the value of her excess demands is zero. Summing over consumers gives the result. $\square$

Walras' law also implies that only *relative* prices are determined: with n goods there are $n - 1$ independent market-clearing conditions, which determine $n - 1$ relative prices. The absolute level of prices—the price level—must be determined by something outside the real economy. In Part VII, that something is money and monetary policy.

22.6.3 Efficiency and the welfare theorems

An allocation is Pareto efficient (Chapter 1) if no one can be made better off without making someone worse off. In the Edgeworth box, at an efficient allocation the two consumers' indifference curves are tangent—their MRSs are equal—for otherwise a trade could benefit both. The locus of efficient allocations is the **contract curve**. In Example 22.10 the MRS for both consumers is y/x , so the efficient allocations are those with $y_A/x_A = y_B/x_B$: the diagonal of the box. The equilibrium allocation (5,5) lies on it.

Proposition 22.12. The fundamental theorems of welfare economics

First welfare theorem: if markets are complete, all agents are price takers, there are no externalities, and information is symmetric, then every competitive equilibrium allocation is Pareto efficient.

Second welfare theorem: under the same conditions plus convexity of preferences and technology, any Pareto-efficient allocation can be achieved as a competitive equilibrium after an appropriate lump-sum redistribution of endowments.

The first theorem is the formal version of Adam Smith's "invisible hand." In equilibrium every consumer sets her MRS equal to the same price ratio, so all MRSs are equal; every firm sets its marginal rate of transformation equal to the price ratio; hence no mutually beneficial reallocation remains. The second theorem says that efficiency and equity can, in principle, be separated: society can choose a distribution by redistributing endowments and then let markets work.

► Common error: reading the welfare theorems as a proof that markets are always efficient

The theorems are conditional. Their assumptions—price-taking, no externalities, complete markets, symmetric information—are precisely what Chapters 24 to 10 relax. The second theorem requires *lump-sum* transfers that do not depend on behaviour; real taxes depend on income or spending and therefore distort choices (Section 22.3). The theorems are best read as a map of what can go wrong: every case of market failure is a failure of one of their assumptions.

22.7 When Markets Fail

The welfare theorems identify the main sources of **market failure**—situations in which market outcomes are not Pareto efficient—and they organize the rest of Part III:

- **Market power** (Chapter 24): firms that are not price takers restrict output to raise prices.
- **Strategic interaction** (Chapter 23): when few agents interact, individually rational choices can produce collectively poor outcomes.
- **Asymmetric information** (Chapter 25): when one side knows more than the other, markets may unravel or function poorly.
- **Externalities and public goods** (Chapter 26): when actions affect others outside the market, or goods are non-rival and non-excludable, prices do not reflect social costs and benefits.

Macroeconomic failures—unemployment, financial crises—can often be traced to the same sources operating at scale, together with nominal rigidities that stop prices from adjusting quickly (Part VI).

Case: Rent control in San Francisco

In 1994 San Francisco voters extended rent control to small multi-family buildings built before 1980, which had previously been exempt. Because the rule applied according to construction date, Diamond, McQuade and Qian (2019) could compare buildings built just before 1980 (newly controlled) with similar buildings built just after (still uncontrolled)—a design combining the logic of difference-in-differences and regression discontinuity from Chapter 17.

Tenants in newly controlled buildings were substantially more likely to stay at their addresses, and the benefits to them were large. But landlords of controlled buildings reduced the supply of rental housing by about 15 per cent relative to the comparison buildings, by converting apartments to condominiums and owner-occupied housing, or redeveloping them. The authors concluded that the reduction in supply likely raised rents elsewhere in the city, offsetting much of the benefit to controlled tenants and transferring costs to future renters.

The result fits the theory of this chapter: a binding ceiling reduces quantity supplied, especially in the long run, when supply is more elastic; benefits go to those who hold the scarce controlled units; and sellers respond along margins (conversion, quality) that the regulation does not control. It also shows the value of a credible research design in settling a question on which strong opinions exist on both sides.

Summary

- Competitive equilibrium equates quantity demanded and supplied. Changes in the good's price move along curves; other changes shift them.
- A shift in one curve is divided between price and quantity according to the elasticity of the other curve. Short-run responses differ from long-run ones.
- Total surplus is maximized at the competitive quantity when there are no externalities, market power or information problems.
- The economic incidence of a tax does not depend on its legal incidence; the more inelastic side bears more. Deadweight loss rises with the square of the tax.
- Binding price ceilings cause shortages and floors cause surpluses; both reduce total surplus, and real-world costs exceed the simple triangle because of misallocation and rent-seeking. Quotas create rents for licence holders.
- Tariffs raise domestic prices, help domestic producers, hurt consumers more, and create production and consumption distortions. Incidence depends on the elasticity of foreign supply.
- In general equilibrium only relative prices are determined, and Walras' law links all markets. The welfare theorems establish efficiency of competitive equilibrium under conditions whose failure defines market failure.

Key terms

competitive equilibrium; excess supply; excess demand; comparative statics; total surplus; deadweight loss; economic incidence; legal incidence; Laffer curve; subsidy; price ceiling; price floor; minimum wage; quota; quota rent; small open economy; tariff; import quota; voluntary export restraint; terms of

trade; partial equilibrium; general equilibrium; endowment; Edgeworth box; Walrasian equilibrium; Walras' law; contract curve; first and second welfare theorems; market failure.

Exercises

Exercise 22.1 (Equilibrium and surplus). Demand is $Q_d = 240 - 4P$ and supply is $Q_s = 6P - 60$. Find the equilibrium price and quantity, consumer surplus and producer surplus.

Exercise 22.2 (Comparative statics). In Exercise 22.1, a rise in income shifts demand to $Q_d = 280 - 4P$. Find the new equilibrium. What fraction of the demand shift (40 units at each price) shows up as increased quantity?

Exercise 22.3 (Tax incidence). In the market of Exercise 22.1, a tax of 5 per unit is levied on sellers. Find P_d , P_s , Q , tax revenue and deadweight loss. Verify buyers' share of the tax using the incidence formula.

Exercise 22.4 (Doubling a tax). Repeat Exercise 22.3 with a tax of 10. By what factor do revenue and deadweight loss increase?

Exercise 22.5 (Subsidy). Instead of a tax, the government pays buyers a subsidy of 5 per unit in the market of Exercise 22.1. Find the new prices and quantity, the cost of the subsidy, and the deadweight loss.

Exercise 22.6 (Price ceiling). A ceiling of 25 is imposed in the market of Exercise 22.1. Find the shortage and the deadweight loss (assuming efficient rationing). Would the deadweight loss be larger or smaller with random rationing? Why?

Exercise 22.7 (Minimum wage). Labour demand is $L_d = 1,000 - 20w$ and labour supply is $L_s = 30w - 250$. Find the competitive wage and employment. A minimum wage of 30 is introduced. Find employment, the number of people who want work but cannot find it, and the deadweight loss.

Exercise 22.8 (Tariff). Domestic demand is $Q_d = 200 - 2P$ and supply $Q_s = 3P - 50$. The world price is 30. Find imports under free trade. Then find the effects of a tariff of 10 on domestic price, domestic production, imports, consumer surplus, producer surplus, revenue and deadweight loss.

Exercise 22.9 (Quota versus tariff). In Exercise 22.8, what import quota would produce the same domestic price as the tariff? Who receives the rectangle that was tariff revenue under (a) auctioned import licences, (b) licences given free to domestic importers, (c) a voluntary export restraint?

Exercise 22.10 (Incidence reasoning). Without calculation, explain who bears most of (a) a tax on land; (b) a tax on a brand of soft drink when many close substitutes are available; (c) a payroll tax when labour supply is very inelastic.

Exercise 22.11 (General equilibrium). Consumer A has $U_A = x^{0.5}y^{0.5}$ and endowment (6, 2); consumer B has $U_B = x^{0.25}y^{0.75}$ and endowment (2, 6). With $p_y = 1$, find the equilibrium price of x and the equilibrium allocation. Check that both markets clear.

Exercise 22.12 (Welfare theorems). For each situation, state which assumption of the first welfare theorem fails: (a) a single firm supplies electricity to a region; (b) a factory's smoke damages nearby crops; (c) used-car sellers know more about their cars' quality than buyers; (d) nobody can be excluded from a city's fireworks display.

Exercise 22.13 (Long-run adjustment). In a constant-cost competitive industry in long-run equilibrium, demand permanently increases. Describe the short-run and long-run effects on price, quantity, the number of firms and economic profits.

Solutions

Solution 22.1. $240 - 4P = 6P - 60$ gives $P^* = 30$, $Q^* = 120$. Inverse demand $P = 60 - Q/4$; choke price 60. Inverse supply $P = 10 + Q/6$; intercept 10. $CS = \frac{1}{2} \times 120 \times 30 = 1,800$. $PS = \frac{1}{2} \times 120 \times 20 = 1,200$.

Solution 22.2. $280 - 4P = 6P - 60$: $P = 34$, $Q = 144$. Quantity rises by 24 of the 40-unit shift, i.e. 60%, which equals $e/(b + e) = 6/10$.

Solution 22.3. $240 - 4(P_s + 5) = 6P_s - 60$: $10P_s = 280$, $P_s = 28$, $P_d = 33$, $Q = 108$. Revenue = 540. $DWL = \frac{1}{2} \times 5 \times 12 = 30$. At the original equilibrium $\epsilon_d = -4 \times 30/120 = -1$, $\epsilon_s = 6 \times 30/120 = 1.5$; buyers' share = $1.5/2.5 = 0.6$, i.e. 3 of the 5. Indeed P_d rose by 3.

Solution 22.4. $10P_s = 260$, $P_s = 26$, $P_d = 36$, $Q = 96$. Revenue = 960 (1.78 times as much). $DWL = \frac{1}{2} \times 10 \times 24 = 120$ (4 times as much).

Solution 22.5. Now $P_d = P_s - 5$: $240 - 4(P_s - 5) = 6P_s - 60$, $10P_s = 320$, $P_s = 32$, $P_d = 27$, $Q = 132$. Cost = $5 \times 132 = 660$. $DWL = \frac{1}{2} \times 5 \times 12 = 30$: units 121–132 cost more to produce than buyers value them.

Solution 22.6. At $P = 25$: $Q_s = 90$, $Q_d = 140$: shortage 50. Demand price at $Q = 90$: $60 - 22.5 = 37.5$; supply price at 90: 25. $DWL = \frac{1}{2} \times (120 - 90) \times (37.5 - 25) = 187.5$. With random rationing the loss is larger, because some units go to buyers valuing them below 37.5 while some buyers valuing them more go without.

Solution 22.7. $1,000 - 20w = 30w - 250$: $w = 25$, $L = 500$. At $w = 30$: $L_d = 400$ (employment), $L_s = 650$; 250 people want work at that wage but cannot find it. Demand wage at $L = 400$: $50 - 20 = 30$; supply wage at 400: $650/30 = 21.67$. $DWL = \frac{1}{2} \times 100 \times (30 - 21.67) = 416.7$.

Solution 22.8. Free trade at 30: $Q_d = 140$, $Q_s = 40$, imports 100. With tariff, price 40: $Q_d = 120$, $Q_s = 70$, imports 50. CS falls by $10 \times (140 + 120)/2 = 1,300$; PS rises by $10 \times (40 + 70)/2 = 550$; revenue = 500; $DWL = 1,300 - 550 - 500 = 250$ (production distortion $\frac{1}{2} \times 10 \times 30 = 150$ plus consumption distortion $\frac{1}{2} \times 10 \times 20 = 100$).

Solution 22.9. A quota of 50 units. (a) The government, if licences are auctioned competitively (each is worth 10 per unit). (b) The domestic importers who receive licences. (c) Foreign exporters, who can charge 40 for goods they would sell at 30; the importing country's loss rises by 500 to 750.

Solution 22.10. (a) Landowners: land supply is perfectly inelastic. (b) The soft drink's producer and its suppliers: demand for one brand is very elastic. (c) Workers: the inelastic side bears the tax, in the form of lower take-home pay, regardless of whether the employer or employee legally pays.

Solution 22.11. A's income $6p + 2$, B's income $2p + 6$. $x_A = 0.5(6p + 2)/p$, $x_B = 0.25(2p + 6)/p$. Clearing x : $(3p + 1 + 0.5p + 1.5)/p = 8$, so $3.5p + 2.5 = 8p$, $p = 5/9 \approx 0.556$. Then $x_A = 4.8$, $x_B = 3.2$; $y_A = 0.5(6p + 2) = 2.667$, $y_B = 0.75(2p + 6) = 5.333$. Totals: x : 8; y : 8. Both clear, as Walras' law guarantees. The price of x is low because B, who has much of the endowment value in y , cares relatively little about x .

Solution 22.12. (a) Price-taking (market power). (b) No externalities. (c) Symmetric information. (d) Complete markets / excludability: the display is a public good for which no market can charge each viewer.

Solution 22.13. Short run: with a fixed number of firms, price rises along the upward-sloping short-run supply curve; each firm expands output where $P = MC$; firms earn positive economic profit. Long run: profits attract entry, shifting short-run supply right until price returns to min $LRAC$. Final result: same price as before, higher quantity, more firms, each producing at minimum efficient scale, zero economic profit.

Game Theory and Strategic Behaviour

After studying this chapter you should be able to

1. Represent strategic situations as games in normal form and in extensive form.
2. Identify dominant and dominated strategies and solve games by iterated elimination.
3. Find pure-strategy Nash equilibria using best responses, including in games with continuous strategies.
4. Compute mixed-strategy equilibria and explain what they mean.
5. Solve sequential games by backward induction, distinguish credible from non-credible threats, and explain the value of commitment.
6. Explain how repetition can sustain cooperation, and derive the critical discount factor for a trigger strategy.
7. Recognize coordination problems and multiple equilibria in economic settings such as bank runs.
8. Explain time inconsistency and why credibility matters for policy.

A motivating question

In November 2014, with oil prices already falling, the members of OPEC met in Vienna and, contrary to many forecasts, decided not to cut production. Within months the price of crude had roughly halved. In 2016 OPEC joined with Russia and other producers to form “OPEC+” and agreed to cut output; prices recovered. In March 2020 the agreement broke down, Saudi Arabia and Russia raised output, and prices collapsed again before a new deal was struck. Why do producers who would all gain from restricting output find it so hard to do so? Why do agreements sometimes hold for years and then suddenly fail?

The competitive model of Chapter 22 cannot answer these questions, because in it no firm’s choice affects any other firm’s payoff. When a few decision makers interact, each must anticipate what the others will do—and each knows that the others are anticipating too. **Game theory** is the study of such strategic interaction. It is used throughout this book: to analyse oligopoly (Chapter 24), contracts and signalling (Chapter 25), public goods and auctions (Chapter 26), bank runs (Chapter 51) and the credibility of central banks (Chapter 50).

23.1 What Is a Game?

Definition 23.1. Game

A **game** consists of

1. a set of *players*;
2. for each player, a set of *strategies*—complete plans of action;
3. for each player, a *payoff* for every combination of strategies chosen by all players.

Payoffs represent players' preferences over outcomes; they may be profits, utilities or anything else a player cares about. Game theory assumes each player chooses rationally given his or her beliefs about the others, and—usually—that the structure of the game and the rationality of all players are *common knowledge*: everyone knows it, everyone knows that everyone knows it, and so on.

23.1.1 Normal form

When players choose simultaneously (or without observing each other's choices), a game can be represented in **normal form** as a table. Each row is a strategy of player 1 (the "row player"), each column a strategy of player 2 (the "column player"), and each cell lists the payoffs, player 1's first.

23.2 Dominance

23.2.1 The prisoner's dilemma

Two firms, A and B, sell competing products. Each can charge a High or a Low price. Their monthly profits (in millions) are:

		Firm B	
		High	Low
Firm A	High	3, 3	0, 5
	Low	5, 0	1, 1

If both charge High, each earns 3. If one undercuts, it captures most of the market and earns 5 while the other earns 0. If both charge Low, each earns 1.

Consider firm A. If B charges High, A earns 3 from High and 5 from Low: Low is better. If B charges Low, A earns 0 from High and 1 from Low: Low is better. Whatever B does, A prefers Low.

Definition 23.2. Dominant strategy

A strategy is **strictly dominant** for a player if it gives a strictly higher payoff than any other strategy, whatever the other players do. A strategy is **strictly dominated** if some other strategy always gives a strictly higher payoff.

By symmetry, Low is dominant for B too. Rational players choose dominant strategies, so the outcome is (Low, Low), with payoffs (1, 1). Yet both firms would be better off at (High, High), with (3, 3). This is the **prisoner's dilemma**, named after a story in which two suspects,

questioned separately, each have an incentive to confess and implicate the other, though both would be better off staying silent.

Key idea. In a prisoner's dilemma, individually rational choices produce an outcome that is worse for everyone. The equilibrium is not Pareto efficient. The dilemma is the structure behind price wars, arms races, overfishing, under-provision of public goods and many failures of international cooperation.

For consumers, the firms' dilemma is good news: competition drives prices down. Whether a prisoner's dilemma is socially bad depends on whose payoffs are left out of the game.

23.2.2 Iterated elimination of dominated strategies

Many games have no dominant strategies, but they may have dominated ones. A rational player never plays a strictly dominated strategy; and if the other player knows this, she can eliminate it from consideration, which may make another strategy dominated, and so on.

Worked Example 23.3. Iterated elimination

		Player 2		
		Left	Centre	Right
Player 1	Up	4, 3	5, 1	6, 2
	Down	2, 1	8, 4	3, 6

Neither of player 1's strategies dominates the other. For player 2, Centre is strictly dominated by Right ($1 < 2$ and $4 < 6$). Eliminate Centre. In the remaining game, Up strictly dominates Down for player 1 ($4 > 2$, $6 > 3$). Eliminate Down. Player 2, facing Up, prefers Left (3) to Right (2). The unique outcome is (Up, Left) with payoffs (4, 3).

Iterated elimination requires more than rationality: it requires each player to know that the others are rational, that they know this, and so on for as many rounds as there are eliminations.

23.3 Nash Equilibrium

23.3.1 Best responses and equilibrium

Most games cannot be solved by dominance. The central solution concept of game theory, due to John Nash (1950), asks only that each player's strategy be the best reply to the others'.

Definition 23.4. Best response and Nash equilibrium

A strategy is a **best response** to the other players' strategies if no other strategy gives the player a higher payoff against them. A **Nash equilibrium** is a combination of strategies, one per player, such that each player's strategy is a best response to the others'.

In a Nash equilibrium no player can gain by changing strategy unilaterally. That matches the general definition of equilibrium in Chapter 1: no one wants to change, given what everyone else is doing. Nash equilibrium is also a condition of *consistent expectations*: if each player correctly anticipates the others' strategies, and each best-responds, the anticipations are confirmed.

Finding pure-strategy equilibria. In a table, underline each player's best payoff for each strategy of the other. A cell in which both payoffs are underlined is a Nash equilibrium. In the prisoner's dilemma the only such cell is (Low, Low). A dominant-strategy outcome is always a Nash equilibrium, but not every Nash equilibrium involves dominant strategies.

23.3.2 Coordination games

Worked Example 23.5. Choosing a technology standard

Two firms must each choose a charging standard, X or Y, for a new product. If they choose the same standard, the market grows; if different, consumers hesitate.

		Firm 2	
		X	Y
Firm 1	X	4, 3	0, 0
	Y	0, 0	3, 4

Both (X,X) and (Y,Y) are Nash equilibria. Firm 1 prefers X and firm 2 prefers Y, but each prefers to match the other rather than mismatch.

Games with multiple equilibria raise a question that the Nash concept alone does not answer: which equilibrium will occur? History, convention, communication, and which choice seems "natural" to both (Thomas Schelling's **focal points**) all matter. Coordination problems are pervasive in economics: which side of the road to drive on, which currency to accept, which technology standard to adopt, and—importantly for Part VII—whether depositors keep their money in a bank.

A particularly important coordination game is the **stag hunt**:

		Hunter 2	
		Stag	Hare
Hunter 1	Stag	4, 4	0, 3
	Hare	3, 0	3, 3

Hunting stag together is best for both, but succeeds only if both participate; hunting hare is safe. Both (Stag, Stag) and (Hare, Hare) are equilibria. The good equilibrium requires *trust*; a mere doubt about the other's choice can tip both players into the safe but inferior one. Bank runs have exactly this structure (Section 23.8).

23.4 Mixed Strategies

23.4.1 When no pure equilibrium exists

Worked Example 23.6. The penalty kick

A penalty taker in football kicks Left or Right; the goalkeeper dives Left or Right (from the kicker's perspective). The table gives the probability that the kick scores; the kicker wants to maximize it and the goalkeeper to minimize it.

		Goalkeeper dives	
		Left	Right
Kicker	Left	0.50	0.90
	Right	0.95	0.60

There is no pure-strategy equilibrium: if the kicker always went Left, the goalkeeper would always dive Left; but then the kicker would switch to Right; and so on. Each player wants to be unpredictable.

Definition 23.7. Mixed strategy

A **mixed strategy** is a probability distribution over a player's pure strategies.

Proposition 23.8. Indifference principle

In a mixed-strategy equilibrium, each player is indifferent among all pure strategies to which he or she assigns positive probability. Each player's mixing probabilities are chosen to make the *other* player indifferent.

Argument. If a player strictly preferred one of the pure strategies in the mix, she could raise her expected payoff by putting all probability on it; so she cannot be mixing optimally. Given that she is indifferent, any mixture is a best response, including the equilibrium one. For her mixture to be part of an equilibrium, it must make the opponent willing to mix as well. $\square$

Worked Example 23.9. Solving the penalty kick

Let the goalkeeper dive Left with probability q . The kicker is indifferent between Left and Right if

$$0.50q + 0.90(1 - q) = 0.95q + 0.60(1 - q) \implies 0.90 - 0.40q = 0.60 + 0.35q \implies q = 0.4.$$

Let the kicker kick Left with probability p . The goalkeeper is indifferent if

$$\begin{aligned} 0.50p + 0.95(1 - p) &= 0.90p + 0.60(1 - p) \\ \implies 0.95 - 0.45p &= 0.60 + 0.30p \implies p \approx 0.467. \end{aligned}$$

In equilibrium the kick scores with probability $0.50(0.4) + 0.90(0.6) = 0.74$. So the kicker goes Right about 53% of the time and the goalkeeper dives Right 60% of the time. Notice a counter-intuitive feature: each player's mixing probabilities are pinned down by the *other* player's payoffs, not his own. The kicker mixes so as to leave the goalkeeper indifferent, and vice versa; if either became predictable, the other would exploit the pattern. Exercise 23.3 shows how an improvement in the kicker's accuracy changes both players' mixes.

Palacios-Huerta (2003), studying over a thousand professional penalty kicks, found that kickers' and goalkeepers' choices were close to the mixed-strategy equilibrium predictions and that players' choices were serially independent, as unpredictability requires—one of the cleanest field confirmations of game theory.

Proposition 23.10. Nash's existence theorem

Every game with a finite number of players, each having a finite number of pure strategies, has at least one Nash equilibrium, possibly in mixed strategies.

Economically, mixed strategies can represent genuine randomization (tax audits, random inspections, sales timed unpredictably), or the uncertainty each player has about the other's choice, or a population in which a fraction of players choose each strategy.

23.5 Games with Continuous Strategies

In many economic games, players choose a quantity, a price or an effort level from a continuum. Each player's best response is then a function of the others' choices, found by maximizing her payoff taking the others' choices as given. A Nash equilibrium is a point where the best-response functions intersect.

Worked Example 23.11. An effort game

Two partners in a firm each choose effort $e_i \geq 0$. Partner 1's payoff is

$$\pi_1 = e_1(12 - e_1 + 0.5e_2),$$

and symmetrically for partner 2. The term $0.5e_2$ means that a partner's effort raises the return to the other's effort: the efforts are **strategic complements**.

Partner 1's best response maximizes π_1 given e_2 : $\partial\pi_1/\partial e_1 = 12 - 2e_1 + 0.5e_2 = 0$, so

$$e_1 = BR_1(e_2) = 6 + 0.25e_2.$$

By symmetry $e_2 = 6 + 0.25e_1$. In a symmetric equilibrium $e = 6 + 0.25e$, so $e^* = 8$ for each partner, and each earns $8 \times (12 - 8 + 4) = 64$.

When the best responses slope upward, as here, a rise in one player's action induces the other to raise hers: **strategic complements**. When they slope downward—a rise in one player's action makes the other want to cut hers—the actions are **strategic substitutes**. Chapter 24 shows that quantities in Cournot oligopoly are typically strategic substitutes and prices in Bertrand oligopoly with differentiated products are strategic complements, and that the distinction shapes how firms compete.

23.6 Sequential Games

23.6.1 Extensive form

When players move in sequence and later movers observe earlier moves, the game is naturally represented in **extensive form**: a *game tree* with a node for each decision, branches for each available action, and payoffs at the end of each path.

Worked Example 23.12. Market entry

An incumbent monopolist earns 10 if no one enters its market. A potential entrant decides whether to enter. If it enters, the incumbent decides whether to accommodate (share the market peacefully) or fight (cut prices aggressively). Payoffs (entrant, incumbent):

- Entrant stays out: (0, 10).
- Entrant enters, incumbent accommodates: (3, 5).
- Entrant enters, incumbent fights: (−2, 2).

23.6.2 Backward induction and credible threats

Solve the game from the end. If the entrant has entered, the incumbent compares 5 (accommodate) with 2 (fight) and accommodates. Anticipating this, the entrant compares 3 (enter)

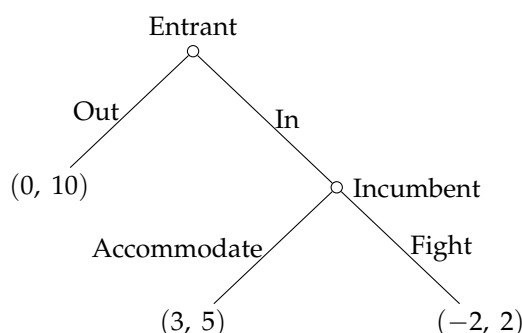


Figure 23.1: The entry game in extensive form. Payoffs are (entrant, incumbent).

with 0 (stay out) and enters. The outcome is (In, Accommodate). This procedure is **backward induction**.

Now consider the incumbent's strategy "Fight if entry occurs." If the entrant believed it, it would stay out (0 is better than -2); and given that the entrant stays out, the threat costs the incumbent nothing, because it is never carried out. So (Out, Fight) is also a Nash equilibrium of this game. But it rests on a **non-credible threat**: if entry actually occurred, the incumbent would not want to fight. A sensible solution concept should rule it out.

Definition 23.13. Subgame-perfect equilibrium

A **subgame** is a part of the game tree that starts at a single decision node and includes everything that follows it. A **subgame-perfect equilibrium** (SPE) is a strategy combination that is a Nash equilibrium in every subgame. In finite games with perfect information, the SPE is found by backward induction.

The concept, due to Reinhard Selten, requires that strategies be optimal not only along the path actually played but at every point where a player might find herself. (In, Accommodate) is the unique subgame-perfect equilibrium of the entry game.

23.6.3 The value of commitment

Suppose the incumbent could, *before* the entrant decides, invest in extra production capacity that would be costly to leave idle. Say the investment costs 2 and changes payoffs so that fighting becomes more profitable than accommodating after entry: accommodating now yields $5 - 2 - 2 = 1$ (the idle capacity is wasted) and fighting yields $2 - 2 + 3 = 3$ (the capacity makes aggressive pricing cheap). Now the threat to fight is credible. The entrant, anticipating a fight, stays out, and the incumbent earns $10 - 2 = 8$, more than the 5 it would earn by accommodating without the investment.

The paradox of commitment. Restricting your own future options can make you better off, because it changes what others expect you to do. A commitment works only if it is visible, understood and costly or impossible to reverse.

Examples abound: a firm that signs long-term supply contracts, a union that announces a strike fund, a general who burns the bridges behind his army, a government that writes a fiscal rule into its constitution (as Germany did with its "debt brake" in 2009), and a country that gives its central bank legal independence. Section 23.9 returns to the last case.

23.7 Repeated Games

23.7.1 The problem with one-shot cooperation

In a one-shot prisoner's dilemma, cooperation is not an equilibrium. But firms, countries and people often interact repeatedly. Repetition allows today's behaviour to be rewarded or punished tomorrow.

23.7.2 Finite repetition

Suppose the pricing game of Section 23.2 is played exactly ten times, and both firms know it. In round 10 there is no future, so each firm charges Low (the dominant strategy). In round 9, since round 10's outcome is fixed regardless of what happens now, there is again no reason to cooperate; each charges Low. By backward induction, the firms charge Low in every round. Known, finite repetition does not rescue cooperation in the prisoner's dilemma. (Experiments with human subjects often find cooperation in early rounds of finitely repeated games, collapsing near the end—a pattern the model predicts only at the end.)

23.7.3 Infinite repetition and trigger strategies

Now suppose the game is repeated indefinitely: after each round there is always a probability that the game continues, or the firms simply have no fixed end date in mind. Let each firm value a payoff one round in the future at a fraction $\delta \in (0, 1)$ of a payoff today. The **discount factor** δ combines impatience (as in Chapter 9, $\delta = 1/(1+i)$) and the probability that the relationship continues.

Consider the **grim trigger strategy**: charge High in the first round and keep charging High as long as both firms have always done so; if anyone ever charges Low, charge Low forever after.

If both firms follow this strategy, each earns 3 per round forever, with present value

$$3 + 3\delta + 3\delta^2 + \dots = \frac{3}{1-\delta}.$$

If a firm deviates by charging Low, it earns 5 this round and then 1 per round forever after (because the other firm punishes):

$$5 + \delta + \delta^2 + \dots = 5 + \frac{\delta}{1-\delta}.$$

Cooperation is sustainable if the first is at least as large as the second:

$$\frac{3}{1-\delta} \geq 5 + \frac{\delta}{1-\delta} \iff 3 \geq 5(1-\delta) + \delta \iff \delta \geq \frac{1}{2}.$$

Proposition 23.14. Sustaining cooperation with trigger strategies

In an infinitely repeated prisoner's dilemma with payoffs T (temptation), R (mutual cooperation), P (mutual punishment), with $T > R > P$, the grim trigger strategy sustains cooperation as a subgame-perfect equilibrium if and only if

$$\delta \geq \frac{T-R}{T-P}.$$

Proof. Cooperating forever gives $R/(1 - \delta)$; deviating gives $T + \delta P/(1 - \delta)$. The first is at least the second iff $R \geq (1 - \delta)T + \delta P$, iff $\delta(T - P) \geq T - R$. The punishment phase is itself a Nash equilibrium in every subgame (each player's Low is a best response to the other's Low forever), so the strategy is subgame-perfect. $\square$

With $T = 5$, $R = 3$, $P = 1$: $\delta \geq 2/4 = 1/2$. Cooperation is easier to sustain when the gain from cheating ($T - R$) is small, when punishment is harsh (P low), and when players are patient or interactions frequent (δ high).

Proposition 23.15. Folk theorem (informal statement)

In an infinitely repeated game with sufficiently patient players, any outcome that gives each player more than the payoff she can guarantee herself in the one-shot game can be sustained as a subgame-perfect equilibrium.

The folk theorem is both encouraging and troubling. Encouraging, because it shows how long-term relationships, reputations and the “shadow of the future” can support cooperation without formal contracts—in trade credit among firms, in relationships between banks and borrowers, in international agreements. Troubling, because it shows that repeated games typically have *many* equilibria, and the theory alone does not say which will occur.

Bridge. Tacit collusion among oligopolists is a repeated game (Chapter 24). A central bank's reputation for keeping inflation low is sustained by the same logic: the public's expectations reward good behaviour and punish lapses (Chapter 50).

23.8 Coordination, Expectations and Multiple Equilibria

23.8.1 Self-fulfilling outcomes

In games with multiple equilibria, what happens depends on what players expect to happen. If every depositor expects every other depositor to leave her money in a bank, each does so, and the bank is safe. If each expects the others to withdraw, each rushes to withdraw first, and the bank fails—even if it was fundamentally sound. Both outcomes are Nash equilibria of the same game; which occurs depends on beliefs.

Worked Example 23.16. A two-depositor bank run

Two depositors have each placed 100 in a bank, which has invested the 200 in a long-term project that yields 240 if left to mature but only 160 if liquidated early. Each depositor may withdraw early or wait. The project cannot be partly liquidated: to pay any early withdrawal, the bank must liquidate all of it. If both wait, each gets 120. If both withdraw early, the bank liquidates and each gets 80. If one withdraws early, the bank liquidates, pays her the full 100, and the other receives what is left, 60.

		Depositor 2	
		Wait	Withdraw
Depositor 1	Wait	120, 120	60, 100
	Withdraw	100, 60	80, 80

Both (Wait, Wait) and (Withdraw, Withdraw) are Nash equilibria. The bank's assets are the same in both; the run is a failure of coordination. This simplified version of the Diamond–Dybvig (1983) model is developed in Chapter 51, which shows how

deposit insurance and a central bank acting as lender of last resort can eliminate the bad equilibrium by making “Wait” a dominant strategy.

Currency crises (Chapter 52), sudden stops in capital flows, and some sovereign-debt crises share this structure: an outcome that no individual wants can be triggered by a shift in expectations, which is why policy-makers pay so much attention to confidence.

23.9 Time Inconsistency and Credibility

A plan is **time-inconsistent** if, when the time comes to carry it out, the planner prefers to deviate from it. Governments face this problem constantly. A government may announce that it will not bail out failing banks, to encourage banks to be prudent; but when a large bank actually fails, it may be optimal *at that moment* to rescue it. Anticipating the rescue, banks take more risk (Chapter 51).

The most influential application is to monetary policy. Kydland and Prescott (1977) and Barro and Gordon (1983) argued that a central bank that cares about both inflation and output faces a temptation: once wages and prices have been set on the expectation of low inflation, a surprise burst of inflation can temporarily raise output. The public, understanding this temptation, expects higher inflation from the start, and the outcome is higher inflation with no gain in output. The problem is the entry game’s non-credible commitment in another form. Solutions are commitment devices: legal independence for the central bank, an explicit inflation target, appointing central bankers known to be averse to inflation, and building a reputation over repeated interaction. Chapter 50 develops the model formally.

23.10 Behavioural Evidence

Laboratory experiments test game-theoretic predictions directly. In the **ultimatum game**, a proposer offers a split of a sum of money; the responder accepts (both are paid as proposed) or rejects (both get nothing). Backward induction with purely selfish players predicts that the responder accepts any positive offer, so the proposer offers the smallest possible amount. In practice, across many countries, typical offers are 40 to 50 per cent, and offers below about 20 per cent are frequently rejected. Responders are willing to pay to punish unfair treatment, and proposers anticipate it. The lesson is not that game theory fails, but that payoffs must include what people actually care about—fairness, reciprocity, reputation—not only money. With those included, strategic reasoning explains behaviour well.

Case: OPEC and the difficulty of cartel discipline

The members of OPEC, joined since 2016 by Russia and other producers as OPEC+, face a repeated prisoner’s dilemma. Each producer gains if all restrict output and prices stay high; each is tempted to produce more than its quota, since its own extra output barely moves the world price. The condition $\delta \geq (T - R)/(T - P)$ identifies what makes the cartel fragile.

Temptation ($T - R$) is large when a producer has spare capacity and when the price is high, since each extra barrel is then very profitable. *Punishment* ($R - P$) is weak if the cartel cannot credibly flood the market in response to cheating, or if a price war hurts the enforcer as much as the cheater. *Patience* (δ) falls when a member expects its oil to become less valuable in future—for instance, because of the energy transition—or when it faces pressing budget needs today. *Monitoring* matters too: trigger strategies require

that cheating be detected, and national production figures are not always transparent. The episodes described at the start of the chapter fit this logic. In 2014, the rise of U.S. shale oil, produced by many small firms outside the cartel, reduced the gains to OPEC from restraint, since cuts would largely hand market share to shale producers. In March 2020 the Saudi–Russian disagreement over deeper cuts at the start of the pandemic produced a brief, deliberate price war—a punishment phase—before a new agreement was reached in April. The pattern of cooperation punctuated by breakdowns is exactly what repeated-game theory predicts when temptation, monitoring and patience vary over time.

Summary

- A game specifies players, strategies and payoffs. Simultaneous-move games are represented in normal form, sequential games in extensive form.
- Rational players do not play strictly dominated strategies; iterated elimination can solve some games. In the prisoner's dilemma, dominant strategies lead to a Pareto-inferior outcome.
- A Nash equilibrium is a profile of mutual best responses. Coordination games have multiple equilibria; which occurs depends on expectations.
- Mixed-strategy equilibria make opponents indifferent. Every finite game has a Nash equilibrium.
- With continuous strategies, equilibria are intersections of best-response functions; actions may be strategic complements or substitutes.
- Backward induction yields subgame-perfect equilibria, which rule out non-credible threats. Commitment can be valuable because it makes threats credible.
- Finite repetition does not sustain cooperation in the prisoner's dilemma; infinite repetition can, if $\delta \geq (T - R)/(T - P)$. The folk theorem shows that many outcomes can be sustained.
- Bank runs are coordination failures. Time inconsistency makes credibility central to policy.

Key terms

game theory; game; normal form; dominant strategy; dominated strategy; prisoner's dilemma; iterated elimination; best response; Nash equilibrium; coordination game; focal point; stag hunt; mixed strategy; indifference principle; strategic complements; strategic substitutes; extensive form; backward induction; non-credible threat; subgame; subgame-perfect equilibrium; commitment; repeated game; discount factor; grim trigger strategy; folk theorem; self-fulfilling equilibrium; time inconsistency; ultimatum game.

Exercises

Exercise 23.1 (Dominance). In the game below, find all strictly dominated strategies and solve by iterated elimination if possible.

	L	M	R
T	3, 1	2, 4	1, 2
B	1, 3	4, 2	0, 1

Exercise 23.2 (Nash equilibrium). Find all pure-strategy Nash equilibria of the game *chicken*: each of two drivers heading toward each other can Swerve or go Straight. Payoffs: (Swerve, Swerve) = (0, 0); (Swerve, Straight) = (−1, 1); (Straight, Swerve) = (1, −1); (Straight, Straight) = (−10, −10). Then find the mixed-strategy equilibrium.

Exercise 23.3 (Mixed strategies). In the penalty-kick game of Example 23.6, the kicker practises and improves his accuracy when kicking Right, so that (Right, Left) scores with probability 1.0 instead of 0.95. Recompute the equilibrium mixes and scoring probability. Which player's mix changes, and why?

Exercise 23.4 (Best responses). Two firms choose advertising levels $a_1, a_2 \geq 0$. Firm 1's profit is $\pi_1 = a_1(20 - a_1 - 0.5a_2)$, and symmetrically for firm 2. Derive the best-response functions, find the Nash equilibrium, and state whether advertising levels are strategic complements or substitutes.

Exercise 23.5 (Backward induction). Modify the entry game so that if the incumbent fights, payoffs are (−2, 6) instead of (−2, 2). Find the subgame-perfect equilibrium. Explain the difference from the text.

Exercise 23.6 (Commitment). In the entry game of Example 23.12, what is the largest amount the incumbent would be willing to pay for a commitment device that made fighting credible and deterred entry, if the device itself did not otherwise change payoffs?

Exercise 23.7 (Repeated game). In a prisoner's dilemma with $T = 10$, $R = 7$, $P = 2$, $S = 0$, what is the minimum discount factor for grim trigger to sustain cooperation? If each round lasts one year and the only reason for discounting is an interest rate i , what is the maximum interest rate at which cooperation is sustainable?

Exercise 23.8 (Probability of continuation). Firms discount at interest rate 5% per period, and after each period there is a probability ρ that the market ceases to exist (because of new technology). Write the effective discount factor. Using the pricing game of the text ($T = 5$, $R = 3$, $P = 1$), find the largest ρ at which collusion is sustainable.

Exercise 23.9 (Finite repetition). Explain why the argument that cooperation unravels in a finitely repeated prisoner's dilemma depends on the one-shot game having a unique Nash equilibrium.

Exercise 23.10 (Bank run). In Example 23.16, suppose deposit insurance guarantees each depositor 100 whatever happens, financed by a tax that is levied only if a run occurs. Rewrite the payoffs (assume the insurer covers any shortfall) and show that Wait becomes a dominant strategy. Would the insurance ever need to pay out?

Exercise 23.11 (Time inconsistency). A government announces that it will never forgive student loans, so that students borrow prudently. Explain why the announcement may not be credible, and suggest a commitment device.

Exercise 23.12 (Ultimatum game). A proposer splits 100. She believes that a responder rejects any offer below 25 with probability 1, and accepts any offer of 25 or more with probability 1. What does a self-interested proposer offer? How would your answer change if she believed that the acceptance probability for an offer x (between 0 and 50) is $x/50$?

Solutions

Solution 23.1. For the column player, R is strictly dominated by M ($2 < 4$, $1 < 2$). Eliminate R. For the row player, neither T nor B dominates (T gives 3 vs 1 against L; B gives 4 vs 2 against M). For the column player in the reduced game, neither L nor M dominates (against T: $1 < 4$; against B: $3 > 2$). Iterated elimination stops. Pure Nash equilibria of the reduced game: (T,L)? Column's best response to T is M, so no. (T,M)? Row's best response to M is B, so no. (B,M)? Column's best response to B is L, so no. (B,L)? Row's best response to L is T, so no. There is no pure-strategy equilibrium; there is a mixed one. Let row play T with probability p : column is indifferent between L and M if $1p + 3(1 - p) = 4p + 2(1 - p)$, so $p = 1/4$. Let column play L with probability q : row is indifferent if $3q + 2(1 - q) = 1q + 4(1 - q)$, so $q = 1/2$.

Solution 23.2. Pure equilibria: (Swerve, Straight) and (Straight, Swerve). Mixed: let player 2 go Straight with probability s . Player 1 is indifferent if Swerve gives $0(1 - s) + (-1)s = -s$ and Straight gives $1(1 - s) + (-10)s = 1 - 11s$: $-s = 1 - 11s$, $s = 0.1$. By symmetry each goes Straight with probability 0.1. The probability of a crash is 0.01.

Solution 23.3. Goalkeeper's mix (making the kicker indifferent): $0.50q + 0.90(1 - q) = 1.0q + 0.60(1 - q)$, so $0.90 - 0.40q = 0.60 + 0.40q$, $q = 0.375$. Kicker's mix (making the goalkeeper indifferent): $0.50p + 1.0(1 - p) = 0.90p + 0.60(1 - p)$, so $1 - 0.5p = 0.6 + 0.3p$, $p = 0.5$. Scoring probability: $0.5(0.375) + 0.9(0.625) = 0.75$. The kicker kicks Left *more* often (0.5 vs 0.467) even though he improved at kicking Right. The reason is the indifference principle: a Right kick now beats a Left dive with certainty, which makes diving Left worse for the goalkeeper; to keep the goalkeeper willing to dive Left, the kicker must kick Left more often. Meanwhile the goalkeeper dives Left less (0.375 vs 0.4). Both mixes change, and each is pinned down by the other player's payoffs.

Solution 23.4. $\partial\pi_1/\partial a_1 = 20 - 2a_1 - 0.5a_2 = 0$, so $a_1 = 10 - 0.25a_2$, and symmetrically. Symmetric equilibrium: $a = 10 - 0.25a$, $a^* = 8$. Best responses slope down: strategic substitutes.

Solution 23.5. After entry, the incumbent compares 5 (accommodate) with 6 (fight) and fights. Anticipating this, the entrant compares -2 with 0 and stays out. SPE: (Out, Fight); the incumbent earns 10. The threat is now credible because fighting is the incumbent's best response after entry.

Solution 23.6. Without the device, the incumbent earns 5 (entry and accommodation). With a device that deters entry, it earns 10 minus the cost. It would pay up to 5.

Solution 23.7. $\delta \geq (10 - 7)/(10 - 2) = 3/8 = 0.375$. With $\delta = 1/(1 + i)$: $1/(1 + i) \geq 0.375$ iff $1 + i \leq 2.667$, i.e. $i \leq 166.7\%$. Cooperation is sustainable at any plausible interest rate; what makes real cartels fragile is uncertainty about continuation, imperfect monitoring and changing payoffs.

Solution 23.8. $\delta = (1 - \rho)/1.05$. Collusion needs $\delta \geq 0.5$: $1 - \rho \geq 0.525$, so $\rho \leq 0.475$. If there is more than a 47.5% chance each period that the market ends, collusion collapses.

Solution 23.9. In the last round, players must play a Nash equilibrium of the one-shot game. If it is unique, the last-round outcome is fixed regardless of history, so nothing done in the penultimate round can be rewarded or punished; the argument continues backward. If the one-shot game had several equilibria, players could threaten to play a bad one in the last round after defection and a good one after cooperation, which could sustain cooperation in earlier rounds.

Solution 23.10. With insurance, each depositor receives at least 100 however the other acts, and a waiting depositor whose bank is not run receives 120. Payoffs: (Wait, Wait) = (120, 120); (Wait, Withdraw) = (100, 100) (the waiter is topped up to 100); (Withdraw, Wait) = (100, 100); (Withdraw, Withdraw) = (100, 100). Wait weakly dominates Withdraw, and strictly so if the other waits. The unique sensible outcome is (Wait, Wait), in which no liquidation occurs and the insurance never pays out. Credible insurance removes the bad equilibrium at no cost—in this model. Chapter 51 adds the moral-hazard cost: insured banks may take more risk.

Solution 23.11. Once borrowers are heavily indebted and struggling, forgiveness may be politically and economically attractive ex post, so the announcement is time-inconsistent; anticipating it, students may borrow more than otherwise. Commitment devices include legislation that makes forgiveness require a

supermajority, contracts with private lenders that the government cannot unilaterally alter, or building a track record of not forgiving.

Solution 23.12. In the first case she offers exactly 25 and keeps 75. In the second, she chooses x to maximize expected payoff $(100 - x)(x/50)$; the derivative is $(100 - 2x)/50$, positive for all $x < 50$, so she offers 50, the maximum in the range, keeping 50 with certainty. Beliefs about the responder's behaviour, not generosity, drive the offer.

Market Power and Industrial Organization

After studying this chapter you should be able to

1. Identify the sources of market power and barriers to entry.
2. Derive the monopolist's pricing rule, the Lerner index and the deadweight loss of monopoly.
3. Analyse natural monopoly and alternative forms of regulation.
4. Analyse first-, second- and third-degree price discrimination and two-part tariffs.
5. Analyse monopsony and explain why a minimum wage can raise employment when employers have market power.
6. Solve the Cournot, Stackelberg and Bertrand models of oligopoly and compare their outcomes.
7. Explain monopolistic competition and the trade-off between variety and cost.
8. Explain network effects and the economics of two-sided platforms.
9. Measure market concentration and describe the aims and tools of competition policy.

A motivating question

In August 2024 a U.S. federal court ruled that Google had unlawfully maintained a monopoly in general internet search, in part by paying device makers and browser developers—Apple above all—to make Google the default search engine. In 2025 the court ordered remedies that prohibited certain exclusive arrangements and required Google to share some of its data with rivals. Why would a firm pay billions of dollars a year simply to be the default? Why does a search engine that charges users nothing have “market power”? And how should courts decide whether a large firm's success reflects superior products or the exclusion of rivals?

Chapter 22 showed that competitive markets, under stated conditions, maximize total surplus. This chapter relaxes the assumption of price-taking. It studies firms that can influence the price they charge—monopolies, oligopolies and firms selling differentiated products—using the optimization tools of Chapters 5 and 6 and the game theory of Chapter 23.

24.1 Sources of Market Power

Definition 24.1. Market power

A firm has **market power** if it can profitably raise its price above marginal cost without losing all its customers—equivalently, if it faces a downward-sloping demand curve for its product.

Market power persists only if something prevents rivals from entering and competing it

away. **Barriers to entry** include:

- **Legal barriers:** patents, copyrights, licences and exclusive franchises. A patent grants a temporary monopoly (typically 20 years from filing) as a reward for invention.
- **Economies of scale:** if minimum efficient scale is large relative to the market (Chapter 21), only one or a few firms can operate at low cost.
- **Control of an essential input:** ownership of a unique resource or facility.
- **Network effects:** when a product becomes more valuable as more people use it, an established network is hard to dislodge (Section 24.8).
- **Switching costs:** customers who have invested in learning one system, or whose data are stored in it, are reluctant to change.
- **Strategic behaviour:** incumbents may deter entry by building excess capacity, cutting prices when entry occurs, or signing exclusive contracts (Chapter 23).

24.2 Monopoly

24.2.1 Marginal revenue

A **monopoly** is the sole seller of a product with no close substitutes. Facing the market inverse demand curve $P(Q)$, its revenue is $R(Q) = P(Q)Q$ and its marginal revenue is

$$MR(Q) = P(Q) + P'(Q)Q.$$

Selling one more unit brings in its price P , but because $P'(Q) < 0$ the monopolist must lower the price on all the units it was already selling, losing $|P'(Q)|Q$. Marginal revenue is therefore less than price. For linear inverse demand $P = a - bQ$, $MR = a - 2bQ$: a line with the same intercept and twice the slope.

Writing marginal revenue in terms of the price elasticity of demand $\varepsilon = (dQ/dP)(P/Q)$:

$$MR = P \left(1 + \frac{P'(Q)Q}{P} \right) = P \left(1 + \frac{1}{\varepsilon} \right).$$

When demand is elastic ($\varepsilon < -1$), MR is positive; when inelastic, MR is negative. This is Theorem 20.11 of Chapter 20 seen from the seller's side.

24.2.2 The monopolist's choice

Profit $\Pi(Q) = R(Q) - C(Q)$ is maximized where

$$MR(Q) = MC(Q).$$

The monopolist then charges the highest price consumers will pay for that quantity, read off the demand curve.

Proposition 24.2. Monopoly markup and the Lerner index

At the profit-maximizing output,

$$\frac{P - MC}{P} = -\frac{1}{\varepsilon}, \quad \text{equivalently} \quad P = \frac{MC}{1 + 1/\varepsilon}.$$

The left side is the **Lerner index** of market power. A monopolist always operates where demand is elastic.

Proof. Set $P(1 + 1/\varepsilon) = MC$ and rearrange. Since $MC > 0$, $1 + 1/\varepsilon > 0$, which requires $\varepsilon < -1$. $\square$

The less elastic the demand, the higher the markup. With $\varepsilon = -2$, price is twice marginal cost; with $\varepsilon = -5$, price is 1.25 times marginal cost; as $\varepsilon \rightarrow -\infty$, price approaches marginal cost, the competitive outcome.

Worked Example 24.3. Monopoly versus competition

Inverse demand is $P = 100 - 2Q$ and marginal cost is constant at 20, with no fixed cost. *Monopoly:* $MR = 100 - 4Q = 20$, so $Q^m = 20$, $P^m = 60$. Profit = $(60 - 20) \times 20 = 800$. Consumer surplus = $\frac{1}{2} \times 20 \times (100 - 60) = 400$.

Competition (price equals marginal cost): $P^c = 20$, $Q^c = 40$. Consumer surplus = $\frac{1}{2} \times 40 \times 80 = 1,600$; producer surplus 0.

Welfare: total surplus is 1,600 under competition and $400 + 800 = 1,200$ under monopoly. The **deadweight loss of monopoly** is 400: the triangle $\frac{1}{2} \times (40 - 20) \times (60 - 20)$. Of the 1,200 fall in consumer surplus, 800 is transferred to the monopolist and 400 is lost to everyone.

Check the Lerner condition: at $Q = 20$, $\varepsilon = (dQ/dP)(P/Q) = (-0.5)(60/20) = -1.5$, and $(60 - 20)/60 = 2/3 = 1/1.5$.

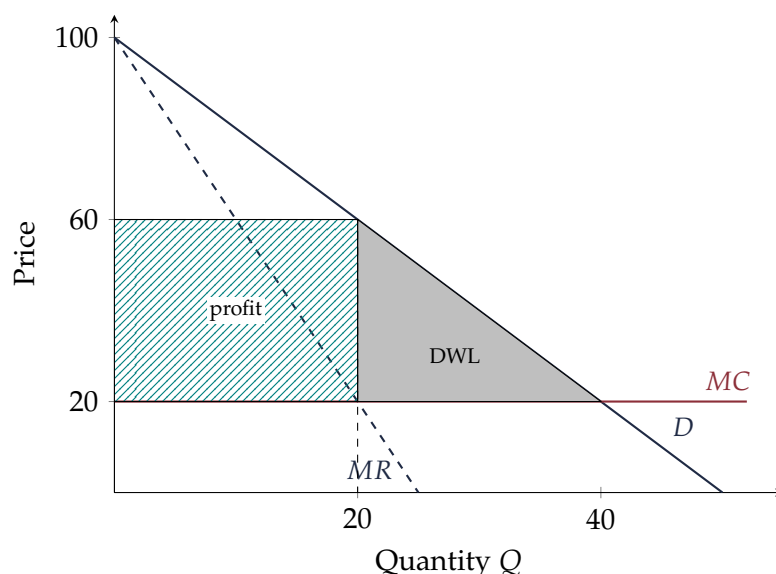


Figure 24.1: Monopoly with $P = 100 - 2Q$ and $MC = 20$. The monopolist produces where $MR = MC$ ($Q = 20$) and charges 60. The grey triangle is the deadweight loss.

► Common error: “a monopolist charges the highest possible price”

A monopolist chooses the price that maximizes profit, not the highest price anyone would pay. Raising the price further loses sales worth more than the extra margin. A monopolist also has no supply curve: the quantity it chooses depends on the shape of demand, not only on price, so there is no one-to-one relation between price and quantity

supplied.

24.2.3 Other costs of monopoly

The deadweight-loss triangle may understate the social cost of monopoly. Firms may spend resources to acquire or defend a monopoly—lobbying for protection, litigating, building excess capacity—dissipating part of the monopoly profit in **rent-seeking**. Monopolists sheltered from competition may also become inefficient (“X-inefficiency”), operating above their cost curves. On the other hand, the prospect of temporary monopoly profits motivates innovation; patents deliberately trade static deadweight loss for dynamic gains, and Joseph Schumpeter argued that large firms with market power are important sources of technical progress. The balance between these effects is a central question in competition policy.

24.3 Natural Monopoly and Regulation

Definition 24.4. Natural monopoly

An industry is a **natural monopoly** if a single firm can supply the entire market demand at lower total cost than two or more firms could.

Natural monopoly typically arises when fixed costs are large and marginal costs low, so that average cost declines over the whole relevant range of output: electricity transmission grids, water pipes, railway track. Duplicating these networks would waste resources, but an unregulated single supplier would charge the monopoly price.

Governments respond in three main ways. *Public ownership*: the state runs the network. *Regulation*: a private firm operates under rules on prices. *Competition for the market*: firms bid for the right to operate for a period.

Regulating price poses a dilemma. **Marginal-cost pricing** achieves efficiency but, because average cost exceeds marginal cost when average cost is falling, it leaves the firm with a loss that must be covered by a subsidy. **Average-cost pricing** allows the firm to break even but leaves some deadweight loss. Traditional **rate-of-return regulation**, common in the United States, sets prices to allow a “fair” return on capital; it gives the firm weak incentives to reduce costs and, if the allowed return exceeds the cost of capital, an incentive to over-invest (the Averch–Johnson effect). **Price-cap regulation**, pioneered in the United Kingdom in the 1980s and now widely used, fixes the price path in advance, typically rising with inflation minus an efficiency factor (“RPI – X”); the firm keeps any savings it achieves, which sharpens incentives, but the regulator must periodically reset the cap and may be tempted to claw back profits, raising time-inconsistency problems of the kind discussed in Chapter 23.

24.4 Price Discrimination

A firm with market power can sometimes charge different prices to different customers, or for different units, for reasons unrelated to cost. This **price discrimination** requires that the firm (i) have market power, (ii) be able to identify or sort customers with different willingness to pay, and (iii) prevent resale between them.

24.4.1 First-degree (perfect) price discrimination

A firm that knows each customer's willingness to pay and charges each exactly that amount captures the entire consumer surplus. It produces until the price paid by the last customer equals marginal cost—the competitive quantity. Perfect discrimination is therefore efficient: there is no deadweight loss, but all surplus goes to the seller. It is rarely achievable, but data-driven personalized pricing moves some markets toward it.

24.4.2 Third-degree price discrimination

The firm divides customers into groups with different demand elasticities and charges each group a different price. Students, seniors, business travellers and customers in different countries are familiar examples.

Proposition 24.5. Third-degree price discrimination

A firm selling in two separable markets with common marginal cost sets $MR_1 = MR_2 = MC$. Consequently it charges the higher price in the market with less elastic demand: $P_i = MC/(1 + 1/\varepsilon_i)$.

Worked Example 24.6. Two markets

A software firm sells to businesses with demand $Q_A = 100 - P_A$ and to students with $Q_B = 60 - 2P_B$. Marginal cost is 10.

Businesses: inverse demand $P_A = 100 - Q_A$, $MR_A = 100 - 2Q_A = 10$, so $Q_A = 45$, $P_A = 55$. *Students:* inverse demand $P_B = 30 - Q_B/2$, $MR_B = 30 - Q_B = 10$, so $Q_B = 20$, $P_B = 20$.

Elasticities at these points: businesses $-1 \times 55/45 = -1.22$; students $-2 \times 20/20 = -2$. The Lerner indexes, $45/55 = 0.82$ and $10/20 = 0.5$, equal the reciprocals of the absolute elasticities. Profits: $45 \times 45 = 2,025$ from businesses and $20 \times 10 = 200$ from students.

Whether third-degree discrimination raises or lowers total welfare compared with a single price is ambiguous. It misallocates output between groups (the marginal buyer in one group values the good more than the marginal buyer in the other), but it may expand total output—for instance, by allowing a market with low willingness to pay to be served at all. A necessary condition for discrimination to raise welfare is that it increase total output.

24.4.3 Second-degree price discrimination and two-part tariffs

When the firm cannot identify customers' types directly, it can offer a menu and let customers sort themselves: quantity discounts, versions of a product (economy and business class; basic and premium software), or bundles. This is **second-degree price discrimination**. The menu must be designed so that high-valuation customers prefer the premium option; this often means making the basic option deliberately less attractive. Airlines' restrictions on cheap tickets (Saturday-night stays, no changes) exist to stop business travellers from buying them. Chapter 25 treats such screening problems formally.

A **two-part tariff** charges an entry fee plus a per-unit price: a gym membership plus a per-visit fee, a phone contract with a monthly charge and per-unit data costs, a printer sold cheaply with expensive ink. With identical customers, the profit-maximizing two-part tariff sets the per-unit price equal to marginal cost (maximizing the surplus created) and captures the entire consumer surplus with the entry fee.

24.5 Monopsony

Market power can also exist on the buying side. A **monopsony** is a sole buyer; more generally, a firm has monopsony power if it faces an upward-sloping supply curve for an input, so that buying more raises the price it must pay. The most important example is the labour market: an employer that is the dominant employer in a town, or whose workers find it costly to change jobs, faces an upward-sloping labour supply curve.

If the employer must pay all workers the same wage $w(L)$, the cost of hiring one more worker is the **marginal factor cost**

$$MFC = w(L) + w'(L)L,$$

which exceeds the wage because the higher wage must be paid to all existing workers. The employer hires until the **marginal revenue product** of labour (the extra revenue from one more worker) equals the marginal factor cost.

Worked Example 24.7. Monopsony and the minimum wage

The marginal revenue product of labour is $MRP = 50 - 0.5L$, and labour supply is $w = 10 + 0.5L$, so $MFC = 10 + L$.

Monopsony: $50 - 0.5L = 10 + L$, so $L = 26.67$, and the wage is read from the supply curve: $w = 10 + 13.33 = 23.33$. Workers are paid less than their marginal revenue product ($50 - 13.33 = 36.67$).

Competitive benchmark: $MRP = w$: $50 - 0.5L = 10 + 0.5L$, so $L = 40$, $w = 30$.

Minimum wage of 30: the employer can now hire any number of workers up to 40 at 30 each, so the marginal factor cost is flat at 30 up to $L = 40$. It hires until $MRP = 30$, i.e. $L = 40$. Employment rises from 26.67 to 40, and wages rise. A minimum wage set between the monopsony wage and the competitive wage raises both wages and employment; one set far above the competitive wage reduces employment, as in Chapter 22.

Evidence that many labour markets have some monopsony power—from studies of employer concentration and of how responsive workers' quits are to wages—has grown considerably since the 2010s and helps explain why moderate minimum-wage increases have often been found to have small employment effects.

24.6 Oligopoly

An **oligopoly** is a market with a few sellers, each large enough that its choices affect the others. Oligopolists must reason strategically, so the tools of Chapter 23 apply. There is no single model of oligopoly; the outcome depends on what firms choose (quantities or prices), whether they move simultaneously or in sequence, and whether the product is homogeneous or differentiated.

24.6.1 Cournot competition: choosing quantities

In the model of Augustin Cournot (1838), firms producing an identical product simultaneously choose quantities; the price then adjusts to clear the market. Let inverse demand be $P = a - b(q_1 + q_2)$ and each firm's marginal cost be c .

Firm 1 takes q_2 as given and maximizes $\Pi_1 = [a - b(q_1 + q_2) - c]q_1$. The first-order condition $a - 2bq_1 - bq_2 - c = 0$ gives the best-response (or *reaction*) function

$$q_1 = BR_1(q_2) = \frac{a - c - bq_2}{2b}.$$

Quantities are strategic substitutes: the more firm 2 produces, the less firm 1 wants to produce. By symmetry $q_2 = (a - c - bq_1)/(2b)$. The Nash equilibrium is the intersection:

Proposition 24.8. Cournot equilibrium

With linear demand $P = a - bQ$ and constant marginal cost c , the symmetric Cournot equilibrium with n firms has

$$q_i = \frac{a - c}{(n + 1)b}, \quad Q = \frac{n}{n + 1} \cdot \frac{a - c}{b}, \quad P = \frac{a + nc}{n + 1}.$$

Proof. With n firms, firm i 's first-order condition is $a - bQ - bq_i - c = 0$. In a symmetric equilibrium $Q = nq_i$, so $a - c = (n + 1)bq_i$. $\square$

As n rises, total output approaches the competitive level $(a - c)/b$ and price approaches marginal cost. With $n = 1$ the formulas give the monopoly outcome.

Worked Example 24.9. Cournot duopoly

Let $P = 120 - Q$ and $c = 30$. With two firms: $q_i = 90/3 = 30$, $Q = 60$, $P = 60$, and each earns $(60 - 30) \times 30 = 900$, for an industry total of 1,800. Under monopoly: $Q = 45$, $P = 75$, profit 2,025. Under competition: $Q = 90$, $P = 30$. The duopoly lies between them.

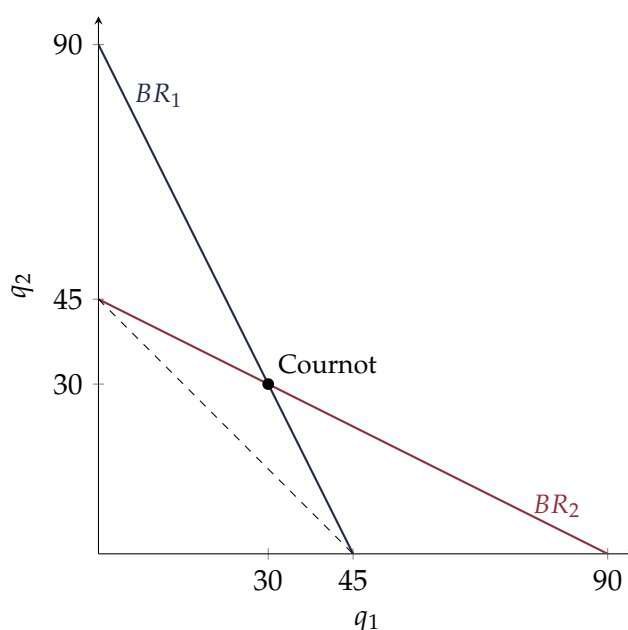


Figure 24.2: Cournot reaction functions for $P = 120 - Q$, $c = 30$. Firm 1's best response is plotted as $q_2 = 90 - 2q_1$ (equivalently $q_1 = 45 - q_2/2$). The equilibrium is at (30, 30). The dashed line shows combinations summing to the monopoly output of 45.

Why do the firms not produce the joint-monopoly output of 22.5 each? Because at that point each has an incentive to expand: its own gain from selling more exceeds its own loss from the lower price, since the other half of the price loss falls on its rival. This is a prisoner's dilemma (Chapter 23). Sustaining the joint-monopoly outcome requires collusion, which the repeated-game logic may or may not support.

Proposition 24.10. Market power in Cournot oligopoly

In a Cournot equilibrium with possibly different marginal costs, each firm's Lerner index is $(P - c_i)/P = s_i/|\varepsilon|$, where s_i is its market share and ε the market elasticity of demand. The share-weighted average Lerner index equals $HHI/|\varepsilon|$, where $HHI = \sum_i s_i^2$ is the Herfindahl–Hirschman index.

Proof. Firm i 's first-order condition is $P + P'(Q)q_i = c_i$. Dividing by P : $(P - c_i)/P = -P'(Q)q_i/P = -(P'(Q)Q/P)(q_i/Q) = s_i/|\varepsilon|$. Multiplying by s_i and summing gives $\sum s_i^2/|\varepsilon|$. $\square$

This result gives a theoretical foundation for using concentration measures to assess market power (Section 24.9).

24.6.2 Stackelberg competition: sequential quantities

In the model of Heinrich von Stackelberg (1934), one firm (the leader) chooses its quantity first; the other (the follower) observes it and responds. Solve by backward induction. The follower's best response is $q_2 = (a - c - bq_1)/(2b)$. The leader anticipates this and maximizes

$$\Pi_1 = \left[a - b \left(q_1 + \frac{a - c - bq_1}{2b} \right) - c \right] q_1 = \frac{a - c - bq_1}{2} q_1,$$

giving $q_1 = (a - c)/(2b)$, the monopoly quantity, and $q_2 = (a - c)/(4b)$.

Worked Example 24.11. Stackelberg

With $P = 120 - Q$ and $c = 30$: the leader produces 45, the follower 22.5; $Q = 67.5$, $P = 52.5$. The leader earns $22.5 \times 45 = 1,012.5$ and the follower $22.5 \times 22.5 = 506.25$. Compared with Cournot (900 each), the leader gains and the follower loses; total output is higher and the price lower.

The leader benefits from *commitment*: by producing a large quantity first, it forces the follower to accept a smaller share. This is the value of commitment of Chapter 23. The leader's quantity must be truly irreversible—a sunk investment in capacity, for instance—or the follower would not believe it.

24.6.3 Bertrand competition: choosing prices

In the model of Joseph Bertrand (1883), firms selling an identical product simultaneously set prices, and consumers buy from the cheapest seller (splitting equally if prices tie).

Proposition 24.12. The Bertrand paradox

With two firms, an identical product, constant and equal marginal cost c , and no capacity limits, the unique Nash equilibrium is $p_1 = p_2 = c$. Both firms earn zero profit.

Proof. If both firms charged the same price above c , either could undercut slightly and capture the whole market, nearly doubling its profit. If one charged more than the other and the lower price exceeded c , the higher-priced firm could undercut. If the lower price were below c , the firm charging it would be making losses. Only $p_1 = p_2 = c$ survives: neither firm can gain by raising its price (it would sell nothing) or lowering it (it would make losses). $\square$

Two firms are enough to produce the competitive outcome. This is called a paradox because it seems implausible that duopolists make no profit. It is resolved in several ways, each of which is itself informative:

- *Capacity constraints*: if firms cannot supply the whole market, undercutting is less attractive. Kreps and Scheinkman (1983) showed that when firms first choose capacities and then compete in prices, the outcome resembles Cournot. Industries with long-lived capacity (steel, cement, airlines on a route) are often well described by Cournot.
- *Product differentiation*: if products differ, a small price cut does not capture the whole market.
- *Repeated interaction*: collusion may be sustained (Chapter 23).

Worked Example 24.13. Bertrand with differentiated products

Two coffee chains face demands $q_1 = 100 - 2p_1 + p_2$ and $q_2 = 100 - 2p_2 + p_1$; marginal cost is 10. Firm 1 maximizes $(p_1 - 10)(100 - 2p_1 + p_2)$. The first-order condition is $100 - 4p_1 + p_2 + 20 = 0$, so

$$p_1 = \frac{120 + p_2}{4}.$$

Prices are strategic complements: if firm 2 raises its price, firm 1 does too. The symmetric equilibrium is $p = (120 + p)/4$, $p^* = 40$. Each sells $100 - 80 + 40 = 60$ and earns $30 \times 60 = 1,800$. Differentiation allows prices well above marginal cost.

24.6.4 Collusion

Oligopolists earn more by colluding than by competing. Explicit agreements to fix prices or divide markets are illegal in most countries (Section 24.10), but tacit collusion, sustained by the threat of future price wars, is possible. Chapter 23's repeated-game analysis shows that it is easier when there are few firms (each firm's share of collusive profit is larger, so cheating is less tempting), when interactions are frequent, when prices are transparent so that cheating is detected quickly, when demand is stable, and when entry is difficult. These factors guide competition authorities in assessing whether a market is prone to collusion.

24.7 Monopolistic Competition

Many markets—restaurants, clothing brands, hairdressers, apps—have many sellers of *differentiated* products and free entry. Edward Chamberlin's (1933) model of **monopolistic competition** combines elements of monopoly and competition. Each firm faces a downward-sloping demand curve for its own variety and sets $MR = MC$, charging a markup. But free entry means that, if firms earn positive profits, new varieties enter, taking customers from existing firms and shifting their demand curves left. In long-run equilibrium, each firm's demand curve is *tangent* to its average cost curve: price equals average cost and economic profit is zero.

Because the demand curve slopes down, the tangency occurs on the downward-sloping part of the average cost curve: each firm produces less than its minimum efficient scale. This **excess capacity** might seem wasteful, but it is the price of variety. Consumers value having many restaurants to choose from; fewer, larger firms would have lower costs but offer less choice. Whether the market provides too much or too little variety is ambiguous in general. Monopolistic competition is also the market structure underlying the "new trade theory" of Paul Krugman, which explains why similar countries—Germany and France, say—trade large volumes of similar goods (German and French cars) with each other.

24.8 Networks and Platforms

24.8.1 Network effects

A product exhibits **network effects** (network externalities) if its value to each user rises with the number of other users. Telephones, messaging apps, social networks, payment systems and operating systems are examples. Network effects create demand-side economies of scale and have three important consequences:

1. *Tipping*: markets may tip toward a single dominant network, since users flock to the largest network, making it larger still.
2. *Multiple equilibria and lock-in*: which network wins can depend on early luck and expectations (a coordination game, Chapter 23), and an inferior standard may persist because no individual user gains by switching alone.
3. *Competition for the market* rather than in the market: firms compete intensely, often at a loss, to become the dominant network, then enjoy durable market power.

24.8.2 Two-sided platforms

A **two-sided platform** serves two distinct groups whose participation benefits each other: a search engine serves users and advertisers; a ride-hailing app serves riders and drivers; a card network serves cardholders and merchants; a marketplace serves buyers and sellers. The platform's pricing must take account of *indirect* network effects: more users on one side attract more on the other.

A striking implication is that the profit-maximizing price on one side can be zero or even negative. A search engine charges users nothing and earns its revenue from advertisers, because each additional user makes the platform more valuable to advertisers. A price below marginal cost on one side is therefore not evidence of predation; conversely, a zero price does not mean the absence of market power. Market power can show up as higher advertising prices, lower quality (more ads, less privacy), or reduced innovation. This complicates competition analysis, because the usual test—can the firm profitably raise its price?—must be adapted when the price to one side is zero.

24.9 Measuring Concentration

Two measures of market concentration are widely used:

- The **concentration ratio** CR_k : the combined market share of the k largest firms.
- The **Herfindahl–Hirschman index** (HHI): the sum of squared market shares, with shares measured in percentage points, so that the index ranges from near 0 (many tiny firms) to 10,000 (monopoly).

A market with four firms with shares of 40, 30, 20 and 10 per cent has $CR_4 = 100$ and $HHI = 1,600 + 900 + 400 + 100 = 3,000$. The HHI gives more weight to large firms and is linked to average market power by Theorem 24.10.

Competition authorities use the HHI to screen mergers. Under the U.S. Merger Guidelines issued by the Department of Justice and Federal Trade Commission in 2023, a market with HHI above 1,800 is considered highly concentrated, and a merger that increases the HHI by more than 100 in such a market is presumed likely to lessen competition substantially. Concentration is only a starting point: the same HHI can mask very different degrees of

competition depending on entry barriers, the closeness of substitutes and whether the firms compete in quantities or prices.

24.10 Competition Policy

Competition (or antitrust) law aims to prevent the creation and abuse of market power. Its three main pillars are similar across jurisdictions:

1. **Prohibition of anti-competitive agreements**, especially cartels that fix prices, restrict output or divide markets. In the United States this is Section 1 of the Sherman Act (1890); in the European Union, Article 101 of the Treaty on the Functioning of the European Union; in China, the Anti-Monopoly Law (2008); in Malaysia, the Competition Act 2010.
2. **Prohibition of abuse of dominance or monopolization**: dominant firms may not exclude rivals by means other than competition on the merits—for example through exclusive dealing, tying, or predatory pricing. In the U.S., Section 2 of the Sherman Act; in the EU, Article 102. Germany's competition authority, the Bundeskartellamt, was given additional powers in 2021 to address large digital firms of "paramount significance for competition across markets."
3. **Merger control**: mergers that would substantially lessen competition may be blocked or approved subject to conditions.

Economics contributes to competition policy by defining relevant markets (using cross-price elasticities, Chapter 20, and the question of whether a hypothetical monopolist of a set of products could profitably raise prices by a small but significant amount), by predicting the effects of mergers (using models like those in this chapter), and by distinguishing exclusionary conduct from vigorous competition.

Case: United States v. Google: defaults, scale and exclusion

The U.S. Department of Justice, joined by many states, sued Google in 2020, alleging that it had unlawfully maintained monopolies in general search and search advertising. In August 2024 the court found that Google was a monopolist in general search services and in search text advertising, and that its exclusive default agreements—with Apple, Android device makers, and browser developers—were unlawful exclusionary conduct. Evidence at trial showed payments to Apple alone reportedly on the order of 20 billion dollars in 2022.

The economics of this chapter explain why defaults were worth so much. Search is a two-sided platform: users attract advertisers. It also has strong data-driven scale economies: more queries provide more data on which results users click, which improves quality, which attracts more queries. Default status matters because most users do not change defaults (a behavioural switching cost). By securing defaults across the major access points, Google denied rivals the query volume they needed to improve—a form of exclusion that raised entry barriers without any change in Google's price to users, which was zero throughout.

In 2025 the court ordered remedies. It prohibited Google from entering or maintaining exclusive contracts for the distribution of its search and related products, and required it to share certain search index and user-interaction data with qualified competitors, but declined the government's request to require the divestiture of the Chrome browser. A separate 2025 decision found that Google had also unlawfully monopolized parts of the advertising-technology market. The case illustrates several themes: market power

without a positive consumer price; the role of scale and network effects; the distinction between winning through superior quality and preserving dominance through exclusion; and the difficulty of designing remedies that restore competition in markets with strong scale economies.

Summary

- Market power is the ability to price above marginal cost; it persists only behind barriers to entry.
- A monopolist sets $MR = MC$; its markup satisfies $(P - MC)/P = -1/\varepsilon$, so it operates where demand is elastic. Monopoly causes a deadweight loss, possibly augmented by rent-seeking, though temporary monopoly can reward innovation.
- Natural monopolies are regulated; marginal-cost pricing needs subsidy, average-cost pricing leaves some deadweight loss, and price caps sharpen efficiency incentives.
- Price discrimination requires market power, sorting and no resale. Third-degree discrimination charges more where demand is less elastic; two-part tariffs can extract surplus efficiently.
- Monopsony depresses wages and employment; a minimum wage in the right range can raise both.
- Cournot oligopoly yields output between monopoly and competition, converging to competition as firms are added; the average Lerner index equals $HHI/|\varepsilon|$. A Stackelberg leader gains from commitment. Bertrand competition with identical products yields marginal-cost pricing; differentiation and capacity constraints restore markups.
- Monopolistic competition combines markups with zero long-run profit and excess capacity, the cost of variety.
- Network effects and two-sided platforms produce tipping, lock-in and zero prices on one side, complicating the assessment of market power.
- Competition policy prohibits cartels and exclusionary conduct and screens mergers, using concentration measures and economic models.

Key terms

market power; barriers to entry; monopoly; marginal revenue; Lerner index; deadweight loss of monopoly; rent-seeking; natural monopoly; marginal-cost pricing; average-cost pricing; rate-of-return regulation; price-cap regulation; price discrimination (first-, second-, third-degree); two-part tariff; monopsony; marginal factor cost; marginal revenue product; oligopoly; Cournot model; reaction function; Stackelberg model; Bertrand model; Bertrand paradox; collusion; monopolistic competition; excess capacity; network effects; tipping; two-sided platform; concentration ratio; Herfindahl–Hirschman index; competition policy; merger control.

Exercises

Exercise 24.1 (Monopoly). A monopolist faces $P = 80 - Q$ and has $TC = 100 + 10Q$. Find the profit-maximizing quantity, price and profit. Compute the Lerner index and verify it equals $-1/\varepsilon$. Compute the deadweight loss.

Exercise 24.2 (Constant elasticity). A monopolist faces demand with constant elasticity -3 and has constant marginal cost 12. What price does it charge? If a per-unit tax of 3 is imposed, by how much does the price rise? Compare with a competitive market with horizontal supply.

Exercise 24.3 (Natural monopoly). A water utility has $TC = 1,200 + 2Q$ and faces demand $P = 20 - 0.1Q$. Find (a) the unregulated monopoly outcome; (b) the outcome under marginal-cost pricing and the required subsidy; (c) the outcome under average-cost pricing (take the larger quantity if two solutions exist).

Exercise 24.4 (Price discrimination). A theatre faces two groups: adults with $Q_A = 200 - 4P_A$ and students with $Q_S = 120 - 4P_S$. Marginal cost is 5 per seat and capacity is not binding. Find the discriminating prices. What single price would maximize profit if discrimination were banned? (Consider whether both groups are served.)

Exercise 24.5 (Two-part tariff). Each of 100 identical gym customers has demand for visits $q = 20 - 2p$. The marginal cost of a visit is 2. Find the profit-maximizing two-part tariff and the gym's profit.

Exercise 24.6 (Monopsony). A firm's marginal revenue product of labour is $MRP = 40 - L$, and labour supply is $w = 4 + 0.5L$. Find employment and the wage under monopsony. Find the minimum wage that maximizes employment.

Exercise 24.7 (Cournot). Two firms face $P = 200 - 2Q$ and have marginal costs $c_1 = 20$ and $c_2 = 40$. Find the Cournot equilibrium quantities, price and profits. Verify the Lerner-share relationship for each firm.

Exercise 24.8 (Number of firms). With $P = 120 - Q$ and $c = 30$, compute the Cournot price and total profit for $n = 1, 2, 3, 5, 10$. Describe the pattern.

Exercise 24.9 (Stackelberg). Using the market of Exercise 24.7 but with both marginal costs equal to 20, find the Stackelberg outcome when firm 1 leads. Compare with Cournot.

Exercise 24.10 (Bertrand). In Example 24.13, suppose firm 1's marginal cost falls to 6 while firm 2's stays at 10. Find the new equilibrium prices.

Exercise 24.11 (Concentration). A market has five firms with shares 35, 25, 20, 10 and 10 per cent. Compute CR_4 and the HHI. The two smallest firms propose to merge. By how much does the HHI rise? Would the merger be presumed anti-competitive under the 2023 U.S. guidelines?

Exercise 24.12 (Platforms). Explain why a platform might rationally set a zero or negative price to one side of the market. Why does this make the standard "small but significant price increase" test for market definition hard to apply?

Solutions

Solution 24.1. $MR = 80 - 2Q = 10$, $Q = 35$, $P = 45$. Profit $= 45 \times 35 - 100 - 350 = 1,575 - 450 = 1,125$. Lerner $= (45 - 10)/45 = 0.778$. $\varepsilon = -1 \times 45/35 = -1.286$; $-1/\varepsilon = 0.778$. Competitive output: $P = 10$, $Q = 70$. DWL $= \frac{1}{2} \times 35 \times 35 = 612.5$.

Solution 24.2. $P = MC / (1 - 1/3) = 1.5MC = 18$. With the tax, effective marginal cost is 15, so $P = 22.5$: the price rises by 4.5, *more* than the tax. Under competition with horizontal supply, the price would rise by exactly 3. With constant-elasticity demand, a monopolist applies a constant percentage markup to marginal cost, so it passes on more than 100% of a cost increase.

Solution 24.3. (a) $MR = 20 - 0.2Q = 2$, $Q = 90$, $P = 11$; profit $= 990 - 1,200 - 180 = -390$. An unregulated firm would not operate at all without some other revenue—in this example demand is too

weak for even a monopolist to cover the fixed cost. (b) $P = 2$: $Q = 180$; revenue 360, cost $1,200 + 360 = 1,560$; subsidy 1,200 (the fixed cost). (c) $P = AC$: $20 - 0.1Q = 1,200/Q + 2$, so $0.1Q^2 - 18Q + 1,200 = 0$, $Q^2 - 180Q + 12,000 = 0$, whose discriminant is $32,400 - 48,000 < 0$: no solution. Average cost is everywhere above demand, so no price allows the firm to break even. The service can be provided only with a subsidy; whether it should be depends on whether total surplus at $P = MC$ (consumer surplus $\frac{1}{2} \times 180 \times 18 = 1,620$) exceeds the fixed cost (1,200)—here it does, so provision with a subsidy is efficient.

Solution 24.4. Adults: $P_A = 50 - Q_A/4$, $MR_A = 50 - Q_A/2 = 5$, $Q_A = 90$, $P_A = 27.5$. Students: $P_S = 30 - Q_S/4$, $MR_S = 30 - Q_S/2 = 5$, $Q_S = 50$, $P_S = 17.5$. Single price: for $P \leq 30$ both groups buy and total demand is $Q = 320 - 8P$, so $P = 40 - Q/8$, $MR = 40 - Q/4 = 5$, $Q = 140$, $P = 22.5$, profit $= 17.5 \times 140 = 2,450$. Serving adults only: $P = 27.5$, profit $= 22.5 \times 90 = 2,025$. So the single price is 22.5 and both groups are served. Discrimination profit: $22.5 \times 90 + 12.5 \times 50 = 2,025 + 625 = 2,650$. Total output is 140 in both cases.

Solution 24.5. Set the per-visit price at marginal cost, 2: each customer makes $20 - 4 = 16$ visits. Consumer surplus per customer: inverse demand $p = 10 - q/2$, so $CS = \frac{1}{2} \times 16 \times (10 - 2) = 64$. Membership fee 64. Profit $= 100 \times 64 = 6,400$ (the per-visit margin is zero).

Solution 24.6. $MFC = 4 + L$. Monopsony: $40 - L = 4 + L$, $L = 18$, $w = 4 + 9 = 13$. Competitive point: $40 - L = 4 + 0.5L$, $L = 24$, $w = 16$. The minimum wage of 16 maximizes employment at 24.

Solution 24.7. Firm 1: $200 - 4q_1 - 2q_2 = 20$, so $q_1 = 45 - q_2/2$. Firm 2: $200 - 2q_1 - 4q_2 = 40$, so $q_2 = 40 - q_1/2$. Solving: $q_1 = 45 - 20 + q_1/4$, $0.75q_1 = 25$, $q_1 = 33.33$, $q_2 = 23.33$, $Q = 56.67$, $P = 86.67$. Profits: $66.67 \times 33.33 = 2,222$ and $46.67 \times 23.33 = 1,089$. Elasticity at equilibrium: $\varepsilon = -(1/2)(86.67/56.67) = -0.765$. Shares 0.588 and 0.412. Lerner: firm 1 $66.67/86.67 = 0.769$; $s_1/|\varepsilon| = 0.588/0.765 = 0.769$. Firm 2 $46.67/86.67 = 0.538$; $0.412/0.765 = 0.538$. Note that the market elasticity is below 1 in absolute value, which is possible in oligopoly (each firm's own elasticity exceeds 1).

Solution 24.8. $P = (120 + 30n)/(n + 1)$; total profit $= n(P - 30)q_i$ with $q_i = 90/(n + 1)$, so $\Pi = n[90/(n + 1)]^2$. $n = 1$: $P = 75$, $\Pi = 2,025$. $n = 2$: $P = 60$, $\Pi = 1,800$. $n = 3$: $P = 52.5$, $\Pi = 1,518.75$. $n = 5$: $P = 45$, $\Pi = 1,125$. $n = 10$: $P = 38.18$, $\Pi = 669.4$. Price and industry profit fall toward the competitive values (30 and 0) as the number of firms rises, quickly at first and then more slowly.

Solution 24.9. Follower's best response: $q_2 = 45 - q_1/2$. Leader maximizes $(200 - 2q_1 - 2(45 - q_1/2) - 20)q_1 = (90 - q_1)q_1$, so $q_1 = 45$, $q_2 = 22.5$, $Q = 67.5$, $P = 65$. Profits $45 \times 45 = 2,025$ and $45 \times 22.5 = 1,012.5$. Cournot with equal costs: $q_i = 180/6 = 30$, $P = 80$, profits 1,800 each. The leader gains, the follower loses, price falls.

Solution 24.10. Firm 1: $\max(p_1 - 6)(100 - 2p_1 + p_2)$: $100 - 4p_1 + p_2 + 12 = 0$, $p_1 = (112 + p_2)/4$. Firm 2: $p_2 = (120 + p_1)/4$. Substituting: $p_1 = (112 + 30 + p_1/4)/4 = 35.5 + p_1/16$, so $p_1 = 37.87$ and $p_2 = 39.47$. The low-cost firm lowers its price; its rival, with prices as strategic complements, also lowers its price slightly.

Solution 24.11. $CR_4 = 90$. $HHI = 1,225 + 625 + 400 + 100 + 100 = 2,450$. After the merger (shares 35, 25, 20, 20): $1,225 + 625 + 400 + 400 = 2,650$, an increase of 200 ($= 2 \times 10 \times 10$). The market is highly concentrated (above 1,800) and the increase exceeds 100, so the merger would be presumed to lessen competition substantially—although the parties could argue that it lets them compete more effectively against the two leaders.

Solution 24.12. Each additional user on the subsidized side raises the platform's revenue from the other side, so the effective marginal cost of attracting a user is the marginal cost minus that indirect benefit, which can be negative. The standard test asks whether a hypothetical monopolist could profitably raise price by 5–10%; a percentage increase on a zero price is meaningless, and a price rise on one side loses users on both sides. Analysts adapt the test by considering quality reductions (more ads, less privacy) or by analysing both sides together.

Risk, Information and Contracts

After studying this chapter you should be able to

1. Explain expected utility and why it differs from expected value.
2. Define risk aversion, the certainty equivalent and the risk premium, and compute them.
3. Show why risk-averse individuals buy full insurance at actuarially fair prices, and explain risk pooling.
4. Distinguish hidden information (adverse selection) from hidden action (moral hazard).
5. Analyse Akerlof's market for lemons and adverse selection in insurance and credit markets.
6. Analyse signalling with Spence's education model, and screening by menus of contracts.
7. Solve a simple principal-agent problem and explain the trade-off between incentives and insurance.
8. Explain how collateral, covenants, monitoring, reputation and regulation respond to information problems.
9. Explain why information problems give banks and other financial intermediaries a reason to exist.

A motivating question

In the years before 2007, American lenders made millions of mortgage loans to borrowers with weak credit histories, and many of those loans were quickly sold and packaged into securities held by investors around the world. When house prices stopped rising, defaults surged and the securities lost much of their value, setting off the global financial crisis. Why were so many poor-quality loans made? One answer, supported by careful empirical work, is that the lenders who screened borrowers did not bear the losses when loans went bad: they had sold them. The information needed to judge each loan was held by the originator; the risk was held by someone else.

Every model in Chapters 20 to 8 assumed that everyone knew the quality of what was being traded and could see what the other party did. This chapter relaxes both assumptions. It first develops a theory of choice under risk, then studies what happens when one side of a transaction knows more than the other. The results explain a remarkable range of institutions: insurance deductibles, warranties, credit scores, collateral, executive bonuses, university degrees, and—as Part VII shows—the existence and regulation of banks.

25.1 Choice under Risk

25.1.1 Lotteries and expected value

A **lottery** is a list of possible outcomes with their probabilities. A person choosing whether to buy insurance, invest in a firm or change jobs is choosing among lotteries. The simplest rule would be to choose the lottery with the highest expected value (Chapter 13). But that rule is inconsistent with obvious behaviour. People buy insurance, although insurance companies must on average pay out less than they collect in premiums, so insurance lowers the buyer's expected wealth. And few people would pay a large sum to play the following game: a coin is tossed until it lands heads; if the first head appears on toss k , you win 2^k . The expected prize is $\frac{1}{2} \cdot 2 + \frac{1}{4} \cdot 4 + \frac{1}{8} \cdot 8 + \dots = 1 + 1 + 1 + \dots$, which is infinite. This is the **St Petersburg paradox**, posed by Nicolaus Bernoulli and resolved by Daniel Bernoulli in 1738: people care not about money itself but about the utility it brings, and the utility of money rises less than proportionally.

25.1.2 Expected utility

Definition 25.1. Expected utility

A person with utility-of-wealth function $u(w)$ evaluates a lottery that yields wealth w_k with probability p_k by its **expected utility**

$$\mathbb{E}[u(w)] = \sum_k p_k u(w_k),$$

and chooses the lottery with the highest expected utility.

John von Neumann and Oskar Morgenstern (1944) showed that if preferences over lotteries satisfy a small set of axioms—completeness, transitivity, continuity, and an “independence” axiom—then they can be represented by expected utility. Unlike the ordinal utility of Chapter 20, this *von Neumann–Morgenstern utility* is cardinal up to positive linear transformations: $a + bu(w)$ with $b > 0$ represents the same preferences, but $\ln u(w)$ generally does not, because curvature now carries meaning—it measures attitudes to risk.

25.1.3 Risk aversion

Definition 25.2. Risk aversion

A person is **risk-averse** if she prefers receiving the expected value of any lottery for certain to facing the lottery itself: $u(\mathbb{E}[w]) > \mathbb{E}[u(w)]$ for every non-degenerate lottery. She is **risk-neutral** if she is indifferent, and **risk-loving** if she prefers the lottery.

By Jensen's inequality (Chapter 13), a person is risk-averse if and only if her utility function is strictly concave: $u'' < 0$. Diminishing marginal utility of wealth is what makes a sure thing better than a gamble with the same average: the utility lost when the gamble turns out badly exceeds the utility gained when it turns out well.

Definition 25.3. Certainty equivalent and risk premium

The **certainty equivalent** (CE) of a lottery is the sure amount of wealth that gives the same utility as the lottery: $u(CE) = \mathbb{E}[u(w)]$. The **risk premium** is the difference between the lottery's expected value and its certainty equivalent: $RP = \mathbb{E}[w] - CE$. It is the amount a person would give up to eliminate the risk.

Worked Example 25.4. Certainty equivalent and the demand for insurance

A homeowner has wealth 10,000 and utility $u(w) = \sqrt{w}$. With probability 0.1 a fire destroys property worth 6,400.

Expected wealth: $0.9 \times 10,000 + 0.1 \times 3,600 = 9,360$. *Expected utility:* $0.9 \times 100 + 0.1 \times 60 = 96$. *Certainty equivalent:* $CE = 96^2 = 9,216$. *Risk premium:* $9,360 - 9,216 = 144$.

The homeowner would be willing to pay up to $10,000 - 9,216 = 784$ for insurance that fully covered the loss, although the expected loss is only $0.1 \times 6,400 = 640$. An insurer charging 700, say, would make an expected profit of 60 per policy, and the homeowner would be better off than uninsured. Both gain: the insurer is better able to bear the risk because it pools many independent policies.

25.1.4 Measuring risk aversion

The concavity of u measures risk aversion, but u'' alone is not a good measure because it changes under linear transformations of u . Kenneth Arrow and John Pratt proposed:

$$\text{absolute risk aversion } A(w) = -\frac{u''(w)}{u'(w)}, \quad \text{relative risk aversion } R(w) = -\frac{w u''(w)}{u'(w)}.$$

For small risks, the risk premium is approximately $\frac{1}{2}A(w)\sigma^2$, where σ^2 is the variance of the gamble: risk premiums rise with risk aversion and with the variance of the risk. The family of utility functions with constant relative risk aversion γ ,

$$u(w) = \frac{w^{1-\gamma}}{1-\gamma} \quad (\gamma \neq 1), \quad u(w) = \ln w \quad (\gamma = 1),$$

is widely used in macroeconomics and finance. With CRRA preferences, a person's attitude to a gamble expressed as a *percentage* of wealth does not depend on wealth. Square-root utility is CRRA with $\gamma = 0.5$. In the Euler equation of Chapter 9, γ also governs how willing a household is to substitute consumption over time: its reciprocal, $1/\gamma$, is the **elasticity of intertemporal substitution**. That the same parameter governs both risk-taking and saving is a restriction of the CRRA form, not a fact about people, and much research in finance relaxes it.

25.2 Insurance and Risk Sharing**25.2.1 Full insurance at fair prices****Proposition 25.5. Full insurance**

A risk-averse person who can buy any amount of insurance at an **actuarially fair price**—a premium per unit of coverage equal to the probability of loss—chooses to insure fully.

Proof. Let wealth be w , the possible loss L with probability p , and coverage x at premium px . Wealth is $w - px$ with no loss and $w - px - L + x$ with a loss. Expected utility is $(1-p)u(w - px) + pu(w - px - L + x)$. The first-order condition with respect to x is

$$-(1-p)p u'(w - px) + p(1-p) u'(w - px - L + x) = 0,$$

so $u'(w - px) = u'(w - px - L + x)$. Since u' is strictly decreasing, the two wealth levels must be equal: $x = L$. $\square$

Full insurance equalizes wealth across outcomes, and at a fair price it costs nothing in expected terms. When the premium includes a loading above the fair price (to cover the insurer's costs and profit), the optimal coverage is less than full: the buyer self-insures part of the loss.

25.2.2 Risk pooling and risk spreading

Why can insurers bear risks that individuals find costly? Two mechanisms, from Chapters 14 and 15, are at work. **Risk pooling**: by the law of large numbers, the average loss across many *independent* policies is very predictable, so the insurer faces little uncertainty per policy. **Risk spreading**: a large risk shared among many shareholders is a small risk for each, and the risk premium, proportional to variance, falls with the square of each person's share. Neither mechanism works for *correlated* risks. A nationwide fall in house prices, a pandemic or a large earthquake hits many policies at once; pooling offers little protection, which is why such risks are often insured with government support or not at all. The same logic reappears in Chapter 51: banks that hold loans exposed to a common shock—a housing market, a sector, the sovereign—are more fragile than their diversified appearance suggests.

25.3 Asymmetric Information

Definition 25.6. Asymmetric information

Asymmetric information exists when one party to a transaction knows something relevant that the other does not. It takes two main forms:

- **Hidden information** (or hidden type) before a contract is signed: the seller knows the quality of the car, the insurance buyer knows her health, the borrower knows the riskiness of his project. This leads to **adverse selection**.
- **Hidden action** after a contract is signed: the insured person's care, the manager's effort, the borrower's choice of how to use the loan. This leads to **moral hazard**.

Both violate an assumption of the first welfare theorem (Chapter 22), and both can cause markets to perform badly or not exist at all.

25.4 Adverse Selection

25.4.1 The market for lemons

George Akerlof (1970) showed how hidden information about quality can destroy a market.

Worked Example 25.7. Unravelling in the used-car market

Used cars vary in quality q , uniformly distributed between 0 and 1,000. Sellers know the quality of their own car and value it at q . Buyers cannot observe quality before buying but know the distribution; they value a car of quality q at $1.5q$. Because every buyer values every car more than its owner does, full information would lead every car to be sold, and every sale would create a gain from trade.

Now suppose a single market price p prevails. An owner sells only if $p \geq q$, so the cars offered have quality uniformly distributed between 0 and p , with average quality $p/2$. A risk-neutral buyer is willing to pay the expected value of a car offered at that price,

$1.5 \times p/2 = 0.75p$. Since $0.75p < p$ for every positive price, buyers are never willing to pay the price at which cars are offered. The only equilibrium is $p = 0$: no cars are traded.

The mechanism is **adverse selection**: at any price, the owners most willing to sell are those whose cars are worst, so the price buyers will pay reflects that below-average quality, which drives the best remaining cars out, and so on. The market “unravels.” The example is extreme—in practice some trade usually survives—but the lesson is general: when one side cannot observe quality, good-quality products may be driven out, and a market that would create large gains under full information may shrink or disappear.

25.4.2 Adverse selection in insurance

People who know they face higher risk are more eager to buy insurance. If an insurer charges a premium based on the average risk, low-risk people find it overpriced and many do not buy; the insured pool becomes riskier than average; premiums must rise; more low-risk people leave. Health insurance markets are the leading example. This is why many countries make health insurance compulsory or provide it publicly, and why the U.S. Affordable Care Act of 2010, which required insurers to accept all applicants at prices that could not depend on pre-existing conditions, also included an individual mandate (a tax penalty for being uninsured, later reduced to zero from 2019) and subsidies designed to keep healthy people in the pool.

25.4.3 Adverse selection in credit markets

Borrowers know more than lenders about the riskiness of their projects. Stiglitz and Weiss (1981) showed that raising the interest rate may reduce a lender’s expected return, because (i) safer borrowers, whose projects have lower payoffs in good states, drop out first when rates rise (adverse selection), and (ii) borrowers who remain have an incentive to choose riskier projects (moral hazard). The lender’s expected return may therefore peak at an interest rate below the market-clearing rate. At that rate, demand for loans exceeds supply, and lenders **ration credit**: some borrowers who would willingly pay more cannot borrow at any price. Credit rationing explains why small firms’ access to credit can dry up abruptly in a downturn and why the credit channel of monetary policy works partly through quantities rather than prices (Chapter 49).

25.5 Responses to Adverse Selection

Markets develop two broad responses: the informed side can *signal*, or the uninformed side can *screen*.

25.5.1 Signalling

A **signal** is a costly action by the informed party that credibly conveys information, because it is less costly for high-quality types than for low-quality types.

Worked Example 25.8. Spence’s education model

Workers are of two types, High and Low, with productivity 2 and 1 respectively. Employers cannot observe productivity but can observe years of education e . Suppose, for the sake of argument, that education does not raise productivity at all; its only role is as a signal. Its cost is e for a Low type and $e/2$ for a High type—able people find study less

burdensome. Employers pay each worker the expected productivity given what they infer.

Consider a *separating equilibrium* in which employers believe that workers with education at least e^* are High (and pay 2) and those with less are Low (and pay 1). Each type then chooses either $e = 0$ or $e = e^*$.

- A Low type must not want to mimic: $1 - 0 \geq 2 - e^*$, i.e. $e^* \geq 1$.
- A High type must want to acquire the signal: $2 - e^*/2 \geq 1$, i.e. $e^* \leq 2$.

Any e^* between 1 and 2 supports a separating equilibrium. The signal works because it is cheaper for High types: education costing 1 to 2 units deters Low types but not High types.

The education model has a sobering implication. If education had no productive value, society would be better off without it: resources are spent purely to sort people. In reality education surely raises productivity, but the signalling view suggests that some of the private return to credentials exceeds the social return. Other signals include product warranties (a firm that offers a long warranty signals confidence in its product, since warranties are costlier for makers of unreliable goods), costly advertising, and, in finance, dividend payments and the decision of insiders to retain a large equity stake in their firm.

25.5.2 Screening

In **screening**, the uninformed party offers a menu of contracts designed so that different types choose different options, revealing themselves. Rothschild and Stiglitz (1976) showed how insurers can offer a full-coverage policy at a high premium alongside a partial-coverage policy (with a large *deductible*) at a low premium per unit of coverage: high-risk customers prefer full coverage, low-risk customers accept the deductible. The low-risk type's contract must be distorted—less than full insurance—to stop the high-risk type from choosing it. The same logic underlies the second-degree price discrimination of Chapter 24. Credit scoring, required down payments and collateral are screening devices in lending: borrowers confident of repaying are more willing to pledge collateral.

25.5.3 Other responses

Reputation lets sellers who interact repeatedly commit to quality (Chapter 23). *Intermediaries and certifiers*—used-car dealers offering inspections, rating agencies, auditors—specialize in acquiring information. *Regulation* can mandate disclosure (securities laws require firms to publish audited accounts), set minimum quality standards, or make participation compulsory (as in social insurance).

25.6 Moral Hazard

25.6.1 The problem

Moral hazard arises when a party whose actions cannot be observed or contracted upon changes its behaviour because it does not bear the full consequences. An insured driver may drive less carefully; an insured homeowner may take fewer precautions against fire; a manager paid a fixed salary may exert less effort; a bank whose depositors are insured may take more risk. In each case, the transfer of risk that makes the contract valuable also weakens incentives.

Insurers respond with **deductibles** (the insured pays the first part of any loss) and **coinsurance** (the insured pays a share of each loss), which restore some incentive to take care at the cost of leaving the insured with some risk. There is an inherent trade-off: more insurance means weaker incentives.

25.6.2 The principal–agent problem

The general framework is the **principal–agent model**. A *principal* (a firm’s owners, an employer, an insurer, a lender) hires an *agent* (a manager, a worker, an insured person, a borrower) to take an action that the principal cannot observe directly but that affects an outcome the principal can observe.

Worked Example 25.9. Paying for performance

A shop owner (principal) hires a manager (agent). Monthly profit, before the manager’s pay, is either high (100) or low (0). If the manager works hard, profit is high with probability 0.8; if she shirks, 0.4. Hard work costs the manager 10 (in money-equivalent terms). The owner cannot observe effort, only profit. Both are risk-neutral, and the manager’s alternative job yields 0. The manager cannot be paid a negative amount (*limited liability*).

The owner offers a bonus b paid only if profit is high. The manager works hard if the expected pay from working hard, net of its cost, is at least that from shirking:

$$0.8b - 10 \geq 0.4b \iff b \geq 25.$$

This is the **incentive-compatibility constraint**. The cheapest bonus that induces effort is 25, with expected pay $0.8 \times 25 = 20$.

Is it worth inducing effort? With effort and $b = 25$: expected profit to the owner $0.8 \times 100 - 20 = 60$. Without effort (pay 0): $0.4 \times 100 = 40$. The owner induces effort.

The cost of hidden action. If the owner could observe effort, she would simply pay 10 for hard work, earning $80 - 10 = 70$. Hidden action costs her 10: the manager earns an **information rent** of $20 - 10 = 10$ above her cost of effort, because limited liability prevents the owner from penalizing low profit.

The example isolates one source of inefficiency, limited liability. When the agent is risk-averse, a second source appears: incentive pay exposes the agent to risk, for which she must be compensated. The optimal contract then balances incentives against insurance, and it generally involves less than full effort-inducing pay. Two further principles follow from the theory:

- The **informativeness principle** (Holmström 1979): pay should depend on any observable variable that carries information about effort. A manager’s bonus should depend on performance *relative* to competitors facing the same market conditions, filtering out luck.
- **Multitasking** (Holmström and Milgrom 1991): if an agent performs several tasks and only some are measurable, strong incentives on the measurable ones divert effort from the others. Teachers paid on test scores may “teach to the test”; bankers paid on loan volume may neglect loan quality.

Bridge. Executive pay, bank bonuses and the incentives of mortgage originators are all principal–agent problems. Chapter 31 examines how reported earnings can be manipulated when pay depends on them, and Chapter 51 examines how bank regulation tries to limit moral hazard created by deposit insurance and implicit government guarantees.

25.7 Contracts, Collateral and Financial Intermediation

25.7.1 Incomplete contracts and hold-up

Real contracts cannot specify every contingency. When one party must make an investment specific to a relationship—a supplier who builds a plant next to a single customer’s factory—the other party can later **hold up** the investor by renegotiating terms, knowing that the investment has no value elsewhere. Anticipating hold-up, the investor under-invests. Vertical integration (merging the two parties, as Coase’s theory of the firm suggested) and long-term contracts are responses. This is one reason why some supply relationships are organized inside firms and others through markets.

25.7.2 Tools of the lender

Lending is shot through with both adverse selection and moral hazard. Lenders use:

- **Collateral:** assets the lender can seize on default. Collateral reduces the lender’s loss, screens borrowers (those confident of repaying are more willing to pledge), and gives the borrower an incentive to avoid default.
- **Net worth requirements:** a borrower with a large equity stake has more to lose from risky behaviour. This is the logic of down payments on mortgages—and of capital requirements on banks (Chapter 51).
- **Covenants:** contract terms restricting the borrower’s actions (limits on further borrowing, dividend payments or asset sales).
- **Monitoring:** checking borrowers’ behaviour and finances.
- **Relationships:** long-term relationships generate information and reputational incentives.

25.7.3 Why banks exist

If individual savers had to screen and monitor each firm they lent to, the effort would be duplicated many times over, or—since each small lender would expect others to monitor—not done at all (a free-rider problem, Chapter 26). Douglas Diamond (1984) showed that a bank can act as a **delegated monitor**: depositors lend to the bank, the bank lends to many firms and monitors them. Diversification across many loans makes the bank’s own return predictable enough that depositors need not monitor the bank closely. Banks also transform illiquid long-term loans into liquid deposits, which exposes them to runs (Chapter 23’s coordination game and Chapter 51). The information problems of this chapter thus explain both why banks exist and why they are fragile.

Case: Securitization and screening in the U.S. subprime mortgage market

Before 2007, a large share of U.S. subprime mortgages was originated by lenders who sold them to be pooled into mortgage-backed securities—the “originate-to-distribute” model. In principle, securitization spreads risk efficiently. But it separates the party who screens the borrower from the party who bears the default risk, creating moral hazard in screening.

Keys, Mukherjee, Seru and Vig (2010) tested this directly. Industry practice made loans to borrowers with a credit (FICO) score just above 620 much easier to securitize than loans to borrowers just below. Borrowers on either side of the threshold were otherwise similar—a regression-discontinuity design (Chapter 17). The authors found that loans

just above 620, which were more likely to be securitized, defaulted at significantly *higher* rates than loans just below, even though the borrowers' observable credit quality was slightly better. The natural interpretation is that lenders screened less carefully the loans they expected to sell. Rating agencies, paid by the issuers of the securities they rated, faced their own incentive problems, and many investors relied on ratings rather than their own analysis.

The crisis thus combined several failures from this chapter: moral hazard in origination, adverse selection in which loans were sold, misaligned incentives among certifiers, and correlated risk (a nationwide fall in house prices) that defeated the diversification the securities appeared to offer. Post-crisis reforms included a requirement, under the U.S. Dodd–Frank Act of 2010 and parallel rules in the European Union, that securitizers retain part of the credit risk—generally 5 per cent—of the loans they securitize (“skin in the game”), a direct application of the principle that agents must bear some consequences of their unobservable actions.

25.8 Limitations

Expected-utility theory is violated in systematic ways in laboratory experiments: people overweight small probabilities, treat losses differently from gains relative to a reference point (Kahneman and Tversky's prospect theory), and are influenced by framing. These departures help explain why many people both gamble and insure, and why demand for insurance against rare catastrophes is often too low. The models of asymmetric information assume that the uninformed party understands the incentives of the informed one—used-car buyers anticipate adverse selection, lenders anticipate moral hazard. When market participants fail to anticipate these problems, as many investors in mortgage securities did, the consequences can be worse than the models predict.

Summary

- Expected utility evaluates lotteries by probability-weighted utility. Concave utility implies risk aversion; the risk premium is expected value minus the certainty equivalent.
- At fair prices, risk-averse people insure fully. Pooling and spreading make risk cheaper to bear, but not for correlated risks.
- Hidden information before contracting causes adverse selection; hidden action after contracting causes moral hazard.
- Adverse selection can unravel markets (lemons), make insurance pools riskier, and cause credit rationing.
- Signals work when they are cheaper for high-quality types; screening menus induce types to reveal themselves at the cost of distorting some contracts.
- In principal–agent problems, incentive pay induces effort but may leave rents or impose risk on agents; pay should use all informative signals, and strong incentives on measured tasks can distort unmeasured ones.
- Collateral, net worth, covenants, monitoring and relationships address information problems in lending; banks exist partly as delegated monitors, which also makes them fragile.

Key terms

lottery; St Petersburg paradox; expected utility; von Neumann–Morgenstern utility; risk aversion; risk neutrality; certainty equivalent; risk premium; absolute and relative risk aversion; CRRA utility; elasticity of intertemporal substitution; actuarially fair insurance; risk pooling; risk spreading; asymmetric information; hidden information; hidden action; adverse selection; market for lemons; credit rationing; signalling; separating equilibrium; screening; deductible; coinsurance; moral hazard; principal–agent model; incentive-compatibility constraint; limited liability; information rent; informativeness principle; multitasking; hold-up; collateral; covenant; delegated monitoring; securitization.

Exercises

Exercise 25.1 (Certainty equivalent). A person with $u(w) = \ln w$ and wealth 100 is offered a gamble that adds 50 or subtracts 50 with equal probability. Compute her expected wealth, expected utility, certainty equivalent and risk premium. Would she accept the gamble for free?

Exercise 25.2 (Insurance). In Example 25.4, the insurer offers full coverage for a premium of 700. What is the homeowner's utility if she buys it? What is the highest premium the insurer can charge? If the insurer's administrative cost is 50 per policy, what range of premiums makes both parties better off?

Exercise 25.3 (Risk aversion measures). Compute the Arrow–Pratt absolute and relative risk aversion for (a) $u = \ln w$; (b) $u = \sqrt{w}$; (c) $u = -e^{-0.01w}$. Which has constant absolute risk aversion?

Exercise 25.4 (Pooling). Each of n independent households faces a loss of 10,000 with probability 0.01. What are the mean and standard deviation of the average loss per household when $n = 1$, $n = 100$ and $n = 10,000$? Explain why this matters for an insurer. What changes if the losses are perfectly correlated?

Exercise 25.5 (Lemons). In Example 25.7, suppose buyers value a car of quality q at $2.5q$ instead of $1.5q$. Find the equilibrium price and the range of qualities traded. Are all gains from trade realized?

Exercise 25.6 (Lemons with a floor). Suppose instead that quality is uniformly distributed between 500 and 1,000, sellers value a car at q and buyers at $1.2q$. Find the equilibrium (assume buyers pay expected value given the cars offered and consider the highest price consistent with equilibrium). Do all cars trade?

Exercise 25.7 (Signalling). In Example 25.8, suppose the cost of education for a High type is $e/3$ rather than $e/2$. Find the range of separating education levels. Which level would High types prefer if they could coordinate employers' beliefs? What is the social waste in that equilibrium?

Exercise 25.8 (Principal–agent). In Example 25.9, suppose the cost of effort rises to 20. Find the minimum bonus that induces effort and determine whether the owner still wants to induce it. What if the probabilities of high profit become 0.9 (effort) and 0.5 (no effort), with cost 20?

Exercise 25.9 (Moral hazard in insurance). Explain why a health insurer might require a co-payment for each doctor's visit even when administrative costs of processing claims are zero. What is the cost of doing so?

Exercise 25.10 (Credit rationing). A bank lends 100 to a borrower, who can choose between two projects: a safe project that returns 130 for certain, and a risky project that returns 150 with

probability 0.5 and 0 otherwise. The borrower has limited liability (repays nothing if the project fails). At what repayment R per 100 lent does the borrower switch from the safe to the risky project? Show that the bank's expected return is maximized at that repayment and falls if the bank charges more.

Exercise 25.11 (Skin in the game). Explain, using the principal–agent model, why requiring securitizers to retain 5% of the credit risk of the loans they sell might improve screening. Why might 5% be too little?

Solutions

Solution 25.1. Expected wealth 100. Expected utility $0.5 \ln 150 + 0.5 \ln 50 = 0.5(5.011) + 0.5(3.912) = 4.461$. $CE = e^{4.461} = 86.6$. Risk premium $= 100 - 86.6 = 13.4$. She would reject the gamble, since $\ln 100 = 4.605 > 4.461$.

Solution 25.2. Wealth $10,000 - 700 = 9,300$ for certain; utility $\sqrt{9,300} = 96.44 > 96$: she buys. Highest premium: 784. The insurer needs at least $640 + 50 = 690$. Premiums between 690 and 784 make both better off.

Solution 25.3. (a) $u' = 1/w$, $u'' = -1/w^2$: $A = 1/w$, $R = 1$. (b) $u' = \frac{1}{2}w^{-1/2}$, $u'' = -\frac{1}{4}w^{-3/2}$: $A = 1/(2w)$, $R = 1/2$. (c) $u' = 0.01e^{-0.01w}$, $u'' = -0.0001e^{-0.01w}$: $A = 0.01$ (constant), $R = 0.01w$. Case (c) has constant absolute risk aversion.

Solution 25.4. Mean loss per household $= 0.01 \times 10,000 = 100$. The standard deviation of one household's loss is $10,000\sqrt{0.01 \times 0.99} = 995$. The average over n independent households has standard deviation $995/\sqrt{n}$: 995, 99.5 and 9.95. With 10,000 policies, the insurer's average claim per policy is almost certainly within about 20 of 100, so it can price accurately and hold modest reserves. With perfect correlation, all households lose together or not at all: the average loss is 10,000 with probability 0.01 and 0 otherwise, standard deviation 995 regardless of n . Pooling does not help.

Solution 25.5. At price p (with $p \leq 1,000$), average quality offered is $p/2$ and buyers pay $2.5p/2 = 1.25p \geq p$. The market can sustain any price; with competition among buyers driving the price up to their willingness to pay for the cars offered, the price rises until all cars are offered: $p = 1,000$, when average quality is 500 and buyers' valuation is 1,250. All cars trade and all gains are realized (buyers earn surplus on average). Adverse selection destroys trade only when the gap between buyers' and sellers' valuations is small relative to the dispersion of quality.

Solution 25.6. At price p between 500 and 1,000, cars with $q \leq p$ are offered, with average quality $(500 + p)/2$. Buyers pay $1.2(500 + p)/2 = 300 + 0.6p$. Equilibrium requires $300 + 0.6p = p$, so $p = 750$. Cars of quality 500 to 750 trade; those above 750 do not, although buyers would value each at 1.2 times its owner's valuation. Adverse selection shrinks the market by half.

Solution 25.7. Low type must not mimic: $e^* \geq 1$. High type must participate: $2 - e^*/3 \geq 1$, $e^* \leq 3$. High types prefer the least costly separating level, $e^* = 1$. The social waste is the High types' cost of education, $1/3$ per High worker, since education is assumed unproductive.

Solution 25.8. IC: $0.8b - 20 \geq 0.4b$, $b \geq 50$; expected pay 40. Owner's profit with effort: $80 - 40 = 40$; without: 40. The owner is indifferent. With probabilities 0.9 and 0.5: $0.9b - 20 \geq 0.5b$, $b \geq 50$; expected pay 45; profit with effort $90 - 45 = 45$ versus 50 without. The owner prefers not to induce effort. Although effort raises expected output by 40 (more than its cost of 20), the information rent ($45 - 20 = 25$) makes it unprofitable to induce—an inefficiency caused by hidden action.

Solution 25.9. Without a co-payment, the insured person's marginal cost of a visit is zero, so she visits whenever the benefit is positive, even if below the true cost of the visit—moral hazard. A co-payment raises her marginal cost toward the social cost, reducing low-value visits. The cost is that she bears some financial risk and may skip valuable visits, especially if she has little income; many schemes exempt preventive care or low-income patients for this reason.

Solution 25.10. Safe project: borrower keeps $130 - R$. Risky: $0.5(150 - R) = 75 - 0.5R$. The borrower chooses risky if $75 - 0.5R > 130 - R$, i.e. $R > 110$. For $R \leq 110$ the borrower chooses safe and the bank receives R , so its return rises to 110 at $R = 110$. For $R > 110$ the borrower chooses risky and the bank receives $0.5R$, at most $0.5 \times 150 = 75$. The bank's expected return peaks at $R = 110$; raising the rate further lowers it, because it induces the borrower to switch to a riskier (and here socially worse) project. If more borrowers want loans at $R = 110$ than the bank can fund, the bank rations credit rather than raising the rate. Collateral or covenants restricting project choice would relax the problem.

Solution 25.11. Retention makes the originator bear part of the loss from poor screening, so its payoff depends on the unobservable action (screening effort), satisfying an incentive-compatibility constraint it previously ignored. Five per cent may be too little if the private cost of careful screening exceeds 5% of the expected losses it prevents, if the retained tranche can be hedged away, or if the originator's limited liability means large losses are not borne in full.

Externalities, Public Goods, Auctions and Mechanism Design

After studying this chapter you should be able to

1. Define externalities and show why markets over-produce goods with negative externalities and under-produce those with positive externalities.
2. Measure the deadweight loss from an externality and design a Pigouvian tax or subsidy to correct it.
3. State the Coase theorem and the conditions under which it holds.
4. Compare standards, taxes and tradable permits as instruments of environmental policy, and show why cost-effective policy equalizes marginal abatement costs.
5. Classify goods by rivalry and excludability and derive the Samuelson condition for public goods.
6. Explain free-riding and the tragedy of the commons.
7. Analyse the main auction formats, prove that truthful bidding is dominant in a second-price auction, and state the revenue-equivalence theorem.
8. Explain the winner's curse and the basic ideas of mechanism design.

A motivating question

Since 2005 firms in the European Union that emit carbon dioxide from power generation and heavy industry have been required to hold a permit for every tonne they emit. The permits are traded in a market, and in 2022 and 2023 their price exceeded 80 euros per tonne. In 2021 China launched its own national emissions-trading system, the world's largest by covered emissions. Why would governments create a market for the right to pollute rather than simply limit each firm's emissions? And why does a problem like climate change require government action at all, when markets allocate most resources without it?

The answer lies in *externalities*: costs or benefits that fall on people outside a transaction. This chapter studies externalities, public goods and common resources—the last of the market failures identified by the welfare theorems in Chapter 22. It then turns to auctions and mechanism design, which ask how to design rules for allocating resources when people have private information, and which underpin the permit markets, spectrum sales and government bond auctions discussed in this and later chapters.

26.1 Externalities

26.1.1 Definition

Definition 26.1. Externality

An **externality** exists when one agent's action directly affects the welfare of another, and the affected party is not compensated through a market transaction. The effect is a **negative externality** (external cost) if it harms, and a **positive externality** (external benefit) if it helps.

Pollution, traffic congestion, antibiotic resistance from overuse of antibiotics, and noise are negative externalities. Vaccination (which protects others from infection), research that others can build on, and the upkeep of a house that raises neighbours' property values are positive externalities. The word "directly" matters: if my purchases raise the price you pay for the same good, that is a *pecuniary* effect working through prices, not an externality in this sense, and it does not cause inefficiency.

26.1.2 Negative externalities and over-production

With a negative externality in production, the firm's private marginal cost (MPC) understates the **marginal social cost**:

$$MSC = MPC + MEC,$$

where MEC is the **marginal external cost**. A competitive market produces where demand (marginal benefit) equals supply (MPC); the efficient quantity is where marginal benefit equals MSC . The market over-produces.

Worked Example 26.2. Pollution from steel

Inverse demand for steel is $P = 100 - Q$; the industry's private marginal cost is $MPC = 10 + Q$. Each unit of steel produces pollution causing damage of 20 to nearby residents, so $MEC = 20$ and $MSC = 30 + Q$.

Market outcome: $100 - Q = 10 + Q$, so $Q_m = 45$, $P_m = 55$. *Efficient outcome:* $100 - Q = 30 + Q$, so $Q^* = 35$, $P^* = 65$.

For each unit between 35 and 45, marginal social cost exceeds marginal benefit. The deadweight loss is the triangle between MSC and demand over that range: at $Q = 45$, $MSC = 75$ and demand is 55, so $DWL = \frac{1}{2} \times (45 - 35) \times (75 - 55) = 100$.

26.1.3 Pigouvian taxes

Arthur Pigou (1920) proposed that the government tax the activity at a rate equal to the marginal external cost at the efficient quantity. The tax makes producers face the full social cost of their output: they *internalize* the externality.

In Example 26.2, a **Pigouvian tax** of 20 per unit shifts the supply curve up to MSC , and the market produces the efficient 35 units. The tax raises revenue of $20 \times 35 = 700$, which the government can use to reduce other, distorting taxes or to compensate those harmed. A Pigouvian tax differs fundamentally from the taxes of Chapter 22: instead of creating a deadweight loss, it removes one.

When the marginal external cost varies with the quantity, the tax should equal the marginal external cost *at the efficient quantity*, which requires the government to know the damage function. For a positive externality, the corresponding remedy is a **Pigouvian subsidy** equal to

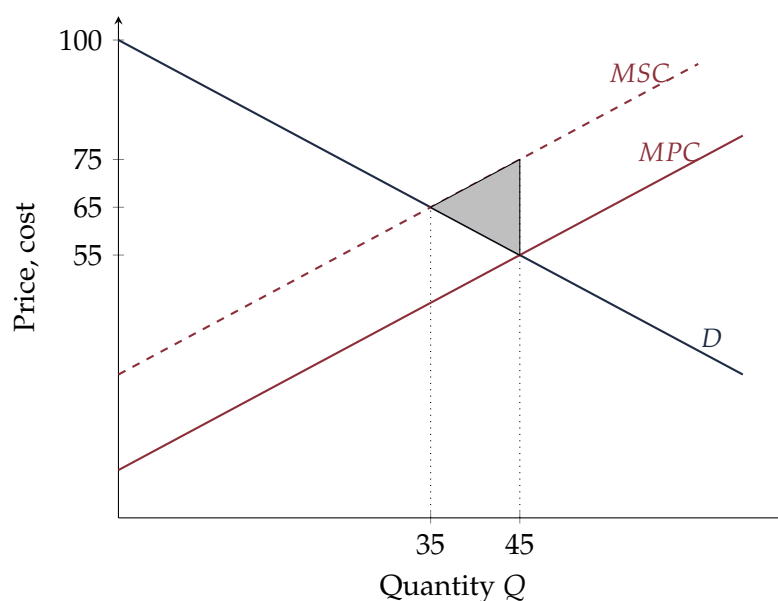


Figure 26.1: A negative production externality. The market produces 45, where demand meets private marginal cost; the efficient output is 35, where demand meets marginal social cost. The shaded triangle is the deadweight loss.

the marginal external benefit. Research subsidies and tax credits, vaccination programmes and support for education are partly justified this way.

► **Common error: “the efficient level of pollution is zero”**

Reducing pollution is costly: it requires cleaner technology, less output, or more expensive inputs. The efficient level of pollution is where the marginal benefit of further reduction (the avoided damage) equals its marginal cost. That is usually not zero. The goal of policy is to make polluters compare these costs, not to eliminate pollution at any price.

26.2 The Coase Theorem

26.2.1 Bargaining over externalities

Ronald Coase (1960) argued that the externality problem is reciprocal: the factory harms the residents by polluting, but restricting the factory harms its owners. Which harm should prevail? Coase pointed out that if property rights are clearly defined and the parties can bargain costlessly, they will reach the efficient outcome whoever holds the rights.

Worked Example 26.3. The factory and the laundry

A factory’s smoke causes 60 of damage per year to a neighbouring laundry. A filter that eliminates the smoke costs the factory 40 per year. Efficiency requires installing the filter ($40 < 60$).

The laundry has the right to clean air. The factory must stop polluting or buy the laundry’s consent. Buying consent would cost at least 60; the filter costs 40. The factory installs the filter.

The factory has the right to pollute. The laundry suffers 60 of damage. It can offer to pay the factory to install the filter; any payment between 40 and 60 benefits both. The filter is installed, paid for by the laundry.

In both cases the filter is installed. The assignment of rights determines who pays, not what is done.

Proposition 26.4. The Coase theorem

If property rights are well defined, transaction costs are negligible, and there are no income effects on the valuation of the externality, private bargaining leads to an efficient outcome regardless of the initial assignment of rights. The assignment affects only the distribution of wealth.

26.2.2 Why Coasian bargaining often fails

The theorem is most valuable for what its assumptions reveal. Bargaining fails when:

- **Transaction costs are high**, especially when many parties are affected. Millions of people suffer from urban air pollution caused by millions of vehicles; negotiating among them is impossible.
- **Free-riding** undermines collective action: each victim hopes others will pay the polluter to stop.
- **Information is asymmetric**: each party has an incentive to overstate its costs or damages to obtain a better deal (Chapter 25).
- **Property rights are unclear or unenforceable**, as for the atmosphere or the high seas.

Coase's insight is therefore both a reason for optimism in small-numbers cases—neighbours often do resolve disputes, and tradable rights can be created where they did not exist—and an explanation of why government action is needed when transaction costs are high.

26.3 Instruments of Environmental Policy

26.3.1 Three instruments

Governments can address a pollution externality with:

1. **Standards** (command and control): limits on emissions per firm, or required technologies.
2. **Taxes**: a charge per unit of emissions.
3. **Tradable permits** (cap and trade): the government sets a cap on total emissions, issues that many permits, and lets firms buy and sell them.

26.3.2 Cost-effectiveness

Suppose the goal is to reduce total emissions by a given amount. Each firm i has a **marginal abatement cost** $MAC_i(a_i)$: the cost of reducing emissions by one more unit when it has already reduced them by a_i .

Proposition 26.5. Least-cost abatement

A given total abatement is achieved at minimum total cost if and only if every firm's marginal abatement cost is equal.

Proof. If $MAC_1 > MAC_2$, shifting one unit of abatement from firm 1 to firm 2 saves MAC_1 and costs MAC_2 , lowering total cost without changing total abatement. (Formally, minimize $\sum_i C_i(a_i)$ subject to $\sum_i a_i = \bar{A}$; the Lagrangian first-order conditions are $C'_i(a_i) = \lambda$ for all i .) $\square$

A uniform tax t achieves this automatically: each firm abates until its marginal abatement cost equals t , since each unit abated saves t in tax. Tradable permits achieve it too: each firm abates until its marginal abatement cost equals the permit price, since a unit not abated requires buying (or not selling) a permit. A uniform standard generally does not, because firms differ in their abatement costs.

Worked Example 26.6. Taxes versus standards

Two firms each emit 50 tonnes. Firm 1's marginal abatement cost is $MAC_1 = 2a_1$ and firm 2's is $MAC_2 = 4a_2$, so their total abatement costs are a_1^2 and $2a_2^2$. The target is a total reduction of 30 tonnes.

Uniform standard: each abates 15. Cost: $15^2 + 2 \times 15^2 = 225 + 450 = 675$.

Cost-effective allocation: $2a_1 = 4a_2$ and $a_1 + a_2 = 30$ give $a_1 = 20$, $a_2 = 10$, with a common marginal cost of 40. Cost: $400 + 200 = 600$. The saving is 75, or 11%.

Implementation: a tax of 40 per tonne induces exactly this allocation. So does a cap of 70 tonnes with 70 tradable permits: whatever the initial allocation, trading continues until the permit price equals both firms' marginal abatement cost, 40.

26.3.3 Taxes or permits?

With full information, a tax and a permit system are equivalent: the government can choose either the price (tax) or the quantity (cap) and obtain the same outcome. Under uncertainty about abatement costs they differ. A tax fixes the marginal cost of abatement but leaves the quantity uncertain; a cap fixes the quantity but leaves the price uncertain. Martin Weitzman (1974) showed that the better instrument depends on the relative slopes of the marginal damage and marginal abatement cost curves: if damages rise steeply with emissions (so the quantity matters a lot), a cap is better; if abatement costs rise steeply (so the cost matters a lot), a tax is better. For long-lived stock pollutants such as carbon dioxide, damage depends on cumulative emissions over decades, so marginal damage from any one year's emissions is nearly flat—an argument often made for carbon taxes. In practice many systems combine features of both, such as caps with price floors or ceilings.

26.4 Public Goods**26.4.1 Rivalry and excludability****Definition 26.7. Rivalry and excludability**

A good is **rival** if one person's consumption reduces the amount available to others. It is **excludable** if people who do not pay can be prevented from consuming it.

	Excludable	Non-excludable
Rival	Private goods (food, clothing)	Common resources (fish stocks, groundwater)
Non-rival	Club goods (streaming services, toll roads when uncongested)	Public goods (national defence, basic research, lighthouses)

A **public good** is non-rival and non-excludable. Once it is provided, everyone benefits and no one can be prevented from benefiting. Markets fail to provide public goods adequately because of **free-riding**: each person, knowing she will benefit whether or not she pays, has an incentive to let others pay.

26.4.2 The efficient quantity: the Samuelson condition

For a private good, the efficient quantity equates each consumer's marginal benefit with marginal cost. For a public good, every unit benefits everyone at once, so the marginal social benefit of one more unit is the *sum* of all individuals' marginal benefits.

Proposition 26.8. Samuelson condition

The efficient quantity of a public good satisfies

$$\sum_i MB_i(G) = MC(G),$$

where MB_i is individual i 's marginal benefit (marginal willingness to pay). Market demand for a public good is obtained by adding individual demand curves *vertically*, not horizontally.

Worked Example 26.9. Street lighting

Three households on a street value lights (G = number of lamps) with marginal benefits $MB_1 = 12 - G$, $MB_2 = 8 - G$, $MB_3 = 6 - G$ (each zero when negative). Each lamp costs 11.

Efficient provision: for $G \leq 6$, $\sum MB_i = 26 - 3G = 11$, so $G^* = 5$. At $G = 5$, the households value the fifth lamp at 7, 3 and 1, which sum to 11.

Voluntary provision: household 1, acting alone, would buy lamps until $12 - G = 11$, i.e. one lamp. Given that lamp, households 2 and 3 value an additional lamp at $8 - 1 = 7$ and $6 - 1 = 5$, less than its cost of 11, so they buy none. Only one lamp is provided, far below the efficient five.

26.4.3 Responses to free-riding

Public goods are typically financed by compulsory taxation and provided by governments. That solves free-riding but raises another problem: the government must know how much people value the good, and people have no incentive to reveal their valuations truthfully if their tax depends on what they report. Section 26.7 returns to this problem. Other responses include privatizing some aspect of the good so that it becomes excludable (encrypted broadcasting turns a public good into a club good), relying on social norms and reciprocity, and, for goods with concentrated beneficiaries, provision by the few who gain most (a large firm may fund research that benefits its whole industry). Laboratory experiments with *public goods games*—in which each player chooses how much to contribute to a common pot that is multiplied and shared equally—typically find that people contribute 40 to 60 per cent of their endowment at

first, well above the zero predicted for self-interested players, but that contributions decline with repetition unless players can punish free-riders.

26.5 Common Resources

26.5.1 The tragedy of the commons

A **common resource** is rival but non-excludable: fish in international waters, groundwater, a congested road, the atmosphere's capacity to absorb carbon dioxide. Each user considers only her own benefit and cost, ignoring the effect of her use on others. The resource is over-used—Garrett Hardin's **tragedy of the commons**.

Worked Example 26.10. An open-access fishery

If n boats fish a lake, the total catch is $F(n) = n(20 - n)$, worth 1 per unit. Each boat costs 4 to operate. The catch per boat is $20 - n$ and the marginal product of a boat is $F'(n) = 20 - 2n$.

Open access: boats enter as long as the catch per boat exceeds the cost, so entry continues until $20 - n = 4$, $n = 16$. Total catch $16 \times 4 = 64$, total cost 64: the resource generates no net value at all.

Efficient use: maximize total catch minus cost, $n(20 - n) - 4n$. The condition $20 - 2n = 4$ gives $n = 8$. Catch $8 \times 12 = 96$, cost 32, net value 64.

Open access doubles the fishing effort and completely dissipates the resource's value. Each entrant compares the *average* catch with its cost, ignoring the fact that its entry reduces every other boat's catch.

The remedies mirror those for externalities: taxes on use (congestion charges in London and Singapore), quotas, and the creation of property rights. Individual transferable quotas in fisheries, used in Iceland and New Zealand among others, assign shares of a total allowable catch that can be traded—a Coasian solution. Elinor Ostrom's research showed that many communities have managed common resources successfully for centuries through local rules, monitoring and graduated sanctions, without either privatization or central control, when users are few, interact repeatedly and can monitor one another—the conditions under which repeated-game cooperation (Chapter 23) is sustainable.

26.6 Auctions

Many resources are allocated by auction: radio spectrum, government bonds, emissions permits, online advertising slots, art, houses in some countries. Auctions are also a laboratory for game theory under private information.

26.6.1 Formats

The four standard formats for selling a single object are:

- **English (ascending):** the price rises until only one bidder remains; she pays the price at which the last rival dropped out.
- **Dutch (descending):** the price falls until a bidder accepts; she pays that price.
- **First-price sealed-bid:** bidders submit sealed bids; the highest bidder wins and pays her bid.

- **Second-price sealed-bid** (Vickrey): the highest bidder wins and pays the *second-highest* bid.

We focus first on **private values**: each bidder knows her own value for the object, values differ across bidders, and one bidder's value does not depend on the others' information.

26.6.2 The second-price auction

Proposition 26.11. Truthful bidding in a second-price auction

In a second-price sealed-bid auction with private values, bidding one's true value is a weakly dominant strategy.

Proof. Let your value be v and let h be the highest of the other bids. If you bid v : you win when $v > h$ and earn $v - h > 0$; you lose when $v < h$ and earn 0. Suppose instead you bid $b > v$. This changes the outcome only when $v < h < b$: you now win and pay $h > v$, a loss. Suppose you bid $b < v$. This changes the outcome only when $b < h < v$: you now lose a profit of $v - h > 0$. Deviating from v can never help and sometimes hurts. $\square$

The price you pay does not depend on your own bid (only on whether you win), so there is no reason to shade your bid. The English auction is strategically similar: with private values, each bidder stays in until the price reaches her value, and the winner pays roughly the second-highest value.

26.6.3 The first-price auction

In a first-price auction, bidding your value guarantees zero profit. Bidders shade their bids below their values, trading a lower probability of winning for a higher profit when they win.

Worked Example 26.12. Equilibrium bidding with uniform values

Suppose n bidders have values drawn independently and uniformly between 0 and 1. It can be shown that in the symmetric equilibrium each bidder bids

$$b(v) = \frac{n-1}{n} v.$$

We verify the case $n = 2$. Suppose your rival bids $v_2/2$. If you bid b , you win when $v_2/2 < b$, i.e. $v_2 < 2b$, which has probability $2b$ (for $b \leq 1/2$). Your expected profit is $(v - b)2b$, maximized at $b = v/2$. So bidding half your value is a best response to your rival doing the same.

26.6.4 Revenue equivalence

Which format raises more revenue for the seller? With two bidders and uniform values, the second-price auction's revenue is the expected second-highest value, $(n-1)/(n+1) = 1/3$. In the first-price auction, the winner is the bidder with the highest value, whose expected value is $n/(n+1) = 2/3$, and she bids half of it: expected revenue $1/3$. The two are equal. This is no coincidence.

Proposition 26.13. Revenue equivalence (Vickrey, Myerson)

With risk-neutral bidders whose private values are independently drawn from a common distribution, any auction format in which the bidder with the highest value always wins, and a bidder with the lowest possible value expects to pay nothing, yields the same expected revenue to the seller.

The theorem is a benchmark whose assumptions indicate when formats matter: when bidders are risk-averse (first-price auctions then raise more, as bidders shade less to reduce the risk of losing), when values are correlated, when bidders are asymmetric, or when collusion is possible (open ascending auctions make collusion easier to enforce because bidders can observe and punish deviations).

26.6.5 Common values and the winner's curse

When the object has a single true value that is unknown—an oil tract, a company being acquired, a batch of government bonds—each bidder has an estimate, and the winner is the bidder with the *most optimistic* estimate. If bidders naively bid their estimates, the winner will on average have over-estimated the value: the **winner's curse**. Rational bidders correct for it by bidding below their estimates, conditioning on the event that they win. The winner's curse has been documented in auctions for oil leases and in corporate takeovers, where acquiring firms have often overpaid.

26.6.6 Auctions in practice

The U.S. Treasury sells its debt through sealed-bid auctions in which all successful bidders pay the same price (a uniform-price, or “single-price,” auction), a format it has used for all its marketable securities since 1998. Central banks, including the European Central Bank and the Federal Reserve, have used auctions to allocate liquidity to banks (Chapter 49). Spectrum auctions designed by economists, beginning with the U.S. Federal Communications Commission in 1994, have raised hundreds of billions of dollars worldwide; the design of these auctions contributed to the 2020 Nobel prize awarded to Paul Milgrom and Robert Wilson. Search advertising is allocated by variants of second-price auctions run billions of times a day.

26.7 Mechanism Design**26.7.1 The problem**

Mechanism design reverses the usual question of game theory. Instead of asking what outcome a given game produces, it asks what game—what rules—will produce a desired outcome when participants have private information and act in their own interest. Auctions are mechanisms; so are voting rules, tax systems, matching procedures for students and schools, and permit markets.

Proposition 26.14. Revelation principle

Any outcome that can be achieved by some mechanism in equilibrium can also be achieved by a *direct* mechanism in which participants simply report their private information and find it optimal to report truthfully.

The revelation principle simplifies design enormously: to find what is achievable, one need

only consider mechanisms that ask people to report their types and make truth-telling optimal (“incentive compatible”).

26.7.2 Truthful provision of public goods

Recall the difficulty with public goods: people have no incentive to reveal their valuations. The **Vickrey–Clarke–Groves** (VCG) mechanism, of which the Clarke “pivot” mechanism is the simplest version, solves it in principle. Each person reports her value; the good is provided if the reported values sum to more than the cost; and each person is charged a tax equal to the net harm her report imposes on the others—which is positive only if her report changes the decision (she is “pivotal”). As in the second-price auction, a person’s payment depends on her report only through whether it changes the outcome, so truthful reporting is dominant. The mechanism has drawbacks: its taxes do not in general balance the budget, and it is vulnerable to collusion. But it shows that incentive problems in public-good provision are not insurmountable in theory.

26.7.3 Limits to what mechanisms can achieve

Mechanism design also establishes impossibility results. The *Myerson–Satterthwaite theorem* shows that when a buyer and a seller each have private information about their valuations, and it is uncertain whether trade would be beneficial, no mechanism can guarantee that trade occurs whenever it is efficient, while being voluntary and balancing the budget. Some efficient trades must be lost. Bargaining breakdowns, strikes and failed negotiations are therefore not necessarily signs of irrationality: they can be the unavoidable cost of private information.

Case: The EU Emissions Trading System and China’s national carbon market

The EU Emissions Trading System (EU ETS), launched in 2005, caps the carbon dioxide emissions of power plants, heavy industry and, since 2012, intra-European aviation, and allows firms to trade allowances. By Theorem 26.5, trading should achieve any given reduction at least cost, because firms with cheap abatement options sell allowances to firms with expensive ones.

The early years illustrated the dangers of setting the cap without good information. Allowances in the first phase (2005–2007) were over-allocated, largely free of charge on the basis of firms’ own estimates of their emissions; when verified emissions turned out lower, the price collapsed toward zero. After the 2008 recession reduced industrial output, a large surplus of allowances built up again and prices stayed low for most of the 2010s. Reforms introduced a Market Stability Reserve in 2019, which withdraws allowances from circulation when the surplus is large, and steepened the rate at which the cap declines. Prices rose substantially, exceeding 80 euros per tonne in 2022 and 2023. The episode shows the distinction between price and quantity instruments: a cap fixes emissions but leaves the price to fluctuate with demand, and a very low price provides little incentive to invest in abatement.

China’s national emissions-trading system, which began trading in 2021, initially covered the power sector. Unlike the EU system, it was designed around emissions *intensity*—allowances are allocated per unit of electricity generated—rather than an absolute cap, so total covered emissions can rise with output. Intensity-based designs reduce the cost of the policy to a fast-growing economy but make total emissions uncertain; in Weitzman’s terms, they sit between a price and a quantity instrument. Both systems illustrate that the details of mechanism design—allocation, caps, price-stabilizing rules and monitoring—determine whether a textbook-efficient instrument works in practice.

Summary

- Externalities are uncompensated effects on others. With external costs, markets over-produce; with external benefits, they under-produce.
- A Pigouvian tax or subsidy equal to the marginal external effect at the efficient quantity restores efficiency and removes the deadweight loss.
- The Coase theorem: with clear rights, negligible transaction costs and no income effects, bargaining is efficient whoever holds the rights. High transaction costs and free-riding explain why it often fails.
- Cost-effective abatement equalizes marginal abatement costs; uniform taxes and tradable permits achieve this, uniform standards generally do not. Under uncertainty, the choice between price and quantity instruments depends on the slopes of marginal damage and abatement cost.
- Public goods are non-rival and non-excludable; efficiency requires $\sum MB_i = MC$, and free-riding leads to under-provision. Common resources are over-used unless access is regulated or rights are created.
- Truthful bidding is dominant in second-price auctions; bidders shade in first-price auctions. Under standard assumptions, all common formats yield the same expected revenue. Common values create the winner's curse.
- Mechanism design asks which rules achieve desired outcomes with private information; the revelation principle focuses the search, VCG mechanisms elicit truthful valuations, and impossibility results bound what can be achieved.

Key terms

externality; negative and positive externality; marginal social cost; marginal external cost; Pigouvian tax; Pigouvian subsidy; Coase theorem; transaction costs; standards; tradable permits; cap and trade; marginal abatement cost; rival; excludable; public good; club good; common resource; free-riding; Samuelson condition; tragedy of the commons; individual transferable quotas; English, Dutch, first-price and second-price auctions; private values; revenue equivalence; common values; winner's curse; mechanism design; revelation principle; incentive compatibility; VCG mechanism; Myerson–Satterthwaite theorem.

Exercises

Exercise 26.1 (Externality). Demand for a good is $P = 60 - 0.5Q$ and private marginal cost is $MPC = 10 + 0.5Q$. Production causes an external cost of $MEC = 0.25Q$ (rising with output). Find the market quantity, the efficient quantity, the Pigouvian tax and the deadweight loss of the market outcome.

Exercise 26.2 (Positive externality). Private marginal benefit from vaccination is $MPB = 50 - Q$, where Q is the number vaccinated (in thousands), and the marginal cost is constant at 20. Each vaccination provides an external benefit of 10 to others. Find the market quantity, the efficient quantity and the subsidy that achieves it.

Exercise 26.3 (Coase). A rancher's cattle damage a farmer's crops. A fence costs 300; crop damage without a fence is 500. The rancher's profit from cattle is 1,000. (a) Who builds the

fence, and who pays, if the farmer has the right to be free of damage? If the rancher has the right to let cattle roam? (b) How would your answer change if the fence cost 700?

Exercise 26.4 (Abatement). Three firms have marginal abatement costs $MAC_1 = a_1$, $MAC_2 = 2a_2$, $MAC_3 = 4a_3$. The regulator wants total abatement of 70. Find the cost-effective allocation, the tax that achieves it, and the total cost. Compare with an equal standard for each firm.

Exercise 26.5 (Permits). In Example 26.6, the government issues 70 permits, giving 50 to firm 1 and 20 to firm 2. Describe the trade that takes place, the permit price, and each firm's net cost (abatement cost plus permit purchases minus sales).

Exercise 26.6 (Public good). Two residents value a park of size G hectares with marginal benefits $MB_1 = 20 - 2G$ and $MB_2 = 10 - G$. Each hectare costs 15. Find the efficient size. If resident 1 alone decides, how large a park is provided?

Exercise 26.7 (Commons). In Example 26.10, the government imposes a licence fee f per boat. What fee achieves the efficient number of boats? How much revenue does it raise?

Exercise 26.8 (Second-price auction). Three bidders have values 40, 55 and 70 for an object sold by a second-price sealed-bid auction. Who wins and what does she pay if all bid truthfully? Show that the winner cannot gain by bidding 60 or 100, and that the bidder with value 55 cannot gain by bidding 80.

Exercise 26.9 (First-price auction). With three bidders whose values are independent and uniform on $[0, 1]$, what does a bidder with value 0.9 bid in the symmetric equilibrium? What is the seller's expected revenue? Compare it with the expected revenue from a second-price auction.

Exercise 26.10 (Winner's curse). Five firms bid for a drilling right whose true value, unknown to all, is 100. Each firm's estimate is the true value plus an independent error that is equally likely to be -20 , -10 , 0 , 10 or 20 . Explain why a firm that bids its estimate in a first-price auction can expect to lose money if it wins.

Exercise 26.11 (Instrument choice). Explain why a government uncertain about firms' abatement costs might prefer a carbon tax to a cap if marginal damages from emissions are nearly constant, but a cap if there is a threshold of emissions beyond which damages rise sharply.

Solutions

Solution 26.1. Market: $60 - 0.5Q = 10 + 0.5Q$, $Q_m = 50$, $P = 35$. $MSC = 10 + 0.75Q$. Efficient: $60 - 0.5Q = 10 + 0.75Q$, $Q^* = 40$, $P = 40$. Pigouvian tax = $MEC(Q^*) = 0.25 \times 40 = 10$. (Check: with the tax, supply is $20 + 0.5Q$, which meets demand at $Q = 40$.) DWL: at $Q = 50$, $MSC = 47.5$, demand 35; triangle $\frac{1}{2} \times 10 \times (47.5 - 35) = 62.5$.

Solution 26.2. Market: $50 - Q = 20$, $Q = 30$ thousand. Marginal social benefit = $60 - Q$; efficient: $60 - Q = 20$, $Q = 40$ thousand. A subsidy of 10 per vaccination makes the private net cost 10, and $50 - Q = 10$ gives $Q = 40$.

Solution 26.3. (a) Efficiency requires the fence ($300 < 500$). If the farmer has the right, the rancher must prevent damage; the cheapest way is the fence, which the rancher builds and pays for. If the rancher has the right, the farmer pays for the fence (or pays the rancher between 300 and 500 to build it). Either way the fence is built. (b) With a fence costing $700 > 500$, efficiency requires no fence and 500 of damage. If the farmer has the right, the rancher compensates the farmer 500 (cheaper than the fence) and keeps the cattle, since $1,000 > 500$. If the rancher has the right, the farmer bears the damage. Either way, no fence.

Solution 26.4. Equalize: $a_1 = 2a_2 = 4a_3 = \lambda$, so $a_1 = \lambda$, $a_2 = \lambda/2$, $a_3 = \lambda/4$; sum $1.75\lambda = 70$, $\lambda = 40$. So $a_1 = 40$, $a_2 = 20$, $a_3 = 10$; tax 40. Costs: $a_1^2/2 + a_2^2 + 2a_3^2 = 800 + 400 + 200 = 1,400$. Equal standard (70/3 each): $(70/3)^2(1/2 + 1 + 2) = 544.4 \times 3.5 = 1,905.6$. The tax saves about 506.

Solution 26.5. Firm 1 holds permits for 50 tonnes, its unabated emissions; firm 2 holds 20, needing to abate 30 if it did not trade. At no trade, firm 1 abates 0 ($MAC_1 = 0$) and firm 2 abates 30 ($MAC_2 = 120$). Firm 2 buys permits from firm 1 until MACs equal 40: firm 1 abates 20 and sells 20 permits; firm 2 abates 10 and buys 20. Price 40. Firm 1: abatement cost 400, revenue 800, net gain 400. Firm 2: abatement cost 200, purchases 800, net cost 1,000. Total abatement cost 600, as in the text; the allocation of permits determines the distribution.

Solution 26.6. For $G \leq 10$, $\sum MB = 30 - 3G = 15$, $G^* = 5$. Resident 1 alone: $20 - 2G = 15$, $G = 2.5$. Resident 2, facing 2.5 hectares, values another at $7.5 < 15$ and contributes nothing.

Solution 26.7. Entry continues until catch per boat equals cost plus fee: $20 - n = 4 + f$. For $n = 8$, $f = 8$. Revenue $= 8 \times 8 = 64$, exactly the resource rent that open access dissipated.

Solution 26.8. The bidder with value 70 wins and pays 55. Bidding 60 or 100 does not change the outcome or the price (she still wins and pays 55). The bidder with value 55 bidding 80 would win and pay 70, a loss of 15.

Solution 26.9. Bid $= (2/3) \times 0.9 = 0.6$. Expected revenue in both formats is the expected second-highest of three uniform values, $(n-1)/(n+1) = 0.5$. (In the first-price auction: expected highest value $3/4$, times the bid fraction $2/3$, equals 0.5.)

Solution 26.10. The firm with the highest estimate wins. With five firms, it is very likely that at least one estimate is +10 or +20 above the true value. Conditional on winning, a firm's estimate is probably too high. Bidding its estimate therefore leads, on average, to paying more than 100 when it wins. A rational firm bids below its estimate, taking into account the information that winning conveys.

Solution 26.11. With nearly constant marginal damage, the social cost of getting the quantity wrong is small, but getting the price wrong (forcing very costly abatement) can be large; a tax set at the marginal damage gets the price right whatever abatement costs turn out to be. With a threshold, exceeding it is very costly, so fixing the quantity is more important than fixing the price; a cap guarantees the threshold is respected, at the cost of uncertain abatement expenditure.

Part IV

Accounting

Accounting Architecture and the Financial Statements

After studying this chapter you should be able to

1. Explain the objective of general-purpose financial reporting and identify its users.
2. Describe the institutions that set accounting standards, including the IASB, the FASB and national regimes in China, Germany and Malaysia.
3. State the qualitative characteristics of useful financial information.
4. Define the elements of financial statements and explain recognition and the main measurement bases, including fair value.
5. Apply the rules of debit and credit, record transactions in a journal, post them to a ledger and prepare a trial balance.
6. Prepare adjusting entries for accruals, deferrals and depreciation.
7. Prepare a full set of financial statements from an adjusted trial balance and close the books.
8. Explain how the statements articulate and compare key features of IFRS and U.S. GAAP.
9. Explain the role of auditing and internal control.

A motivating question

In June 2020 Wirecard AG, a German payments company that had joined the DAX index of Germany's largest listed companies two years earlier, announced that 1.9 billion euros of cash—about a quarter of its balance sheet—probably did not exist. The money was supposedly held in trust accounts at two banks in the Philippines; both banks said they had never had a relationship with Wirecard. Within a week the company filed for insolvency. Its auditor had signed off on its accounts for more than a decade.

How can a large listed company report cash that does not exist? What does it mean for an asset to “exist” in accounting terms, who decides what goes into financial statements, and what is an auditor supposed to check? This chapter builds the architecture of financial accounting from its foundations: why financial reports exist, the concepts that define their contents, and the mechanics—double entry, journals, ledgers, adjustments—that turn thousands of transactions into a balance sheet, an income statement and a cash-flow statement. Chapter 10 introduced the statements through the effects of transactions on the accounting equation; this chapter shows how accountants actually record and process those transactions, and why.

27.1 The Purpose of Financial Reporting

27.1.1 Users and their needs

Modern corporations separate ownership from control. Shareholders provide equity but do not run the firm; lenders provide debt but cannot see inside the business; employees, suppliers, customers and governments all have stakes in its survival. Chapter 25 showed that when one party knows more than another, markets may fail. Financial reporting is an institution that reduces that asymmetry.

Definition 27.1. Objective of general-purpose financial reporting

The objective of general-purpose financial reporting is to provide financial information about the reporting entity that is useful to existing and potential investors, lenders and other creditors in making decisions about providing resources to the entity.

This definition, from the IASB's Conceptual Framework for Financial Reporting (revised in 2018), has two important features. First, it identifies the *primary users*: providers of capital. Other groups, such as regulators, employees and the public, may find the reports useful, but they are not the audience the reports are designed for. Second, it is *decision-oriented*: information is useful if it helps users assess the amount, timing and uncertainty of the entity's future net cash inflows, and management's stewardship of the entity's resources.

27.1.2 Financial versus managerial accounting

Financial accounting produces general-purpose reports for external users, according to standards. **Managerial accounting** produces information for internal decision makers—costs by product, budgets, performance measures—in whatever form managers find useful, without external standards. Chapters 27 to 30 concern financial accounting; Chapter 31 turns to managerial accounting.

27.2 Who Sets the Rules

27.2.1 Standard setters

Accounting standards are set by bodies that, in most countries, are independent of government but whose standards are given legal force by regulators.

- The **International Accounting Standards Board (IASB)**, based in London and overseen by the IFRS Foundation, issues **International Financial Reporting Standards (IFRS)**. IFRS are required or permitted for listed companies in the large majority of the world's jurisdictions, including the European Union (for the consolidated accounts of listed groups since 2005), the United Kingdom, Australia, Canada and many countries in Asia, Africa and Latin America.
- The **Financial Accounting Standards Board (FASB)** sets **U.S. GAAP**. Since 2009 its standards have been organized in a single *Accounting Standards Codification (ASC)*. The U.S. Securities and Exchange Commission (SEC) has legal authority over reporting by public companies and recognizes FASB standards as authoritative. Foreign companies listed in the United States may file IFRS accounts without reconciliation to U.S. GAAP.

Three national regimes relevant to this book:

- **Germany:** listed groups prepare consolidated accounts under IFRS as adopted by the EU. Individual company accounts, which determine distributable profit and form the starting point for taxation, are prepared under the German Commercial Code (*Handelsgesetzbuch, HGB*), a more creditor-oriented and conservative tradition.
- **China:** the Ministry of Finance issues Chinese Accounting Standards for Business Enterprises (*CAS*), which since 2006 have been substantially converged with IFRS, though not identical to them.
- **Malaysia:** the Malaysian Accounting Standards Board issues Malaysian Financial Reporting Standards (*MFRS*), which are word-for-word IFRS for non-private entities and have been applied since 2012.

IFRS and U.S. GAAP have converged in many areas—revenue recognition (IFRS 15 and ASC 606, both effective from 2018) and leases (IFRS 16 and ASC 842, from 2019) were joint or parallel projects—but important differences remain. This book treats both in parallel.

27.2.2 Principles and rules

IFRS is often described as more *principles-based* and U.S. GAAP as more *rules-based*. The distinction is one of degree. Principles-based standards state objectives and require judgement; rules-based standards contain more bright lines, detailed guidance and industry-specific rules. Principles give preparers flexibility to reflect economic substance but make comparisons harder; bright-line rules improve comparability but invite “structuring” of transactions to fall just on the preferred side of a line. The history of lease accounting, discussed in Chapter 29, is a classic example.

27.3 The Conceptual Framework

27.3.1 Qualitative characteristics

What makes financial information useful? The Conceptual Framework identifies two *fundamental* qualitative characteristics:

Definition 27.2. Relevance and faithful representation

Information is **relevant** if it is capable of making a difference to users’ decisions—because it has predictive value, confirmatory value, or both. **Materiality** is an entity-specific aspect of relevance: information is material if omitting, misstating or obscuring it could reasonably be expected to influence users’ decisions.

Information is a **faithful representation** if it depicts the substance of the economic phenomenon it purports to represent, and is, to the maximum extent possible, *complete*, *neutral* (without bias in selection or presentation) and *free from error*.

Four *enhancing* characteristics make relevant and faithfully represented information more useful: *comparability* (across entities and over time), *verifiability* (independent observers could reach consensus), *timeliness*, and *understandability*. The whole framework is subject to a *cost constraint*: the benefits of reporting information must justify its costs.

The 2018 revision reintroduced **prudence**, defined as the exercise of caution when making judgements under uncertainty, as a support for neutrality. Prudence does not permit deliberately understating assets or overstating liabilities; that would violate neutrality. It means not being overly optimistic.

There is often a tension between relevance and faithful representation. The current market value of a building is highly relevant to a lender, but if there is no active market for similar buildings, any estimate is uncertain. Historical cost is verifiable but may be decades out of date. Much of accounting consists of managing this trade-off.

27.3.2 Underlying assumption: going concern

Financial statements are normally prepared on the assumption that the entity is a **going concern**: that it will continue in operation for the foreseeable future. That is why equipment is reported at cost less depreciation rather than at what it would fetch in a forced sale. If management intends or is forced to liquidate, a different basis is required, and auditors must report if there is material uncertainty about the entity's ability to continue.

27.3.3 The elements

Definition 27.3. Elements of financial statements (IASB, 2018)

- An **asset** is a present economic resource controlled by the entity as a result of past events. An *economic resource* is a right that has the potential to produce economic benefits.
- A **liability** is a present obligation of the entity to transfer an economic resource as a result of past events.
- **Equity** is the residual interest in the assets of the entity after deducting all its liabilities.
- **Income** is increases in assets, or decreases in liabilities, that result in increases in equity, other than those relating to contributions from holders of equity claims.
- **Expenses** are decreases in assets, or increases in liabilities, that result in decreases in equity, other than those relating to distributions to holders of equity claims.

Several points deserve emphasis.

Assets are rights, controlled. A firm's workforce is valuable but is not an asset, because the firm does not control its employees. A firm's reputation built internally is valuable but is generally not recognized, because it is not a separable right that can be reliably measured. A firm need not *own* an asset: under a lease, the lessee controls the right to use the asset, and since 2019 recognizes a "right-of-use" asset (Chapter 29).

Past events. An obligation to pay for goods already received is a liability; a plan to buy goods next year is not. A firm that has signed a contract to buy a machine next year has, in general, no liability until the machine is delivered.

Income and expenses are defined through the balance sheet. This is the **asset–liability view**: income and expenses are changes in assets and liabilities. It contrasts with an older *revenue–expense view* in which the income statement was primary and the balance sheet contained whatever was left over. The asset–liability view is why standard setters insist that items such as "deferred charges" appear on the balance sheet only if they meet the definition of an asset.

27.3.4 Recognition

Definition 27.4. Recognition

Recognition is the process of capturing an item that meets the definition of an element for inclusion in the balance sheet or income statement. An item is recognized only if recognition provides users with relevant information and a faithful representation.

Recognition may be inappropriate when it is uncertain whether an asset or liability exists, when the probability of an inflow or outflow of benefits is low, or when measurement uncertainty is so great that no number would be a faithful representation. Items that are not recognized may still require *disclosure* in the notes—a lawsuit whose outcome is uncertain, for instance (Chapter 30).

27.3.5 Measurement

Once an item is recognized, it must be measured. The Conceptual Framework describes two families of measurement bases:

- **Historical cost:** the price paid (for an asset) or received (for a liability) in the original transaction, adjusted for subsequent consumption (depreciation), impairment and, for financial instruments, interest (“amortized cost”).
- **Current value**, of which there are three types:
 - **Fair value:** the price that would be received to sell an asset, or paid to transfer a liability, in an orderly transaction between market participants at the measurement date.
 - *Value in use* (for assets) or *fulfilment value* (for liabilities): the present value of the cash flows the entity expects to derive from using the asset or to incur in fulfilling the liability.
 - *Current cost:* the cost of an equivalent asset today.

Financial statements use a *mixed measurement model*: different items are measured on different bases. Inventory is generally at the lower of cost and net realizable value; property, plant and equipment at cost less depreciation (IFRS permits revaluation to fair value; U.S. GAAP does not); many financial assets at fair value or amortized cost depending on their classification (Chapter 30).

Definition 27.5. The fair-value hierarchy

IFRS 13 and ASC 820 classify fair-value measurements by the observability of their inputs:

- **Level 1:** quoted prices in active markets for identical assets (a listed share).
- **Level 2:** other observable inputs (quoted prices for similar assets, observable interest rates).
- **Level 3:** unobservable inputs based on the entity’s own assumptions (a model valuation of an unlisted investment).

Level 3 measurements are the most judgemental and the most vulnerable to bias. In 2007–2008, as markets for mortgage-backed securities froze, many banks’ holdings migrated from

Level 2 to Level 3, and investors discounted the reported values heavily. The fair-value hierarchy is therefore also a map of where to look for measurement risk.

27.4 Double-Entry Bookkeeping

27.4.1 Why double entry

Chapter 10 recorded transactions directly as changes in the accounting equation. Real businesses record thousands of transactions, and they need a system that (i) records each transaction once, in chronological order; (ii) accumulates the effects on each item separately; and (iii) contains a built-in check. Double-entry bookkeeping, codified by Luca Pacioli in 1494 but practised by Italian merchants long before, provides all three.

27.4.2 Accounts, debits and credits

An **account** records increases and decreases in a single item—cash, receivables, sales revenue. Each account has two sides: the left side is the **debit** side and the right side the **credit** side. “Debit” and “credit” mean nothing more than left and right; they do not mean good or bad, plus or minus.

The rules are set so that the accounting equation always balances:

Account type	Increase recorded as	Normal balance
Assets	Debit	Debit
Expenses (and dividends)	Debit	Debit
Liabilities	Credit	Credit
Equity (share capital, retained earnings)	Credit	Credit
Income (revenue, gains)	Credit	Credit

Why do expenses behave like assets? Because expenses *reduce* equity, and equity increases are credits, so equity decreases are debits. Likewise income increases equity and is credited. The logic follows from the expanded accounting equation:

$$\begin{aligned} \text{Assets} + \text{Expenses} + \text{Dividends} \\ \equiv \text{Liabilities} + \text{Share capital} + \text{Retained earnings (opening)} + \text{Income,} \end{aligned}$$

with every item on the left increased by debits and every item on the right by credits.

The double-entry rule. Every transaction is recorded with total debits equal to total credits. Because the rules for debits and credits mirror the two sides of the accounting equation, equal debits and credits guarantee that the equation continues to balance.

27.4.3 The journal and the ledger

Transactions are first recorded chronologically in a **journal**, as **journal entries**. By convention, debits are listed first and credits indented beneath them. The entries are then *posted* to individual accounts in the **ledger**, which accumulates each account’s balance. A simplified ledger account is drawn as a **T-account**, with debits on the left and credits on the right.

Worked Example 27.6. Journal entries for Harbour Coffee

Three of Harbour Coffee's January transactions from Chapter 10, as journal entries:

	DR	CR
Cash	50,000	
Share capital		50,000
<i>Issue of shares for cash</i>		
Inventory	8,000	
Accounts payable		8,000
<i>Purchase of supplies on credit</i>		
Cash	15,000	
Sales revenue		15,000
Cost of goods sold	5,000	
Inventory		5,000
<i>Cash sales and cost of supplies used</i>		

27.4.4 The trial balance

After posting, the accountant lists every account's balance in a **trial balance**. Because every entry had equal debits and credits, total debit balances must equal total credit balances. If they do not, an error has been made. The converse is not true: a balanced trial balance does not prove the records are correct. An entry posted to the wrong account, recorded at the wrong amount on both sides, or omitted entirely leaves the trial balance in balance.

27.5 The Accounting Cycle: A Full Year

We now follow a company through a complete year. Kestrel Bicycles Ltd is formed on 1 January 2026 to import and sell bicycles. Its tax rate is 25%.

27.5.1 Transactions during the year

- (1) 1 Jan: shareholders subscribe 100,000 cash for shares.
- (2) 1 Jan: the company buys workshop equipment for 80,000 cash. It expects to use the equipment for 8 years, with no residual value.
- (3) 1 Apr: pays 6,000 for a 12-month insurance policy covering 1 April 2026 to 31 March 2027.
- (4) 1 Jul: borrows 60,000 from a bank at 5% per year; interest is payable annually on 30 June.
- (5) During the year: buys bicycles on credit for 200,000.
- (6) During the year: sells bicycles for 180,000 cash and 90,000 on credit. The bicycles sold had cost 160,000.
- (7) During the year: collects 70,000 from credit customers.
- (8) During the year: pays suppliers 170,000.
- (9) During the year: pays wages of 50,000.
- (10) During the year: pays rent of 24,000 for the year.
- (11) 15 Dec: a sports club pays 12,000 in advance for bicycles to be delivered in February 2027.
- (12) 20 Dec: the company declares and pays a dividend of 10,000.

27.5.2 Journal entries

		DR	CR
(1)	Cash	100,000	
	Share capital		100,000
(2)	Equipment	80,000	
	Cash		80,000
(3)	Prepaid insurance	6,000	
	Cash		6,000
(4)	Cash	60,000	
	Bank loan		60,000
(5)	Inventory	200,000	
	Accounts payable		200,000
(6)	Cash	180,000	
	Accounts receivable	90,000	
	Sales revenue		270,000
	Cost of goods sold	160,000	
	Inventory		160,000
(7)	Cash	70,000	
	Accounts receivable		70,000
(8)	Accounts payable	170,000	
	Cash		170,000
(9)	Wages expense	50,000	
	Cash		50,000
(10)	Rent expense	24,000	
	Cash		24,000
(11)	Cash	12,000	
	Unearned revenue (contract liability)		12,000
(12)	Dividends	10,000	
	Cash		10,000

Three entries deserve comment. Entry (3) records the insurance payment as an *asset*, prepaid insurance, because on 1 April the company acquired a right to twelve months of cover, none of which had yet been used. Entry (11) records the advance payment as a *liability*, because Kestrel owes the club bicycles; no revenue has been earned. Entry (12) debits a Dividends account rather than an expense: dividends are distributions to owners, not costs of earning income, and they never appear in the income statement.

27.5.3 The unadjusted trial balance

Posting the entries to the ledger gives the following balances at 31 December 2026, before adjustments. (Cash: $100,000 - 80,000 - 6,000 + 60,000 + 180,000 + 70,000 - 170,000 - 50,000 - 24,000 + 12,000 - 10,000 = 82,000$.)

Account	Debit	Credit
Cash	82,000	
Accounts receivable	20,000	
Inventory	40,000	
Prepaid insurance	6,000	
Equipment	80,000	
Accounts payable		30,000
Unearned revenue		12,000
Bank loan		60,000
Share capital		100,000
Dividends	10,000	
Sales revenue		270,000
Cost of goods sold	160,000	
Wages expense	50,000	
Rent expense	24,000	
Totals	472,000	472,000

27.5.4 Adjusting entries

The unadjusted balances do not yet reflect several economic events that occurred during the year without a transaction to trigger an entry. **Adjusting entries** at the end of the period bring the accounts into line with accrual accounting. They fall into four types:

Type	Situation
Accrued expense	Expense incurred but not yet paid (wages earned by staff, interest on loans).
Accrued revenue	Revenue earned but not yet billed or received.
Prepaid expense (deferral)	Cash paid in advance; part has now been used up (insurance, rent in advance).
Unearned revenue (deferral)	Cash received in advance; part has now been earned.

Depreciation is a special case of a deferral: the cost of equipment, paid in advance, is allocated to the periods that use it. Estimates such as expected credit losses and tax are also recorded at the period end.

For Kestrel, the adjustments at 31 December are:

- Depreciation:** $80,000 / 8 = 10,000$ for the year.
- Insurance:** 9 of the 12 months (April to December) have expired, so $6,000 \times 9/12 = 4,500$ is an expense; 1,500 remains prepaid.
- Accrued wages:** employees have earned 3,000 for the last week of December, to be paid in January.
- Accrued interest:** six months' interest on the loan, $60,000 \times 5\% \times 6/12 = 1,500$, has accrued.
- Income tax:** computed after the other adjustments.

	DR	CR
(a) Depreciation expense	10,000	
Accumulated depreciation		10,000
(b) Insurance expense	4,500	
Prepaid insurance		4,500
(c) Wages expense	3,000	
Wages payable		3,000
(d) Interest expense	1,500	
Interest payable		1,500

Depreciation is credited to **accumulated depreciation**, a **contra-asset** account that is deducted from equipment on the balance sheet. The equipment account continues to show the original cost, so users can see both the cost and how much of it has been consumed.

After adjustments (a)–(d), profit before tax is

$$270,000 - 160,000 - 53,000 - 24,000 - 4,500 - 10,000 - 1,500 = 17,000.$$

Assuming taxable profit equals accounting profit (Chapter 30 examines differences), tax is $25\% \times 17,000 = 4,250$:

(e) Income tax expense	4,250	
Income tax payable		4,250

27.5.5 The financial statements

Kestrel Bicycles Ltd — Statement of profit or loss, year to 31 Dec 2026

Revenue	270,000
Cost of sales	(160,000)
Gross profit	110,000
Wages	(53,000)
Rent	(24,000)
Insurance	(4,500)
Depreciation	(10,000)
Operating profit	18,500
Finance costs (interest)	(1,500)
Profit before tax	17,000
Income tax expense	(4,250)
Profit for the year	12,750

Statement of changes in equity, year to 31 Dec 2026

Share capital issued	100,000
Profit for the year	12,750
Dividends	(10,000)
Equity at 31 December	102,750
<i>of which: share capital 100,000; retained earnings 2,750</i>	

Statement of financial position at 31 December 2026

Non-current assets		Equity	
Equipment at cost	80,000	Share capital	100,000
Less accumulated depreciation	(10,000)	Retained earnings	2,750
	70,000		102,750
Current assets		Non-current liabilities	
Inventory	40,000	Bank loan	60,000
Accounts receivable	20,000		
Prepaid insurance	1,500	Current liabilities	
Cash	82,000	Accounts payable	30,000
	143,500	Unearned revenue	12,000
		Wages payable	3,000
		Interest payable	1,500
		Income tax payable	4,250
			50,750
Total assets	213,500	Total equity and liabilities	213,500

Statement of cash flows, year to 31 Dec 2026 (indirect method)

Profit before tax	17,000
Add: depreciation	10,000
Add: interest expense (shown separately if paid)	1,500
Increase in accounts receivable	(20,000)
Increase in inventory	(40,000)
Increase in prepaid insurance	(1,500)
Increase in accounts payable	30,000
Increase in unearned revenue	12,000
Increase in wages payable	3,000
Cash generated from operations	12,000
Interest paid	0
Income tax paid	0
Net cash from operating activities	12,000
Purchase of equipment	(80,000)
Net cash used in investing activities	(80,000)
Proceeds from issue of shares	100,000
Proceeds from bank loan	60,000
Dividends paid	(10,000)
Net cash from financing activities	150,000
Net increase in cash	82,000
Cash at 1 January	0
Cash at 31 December	82,000

The cash-flow statement here follows the common IFRS presentation that starts from profit before tax, adds back interest expense, and then deducts interest and tax actually paid, so that these appear as separate lines. Starting instead from profit for the year (12,750) and adding back the increases in tax payable (4,250) and interest payable (1,500) gives the same 12,000. Under U.S. GAAP, dividends paid must be classified as financing; under IFRS (until IFRS 18 takes effect—see below), a company may choose operating or financing, and must apply the choice consistently.

Check your reasoning

1. Kestrel's operating cash flow (12,000) is below its profit (12,750), even though depreciation of 10,000 was added back. Which items explain this?
2. The club's 12,000 advance increased cash and a liability. When will it become revenue, and what entry will record that?

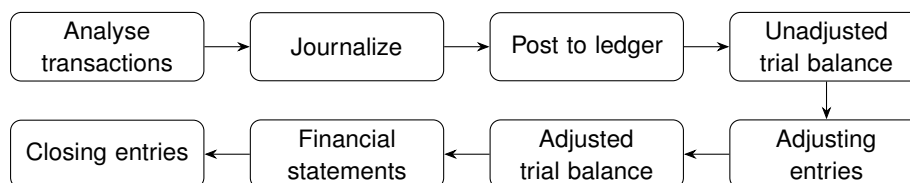
Answers. (1) The build-up of working capital: inventory rose by 40,000 and receivables by 20,000, only partly offset by more credit from suppliers (30,000), the advance (12,000) and accrued items. A growing trading business ties up cash in stock and customer credit. (2) In February 2027, on delivery: debit Unearned revenue 12,000, credit Sales revenue 12,000 (and record the cost of the bicycles delivered).

27.5.6 Closing the books

Income, expense and dividend accounts are **temporary** (nominal) accounts: they measure flows for one period and start the next period at zero. At the year end, **closing entries** transfer their balances to retained earnings:

Sales revenue	270,000	
Retained earnings		270,000
Retained earnings	257,250	
Cost of goods sold		160,000
Wages expense		53,000
Rent expense		24,000
Insurance expense		4,500
Depreciation expense		10,000
Interest expense		1,500
Income tax expense		4,250
Retained earnings	10,000	
Dividends		10,000

After closing, retained earnings is $270,000 - 257,250 - 10,000 = 2,750$, as in the balance sheet, and only **permanent** (real) accounts—assets, liabilities and equity—carry balances into 2027. The stock-flow identity of Chapter 1 is at work: the flow of profit for the period, less distributions, becomes part of the stock of equity.

27.5.7 The cycle summarized

Modern accounting software performs most of these steps automatically, but the logic is unchanged, and understanding it is essential for interpreting what the software produces—and for recognizing when it has been manipulated.

27.6 Presentation of Financial Statements

27.6.1 A complete set

Under IFRS a complete set of financial statements comprises a statement of financial position; a statement of profit or loss and other comprehensive income; a statement of changes in equity; a statement of cash flows; and notes, including material accounting policy information and comparative figures for the previous period. U.S. GAAP requires equivalent statements.

27.6.2 Current and non-current classification

Assets and liabilities are normally classified as **current** (expected to be realized or settled within twelve months or within the normal operating cycle) or **non-current**. The distinction matters for assessing **liquidity**: a firm whose current liabilities greatly exceed its current assets may struggle to meet its obligations even if it is profitable. Banks, whose assets and liabilities do not fit this classification well, instead present items broadly in order of liquidity (Chapter 48).

27.6.3 Other comprehensive income

Some gains and losses bypass profit or loss and are recorded in **other comprehensive income** (OCI), accumulating in a separate component of equity. Examples include gains on revaluing property under IFRS, some changes in the fair value of investments, remeasurements of pension obligations, and foreign-currency translation differences on foreign subsidiaries. **Total comprehensive income** is profit plus OCI. The boundary between profit and OCI is one of the least principled parts of accounting: it reflects a historical compromise over items that standard setters considered too volatile or too far from the entity's operating performance to include in profit. In Chapter 51 we shall see that the treatment of unrealized losses on banks' bond holdings—some of which flowed through OCI rather than profit, and some of which were not recognized at all—mattered greatly in the bank failures of 2023.

27.6.4 IFRS 18

In April 2024 the IASB issued IFRS 18, *Presentation and Disclosure in Financial Statements*, which replaces IAS 1 for annual periods beginning on or after 1 January 2027. It requires income and expenses in the statement of profit or loss to be classified into operating, investing and financing categories, with defined subtotals including "operating profit"; requires disclosure and reconciliation of management-defined performance measures (such as "adjusted EBITDA") used outside the financial statements; and, with related amendments, removes most of the classification choices for interest and dividends in the cash-flow statement. The aim is greater comparability of the income statement across companies.

27.7 IFRS and U.S. GAAP Compared

Topic	IFRS	U.S. GAAP
Style	More principles-based	More rules-based, with industry guidance
Inventory cost formulas	FIFO or weighted average; LIFO prohibited	FIFO, weighted average or LIFO
Property, plant and equipment	Cost or revaluation model	Cost model only
Impairment reversals	Permitted (except goodwill)	Prohibited for assets held for use
Development costs	Capitalized when criteria met	Generally expensed (software exceptions)
Interest paid (cash flows)	Operating or financing (choice, until IFRS 18 effects)	Operating
Dividends paid (cash flows)	Operating or financing (choice, until IFRS 18 effects)	Financing
Extraordinary items	Prohibited	Prohibited (since 2015)

Chapters 28 to 14 explain each of these differences in context.

27.8 Auditing and Internal Control

27.8.1 The auditor's role

Financial statements are prepared by management, who have incentives to present results favourably (Chapter 25's principal-agent problem). An independent **external audit** provides reasonable assurance that the statements are free from material misstatement, whether due to error or fraud. The auditor's report states an opinion; a *clean* (unmodified) opinion says the statements present fairly, in all material respects, the entity's position and performance in accordance with the applicable framework. Auditing is itself a response to asymmetric information: an auditor acts as a certifier, like the rating agencies and inspectors of Chapter 25.

Audits are sample-based and rely heavily on *evidence*. Some evidence is more reliable than others: evidence obtained directly by the auditor from an independent third party—for instance, a confirmation of a bank balance sent by the bank directly to the auditor—is more reliable than documents supplied by the client.

27.8.2 Internal control

Internal controls are the processes a company uses to ensure reliable reporting, compliance and efficient operations: segregation of duties (the person who approves payments does not also record them), authorization limits, reconciliations of bank statements to the ledger, and physical controls over assets. After the accounting scandals at Enron and WorldCom, the U.S. Sarbanes-Oxley Act of 2002 required chief executives and chief financial officers of public companies to certify their financial reports personally, required management to assess the effectiveness of internal control over financial reporting, required auditors of larger public companies to attest to that assessment, and created the Public Company Accounting Oversight Board (PCAOB) to regulate auditors. The EU introduced audit reforms in 2014, including mandatory rotation of audit firms for public-interest entities; Germany tightened its rules further after Wirecard.

Case: Wirecard: when the cash is not there

Wirecard processed card payments for merchants. From the mid-2010s, a growing share of its reported revenue and profit came from “third-party acquiring” partners in Asia and the Middle East, to whom Wirecard said it outsourced business in countries where it lacked licences. The partners’ payments to Wirecard were, it said, held in escrow accounts managed by a trustee. By 2019 these escrow balances had reached about 1.9 billion euros.

The accounting architecture of this chapter shows why the case turned on a single question: *existence*. Cash is the simplest asset to verify: it either is or is not in a bank account under the entity’s control. For years, Wirecard’s auditor accepted confirmations and documents obtained via the trustee rather than obtaining confirmations directly from the banks. Journalists at the *Financial Times* published detailed allegations from 2019 onwards; Germany’s financial regulator, BaFin, responded by filing a criminal complaint against the journalists and by temporarily banning short-selling of Wirecard shares. A special audit by a different firm, commissioned under pressure from investors, reported in April 2020 that it could not verify the existence of the third-party business. In June 2020 the two Philippine banks said the documents purporting to show Wirecard’s accounts were false; the company withdrew its accounts, its chief executive resigned and was arrested, and it filed for insolvency.

The episode prompted a German parliamentary inquiry and the Financial Market Integrity Strengthening Act (FISG) of 2021, which strengthened BaFin’s powers, tightened auditor-rotation rules and increased auditors’ liability. For our purposes it illustrates several points. The double-entry system guarantees that the books balance, not that they are true: fictitious revenue credited to income was matched by fictitious receivables and then fictitious cash debited to assets. Internal consistency is not evidence of existence. The quality of audit evidence matters more than its quantity. And certification institutions—auditors, regulators—are themselves subject to the incentive and information problems they are meant to solve.

Summary

- Financial reporting provides information useful to investors and lenders; it is an institutional response to asymmetric information.
- IFRS (IASB) and U.S. GAAP (FASB, with SEC oversight) are the two main frameworks; Germany uses IFRS for listed groups and HGB for individual accounts, China uses IFRS-converged CAS, Malaysia uses MFRS (IFRS).
- Useful information is relevant and faithfully represented, and ideally comparable, verifiable, timely and understandable.
- Assets are present economic resources controlled as a result of past events; liabilities are present obligations to transfer resources; income and expenses are changes in these other than transactions with owners.
- Recognition requires relevance and faithful representation; measurement uses historical cost or current values, with fair values classified in a three-level hierarchy.
- Double entry records every transaction with equal debits and credits; journals, ledgers and trial balances process transactions.
- Adjusting entries record accruals, deferrals, depreciation and taxes; closing entries transfer temporary accounts to retained earnings.

- Auditing and internal control provide assurance, but a balanced set of books is not evidence that the recorded assets exist.

Key terms

financial accounting; managerial accounting; IASB; IFRS; FASB; U.S. GAAP; HGB; CAS; MFRS; relevance; materiality; faithful representation; prudence; going concern; asset; liability; equity; income; expenses; asset–liability view; recognition; historical cost; fair value; fair-value hierarchy; account; debit; credit; journal; ledger; T-account; trial balance; adjusting entries; accumulated depreciation; contra-asset; temporary and permanent accounts; closing entries; current and non-current; liquidity; other comprehensive income; total comprehensive income; external audit; internal control; Sarbanes–Oxley Act.

Exercises

Exercise 27.1 (Elements). State whether each item is an asset, liability, equity, income, expense or none, and explain: (a) a firm’s highly skilled workforce; (b) a signed contract to buy raw materials next month; (c) a customer’s deposit for goods not yet delivered; (d) a patent purchased from an inventor; (e) a brand developed internally over decades; (f) cash received from issuing new shares; (g) the cost of electricity used this month but billed next month.

Exercise 27.2 (Debits and credits). For each transaction, give the journal entry: (a) buy a delivery van for 30,000, paying 10,000 cash and signing a note for the rest; (b) receive 5,000 from a customer for an invoice issued last month; (c) pay 2,400 for a two-year software subscription in advance; (d) declare and pay a dividend of 8,000; (e) provide consulting services worth 7,000 on credit.

Exercise 27.3 (Trial balance errors). Which of the following errors would cause the trial balance not to balance? (a) A 500 cash sale is recorded as a debit to cash of 500 and a credit to sales of 50. (b) A payment of 300 to a supplier is omitted entirely. (c) Rent of 1,000 is debited to wages expense instead of rent expense. (d) A receipt of 800 from a customer is debited to cash and debited to receivables.

Exercise 27.4 (Adjusting entries). Prepare adjusting entries at 31 December for: (a) a one-year rent payment of 18,000 made on 1 October and debited to prepaid rent; (b) a 12,000 fee received on 1 November for six months of maintenance services starting that day, credited to unearned revenue; (c) a 100,000 loan taken on 1 September at 6% per year, interest payable annually; (d) services worth 4,000 performed in December but not yet billed.

Exercise 27.5 (Effect of omitted adjustments). If Kestrel had forgotten adjustments (a) to (d), what would its profit before tax have been? By how much would total assets and total liabilities have been misstated?

Exercise 27.6 (Full cycle). Nordlicht Design GmbH begins on 1 March. During March: (1) owners invest 40,000 cash; (2) it buys computers for 12,000 cash (useful life 3 years, no residual value); (3) it pays 3,000 for three months’ office rent in advance; (4) it bills clients 25,000 for completed work, of which 15,000 is received in cash; (5) it pays staff 9,000; (6) it receives a telephone bill for 400, unpaid at month end. Prepare journal entries, adjusting entries at 31 March (depreciation, rent), an income statement for March (ignore tax), and a balance sheet at 31 March.

Exercise 27.7 (Measurement). A company holds (a) 10,000 shares of a listed company; (b) an interest-rate swap valued using quoted swap rates; (c) a 5% stake in an unlisted start-up valued using a discounted cash-flow model. Assign each to a level of the fair-value hierarchy. Which valuation should an analyst scrutinize most, and why?

Exercise 27.8 (Qualitative characteristics). A retailer's land was bought in 1985 for 2 million and is reported at that cost. A recent independent appraisal values it at 30 million. Discuss the relevance and faithful representation of each figure. Which accounting treatments are available under IFRS and U.S. GAAP?

Exercise 27.9 (Audit evidence). Rank the following as audit evidence for the existence of a bank balance, from most to least reliable, and explain: (a) a bank statement provided by the client's finance department; (b) a confirmation sent by the bank directly to the auditor; (c) a screenshot of an online banking portal emailed by the CFO; (d) a letter from a third-party trustee stating that it holds the funds.

Solutions

Solution 27.1. (a) None: the firm does not control its employees. (b) None yet: an executory contract with no performance by either side; no present obligation arises until delivery (unless the contract is onerous). (c) Liability (contract liability): an obligation to deliver goods. (d) Asset: a purchased, controlled right with expected benefits. (e) Generally not recognized: internally generated brands are not recognized as assets under IFRS or U.S. GAAP because their cost cannot be distinguished from the cost of developing the business as a whole. (f) Equity (share capital), and an asset (cash); not income, since it is a contribution from owners. (g) Expense for this month, with a liability (accrued expense).

Solution 27.2. (a) Dr Vehicles 30,000; Cr Cash 10,000; Cr Notes payable 20,000. (b) Dr Cash 5,000; Cr Accounts receivable 5,000. (c) Dr Prepaid software 2,400; Cr Cash 2,400. (d) Dr Dividends 8,000; Cr Cash 8,000. (e) Dr Accounts receivable 7,000; Cr Service revenue 7,000.

Solution 27.3. (a) Yes: debits exceed credits by 450. (b) No: both sides omitted. (c) No: debit is in the wrong account but total debits still equal credits. (d) Yes: two debits and no credit, so debits exceed credits by 1,600.

Solution 27.4. (a) Three months used: Dr Rent expense 4,500; Cr Prepaid rent 4,500. (b) Two months earned: Dr Unearned revenue 4,000; Cr Service revenue 4,000. (c) Four months' interest, $100,000 \times 6\% \times 4/12 = 2,000$: Dr Interest expense 2,000; Cr Interest payable 2,000. (d) Dr Accounts receivable (or accrued revenue) 4,000; Cr Service revenue 4,000.

Solution 27.5. The omitted adjustments total $10,000 + 4,500 + 3,000 + 1,500 = 19,000$ of expenses, so profit before tax would have been $17,000 + 19,000 = 36,000$ (and tax would have been overstated too). Assets would have been overstated by 14,500 (equipment 10,000 and prepaid insurance 4,500) and liabilities understated by 4,500 (wages and interest payable), before considering tax.

Solution 27.6. Entries: (1) Dr Cash 40,000; Cr Share capital 40,000. (2) Dr Computers 12,000; Cr Cash 12,000. (3) Dr Prepaid rent 3,000; Cr Cash 3,000. (4) Dr Cash 15,000; Dr Receivables 10,000; Cr Revenue 25,000. (5) Dr Wages 9,000; Cr Cash 9,000. (6) Dr Telephone expense 400; Cr Accounts payable 400. Adjustments: depreciation $12,000/36 = 333$ (rounded): Dr Depreciation 333; Cr Accumulated depreciation 333. Rent: Dr Rent expense 1,000; Cr Prepaid rent 1,000.

Income statement: revenue 25,000; wages 9,000; rent 1,000; telephone 400; depreciation 333; profit 14,267.

Balance sheet: cash $40,000 - 12,000 - 3,000 + 15,000 - 9,000 = 31,000$; receivables 10,000; prepaid rent 2,000; computers $12,000 - 333 = 11,667$. Total assets 54,667. Liabilities: accounts payable 400. Equity: share capital 40,000; retained earnings 14,267. Total 54,667.

Solution 27.7. (a) Level 1. (b) Level 2: observable market inputs, but not a quoted price for the identical instrument. (c) Level 3: unobservable inputs (projected cash flows, discount rate). The Level 3 valuation

deserves the most scrutiny, because management's assumptions drive it and small changes in them can produce large changes in value.

Solution 27.8. The historical cost of 2 million is verifiable and free from error as a record of the transaction, but it is not relevant to assessing the land's current value or the firm's collateral. The appraisal of 30 million is relevant but less verifiable; it depends on the appraiser's judgement. Under IFRS (IAS 16) the retailer may adopt the revaluation model for the class of land, reporting it at fair value with increases credited to OCI (a revaluation surplus), provided revaluations are kept up to date. Under U.S. GAAP, land must remain at cost; the appraisal could be disclosed voluntarily but not recognized.

Solution 27.9. Most reliable: (b), obtained directly by the auditor from an independent source. Then (d), from a third party but not the bank itself and routed through an intermediary whose independence and existence must also be verified. Then (a), a document from the client, which could be altered. Least reliable: (c), easily fabricated and provided by an interested party. The Wirecard audit relied on evidence closer to (d) than to (b).

Revenue, Inventory and Working Capital

After studying this chapter you should be able to

1. Apply the five-step model of IFRS 15 and ASC 606 to recognize revenue.
2. Allocate a transaction price among performance obligations and recognize revenue over time on long-term contracts.
3. Distinguish a principal from an agent and explain why it matters for platforms.
4. Account for receivables and estimate expected credit losses under IFRS 9 and CECL.
5. Apply FIFO, weighted-average and LIFO cost formulas and explain their effects on profit, taxes and the balance sheet when prices change.
6. Apply the lower of cost and net realizable value rule.
7. Compute the operating cycle, the cash conversion cycle and liquidity ratios, and interpret them.
8. Explain supplier finance and why it matters for analysing working capital.
9. Connect inventory investment to the business cycle.

A motivating question

When a customer signs a two-year mobile-phone contract that includes a “free” handset, how much revenue does the phone company earn on the day it hands over the phone? When a construction firm builds a bridge over three years, when does it earn its profit—at the end, or bit by bit? When a ride-hailing app collects a fare, is the whole fare its revenue, or only its commission? These are not academic questions. Revenue is the largest number in most income statements, it drives profit, and it has been at the centre of a large share of accounting frauds. Enforcement actions by the U.S. Securities and Exchange Commission have, for decades, involved revenue recognition more than any other single area.

This chapter studies the operating cycle of a business: selling goods and services, collecting cash from customers, buying and holding inventory, and paying suppliers. These activities determine not only reported profit but also how much cash a business needs to operate—its *working capital*. The chapter ends by connecting firms’ inventory decisions to macroeconomic fluctuations.

28.1 Revenue Recognition

28.1.1 The five-step model

Since 2018, IFRS and U.S. GAAP have applied essentially identical standards on revenue from contracts with customers: IFRS 15 and ASC 606, the product of a joint project between the IASB and the FASB. Their core principle is:

Core principle of revenue recognition. An entity recognizes revenue to depict the transfer of promised goods or services to customers in an amount that reflects the consideration to which it expects to be entitled in exchange for those goods or services.

The principle is applied in five steps:

1. **Identify the contract** with a customer: an enforceable agreement with commercial substance, in which collection of the consideration is probable.
2. **Identify the performance obligations:** the distinct goods or services promised. A good or service is distinct if the customer can benefit from it on its own (or with readily available resources) and it is separately identifiable within the contract.
3. **Determine the transaction price:** the consideration the entity expects to be entitled to, including estimates of *variable consideration* (discounts, rebates, bonuses, returns), which are included only to the extent that a significant reversal of revenue is highly probable (“probable” under U.S. GAAP) not to occur—the **variable consideration constraint**.
4. **Allocate the transaction price** to the performance obligations in proportion to their **standalone selling prices**—the prices at which the entity would sell each good or service separately.
5. **Recognize revenue** when (or as) each performance obligation is satisfied, that is, when control of the good or service passes to the customer. This may be at a point in time or over time.

28.1.2 Bundles and allocation

Worked Example 28.1. A phone and a service plan

A telecoms company sells a two-year contract for 60 per month (total 1,440), including a smartphone delivered on day one. The company sells the same phone on its own for 800 and a two-year SIM-only plan with the same service for 40 per month (total 960).

Step 2: two performance obligations—the phone (satisfied at a point in time, on delivery) and the network service (satisfied over 24 months). *Step 3:* transaction price 1,440. *Step 4:* the standalone prices sum to 1,760, so the bundle is sold at a discount of 320. Allocate proportionally:

$$\text{phone: } 1,440 \times \frac{800}{1,760} = 654.55; \quad \text{service: } 1,440 \times \frac{960}{1,760} = 785.45.$$

Step 5: on delivery, recognize revenue of 654.55 for the phone. Recognize service revenue of $785.45/24 = 32.73$ per month.

In month 1 the customer pays 60, but the company recognizes $654.55 + 32.73 = 687.28$ of revenue. The difference, 627.28, is a **contract asset**: a right to consideration for goods already transferred, conditional on the customer continuing to pay. Over the contract, the monthly cash of 60 exceeds monthly service revenue of 32.73, and the contract asset unwinds to zero.

Before IFRS 15, many telecoms companies recognized almost no revenue for handsets given “free” with contracts, and recognized all revenue as monthly service charges. The new standard shifted revenue toward the start of contracts. Total revenue over the contract is unchanged; its timing is not.

28.1.3 Point in time or over time

A performance obligation is satisfied *over time* if any of the following holds: the customer simultaneously receives and consumes the benefits as the entity performs (a cleaning service); the entity's work creates or enhances an asset that the customer controls as it is created (building on the customer's land); or the entity's work creates an asset with no alternative use to the entity and the entity has an enforceable right to payment for performance to date. Otherwise, it is satisfied at a *point in time*, when control passes—indicated by legal title, physical possession, the transfer of risks and rewards, customer acceptance, and a present right to payment.

28.1.4 Long-term contracts

For obligations satisfied over time, revenue is recognized by measuring progress toward completion. A common input method is **cost-to-cost**: progress equals costs incurred to date divided by total expected costs.

Worked Example 28.2. A three-year construction contract

A firm agrees to build a bridge for 10 million. It initially expects total costs of 8 million. Costs incurred: year 1, 3.2 million; year 2, 3.6 million. At the end of year 2, the firm revises its estimate of total costs to 8.5 million. Year 3 costs are 1.7 million.

	Year 1	Year 2	Year 3
Cumulative cost	3.2	6.8	8.5
Estimated total cost	8.0	8.5	8.5
Progress (cumulative cost / total)	40%	80%	100%
Cumulative revenue ($\times 10$)	4.0	8.0	10.0
Revenue for the year	4.0	4.0	2.0
Cost for the year	3.2	3.6	1.7
Profit for the year	0.8	0.4	0.3

Total profit is 1.5 million, equal to the contract price less actual total cost. The revision of estimated cost in year 2 is a *change in estimate*: it affects the current and future years but does not restate year 1. If at any point the firm expected total costs to exceed 10 million, it would recognize the entire expected loss immediately (an *onerous contract*), an application of prudence.

Percentage-of-completion accounting is informative—it shows profits being earned as work progresses—but it depends heavily on estimates of total cost, which management controls. Chapter 31's analysis of earnings quality returns to this risk; the troubled aircraft programmes of Boeing (Chapter 29's case) illustrate how large revisions of estimates can be.

28.1.5 Principal or agent

When another party is involved in providing goods or services to a customer, the entity must decide whether it is a **principal** (it controls the good or service before transfer and recognizes revenue for the gross amount the customer pays) or an **agent** (it arranges for another party to provide the good or service and recognizes only its fee or commission, the net amount). Indicators of a principal include primary responsibility for fulfilling the promise, inventory risk, and discretion in setting the price.

The distinction matters enormously for platforms. An online retailer selling its own inventory reports the full sale price as revenue; the same company acting as a marketplace for

third-party sellers reports only its commission and fees. Two businesses with identical profits can report revenues that differ by a factor of five or more. For this reason, analysts compare platforms using *gross merchandise value* or *gross bookings* alongside reported revenue, and pay close attention to principal–agent judgements. Ride-hailing companies, for example, have generally concluded that drivers provide the transport service and that the platform acts as an agent, reporting net revenue; some have faced regulatory and legal challenges on precisely these questions of control.

28.1.6 Other issues

Returns and rebates. Expected returns and volume rebates are variable consideration: revenue is reduced by the amount expected to be refunded, and a refund liability is recognized.

Contract liabilities. Cash received before performance is a **contract liability** (deferred or unearned revenue), as in Chapter 27’s advance from the sports club. Subscription software companies often report large contract liabilities, which analysts watch as an indicator of future revenue.

Warranties. A standard warranty that assures the product works as specified is a cost, provided for when the product is sold (Chapter 30). A warranty that provides an extra service beyond that assurance is a separate performance obligation, with revenue allocated to it.

► Common error: revenue equals cash received

Revenue follows the transfer of control of goods or services, not the receipt of cash. Cash can precede revenue (advance payments, subscriptions), coincide with it (a shop sale), or follow it (credit sales, contract assets). Many revenue frauds exploit this gap by recognizing revenue before control has really passed—for instance, “bill-and-hold” arrangements in which goods are invoiced but kept in the seller’s warehouse, or “channel stuffing,” in which distributors are pressed to take excess stock with generous return rights.

28.2 Receivables and Expected Credit Losses

28.2.1 Recognizing receivables

A **trade receivable** arises when a customer is given credit. It is initially measured at the transaction price. The critical accounting question is how much of it will be collected.

28.2.2 The allowance method

Accounting standards require firms to estimate expected losses on receivables and recognize them in advance, rather than waiting until specific customers fail to pay. The estimate is recorded in an **allowance for credit losses** (a contra-asset), and the charge is an expense (called impairment loss or credit-loss expense):

Credit-loss expense	X
Allowance for credit losses	X

When a specific receivable is judged uncollectible, it is written off against the allowance, with no further effect on profit:

Allowance for credit losses	Y
Accounts receivable	Y

Receivables are reported net of the allowance. Profit is affected when the estimate is made, not when the write-off occurs: that is the point of accrual accounting.

28.2.3 Expected-loss models

Before the global financial crisis, both IFRS and U.S. GAAP used an *incurred-loss* model: losses were recognized only when there was objective evidence that they had already been incurred. Critics argued that this recognized losses “too little, too late,” especially for banks entering a downturn. Both standard setters moved to *expected-loss* models, but not identical ones:

- **IFRS 9** (effective 2018) uses a three-stage model. For financial assets whose credit risk has not increased significantly since initial recognition (stage 1), the allowance equals *12-month* expected credit losses; after a significant increase in credit risk (stage 2) or on impairment (stage 3), it equals *lifetime* expected credit losses. For trade receivables without a significant financing component, a simplified approach requires lifetime expected losses from the start.
- **CECL** (Current Expected Credit Losses, ASC 326), effective for large U.S. public companies from 2020, requires *lifetime* expected credit losses for all loans and receivables from the date they are recognized.

For ordinary trade receivables the two approaches give similar answers. For banks they differ substantially: CECL front-loads provisions for new loans. Chapter 51 discusses concerns that expected-loss provisioning may amplify credit cycles if provisions rise sharply at the start of downturns.

Worked Example 28.3. An ageing schedule

A wholesaler estimates expected credit losses on its receivables using historical loss rates adjusted for current conditions:

Age of receivable	Balance	Expected loss rate	Expected loss
Not yet due	100,000	1%	1,000
1–30 days overdue	40,000	3%	1,200
31–90 days overdue	20,000	10%	2,000
Over 90 days overdue	10,000	40%	4,000
Total	170,000		8,200

The allowance should be 8,200. If the allowance currently has a credit balance of 3,000 (after write-offs during the year), the adjusting entry is: Dr Credit-loss expense 5,200; Cr Allowance 5,200. Receivables are reported at $170,000 - 8,200 = 161,800$.

The loss rates are estimates, and they are among the most judgemental numbers in the accounts. A firm that wishes to report higher profit can lower them; one that wishes to build a “cushion” for bad years can raise them. Analysts compare the allowance with overdue balances and with actual write-offs over time.

28.3 Inventory

28.3.1 What is included in cost

Inventory is measured at *cost*, which includes all costs of purchase (the price, import duties, transport, less discounts), costs of conversion (for manufacturers, direct labour and a systematic allocation of production overheads) and other costs incurred to bring the inventory to its present location and condition. Storage, selling and administrative costs are expenses when incurred. When a good is sold, its cost is transferred from inventory (an asset) to cost of goods sold (an expense), matching the cost to the revenue.

28.3.2 Cost formulas

When identical items are bought at different prices, the firm must decide which costs to assign to the units sold and which to the units remaining. Specific identification is used only for unique, high-value items. Otherwise, a **cost formula** is applied:

- **First-in, first-out (FIFO)**: the earliest costs go to cost of goods sold; ending inventory reflects the most recent costs.
- **Weighted average cost**: each unit carries the average cost of the units available.
- **Last-in, first-out (LIFO)**: the most recent costs go to cost of goods sold; ending inventory reflects the oldest costs. LIFO is permitted under U.S. GAAP but prohibited under IFRS.

Worked Example 28.4. Three cost formulas

A retailer starts the year with 100 units costing 10 each. It buys 200 units at 12 and then 300 units at 14. It sells 400 units at 20 each. Goods available: 600 units costing $1,000 + 2,400 + 4,200 = 7,600$.

	FIFO	Weighted average	LIFO
Cost of goods sold (400 units)	4,800	5,067	5,400
Ending inventory (200 units)	2,800	2,533	2,200
Revenue	8,000	8,000	8,000
Gross profit	3,200	2,933	2,600

FIFO: $100 \times 10 + 200 \times 12 + 100 \times 14 = 4,800$; ending 200×14 . Average cost per unit $7,600/600 = 12.67$. LIFO (applied at the period end): $300 \times 14 + 100 \times 12 = 5,400$; ending $100 \times 10 + 100 \times 12$. In each case, cost of goods sold plus ending inventory equals the 7,600 available.

When prices are rising, as here:

- FIFO gives the highest profit and the most current balance-sheet value for inventory, but its cost of goods sold uses old, lower costs and so overstates the “sustainable” margin: replacing the units sold will cost more.
- LIFO gives the lowest profit, matching recent costs against current revenue, but leaves inventory on the balance sheet at costs that may be decades old.
- Weighted average lies in between.

LIFO and taxes. In the United States, LIFO lowers taxable income when prices rise, deferring tax. The *LIFO conformity rule* requires firms that use LIFO for tax to use it in their financial statements too. U.S. firms using LIFO must disclose the **LIFO reserve**—the difference between inventory at FIFO (or current cost) and at LIFO—which lets analysts restate inventory and

profit on a FIFO basis for comparison with IFRS reporters. Inflation in 2021–2023 raised LIFO reserves sharply for firms with long-held inventory layers.

LIFO liquidation. If a LIFO firm sells more than it buys, old, low-cost layers flow into cost of goods sold, inflating profit in a way that is not repeatable.

28.3.3 Lower of cost and net realizable value

Inventory must not be carried above the amount expected to be recovered from selling it. Under IFRS (and U.S. GAAP for FIFO and average-cost inventories), inventory is measured at the **lower of cost and net realizable value** (NRV), where NRV is the expected selling price less costs to complete and sell. If fashion changes, technology becomes obsolete, or commodity prices fall, inventory is written down and the loss recognized as an expense. IFRS requires a write-down to be reversed if NRV later recovers (up to original cost); U.S. GAAP prohibits reversal.

28.3.4 Inventory errors

An overstatement of ending inventory understates cost of goods sold and overstates profit in the current year. Because this year's ending inventory is next year's opening inventory, the error reverses next year: cost of goods sold is overstated and profit understated. Inventory errors are “self-correcting” over two years, which is precisely what makes them attractive to managers wishing to shift profit between periods—and why auditors attend physical stock counts.

28.4 Payables, Prepayments and Accruals

Accounts payable are amounts owed to suppliers for goods and services received on credit; they are the mirror image of receivables. Prepaid expenses (Chapter 27's insurance) and accrued expenses (wages, interest, utilities) complete the operating-cycle accounts. Together with inventory, receivables and contract balances, they make up **working capital**.

Definition 28.5. Working capital

Working capital is current assets minus current liabilities. **Operating working capital** (sometimes “non-cash working capital”) excludes cash and interest-bearing debt: it is operating current assets (receivables, inventory, prepayments, contract assets) minus operating current liabilities (payables, accruals, contract liabilities).

An increase in operating working capital absorbs cash; a decrease releases it. That is why the indirect cash-flow statement subtracts increases in receivables and inventory and adds increases in payables (Chapters 10 and 27).

28.5 The Operating Cycle and the Cash Conversion Cycle

28.5.1 Definitions

A trading firm pays for inventory, holds it, sells it on credit, and collects the cash. The **operating cycle** is the time from acquiring inventory to collecting cash from its sale. The firm does not

need to finance the whole cycle, because suppliers extend credit. The **cash conversion cycle** (CCC) measures the days for which the firm itself must finance its operations:

$$\text{CCC} = \underbrace{\text{DIO}}_{\text{days inventory outstanding}} + \underbrace{\text{DSO}}_{\text{days sales outstanding}} - \underbrace{\text{DPO}}_{\text{days payables outstanding}},$$

where, using annual figures,

$$\text{DIO} = \frac{\text{Inventory}}{\text{COGS}} \times 365, \quad \text{DSO} = \frac{\text{Receivables}}{\text{Revenue}} \times 365, \quad \text{DPO} = \frac{\text{Payables}}{\text{COGS}} \times 365.$$

(Purchases rather than COGS is the more precise denominator for DPO, but COGS is commonly used because purchases are rarely disclosed. Average balances over the year are preferable to year-end balances when available.)

Worked Example 28.6. Kestrel's cash conversion cycle

From Chapter 27: revenue 270,000, COGS 160,000, year-end inventory 40,000, receivables 20,000, payables 30,000.

$$\begin{aligned} \text{DIO} &= \frac{40,000}{160,000} \times 365 = 91.3, & \text{DSO} &= \frac{20,000}{270,000} \times 365 = 27.0, \\ \text{DPO} &= \frac{30,000}{160,000} \times 365 = 68.4. \end{aligned}$$

$\text{CCC} = 91.3 + 27.0 - 68.4 = 49.9$ days. Kestrel must finance about 50 days of its costs itself. If it grows its sales by 20,000 a month, it will need extra funding for its working capital, even if every sale is profitable.

Two liquidity ratios complement the CCC. The **current ratio**, current assets divided by current liabilities, is $143,500/50,750 = 2.83$. The **quick ratio** (acid test) excludes inventory and prepayments: $(82,000 + 20,000)/50,750 = 2.01$. Both are high, because Kestrel has just raised equity and a loan and holds a large cash balance.

28.5.2 Interpreting the cycle

A shorter CCC means less capital tied up in operations. Firms shorten it by holding less inventory (just-in-time production), collecting faster (discounts for early payment, stricter credit terms, factoring receivables) and paying suppliers later. Some firms achieve a *negative* CCC: customers pay before the firm pays its suppliers, so growth *generates* cash. Supermarkets, which sell goods for cash within days of receiving them but pay suppliers after weeks, often operate this way; so do some technology hardware firms with strong bargaining power over suppliers.

A lengthening CCC is often an early warning. Rising DSO may mean customers are paying more slowly—or that revenue has been recognized too aggressively on sales to customers who will never pay. Rising DIO may mean inventory is becoming obsolete. Falling DPO may mean suppliers are losing confidence and demanding faster payment.

28.5.3 Supplier finance

In a **supplier finance** (reverse factoring) arrangement, a bank pays the firm's suppliers early at a discount, and the firm pays the bank later, often long after the original due date. The firm's DPO rises and its operating cash flow improves; economically, however, the firm has borrowed from the bank. If the obligation remains classified as a trade payable rather than as debt, the

firm's leverage and cash flows can be flattered. The collapse of the British construction group Carillion in 2018 and the failure of Greensill Capital, a major provider of supply-chain finance, in 2021 drew attention to the problem. The FASB (from 2023) and the IASB (from 2024) now require companies to disclose the terms and amounts of supplier finance programmes, so that analysts can adjust working capital and debt accordingly.

28.6 Working Capital and the Macroeconomy

Bridge to Part VI. Inventory investment—the change in inventories—is a small part of GDP on average but a large part of its *fluctuations*. When firms expect sales to fall, they cut orders to reduce inventories, so production falls by more than sales; when demand recovers, restocking amplifies the rebound. In many U.S. recessions, the fall in inventory investment has accounted for a large share of the decline in GDP. The macroeconomic “inventory cycle” is the aggregate of the working-capital decisions studied in this chapter. Similarly, a tightening of credit forces firms to shrink working capital—collecting from customers faster and paying suppliers more slowly—which transmits financial stress along supply chains (the “trade credit channel”). Chapters 43 and 46 return to these links, and Chapter 49 shows how central-bank rates affect the cost of financing working capital.

Case: Apple: working capital as a competitive advantage

Apple's annual reports, available on the SEC's EDGAR system, illustrate how a company with strong bargaining power can turn working capital into a source of funds. Apple outsources most manufacturing to contract manufacturers, holds very little finished inventory relative to its sales, collects quickly from retail customers and carriers, and pays its suppliers on relatively long terms. For many years this combination produced a cash conversion cycle that was strongly negative: Apple was, in effect, financed by its suppliers.

The analyst's lesson is to decompose working capital rather than take it at face value. Apple's balance sheet also reports large “vendor non-trade receivables”—amounts due from contract manufacturers for components Apple buys and resells to them for assembly—which are excluded from a trade-receivables DSO calculation but matter for understanding the flows. Revenue recognition also requires judgement: when Apple sells a device bundled with software updates and cloud services, part of the price is allocated to those services and deferred, as in Example 28.1. And the contrast with a manufacturer that holds months of inventory and extends long credit to customers shows why two firms with identical profit margins can have very different needs for external finance. In an exercise, reconstruct Apple's DIO, DSO and DPO from its most recent 10-K and compare them with a retailer and an industrial manufacturer.

Summary

- IFRS 15 and ASC 606 recognize revenue as control of goods or services passes, in five steps: contract, performance obligations, transaction price, allocation, and recognition.
- Bundles are allocated by standalone selling prices; long-term contracts recognize revenue over time by measuring progress; expected losses on contracts are recognized immediately.

- Principals report gross revenue and agents net; the judgement drives platforms' reported revenue.
- Receivables are reported net of an allowance for expected credit losses (IFRS 9's staged model; CECL's lifetime model).
- FIFO, weighted average and LIFO allocate costs differently; in inflation, FIFO raises reported profit and LIFO lowers profit and taxes. Inventory is written down to net realizable value.
- Operating working capital absorbs cash when it grows. The cash conversion cycle ($\text{DIO} + \text{DSO} - \text{DPO}$) measures the days of operations the firm must finance itself.
- Supplier finance can disguise borrowing as payables; new disclosures address this.
- Firms' inventory decisions aggregate into the inventory cycle, a major contributor to recessions.

Key terms

five-step model; performance obligation; transaction price; variable consideration; standalone selling price; contract asset; contract liability; point in time; over time; cost-to-cost method; onerous contract; principal; agent; trade receivable; allowance for credit losses; expected credit losses; CECL; ageing schedule; FIFO; weighted average cost; LIFO; LIFO reserve; LIFO conformity rule; net realizable value; working capital; operating working capital; operating cycle; cash conversion cycle; DIO; DSO; DPO; current ratio; quick ratio; supplier finance.

Exercises

Exercise 28.1 (Allocation). A software company sells a licence, one year of support and on-site installation for a total of 100,000. Standalone selling prices are: licence 80,000, support 20,000, installation 10,000. The licence is delivered and installed on 1 October; support runs from 1 October for 12 months. How much revenue is recognized in the year ending 31 December?

Exercise 28.2 (Variable consideration). A manufacturer sells 10,000 units to a distributor at 50 each, with a right to return unsold units within 90 days. Based on extensive experience, it expects 5% to be returned. How much revenue is recognized on delivery, and what liability?

Exercise 28.3 (Over time). A shipbuilder signs a contract for 50 million with expected total cost of 40 million. Costs are 12 million in year 1 and 20 million in year 2, when the estimate of total cost rises to 45 million. Year 3 costs are 13 million. Compute revenue and profit each year using cost-to-cost. What if, at the end of year 2, total cost were expected to be 52 million?

Exercise 28.4 (Principal or agent). An online travel company sells hotel rooms. In model A it books rooms at hotels' prices and earns a 15% commission; in model B it buys blocks of rooms in advance at a fixed price, bears the risk that they go unsold, and sets its own resale prices. For a room sold to a customer for 200, what revenue does it report in each model? Why might two firms with identical profits report very different revenue growth?

Exercise 28.5 (Credit losses). At year end, a firm's receivables are 500,000, and its ageing analysis implies expected credit losses of 18,000. The allowance had an opening credit balance of 15,000; during the year, 12,000 of receivables were written off. Record the write-offs and the year-end adjustment. What is the credit-loss expense for the year?

Exercise 28.6 (Cost formulas). Opening inventory is 50 units at 20. Purchases: 150 at 22, then 100 at 25. Sales: 220 units at 40. Compute cost of goods sold, ending inventory and gross profit under FIFO, weighted average and (periodic) LIFO. What is the LIFO reserve?

Exercise 28.7 (Tax effect of LIFO). In Exercise 28.6, if the tax rate is 21% and LIFO is used for tax, how much tax does the firm defer compared with FIFO? Is this a permanent saving?

Exercise 28.8 (NRV). A fashion retailer holds 1,000 jackets that cost 60 each. At year end the expected selling price has fallen to 70 and selling costs are 15 per jacket. What write-down is required? If next year the expected selling price recovers to 85, what does IFRS require, and what does U.S. GAAP require?

Exercise 28.9 (Inventory error). Ending inventory for 2025 was overstated by 30,000. What are the effects on 2025 and 2026 profit before tax, and on retained earnings at the end of each year?

Exercise 28.10 (Cash conversion cycle). Firm X has revenue 1,200,000, COGS 800,000, inventory 160,000, receivables 150,000 and payables 60,000. Firm Y has the same revenue and COGS, but inventory 40,000, receivables 20,000 and payables 140,000. Compute each firm's CCC. If each grows its revenue by 10% with unchanged ratios, how much additional operating working capital does each need?

Exercise 28.11 (Supplier finance). A firm with payables of 100 million on 60-day terms joins a supplier finance programme under which its bank pays suppliers after 10 days and the firm pays the bank after 120 days. If annual COGS is 600 million and the balances remain classified as trade payables, what happens to reported DPO and operating cash flow in the year the programme starts? How should an analyst adjust?

Solutions

Solution 28.1. Standalone prices sum to 110,000. Allocation: licence $100,000 \times 80/110 = 72,727$; support 18,182; installation 9,091. Recognized by 31 December: licence 72,727 (point in time), installation 9,091 (performed), support $18,182 \times 3/12 = 4,545$. Total 86,364.

Solution 28.2. Revenue = $10,000 \times 50 \times 0.95 = 475,000$. A refund liability of 25,000 is recognized, together with an asset for the right to recover the returned goods (measured at their former carrying amount less recovery costs), and cost of sales is reduced accordingly.

Solution 28.3. Year 1: progress $12/40 = 30\%$; revenue 15; profit $15 - 12 = 3$. Year 2: cumulative cost 32 of 45, progress 71.1%; cumulative revenue 35.56; year-2 revenue 20.56 ; year-2 profit $20.56 - 20 = 0.56$. Year 3: revenue $50 - 35.56 = 14.44$; profit $14.44 - 13 = 1.44$. Total profit $3 + 0.56 + 1.44 = 5$ ($50 - 45$). If the total cost estimate were 52 at the end of year 2, the contract would be loss-making by 2 overall. Having already recognized profit of 3 in year 1, the firm must recognize in year 2 whatever makes the cumulative result equal to the full expected loss of -2 : a year-2 loss of 5.

Solution 28.4. Model A: the company is an agent; revenue is the commission, $0.15 \times 200 = 30$. Model B: it controls the rooms before transfer and bears inventory risk, so it is a principal; revenue is 200 (with the room cost in cost of sales). A firm shifting from model A to model B would report sharply higher revenue growth without any change in profit; comparisons must use gross bookings and margins.

Solution 28.5. Write-offs: Dr Allowance 12,000; Cr Receivables 12,000. The allowance falls to $15,000 - 12,000 = 3,000$. Required balance 18,000: Dr Credit-loss expense 15,000; Cr Allowance 15,000. The expense for the year is 15,000.

Solution 28.6. Available: 300 units costing $1,000 + 3,300 + 2,500 = 6,800$. Revenue $220 \times 40 = 8,800$. FIFO: COGS = $1,000 + 3,300 + 20 \times 25 = 4,800$; ending $80 \times 25 = 2,000$; gross profit 4,000. Weighted average: $6,800/300 = 22.667$; COGS 4,986.67; ending 1,813.33; gross profit 3,813.33. LIFO: COGS

$= 2,500 + 120 \times 22 = 5,140$; ending $30 \times 22 + 50 \times 20 = 1,660$; gross profit 3,660. LIFO reserve $= 2,000 - 1,660 = 340$.

Solution 28.7. Taxable profit is lower by 340 under LIFO, so tax is lower by $0.21 \times 340 = 71.40$. It is a deferral: if inventory layers are later liquidated, or prices fall, the tax becomes payable. But as long as the firm maintains its inventory and prices keep rising, the deferral can persist indefinitely, which gives it real present value.

Solution 28.8. $NRV = 70 - 15 = 55$ per jacket, below cost of 60. Write-down $1,000 \times 5 = 5,000$. Next year $NRV = 85 - 15 = 70$, above original cost. IFRS requires reversal of the write-down, up to original cost: inventory returns to 60 per jacket, a gain of 5,000. U.S. GAAP prohibits reversal; the jackets remain at 55.

Solution 28.9. 2025: COGS understated by 30,000, so profit overstated by 30,000 and retained earnings at end-2025 overstated by 30,000. 2026: opening inventory overstated, so COGS overstated and profit understated by 30,000; retained earnings at end-2026 are correct. (Taxes would shift correspondingly.)

Solution 28.10. X: $DIO = 160/800 \times 365 = 73$; $DSO = 150/1,200 \times 365 = 45.6$; $DPO = 60/800 \times 365 = 27.4$; $CCC = 91.3$ days. Y: $DIO 18.3$; $DSO 6.1$; $DPO 63.9$; $CCC = -39.5$ days. Operating working capital: $X = 160 + 150 - 60 = 250$ thousand; $Y = 40 + 20 - 140 = -80$ thousand. With 10% growth and unchanged ratios, X needs an extra 25,000, while Y's working capital becomes -88 thousand, releasing 8,000 of cash.

Solution 28.11. If the firm now pays after 120 days instead of 60, payables roughly double to about 200 million; reported DPO rises from $100/600 \times 365 = 61$ to about 122 days, and operating cash flow is boosted by about 100 million in the first year as payments are delayed. Economically, about 100 million (and arguably all 200 million owed to the bank) is short-term bank borrowing. An analyst should reclassify the amounts owed under the programme as debt, move the associated cash flows from operating to financing, and recompute DPO and leverage.

Long-Lived Assets, Leases and Intangibles

After studying this chapter you should be able to

1. Determine which costs are capitalized as property, plant and equipment and which are expensed.
2. Compute depreciation by straight-line, declining-balance and units-of-production methods, and account for changes in estimates and disposals.
3. Apply the IFRS revaluation model.
4. Test assets for impairment under IFRS and U.S. GAAP and explain why the two can give different answers.
5. Account for intangible assets, including research and development, and explain why internally generated intangibles are largely unrecognized.
6. Measure goodwill in a business combination and explain how it is tested for impairment.
7. Account for leases as a lessee under IFRS 16 and ASC 842, and analyse the effects on profit, EBITDA, leverage and cash flows.
8. Explain how the capitalization decision can be used to manipulate earnings, using the WorldCom case.
9. Relate accounting book values to economic values and to Tobin's q .

A motivating question

In June 2002 WorldCom, then the second-largest long-distance telephone company in the United States, announced that it had improperly accounted for billions of dollars of ordinary operating costs. The Securities and Exchange Commission alleged that the company had transferred about 3.8 billion dollars of “line costs”—fees paid to other carriers for the use of their networks—from expenses to capital accounts, overstating pretax income by roughly 3.1 billion dollars in 2001 and about 0.8 billion in the first quarter of 2002. WorldCom filed for bankruptcy the following month, then the largest bankruptcy in U.S. history.

The fraud exploited a single decision that every business makes thousands of times a year: whether a payment creates an asset, to be expensed gradually over future periods, or is an expense of the current period. This chapter studies that decision and its consequences for the long-lived resources on which a business depends: plant and equipment, leased assets, intangible assets such as patents and software, and the goodwill that arises when one company buys another.

29.1 Property, Plant and Equipment

29.1.1 Recognition and initial measurement

Property, plant and equipment (PPE) comprises tangible items held for use in production or supply, for rental or for administrative purposes, and expected to be used for more than one period. It is initially measured at cost, which includes:

- the purchase price, including import duties and non-refundable taxes, less discounts;
- costs directly attributable to bringing the asset to the location and condition necessary for it to operate as intended: delivery, site preparation, installation, testing, professional fees;
- under IFRS, the initial estimate of the costs of dismantling and removing the item and restoring the site, if the entity has an obligation to incur them;
- **borrowing costs** directly attributable to constructing an asset that takes a substantial period to get ready (IAS 23; ASC 835-20 is similar).

Costs of training staff, general overheads, and costs of opening a new facility (advertising, relocation) are expensed.

29.1.2 Capitalize or expense?

A cost is **capitalized**—recorded as an asset—if it is expected to generate economic benefits over more than one period; otherwise it is **expensed**. After an asset is in use, *repairs and maintenance* that restore it to its original condition are expenses; *improvements* that extend its life or increase its capacity are capitalized. The line is not always sharp. Replacing a building's roof restores it; adding a floor enlarges it; replacing an aircraft engine after a set number of flight hours may be capitalized as a separate component (IFRS requires separate depreciation of significant components with different lives).

Key idea. Capitalizing a cost rather than expensing it raises current profit and assets, and lowers future profit through depreciation. Total expense over the asset's life is the same; its timing differs. Operating cash flow is also higher, because capital expenditure is classified as an investing outflow—so capitalization flatters both profit and operating cash flow.

29.2 Depreciation

29.2.1 The concept

Definition 29.1. Depreciation

Depreciation is the systematic allocation of the **depreciable amount** of an asset—its cost less its **residual value**—over its **useful life**.

Accounting depreciation is an *allocation of cost*, not a measurement of the decline in the asset's market value. A machine may fall in resale value by half the moment it is installed, because it has become specialized to the firm, while its accounting carrying amount falls gradually. The depreciation pattern should reflect the pattern in which the asset's benefits are consumed.

29.2.2 Methods

Consider a machine costing 100,000 with residual value 10,000 and a useful life of five years.

Straight-line. An equal charge each year: $(100,000 - 10,000) / 5 = 18,000$.

Declining balance. A fixed percentage of the opening carrying amount (**net book value**) each year. The *double-declining-balance* method uses twice the straight-line rate: $2 \times 1/5 = 40\%$. Residual value is not deducted from the base, but depreciation stops when the carrying amount reaches the residual value.

Units of production. Depreciation per unit of output = depreciable amount / total expected output. If the machine is expected to produce 450,000 units, the rate is 0.20 per unit.

Year	Straight-line		Double-declining balance	
	Charge	Book value	Charge	Book value
0		100,000		100,000
1	18,000	82,000	40,000	60,000
2	18,000	64,000	24,000	36,000
3	18,000	46,000	14,400	21,600
4	18,000	28,000	8,640	12,960
5	18,000	10,000	2,960	10,000
Total	90,000		90,000	

Both methods charge the same total of 90,000; declining balance front-loads it. In year 5 the declining-balance charge is limited to the amount that brings the carrying amount down to the residual value.

29.2.3 Book and tax depreciation

Tax law sets its own depreciation rules, often more accelerated than those used in financial statements, to encourage investment: the U.S. Modified Accelerated Cost Recovery System (MACRS) and “bonus depreciation” (immediate expensing of a large share of qualifying equipment); German tax depreciation tables (AfA-Tabellen); and capital allowances in Malaysia. When tax depreciation exceeds book depreciation, taxable income is below accounting income, and tax is deferred; Chapter 30 shows how this creates a *deferred tax liability*. Accelerated tax depreciation is a direct instrument of macroeconomic policy: by lowering the user cost of capital (Chapter 21), it aims to raise investment (Chapter 45).

29.2.4 Changes in estimates

Useful lives and residual values are estimates. When they change, the change is applied *prospectively*: the remaining carrying amount, less the new residual value, is depreciated over the new remaining life. Prior years are not restated. Suppose that, after two years of straight-line depreciation, the machine’s remaining life is re-estimated at five more years (seven in total) with residual value unchanged. The carrying amount of 64,000 less 10,000 is depreciated over five years: 10,800 per year.

Lengthening useful lives is a quiet way to raise reported profit. Airlines, cloud-computing firms and railways have all extended the estimated lives of their assets at times; such changes are disclosed and deserve scrutiny, especially when they coincide with pressure on profits.

29.2.5 Disposals

When an asset is sold, it is removed from the accounts at its carrying amount (cost less accumulated depreciation), and any difference from the proceeds is a gain or loss. If the machine is sold at the end of year 3 for 40,000 after straight-line depreciation (carrying amount 46,000), the entry is:

Cash	40,000	
Accumulated depreciation	54,000	
Loss on disposal	6,000	
Machine (cost)		100,000

A loss on disposal indicates that past depreciation was, with hindsight, too low; a gain, that it was too high.

29.3 The Revaluation Model

Under IFRS, an entity may choose, for an entire class of PPE, the **revaluation model**: the assets are carried at fair value at the revaluation date less subsequent depreciation, and revaluations must be made regularly enough that the carrying amount does not differ materially from fair value. An increase in value is credited to other comprehensive income and accumulated in a *revaluation surplus* within equity (unless it reverses a previous decrease recognized in profit). A decrease is recognized in profit, except to the extent that it reverses a previous surplus. Depreciation is then based on the revalued amount. U.S. GAAP does not permit revaluation of PPE.

Revaluation improves relevance for assets whose values change substantially—land and buildings especially—at some cost in verifiability. Many IFRS reporters, including most large listed companies in continental Europe, nevertheless choose the cost model, for comparability and simplicity. A separate standard, IAS 40, allows *investment property* (property held to earn rent or for capital appreciation) to be measured at fair value with changes recognized in profit, a model widely used by real-estate companies.

29.4 Impairment

29.4.1 The idea

An asset should not be carried at more than the amount the entity can recover from it. When circumstances indicate that an asset's carrying amount may not be recoverable—a fall in market prices, technological obsolescence, physical damage, a restructuring, or persistent losses—the asset must be tested for **impairment**.

29.4.2 IFRS: IAS 36

Under IAS 36, an asset is impaired if its carrying amount exceeds its **recoverable amount**, defined as the *higher* of:

- *fair value less costs of disposal* (what it could be sold for), and
- *value in use*: the present value of the future cash flows expected from continuing to use it and eventually disposing of it.

The logic is that a rational owner will either sell or keep the asset, whichever is worth more. The impairment loss is the excess of carrying amount over recoverable amount. If cash flows cannot be attributed to an individual asset, the test is applied to the smallest group that generates largely independent cash flows, a **cash-generating unit**. Impairment losses on assets other than goodwill are reversed if the recoverable amount later increases (but not above the carrying amount that would have existed without the impairment).

29.4.3 U.S. GAAP: ASC 360

U.S. GAAP applies a two-step test to long-lived assets held for use:

1. *Recoverability*: is the carrying amount greater than the *undiscounted* future cash flows expected from the asset (or asset group)? If not, there is no impairment.
2. *Measurement*: if so, the asset is written down to its *fair value*.

Reversals are prohibited.

Worked Example 29.2. The same asset, two answers

A factory has a carrying amount of 500 million. Management expects it to generate cash flows over its remaining life that sum to 520 million undiscounted, with a present value (value in use) of 420 million. Its fair value less costs of disposal is 400 million; its fair value is 405 million.

IFRS: recoverable amount = $\max(400, 420) = 420$. Impairment loss = $500 - 420 = 80$ million.

U.S. GAAP: undiscounted cash flows (520) exceed the carrying amount (500), so the asset passes the recoverability test. No impairment is recognized.

The U.S. test uses undiscounted flows as a screening device, which delays recognition of impairments. If conditions worsened so that undiscounted flows fell below 500, the U.S. write-down would go to fair value (405), a loss of 95—larger than under IFRS, which uses the higher value in use.

Impairment tests rely on forecasts prepared by management, discounted at rates management chooses. They are therefore among the most judgemental areas of accounting. Impairments tend to be recognized late and in large amounts, often after a change of management (a “big bath” that clears the way for later profit growth).

29.5 Intangible Assets

29.5.1 Definition and recognition

Definition 29.3. Intangible asset

An **intangible asset** is an identifiable non-monetary asset without physical substance. It is *identifiable* if it is separable (can be sold, licensed or transferred) or arises from contractual or legal rights.

Patents, copyrights, trademarks, licences, customer lists and software are intangible assets. Whether they appear on the balance sheet depends heavily on how they were obtained:

- **Purchased separately**: recognized at cost.

- **Acquired in a business combination:** recognized at fair value, separately from goodwill, if identifiable.
- **Internally generated:** largely not recognized.

29.5.2 Research and development

Under IFRS (IAS 38), expenditure in the *research* phase is always expensed. Expenditure in the *development* phase is capitalized if the entity can demonstrate six criteria: technical feasibility; the intention to complete the asset; the ability to use or sell it; how it will generate probable future benefits; the availability of resources to complete it; and the ability to measure the expenditure reliably. German car makers, for example, capitalize a substantial share of their vehicle development costs under IFRS.

Under U.S. GAAP (ASC 730), research and development costs are generally expensed as incurred. The main exception is software: costs of developing software for sale are capitalized after technological feasibility is established, and certain costs of developing internal-use software are capitalized during the application-development stage.

29.5.3 Amortization

Intangibles with finite useful lives are **amortized**, usually on a straight-line basis; amortization is simply depreciation applied to an intangible asset. Intangibles with indefinite useful lives (some trademarks and licences that can be renewed indefinitely at little cost) are not amortized but are tested for impairment at least annually.

29.5.4 The missing intangibles

Because internally generated brands, customer relationships, organizational know-how and most research are expensed, the balance sheets of knowledge-intensive firms omit much of their value. A pharmaceutical company that develops a blockbuster drug in its own laboratories reports little or no asset for it; a rival that buys the same drug by acquiring the developer reports a large one. Two economically similar companies can therefore have very different reported assets, returns on assets and book-to-market ratios. The growing importance of intangible investment in modern economies—software, data, research, design, brands, training—is one reason why accounting book values have become less informative about market values over time, a theme taken up in Section 29.9.

29.6 Goodwill and Business Combinations

29.6.1 The acquisition method

When one company obtains control of another, both IFRS 3 and ASC 805 require the **acquisition method**. The acquirer recognizes the target's identifiable assets and liabilities at their fair values at the acquisition date—including intangibles, such as customer relationships and technology, that the target itself had never recognized. The excess of the consideration paid over the fair value of the net identifiable assets is **goodwill**.

Worked Example 29.4. Measuring goodwill

Alpha pays 900 million in cash for all the shares of Beta. At the acquisition date, the fair values of Beta's identifiable assets are 1,000 million (including 150 million of customer

relationships and patented technology not previously on Beta's balance sheet), and of its liabilities 400 million.

$$\text{Goodwill} = 900 - (1,000 - 400) = 300 \text{ million.}$$

In Alpha's consolidated balance sheet, Beta's assets and liabilities appear line by line at fair value, together with goodwill of 300 million; the 150 million of intangibles is amortized over its useful life.

If the consideration had been less than the fair value of the net assets (say 550 million), the acquirer would first re-check its measurements and then recognize the 50 million difference as a gain on a "bargain purchase".

29.6.2 What goodwill represents

Goodwill captures everything the acquirer paid for that cannot be separately identified: synergies with its existing business, the target's assembled workforce, future customer relationships, market position—and any overpayment. Evidence on acquisitions suggests that overpayment is common: acquirers' share prices often fall when large acquisitions are announced, consistent with the winner's curse of Chapter 26.

29.6.3 Subsequent accounting

Goodwill is not amortized under IFRS or for U.S. public companies. Instead it is allocated to cash-generating units (IFRS) or reporting units (U.S. GAAP) and tested for impairment at least annually. Goodwill impairments cannot be reversed. U.S. private companies may elect to amortize goodwill over up to ten years. The absence of amortization means that goodwill stays on the balance sheet until an impairment is recognized, which critics argue happens too late. The IASB considered reintroducing amortization and decided against it, choosing instead to require better disclosures about the performance of acquisitions.

29.7 Leases

29.7.1 The old problem

Before 2019, lessees under both IFRS and U.S. GAAP classified leases as *finance* (capital) leases, which were on the balance sheet, or *operating* leases, which were not: the lessee simply expensed the rental payments. The classification depended on tests such as whether the lease term covered 75% or more of the asset's economic life, or the present value of payments was 90% or more of its fair value—bright lines that firms structured leases to avoid. Airlines, retailers and many other firms carried vast lease obligations off their balance sheets. Before the new standards, the IASB estimated that listed companies using IFRS or U.S. GAAP had around three trillion dollars of lease commitments, most of which were not reported as liabilities.

29.7.2 IFRS 16: a single model

Since 2019, IFRS 16 requires a lessee to recognize almost all leases on the balance sheet (exemptions exist for short-term leases of twelve months or less and leases of low-value assets):

- a **lease liability**, initially the present value of the lease payments, discounted at the rate implicit in the lease or, if that cannot be determined, the lessee's incremental borrowing

rate;

- a **right-of-use asset**, initially equal to the lease liability plus initial direct costs and prepayments.

Subsequently, the right-of-use asset is depreciated (usually straight-line) and the liability is accounted for like a loan: interest accrues on the outstanding balance, and each payment reduces it.

Worked Example 29.5. A five-year lease under IFRS 16

A retailer leases a shop for five years, paying 10,000 at the end of each year. Its incremental borrowing rate is 6%.

Initial lease liability and right-of-use asset:

$$10,000 \times \frac{1 - 1.06^{-5}}{0.06} = 42,124.$$

Annual depreciation of the right-of-use asset: $42,124/5 = 8,425$.

Year	Opening liability	Interest (6%)	Payment	Closing liability	Total expense
1	42,124	2,527	10,000	34,651	10,952
2	34,651	2,079	10,000	26,730	10,504
3	26,730	1,604	10,000	18,334	10,029
4	18,334	1,100	10,000	9,434	9,525
5	9,434	566	10,000	0	8,991
Total		7,876	50,000		50,000

(Total expense is depreciation of 8,425 plus interest; totals differ from row sums by rounding.) Over the lease, total expense equals total payments, 50,000, but expense is higher than the payment in early years and lower later, because interest is charged on a declining balance.

29.7.3 Effects on the financial statements

Compared with the old off-balance-sheet treatment, IFRS 16:

- increases assets and liabilities, and so *leverage* ratios;
- replaces a rent expense within operating costs with depreciation (operating) and interest (financing), raising operating profit and *EBITDA* (earnings before interest, taxes, depreciation and amortization) substantially;
- front-loads total expense;
- in the cash-flow statement, moves the principal part of lease payments from operating to financing activities (interest may be operating or financing), so *operating cash flow rises* and financing outflows rise by the same amount. Total cash flow is unchanged.

Analysts comparing firms across the 2019 transition, or across firms that lease with firms that own, must take these effects into account. An airline that leases its fleet and one that owns it now report broadly comparable balance sheets—which was the purpose of the standard.

29.7.4 ASC 842: two models

U.S. GAAP (ASC 842) also puts almost all leases on the balance sheet, with a lease liability and right-of-use asset measured in the same way. But it retains a distinction between *finance* leases, accounted for like IFRS 16 (depreciation plus interest), and *operating* leases, for which the lessee recognizes a single, straight-line lease cost in operating expenses (10,000 a year in the example), with the right-of-use asset adjusted to make this work. Operating-lease payments remain in operating cash flow. Consequently, EBITDA and operating cash flow of U.S. firms with operating leases are not directly comparable with those of IFRS reporters. The FASB judged that a straight-line expense better reflects the economics of a lease of part of an asset's life; the IASB judged that every lease is a financing of the right to use an asset.

29.8 Capitalization and Earnings Manipulation

Case: WorldCom: turning expenses into assets

WorldCom grew in the 1990s through a series of acquisitions into a telecommunications giant, and its share price was highly sensitive to its reported profit margins. Its largest operating cost was "line costs": fees paid to local telephone companies and other carriers to connect calls to their networks. These were clearly operating expenses of the period in which the connections were used; they created no asset controlled by WorldCom with future economic benefits.

As the telecoms boom ended in 2000–2001, revenue growth slowed while line costs remained high relative to revenue. According to the SEC's complaint, senior finance executives directed that line costs be reclassified from expense into capital accounts—recorded as additions to property, plant and equipment—beginning in 2001. The effect, as this chapter predicts, was threefold. Operating expenses fell, so reported profit rose: the SEC alleged that pretax income was overstated by about 3.055 billion dollars in 2001 and 797 million dollars in the first quarter of 2002. Assets rose by the same amount. And because capital expenditure is an investing cash flow, operating cash flow was overstated as well—so the two measures an analyst might use to cross-check each other both looked healthier.

The fraud was uncovered in 2002 by WorldCom's own internal audit department, led by Cynthia Cooper, which found capital entries without supporting documentation. The company restated its accounts, and its eventual restatement of prior financial statements was much larger than the initial announcement, as further improper entries came to light. WorldCom filed for bankruptcy in July 2002; its chief executive and chief financial officer were later convicted and imprisoned. Together with Enron, the case led directly to the Sarbanes–Oxley Act (Chapter 27).

The case teaches an analytical habit. When a company's capital expenditure rises sharply while its revenue does not, when capitalized amounts appear in accounts that normally hold expenses, or when operating cash flow improves while investing outflows grow in step, the capitalization decision deserves scrutiny.

29.9 Book Value, Economic Value and Tobin's q

The carrying amounts of long-lived assets are not estimates of their economic value. PPE at cost less depreciation reflects historical prices; most internally generated intangibles are missing; goodwill reflects past acquisitions, not current value. The ratio of a company's market

value to the book value of its equity (the *market-to-book* ratio) is therefore well above 1 for most successful companies, and very high for companies whose value rests on unrecorded intangibles.

Bridge to Chapter 45. James Tobin proposed that investment depends on q , the ratio of the market value of a firm's capital to its replacement cost. If $q > 1$, the market values an additional unit of capital at more than it costs to install, so firms should invest; if $q < 1$, they should not. Measuring q requires estimates of replacement cost, which accounting book values provide only roughly. As intangible investment has grown—and most of it is expensed—measured q and market-to-book ratios have risen, and the relationship between measured q and investment has weakened. Economists have responded by estimating firms' intangible capital from their past spending on R&D and on selling, general and administrative activities, and adding it to measured capital. The national accounts made a related change: since 2013 the U.S. national accounts, and since the 2008 System of National Accounts most other countries' accounts, treat research and development as investment rather than intermediate consumption, raising measured GDP and investment (Chapter 43).

Summary

- PPE is measured at cost, including all costs to bring it into use; costs that yield future benefits are capitalized, others expensed. Capitalization raises current profit, assets and operating cash flow.
- Depreciation allocates cost less residual value over useful life; methods differ in timing, not total. Changes in estimates are prospective; disposals produce gains or losses.
- IFRS permits revaluation of PPE; U.S. GAAP does not.
- Impairment under IFRS compares carrying amount with the higher of fair value less costs of disposal and value in use, and permits reversal; U.S. GAAP screens with undiscounted cash flows, measures at fair value, and prohibits reversal.
- Internally generated intangibles are mostly unrecognized; IFRS capitalizes development costs meeting criteria; U.S. GAAP expenses R&D except some software.
- Goodwill is the excess of consideration over the fair value of net identifiable assets acquired; it is tested annually for impairment and not amortized (except by U.S. private companies electing to).
- IFRS 16 puts leases on balance sheet with depreciation and interest; ASC 842 does too but keeps a straight-line cost for operating leases. The changes affect leverage, EBITDA and operating cash flow.
- Accounting book values diverge from economic values, especially for intangible-intensive firms, which affects measured q .

Key terms

property, plant and equipment; capitalize; expense; borrowing costs; depreciation; depreciable amount; residual value; useful life; straight-line; declining balance; units of production; net book value; change in estimate; gain or loss on disposal; revaluation model; impairment; recoverable amount; value in use; cash-generating unit; recoverability test; intangible asset; amortization; research and development;

acquisition method; goodwill; bargain purchase; lease liability; right-of-use asset; EBITDA; finance lease; operating lease; market-to-book ratio; Tobin's q .

Exercises

Exercise 29.1 (Cost of PPE). A factory buys a machine: list price 250,000, trade discount 5%, delivery 6,000, installation 9,000, staff training 4,000, a fee for a consultant who supervised testing 3,000, and repair of damage caused by the firm's own employees while unloading 2,500. What is the machine's cost?

Exercise 29.2 (Depreciation methods). Equipment costs 60,000, has a residual value of 6,000 and a useful life of 4 years (expected output 27,000 units: 9,000, 8,000, 6,000 and 4,000 in successive years). Prepare depreciation schedules under straight-line, double-declining-balance and units-of-production methods.

Exercise 29.3 (Change in estimate). A building bought for 2,000,000 with no residual value was depreciated straight-line over 40 years. At the start of year 11 management extends the remaining useful life to 50 years. What is the depreciation charge in year 11? By how much does this change increase profit before tax in year 11?

Exercise 29.4 (Disposal). The equipment in Exercise 29.2, depreciated straight-line, is sold at the end of year 3 for 15,000. Record the disposal.

Exercise 29.5 (Impairment). An asset group has a carrying amount of 80 million. Undiscounted expected cash flows are 75 million; value in use is 60 million; fair value is 62 million and costs of disposal 2 million. Compute the impairment loss under IFRS and under U.S. GAAP. Two years later, value in use rises to 70 million and the carrying amount (after depreciation of the impaired amount) is 50 million; the carrying amount without impairment would have been 64 million. What does each framework require?

Exercise 29.6 (R&D). A pharmaceutical company spends 300 million on research and 200 million on development in a year; of the development spending, 120 million meets IAS 38's capitalization criteria. Compare profit before tax and total assets under IFRS and U.S. GAAP for the year. Which framework gives the more relevant information to investors, in your view?

Exercise 29.7 (Goodwill). Gamma acquires 100% of Delta for 1.2 billion. Delta's book equity is 500 million. Fair-value adjustments: property up 100 million; identifiable brand not previously recognized 250 million; contingent liability for litigation 50 million; deferred tax on the adjustments 75 million. Compute goodwill.

Exercise 29.8 (Lease). A company leases a machine for 4 years, paying 25,000 at the end of each year; the discount rate is 8%. Compute the initial lease liability, and prepare the liability schedule and total expense for each year under IFRS 16. What is the annual expense under ASC 842 if it is classified as an operating lease?

Exercise 29.9 (EBITDA effect). Before IFRS 16, an airline reported EBITDA of 1,000 million after operating-lease rental expense of 400 million. Under IFRS 16, the rentals are replaced by depreciation of 330 million and interest of 110 million. What is the new EBITDA? What is the effect on operating profit? Why might lenders who use a covenant based on debt/EBITDA need to rewrite it?

Exercise 29.10 (Capitalization analysis). A telecoms company's revenue is flat at 30 billion for three years. Its operating expenses fall from 26 to 24 billion while capital expenditure rises from 4 to 6 billion. List the explanations an analyst should consider and the evidence that would distinguish them.

Solutions

Solution 29.1. $250,000 \times 0.95 = 237,500$, plus delivery 6,000, installation 9,000 and testing supervision 3,000: 255,500. Training (4,000) is expensed, and the damage repair (2,500) is expensed because it does not bring the asset to its intended condition; it results from an error.

Solution 29.2. Depreciable amount 54,000. Straight-line: 13,500 a year; book values 46,500, 33,000, 19,500, 6,000. DDB (50%): year 1 30,000 (BV 30,000); year 2 15,000 (BV 15,000); year 3 7,500 (BV 7,500); year 4 limited to 1,500 (BV 6,000). Units of production: rate $54,000/27,000 = 2$ per unit; charges 18,000, 16,000, 12,000, 8,000; BVs 42,000, 26,000, 14,000, 6,000.

Solution 29.3. After 10 years the carrying amount is $2,000,000 - 10 \times 50,000 = 1,500,000$. New charge $= 1,500,000/50 = 30,000$, against 50,000 before. Profit before tax rises by 20,000 in year 11 and each later year.

Solution 29.4. Accumulated depreciation $3 \times 13,500 = 40,500$; carrying amount 19,500. Dr Cash 15,000; Dr Accumulated depreciation 40,500; Dr Loss on disposal 4,500; Cr Equipment 60,000.

Solution 29.5. IFRS: recoverable amount $= \max(62 - 2, 60) = 60$; impairment 20 million. U.S. GAAP: undiscounted flows $75 < 80$, so impaired; write down to fair value 62: loss 18 million. Two years later: IFRS requires a reversal to the lower of recoverable amount (70) and the carrying amount without impairment (64): the asset is written up from 50 to 64, a gain of 14 million. U.S. GAAP prohibits reversal.

Solution 29.6. IFRS: expense $300 + 80 = 380$ million; capitalize 120 million as an intangible asset. U.S. GAAP: expense all 500 million. IFRS profit before tax is 120 million higher and total assets 120 million higher. IFRS arguably gives more relevant information by recognizing development that is likely to produce benefits; U.S. GAAP gives more verifiable and comparable information, avoiding management judgement about feasibility. Reasonable people disagree.

Solution 29.7. Fair value of net identifiable assets $= 500 + 100 + 250 - 50 - 75 = 725$ million. Goodwill $= 1,200 - 725 = 475$ million.

Solution 29.8. Liability $= 25,000 \times (1 - 1.08^{-4})/0.08 = 82,803$. Depreciation $82,803/4 = 20,701$. Year 1: interest 6,624, closing liability 64,427, expense 27,325. Year 2: interest 5,154, closing 44,582, expense 25,855. Year 3: interest 3,567, closing 23,148, expense 24,267. Year 4: interest 1,852, closing 0, expense 22,553. Total interest 17,197; total expense 100,000 (with rounding). Under ASC 842 as an operating lease, the expense is 25,000 each year.

Solution 29.9. EBITDA rises by the 400 million of rentals no longer deducted: 1,400 million. Operating profit (EBIT) rises by $400 - 330 = 70$ million, since depreciation replaces rent but interest is below the operating line. Debt now includes lease liabilities and EBITDA is higher, so a covenant defined under the old standard would change meaning; lenders and borrowers typically either freeze the old definitions ("frozen GAAP") or recalibrate the thresholds.

Solution 29.10. Possible explanations: (i) genuine efficiency gains funded by investment in new network equipment that substitutes for operating costs; (ii) reclassification of costs from operating expense to capital (legitimate if the costs meet capitalization criteria, fraudulent otherwise); (iii) shifting from leasing capacity (expense) to owning it (capex). Evidence: the composition of capex and of fixed-asset additions in the notes; whether depreciation rises as the new assets come into use; whether physical measures of the network (capacity, fibre miles) grow in line with capex; changes in accounting policies disclosed; and whether operating cash flow plus investing cash flow—free cash flow—has actually improved.

Debt, Equity, Taxes and Obligations

After studying this chapter you should be able to

1. Account for bonds issued at par, at a discount and at a premium using the effective-interest method.
2. Classify and measure financial assets under IFRS 9 and U.S. GAAP, and explain the consequences for reported profit and equity when interest rates change.
3. Account for share issues, buybacks, dividends and stock splits, and compute basic and diluted earnings per share.
4. Distinguish current from deferred tax, compute deferred tax liabilities and assets, and reconcile the effective tax rate.
5. Recognize and measure provisions and distinguish them from contingent liabilities.
6. Explain the accounting for defined-benefit pensions and why pension obligations are sensitive to interest rates.
7. State the Modigliani–Miller propositions and explain how taxes and financial distress affect capital structure.
8. Analyse the failure of Silicon Valley Bank using the accounting for securities and interest-rate risk.

A motivating question

On 8 March 2023, Silicon Valley Bank, the sixteenth-largest bank in the United States, announced that it had sold 21 billion dollars of securities at a loss of about 1.8 billion dollars and would raise new equity. The next day depositors withdrew about 42 billion dollars. On 10 March the bank was closed by regulators. It had not made bad loans. Its problem was that it had invested a large share of its deposits in high-quality, long-term bonds bought when interest rates were very low; when the Federal Reserve raised rates rapidly in 2022, the market value of those bonds fell far below what the bank had paid. Much of that loss did not appear in its reported profit, and a large part did not appear in its reported equity at all, because of how the securities were classified.

This chapter studies the right-hand side of the balance sheet—debt and equity—together with the tax and other obligations that firms incur, and the accounting for financial assets such as the bonds SVB held. It connects accounting directly to the interest-rate mathematics of Chapter 9 and to the monetary economics of Part VII.

30.1 Liabilities and Debt

30.1.1 Classification

Liabilities are classified as current or non-current. A liability is current if the entity expects to settle it within twelve months (or its operating cycle), holds it primarily for trading, or does not have the right at the reporting date to defer settlement for at least twelve months. The last condition matters for loans with covenants: if a covenant has been breached and the lender could demand repayment, the loan may have to be reclassified as current, which can itself trigger a crisis of confidence.

30.1.2 Bonds and amortized cost

When a company issues a bond, it receives cash and records a liability. If the bond's coupon rate equals the market yield demanded by investors, it is issued at face value (*par*). If the coupon rate is below the market yield, investors pay less than face value (a *discount*); if above, more (a *premium*). Chapter 9 showed how to compute the price.

The liability is then measured at **amortized cost** using the **effective-interest method**: each period, interest expense equals the carrying amount at the start of the period multiplied by the effective (market) interest rate at issue. The difference between interest expense and the cash coupon adjusts the carrying amount, which converges to face value at maturity.

Worked Example 30.1. A bond issued at a discount

On 1 January a company issues three-year bonds with a face value of 1,000,000 and an annual coupon of 5% paid each 31 December. The market yield for such bonds is 6%, so the issue price is

$$\frac{50,000}{1.06} + \frac{50,000}{1.06^2} + \frac{1,050,000}{1.06^3} = 973,270.$$

(This is Example 9.6 of Chapter 9 scaled up.) On issue: Dr Cash 973,270; Cr Bonds payable 973,270.

Year	Opening amount	Interest (6%)	Coupon	Closing amount
1	973,270	58,396	50,000	981,666
2	981,666	58,900	50,000	990,566
3	990,566	59,434	50,000	1,000,000

Year 1 entry: Dr Interest expense 58,396; Cr Cash 50,000; Cr Bonds payable 8,396. At maturity the company repays 1,000,000. Total interest expense over three years is 176,730 = 150,000 of coupons plus the 26,730 discount: the discount is additional interest, spread over the bond's life.

The effective-interest method makes the interest expense a constant percentage of the outstanding liability, exactly as the present-value mathematics requires. Under amortized cost, subsequent changes in market interest rates do *not* change the carrying amount of the liability. If market yields later rise to 8%, the bond's market value falls, but the company continues to report it at amortized cost.

30.1.3 The fair value option and own credit risk

Both IFRS 9 and U.S. GAAP allow entities, in certain circumstances, to elect to measure financial liabilities at fair value. This produces a well-known paradox: if the company's own credit quality deteriorates, the market value of its debt falls, and a fair-value liability would

decrease, producing a *gain*. To avoid reporting profits from becoming less creditworthy, IFRS 9 requires the part of the fair-value change caused by changes in the entity's own credit risk to be recognized in other comprehensive income rather than profit; U.S. GAAP has a similar requirement.

30.1.4 Covenants

Debt contracts contain **covenants**—promises by the borrower—often expressed in accounting numbers: a maximum ratio of debt to EBITDA, a minimum interest coverage ratio (EBIT or EBITDA divided by interest expense), or a minimum net worth. Covenants are a response to the moral-hazard problems of Chapter 25: they restrict borrowers from taking actions that transfer value from lenders to shareholders. Because they are written in accounting terms, changes in accounting standards (such as IFRS 16's effect on EBITDA and debt, Chapter 29) can change their meaning, and managers close to a covenant threshold have incentives to manage the reported numbers.

30.2 Financial Assets

30.2.1 Classification under IFRS 9

For a holder of financial assets, measurement depends on classification. IFRS 9 classifies debt instruments by two tests: the *business model* in which the asset is held, and whether its contractual cash flows are *solely payments of principal and interest* (SPPI):

- **Amortized cost:** held to collect contractual cash flows, and SPPI.
- **Fair value through other comprehensive income (FVOCI):** held both to collect and to sell, and SPPI. Fair-value changes go to OCI; interest income and credit losses go to profit; accumulated OCI is reclassified to profit on sale.
- **Fair value through profit or loss (FVTPL):** everything else, including trading assets.

Equity investments are at FVTPL, unless the holder irrevocably elects FVOCI for a non-trading equity investment, in which case gains and losses are never reclassified to profit.

30.2.2 Classification under U.S. GAAP

U.S. GAAP classifies debt securities into three categories:

- **Held-to-maturity (HTM):** securities the entity has the positive intent and ability to hold to maturity; measured at amortized cost. Unrealized gains and losses are not recognized in profit or in equity (they are disclosed in the notes).
- **Available-for-sale (AFS):** measured at fair value, with unrealized gains and losses in OCI.
- **Trading:** fair value through net income.

Selling or transferring more than an insignificant amount of HTM securities before maturity "taints" the portfolio, generally requiring all remaining HTM securities to be reclassified as AFS. Equity securities with readily determinable fair values are measured at fair value through net income.

Key idea. The same bond, held by two banks, can be reported at very different amounts, with its losses flowing through profit, through equity only, or through neither, depending on classification. Interest-rate risk that has materialized in market prices may be invisible in reported capital.

30.3 Equity

30.3.1 Components

Equity typically comprises:

- **Contributed capital:** *share capital* (the nominal or par value of shares issued) and *share premium* or *additional paid-in capital* (the excess of the issue price over par). In many jurisdictions, including Germany, the nominal share capital (*Grundkapital* of an AG) is subject to legal protection for creditors.
- **Retained earnings:** accumulated profits not distributed.
- **Accumulated other comprehensive income** (and IFRS reserves such as the revaluation surplus).
- **Treasury shares:** the company's own shares repurchased and held, shown as a deduction.
- **Non-controlling interests:** in consolidated accounts, the share of subsidiaries' equity not owned by the parent.

30.3.2 Distributions

Cash dividends. On declaration, retained earnings are debited and a dividend payable credited; on payment, the payable is settled with cash. Dividends are distributions of profit, not expenses. In Germany, distributions by an AG are limited to the distributable profit shown in its HGB individual accounts, a creditor-protection rule.

Share repurchases (buybacks). A company buys its own shares from shareholders. Under U.S. GAAP and IFRS, the cost is deducted from equity, often as treasury shares. Buybacks return cash to shareholders like dividends, but only to those who choose to sell, and they reduce the number of shares outstanding, raising earnings per share. U.S. companies have used buybacks heavily since the 1980s; Apple, for instance, has repurchased hundreds of billions of dollars of its own shares over the past decade. Since 2023 the United States has levied a 1% excise tax on net repurchases by listed corporations.

Stock dividends and splits. A stock split (for example, two shares for one) increases the number of shares and proportionally reduces the price per share; it transfers no value and has no effect on total equity. A stock dividend is similar in substance.

30.3.3 Earnings per share

Definition 30.2. Earnings per share

Basic earnings per share (EPS) is profit attributable to ordinary shareholders (net income less preferred dividends) divided by the weighted average number of ordinary shares outstanding during the period. **Diluted EPS** adjusts both numerator and denominator for the effects of all dilutive potential ordinary shares: options, warrants, convertible bonds and convertible preferred shares.

Worked Example 30.3. Basic and diluted EPS

Net income is 12 million; preferred dividends are 2 million. On 1 January there were 4.5 million ordinary shares; on 1 July 1 million more were issued. Weighted average shares $= 4.5 + 1 \times 6/12 = 5.0$ million. Basic EPS $= (12 - 2)/5.0 = 2.00$.

Employees hold options on 1 million shares at an exercise price of 20; the average market price during the year was 25. Under the **treasury stock method**, exercise would bring in $1 \times 20 = 20$ million, enough to repurchase $20/25 = 0.8$ million shares at market, so the net increase in shares is 0.2 million. Diluted EPS $= 10/5.2 = 1.92$.

30.3.4 Share-based payment

Employee share options and restricted stock are compensation. Since 2005 (IFRS 2 and ASC 718), their fair value at the grant date is recognized as an expense over the vesting period, with a corresponding credit to equity. Before these standards, most U.S. companies recognized no expense for options granted at the money, which flattered the profits of companies—especially technology firms—that paid employees heavily in options. Many companies still exclude “stock-based compensation” from the adjusted profit measures they emphasize to investors, a practice that IFRS 18’s disclosure requirements (Chapter 27) aim to make more transparent.

30.4 Income Taxes

30.4.1 Current and deferred tax

Taxable profit is determined by tax law, which differs from accounting standards in the timing and sometimes the amount of income and expenses. **Current tax** is the tax payable on taxable profit for the period. **Deferred tax** accounts for the future tax consequences of **temporary differences**: differences between the carrying amount of an asset or liability in the balance sheet and its *tax base* (the amount attributed to it for tax purposes) that will reverse in future periods.

- A **deferred tax liability** arises when taxable profit will be *higher* in future than accounting profit because of the reversal—for example, when tax depreciation has been faster than book depreciation, so the asset’s carrying amount exceeds its tax base.
- A **deferred tax asset** arises when future taxable profit will be *lower*—for example, when an expense (a warranty provision) is recognized in the accounts before it is deductible for tax, or when the entity has unused tax losses to carry forward. Deferred tax assets are recognized only to the extent that it is probable that future taxable profit will be available to use them.

Total income tax expense = current tax + deferred tax expense (the change in net deferred tax liabilities).

Worked Example 30.4. Accelerated tax depreciation

A company buys equipment for 100,000 at the start of year 1. For accounting it depreciates straight-line over 5 years (20,000 a year). For tax it may deduct 40,000 in year 1, then 15,000 a year for four years. Accounting profit before depreciation is 150,000 each year; the tax rate is 25%.

Year 1: accounting profit = 130,000; taxable profit = 110,000; current tax = 27,500. At year end the equipment's carrying amount is 80,000 and its tax base 60,000: a taxable temporary difference of 20,000, requiring a deferred tax liability of $25\% \times 20,000 = 5,000$. Tax expense = $27,500 + 5,000 = 32,500 = 25\% \times 130,000$.

Year 2: accounting profit 130,000; taxable profit 135,000; current tax 33,750. Carrying amount 60,000, tax base 45,000: difference 15,000, deferred tax liability 3,750, a decrease of 1,250. Tax expense = $33,750 - 1,250 = 32,500$. The liability unwinds to zero by the end of year 5.

Deferred tax accounting makes the tax expense match accounting profit at the statutory rate when differences are temporary. The **effective tax rate**—tax expense divided by profit before tax—then departs from the statutory rate only because of *permanent* differences (expenses never deductible, income never taxable), different rates in different jurisdictions, tax credits, and changes in tax rates or in the recognition of deferred tax assets. Companies disclose a reconciliation between the statutory and effective rates, which analysts use to judge whether a low tax rate is sustainable.

30.4.2 International taxation and the global minimum tax

Multinational groups have long reduced their tax bills by locating profits—often profits attributable to intellectual property—in low-tax jurisdictions (Chapter 10's Irish case). Under the OECD/G20 "Pillar Two" agreement, large multinational groups (with revenue of at least 750 million euros) are subject to a global minimum effective tax rate of 15% in each jurisdiction where they operate, enforced through "top-up" taxes. The European Union adopted the rules from 2024, so they apply in Germany; Malaysia introduced its implementation from 2025; the United States has not adopted them. Groups now disclose their expected exposure to the top-up taxes.

30.5 Provisions and Contingencies

30.5.1 Provisions

Definition 30.5. Provision

A **provision** is a liability of uncertain timing or amount. Under IAS 37 a provision is recognized when (i) the entity has a present obligation (legal or constructive) as a result of a past event; (ii) it is probable (more likely than not) that an outflow of resources will be required; and (iii) a reliable estimate can be made.

A *constructive* obligation arises from an established pattern of practice or a public statement that creates a valid expectation—for example, a retailer's well-known policy of refunding unwanted goods even beyond its legal obligation. Provisions are measured at the best estimate

of the expenditure required to settle the obligation, discounted when the time value of money is material. Common provisions include warranties, legal claims, restructuring costs (only once a detailed plan has been announced), environmental clean-up and decommissioning (dismantling oil platforms or nuclear plants).

Worked Example 30.6. A warranty provision

A manufacturer sells 10,000 washing machines with a one-year warranty. Experience suggests that 90% will need no repair, 8% minor repairs costing 100 each, and 2% major repairs costing 500 each. The expected cost is $10,000 \times (0.08 \times 100 + 0.02 \times 500) = 10,000 \times 18 = 180,000$. At the time of sale: Dr Warranty expense 180,000; Cr Warranty provision 180,000. As repairs are made, the provision is used up. For tax purposes, warranty costs are usually deductible only when incurred, which creates a deferred tax asset.

30.5.2 Contingent liabilities

A **contingent liability** is a possible obligation whose existence will be confirmed only by uncertain future events, or a present obligation that is not recognized because an outflow is not probable or cannot be reliably measured. Contingent liabilities are not recognized but are disclosed unless the possibility of an outflow is remote. Contingent assets (a lawsuit the entity expects to win) are disclosed only when an inflow is probable, and recognized only when virtually certain—an asymmetry that reflects prudence.

U.S. GAAP (ASC 450) recognizes a loss contingency when a loss is “probable”, a term generally interpreted as a higher threshold than IFRS’s “more likely than not”; and when a range of outcomes is equally likely, U.S. GAAP uses the low end of the range while IFRS uses the midpoint. So U.S. companies may recognize litigation liabilities later and at lower amounts than IFRS reporters.

30.6 Pensions

In a **defined-contribution** (DC) plan, the employer pays agreed contributions into a fund, and the employee’s pension depends on the fund’s investment performance. The employer’s expense is simply the contribution; it bears no further risk.

In a **defined-benefit** (DB) plan, the employer promises a pension based on salary and years of service. The employer bears the investment and longevity risk. The accounting obligation—the *defined benefit obligation*—is the present value of benefits earned to date, discounted at a rate based on high-quality corporate bond yields. The *funded status* is plan assets minus the obligation; a deficit is a liability on the balance sheet.

Because the obligation is a present value of payments stretching decades into the future, it is highly sensitive to the discount rate. When long-term interest rates fell to very low levels in the 2010s, reported pension deficits ballooned; when rates rose sharply in 2022, many deficits shrank or turned into surpluses. Under IFRS (IAS 19), remeasurements from changes in actuarial assumptions and asset returns are recognized in OCI; the profit-or-loss charge comprises service cost and net interest on the net liability.

German companies have traditionally provided DB pensions through book reserves (*Pensionsrückstellungen*) rather than external funds, so their balance sheets show large pension provisions with few segregated assets—an unfunded obligation financed by the company itself. Analysts treat unfunded pension obligations as debt-like when assessing leverage.

30.7 Capital Structure: The Economics behind the Accounts

The accounts record how a firm is financed; economics asks whether the mix matters. Franco Modigliani and Merton Miller (1958) gave a famous benchmark.

Proposition 30.7. Modigliani–Miller propositions (no taxes)

In a market with no taxes, no costs of financial distress, no transaction costs and symmetric information:

1. The value of a firm does not depend on its capital structure: $V_L = V_U$, where V_L is the value of a levered firm and V_U that of an otherwise identical unlevered firm.
2. The expected return on equity rises linearly with leverage: $r_E = r_A + (r_A - r_D) D/E$, where r_A is the return on the firm's assets and r_D the return on debt.

The first proposition says that slicing a firm's cash flows differently between debt holders and shareholders cannot change their total value. If it did, investors could make arbitrage profits by combining shares and personal borrowing. The second says that debt, though it has a lower required return, makes equity riskier, and the higher required return on equity exactly offsets the apparent cheapness of debt.

The benchmark is valuable because its assumptions identify what *does* make capital structure matter:

- **Taxes:** interest is deductible for corporate tax in most countries, while dividends are not. Debt then creates a *tax shield* worth, in the simplest case, the tax rate times the amount of permanent debt: $V_L = V_U + \tau D$. (Several countries, including Germany with its "interest barrier" and the United States since 2018, limit the deductibility of net interest to a percentage of EBITDA or EBIT.)
- **Costs of financial distress:** bankruptcy is costly, and the risk of it disrupts relationships with customers, suppliers and employees. The *trade-off theory* balances the tax benefits of debt against these costs.
- **Asymmetric information:** managers know more than investors, so issuing equity may signal that shares are overvalued (Chapter 25). The *pecking-order theory* (Myers and Majluf, 1984) predicts that firms prefer internal funds, then debt, then equity.
- **Agency problems:** debt disciplines managers by committing cash to interest payments (reducing free cash flow they might waste), but shareholders of a highly levered firm may prefer excessively risky projects, since they capture the upside while creditors bear the downside.

Bridge to Part VII. Banks are extremely levered firms: equity is typically less than a tenth of assets. The Modigliani–Miller logic implies that requiring banks to hold more equity would raise the required return on bank equity less than bankers often claim, because the equity would become safer. Bank regulators, and the debate over capital requirements after 2008, lean heavily on this reasoning (Chapter 51). The tax advantage of debt, and implicit government guarantees for large banks, are the main reasons why private incentives favour more leverage than is socially optimal.

Case: Silicon Valley Bank: accounting classification and interest-rate risk

Silicon Valley Bank (SVB) served technology start-ups and venture-capital firms. In 2020 and 2021, as its clients raised record amounts of venture funding, its deposits roughly tripled, to around 190 billion dollars. Loan demand did not grow as fast, so SVB invested much of the inflow in securities—mainly U.S. Treasury bonds and government-agency mortgage-backed securities, with little credit risk but long maturities—bought when yields were near historic lows. By the end of 2021, investment securities made up more than half of its assets.

SVB classified the larger part of these securities, about 91 billion dollars at the end of 2022, as held-to-maturity. Under U.S. GAAP, HTM securities are carried at amortized cost; their unrealized losses do not affect net income or equity. As the Federal Reserve raised its policy rate from near zero to over 4% during 2022, the market value of long-term bonds fell sharply (the inverse relation between yields and prices from Chapter 9). SVB's own disclosures showed unrealized losses on its HTM portfolio of about 15 billion dollars at the end of 2022—roughly equal to its total shareholders' equity of about 16 billion dollars. Its smaller AFS portfolio also showed losses, which did flow through OCI and reduced equity.

Meanwhile, deposits began to fall as start-ups burned cash and venture funding slowed. To meet withdrawals and reposition its balance sheet, SVB sold its entire AFS portfolio in early March 2023, realizing a 1.8 billion dollar loss, and announced a plan to raise capital. Selling HTM securities would have tainted the whole HTM portfolio and forced its losses into equity. The announcement alerted depositors—more than 90% of whose deposits exceeded the 250,000-dollar insurance limit—to the scale of the unrealized losses. Coordinated through social media and a concentrated network of venture investors, they attempted to withdraw about 42 billion dollars on 9 March. The bank was closed the next day. (The stag hunt and bank-run game of Chapter 23 describe the dynamic exactly.) The case links several threads of this book. The accounting classification determined how much of a real economic loss appeared in reported profit and capital, but it did not change the loss itself: market participants who read the notes could calculate it. Interest-rate risk—a mismatch between long-duration assets and short-term, flighty liabilities—can be as dangerous as credit risk. U.S. regulatory rules at the time allowed banks of SVB's size to exclude AOCI from regulatory capital, so even the AFS losses did not reduce its regulatory capital ratios. And monetary policy tightening (Part VII) transmits through the balance sheets of banks in ways that depend on their accounting and their funding. Chapter 51 returns to the policy response, including the Federal Reserve's emergency lending facility that valued banks' collateral at par rather than market value.

Summary

- Bonds are measured at amortized cost using the effective-interest method; discounts and premiums adjust interest expense over the bond's life. Later changes in market rates do not change amortized cost.
- Financial assets are measured at amortized cost, FVOCI or FVTPL (IFRS 9), or as HTM, AFS or trading (U.S. GAAP); classification determines whether market-value changes affect profit, equity, or neither.
- Equity includes contributed capital, retained earnings, accumulated OCI and treasury shares. Dividends and buybacks are distributions; EPS is computed on weighted average shares, and diluted EPS includes potential shares.

- Tax expense combines current tax and deferred tax on temporary differences; the effective rate differs from the statutory rate because of permanent differences, jurisdictions and credits. Pillar Two sets a 15% global minimum.
- Provisions are recognized for probable, measurable obligations; contingent liabilities are disclosed. U.S. GAAP sets a higher threshold.
- Defined-benefit pension obligations are present values highly sensitive to discount rates.
- Modigliani–Miller: capital structure is irrelevant without taxes and frictions; taxes, distress costs, asymmetric information and agency problems make it matter.
- SVB showed how accounting classification can hide interest-rate losses from reported capital without making them less real.

Key terms

amortized cost; effective-interest method; discount; premium; fair value option; own credit risk; covenant; SPPI test; FVOCI; FVTPL; held-to-maturity; available-for-sale; share capital; share premium; retained earnings; treasury shares; non-controlling interest; dividend; share repurchase; stock split; basic EPS; diluted EPS; treasury stock method; share-based payment; current tax; deferred tax; temporary difference; tax base; deferred tax liability; deferred tax asset; effective tax rate; Pillar Two; provision; constructive obligation; contingent liability; defined-contribution plan; defined-benefit plan; funded status; Modigliani–Miller propositions; tax shield; trade-off theory; pecking-order theory.

Exercises

Exercise 30.1 (Bond at a premium). A company issues two-year bonds with face value 500,000 and an 8% annual coupon when the market yield is 6%. Compute the issue price and prepare the amortization schedule. Record the entries for year 1.

Exercise 30.2 (Market value versus amortized cost). One year after issue, market yields on the bonds in Example 30.1 rise to 9%. What is the market value of the bonds (two coupons remaining)? At what amount are they reported? Is the company better or worse off economically, and how would you describe the difference?

Exercise 30.3 (Securities classification). A bank holds a 10-year Treasury bond bought at par for 100 million with a 2% coupon. Market yields rise so that its fair value falls to 80 million. Describe the effect on net income, OCI and total equity if the bond is classified (a) HTM, (b) AFS, (c) trading. What happens in each case if the bank then sells the bond?

Exercise 30.4 (EPS). A company has net income of 30 million, no preferred shares, 10 million shares outstanding on 1 January, issues 2 million shares on 1 April and buys back 1 million on 1 October. It has convertible bonds (face value 20 million, 5% coupon) convertible into 1 million shares; the tax rate is 25%. Compute basic EPS and diluted EPS (the converted bonds would save interest net of tax).

Exercise 30.5 (Deferred tax). Continue Example 30.4 through years 3 to 5. Show the deferred tax liability at each year end and the tax expense each year.

Exercise 30.6 (Deferred tax asset). A company records a warranty provision of 200,000 in year 1; warranty costs are deductible for tax only when paid. It pays 120,000 of warranty costs in year 2 and 80,000 in year 3. With a 30% tax rate, what deferred tax balance appears at the end of years 1 and 2?

Exercise 30.7 (Provision or contingency). Classify and explain: (a) a lawsuit in which the company's lawyers estimate a 70% chance of losing and a best estimate of damages of 5 million; (b) a lawsuit with a 20% chance of losing; (c) a board decision, not yet announced, to close a factory next year; (d) a legal obligation to dismantle a drilling platform in 20 years at an estimated cost of 100 million (discount rate 5%).

Exercise 30.8 (Pension sensitivity). A DB plan will pay 10 million a year for 20 years starting in one year. Compute the obligation at discount rates of 2% and 5%. By what percentage does the obligation fall when the rate rises?

Exercise 30.9 (Modigliani–Miller). An unlevered firm's assets earn an expected return of 10% and it is worth 1,000. It issues 400 of debt at 5% and uses the proceeds to buy back shares. Ignoring taxes, what is the expected return on equity after the recapitalization? With a 25% corporate tax rate and permanent debt, what is the value of the tax shield?

Exercise 30.10 (SVB). Using the figures in the case, explain why SVB's reported equity of about 16 billion dollars was a misleading guide to its ability to absorb losses. What information in the financial statements would have revealed the problem? Why did selling the AFS portfolio make matters worse rather than better?

Solutions

Solution 30.1. $\text{Price} = 40,000/1.06 + 540,000/1.06^2 = 37,736 + 480,598 = 518,334$. Year 1: interest expense $0.06 \times 518,334 = 31,100$; coupon 40,000; carrying amount falls by 8,900 to 509,434. Year 2: interest 30,566; carrying amount falls by 9,434 to 500,000, then repaid. Year 1 entries: at issue, Dr Cash 518,334, Cr Bonds payable 518,334; at year end, Dr Interest expense 31,100, Dr Bonds payable 8,900, Cr Cash 40,000.

Solution 30.2. $\text{Market value} = 50,000/1.09 + 1,050,000/1.09^2 = 45,872 + 883,764 = 929,636$. Reported at amortized cost, 981,666. The company is economically better off by about 52,000: it has locked in borrowing at 6% while the market now demands 9%. This gain is unrecognized under amortized cost. (Its lenders have suffered the mirror-image loss.)

Solution 30.3. (a) HTM: no effect on net income, OCI or equity; the 20 million loss is disclosed only. (b) AFS: 20 million loss in OCI; equity falls by 20 million; net income unaffected. (c) Trading: 20 million loss in net income; equity falls by 20 million. On sale: HTM—a realized loss of 20 million in net income (and possible tainting of the portfolio); AFS—the 20 million accumulated in OCI is reclassified to net income (equity unchanged by the reclassification, since it was already reduced); trading—no further effect. Total equity ends 20 million lower in every case once the bond is sold.

Solution 30.4. Weighted shares: $10 + 2 \times 9/12 - 1 \times 3/12 = 10 + 1.5 - 0.25 = 11.25$ million. Basic EPS = $30/11.25 = 2.667$. Diluted: interest saved after tax = $20 \times 5\% \times 0.75 = 0.75$ million; shares +1 million. Diluted EPS = $30.75/12.25 = 2.510$. Since $2.510 < 2.667$, the bonds are dilutive and included.

Solution 30.5. Year 3: carrying amount 40,000, tax base 30,000, DTL 2,500 (decrease 1,250). Year 4: 20,000 vs 15,000, DTL 1,250. Year 5: 0 vs 0, DTL 0. Each year current tax is 33,750 (taxable profit 135,000) and deferred tax is $-1,250$, so tax expense is $32,500 = 25\%$ of accounting profit of 130,000.

Solution 30.6. End of year 1: provision 200,000, tax base 0; deductible temporary difference 200,000; deferred tax asset $0.3 \times 200,000 = 60,000$ (if future taxable profit is probable). End of year 2: provision 80,000; DTA 24,000. End of year 3: 0.

Solution 30.7. (a) Provision of 5 million: present obligation, probable outflow, reliable estimate. Under U.S. GAAP, also likely recognized if 70% meets "probable". (b) Contingent liability: disclosed, not recognized (not probable, not remote). (c) Neither yet: no constructive obligation until a detailed plan is announced or implementation begins; no provision. (d) Provision measured at present value

$100/1.05^{20} = 37.7$ million, added to the cost of the platform; the discount unwinds as a finance cost each year.

Solution 30.8. At 2%: $10 \times (1 - 1.02^{-20})/0.02 = 163.5$ million. At 5%: $10 \times (1 - 1.05^{-20})/0.05 = 124.6$ million. The obligation falls by about 24%.

Solution 30.9. Equity after recapitalization = 600. $r_E = 10\% + (10\% - 5\%) \times 400/600 = 13.33\%$. Check: asset return 100 minus interest 20 leaves 80 for equity of 600, i.e. 13.33%. Tax shield = $\tau D = 0.25 \times 400 = 100$.

Solution 30.10. Reported equity excluded the roughly 15 billion of unrealized losses on HTM securities, which were economically real: had the bank needed to sell, it would have realized them. On a fair-value basis its equity was close to zero. The HTM fair-value disclosure in the notes, the maturity profile of the securities, the composition and concentration of deposits (largely uninsured, from a single industry) and the sensitivity of economic value to interest rates would all have revealed the problem. Selling the AFS portfolio realized a loss, required new capital, and—by publicizing the problem and the need for funding—triggered a run by uninsured depositors who feared being last in line.

Cash Flow, Profitability, Earnings Quality and Managerial Decisions

After studying this chapter you should be able to

1. Interpret the three sections of the cash-flow statement and compute free cash flow.
2. Decompose return on equity with the DuPont framework and compute return on invested capital and economic value added.
3. Value a firm from its financial statements using dividend-discount, discounted-cash-flow and residual-income models, and show their equivalence.
4. Measure accruals and assess earnings quality; recognize common red flags of earnings management.
5. Distinguish fixed and variable costs; compute contribution margins, break-even points and operating leverage.
6. Identify relevant costs for decisions such as special orders and make-or-buy.
7. Explain activity-based costing and why traditional overhead allocation can distort product costs.
8. Prepare budgets and compute price and quantity variances.

A motivating question

For its first several years as a public company, Amazon reported losses or very small profits, and for most of the following two decades its reported profit margins remained thin. Yet over the same period its market value rose to rank among the largest in the world. In his 2004 letter to shareholders, Jeff Bezos wrote that Amazon's ultimate financial measure, and the one it most wanted to drive over the long term, was free cash flow per share. Why would a company emphasize cash flow over reported earnings? How do investors translate financial statements into a valuation? And how can they tell when reported profits are a reliable guide to a firm's economic performance and when they are not?

This chapter completes Part IV. Its first half turns from *preparing* financial statements to *analysing* them: cash flows, profitability, valuation and earnings quality. Its second half turns inward, to the managerial accounting that firms use to make decisions—about prices, products and budgets—which connects back to the cost theory of Chapter 21.

31.1 Reading the Cash-Flow Statement

31.1.1 The three sections revisited

Chapters 10 and 27 introduced the cash-flow statement. Its three sections tell different stories:

- **Operating cash flow (CFO)** shows whether the core business generates cash. A healthy, mature firm usually has CFO above net income, because depreciation is a non-cash expense.
- **Investing cash flow (CFI)** is usually negative for a firm maintaining or growing its productive capacity. Large positive CFI may mean the firm is selling assets, possibly to raise cash.
- **Financing cash flow (CFF)** shows how the firm raises funds and returns them to investors.

Typical patterns emerge over a firm's life. A young, fast-growing company often has negative CFO (losses and working-capital build-up), large negative CFI (investment) and large positive CFF (raising capital). A mature company typically has positive CFO, moderate negative CFI and negative CFF (paying dividends, repurchasing shares, repaying debt). A declining company may have positive CFO, positive CFI (asset sales) and negative CFF. Reading the three together is more informative than reading any one.

31.1.2 Free cash flow

Definition 31.1. Free cash flow

Free cash flow to the firm (FCFF) is the cash generated by operations after the investment needed to sustain and grow the business, available to all providers of capital:

$$\text{FCFF} = \text{EBIT}(1 - \tau) + \text{Depreciation} \\ - \text{Capital expenditure} - \Delta \text{Operating working capital},$$

where τ is the tax rate. **Free cash flow to equity (FCFE)** is what remains for shareholders after payments to and from debt holders:

$$\text{FCFE} = \text{FCFF} - \text{Interest}(1 - \tau) + \text{Net borrowing}.$$

A simpler, commonly reported version is “free cash flow” = CFO – capital expenditure. Definitions vary across companies, which is one reason IFRS 18 requires management-defined performance measures to be reconciled to standard subtotals.

Free cash flow matters because it is what a firm can pay to its investors, now or in future, without shrinking. Accrual profit is a better measure of a single period's performance; free cash flow is closer to what ultimately determines value. The two converge over a firm's whole life but can diverge for many years, especially for a fast-growing firm investing heavily.

31.2 Profitability Analysis

31.2.1 Return on equity and the DuPont decomposition

Return on equity (ROE) is net income divided by (average) shareholders' equity. It measures the return the accounts attribute to the owners' investment. The DuPont Corporation's managers developed a decomposition in the 1920s that remains the backbone of ratio analysis:

$$\text{ROE} = \frac{\text{Net income}}{\text{Equity}} = \underbrace{\frac{\text{Net income}}{\text{Sales}}}_{\text{profit margin}} \times \underbrace{\frac{\text{Sales}}{\text{Assets}}}_{\text{asset turnover}} \times \underbrace{\frac{\text{Assets}}{\text{Equity}}}_{\text{financial leverage}}.$$

The first two factors multiply to **return on assets (ROA)**. The identity shows three distinct routes to a high ROE: high margins, efficient use of assets, or more debt.

Worked Example 31.2. Two routes to the same ROE

A discount retailer has a net margin of 3%, asset turnover of 3.0 and leverage of 2.0: $ROE = 0.03 \times 3.0 \times 2.0 = 18\%$. A luxury-goods maker has a margin of 15%, turnover of 0.8 and leverage of 1.5: $ROE = 0.15 \times 0.8 \times 1.5 = 18\%$. The two earn the same return for their owners through entirely different business models. The retailer's return is vulnerable to small falls in margin; the luxury firm's to falls in demand that reduce turnover. The retailer relies more on leverage, which amplifies both.

An extended five-factor DuPont analysis separates the tax burden (net income / pre-tax income) and interest burden (pre-tax income / EBIT) from the operating margin (EBIT / sales), isolating the effects of tax and financing on ROE.

► Common error: higher ROE is always better

ROE can be raised by borrowing more, by buying back shares with debt, or by writing down assets (which reduces equity). None of these necessarily creates value. A firm with ROE of 25% achieved through leverage of 5 is riskier than one with ROE of 15% and no debt. ROE must be read alongside leverage and risk.

31.2.2 Return on invested capital and economic value added

Because ROE mixes operating performance with financing, analysts also compute **return on invested capital (ROIC)**:

$$ROIC = \frac{NOPAT}{\text{Invested capital}}, \quad NOPAT = EBIT(1 - \tau),$$

where invested capital is operating working capital plus net long-lived operating assets (equivalently, debt plus equity minus excess cash). ROIC measures how well the business itself uses capital, independent of how it is financed.

A firm creates value only if its ROIC exceeds its **weighted average cost of capital (WACC)**—the return investors require, averaged across debt and equity. **Economic value added (EVA)**, a measure popularized by consultants in the 1990s, makes the comparison explicit:

$$EVA = (ROIC - WACC) \times \text{Invested capital} = NOPAT - WACC \times \text{Invested capital}.$$

A firm with NOPAT of 150, invested capital of 1,000 and WACC of 9% has EVA of $150 - 90 = 60$. EVA is an accounting approximation to the *economic profit* of Chapters 1 and 21: it charges for the cost of all capital, including equity, which ordinary accounting profit omits.

31.2.3 Other ratios

Category	Ratios
Liquidity	Current ratio; quick ratio; cash conversion cycle (Chapter 28)
Solvency and leverage	Debt/equity; net debt/EBITDA; interest coverage (EBIT/interest)
Efficiency	Asset turnover; inventory turnover (COGS/inventory); receivables turnover
Profitability	Gross margin; operating (EBIT) margin; net margin; ROA; ROE; ROIC
Market	Price/earnings (P/E); price/book (P/B); enterprise value/EBITDA; dividend yield

Ratios are meaningful only in comparison: with the same firm's history, with competitors, and with the economics of the industry. They depend on accounting choices—FIFO or LIFO, capitalizing or expensing development, IFRS or U.S. GAAP lease treatment—which must be adjusted before comparisons are made.

31.3 From Accounts to Value

31.3.1 Three valuation models

What is a share worth? In finance, the value of any asset is the present value of the cash flows it will deliver. For a share, those cash flows are dividends (including, in effect, buybacks). Three models express this idea in different but equivalent ways.

Dividend discount model. With cost of equity r and dividends D_t ,

$$V_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t}.$$

If dividends grow at a constant rate $g < r$ from D_1 , the growing-perpetuity formula of Chapter 9 gives the **Gordon growth model**, $V_0 = D_1/(r - g)$.

Discounted cash flow. The value of the firm is the present value of FCFF discounted at the WACC; equity value is firm value minus net debt.

Residual income. Accounting numbers can be substituted for dividends using the *clean-surplus relation*: if all changes in equity other than transactions with owners pass through income, then $B_t = B_{t-1} + NI_t - D_t$, where B_t is book equity. Substituting into the dividend discount model gives the **residual income model**:

$$V_0 = B_0 + \sum_{t=1}^{\infty} \frac{NI_t - r B_{t-1}}{(1+r)^t}.$$

Value equals current book value plus the present value of future *residual income*: earnings in excess of a normal return on the book value invested.

Proposition 31.3. Equivalence of the dividend and residual-income models

Under the clean-surplus relation, and provided $B_t/(1+r)^t \rightarrow 0$, the residual-income value equals the dividend-discount value.

Proof. From clean surplus, $D_t = NI_t - (B_t - B_{t-1})$. Then $\sum_t D_t/(1+r)^t = \sum_t [NI_t - rB_{t-1}]/(1+r)^t + \sum_t [(1+r)B_{t-1} - B_t]/(1+r)^t$. The second sum telescopes: $\sum_{t=1}^T [B_{t-1}/(1+r)^{t-1} - B_t/(1+r)^t] = B_0 - B_T/(1+r)^T$, which tends to B_0 . $\square$

Worked Example 31.4. Valuing a firm with constant ROE

A firm has book equity of 100, earns a constant ROE of 15% on opening equity, pays out 60% of earnings as dividends, and has a cost of equity of 10%. Retained earnings grow book equity at $g = 0.4 \times 15\% = 6\%$ a year.

Dividend discount: $D_1 = 0.6 \times 15 = 9$; $V_0 = 9/(0.10 - 0.06) = 225$.

Residual income: $RI_1 = 15 - 0.10 \times 100 = 5$, growing at 6%. $V_0 = 100 + 5 / (0.10 - 0.06) = 100 + 125 = 225$.

The price-to-book ratio is 2.25, above 1 because ROE exceeds the cost of equity. Had ROE equalled 10%, residual income would be zero and the firm would be worth exactly its book value, however fast it grew: growth creates value only when the return on new investment exceeds its cost.

The residual income model makes the link between accounting and value explicit. It also shows why conservative accounting (expensing R&D, Chapter 29) does not by itself mislead a careful valuation: lower book value today is offset by higher future residual income. What matters is that the accounting is consistent and that the analyst forecasts future profitability well.

31.4 Earnings Quality

31.4.1 Accruals

Net income equals operating cash flow plus **accruals**: the non-cash adjustments that accrual accounting makes (changes in receivables, inventory, payables, provisions, depreciation and so on):

$$\text{Accruals} = \text{Net income} - \text{CFO}.$$

Accruals improve the measurement of performance, but they rest on estimates, and estimates can be biased. The larger the accruals, the more reported earnings depend on judgement.

Worked Example 31.5. Two firms, same profit

Firms A and B each report net income of 100 and have average total assets of 1,000. Firm A's CFO is 120; Firm B's is 40. Accruals are -20 for A and $+60$ for B, and the *accrual ratio* (accruals divided by average assets) is -2% for A and $+6\%$ for B. B's profit depends much more on estimates—perhaps rising receivables, inventories or capitalized costs. Other things equal, A's earnings are of higher quality.

Richard Sloan (1996) showed that U.S. firms with high accruals subsequently had lower earnings and lower stock returns than firms with low accruals, and that the stock market appeared to overestimate the persistence of the accrual component of earnings. This *accrual anomaly* weakened after it was published, as investors learned to price it, but the underlying lesson remains: earnings supported by cash flows are more persistent than earnings supported by accruals.

31.4.2 Earnings management

Managers may manage earnings to meet analysts' forecasts, avoid reporting losses, satisfy debt covenants or maximize bonuses (the principal-agent problem of Chapter 25). Two broad forms exist:

- **Accrual management:** adjusting estimates within the rules—allowances for credit losses, warranty provisions, useful lives, impairment assumptions, percentage of completion—or, beyond the rules, fabricating entries.
- **Real earnings management:** changing real decisions to affect reported numbers—cutting research or advertising, offering discounts to pull sales forward, over-producing to spread

fixed overhead over more units in inventory. Real management is legal but may destroy value.

31.4.3 Red flags

Analysts and auditors look for patterns that often accompany low-quality earnings:

- Net income persistently exceeding operating cash flow; rising accrual ratios.
- Receivables or inventory growing much faster than revenue (rising DSO or DIO).
- Capitalized costs rising, or depreciation lives lengthening, when profits are under pressure (Chapter 29).
- Frequent “one-off” restructuring charges that recur; large write-offs by new management.
- Revenue recognized near period ends in unusual amounts; complex transactions with related parties or special-purpose entities.
- Growing gaps between reported profit and management’s “adjusted” profit, or between accounting profit and taxable profit.
- Changes of auditor, late filings, or the resignation of the chief financial officer.

Statistical models combine such signals. The Beneish “M-score” uses eight ratios to estimate the probability that a firm is manipulating its earnings; it was built on firms later subject to SEC enforcement.

Case: Enron: accounting for what has not happened yet

Enron, an energy company that transformed itself in the 1990s into a trader of energy contracts and other commodities, was ranked for six consecutive years by *Fortune* as America’s most innovative company. It filed for bankruptcy in December 2001. Two accounting practices were central. First, Enron applied mark-to-market accounting to long-term energy contracts, recognizing at inception the estimated present value of profits expected over many years, based on its own models of future prices for which there were often no observable markets—Level 3 fair values before the term existed (Chapter 27). Reported earnings therefore ran far ahead of cash. Second, Enron used special-purpose entities, some managed by its own chief financial officer, to move debt and poorly performing assets off its balance sheet and to create apparent profits on transactions with itself. When these entities were consolidated, Enron’s reported equity fell by more than a billion dollars and its debt rose sharply; confidence collapsed.

Several red flags were visible before the collapse: operating cash flow that lagged far behind reported earnings, extensive related-party transactions disclosed in the notes, and financial statements that sophisticated analysts found impossible to understand. The case, together with WorldCom (Chapter 29), led to the Sarbanes–Oxley Act and to tighter rules on consolidating special-purpose entities.

31.5 Managerial Accounting

The remainder of the chapter concerns information for decisions *within* the firm. It connects the accounting records to the cost theory of Chapter 21, but with a practical focus: managers must work with the data they have.

31.5.1 Cost behaviour

A **variable cost** changes in proportion to activity (materials, sales commissions); a **fixed cost** does not change within a relevant range of activity (rent, salaried staff, depreciation). Many costs are *mixed*. The distinction depends on the time horizon and the decision, exactly as short-run and long-run costs did in Chapter 21.

31.5.2 Contribution margin and cost–volume–profit analysis

The **contribution margin** per unit is price minus variable cost per unit: the amount each unit sold contributes toward covering fixed costs and then to profit. Profit is

$$\Pi = (p - v)Q - F,$$

where p is price, v variable cost per unit, Q volume and F fixed costs. The **break-even** volume is $Q_{BE} = F/(p - v)$, and the volume needed for a target profit Π^* is $(F + \Pi^*)/(p - v)$.

Worked Example 31.6. Cost–volume–profit

A company sells a product for 50; variable cost is 30 per unit and fixed costs are 200,000 a year. The contribution margin is 20. Break-even volume is $200,000/20 = 10,000$ units. To earn a profit of 60,000 it must sell $(200,000 + 60,000)/20 = 13,000$ units.

31.5.3 Operating leverage

The **degree of operating leverage** (DOL) measures how sensitive operating profit is to changes in sales:

$$\text{DOL} = \frac{\% \Delta \text{Operating profit}}{\% \Delta \text{Sales}} = \frac{\text{Total contribution margin}}{\text{Operating profit}}.$$

In Example 31.6, at 15,000 units the total contribution margin is 300,000 and profit 100,000, so $\text{DOL} = 3$: a 10% rise in sales raises profit by 30%, and a 10% fall reduces it by 30%. Firms with high fixed costs—airlines, semiconductor makers, software companies—have high operating leverage, so their profits swing sharply over the business cycle. Operating leverage compounds with financial leverage: a firm with both is doubly exposed to a fall in demand. This helps explain why investment and profits are so much more volatile than GDP (Chapter 46).

31.5.4 Relevant costs for decisions

The principle of Chapter 1 applies directly: only costs and revenues that differ between alternatives are relevant. Sunk costs and costs that will be incurred whatever is decided are irrelevant; opportunity costs are relevant even though they appear nowhere in the accounts.

Worked Example 31.7. Special order and make-or-buy

Special order. The company in Example 31.6 has spare capacity and receives a one-off order for 2,000 units at 38, below its full cost per unit (at 12,000 units of regular sales, $30 + 200,000/12,000 = 46.67$). Fixed costs will not change. The order's relevant revenue is 38 per unit and its relevant cost 30, so accepting it raises profit by $2,000 \times 8 = 16,000$. Full cost is irrelevant: the fixed costs are incurred anyway. (If the order would displace regular sales at 50, the forgone contribution of 20 per unit would be an opportunity cost, and the order should be rejected.)

Make or buy. A component can be bought for 22 per unit or made internally at a variable cost of 18 plus 30,000 of annual fixed costs that could be avoided by buying (a supervisor's salary and equipment rental), for 10,000 units a year. The relevant cost of making is $18 + 30,000/10,000 = 21$ per unit, less than 22: make. An allocated share of factory-wide overhead that would continue either way is irrelevant.

31.5.5 Activity-based costing

Firms need product costs for pricing, for inventory valuation and for deciding which products to keep. Direct materials and labour are easily traced to products; *overhead*—supervision, machine set-ups, quality control, purchasing—is not. Traditional systems allocate overhead in proportion to a single volume measure, such as direct labour hours or machine hours. When products differ in how much they use overhead activities, this can badly distort product costs. **Activity-based costing** (ABC) assigns overhead to activities and then to products according to their use of each activity's *cost driver*.

Worked Example 31.8. Activity-based costing

A factory makes a Standard product (10,000 units a year, in batches of 1,000, using 20,000 machine hours) and a Custom product (1,000 units a year, in batches of 50, using 5,000 machine hours). Overhead is 400,000: 250,000 related to machine operation and 150,000 to set-ups. Standard needs 10 set-ups and Custom 20.

Traditional (all overhead on machine hours): rate $400,000/25,000 = 16$ per hour. Standard: 320,000, or 32 per unit. Custom: 80,000, or 80 per unit.

ABC: machine rate $250,000/25,000 = 10$ per hour; set-up rate $150,000/30 = 5,000$ per set-up. Standard: $200,000 + 50,000 = 250,000$, or 25 per unit. Custom: $50,000 + 100,000 = 150,000$, or 150 per unit.

The traditional system overcosts the high-volume Standard product and undercosts the Custom product by 70 per unit, because Custom's many small batches consume a large share of set-up activity. A company pricing on traditional costs might underprice Custom orders and lose money on them without knowing it, while losing Standard business to competitors with better cost information.

31.5.6 Budgets and variance analysis

A **budget** is a quantitative plan for a future period. Budgets coordinate activities across a firm, allocate resources, and set benchmarks for evaluating performance. *Standard costs* specify how much of each input a unit of output should use and at what price. Comparing actual results with standards produces **variances**, which management investigates.

For a direct material, with standard price SP , actual price AP , standard quantity allowed for actual output SQ and actual quantity used AQ :

$$\text{Price variance} = (AP - SP) \times AQ, \quad \text{Quantity variance} = (AQ - SQ) \times SP.$$

Positive values are unfavourable (costs above standard); negative values favourable.

Worked Example 31.9. Material variances

The standard is 2 kg of material per unit at 5 per kg. In a month, 1,000 units are produced using 2,100 kg bought at 4.80 per kg. Standard cost for actual output: $1,000 \times 2 \times 5 =$

10,000. Actual cost: $2,100 \times 4.80 = 10,080$.

Price variance = $(4.80 - 5.00) \times 2,100 = -420$ (favourable),

Quantity variance = $(2,100 - 2,000) \times 5 = 500$ (unfavourable).

The total variance is 80 unfavourable. The purchasing department bought cheaply, but production used more material than standard—perhaps because the cheaper material was of lower quality. Variances are starting points for questions, not verdicts.

Budgets create incentive problems of their own (Chapter 25): managers who are evaluated against budgets have reasons to negotiate easy targets (“budgetary slack”) and to spend any remaining budget at year end. Some firms have moved toward rolling forecasts and relative performance evaluation to reduce these distortions.

Case: Amazon: thin margins, large cash flows

Amazon’s financial statements, available on EDGAR, illustrate nearly every theme of this chapter. For many years its reported net margins were low, often a few per cent or less, while its operating cash flow was large and growing. Three features of its business explain the gap.

Working capital. In its retail business Amazon has generally collected from customers immediately, turned over its inventory quickly, and paid suppliers on longer terms, giving it a negative cash conversion cycle (Chapter 28). As sales grew, payables grew faster than receivables and inventory, and growth itself generated cash.

Investment. Amazon invested heavily in fulfilment centres, logistics, data centres and technology. Much of this appeared as capital expenditure (investing cash flow), depreciated over future years; much else—software development, content, experimentation—was expensed, depressing current profit. Leases of warehouses and equipment added further large obligations, on the balance sheet since 2019 (Chapter 29).

Segments. Segment disclosures show that Amazon Web Services, its cloud-computing business, has in many recent years generated the majority of Amazon’s operating income despite providing a minority of its revenue. Aggregated ratios mix a low-margin, high-turnover retail business with a high-margin, capital-intensive cloud business; DuPont analysis by segment is far more informative than for the company as a whole.

The case shows why valuation models built on free cash flow or on residual income, rather than on current earnings alone, were necessary to understand the company, and why investors who focused on its low reported profit margins misjudged it. It also shows the limits of any single ratio: the same low margin that signals weakness in one business model reflects deliberate reinvestment in another.

Summary

- The three sections of the cash-flow statement together describe a firm’s life cycle; free cash flow measures cash available to investors after necessary investment.
- DuPont decomposes ROE into margin, turnover and leverage. ROIC measures operating returns; value is created only when ROIC exceeds WACC, as EVA makes explicit.
- Dividend-discount, discounted-cash-flow and residual-income models are equivalent in principle; the residual-income model expresses value as book value plus the present value of excess returns.

- Accruals connect profit and cash flow; high accruals signal lower-quality, less persistent earnings. Red flags include profit outrunning cash, rising DSO or DIO, and aggressive capitalization.
- Contribution margin analysis yields break-even points and operating leverage; only relevant (differential) costs matter for decisions.
- Activity-based costing assigns overhead by cost drivers, correcting distortions from volume-based allocation. Variance analysis compares actual with standard costs.

Key terms

operating cash flow; free cash flow to the firm; free cash flow to equity; return on equity; DuPont decomposition; return on assets; financial leverage; return on invested capital; NOPAT; weighted average cost of capital; economic value added; Gordon growth model; clean-surplus relation; residual income model; accruals; accrual ratio; accrual anomaly; earnings management; real earnings management; M-score; variable cost; fixed cost; contribution margin; break-even; degree of operating leverage; relevant cost; activity-based costing; cost driver; budget; standard cost; price variance; quantity variance.

Exercises

Exercise 31.1 (Free cash flow). A company has EBIT of 400, a tax rate of 25%, depreciation of 120, capital expenditure of 200, and an increase in operating working capital of 50. It pays interest of 60 and borrows a net 30. Compute FCFF and FCFE.

Exercise 31.2 (DuPont). A company has sales of 2,000, net income of 100, total assets of 1,600 and equity of 800. Compute ROE and its three DuPont components. If it borrows 400 and uses the money to buy back shares (assets unchanged), and after-tax interest cost reduces net income by 15, what is the new ROE? Has the firm become more valuable?

Exercise 31.3 (EVA). A division has invested capital of 500, EBIT of 80 and a tax rate of 25%. The WACC is 10%. Compute ROIC and EVA. Should the division undertake a project requiring 100 of additional capital that would raise EBIT by 12?

Exercise 31.4 (Residual income). A firm has book equity of 200, expected ROE of 12% forever, a payout ratio of 50%, and a cost of equity of 9%. Value the equity using the residual income model and check with the Gordon model. What is the P/B ratio? What happens to value if the payout ratio falls to 25%? Explain.

Exercise 31.5 (Accruals). Firm X reports net income of 50 and CFO of 10; its receivables rose by 35 and inventory by 15 while payables were unchanged; depreciation was 10. Reconcile net income to CFO. What questions would you ask management?

Exercise 31.6 (Break-even). A café sells coffee for 3.50; variable cost per cup is 1.10. Monthly fixed costs are 12,000. Find the break-even volume and the volume needed for monthly profit of 6,000. If rent rises by 1,200 a month, by how much must volume rise to keep the same profit?

Exercise 31.7 (Operating leverage). Two firms each earn operating profit of 100 on sales of 1,000. Firm A has fixed costs of 100; Firm B has fixed costs of 500. Compute each firm's DOL and the change in operating profit if sales fall 20%.

Exercise 31.8 (Special order). A plant with capacity of 50,000 units currently sells 45,000 at 20; variable cost is 12, fixed cost 250,000. A customer offers to buy 8,000 units at 15. Should the firm accept? Consider both the capacity constraint and the possibility of accepting only part of the order.

Exercise 31.9 (ABC). In Example 31.8, suppose the Custom product is sold at 120 per unit and has direct costs of 40 per unit. Is it profitable under the traditional allocation? Under ABC? What would you recommend?

Exercise 31.10 (Variances). Standard labour is 0.5 hours per unit at 24 per hour. In a month, 4,000 units are produced using 2,150 hours at a cost of 50,525. Compute the labour rate and efficiency variances.

Solutions

Solution 31.1. $FCFF = 400 \times 0.75 + 120 - 200 - 50 = 170$. $FCFE = 170 - 60 \times 0.75 + 30 = 155$.

Solution 31.2. Margin = 5%; turnover = 1.25; leverage = 2.0; ROE = 12.5%. After the recapitalization, equity is 400, leverage 4.0, net income 85, margin 4.25%: $ROE = 0.0425 \times 1.25 \times 4.0 = 21.25\%$. ROE rises, but by Modigliani–Miller logic (Chapter 30) the firm is not more valuable merely because of this; equity is now riskier and its required return higher. With the interest tax shield, value may rise somewhat, but so does the risk of financial distress.

Solution 31.3. NOPAT = 60; ROIC = 12%; EVA = $60 - 50 = 10$. The project adds NOPAT of $12 \times 0.75 = 9$ on capital of 100: a 9% return, below the 10% WACC. It would reduce EVA by 1; reject it (it would lower the division's average ROIC and destroy value).

Solution 31.4. $g = 0.5 \times 12\% = 6\%$. $RI_1 = 24 - 18 = 6$; $V = 200 + 6 / (0.09 - 0.06) = 400$. Gordon: $D_1 = 12$; $V = 12 / 0.03 = 400$. P/B = 2. With payout 25%, $g = 9\%$ equals r and the formulas break down: value becomes unbounded because the firm reinvests ever more at 12% when investors require 9%. Realistically, ROE cannot remain above the cost of equity forever on ever-growing reinvestment; competition erodes excess returns, and a proper valuation would model ROE fading toward r .

Solution 31.5. $CFO = 50 + 10 - 35 - 15 + 0 = 10$. Accruals = 40. Questions: why receivables grew so much faster than sales (are customers paying? was revenue recognized early?), why inventory built up (slowing sales, obsolescence?), and whether credit terms or revenue policies changed.

Solution 31.6. Contribution 2.40 per cup. Break-even = $12,000 / 2.40 = 5,000$ cups. For profit 6,000: $18,000 / 2.40 = 7,500$ cups. Extra rent of 1,200 requires 500 more cups a month.

Solution 31.7. A: contribution margin = 200; DOL = 2. B: contribution = 600; DOL = 6. With sales down 20%: A's contribution falls to 160, profit to 60 (down 40%). B's contribution falls to 480, profit to -20 (down 120%).

Solution 31.8. Spare capacity is 5,000 units. Accepting all 8,000 would require giving up 3,000 regular sales. Gain from 5,000 spare units: $5,000 \times (15 - 12) = 15,000$. Loss from displacing 3,000 regular sales and filling them with the order: each such unit earns 3 instead of 8, a loss of $3,000 \times 5 = 15,000$. Net effect of the full order: 0. Accepting only 5,000 units (if the customer agrees) raises profit by 15,000.

Solution 31.9. Traditional: cost $40 + 80 = 120$, zero margin—apparently break-even. ABC: cost $40 + 150 = 190$: a loss of 70 per unit, 70,000 a year. Recommend raising Custom prices, reducing its set-up costs (larger batches, faster changeovers), or dropping it if neither is feasible—after checking which of its allocated overheads would actually be avoided if it were dropped (only avoidable costs are relevant).

Solution 31.10. Actual rate = $50,525 / 2,150 = 23.50$. Standard hours for actual output = 2,000. Rate variance = $(23.50 - 24) \times 2,150 = -1,075$ (favourable). Efficiency variance = $(2,150 - 2,000) \times 24 = 3,600$ (unfavourable). Total = 2,525 unfavourable ($50,525 - 48,000$).

Part V

Finance

Capital Budgeting and Real Options

After studying this chapter you should be able to

1. Identify the incremental cash flows of a project: including opportunity costs, working capital, taxes and terminal values, and excluding sunk costs and allocated overheads.
2. Build a project cash-flow forecast and compute its NPV, IRR and payback.
3. Explain when the IRR rule fails—multiple roots, scale and timing—and why NPV is the reliable criterion.
4. Allocate a limited budget among projects, and compare projects of unequal lives with equivalent annual amounts.
5. Analyse uncertainty with sensitivity, scenario, break-even and simulation analysis.
6. Value managerial flexibility—options to delay, expand and abandon—with decision trees.
7. Recognize optimism bias in project appraisal and the remedies for it.

A motivating question

When the Channel Tunnel between England and France opened in 1994, it was a triumph of engineering and a financial disaster. Its construction cost far exceeded the original estimate, its financing costs even more so, and its traffic in the early years fell far short of forecasts. The company that owned it went through repeated debt restructurings, and its original shareholders lost most of their investment. Yet the decision to build rested on detailed financial projections showing a large positive net present value.

Capital budgeting—deciding which long-term investments to make—is the most consequential financial decision most firms and governments take. Chapter 9 introduced the tool: accept projects with positive net present value. This chapter shows how to apply it in practice: which cash flows to count, how to compare projects, how to deal with uncertainty, and how to value the flexibility to change course. It ends with the most common reason projects disappoint—not errors in discounting, but systematically optimistic forecasts.

32.1 Relevant Cash Flows

32.1.1 The incremental principle

A project should be evaluated by the *difference* it makes to the firm's future cash flows: cash flows with the project minus cash flows without it. This **incremental principle** settles most questions about what to include:

- **Cash, not accounting profit.** Depreciation is not a cash outflow (the cash went out when the asset was bought); but because it reduces taxable income, it creates a *depreciation tax*

shield (Chapter 29).

- **Sunk costs are excluded.** Money already spent, and unrecoverable whatever is decided—last year's feasibility study, research already done—is irrelevant to the decision (Chapter 1).
- **Opportunity costs are included.** If the project uses a warehouse the firm owns and could otherwise rent out, the forgone rent is a cost of the project.
- **Side effects are included.** Sales a new product takes from the firm's existing products (*cannibalization*) reduce the project's incremental revenue; sales it adds to complementary products increase it.
- **Working capital is included.** A project that requires inventory and extends credit to customers ties up cash in operating working capital (Chapter 28), usually recovered at the end.
- **Allocated overheads are excluded** unless the project actually changes them. Charging a project a share of head-office costs that would be incurred anyway misstates its incremental effect.
- **Financing flows are excluded** from the project's cash flows, because their cost is captured in the discount rate (Chapter 35). Including interest payments and also discounting at a rate that reflects the cost of debt would count financing costs twice.

32.1.2 A project forecast

Operating cash flow each year is after-tax operating profit plus depreciation:

$$OCF = (R - C - Dep)(1 - \tau) + Dep = (R - C)(1 - \tau) + \tau Dep,$$

where R is revenue, C cash operating costs and τ the tax rate. The second form shows the depreciation tax shield τDep explicitly. Adding capital spending, working-capital investment and terminal values gives the project's free cash flow—the project-level counterpart of the free cash flow to the firm of Chapter 31.

Worked Example 32.1. Evaluating a machine

A machine costs 1,000 and will be depreciated straight-line to zero over five years (200 a year). It will generate revenue of 800 and cash costs of 400 a year, and require working capital of 100 at the start, recovered at the end. It can be sold for 100 at the end of year 5; since its book value is then zero, the gain is taxed at the 25 per cent rate, leaving 75. The discount rate is 10 per cent. A feasibility study costing 50 has already been paid. Operating cash flow is $(800 - 400)(0.75) + 0.25 \times 200 = 300 + 50 = 350$ a year. The cash flows are:

Year	0	1	2	3	4	5
Machine	-1,000					75
Working capital	-100					100
Operating cash flow		350	350	350	350	350
Free cash flow	-1,100	350	350	350	350	525

The NPV at 10 per cent is 335.4; the IRR is 20.7 per cent. The study cost of 50 is sunk and excluded. If the machine would occupy space that could otherwise be rented for 30 a year after tax, the present value of the forgone rent, $30 \times 3.791 = 113.7$, must be subtracted, reducing the NPV to 221.7.

Inflation. Cash flows and discount rates must be consistently nominal or real (Chapter 9). A common error is to forecast revenue in today's prices while discounting at a nominal rate, which understates value; another is to hold depreciation tax shields fixed in nominal terms but treat them as real.

32.2 Decision Rules

32.2.1 NPV and IRR

The NPV rule (Theorem 9.4 of Chapter 9) accepts projects with positive NPV. The IRR rule accepts projects whose internal rate of return exceeds the cost of capital. For a conventional project—an outflow followed by inflows—the two agree, because NPV falls steadily as the discount rate rises and crosses zero exactly at the IRR. They can disagree in three situations.

Multiple IRRs. If cash flows change sign more than once, NPV can be zero at several discount rates. A project with cash flows $-100, +230, -132$ —a mine that requires clean-up spending at the end, say—has IRRs of both 10 and 20 per cent: substituting $x = 1 + r$, $100x^2 - 230x + 132 = 0$ has roots 1.1 and 1.2. Its NPV is positive only for discount rates between them. No single IRR summarizes it.

Scale. Project A costs 100 and returns 130 a year later (IRR 30 per cent, NPV at 10 per cent of 18.2); project B costs 1,000 and returns 1,200 (IRR 20 per cent, NPV 90.9). If only one can be chosen, the IRR prefers A, but B adds five times as much value. A percentage return says nothing about how much wealth is created.

Timing. Of two mutually exclusive projects, one with early cash flows and one with late ones, the IRR can rank them in the opposite order to NPV at the actual cost of capital, because the IRR implicitly assumes that cash flows are reinvested at the IRR itself rather than at the cost of capital.

Key idea. NPV measures value created, in currency, at the firm's opportunity cost of capital. Other criteria are useful summaries—the IRR communicates a margin of safety on the discount rate—but where they disagree with NPV, NPV is the rule that maximizes shareholder value.

32.2.2 Payback

The **payback period** is the time until cumulative cash flows turn positive. In Example 32.1, cumulative cash flow is -50 after three years and $+300$ after four, so payback is about 3.1 years. Payback ignores the time value of money within the period, ignores everything after it, and has no economic cutoff. It survives because it is simple, because it is a rough measure of exposure to risks that grow with time, and because firms short of liquidity care how quickly cash returns. The *discounted payback* period, which uses discounted cash flows, corrects the first defect only.

32.2.3 Capital rationing

If a firm can raise only a fixed budget this year, it should choose the set of projects with the highest total NPV that fits within the budget. The **profitability index**—the present value of future cash flows divided by the initial investment, or NPV per unit invested—ranks projects

by value created per unit of the scarce resource. Ranking by the index works when projects are divisible; with indivisible projects, the problem is an integer linear program (Chapter 6).

Worked Example 32.2. Allocating a budget

A firm has 1,000 to invest. Project P1 costs 600 with NPV 180; P2 costs 400 with NPV 100; P3 costs 500 with NPV 140. Their NPVs per unit invested are 0.30, 0.25 and 0.28. The feasible combinations are P1 and P2 (cost 1,000, NPV 280), P2 and P3 (900, NPV 240), or any single project. The best choice is P1 and P2. Taking the two highest-ranked projects by the index, P1 and P3, would cost 1,100 and not fit. With genuinely rationed capital, the shadow price of the budget constraint (Chapter 6) is the return the marginal project would have earned—which is why rationed firms in effect use a hurdle rate above their cost of capital.

Genuine capital rationing is rarer than it appears: a firm with positive-NPV projects can usually raise funds. Budgets are often imposed internally to discipline managers' optimism, a point to which the chapter returns.

32.2.4 Projects with unequal lives

When mutually exclusive assets have different lives and will be replaced, comparing their NPVs over their own lives is misleading. The **equivalent annual amount** converts each asset's present value into a constant annual amount over its life, using the annuity formula (Chapter 9).

Worked Example 32.3. Choosing a machine

Machine A costs 500, lasts three years and costs 100 a year to run; machine B costs 700, lasts five years and costs 80 a year. At 10 per cent, the three- and five-year annuity factors are 2.487 and 3.791. A's present cost is $500 + 100 \times 2.487 = 748.7$, an equivalent annual cost of $748.7/2.487 = 301.1$. B's is $700 + 80 \times 3.791 = 1,003.3$, or $1,003.3/3.791 = 264.7$ a year. Although B costs more in total, it is cheaper per year of service, and should be chosen if the service will be needed indefinitely.

32.3 Analysing Uncertainty

A single NPV built on best estimates hides how much the answer could change. Four tools reveal it:

- **Sensitivity analysis** changes one input at a time—price, volume, cost, discount rate, salvage value—and records the effect on NPV. It identifies the inputs that matter most, where further research is most valuable (Chapter 19). In Example 32.1, a 10 per cent fall in revenue reduces operating cash flow by 60 a year and NPV by about 230.
- **Break-even analysis** finds the value of an input at which NPV is zero. Each unit fall in annual revenue lowers NPV by $0.75 \times 3.791 = 2.84$, so revenue could fall by about 118, to about 682 a year, before the machine's NPV reached zero: a margin of safety of about 15 per cent.
- **Scenario analysis** changes several inputs together in internally consistent combinations—a recession scenario with lower volume, lower price and higher bad debts—rather than one at a time.

- **Monte Carlo simulation** assigns distributions to the uncertain inputs, including their correlations, and computes the distribution of NPV (Chapter 19).

The value of these tools lies less in the numbers they produce than in the questions they force: what would have to be true for this project to succeed, and how cheaply can we find out whether it is?

32.4 Real Options

32.4.1 Flexibility has value

A standard NPV calculation assumes a fixed plan: invest now, operate for a set life, stop. Real managers can adapt as information arrives. They can wait before investing, expand if demand is strong, switch inputs or outputs, or abandon a failing project and sell its assets. These possibilities are **real options**: rights, but not obligations, to take future actions. Because an option is exercised only when doing so is beneficial, flexibility adds value—value that a fixed-plan NPV omits.

32.4.2 The option to delay

Worked Example 32.4. Waiting for information

A project costs 1,000 whether undertaken now or next year. Once built, it produces a perpetual cash flow that will be either 150 or 60 a year with equal probability; which one will be known in a year. The discount rate is 10 per cent.

Invest now: the expected cash flow is 105 a year, worth $105/0.10 = 1,050$, so $\text{NPV} = 50$.

Wait a year: if the cash flow turns out to be 150, invest then, earning $150/0.10 - 1,000 = 500$ at year 1; if it turns out to be 60, the project would be worth $600 - 1,000 = -400$, so do not invest. The expected value at year 1 is $0.5 \times 500 = 250$, worth $250/1.1 = 227.3$ today.

Waiting is better by $227.3 - 50 = 177.3$: that is the value of the option to delay. A positive NPV today is not sufficient reason to invest now if waiting reveals information that could prevent a costly mistake. The value of waiting comes from avoiding the bad outcome, and it rises with uncertainty.

The example is the logic of Dixit and Pindyck's analysis of irreversible investment (Chapter 45): uncertainty raises the value of waiting, so firms facing more uncertainty invest less today, even if they are risk-neutral.

32.4.3 Options to expand and abandon

An *option to expand* makes a small initial project valuable as a platform: a pilot plant, a first store in a new country or an early-stage drug trial may have negative NPV on its own but positive value once the right to scale up after success is counted. An *option to abandon* limits losses: a project that can be stopped and its equipment sold if demand disappoints is worth more than one that cannot, and assets with good resale value (general-purpose buildings, aircraft) carry more of this option than specialized ones.

► Option value as a plug

Because options add value, analysts sometimes invoke “strategic option value” to justify projects with negative NPV. A real option is worth something only if the decision, its timing, its cost and the information that would trigger it can be specified—and if competitors cannot take the opportunity first. Valuing options formally requires the tools of Chapter 41; at a minimum, a decision tree should show exactly when and why the option would be exercised.

32.5 Optimism Bias

Most project disappointments do not arise from errors in discounting. They arise because the cash-flow forecasts were too optimistic. Bent Flyvbjerg and co-authors, studying hundreds of large infrastructure projects around the world—rail lines, bridges, tunnels, dams, information-technology systems—found that the large majority ran over budget and over schedule, and many delivered fewer benefits than forecast, with little improvement over decades. Two explanations reinforce each other: psychological *optimism bias* (planners underestimate costs and overestimate benefits, focusing on the specific plan rather than on how similar projects fared) and strategic *misrepresentation* (sponsors competing for approval have incentives to present rosy numbers; an agency problem of the kind analysed in Chapter 25).

Remedies include *reference-class forecasting*—adjusting a project’s forecast by the distribution of outcomes of comparable past projects—*independent review*, holding sponsors accountable for forecasts, and *staged funding* that releases money as milestones are met, which in effect builds abandonment options into the process.

Case: The Channel Tunnel

The project. The Channel Tunnel, linking Folkestone in England with Coquelles near Calais, was built by a private consortium, Eurotunnel, under a concession from the British and French governments, with no government financial guarantees. Construction began in 1988, and the tunnel opened in 1994, about a year later than planned.

The numbers. Flyvbjerg’s analysis of the project reports that construction costs overran their estimate by around 80 per cent in real terms, and financing costs by considerably more, as delays extended the period in which interest accumulated before any revenue arrived. Traffic and revenue in the early years were well below forecasts, as ferry operators responded with lower fares and faster ships, and budget airlines later competed on some routes. Eurotunnel was unable to service its debt from operating cash flows; it restructured its debt in the late 1990s and again under French court protection in 2007, with creditors taking large losses and original shareholders seeing most of their investment wiped out. Economists have calculated that, taking all costs and benefits together, Britain might have been better off had the tunnel never been built.

In terms of this chapter. The project combined every source of error: optimistic cost and schedule estimates; revenue forecasts that did not anticipate competitors’ responses (Chapter 23); high leverage, which turned a cost overrun into a solvency crisis (Chapter 39); and extreme irreversibility, with no option to abandon a half-built tunnel. The tunnel is now an indispensable piece of European infrastructure, operating profitably—for owners who acquired it after the losses had been written off. A project can be socially valuable and a financial failure for those who financed it; the NPV depends on the price paid.

Summary

- Evaluate projects by incremental after-tax cash flows: include opportunity costs, side effects, working capital and terminal values; exclude sunk costs, unchanged overheads and financing flows.
- Operating cash flow is $(R - C)(1 - \tau) + \tau Dep$.
- NPV is the reliable criterion; IRR can mislead with multiple sign changes, differing scales and timing; payback ignores value after the cutoff.
- Under capital rationing, maximize total NPV within the budget; the profitability index ranks divisible projects. Equivalent annual amounts compare assets with different lives.
- Sensitivity, break-even, scenario and simulation analysis reveal which assumptions drive the decision.
- Real options—to delay, expand, abandon—add value that fixed-plan NPV omits; the value of waiting rises with uncertainty.
- Optimism bias and strategic misrepresentation make forecasts too rosy; reference-class forecasting and staged funding counter them.

Key terms

capital budgeting; incremental principle; sunk cost; opportunity cost; cannibalization; operating working capital; depreciation tax shield; operating cash flow; terminal value; salvage value; multiple IRRs; scale problem; payback period; discounted payback; capital rationing; profitability index; equivalent annual amount; sensitivity analysis; break-even analysis; scenario analysis; real option; option to delay; option to expand; option to abandon; optimism bias; strategic misrepresentation; reference-class forecasting; staged funding.

Exercises

Exercise 32.1 (Relevant cash flows). A firm considers launching a new drink. Classify each item as relevant or not, and explain: (a) 2 million already spent on market research; (b) sales of the firm's existing drinks expected to fall by 1 million a year; (c) a share of head-office costs of 0.5 million a year, which will not change; (d) an unused bottling line that could be sold for 3 million; (e) interest on a loan taken to finance the launch.

Exercise 32.2 (Operating cash flow). A project has revenue 2,000, cash costs 1,200 and depreciation 300 a year; the tax rate is 30 per cent. Compute operating cash flow in two ways.

Exercise 32.3 (Project NPV). A machine costs 100 now, yields 55 at the end of each of two years and has salvage value 10 at the end of year 2 (ignore taxes). Compute its NPV at 10 per cent. What if salvage is zero? What does this suggest about which assumption to check first?

Exercise 32.4 (Multiple IRRs). A project has cash flows $-100, +250, -154$. Find its IRRs and the range of discount rates over which its NPV is positive.

Exercise 32.5 (Scale and timing). Project X: $-1,000$ now, $+1,500$ in three years. Project Y: $-1,000$ now, $+600$ in each of years 1 and 2. Compute each project's IRR and NPV at 8 per cent. Which is preferred if they are mutually exclusive?

Exercise 32.6 (Capital rationing). With a budget of 700, choose among: A (cost 300, NPV 60), B (cost 400, NPV 90), C (cost 350, NPV 80), D (cost 250, NPV 45). Projects are indivisible.

Exercise 32.7 (Equivalent annual cost). A delivery van costing 40,000 lasts four years with running costs of 6,000 a year; an electric van costing 55,000 lasts six years with running costs of 3,000 a year. At 8 per cent, which is cheaper per year?

Exercise 32.8 (Break-even). In Example 32.1, verify the break-even annual revenue (holding cash costs at 400). What is the break-even level of cash costs, holding revenue at 800?

Exercise 32.9 (Option to abandon). A project costs 500 and in one year will be worth either 800 or 300 (equal probabilities), with the discount rate 10 per cent. Its equipment can be sold for 400 at the end of year 1. Find the NPV without and with the option to abandon, and the value of the option.

Exercise 32.10 (Funding the project). A project requires 80 now and produces free cash flows of 50 and 60 at the end of years 1 and 2. Its appropriate discount rate is 10 per cent. Compute its NPV. A finance proposal would fund it with 60 per cent equity costing 12 per cent and 40 per cent debt costing 6 per cent, with a 25 per cent tax rate. Compute the resulting weighted average and explain whether it should replace the project's 10 per cent rate.

Solutions

Solution 32.1. (a) Sunk: exclude. (b) Cannibalization: include as a cost of 1 million a year. (c) Unchanged overhead: exclude. (d) Opportunity cost: include the 3 million forgone sale proceeds (after tax) at the start. (e) Financing flow: exclude; the cost of financing enters through the discount rate.

Solution 32.2. $(2,000 - 1,200 - 300)(0.7) + 300 = 350 + 300 = 650$. Equivalently $800 \times 0.7 + 0.3 \times 300 = 560 + 90 = 650$.

Solution 32.3. $-100 + 55/1.1 + 65/1.21 = -100 + 50 + 53.72 = 3.72$. Without salvage: $-100 + 50 + 45.45 = -4.55$. The decision flips on the salvage value, so its realizability is the assumption to verify first.

Solution 32.4. With $x = 1 + r$: $100x^2 - 250x + 154 = 0$, $x = (250 \pm \sqrt{62,500 - 61,600})/200 = (250 \pm 30)/200 = 1.1$ or 1.4 . IRRs 10 and 40 per cent; NPV (a downward-opening parabola in x after multiplying through) is positive for discount rates between 10 and 40 per cent.

Solution 32.5. X: IRR = $1.5^{1/3} - 1 = 14.5$ per cent; NPV = $-1,000 + 1,500/1.08^3 = 190.7$. Y: IRR solves $600/(1+r) + 600/(1+r)^2 = 1,000$; $r = 13.1$ per cent; NPV = $-1,000 + 600 \times 1.7833 = 70.0$. X is preferred (higher NPV, and here also higher IRR).

Solution 32.6. Feasible pairs within 700: A+B (700, 150), A+C (650, 140), A+D (550, 105), B+D (650, 135), C+D (600, 125); B+C costs 750. Triples all exceed 700 except none (A+C+D = 900). Best: A and B, NPV 150.

Solution 32.7. Annuity factors at 8 per cent: four years 3.3121, six years 4.6229. Diesel: $(40,000 + 6,000 \times 3.3121)/3.3121 = 12,077 + 6,000 = 18,077$ a year. Electric: $55,000/4.6229 + 3,000 = 11,897 + 3,000 = 14,897$ a year. The electric van is cheaper per year.

Solution 32.8. NPV falls by $0.75 \times 3.7908 = 2.843$ per unit fall in annual revenue. Setting $335.44 - 2.843 \Delta R = 0$ gives $\Delta R = 118$: break-even revenue ≈ 682 a year, as in the text. By the same logic, cash costs could rise by 118, to about 518 a year.

Solution 32.9. Without abandonment: $-500 + (0.5 \times 800 + 0.5 \times 300)/1.1 = -500 + 500 = 0$. With it, the bad outcome becomes 400: $-500 + (400 + 200)/1.1 = 45.5$. Option value 45.5.

Solution 32.10. NPV = $-80 + 50/1.1 + 60/1.21 = -80 + 45.45 + 49.59 = 15.04$. The proposal's weighted average is $0.6 \times 12 + 0.4 \times 6 \times 0.75 = 7.2 + 1.8 = 9.0$ per cent. It should not automatically replace 10 per cent: the discount rate should reflect the risk of the project's cash flows, not the particular way it happens to be financed; Chapter 35 shows how financing policy enters.

Fixed Income: Yields, Duration and the Term Structure

After studying this chapter you should be able to

1. Distinguish clean and dirty prices, and compute accrued interest.
2. Define discount factors, spot rates and forward rates, and extract spot rates from bond prices by bootstrapping.
3. Interpret yield to maturity and par yields, and explain their limitations as measures of return.
4. Compute Macaulay and modified duration, DV01 and convexity, and use them to approximate price changes.
5. Immunize a liability by matching durations, and explain why immunization needs rebalancing.
6. State the expectations, term-premium and segmentation theories of the term structure.
7. Decompose credit spreads into expected loss and risk premia, and describe inflation-linked, floating-rate and callable bonds.

A motivating question

On 23 September 2022 the British government announced large unfunded tax cuts. Over the next few days, yields on long-term government bonds—gilts—rose by more than a percentage point, an extraordinary move for the bonds of a major economy. The rise threatened to bankrupt a corner of the pension industry that most people had never heard of: funds that had used borrowed money to buy long-term gilts in order to match the long-dated pension promises they had made. Within five days the Bank of England intervened to buy gilts and stop a spiral of forced selling.

Why can a one-point rise in interest rates be a catastrophe? The answer lies in how sensitive the prices of long-term bonds are to yields—their *duration*—and in what happens when that sensitivity is multiplied by leverage. Chapter 9 showed how to price a bond and that prices fall when yields rise. This chapter measures by how much, builds the term structure of interest rates from bond prices, explains what shapes it, and analyses the credit risk that separates corporate from government bonds. The tools are used again in accounting for debt and financial assets (Chapter 30), in banking and central banking (Chapters 48 and 49), and in derivatives (Chapter 41).

33.1 Bond Conventions

A bond is a promise to pay specified cash flows on specified dates. Government bonds (U.S. Treasuries, German Bunds, U.K. gilts, Malaysian Government Securities) are the benchmarks against which other bonds are priced. Corporate bonds carry the additional risk that the issuer defaults.

Clean and dirty prices. Coupons are paid periodically, but bonds trade every day. A buyer between coupon dates must compensate the seller for the interest earned since the last coupon: the **accrued interest**, equal to the coupon times the fraction of the coupon period that has elapsed (measured by a *day-count convention* specific to each market). Prices are quoted *clean*, excluding accrued interest; the buyer pays the *dirty* (full) price, clean price plus accrued interest. If a bond with an annual coupon of 6 is bought four months after its last coupon, accrued interest is $6 \times 4/12 = 2$, and a clean price of 101 means a payment of 103. Quoting clean prices avoids the saw-tooth pattern that dirty prices follow as coupons accrue and are paid.

33.2 Discount Factors, Spot Rates and Forward Rates

33.2.1 Zero-coupon bonds and spot rates

A *zero-coupon bond* pays a single amount at maturity. Its price per unit of face value is the **discount factor** d_t for that date, and the **spot rate** s_t is the annual rate that discounts the payment to its price:

$$d_t = \frac{1}{(1 + s_t)^t}.$$

The set of spot rates for different maturities is the **term structure** of interest rates, or *yield curve*. Any fixed cash-flow stream can be valued by discounting each payment at the spot rate for its own date. Using a single rate for all dates, as in Chapter 9, is a simplification that is exact only when the term structure is flat.

33.2.2 Bootstrapping

Few zero-coupon bonds trade at every maturity, but spot rates can be extracted from the prices of coupon bonds by **bootstrapping**: solving for discount factors one maturity at a time.

Worked Example 33.1. Bootstrapping a yield curve

Three government bonds with annual coupons and face value 100 trade at:

Maturity	1 year	2 years	3 years
Coupon	0	4	5
Price	97.087	100.50	101.20

From the one-year bond, $d_1 = 0.970874$, so $s_1 = 3.00$ per cent. The two-year bond pays 4 in a year and 104 in two years: $100.50 = 4d_1 + 104d_2$, so $d_2 = (100.50 - 3.883)/104 = 0.929005$ and $s_2 = d_2^{-1/2} - 1 = 3.75$ per cent. The three-year bond gives $101.20 = 5d_1 + 5d_2 + 105d_3$, so $d_3 = 0.873339$ and $s_3 = 4.62$ per cent. The yield curve slopes upward.

33.2.3 Forward rates

The discount factors also imply rates for borrowing and lending between future dates. The **forward rate** $f_{t-1,t}$ for the year from $t-1$ to t satisfies

$$(1 + s_t)^t = (1 + s_{t-1})^{t-1}(1 + f_{t-1,t}), \quad \text{so} \quad 1 + f_{t-1,t} = \frac{d_{t-1}}{d_t}.$$

Investing for two years at the two-year spot rate must give the same result as investing for one year and locking in the one-year forward rate for the second year; otherwise there would be an arbitrage. In Example 33.1, $f_{1,2} = 0.970874/0.929005 - 1 = 4.51$ per cent and $f_{2,3} = 0.929005/0.873339 - 1 = 6.37$ per cent. When the yield curve slopes upward, forward rates lie above spot rates; each spot rate is approximately the average of the forward rates up to its maturity.

33.3 Yields

33.3.1 Yield to maturity

The **yield to maturity** of a coupon bond is the single rate that equates the present value of its cash flows to its dirty price—its internal rate of return (Chapter 9). It is a convenient summary, but it is *not* the return an investor will earn, unless the bond is held to maturity, does not default, and every coupon is reinvested at the yield itself. A bond sold before maturity earns a return that depends on the yield at the time of sale. The yield to maturity is also a blend of spot rates weighted toward the dates of the largest payments, so two bonds of the same maturity but different coupons can have different yields on the same curve.

Par yields. The *par yield* for a maturity is the coupon rate at which a bond would trade at face value. From discount factors, the n -year par yield is $c_n = (1 - d_n) / \sum_{t=1}^n d_t$. In Example 33.1, the three-year par yield is $(1 - 0.873339) / (0.970874 + 0.929005 + 0.873339) = 4.57$ per cent. Published government yield curves are often par curves.

33.4 Interest-Rate Risk

33.4.1 Duration

How much does a bond's price change when yields change? The first-order Taylor approximation (Chapter 5) gives

$$\frac{\Delta P}{P} \approx -D_{\text{mod}} \Delta y,$$

where the **modified duration** $D_{\text{mod}} = -\frac{1}{P} \frac{dP}{dy}$. For a bond with cash flows CF_t and annual yield y , differentiating the price formula gives $D_{\text{mod}} = D_{\text{Mac}} / (1 + y)$, where the **Macaulay duration**

$$D_{\text{Mac}} = \frac{1}{P} \sum_t t \frac{CF_t}{(1 + y)^t}$$

is the weighted average time to the bond's payments, each weighted by its share of the bond's present value. A zero-coupon bond's Macaulay duration equals its maturity; a coupon bond's is shorter, because some value arrives earlier. Duration rises with maturity and falls with the coupon and the yield.

Worked Example 33.2. Duration of a three-year bond

The three-year bond of Example 9.6 (Chapter 9) pays 50, 50 and 1,050 and yields 6 per cent, with price 973.27. The present values of its payments are 47.17, 44.50 and 881.60, so

$$D_{\text{Mac}} = \frac{1 \times 47.17 + 2 \times 44.50 + 3 \times 881.60}{973.27} = 2.857 \text{ years}, \quad D_{\text{mod}} = \frac{2.857}{1.06} = 2.696.$$

A one-point rise in yield is predicted to lower the price by about 2.70 per cent, to 947.0. The exact price at 7 per cent is 947.51, a fall of 2.65 per cent.

DV01. Traders measure risk by the *dollar value of a basis point* (DV01): the change in price for a one-basis-point change in yield, $D_{\text{mod}} \times P \times 0.0001$. For the bond above, it is 0.26 per 1,000 of face value.

33.4.2 Convexity

The duration approximation is linear, but the price–yield relationship curves: bond prices are convex in yields (Chapter 5). Adding the second-order term,

$$\frac{\Delta P}{P} \approx -D_{\text{mod}} \Delta y + \frac{1}{2} C (\Delta y)^2, \quad C = \frac{1}{P} \frac{d^2 P}{dy^2}.$$

For the bond above, $C = 10.0$, and the approximation for a one-point rise becomes $-2.696 + 0.5 \times 10.0 \times 0.0001 \times 100 = -2.646$ per cent, essentially the exact answer. Convexity benefits holders of ordinary bonds: prices rise more when yields fall than they fall when yields rise by the same amount. It matters most for long bonds and large yield changes.

33.4.3 Long bonds and leverage

Duration explains why long bonds are so risky. A 30-year zero-coupon bond has a duration of 30: a one-point rise in its yield lowers its price by roughly 25 to 30 per cent. A bank that funds long-duration assets with short-duration deposits bears this risk on its balance sheet (Chapter 30's Silicon Valley Bank case); an investor who buys long bonds with borrowed money multiplies it.

33.4.4 Immunization

A pension fund or insurer with a fixed liability at a future date can protect itself against changes in interest rates by holding assets whose present value equals the liability's and whose duration matches the liability's. Then a small parallel shift in yields changes assets and liabilities by the same amount: the position is **immunized**. For example, a liability of 1,000 due in 5 years has a duration of 5; holding bonds worth its present value with a portfolio duration of 5 hedges it. Two cautions apply. Durations drift as time passes and yields move, so the portfolio must be rebalanced. And duration matching protects against small *parallel* shifts; changes in the *shape* of the curve require matching exposures at several maturities (key-rate durations) or cash-flow matching.

33.5 The Term Structure

Why do spot rates differ by maturity? Three theories each capture part of the answer.

The expectations hypothesis. If investors care only about expected returns, the n -year yield must equal (approximately) the average of the short rates expected over the next n years, so that holding a long bond and rolling over short ones offer the same expected return. Forward rates are then forecasts of future short rates. An upward-sloping curve signals that rates are expected to rise; an inverted curve that they are expected to fall.

The term premium. Long bonds are riskier than short ones for investors who might need to sell before maturity, because their prices move more when rates change (duration). If investors require compensation for this risk, long yields exceed the average of expected short rates by a **term premium**:

$$y_{n,t} \approx \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{E}_t i_{1,t+k} + TP_{n,t}.$$

Estimates suggest that term premia vary over time—they were high in the 1980s, fell to around zero or below in the 2010s partly because of central-bank bond purchases (Chapter 51), and rose again from 2022—so forward rates are biased forecasts of future short rates.

Segmentation and preferred habitat. Some investors have strong preferences for particular maturities—pension funds and insurers for long bonds that match their liabilities, banks and money-market funds for short ones—so the supply of and demand for bonds at each maturity affect yields there. This view explains why central-bank purchases concentrated in long bonds, and changes in government debt management, can move long yields.

Chapter 49 uses these ideas to explain how central banks influence long-term rates by shaping expectations of future policy.

33.6 Credit Risk

33.6.1 Credit spreads

A corporate bond's yield exceeds that of a government bond of the same maturity by its **credit spread**. In a simple one-period model, an investor who expects a default probability p and a loss given default LGD (one minus the recovery rate) requires, if risk-neutral, a spread of approximately

$$\text{spread} \approx p \times \text{LGD}.$$

If a five-year corporate bond yields 6.2 per cent and the government bond 4.2 per cent, the spread of 200 basis points is consistent, with an LGD of 60 per cent, with an annual default probability of about 3.3 per cent. Observed spreads, however, are typically several times what historical default losses would justify, especially for high-grade bonds: spreads also compensate for the tendency of defaults to cluster in recessions (a risk that cannot be diversified away, Chapter 14), for lower liquidity and for taxes. Spreads widen sharply in crises.

33.6.2 Ratings

Credit-rating agencies (S&P Global, Moody's, Fitch) classify issuers from AAA down to D. Bonds rated BBB– (Baa3) or above are *investment grade*; below that, *high yield* or speculative. Historical default rates rise steeply as ratings fall. Ratings migrate over time, and agencies publish *transition matrices* giving the probability of moving from each rating to each other in a year—Markov chains of the kind described in Chapter 8. Ratings matter beyond information because regulations and investment mandates refer to them; a downgrade below investment

grade can force sales. Their failure to anticipate losses on mortgage securities before 2007 is discussed in Chapter 51. Chapter 39 develops credit analysis in detail.

33.7 Other Instruments

- **Inflation-linked bonds** (U.S. TIPS, U.K. index-linked gilts) pay coupons and principal indexed to a price index, so they pay a real yield. The difference between nominal and real yields is the *breakeven inflation rate* (Chapter 50).
- **Floating-rate notes** pay coupons that reset periodically to a reference rate plus a fixed spread (Chapter 49). Because the coupon keeps pace with market rates, their price stays near par and their duration is short—roughly the time to the next reset.
- **Callable bonds** give the issuer the right to repay early, which it exercises when rates fall. The investor loses the price gains that falling rates would bring, so a callable bond's price curve flattens at low yields: *negative convexity*. Mortgage-backed securities behave similarly, because homeowners refinance when rates fall.

Debt capital markets. Governments issue bonds through regular auctions; companies usually issue through investment banks that underwrite and distribute the bonds to investors, with a prospectus setting out the terms and covenants (Chapter 30). Many firms borrow instead from banks or, increasingly, from private credit funds (Chapter 39).

Case: The 2022 gilt crisis and liability-driven investment

The strategy. British defined-benefit pension schemes promise pensions far into the future, so the present value of their liabilities rises sharply when long-term interest rates fall. Many adopted *liability-driven investment* (LDI): holding long-dated gilts and interest-rate swaps to match the duration of their liabilities. To keep money in higher-returning assets such as equities, many used leverage—borrowing through repurchase agreements and derivatives—to obtain more interest-rate exposure than their capital alone would buy. With leverage, a pension scheme could hedge most of its interest-rate risk using only part of its assets.

The shock. After the government's "mini-budget" of 23 September 2022, long gilt yields rose from around 3.7 per cent to around 5 per cent within days. For a long bond with a duration of about 20, a rise of that size cuts the price by roughly a fifth. The schemes' liabilities also fell in value, so their funding positions were not necessarily worse; but their leveraged hedges required immediate cash collateral to cover the losses on repos and swaps. To raise it, LDI funds sold gilts, pushing yields higher and triggering more collateral calls—a fire-sale spiral of the kind described in Chapter 51.

The response. On 28 September the Bank of England announced temporary purchases of long-dated gilts on whatever scale was needed to restore orderly markets, explicitly as a financial-stability operation rather than monetary policy. Yields fell back, and the Bank bought about 19 billion pounds of gilts before the programme ended on 14 October. The government reversed most of the tax cuts, and the prime minister resigned shortly afterwards. Regulators later required LDI funds to hold much larger buffers against yield rises.

In terms of this chapter. The crisis combined long duration, leverage and liquidity. Duration made a one-point yield move large in price terms; leverage turned price losses into cash demands; and the need for cash forced sales into a falling market. Immunization had worked on paper—liabilities fell with assets—but a hedge that requires cash when rates

rise can fail through liquidity rather than solvency, as depositors' runs do for banks (Chapter 48).

Summary

- Bonds are quoted at clean prices; buyers pay the dirty price, including accrued interest.
- Discount factors $d_t = (1 + s_t)^{-t}$ define spot rates; bootstrapping extracts them from coupon bond prices; forward rates satisfy $1 + f_{t-1,t} = d_{t-1}/d_t$.
- Yield to maturity is a bond's IRR, not its realized return; par yields are coupons that price bonds at par.
- Modified duration $D_{\text{Mac}}/(1 + y)$ gives $\Delta P/P \approx -D_{\text{mod}}\Delta y$; convexity adds $\frac{1}{2}C(\Delta y)^2$. Long bonds are very rate-sensitive, and leverage multiplies the sensitivity.
- Matching present values and durations immunizes against small parallel shifts, with rebalancing.
- The term structure reflects expected future short rates, time-varying term premia and maturity-specific demand and supply.
- Credit spreads compensate for expected default losses ($p \times \text{LGD}$) and for systematic default risk and illiquidity. Ratings summarize credit quality and migrate as Markov chains.
- Inflation-linked bonds pay real yields; floating-rate notes have short durations; callable bonds and mortgage securities have negative convexity.

Key terms

accrued interest; day-count convention; clean price; dirty price; zero-coupon bond; discount factor; spot rate; term structure; yield curve; bootstrapping; forward rate; yield to maturity; par yield; modified duration; Macaulay duration; DV01; convexity; immunization; key-rate duration; expectations hypothesis; term premium; preferred habitat; credit spread; loss given default; recovery rate; credit rating; investment grade; high yield; transition matrix; inflation-linked bond; floating-rate note; callable bond; negative convexity; liability-driven investment.

Exercises

Exercise 33.1 (Accrued interest). A bond with face value 100 pays a 4.5 per cent coupon semi-annually. It is bought two months after the last coupon at a clean price of 98.40. Find the accrued interest and the dirty price (use months as the day count).

Exercise 33.2 (Bootstrapping). One-year and two-year zero-coupon bonds trade at 96.15 and 91.57 per 100 face value. A three-year bond with a 6 per cent annual coupon trades at 102.30. Find the three spot rates and the forward rates for years 2 and 3.

Exercise 33.3 (Pricing off the curve). Using the spot rates of Example 33.1, price a three-year bond with a 3 per cent annual coupon. Is its yield to maturity above or below that of the 5 per cent bond? Why?

Exercise 33.4 (Duration). Compute the Macaulay and modified duration of a two-year bond with an 8 per cent annual coupon at a yield of 10 per cent (price 965.29 per 1,000). Estimate the price change for a yield fall of 50 basis points.

Exercise 33.5 (Long bonds). Compare the price change of a 2-year and a 30-year zero-coupon bond, each yielding 4 per cent, when yields rise to 5 per cent. Use exact prices.

Exercise 33.6 (Leverage). A fund holds 500 million of long gilts with modified duration 18, financed with 350 million of repo borrowing and 150 million of its own capital. Yields rise by 1.2 percentage points. Estimate the loss and the fall in the fund's capital. What happens if the repo lenders demand collateral equal to the loss within a day?

Exercise 33.7 (Immunization). An insurer must pay 10 million in 6 years; the yield curve is flat at 5 per cent. It can buy a 2-year zero and a 10-year zero. How much present value should it place in each to match present value and duration?

Exercise 33.8 (Credit spread). A four-year BBB bond yields 6.5 per cent; the government bond yields 4.0 per cent. With a recovery rate of 40 per cent, what annual default probability would a risk-neutral investor infer? Why might the true expected default probability be lower?

Exercise 33.9 (Term structure). One-year rates are 3 per cent now, and the two-year spot rate is 3.8 per cent. What forward rate is implied for year two? If the expectations hypothesis held, what would the market be forecasting? If investors demand a term premium of 0.5 points on two-year bonds, what is the expected one-year rate in a year?

Solutions

Solution 33.1. Semi-annual coupon 2.25; two of six months elapsed: accrued $2.25 \times 2/6 = 0.75$. Dirty price $98.40 + 0.75 = 99.15$.

Solution 33.2. $d_1 = 0.9615$, $s_1 = 4.00$ per cent; $d_2 = 0.9157$, $s_2 = 0.9157^{-1/2} - 1 = 4.50$ per cent. $102.30 = 6(0.9615 + 0.9157) + 106d_3$, so $d_3 = (102.30 - 11.263)/106 = 0.85884$, $s_3 = 5.20$ per cent. Forwards: $f_{1,2} = 0.9615/0.9157 - 1 = 5.00$ per cent; $f_{2,3} = 0.9157/0.85884 - 1 = 6.62$ per cent.

Solution 33.3. $3(0.970874 + 0.929005) + 103 \times 0.873339 = 5.700 + 89.954 = 95.65$. Its yield to maturity is slightly higher than the 5 per cent bond's: with a lower coupon, more of its value comes from the final payment, discounted at the highest (three-year) spot rate, so its blend of spot rates is weighted toward 4.62 per cent.

Solution 33.4. PVs: $80/1.1 = 72.73$, $1,080/1.21 = 892.56$. $D_{\text{Mac}} = (72.73 + 2 \times 892.56)/965.29 = 1.925$; $D_{\text{mod}} = 1.750$. Price change $\approx +1.750 \times 0.5 = +0.875$ per cent, about +8.45.

Solution 33.5. 2-year: $100/1.04^2 = 92.46$ to $100/1.05^2 = 90.70$: -1.9 per cent. 30-year: $100/1.04^{30} = 30.83$ to $100/1.05^{30} = 23.14$: -24.9 per cent.

Solution 33.6. Loss $\approx 18 \times 0.012 \times 500 = 108$ million (convexity would reduce it slightly). Capital falls from 150 to 42 million, a 72 per cent loss. If collateral of 108 million is demanded within a day, the fund must sell about a fifth of its gilts or find cash elsewhere; if many funds sell at once, prices fall further, raising losses and collateral calls—the spiral of the case.

Solution 33.7. PV of liability $= 10/1.05^6 = 7.462$ million, duration 6. Weights w (2-year) and $1 - w$ (10-year): $2w + 10(1 - w) = 6$, $w = 0.5$. Place 3.731 million in each zero (face values $3.731 \times 1.05^2 = 4.113$ and $3.731 \times 1.05^{10} = 6.077$ million).

Solution 33.8. Spread 2.5 per cent; LGD 60 per cent: $p \approx 0.025/0.6 = 4.2$ per cent a year. The true expected probability is likely lower because the spread also compensates for bearing default risk that rises in recessions, for illiquidity and for taxes.

Solution 33.9. $f_{1,2} = 1.038^2 / 1.03 - 1 = 4.60$ per cent. Under the expectations hypothesis, the market expects the one-year rate to be 4.6 per cent next year. With a term premium of 0.5 points in the two-year yield, the expected average of short rates is 3.3 per cent, so the expected one-year rate next year is about $2 \times 3.3 - 3 = 3.6$ per cent.

Risk, Return and Asset Pricing

After studying this chapter you should be able to

1. Measure returns and excess returns, and describe the historical rewards to bearing risk.
2. Construct the efficient frontier with a safe asset, find the tangency portfolio, and explain the separation theorem.
3. Derive the capital asset pricing model from the conditions for an optimal portfolio, and use beta to compute required returns.
4. Estimate beta by regression and judge the precision of the estimate.
5. Describe multifactor models and the arbitrage pricing theory, and the evidence that led to them.
6. Distinguish the forms of market efficiency, run a simple event study, and explain the behavioural challenge and the limits to arbitrage.
7. Compare value at risk with expected shortfall, and evaluate investment performance while allowing for luck.

A motivating question

In 1976 a small American fund company launched a mutual fund that made no attempt to pick winning shares. It simply bought every share in the Standard & Poor's 500 index in proportion to its market value. Critics called it un-American and a recipe for mediocrity; its launch raised a fraction of what its sponsors had hoped. Half a century later, funds of this kind manage trillions of dollars, and most professional managers who try to beat them fail to do so over long periods.

Why should doing nothing beat trying hard? The answer lies in how financial markets price risk. Chapter 14 showed that diversification removes risk specific to individual assets but not the risk they share. This chapter builds on that result. It asks which risks investors are paid to bear, derives the most widely used model of required returns, examines the evidence for and against it, and shows how to tell skill from luck in investment performance. The required returns derived here become the discount rates of Chapter 35.

34.1 Returns and the Reward for Risk

34.1.1 Measuring returns

The return on an asset over a period combines income and the change in price. For a share bought at P_0 , paying a dividend D_1 and sold at P_1 , the **holding-period return** is

$$R = \frac{D_1 + P_1 - P_0}{P_0} = \underbrace{\frac{D_1}{P_0}}_{\text{dividend yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{capital gain}}.$$

The **excess return** is the return minus the return on a safe asset over the same period, $R - r_f$; its expected value is the asset's **risk premium**. Returns over several periods compound: the average annual return that an investor actually earns is the geometric mean, which falls short of the arithmetic mean by roughly half the variance (Chapter 13). The arithmetic mean is the right estimate of the expected return for a single future period; the geometric mean describes realized growth.

34.1.2 The historical record

Long historical studies, such as the international database of Elroy Dimson, Paul Marsh and Mike Staunton, find that over the twentieth and early twenty-first centuries shares earned substantially more than government bills in almost every country with continuous markets: a world average of the order of 4 to 5 per cent a year in geometric terms, with the United States somewhat higher. Government bonds earned less than shares and more than bills. The premium came with risk: shares lost more than half their real value at least once in many countries, and there were periods of a decade or more in which they underperformed bills.

Two cautions attach to such figures. First, averages over a century are estimated imprecisely: with an annual standard deviation of about 20 per cent, the standard error of a 100-year mean is $20/\sqrt{100} = 2$ percentage points (Chapter 15). Second, the countries whose records survive are, in part, the ones whose markets did well; markets that were closed by war or revolution are harder to include. The historical premium is evidence about the future premium, not a measurement of it.

34.2 Optimal Portfolios

34.2.1 The efficient frontier

Chapter 14 computed the expected return $\mathbf{w}'\boldsymbol{\mu}$ and variance $\mathbf{w}'\boldsymbol{\Sigma}\mathbf{w}$ of a portfolio with weights $\mathbf{w}$. Plotting every attainable combination of standard deviation and expected return traces a region whose upper-left boundary is the **efficient frontier**: the portfolios with the highest expected return for each level of risk. Its leftmost point is the minimum-variance portfolio. Portfolios below the frontier are *inefficient*: another portfolio has the same risk and a higher expected return.

34.2.2 Adding a safe asset

Now let the investor also lend or borrow at the safe rate r_f . Combining the safe asset with a risky portfolio P , with a fraction w in P , gives expected return $r_f + w(\mu_P - r_f)$ and standard deviation $w\sigma_P$. Eliminating w :

$$\mathbb{E}[R] = r_f + \frac{\mu_P - r_f}{\sigma_P} \sigma.$$

The combinations lie on a straight line from the safe asset whose slope is the **Sharpe ratio** of P ,

$$S_P = \frac{\mu_P - r_f}{\sigma_P},$$

the expected excess return per unit of risk. Every investor, whatever her risk aversion, prefers the line with the steepest slope, because it offers more return for any given risk. The steepest line touches the frontier at the **tangency portfolio** T : the risky portfolio with the highest Sharpe ratio.

Proposition 34.1. Separation

If investors care about the mean and variance of their returns and can lend and borrow at r_f , all of them hold the same risky portfolio T , combined with the safe asset. Risk aversion determines only how much is placed in T and how much in the safe asset, not the composition of T .

The choice of *which* risky assets to hold is thus separated from the choice of *how much* risk to take. The second choice is the risky-share problem of Example 14.6 in Chapter 14, with T as the risky fund.

34.2.3 Finding the tangency portfolio

Write $\tilde{\mu} = \mu - r_f \mathbf{1}$ for the vector of expected excess returns. Because the weights of a fully invested portfolio sum to one, its expected excess return is $\mathbf{w}'\tilde{\mu}$, and its Sharpe ratio is

$$S(\mathbf{w}) = \frac{\mathbf{w}'\tilde{\mu}}{(\mathbf{w}'\Sigma\mathbf{w})^{1/2}}.$$

Two facts are needed to maximize it. First, S is homogeneous of degree zero in $\mathbf{w}$ (Definition 5.13): doubling every weight doubles the numerator and the denominator. So we may maximize without the constraint that weights sum to one and rescale at the end. Second, differentiating the double sum $\sigma_p^2 = \sum_i \sum_j w_i w_j \sigma_{ij}$ with respect to w_i gives

$$\frac{\partial \sigma_p^2}{\partial w_i} = 2 \sum_j \sigma_{ij} w_j = 2 \text{Cov}(R_i, R_p) :$$

an asset's marginal contribution to portfolio variance is twice its covariance with the portfolio, not its own variance.

Proposition 34.2. The tangency condition

At the tangency portfolio T , every asset's expected excess return is proportional to its covariance with T :

$$\mu_i - r_f = \frac{\mu_T - r_f}{\sigma_T^2} \text{Cov}(R_i, R_T) \quad \text{for every } i,$$

and the weights solve $\Sigma\mathbf{w} \propto \tilde{\mu}$, that is, $\mathbf{w}_T \propto \Sigma^{-1}\tilde{\mu}$.

Proof. Write $\sigma = (\mathbf{w}'\Sigma\mathbf{w})^{1/2}$. By the quotient rule and the second fact, the partial derivative of S with respect to w_i is

$$\frac{\tilde{\mu}_i}{\sigma} - \frac{(\mathbf{w}'\tilde{\mu}) \text{Cov}(R_i, R_p)}{\sigma^3}.$$

Setting it to zero for every i gives $\tilde{\mu}_i = (\mathbf{w}'\tilde{\mu}/\sigma^2) \text{Cov}(R_i, R_p)$. At the maximum, $p = T$, and $\mathbf{w}'\tilde{\mu}/\sigma^2 = (\mu_T - r_f)/\sigma_T^2$. In vector form the conditions say $\tilde{\mu}$ is a multiple of $\Sigma\mathbf{w}$. $\square$

The condition has an economic reading. If some asset offered more expected excess return per unit of its contribution to portfolio risk than the others, shifting a little money into it would raise the Sharpe ratio; at the optimum, no such shift helps. It is the equal-marginal-benefit-per-unit-cost rule of Chapter 6 applied to risk.

Worked Example 34.3. A tangency portfolio

A defensive fund A has expected return 6 per cent and standard deviation 10 per cent; a growth fund B has 11 per cent and 20 per cent; their returns are uncorrelated; $r_f = 3$ per cent. With zero correlation, Σ is diagonal and $w_i \propto \tilde{\mu}_i/\sigma_i^2$: $0.03/0.01 = 3$ for A and $0.08/0.04 = 2$ for B . Rescaling, $\mathbf{w}_T = (0.6, 0.4)$. Then

$$\mu_T = 0.6 \times 6 + 0.4 \times 11 = 8\%, \quad \sigma_T = \sqrt{0.36 \times 0.01 + 0.16 \times 0.04} = 10\%, \quad S_T = 0.5,$$

higher than the Sharpe ratios of A (0.3) and B (0.4) alone. An investor who wants 6 per cent volatility should place 60 per cent in T and 40 per cent in the safe asset, for an expected return of $3 + 0.6 \times 5 = 6$ per cent. Fund A alone offers the same expected return with 10 per cent volatility: it is inefficient.

► Optimizers amplify errors

Theorem 34.2 takes μ and Σ as known. In practice they are estimated, and expected returns especially are estimated with large errors. An optimizer treats every estimate as exact and piles into the assets whose expected returns happen to be overestimated, producing extreme weights that perform poorly out of sample. Practitioners constrain the optimization (no short sales, position limits, sector limits—a linear or quadratic programme, Chapter 6), shrink estimates toward simple benchmarks (Chapter 19), or start from the weights implied by market values, as the next section suggests.

34.3 The Capital Asset Pricing Model

34.3.1 From portfolio choice to equilibrium

Theorem 34.2 describes one investor's optimal portfolio. The **capital asset pricing model** (CAPM), developed by William Sharpe (1964), John Lintner (1965) and Jan Mossin (1966), asks what happens when *every* investor behaves this way. Suppose investors care about mean and variance, agree about μ and Σ , and can lend and borrow at r_f . By Theorem 34.1 they all hold the same risky portfolio T . But the shares that investors hold in aggregate are, by definition, all the shares in existence, in proportion to their market values. In equilibrium, therefore, T must be the **market portfolio** M . Substituting M for T in Theorem 34.2 and defining the asset's **beta**,

$$\beta_i = \frac{\text{Cov}(R_i, R_M)}{\text{Var}(R_M)},$$

gives the model.

Proposition 34.4. Capital asset pricing model

Under the assumptions above, the expected return on every asset satisfies

$$\mathbb{E}[R_i] = r_f + \beta_i (\mathbb{E}[R_M] - r_f).$$

An asset's risk premium equals its beta times the **market risk premium**.

The model makes precise the intuition of Theorem 14.4. Total risk does not matter; only the part of an asset's risk that moves with the market—the risk that diversified investors cannot shed—is rewarded. The market itself has $\beta = 1$; the safe asset has $\beta = 0$. A share with $\beta = 1.5$ tends to rise 1.5 per cent when the market rises 1 per cent and demands a risk premium 1.5 times the market's. A share whose returns are highly volatile but uncorrelated with the market has $\beta = 0$ and should earn only the safe rate: its risk can be diversified away. Because covariance is linear, the beta of a portfolio is the weighted average of the betas of its holdings.

Worked Example 34.5. A required return

A share's returns have covariance 0.06 with the market's, and the market's variance is 0.04 (a standard deviation of 20 per cent). Its beta is $0.06/0.04 = 1.5$. With $r_f = 3$ per cent and a market risk premium of 5 per cent, the CAPM required return is $3 + 1.5 \times 5 = 10.5$ per cent. A portfolio holding 60 per cent in this share and 40 per cent in bills has beta $0.6 \times 1.5 = 0.9$ and required return $3 + 0.9 \times 5 = 7.5$ per cent.

In Example 34.3, T plays the role of M : $\text{Cov}(R_A, R_T) = 0.6 \times 0.01 = 0.006$, so $\beta_A = 0.006/0.01 = 0.6$, and $\beta_B = 0.4 \times 0.04/0.01 = 1.6$. The CAPM gives $3 + 0.6 \times 5 = 6$ and $3 + 1.6 \times 5 = 11$ per cent: exactly the funds' expected returns, as Theorem 34.2 requires.

34.3.2 The security market line and alpha

Plotting expected return against beta, the CAPM says every asset lies on a straight line, the **security market line** (SML), with intercept r_f and slope equal to the market risk premium. An asset expected to return more than the SML predicts has positive **alpha**,

$$\alpha_i = \mathbb{E}[R_i] - [r_f + \beta_i(\mathbb{E}[R_M] - r_f)],$$

and would be underpriced: investors buying it would push up its price and reduce its expected return until $\alpha_i = 0$. Figure 34.1 contrasts the SML with the capital market line of the previous section. The capital market line relates expected return to *total* risk and contains only efficient portfolios; the SML relates it to *systematic* risk and contains every asset.

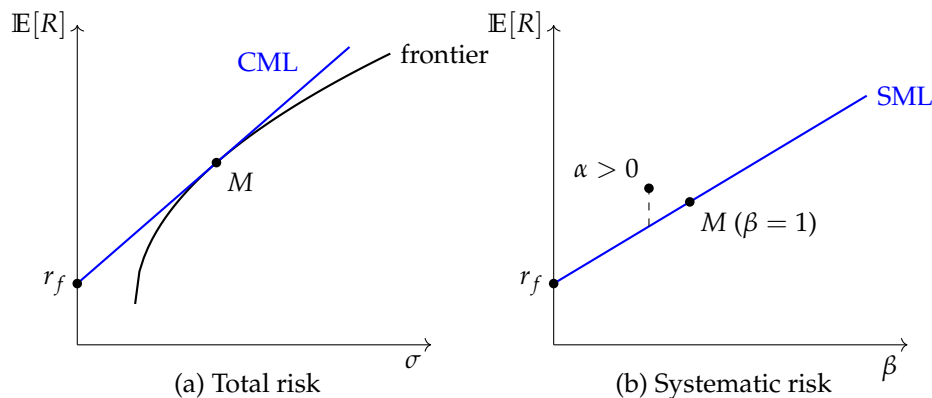


Figure 34.1: The capital market line and the security market line. Efficient portfolios lie on the CML; under the CAPM every asset lies on the SML, and an asset above it has positive alpha.

34.3.3 Estimating beta

Beta is estimated by regressing an asset's excess returns on the market's (Chapter 16):

$$R_{it} - r_{ft} = \alpha_i + \beta_i(R_{Mt} - r_{ft}) + \varepsilon_{it}.$$

The OLS slope is the sample covariance over the sample variance: beta's definition, estimated (Theorem 16.1). The regression also splits the asset's variance into a systematic part and an idiosyncratic part, $\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma_\varepsilon^2$, and its R^2 is the systematic share. Practitioners typically use 60 monthly or 104 weekly returns against a broad index.

Worked Example 34.6. How precise is a beta?

A share has an annual standard deviation of 40 per cent and a true beta of 1.5; the market's standard deviation is 20 per cent. The systematic variance is $1.5^2 \times 0.04 = 0.09$ of a total of 0.16, so $R^2 = 0.56$ and the residual variance is 0.07. With 60 monthly observations, the standard error of the slope is approximately $\sigma_\varepsilon / (\sigma_M \sqrt{T})$ in monthly units:

$$\frac{\sqrt{0.07/12}}{(0.20/\sqrt{12})\sqrt{60}} = \frac{0.0764}{0.0577 \times 7.75} = 0.17.$$

The 95 per cent confidence interval is about 1.17 to 1.83, and with $r_f = 3$ and a market premium of 5 per cent, the implied cost of equity runs from 8.8 to 12.2 per cent. Five years of data pin beta down only roughly.

Estimated betas are noisy, and extreme estimates are more likely to be extreme by chance. Because betas across firms average about one, a common adjustment shrinks each estimate toward one, for example $\beta_{\text{adj}} = \frac{2}{3}\hat{\beta} + \frac{1}{3}$, an application of the shrinkage idea of Chapter 19. Industry betas, averaging over many firms, are more precise than individual ones; Chapter 35 shows how to use them. Factor structure also helps with the estimation problem of Chapter 14: if every covariance runs through the market, $\sigma_{ij} = \beta_i \beta_j \sigma_M^2$, a 500-share covariance matrix needs $2 \times 500 + 1 = 1,001$ parameters (betas, residual variances and the market variance) rather than more than 125,000.

Advanced Analysis: Consumption, the stochastic discount factor and the equity premium puzzle

The CAPM is a special case of a more general principle. Extend the saving problem of Chapter 9 to a household that can buy an asset at price p paying an uncertain x next period. Giving up p units of consumption now costs $p u'(c_0)$; the payoff brings $\beta \mathbb{E}[u'(c_1) x]$. At the optimum they are equal, so

$$p = \mathbb{E}[m x], \quad m = \beta \frac{u'(c_1)}{u'(c_0)},$$

where m is the *stochastic discount factor*. For a return $R = x/p$, $\mathbb{E}[mR] = 1$, and a little algebra gives $\mathbb{E}[R] - r_f = -(1 + r_f) \text{Cov}(m, R)$. Because u' falls as consumption rises, an asset that pays off in good times (low m) has negative covariance with m and must offer a premium; an asset that pays off in bad times, like insurance, can offer a low or negative one. Risk is priced by what it does in the states in which consumption is scarce. The CAPM follows if m is linear in the market return.

Rajnish Mehra and Edward Prescott (1985) confronted this theory with the data. Aggregate consumption in the United States is smooth, so its covariance with share returns is small. Explaining a historical equity premium of about 6 per cent then requires a

coefficient of relative risk aversion far above the values that seem plausible from individual behaviour—the *equity premium puzzle*. Proposed resolutions include rare disasters, habit formation, long-run consumption risk and limited participation; none commands consensus.

34.4 Testing the CAPM: Factors and Anomalies

34.4.1 The evidence

The CAPM's prediction is sharp: expected returns rise linearly with beta, with slope equal to the market risk premium, and nothing else matters. Early tests, such as those of Fischer Black, Michael Jensen and Myron Scholes (1972), found that returns did rise with beta, but less steeply than the model predicts: low-beta shares earned more, and high-beta shares less, than the SML implies. Eugene Fama and Kenneth French (1992) found that, once firm size and the ratio of book value to market value were accounted for, beta had little power to explain average US returns from 1963 to 1990. Small firms and *value* firms (high book-to-market) had earned more than the CAPM predicted. Narasimhan Jegadeesh and Sheridan Titman (1993) documented *momentum*: shares that had done well over the previous three to twelve months continued to outperform over the following months.

A caution runs through all such tests. Richard Roll (1977) pointed out that the true market portfolio includes every risky asset—property, private businesses, human capital—and cannot be observed. A test that uses a share index as a proxy tests the CAPM and the proxy jointly.

34.4.2 Multifactor models

The response was to add factors. In the **Fama–French three-factor model**, an asset's excess return depends on its exposures to the market, to a size factor (SMB, the return on small firms minus big firms) and to a value factor (HML, high minus low book-to-market):

$$E[R_i] - r_f = \beta_{iM} \lambda_M + \beta_{i,SMB} \lambda_{SMB} + \beta_{i,HML} \lambda_{HML},$$

where each β is a coefficient from a multiple regression of the asset's excess returns on the factor returns and each λ is the factor's average return. Mark Carhart (1997) added momentum; Fama and French (2015) added profitability and investment factors.

Worked Example 34.7. A factor-model required return

A share has loadings $\beta_M = 1.1$, $\beta_{SMB} = 0.4$ and $\beta_{HML} = -0.2$ (a smallish growth firm). With illustrative factor premia of 5, 2 and 3 per cent and $r_f = 3$ per cent, its required return is $3 + 1.1 \times 5 + 0.4 \times 2 - 0.2 \times 3 = 3 + 5.5 + 0.8 - 0.6 = 8.7$ per cent. The single-factor CAPM would give $3 + 1.1 \times 5 = 8.5$ per cent. For this share the models nearly agree; for a small value firm they can differ by several points.

The theoretical basis for multiple factors is the **arbitrage pricing theory** (APT) of Stephen Ross (1976). If returns are driven by a few common factors plus idiosyncratic noise, a well-diversified portfolio has almost no idiosyncratic risk. Two such portfolios with the same factor exposures must then offer the same expected return, or investors could buy one and sell the other for a nearly riskless profit. Absence of such arbitrage implies that expected excess returns are linear in factor exposures, as in the equation above. The APT does not say which factors are priced or how large their premia are; those are empirical questions.

Whether the size, value and momentum premia are rewards for risk or the results of mispricing is disputed. Value firms may be riskier in ways beta misses (they tend to be distressed and inflexible); or investors may extrapolate past growth, overpricing glamorous growth firms. The size premium has been weak in US data since its discovery, and the value premium was negative for much of the 2010s. The factor-zoo case of Chapter 15 explains why many published factors are likely to be false discoveries.

34.5 Market Efficiency and Its Critics

34.5.1 The efficient markets hypothesis

A market is **informationally efficient** if prices fully reflect available information, so that no investor can earn abnormal returns, adjusted for risk, by trading on that information. Fama (1970) distinguished three forms by the information set:

- **Weak form:** prices reflect all information in past prices. Technical trading rules should not earn abnormal returns.
- **Semi-strong form:** prices reflect all public information, including accounts, announcements and news. Fundamental analysis of public data should not earn abnormal returns.
- **Strong form:** prices reflect all information, including private information. Even insiders should not profit—a form that evidence on insider trading rejects.

Efficiency does not mean that prices are always right, only that errors are not predictable from the information set. If prices react to news, and news is by definition unpredictable, price changes should be close to unpredictable: prices should behave roughly like a random walk (Chapter 18), with a drift equal to the expected return.

The joint-hypothesis problem. Abnormal returns are defined relative to a model of normal returns, such as the CAPM. A test that finds abnormal returns may therefore be evidence against efficiency or against the model of expected returns; the two cannot be tested separately. This is why the factor debate above is also a debate about efficiency.

The Grossman–Stiglitz paradox. If prices already reflected all information, nobody would be paid to gather it, and then prices could not reflect it. Sanford Grossman and Joseph Stiglitz (1980) concluded that markets must be inefficient enough to reward those who acquire information—but perhaps only by enough to cover their costs.

34.5.2 Event studies

The semi-strong form is tested by **event studies**, which measure how prices respond to announcements. The **abnormal return** on a day is the actual return minus the return predicted by a model estimated on earlier data, such as $\hat{\alpha} + \hat{\beta}R_{Mt}$. Averaging abnormal returns across many events, and cumulating them over days around the announcement, shows how quickly and fully prices adjust.

Worked Example 34.8. An abnormal return

On the day a firm announces a large contract, its shares rise 6 per cent and the market rises 2 per cent. Its estimated market model is $R = 0 + 1.2R_M$, with residual standard deviation 2 per cent a day. The expected return given the market is $1.2 \times 2 = 2.4$ per cent,

so the abnormal return is $6 - 2.4 = 3.6$ per cent, a t -statistic of $3.6/2 = 1.8$. For one firm the evidence is modest; averaged over hundreds of similar announcements, the noise shrinks with $\sqrt{n}$ and the average reaction can be measured precisely.

Event studies generally find that prices react within minutes to public announcements, with little predictable drift afterwards—strong support for semi-strong efficiency. Well-documented exceptions include the drift of prices in the direction of earnings surprises for weeks afterwards.

34.5.3 Behavioural finance and the limits to arbitrage

The behavioural critique makes two claims. First, many investors are not fully rational: they are overconfident, trade too much, hold losing shares too long and sell winners too soon (the *disposition effect*), extrapolate recent returns, and neglect base rates (Chapter 12). Robert Shiller (1981) argued that share prices move far more than subsequent dividends could justify. Second, and crucially, rational investors cannot always correct the resulting mispricing. Arbitrage is costly and risky: mispricing can widen before it narrows, and professional arbitrageurs, who manage other people's money, face withdrawals precisely when their positions move against them. Andrei Shleifer and Robert Vishny (1997) called these the **limits to arbitrage**.

The clearest evidence comes from cases where the correct relative price is known. Royal Dutch and Shell Transport, which shared the profits of one group in a fixed 60:40 ratio, traded for years at prices that deviated from that ratio, at times by more than 10 per cent. In March 2000, 3Com sold about 5 per cent of its subsidiary Palm to the public and announced that it would distribute the rest to its own shareholders, about 1.5 Palm shares per 3Com share. On Palm's first trading day its shares closed at a price that valued 3Com's stake in Palm at more than the whole of 3Com, implying, on Owen Lamont and Richard Thaler's calculation, that the rest of 3Com's business was worth roughly minus 22 billion dollars. Short-selling Palm, the obvious arbitrage, was difficult and expensive, and the anomaly persisted for about two months.

Key idea. Markets are efficient enough that beating them after costs is hard, and inefficient enough that prices sometimes stray far from any reasonable value. Both statements are consistent with the evidence. The practical implication for most investors is the same: diversify, keep costs low, and treat claims of consistent outperformance with scepticism.

34.6 Measuring Risk and Performance

34.6.1 Value at risk and expected shortfall

Chapter 13 defined the value at risk (VaR): the loss exceeded with a given probability over a given horizon. VaR has two weaknesses. It says nothing about the size of losses beyond the threshold. And it can penalize diversification.

Worked Example 34.9. VaR can punish diversification

A lender makes loans of 100, each defaulting with probability 4 per cent and a total loss if it does. For one loan, the probability of any loss is 4 per cent, below 5 per cent, so its 5 per cent VaR is zero. For two independent loans, the probability of at least one default is $1 - 0.96^2 = 7.84$ per cent, above 5 per cent, so the portfolio's VaR is 100. By this measure, the diversified portfolio (VaR 100) is riskier than the two loans held separately (VaR

$0 + 0$).

The **expected shortfall** (ES) at level α is the average loss in the worst α share of outcomes. For one loan, the worst 5 per cent of outcomes consist of the 4 per cent with a loss of 100 and 1 per cent with none: $ES = (0.04 \times 100)/0.05 = 80$. For two loans, the worst 5 per cent consist of both defaulting (probability 0.16 per cent, loss 200) and 4.84 per cent of the single-default outcomes (loss 100): $ES = (0.0016 \times 200 + 0.0484 \times 100)/0.05 = 103.2$, less than $80 + 80$. Expected shortfall is *subadditive*—the risk of a combination never exceeds the sum of the risks—and it looks into the tail. Partly for these reasons, the Basel Committee's revised market-risk framework (the Fundamental Review of the Trading Book) replaced 99 per cent VaR with 97.5 per cent expected shortfall for banks' trading books.

For a normal distribution of returns with mean μ and standard deviation σ , the expected shortfall at level α is $-\mu + \sigma \phi(z_\alpha)/\alpha$, where ϕ is the standard normal density and z_α the critical value. For the portfolio of Chapter 13, with monthly mean 0.5 per cent and standard deviation 4 per cent, the 5 per cent VaR was 6.08 per cent; the 5 per cent ES is $-0.5 + 4 \times 0.103/0.05 = 7.75$ per cent.

► Normal tails are too thin

Normal-based risk measures understate the probability of extreme losses, because actual returns have heavy tails and volatility clusters (Chapters 11 and 18). In August 2007, as quantitative funds suffered heavy losses, Goldman Sachs's chief financial officer said that the firm was seeing moves of 25 standard deviations several days in a row. Under a normal distribution a single 25-standard-deviation day has a probability below 10^{-137} ; such days should never occur in the life of the universe. The moves were not freakish; the model was wrong. Risk managers supplement normal models with historical simulation, heavy-tailed distributions and stress tests of named scenarios.

34.6.2 Evaluating performance

A fund's raw return says little on its own: a fund that takes more risk should earn more. Standard measures adjust for risk in different ways:

- The **Sharpe ratio** $(\bar{R}_p - r_f)/\sigma_p$ uses total risk. It is the right measure for a portfolio that is an investor's whole risky wealth.
- **Jensen's alpha** $\bar{R}_p - [r_f + \beta_p(\bar{R}_M - r_f)]$ is the intercept of the beta regression: the return beyond what the fund's market exposure explains. With factor models, alpha is the intercept of the multifactor regression.
- The **Treynor ratio** $(\bar{R}_p - r_f)/\beta_p$ uses systematic risk; it suits a fund that will be a small part of a diversified portfolio.
- The **information ratio** $\alpha_p/\sigma_\varepsilon$ divides alpha by the fund's **tracking error**—the standard deviation of its residual return—measuring alpha per unit of active risk.

Worked Example 34.10. Is the manager skilled?

Over a period in which the market returned 9 per cent with standard deviation 15 per cent and $r_f = 3$ per cent, a fund returned 11 per cent with standard deviation 18 per cent and beta 1.2. Its Sharpe ratio is $8/18 = 0.44$ against the market's $6/15 = 0.40$. Its alpha is $11 - (3 + 1.2 \times 6) = 0.8$ per cent a year. If its tracking error is 4 per cent, its information ratio is $0.8/4 = 0.2$. The t -statistic for alpha after T years is approximately

the information ratio times $\sqrt{T}$. To reach $t = 2$, the manager needs $T = (2/0.2)^2 = 100$ years of data. A decade of such performance is entirely consistent with luck.

Two further results temper enthusiasm for active management. The first is arithmetic.

Proposition 34.11. The arithmetic of active management

Let passive investors hold the market portfolio. Then, before costs, the value-weighted average return of active investors equals the market return; after costs, it is lower, by the difference in costs.

Proof. The market return is the value-weighted average of the returns of all investors' holdings, because together they hold the market. Passive investors earn the market return. A weighted average of two groups equals the whole only if the second group's average also equals the whole (Chapter 2). So active investors, as a group, earn the market return before costs. Their costs—fees, trading costs—are higher than passive investors'. $\square$

William Sharpe (1991), who stated the result, stressed that it requires no assumption about efficiency: it holds however skilled some active managers are, because every gain by one active investor relative to the market is another active investor's loss. The second result concerns costs over time. A difference of one percentage point a year in fees, compounded over 30 years at 7 per cent gross, reduces terminal wealth from $1.07^{30} = 7.61$ to $1.06^{30} = 5.74$ times the initial investment: a quarter of the final value.

Backtests. A strategy that looks profitable on historical data may have been found by searching. If a researcher tries a thousand variations of a trading rule with no true predictive power and tests each at the 5 per cent level, about fifty will look significant (Chapter 15). Credible backtests fix the rule before looking at the data, test it out of sample and in other markets, include realistic transaction costs, and report how many variations were tried.

Case: The index fund: from “folly” to the mainstream

The idea. In 1974 Paul Samuelson challenged the investment industry to show that it could beat the market after costs, and suggested that someone should launch a fund that simply tracked a broad index. John Bogle, who had founded the Vanguard Group in 1975 with a structure in which the funds' investors owned the management company, took up the suggestion. In 1976 Vanguard launched the First Index Investment Trust, tracking the S&P 500. Its initial offering raised only about 11 million dollars, far short of its target, and rival firms derided it as “Bogle's folly”.

What followed. The fund, later renamed the Vanguard 500 Index Fund, grew slowly and then rapidly, and index mutual funds and exchange-traded funds spread across markets and asset classes. Around the start of 2024, the assets of passive funds in the United States overtook those of actively managed funds, according to Morningstar. Meanwhile, S&P Dow Jones Indices' regular SPIVA reports have found, year after year, that most actively managed US large-company equity funds underperform the S&P 500 over long horizons—typically in the region of 85 to 95 per cent over fifteen years—and that funds in the top quartile in one period show little tendency to stay there.

In terms of this chapter. Theorem 34.11 explains the SPIVA results without any claim that managers lack skill: active managers as a group must earn the market return before costs, and their higher costs push the average below it. Example 34.10 explains why the few who outperform are hard to identify in advance. The CAPM gives the index fund its

theoretical justification—the market portfolio is the efficient risky portfolio—while the Grossman–Stiglitz paradox marks its limit: if nearly everyone indexed, no one would be paid to make prices informative, and active management would again become profitable. Where between these poles the market settles is one of the open questions in finance. The growth of index funds has also concentrated share ownership in a few large asset managers, raising questions of corporate governance and competition (Chapter 24).

Summary

- The holding-period return is dividend yield plus capital gain; the risk premium is the expected excess return over the safe rate. Historically, shares have earned a substantial premium over bills, estimated imprecisely.
- With a safe asset, every mean–variance investor holds the tangency portfolio, the risky portfolio with the highest Sharpe ratio, combined with lending or borrowing (separation).
- At the tangency portfolio, each asset's expected excess return is proportional to its covariance with the portfolio; the weights are proportional to $\Sigma^{-1}(\boldsymbol{\mu} - r_f \mathbf{1})$.
- In equilibrium the tangency portfolio is the market, giving the CAPM: $\mathbb{E}[R_i] = r_f + \beta_i(\mathbb{E}[R_M] - r_f)$, with $\beta_i = \text{Cov}(R_i, R_M) / \text{Var}(R_M)$. Only systematic risk is priced.
- Beta is estimated by regression, with substantial error; adjusted and industry betas help.
- Empirically, the SML is too flat, and size, value and momentum help explain returns; multifactor models and the APT price several factors.
- Efficient markets make prices unpredictable from available information; tests face the joint-hypothesis problem. Behavioural biases and the limits to arbitrage allow persistent mispricing.
- Expected shortfall, unlike VaR, is subadditive and measures the tail; normal models understate extreme losses.
- Performance must be judged against risk and luck: alpha is hard to distinguish from noise, active management as a whole cannot beat the market after costs, and backtests must allow for data mining.

Key terms

holding-period return; excess return; risk premium; efficient frontier; Sharpe ratio; tangency portfolio; capital market line; separation theorem; capital asset pricing model; market portfolio; beta; market risk premium; security market line; alpha; adjusted beta; stochastic discount factor; equity premium puzzle; size and value effects; momentum; Fama–French three-factor model; arbitrage pricing theory; informational efficiency; weak, semi-strong and strong forms; joint-hypothesis problem; event study; abnormal return; behavioural finance; disposition effect; limits to arbitrage; expected shortfall; subadditivity; Jensen's alpha; Treynor ratio; information ratio; tracking error; arithmetic of active management; backtest overfitting; index fund.

Exercises

Exercise 34.1 (Returns). A share bought at 50 pays a dividend of 1.50 and is sold a year later at 54. The one-year safe rate is 3 per cent. Find the dividend yield, capital gain, holding-period

return and excess return.

Exercise 34.2 (Tangency portfolio). Asset A has expected return 4 per cent and standard deviation 10 per cent; asset B has 10 per cent and 20 per cent; they are uncorrelated; $r_f = 2$ per cent. Find the tangency portfolio's weights, expected return, standard deviation and Sharpe ratio, and compare the Sharpe ratio with those of A and B .

Exercise 34.3 (CAPM). A share's covariance with the market is 0.048 and the market's standard deviation is 20 per cent. The safe rate is 4 per cent and the market risk premium 5.5 per cent. Find its beta and required return. Analysts expect it to return 12 per cent. What is its alpha, and what does the CAPM suggest an investor should do?

Exercise 34.4 (Portfolio beta). A portfolio holds 40 per cent in a share with beta 0.8, 35 per cent in a share with beta 1.3 and 25 per cent in Treasury bills. With $r_f = 3$ per cent and a market premium of 6 per cent, find the portfolio's beta and required return.

Exercise 34.5 (Precision of beta). A beta regression on 36 monthly returns gives $\hat{\beta} = 1.4$. The residual standard deviation is 8 per cent a month and the market's standard deviation 4.5 per cent a month. Find the approximate standard error and 95 per cent confidence interval for beta, and the adjusted beta $\frac{2}{3}\hat{\beta} + \frac{1}{3}$.

Exercise 34.6 (Factor model). A share has loadings 0.9 on the market, -0.3 on SMB and 0.6 on HML. With factor premia of 6, 2 and 3.5 per cent and $r_f = 3$ per cent, find its required return. What kind of firm is it likely to be?

Exercise 34.7 (Event study). On the day a firm announces a profit warning, its shares fall 8 per cent while the market falls 1 per cent. Its market model is $R = 0.02 + 0.9R_M$ (in per cent a day), with residual standard deviation 1.8 per cent. Find the abnormal return and its t -statistic. What would semi-strong efficiency predict about returns over the following days?

Exercise 34.8 (Performance measures). The market returned 10 per cent (standard deviation 16 per cent) and $r_f = 3$ per cent. Fund X returned 12 per cent with standard deviation 20 per cent and beta 1.4; fund Y returned 9 per cent with standard deviation 12 per cent and beta 0.7. Compute each fund's Sharpe ratio, alpha and Treynor ratio. Which would you prefer as your only investment, and which as a small addition to a diversified portfolio?

Exercise 34.9 (VaR and expected shortfall). A 10 million portfolio has normally distributed daily returns with mean zero and standard deviation 1.2 per cent. Find its one-day 1 per cent VaR and expected shortfall ($z = 2.326$, $\phi(2.326) = 0.0267$). Why might both understate the true risk?

Exercise 34.10 (Active management). Active investors hold 60 per cent of a market that returns 8 per cent in a year. Passive funds charge 0.05 per cent and active funds 1 per cent a year. What is the average net return of passive and of active investors? If one group of active managers, holding half of the actively managed money, earned 10 per cent before fees, what must the other half have earned?

Solutions

Solution 34.1. Dividend yield $1.5/50 = 3$ per cent; capital gain $4/50 = 8$ per cent; holding-period return 11 per cent; excess return 8 per cent.

Solution 34.2. $w_i \propto \tilde{\mu}_i/\sigma_i^2$: $0.02/0.01 = 2$ and $0.08/0.04 = 2$, so $\mathbf{w}_T = (0.5, 0.5)$. $\mu_T = 7$ per cent; $\sigma_T = \sqrt{0.25 \times 0.01 + 0.25 \times 0.04} = 11.18$ per cent; $S_T = 5/11.18 = 0.447$, above A 's 0.2 and B 's 0.4. Diversifying between uncorrelated assets raises the reward per unit of risk.

Solution 34.3. $\beta = 0.048/0.04 = 1.2$; required return $4 + 1.2 \times 5.5 = 10.6$ per cent. Alpha $= 12 - 10.6 = 1.4$ per cent: if the analysts are right, the share lies above the SML and is underpriced, so investors should buy it (tilting their portfolios toward it), which would raise its price until the alpha disappeared. The alpha is only as good as the forecast.

Solution 34.4. $\beta_p = 0.4 \times 0.8 + 0.35 \times 1.3 + 0.25 \times 0 = 0.32 + 0.455 = 0.775$; required return $3 + 0.775 \times 6 = 7.65$ per cent.

Solution 34.5. Standard error $\approx 0.08/(0.045\sqrt{36}) = 0.08/0.27 = 0.30$. The 95 per cent interval is $1.4 \pm 1.96 \times 0.30$, about 0.82 to 1.98. Adjusted beta $= 0.933 + 0.333 = 1.27$.

Solution 34.6. $3 + 0.9 \times 6 - 0.3 \times 2 + 0.6 \times 3.5 = 3 + 5.4 - 0.6 + 2.1 = 9.9$ per cent. The negative SMB loading suggests a large firm; the positive HML loading, a value firm (high book-to-market)—for example, a mature bank or utility.

Solution 34.7. Expected return $0.02 + 0.9 \times (-1) = -0.88$ per cent; abnormal return $-8 - (-0.88) = -7.12$ per cent; $t = -7.12/1.8 = -3.96$, clearly significant. Semi-strong efficiency predicts no predictable abnormal returns on the following days: the price should have adjusted fully on the announcement. (A continued drift would be evidence of underreaction.)

Solution 34.8. Sharpe: $X\ 9/20 = 0.45$, $Y\ 6/12 = 0.50$, market $7/16 = 0.44$. Alpha: $X\ 12 - (3 + 1.4 \times 7) = -0.8$ per cent; $Y\ 9 - (3 + 0.7 \times 7) = 1.1$ per cent. Treynor: $X\ 9/1.4 = 6.4$, $Y\ 6/0.7 = 8.6$, market 7. As a sole investment, compare Sharpe ratios: Y . As a small addition to a diversified portfolio, what matters is the return relative to systematic risk: Y again, with positive alpha; X 's high raw return is more than explained by its beta. (With these numbers the measures agree; they need not.)

Solution 34.9. VaR $= 2.326 \times 1.2 = 2.79$ per cent, 279,000. ES $= 1.2 \times 0.0267/0.01 = 3.20$ per cent, 320,000. Both assume normality and a stable volatility; actual daily returns have heavier tails and volatility clusters, so extreme losses are more likely than the normal model implies, especially in a crisis, when correlations also rise.

Solution 34.10. Passive: $8 - 0.05 = 7.95$ per cent. Active before fees: 8 per cent (Theorem 34.11); after fees 7 per cent. If half of active money earned 10 per cent before fees, the other half must have earned r with $0.5 \times 10 + 0.5r = 8$, so $r = 6$ per cent before fees, 5 per cent after.

The Cost of Capital, Financing and Payout

After studying this chapter you should be able to

1. Explain why the cost of capital is an opportunity cost that depends on the risk of the investment, not on who invests or how it is financed.
2. Estimate the cost of equity with the CAPM, choosing the safe rate, the market risk premium and beta with care.
3. Estimate the cost of debt as an expected return, not a promised yield, and compute its after-tax cost.
4. Compute the weighted average cost of capital with market-value weights, and explain how it varies with leverage.
5. Unlever and relever betas, and use divisional and project costs of capital.
6. Value financing side effects with adjusted present value.
7. Analyse payout policy: dividends versus buybacks, and why earnings-per-share effects mislead.

A motivating question

In April 2013, Apple, then holding about 145 billion dollars of cash and securities, borrowed 17 billion dollars in what was at the time the largest corporate bond issue ever made by a company outside banking. It used the money to help pay for share repurchases and dividends. Why would a company with so much cash borrow? And did borrowing to buy back shares make Apple's shareholders richer?

Answering requires the concepts of this chapter. Chapter 32 discounted project cash flows at a given rate; Chapter 34 explained how risk determines the return investors require. This chapter turns required returns into the discount rate a firm should use—its cost of capital—and asks how financing and payout decisions affect value. It builds on the Modigliani–Miller propositions of Chapter 30, which it assumes rather than repeats.

35.1 The Cost of Capital as an Opportunity Cost

The **cost of capital** for an investment is the expected return that investors could earn on other investments of the same risk. It is an opportunity cost (Chapter 1): by putting money into the project, investors forgo the return available elsewhere in the capital market. Three consequences follow.

- **It depends on the investment, not the investor.** A safe project should be discounted at a low rate even by a risky firm, and a risky project at a high rate even by a safe one.
- **It is set by the market, not by the firm's past.** The historical interest rate on a firm's existing loans, or the return shareholders happen to have earned, is not the cost of new

capital.

- **It is not, in the first instance, a matter of financing.** Under the conditions of the Modigliani–Miller theorem (Theorem 30.7), how an investment is financed does not change its value. Financing matters only through the frictions the theorem assumes away—above all taxes and the costs of financial distress—and these are best valued separately.

A firm's securities are claims on its assets. The expected return on its assets, r_A —the cost of capital for the firm's existing business as a whole—is the weighted average of the expected returns on those claims. For a firm financed by equity E and debt D , with value $V = D + E$,

$$r_A = \frac{E}{V} r_E + \frac{D}{V} r_D,$$

a weighted mean (Chapter 2) with weights equal to each claim's share of the firm's market value. Rearranged, this is the second Modigliani–Miller proposition. The rest of the chapter estimates r_E and r_D and adds taxes.

35.2 The Cost of Equity

35.2.1 The CAPM in practice

The cost of equity is not observable: shareholders are promised nothing. The most common estimate uses the CAPM (Theorem 34.4),

$$r_E = r_f + \beta_E (\text{MRP}),$$

which needs three inputs.

The safe rate. Because corporate cash flows extend over decades, practitioners usually take the yield on long-term government bonds—commonly 10-year—in the currency of the cash flows, rather than the Treasury-bill rate. The choice should match the market risk premium: a premium measured over bills should be added to a bill rate, one measured over bonds to a bond yield. For cash flows in euros, a German government yield is a natural choice; for currencies whose governments carry default risk, analysts subtract an estimate of the sovereign's default spread.

The market risk premium. There is no agreed figure. Historical averages (Chapter 34) are imprecise and depend on the period and country. An alternative infers the premium from current prices: applying the growing-perpetuity formula of Chapter 9 to the whole market, the expected market return is the cash yield plus expected growth.

Worked Example 35.1. An implied market risk premium

A share index stands at 5,000. Its constituents are expected to pay 175 in dividends and buybacks over the next year, a yield of 3.5 per cent, and these payouts are expected to grow at 4.5 per cent a year in the long run. The implied expected market return is $3.5 + 4.5 = 8.0$ per cent. With a 10-year government yield of 4.2 per cent, the implied premium is 3.8 per cent.

The implied premium varies with market conditions—it rises when prices fall—and depends on the growth assumption. Surveys of practitioners and academics have typically reported

premia between about 4 and 6 per cent for developed markets. Whatever figure is chosen, it should be used consistently across projects and years, so that differences in discount rates reflect differences in risk rather than changes of analyst.

Beta. A regression beta for a single company is imprecise (Example 34.6), and it measures the risk of the firm's equity, which depends on the firm's leverage as well as on its business. Both problems are addressed by working with asset betas.

35.2.2 Leverage and beta

The beta of a portfolio is the weighted average of the betas of its holdings (Chapter 34). A firm's assets are a portfolio of its debt and equity, so

$$\beta_A = \frac{E}{V} \beta_E + \frac{D}{V} \beta_D.$$

Debt of healthy firms has a beta close to zero. Setting $\beta_D = 0$,

$$\beta_E = \beta_A \left(1 + \frac{D}{E} \right).$$

Leverage amplifies the equity's exposure to business risk, just as borrowing to buy shares amplifies an investor's. The formula lets an analyst *unlever* the equity betas of comparable firms to estimate the beta of the business, and *relever* it at the capital structure of the firm being analysed.

Worked Example 35.2. A pure-play beta

A firm is considering entering a new industry. Three listed firms operate only in that industry:

	Firm 1	Firm 2	Firm 3
Equity beta	1.20	1.125	1.53
Debt-to-equity (market values)	0.50	0.25	0.80
Asset beta = $\beta_E / (1 + D/E)$	0.80	0.90	0.85

The average asset beta is 0.85. The equity betas differ mostly because of leverage; the businesses are similarly risky. If the new division is financed with a debt-to-equity ratio of 0.6, its equity beta is $0.85 \times 1.6 = 1.36$. With $r_f = 4$ per cent and a premium of 5 per cent, the cost of equity is $4 + 1.36 \times 5 = 10.8$ per cent, and the cost of capital of the business, r_A , is $4 + 0.85 \times 5 = 8.25$ per cent.

► Which relevering formula?

The formula above assumes that the firm keeps its debt ratio roughly constant as its value changes, so that the tax shields from debt are as risky as the business. Many textbooks and data services instead use the formula of Robert Hamada (1972), $\beta_E = \beta_A [1 + (1 - \tau) D/E]$, which assumes that the *amount* of debt is fixed forever, making tax shields as safe as the debt. The two give different asset betas from the same data (Exercise 35.4). Either can be defended; what matters is using the same formula to unlever and to relever, and choosing the one that matches the firm's financing policy.

35.3 The Cost of Debt

The cost of debt is the expected return that lenders require. For a firm with traded bonds, the yield to maturity (Chapter 33) is a starting point; for one without, analysts add to the government yield a spread typical of the firm's credit rating, or of the rating its interest coverage would imply.

But a yield is a *promised* return, achieved only if the borrower pays in full. For safe borrowers the difference is negligible. For risky ones it is not: the expected return is roughly the promised yield minus the expected loss from default, the probability of default times the loss given default (Chapter 33). A bond rated B yielding 9 per cent, with an annual default probability of 4 per cent and a loss given default of 60 per cent, offers an expected return of about $9 - 0.04 \times 60 = 6.6$ per cent. Using 9 per cent in the cost of capital would overstate the cost of debt.

Because interest is deductible for corporate tax in most jurisdictions, the *after-tax* cost of debt is $r_D(1 - \tau)$, where τ is the marginal tax rate the firm expects to pay on the margin. A firm with tax losses that it cannot use soon, or one whose interest deductions are capped (Chapter 30), gains less.

35.4 The Weighted Average Cost of Capital

35.4.1 Definition

Standard discounted-cash-flow valuation discounts the firm's free cash flow to the firm—computed as if the firm had no debt, so that its taxes are those of an all-equity firm (Chapter 31)—at a rate that captures the tax benefit of debt. That rate is the **weighted average cost of capital**,

$$\text{WACC} = \frac{E}{V} r_E + \frac{D}{V} r_D(1 - \tau).$$

It is the weighted mean of the costs of the firm's sources of finance, with the cost of debt taken after tax.

Worked Example 35.3. Computing the WACC

A company has 50 million shares trading at 24, so its equity is worth 1,200 million. Its bonds, with a book value of 850 million, trade at 800 million and yield 6.5 per cent; their default risk is small. Its equity beta is 1.2, the government yield 4 per cent, the market premium 5 per cent and the tax rate 25 per cent. Then $r_E = 4 + 1.2 \times 5 = 10$ per cent; the after-tax cost of debt is $6.5 \times 0.75 = 4.875$ per cent; $V = 2,000$, with weights 0.6 and 0.4. The WACC is

$$0.6 \times 10 + 0.4 \times 4.875 = 6.00 + 1.95 = 7.95 \text{ per cent.}$$

Had the analyst used book values (book equity 500, book debt 850), the weights would have been 0.37 and 0.63 and the WACC 6.8 per cent—more than a point lower, and wrong.

The weights must be *market* values, because the costs are market returns: r_E is the return investors require on the price they would pay for the shares today, not on the historical book value of equity. Book equity reflects past issues and retained earnings and the accounting conventions of Chapters 27–30; it can even be negative for a valuable firm that has bought back many shares. When a firm plans to move toward a different capital structure, the weights should be the *target* proportions it will maintain.

35.4.2 Leverage and the WACC

How does the WACC change with leverage? Suppose the firm keeps its debt ratio constant, so that the Modigliani–Miller relation $r_E = r_A + (r_A - r_D)D/E$ holds. Then the pre-tax weighted average $\frac{E}{V}r_E + \frac{D}{V}r_D$ equals r_A , whatever the leverage, and subtracting the tax saving on interest gives

$$\text{WACC} = r_A - \tau r_D \frac{D}{V}.$$

More debt lowers the WACC only through the tax deduction; the cheapness of debt before tax is an illusion, offset by the rising cost of equity.

Worked Example 35.4. WACC and leverage

Take $r_A = 9$ per cent and $\tau = 25$ per cent, with the cost of debt rising with leverage:

D/V	r_D	$r_E = r_A + (r_A - r_D)D/E$	WACC	$r_A - \tau r_D D/V$
0	–	9.00%	9.00%	9.00%
0.2	5.0%	10.00%	8.75%	8.75%
0.4	5.5%	11.33%	8.45%	8.45%
0.6	7.0%	12.00%	7.95%	7.95%

The WACC falls steadily. It does not show the costs of financial distress, which reduce expected cash flows rather than the discount rate. In the trade-off theory (Chapter 30), the value of the firm rises with leverage as long as the marginal tax benefit exceeds the marginal expected cost of distress, then falls.

35.4.3 Divisional and project costs of capital

The WACC of a company is the right discount rate only for projects with the same risk as the company's existing business, financed in the same proportions. A company that uses one rate for everything will accept too many risky projects and too few safe ones.

Worked Example 35.5. The company-wide rate

A company has a utility division with asset beta 0.5 and a technology division with asset beta 1.5; overall its asset beta is 1.0. With $r_f = 4$ per cent and a premium of 5 per cent, the company's cost of capital is 9 per cent, the utility's 6.5 per cent and the technology division's 11.5 per cent. A utility project expected to return 8 per cent is worth doing—it beats the 6.5 per cent available on investments of similar risk—but is rejected at the company-wide 9 per cent. A technology project expected to return 10 per cent is accepted at 9 per cent, although investors could earn 11.5 per cent elsewhere at the same risk. Over time, the company channels capital to its riskiest activities and becomes riskier.

The remedy is a cost of capital for each division, estimated from pure-play comparables as in Example 35.2. For projects that differ from any division—a venture into a new country, a new technology—the same method applies. What should *not* be done is to add arbitrary premia for risks that diversified investors do not bear, or to raise the discount rate to compensate for optimistic forecasts. Diversifiable risks and forecast bias should be handled in the cash flows (Chapter 32), where their effects can be seen and argued about.

35.5 Adjusted Present Value

The WACC folds the tax benefit of debt into the discount rate. That works well when the debt ratio is stable. When financing is lumpy or special—a leveraged buyout that pays debt down quickly, a subsidized loan, a project with large issue costs—it is clearer to value the financing effects separately. The **adjusted present value** is

$$\text{APV} = \underbrace{\text{NPV at } r_A}_{\text{base case, all-equity}} + \text{PV}(\text{interest tax shields}) \\ - \text{PV}(\text{issue and distress costs}) + \text{PV}(\text{subsidies}).$$

The base case discounts the unlevered free cash flows at the cost of capital of the assets; each financing effect is then valued at a rate that reflects its own risk.

Worked Example 35.6. APV and WACC agree

A project costs 1,000 and produces after-tax unlevered free cash flows of 95 a year forever; $r_A = 10$ per cent. The base-case value is $95/0.10 = 950$ and the base NPV -50 : financed with equity, the project destroys value. Suppose it supports 400 of permanent debt at 6 per cent, with $\tau = 25$ per cent. The annual tax saving is $0.25 \times 0.06 \times 400 = 6$; as a perpetuity at the debt's rate, it is worth $6/0.06 = \tau D = 100$. The APV is $-50 + 100 = 50$: the project is worth doing, but only because of its tax shield.

The WACC route gives the same answer if done consistently. The levered value is $950 + 100 = 1,050$. With permanent debt, the WACC is $r_A(1 - \tau D/V) = 10 \times (1 - 0.25 \times 400/1,050) = 9.048$ per cent, and $95/0.09048 = 1,050$.

In the example the debt is fixed in amount, so its tax shields are as safe as the debt and are discounted at r_D . If instead debt is rebalanced to a constant ratio, the tax shields rise and fall with the firm's value and should be discounted at r_A , giving the WACC formula of the previous section. APV makes such assumptions explicit, which is its main virtue.

35.6 Payout Policy

35.6.1 Irrelevance and its limits

Once a firm has decided its investments and its borrowing, the cash left over—free cash flow to equity—must be retained or paid out. Miller and Modigliani (1961) showed that in a frictionless market payout policy does not affect value: a shareholder who wants more cash than the firm pays can sell shares; one who wants less can reinvest dividends. Whether cash is paid as dividends or as buybacks is equally irrelevant.

Worked Example 35.7. Dividend or buyback?

A firm has operating assets worth 9,000 and cash of 1,000; it has 100 shares worth 100 each. If it pays a dividend of 10 per share, each shareholder receives 10 and each share falls to 90: wealth per original share remains 100. If instead it buys back 10 shares at 100, the remaining 90 shares are claims on 9,000 of assets, still 100 each; sellers receive cash, holders keep shares worth the same as before. In both cases shareholders' total wealth is unchanged at 10,000.

Earnings per share tell a different story. Suppose the operating assets earn 900 a year and the cash earns 4 per cent. Before the payout, earnings are 940 and EPS 9.40. After the buyback, earnings are 900 and EPS 10.00, a rise of 6.4 per cent. But the price-to-earnings

ratio falls from 10.6 to 10.0: the remaining earnings are riskier, because the safe cash that used to dilute them is gone. The higher EPS is not a gain.

In practice, payout choices matter through the frictions:

- **Taxes.** Where dividends are taxed more heavily than capital gains, or gains can be deferred until shares are sold, buybacks are more tax-efficient for many shareholders.
- **Signalling.** Managers smooth dividends and are reluctant to cut them; John Lintner's (1956) interviews found that firms raised dividends only when they were confident the increase could be sustained. A dividend increase therefore conveys information, and a cut is read as bad news (Chapter 25). Buybacks, which carry no such commitment, suit fluctuating cash flows.
- **Agency.** Paying out cash that managers might otherwise spend on unprofitable empire-building disciplines them; Michael Jensen (1986) called this the problem of *free cash flow*.
- **Management incentives.** Because buybacks raise EPS and executives' pay is often tied to it, buybacks can be chosen for the wrong reasons.

Key idea. Financing and payout decisions create value only through frictions: taxes, the costs of distress, information asymmetries and agency problems. Analyses that attribute value to cheap debt, higher EPS or a higher ROE, without identifying which friction is at work, are usually confusing risk with return.

Case: Apple's 2013 bond issue

The situation. By early 2013 Apple had accumulated about 145 billion dollars of cash and marketable securities, and its shares had fallen sharply from their 2012 peak. Investors, including the hedge-fund manager David Einhorn, who sued Apple that February over a proposal restricting the issue of preferred shares, were pressing it to return more cash. But roughly two-thirds of the cash was held by foreign subsidiaries. Under the US tax rules of the time, bringing it home would have triggered US corporate tax at up to 35 per cent, less credit for foreign taxes paid.

The decision. In April 2013 Apple announced a capital-return programme of 100 billion dollars through 2015, including a 60 billion dollar share repurchase authorization, and financed part of it by issuing 17 billion dollars of bonds in six tranches, with maturities of up to 30 years. Demand was heavy, and the coupons were low—the 10-year notes paid 2.4 per cent. Apple went on to issue bonds repeatedly over the following years while continuing to hold very large cash balances abroad. After the Tax Cuts and Jobs Act of 2017 imposed a one-off tax on accumulated foreign earnings and largely removed the tax on future repatriation, Apple brought much of the cash home.

In terms of this chapter. Borrowing while holding cash made sense because of a friction: the repatriation tax. Borrowing 17 billion at an after-tax cost of well under 2 per cent was cheaper than paying tax on repatriated cash, and the interest was deductible. Did the buyback itself make shareholders richer? Example 35.7 says that swapping cash for shares does not, by itself, create value; what could create value was the tax saving, a modest interest tax shield, and perhaps a signal from management that it considered the shares undervalued. The rise in EPS that followed was a mechanical consequence of fewer shares, not evidence of value created.

Summary

- The cost of capital is the expected return on investments of the same risk: an opportunity cost set by the market and specific to the investment.
- The cost of equity is usually estimated with the CAPM, using a long-term government yield, a consistently chosen market risk premium (historical or implied) and an asset beta from comparables, relevered as $\beta_E = \beta_A(1 + D/E)$ (or Hamada's variant).
- The cost of debt is an expected return; for risky debt, subtract expected default losses from the yield. Interest deductibility gives an after-tax cost $r_D(1 - \tau)$.
- $WACC = (E/V)r_E + (D/V)r_D(1 - \tau)$, with market-value or target weights. With a constant debt ratio, $WACC = r_A - \tau r_D D/V$: leverage lowers it only through taxes.
- Using one company-wide rate misallocates capital; use divisional or project rates, and handle diversifiable risk and optimism in the cash flows.
- APV values a project as all-equity and adds financing side effects; it agrees with the WACC when both are applied consistently.
- Payout policy is irrelevant without frictions; taxes, signalling and agency explain real choices. Buybacks raise EPS without, in themselves, creating value.

Key terms

cost of capital; opportunity cost of capital; cost of equity; safe rate; market risk premium; implied equity risk premium; asset beta; equity beta; unlevering and relevering; pure-play comparable; Hamada formula; cost of debt; promised versus expected return; after-tax cost of debt; weighted average cost of capital; market-value weights; target capital structure; divisional cost of capital; adjusted present value; interest tax shield; issue costs; payout policy; dividend irrelevance; share repurchase; dividend smoothing; signalling; free cash flow problem; repatriation tax.

Exercises

Exercise 35.1 (WACC). A company has 20 million shares at 15. Its bonds have a face value of 200 million, trade at 95 per cent of face and yield 7 per cent. Its equity beta is 1.1, the government yield 4 per cent, the market premium 6 per cent and the tax rate 25 per cent. Compute its WACC.

Exercise 35.2 (Book weights). In Exercise 35.1, the book value of equity is 90 million. Recompute the WACC with book weights and explain which figure should be used, and why.

Exercise 35.3 (Relevering). A comparable firm has an equity beta of 1.4 and a debt-to-equity ratio of 0.75. Your firm plans a debt-to-equity ratio of 0.25. Assuming debt betas of zero and a constant debt ratio, find the asset beta, your equity beta and your cost of equity ($r_f = 4$ per cent, premium 5 per cent).

Exercise 35.4 (Hamada). Repeat Exercise 35.3 with Hamada's formula and a tax rate of 25 per cent. Why do the answers differ?

Exercise 35.5 (Expected cost of debt). A firm's bonds yield 8.5 per cent. Its annual probability of default is estimated at 3 per cent, with recovery of 40 per cent. Estimate the expected return on the debt and its after-tax cost at a 25 per cent tax rate.

Exercise 35.6 (Adjusted present value). A project costs 500 and generates unlevered after-tax free cash flow of 48 a year forever; $r_A = 10$ per cent. It can support 300 of permanent debt at 5 per cent; the tax rate is 30 per cent; issuing the debt costs 2 per cent of the amount raised. Compute the base NPV and the APV.

Exercise 35.7 (WACC and leverage). A firm has $r_A = 10$ per cent and keeps $D/V = 0.3$, borrowing at 6 per cent; the tax rate is 25 per cent. Find r_E and the WACC in two ways.

Exercise 35.8 (One rate for all). A conglomerate uses a 9 per cent hurdle rate everywhere. Projects: P in its food division (appropriate rate 7 per cent) with IRR 8 per cent; Q in its biotech division (appropriate rate 13 per cent) with IRR 11 per cent; R in its logistics division (appropriate rate 9 per cent) with IRR 10 per cent. Which decisions does the single rate get wrong?

Exercise 35.9 (Implied premium). A share index stands at 4,000; expected dividends and buybacks next year are 150, and long-run growth is expected to be 4 per cent. The 10-year government yield is 3.5 per cent. Find the implied market return and risk premium. What happens to the implied premium if the index falls 20 per cent with unchanged payouts and growth?

Exercise 35.10 (Debt-financed buyback). A firm earns 500 after tax, has 100 shares priced at 60 and no debt. It borrows 600 at 5 per cent (tax rate 25 per cent) to buy back 10 shares. Find EPS before and after. Explain why the rise in EPS does not by itself show that shareholders are better off, and identify what could make them better off.

Solutions

Solution 35.1. $E = 300$, $D = 190$, $V = 490$: weights 0.612 and 0.388. $r_E = 4 + 1.1 \times 6 = 10.6$ per cent; after-tax $r_D = 7 \times 0.75 = 5.25$ per cent. WACC = $0.612 \times 10.6 + 0.388 \times 5.25 = 6.49 + 2.04 = 8.53$ per cent.

Solution 35.2. Book weights: $E = 90$, $D = 200$: 0.310 and 0.690. WACC = $0.310 \times 10.6 + 0.690 \times 5.25 = 3.29 + 3.62 = 6.91$ per cent. Market weights are right: the costs are returns required on current market values, and investors could buy the claims today only at those values.

Solution 35.3. $\beta_A = 1.4/1.75 = 0.80$; $\beta_E = 0.80 \times 1.25 = 1.00$; $r_E = 4 + 5 = 9$ per cent.

Solution 35.4. $\beta_A = 1.4/(1 + 0.75 \times 0.75) = 0.896$; $\beta_E = 0.896 \times (1 + 0.75 \times 0.25) = 1.064$; $r_E = 9.32$ per cent. Hamada assumes fixed debt, so the tax shields are riskless. All of the levered firm's risk is then attributed to the business assets, which are only part of its value (the rest being the safe tax shields), so their beta must be higher.

Solution 35.5. Expected loss $\approx 0.03 \times 0.60 = 1.8$ per cent; expected return $\approx 8.5 - 1.8 = 6.7$ per cent; after tax $6.7 \times 0.75 = 5.0$ per cent.

Solution 35.6. Base value $48/0.10 = 480$; base NPV -20 . Tax shield $\tau D = 0.30 \times 300 = 90$; issue costs 6. APV = $-20 + 90 - 6 = 64$: accept, on the strength of the financing.

Solution 35.7. $D/E = 0.3/0.7$; $r_E = 10 + (10 - 6) \times 0.4286 = 11.71$ per cent. WACC = $0.7 \times 11.71 + 0.3 \times 6 \times 0.75 = 8.20 + 1.35 = 9.55$ per cent; check: $10 - 0.25 \times 6 \times 0.3 = 9.55$ per cent.

Solution 35.8. P is rejected at 9 per cent although its 8 per cent IRR exceeds its 7 per cent cost: a good project lost. Q is accepted at 9 per cent although its 11 per cent IRR is below its 13 per cent cost: a bad project taken. R is correctly accepted.

Solution 35.9. Yield $150/4,000 = 3.75$ per cent; implied return 7.75 per cent; premium 4.25 per cent. After a 20 per cent fall, the yield is $150/3,200 = 4.69$ per cent, the implied return 8.69 per cent and the premium 5.19 per cent: implied premia rise when prices fall.

Solution 35.10. Before: $\text{EPS} = 500/100 = 5.00$. After-tax interest $= 600 \times 0.05 \times 0.75 = 22.5$; after: $\text{EPS} = 477.5/90 = 5.31$, up 6.1 per cent. But the equity is now levered and riskier, so its required return is higher and its P/E lower (Example 35.7). Shareholders gain only through frictions: the present value of the interest tax shield (up to $\tau D = 150$ if the debt is permanent), less any expected costs of financial distress, and any information the buyback conveys—not through the arithmetic of EPS.

Financial Modelling and Forecasting

After studying this chapter you should be able to

1. Design a financial model whose structure separates inputs, calculations, checks and outputs, with explicit units, signs and timing.
2. Forecast revenue, costs, working capital and fixed assets from economic drivers, anchored to base rates.
3. Build an integrated three-statement forecast that balances without plugs, using a revolving credit facility as the balancing item.
4. Recognize circular references, and solve them algebraically or by controlled iteration.
5. Use sensitivity, scenario and simulation analysis to show which assumptions matter.
6. Audit a model: design checks, test reasonableness, and recognize the most common spreadsheet errors.

A motivating question

In January 2012 the Chief Investment Office of JPMorgan Chase, the largest bank in the United States, adopted a new model for measuring the risk of a large portfolio of credit derivatives. The model reported that the portfolio's risk had roughly halved. Over the following months the portfolio lost about 6 billion dollars. An internal investigation later found, among many other failings, that the model ran on spreadsheets fed by manual copying and pasting, and that one formula divided by a sum where it should have divided by an average.

Nearly every financial decision in this Part—a capital budget, a valuation, a buyout, a loan—rests on a model, usually a spreadsheet, that turns assumptions into forecasts of accounts and cash flows. The mathematics is the easy part; Chapters 2 to 9 supply all of it. The hard parts are forecasting honestly, linking the statements so that every number has a reason, and keeping errors out. This chapter shows how.

36.1 Architecture

36.1.1 Purpose first

A model is built to inform a decision: whether to invest, at what price to bid, how much to lend, whether a covenant will hold. Its purpose determines what it must show, how detailed it should be, and what can be left out. Before the first formula, a model should state:

- its **purpose** and the outputs the decision needs;
- its **perimeter**: which legal entities, businesses, assets and periods it covers;
- its **conventions**: currency, units (thousands or millions), period length, whether flows occur at the end or the middle of periods, and signs (for example, inflows positive and

outflows negative throughout);

- its **valuation or transaction date**, and the date at which each piece of information was known.

Unit and sign errors are the commonest serious mistakes, precisely because formulas remain internally consistent when they are made. The Mars Climate Orbiter of Chapter 2 was lost to a units error; a model that mixes thousands with millions is wrong by a factor of a thousand with every formula “working”.

36.1.2 Inputs, calculations, checks, outputs

Good models follow a one-directional flow (Figure 36.1). Inputs—historical data and forecast assumptions—are entered once, in clearly marked places, with sources and dates. Calculations transform them in causal order: operating schedules, then the statements, then financing, then valuation. Checks test that identities hold. Outputs present what the decision needs. Three rules follow.

1. **Enter each assumption once.** A growth rate typed into twenty formulas will be changed in nineteen.
2. **Never overwrite a formula with a number.** A “hard-coded” figure inside a calculation block silently stops responding to inputs.
3. **Let calculations flow forward.** An output sheet should never feed a hidden adjustment back into a calculation.

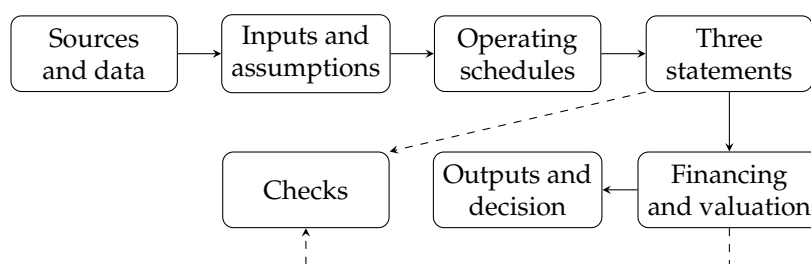


Figure 36.1: The flow of a well-structured model. Each arrow can be traced; checks draw on the statements and financing without feeding back into them.

36.1.3 Timing

A model’s timeline should consist of real dates, with flags marking historical, forecast and partial (“stub”) periods. Timing affects values: interest accrues day by day, according to the contract’s day-count convention (Chapter 33); a transaction that closes mid-year needs a stub period; cash flows assumed to arrive at the end of each year are worth less than the same cash arriving evenly through the year. A common compromise in valuation is the *mid-year convention*, which discounts each year’s cash flow by $(1 + r)^{t-1/2}$ rather than $(1 + r)^t$.

36.2 Forecasting from Drivers

36.2.1 Revenue

A forecast should be an explicit argument about the business, not an extrapolation of past totals. Revenue is usually built from drivers:

- **price** × **volume**, with separate views on each;
- **market size** × **market share**, useful for new products and for checking that forecasts do not imply an implausible share;
- **customers** × **revenue per customer**, with separate rates for acquiring and losing customers (cohorts), for subscription businesses;
- **capacity** × **utilization** × **price**, for hotels, airlines, power stations and factories (Chapter 40).

36.2.2 Costs, working capital and fixed assets

Costs are forecast according to their behaviour (Chapter 31): variable costs as a share of revenue or per unit, fixed costs from headcount, wages and rents, with inflation. Working capital follows from days ratios (Chapter 28): receivables = revenue × DSO/360, inventory = cost of sales × DIO/360, payables = cost of sales × DPO/360. Fixed assets follow a **roll-forward**:

$$\text{Closing PP\&E} = \text{Opening PP\&E} + \text{Capital expenditure} - \text{Depreciation} - \text{Disposals}.$$

Every balance-sheet line should have such a schedule, from opening to closing balance, with each movement linked to the income statement or the cash-flow statement.

36.2.3 Base rates and mean reversion

The most important discipline in forecasting is to compare the forecast with what usually happens. Studies of large samples of companies find that very high growth rarely persists: Louis Chan, Jason Karceski and Josef Lakonishok (2003) found little more persistence in firms' long-term earnings growth than chance would produce. Profit margins and returns on capital also tend to revert toward industry and economy-wide averages, as competition responds to high profits (Chapter 22). A forecast of 25 per cent annual growth for ten years, or of margins far above the industry's best, should be tested against the *reference class* of firms that have attempted the same thing (Chapter 32). The inside view—the detailed plan for this business—should be adjusted toward the outside view.

► Growth consumes cash

Growing firms need more working capital and more fixed assets before the extra revenue arrives. A model that forecasts profit without tracking the balance sheet will miss the cash that growth absorbs—and many profitable, fast-growing firms have failed for lack of cash. The integrated model of the next section exists to catch this.

36.3 The Integrated Three-Statement Model

36.3.1 How the statements link

The three statements of Chapter 27 are one system. Net income flows from the income statement to retained earnings. The cash-flow statement explains the change in cash on the balance sheet: operating cash flow starts from net income, adds back non-cash expenses such as depreciation, and subtracts increases in working capital; investing and financing flows follow from the fixed-asset and debt schedules. If every balance-sheet line is driven by a schedule and every movement is recorded in the cash-flow statement, the balance sheet must balance: assets equal liabilities plus equity by the double-entry identity. A model that balances only because a number has been forced to make it balance—a **plug**—hides an error.

36.3.2 The balancing item

One problem remains: the forecast may produce too little cash, or too much. Real firms fill shortfalls by drawing on a **revolving credit facility** (revolver), a committed line of credit that can be drawn and repaid at will, and use surpluses to repay it. A model does the same. It first computes cash *before* revolver flows; if that falls below the firm's minimum cash balance, the revolver is drawn by the difference; if cash exceeds the minimum and the revolver is drawn, the excess repays it. Cash and the revolver balance are then outputs of an explicit rule, not plugs. If the shortfall exceeds the facility's limit, the model should report a funding gap rather than show negative cash.

Worked Example 36.1. A three-statement forecast

At the end of year 0, a manufacturer has the balance sheet in the left column of Table 36.1. Assumptions for year 1: revenue grows 10 per cent from 400; cost of sales is 60 per cent of revenue and overheads 15 per cent; depreciation is 10 per cent of opening PP&E; a new plant requires capital expenditure of 90; DSO 45 days, DIO 60 days and DPO 30 days (360-day year); the term loan pays 6 per cent on its opening balance and amortizes 10 a year; the revolver costs 7 per cent; the tax rate is 25 per cent; 40 per cent of net income is paid as dividends; minimum cash is 20.

Income statement. Revenue 440; cost of sales 264; overheads 66; depreciation 20; EBIT 90; interest $0.06 \times 100 = 6$; pre-tax profit 84; tax 21; net income 63; dividends 25.2.

Working capital. Receivables $440 \times 45/360 = 55$; inventory $264 \times 60/360 = 44$; payables $264 \times 30/360 = 22$. The increase in operating working capital is $(55 - 50) + (44 - 40) - (22 - 20) = 7$.

Cash flow before the revolver. Operating: $63 + 20 - 7 = 76$. Investing: -90 . Financing before the revolver: term-loan repayment -10 , dividends -25.2 . Cash before the revolver: $20 + 76 - 90 - 10 - 25.2 = -29.2$.

Revolver. To restore minimum cash of 20, the firm draws $20 - (-29.2) = 49.2$.

Balance sheet. PP&E = $200 + 90 - 20 = 270$; retained earnings = $110 + 63 - 25.2 = 147.8$. The right column of Table 36.1 shows that assets and claims both total 389.0: the check is zero.

Table 36.1: Balance sheets for Example 36.1.

	Year 0	Year 1
Cash	20.0	20.0
Receivables	50.0	55.0
Inventory	40.0	44.0
PP&E (net)	200.0	270.0
Total assets	310.0	389.0
Payables	20.0	22.0
Revolver	0.0	49.2
Term loan	100.0	90.0
Share capital	80.0	80.0
Retained earnings	110.0	147.8
Total liabilities and equity	310.0	389.0
Check (assets – claims)	0.0	0.0

The example shows the pitfall above at work. The firm is profitable—net income of 63 on

revenue of 440—yet it needs to borrow nearly 50, because its investment and dividend exceed its operating cash flow. A lender reading only the income statement would miss the need; a model that balances reveals it, and shows when it will be repaid.

36.3.3 Circularity

In Example 36.1, interest was charged on *opening* balances. If it is charged instead on the *average* balance during the year, the model becomes circular: interest depends on the revolver balance, which depends on cash, which depends on net income, which depends on interest. Spreadsheets flag such **circular references**, and they should not be ignored or suppressed.

Worked Example 36.2. Solving a circular reference

In Example 36.1, suppose the revolver's interest is 7 per cent of the average of its opening (zero) and closing balances R . Extra interest of $0.035R$ lowers net income by $0.75 \times 0.035R$ and retained cash by 60 per cent of that (the rest would have been paid as dividends), so it raises the required draw by $0.45 \times 0.035R = 0.01575R$. The closing balance satisfies

$$R = 49.2 + 0.01575R \implies R = \frac{49.2}{1 - 0.01575} = 49.99.$$

Iterating instead—start at $R = 49.2$, compute interest, recompute R —gives 49.975, then 49.987, then 49.9873: the sequence converges quickly because each step multiplies the error by 0.01575.

The iteration is a first-order difference equation, $R_{n+1} = 49.2 + 0.01575R_n$, which converges because the coefficient is less than one in absolute value (Theorem 4.4). Circular models converge when feedback is weak, as it usually is. But if an input error or a rogue formula makes the feedback strong, iteration can diverge or produce nonsense, and a spreadsheet left in iterative mode will not say so. Good practice is to break the circle where possible (use opening balances, or solve the algebra), or to control the iteration with a visible switch and a convergence check.

36.4 Uncertainty: Sensitivity, Scenarios and Simulation

A model's outputs are only as good as its assumptions, and it should show which assumptions matter. Chapter 32 introduced the tools for a project; they apply to any model.

Sensitivity analysis changes one input at a time and records the output. Arranged from the widest range to the narrowest, the results form a *tornado diagram*, which directs attention to the inputs that deserve the most research.

Worked Example 36.3. What drives the funding need?

In Example 36.1, the year-1 revolver draw of 49.2 responds to four inputs as follows:

Input (base value)	Range tested	Revolver draw
Capital expenditure (90)	70 to 110	29.2 to 69.2
DSO (45 days)	30 to 60 days	30.9 to 67.5
Cost of sales (60% of revenue)	56% to 64%	39.8 to 58.6
Revenue growth (10%)	0% to 20%	46.7 to 51.7

Capital expenditure and customers' payment terms dominate. Revenue growth matters surprisingly little, and in the "wrong" direction: faster growth *raises* the funding need, because extra working capital outweighs extra profit in the first year.

Scenario analysis changes several inputs together in internally consistent combinations: a recession scenario lowers volume, prices *and* customers' willingness to pay promptly. In the example, a downside case with no growth, a cost-of-sales ratio of 64 per cent and DSO of 55 days raises the draw to about 66. A scenario selector should switch a whole coherent set of drivers, not whichever cells the author remembered to connect.

Monte Carlo simulation (Chapter 19) draws the inputs from probability distributions, with correlations, and reports the distribution of outputs—for example, the probability that the funding need exceeds the revolver's limit. Its results are only as good as the distributions assumed, and correlations in bad times tend to be higher than in calm periods (Chapter 14).

36.5 Model Risk and Audit

36.5.1 Checks

A **check** is an independent relationship that should hold: a difference that should be zero, a total that should agree with a source, a ratio that should stay within bounds. The balance-sheet check is necessary but not sufficient: a model can balance and still be wrong, for instance if an error affects an asset and a liability equally. Good models also check that the cash-flow statement explains the change in cash, that retained earnings roll forward correctly, that debt schedules reconcile, and that sources equal uses in a transaction. Checks should fail visibly—a model-wide flag, not a cell formatted to hide the error.

36.5.2 Reasonableness

Arithmetic checks cannot catch bad assumptions. For that, a model should display the ratios implied by its forecasts—growth, margins, return on invested capital, working-capital days, capital expenditure relative to depreciation, market share—and compare them with history, with peers and with base rates. A forecast in which return on capital rises forever, or capital expenditure stays below depreciation while revenue grows, is almost certainly wrong.

36.5.3 Common errors

Audits of spreadsheets used in business, surveyed by Raymond Panko and others, have found errors in most large spreadsheets examined. The usual culprits are mundane:

- formulas that cover the wrong range—omitting rows added later, or the last rows of a list;
- formulas copied across rows or columns with references that should have been fixed but were not;
- hard-coded numbers left inside calculations;
- unit, sign and timing inconsistencies;
- links to external files that have moved or changed;

- manual steps—copying and pasting between files—that are forgotten or done wrongly.

A widely discussed example came from academic economics: in 2013 Thomas Herndon, Michael Ash and Robert Pollin reported that an influential study by Carmen Reinhart and Kenneth Rogoff of growth and public debt had omitted five countries from an average through a spreadsheet range error. The error itself changed the headline result only modestly; larger differences came from data exclusions and the weighting of countries, which Reinhart and Rogoff defended as choices. But the episode showed how rarely even influential spreadsheets are independently checked.

Key idea. A model is a chain of transformations from evidence to decision. Its value depends on the quality of its weakest link, which is usually not its mathematics but its assumptions, its units, or a formula no one checked. Keep assumptions visible, flows one-directional, checks independent, and versions archived with a record of what changed and why.

Case: The London Whale

The portfolio. JPMorgan Chase's Chief Investment Office (CIO), which invested the bank's surplus deposits, ran a large "synthetic credit portfolio" of credit-default-swap indices, originally intended as a hedge against credit losses. By early 2012 the portfolio had grown enormously, and one of its traders, whose positions were big enough to move prices in the indices, was nicknamed the "London Whale" by other market participants.

The model. In January 2012, as the portfolio's measured risk exceeded its limits, the CIO adopted a new value-at-risk model (Chapter 34) that cut the reported VaR roughly in half. The bank's own task force report, published in January 2013, found that the model "operated through a series of Excel spreadsheets, which had to be completed manually, by a process of copying and pasting data from one spreadsheet to another". In one calculation, after subtracting an old rate from a new one, the spreadsheet divided by their sum instead of their average, as the modeller had intended—an error that likely muted measured volatility by a factor of two and lowered the VaR. The model had been approved under time pressure and without the testing that the bank's procedures required.

What followed. As the positions moved against the CIO in the spring of 2012, losses mounted to about 6.2 billion dollars. The bank paid about 920 million dollars in fines to US and UK regulators in 2013 and admitted to failures of oversight. Reports by the bank's task force and by a US Senate subcommittee found that risk limits had been breached repeatedly, that warnings had been dismissed, and that the new model was welcomed partly because it produced lower numbers.

In terms of this chapter. The spreadsheet error was not the cause of the losses—the positions were. But it illustrates every failing of this section: manual steps, no independent recalculation, no reasonableness test (a model change that halves measured risk overnight deserves suspicion, not relief), and a model whose output was accepted because it was convenient. The lessons for any model are the same: automate data flows, check independently, and treat a result that conveniently solves a problem as a reason to look harder.

Summary

- A model starts from its purpose, perimeter and conventions. Units, signs and timing must be stated and applied consistently.

- Structure flows one way: sources, inputs, calculations, checks, outputs. Each assumption is entered once; formulas are never overwritten with numbers.
- Forecasts are built from drivers—price and volume, share and market size, customers, capacity—and compared with base rates; growth and margins tend to revert.
- Every balance-sheet line needs a roll-forward schedule. The three statements then balance by construction; the revolver, drawn or repaid by an explicit rule, is the balancing item, never a plug.
- Circular references arise when interest depends on balances within the period; solve them algebraically or by controlled iteration, which converges when feedback is weak.
- Sensitivity (tornado), scenario and simulation analysis show which assumptions matter; scenarios must change drivers coherently.
- Checks, reasonableness tests and version control guard against error. Most spreadsheet errors are mundane, and convenient results deserve the most scrutiny.

Key terms

model purpose; model perimeter; conventions; hard-coding; inputs–calculations–outputs; stub period; mid-year convention; driver-based forecast; cohort; roll-forward; base rate; reference class; mean reversion; three-statement model; plug; revolving credit facility; minimum cash; funding gap; circular reference; iteration; convergence; sensitivity analysis; tornado diagram; scenario analysis; Monte Carlo simulation; check; reasonableness test; version control; model risk.

Exercises

Exercise 36.1 (Units). A source reports revenue of 125,000 in thousands of euros. The model works in millions of dollars at 1.08 dollars per euro. What figure should be entered, and what should be recorded alongside it? What happens if the analyst multiplies the raw figure by 1.08 and enters it in a sheet labelled “millions”?

Exercise 36.2 (Time-weighted interest). A loan of 100 is outstanding for the first half of a year and 60 for the second half, at 8 per cent a year. Compute the year’s interest. What would opening-balance and closing-balance conventions give?

Exercise 36.3 (Working capital). Revenue is forecast at 730 and cost of sales at 438 (365-day year). DSO is 40 days, DIO 50 and DPO 35. Opening receivables, inventory and payables are 70, 55 and 40. Find the closing balances and the change in operating working capital, and its effect on operating cash flow.

Exercise 36.4 (Roll-forwards). Opening PP&E is 500; capital expenditure 80; depreciation 60; an asset with net book value 15 is sold for 20. Find closing PP&E and the gain on disposal. Where does each figure appear in the three statements?

Exercise 36.5 (Revolver). Cash before revolver flows is -8 , minimum cash is 12, and the revolver has 30 of undrawn capacity. Find the draw and closing cash. What should the model show if only 15 were available?

Exercise 36.6 (Year 2). Continue Example 36.1 into year 2, with revenue growth of 10 per cent, capital expenditure of 30, and all other assumptions unchanged (interest on opening balances). Find net income, cash before the revolver, the revolver balance at the end of year 2, and total assets. Check that the balance sheet balances.

Exercise 36.7 (Circularity). A firm's closing debt D satisfies $D = 200 + 0.5 \times (1 - \tau) \times r \times (D_0 + D)/2$, where opening debt $D_0 = 150$, $r = 8$ per cent and $\tau = 25$ per cent. (Half of every unit of after-tax interest must be borrowed.) Solve for D algebraically and by two rounds of iteration from $D = 200$.

Exercise 36.8 (Reasonableness). A model forecasts revenue growth of 30 per cent a year for ten years, an EBIT margin rising from 8 to 30 per cent, and capital expenditure equal to 60 per cent of depreciation throughout. Identify three reasons for scepticism and the checks you would run.

Exercise 36.9 (Balance check). A model shows total assets of 512.4 and total liabilities and equity of 512.3. The author says the difference is rounding. What would you check before accepting that explanation? Why is a zero balance check not proof that a model is correct?

Solutions

Solution 36.1. 125,000 thousand euros = 125 million euros = 135 million dollars at 1.08. Record the source, date, original units and currency, and the exchange rate used. Multiplying the raw figure gives 135,000, which in a sheet labelled "millions" overstates revenue a thousandfold, while every formula still "works".

Solution 36.2. $100 \times 0.08 \times 0.5 + 60 \times 0.08 \times 0.5 = 4.0 + 2.4 = 6.4$. Opening balance: 8.0 (overstated); closing balance: 4.8 (understated).

Solution 36.3. Receivables $730 \times 40/365 = 80$; inventory $438 \times 50/365 = 60$; payables $438 \times 35/365 = 42$. Change in operating working capital $(80 - 70) + (60 - 55) - (42 - 40) = 13$, which reduces operating cash flow by 13.

Solution 36.4. Closing PP&E = $500 + 80 - 60 - 15 = 505$; gain = $20 - 15 = 5$. Capital expenditure (-80) and proceeds ($+20$) are investing cash flows; depreciation (60) is an expense added back in operating cash flow; the gain (5) raises net income but is deducted in operating cash flow because the cash is in investing; PP&E appears on the balance sheet.

Solution 36.5. Draw = $12 - (-8) = 20$, within the 30 available; closing cash 12. With only 15 available, the draw is 15, closing cash would be 7, below the minimum: the model should flag a funding shortfall of 5 rather than force cash to 12.

Solution 36.6. Revenue 484; cost of sales 290.4; overheads 72.6; depreciation $0.1 \times 270 = 27$; EBIT 94.0; interest $0.06 \times 90 + 0.07 \times 49.2 = 8.84$; pre-tax 85.16; tax 21.29; net income 63.87; dividends 25.55. Receivables 60.5, inventory 48.4, payables 24.2: working capital rises by $5.5 + 4.4 - 2.2 = 7.7$. Operating cash flow $63.87 + 27 - 7.7 = 83.17$; cash before revolver $20 + 83.17 - 30 - 10 - 25.55 = 37.62$. The excess of 17.62 over minimum cash repays the revolver, leaving 31.58. PP&E = $270 + 30 - 27 = 273$. Total assets = $20 + 60.5 + 48.4 + 273 = 401.9$; claims = $24.2 + 31.58 + 80 + 80 + 186.12 = 401.9$, where retained earnings = $147.8 + 63.87 - 25.55 = 186.12$. The check is zero.

Solution 36.7. The coefficient on D is $0.5 \times 0.75 \times 0.08/2 = 0.015$; the constant is $200 + 0.015 \times 150 = 202.25$. So $D = 202.25/(1 - 0.015) = 205.33$. Iteration: $D_1 = 202.25 + 0.015 \times 200 = 205.25$; $D_2 = 202.25 + 0.015 \times 205.25 = 205.33$.

Solution 36.8. (i) Thirty per cent growth for ten years multiplies revenue by $1.3^{10} = 13.8$; compare with the base rate of firms that have sustained such growth (very few) and with the size of the market. (ii) A margin of 30 per cent is likely to attract competition; compare with the best firms in the industry. (iii) Capital expenditure below depreciation while revenue grows fourteenfold implies a shrinking asset base supporting a far larger business: check implied asset turnover and return on invested capital, which would rise implausibly.

Solution 36.9. Check whether the difference is stable or changes with inputs (rounding differences do not grow when inputs change; link errors do), whether sub-totals reconcile, and whether the cash-flow statement explains the change in cash. A zero balance check shows only that double entry has been respected; an error that affects both sides equally, or a wrong assumption, leaves it at zero.

Valuation

After studying this chapter you should be able to

1. Distinguish enterprise value from equity value and move between them with a complete bridge.
2. Value a business by discounting free cash flow, with an explicit forecast and a terminal value, and explain why the terminal value dominates.
3. Derive the value-driver formula and show that growth creates value only when the return on new investment exceeds the cost of capital.
4. Use trading and transaction multiples consistently, and derive justified multiples from fundamentals.
5. Value a group by the sum of its parts, and a young firm by scenarios and the venture-capital method.
6. Recognize the common errors that make valuations wrong.

A motivating question

In January 2019, an investment by SoftBank valued The We Company, the parent of the office-sharing business WeWork, at about 47 billion dollars. In August it published its prospectus for a stock-market listing. Within six weeks the listing had been abandoned, its founder had stepped down as chief executive, and a rescue in October valued the company at about 8 billion dollars. Nothing about the business had changed in those months except the information available to investors and the way they used it.

How can the value of a business be estimated, and how can estimates differ by a factor of six? Chapter 31 introduced three equivalent models of value—discounted dividends, discounted free cash flow and residual income. This chapter turns them into practical tools. It begins with what exactly is being valued, develops discounted-cash-flow valuation and its economic logic, and then treats the multiples that dominate practice. Throughout, the aim is to see what assumptions each number embodies.

37.1 What Is Being Valued?

37.1.1 Enterprise value and equity value

A company's operating business is financed by several kinds of claim. **Enterprise value** (EV) is the value of the operating business to all of them; **equity value** is the value of the ordinary shares alone. The **EV bridge** links the two:

$$\begin{aligned} \text{Enterprise value} &= \text{Equity value} + \text{Debt} - \text{Cash} \\ &\quad + \text{Other claims (preferred shares, non-controlling)} \end{aligned}$$

- interests, leases, pension deficits)
- Non-operating assets (investments in associates, surplus property).

The logic is that of a house bought with a mortgage: the house's value equals the owner's equity plus the mortgage. Cash is subtracted because it is not needed to run the business (beyond an operating minimum) and could be used to repay debt. Non-controlling interests are added because consolidated accounts include 100 per cent of subsidiaries' profits, part of which belongs to other shareholders (Chapter 30). Investments in associates are subtracted because their profits are not in consolidated EBITDA.

Worked Example 37.1. An EV bridge

A company's shares are worth 800 in total. It has debt of 300, cash of 80, lease liabilities of 50, a pension deficit of 20 and non-controlling interests of 30, and holds a stake in an associate worth 40. Its enterprise value is

$$800 + 300 - 80 + 50 + 20 + 30 - 40 = 1,080.$$

Going the other way, an analyst who estimates EV at 1,200 would derive equity of $1,200 - 300 + 80 - 50 - 20 - 30 + 40 = 920$, or 9.20 a share if there are 100 shares.

37.1.2 Consistency

The central rule of valuation is *consistency*: cash flows, discount rates and multiples must refer to the same claims. Free cash flow to the firm (Definition 31.1), before interest, belongs to all capital providers and is discounted at the WACC (Chapter 35) to give enterprise value. Free cash flow to equity, after interest and debt flows, is discounted at the cost of equity to give equity value directly. EBITDA and EBIT, earned before interest, are compared with enterprise value; net income, after interest, with equity value. If leases are treated as debt in the bridge, lease payments must be excluded from operating costs, as IFRS 16 does (Chapter 29); if they are treated as operating costs, lease liabilities must be left out of the bridge. Most valuation errors are breaches of this rule.

37.2 Discounted Cash Flow

37.2.1 Explicit forecast and terminal value

A discounted-cash-flow (DCF) valuation forecasts free cash flow year by year for an *explicit period*—typically five to ten years, long enough for the business to reach a steady state—and summarizes everything after that in a **terminal value**. If free cash flow grows at a constant rate g after year T , the growing-perpetuity formula (Chapter 9) gives

$$TV_T = \frac{FCFF_{T+1}}{WACC - g} = \frac{FCFF_T(1 + g)}{WACC - g}, \quad EV_0 = \sum_{t=1}^T \frac{FCFF_t}{(1 + WACC)^t} + \frac{TV_T}{(1 + WACC)^T}.$$

Worked Example 37.2. A two-year DCF

A business is forecast to generate free cash flow of 10 in year 1 and 11 in year 2, after which it grows at 2 per cent a year forever. The WACC is 10 per cent and net debt 30.

The terminal value at the end of year 2 is $11 \times 1.02 / (0.10 - 0.02) = 140.25$. Then

$$EV = \frac{10}{1.10} + \frac{11 + 140.25}{1.10^2} = 9.09 + 125.00 = 134.09,$$

and equity value is $134.09 - 30 = 104.09$. The terminal value contributes $140.25 / 1.21 = 115.91$, or 86 per cent of enterprise value.

The share of value in the terminal value is typically large—often more than two-thirds—and for young, growing firms it can exceed 100 per cent, because near-term cash flows are negative. A DCF is therefore mostly a statement about the long run. That is not a flaw of the method; it reflects the fact that most of the value of a going concern lies in the distant future.

37.2.2 The sensitivity of the terminal value

The terminal value is a multiple $(1 + g) / (\text{WACC} - g)$ of the final year's cash flow. As g approaches the WACC, the denominator approaches zero and the multiple explodes. With a WACC of 10 per cent:

g	0%	2%	4%	6%	8%	9%
$(1 + g) / (0.10 - g)$	10.0	12.8	17.3	26.5	54.0	109.0

For $g \geq \text{WACC}$ the formula is meaningless: the geometric series of Theorem 4.3 does not converge, and the “value” is infinite. No business can grow faster than the economy forever, since it would eventually become larger than the economy. Long-run nominal growth assumptions are therefore usually capped near the expected long-run growth of nominal GDP, and often set below it.

Because terminal values are so sensitive, a DCF should always be presented with a sensitivity table. For Example 37.2, enterprise value varies as follows:

	$g = 1\%$	$g = 2\%$	$g = 3\%$
WACC 9%	135.3	153.3	177.4
WACC 10%	120.2	134.1	151.9
WACC 11%	108.1	119.1	132.9

Plausible ranges for two inputs produce values from 108 to 177. A single-point DCF conceals this uncertainty; a sensitivity table displays it.

37.2.3 Where growth comes from

The terminal value should not treat growth as free. To grow, a firm must invest: in working capital, plant, software, brands. Let NOPAT be net operating profit after tax, $\text{EBIT}(1 - \tau)$, and let the firm reinvest a fraction b of it in net new capital, earning a **return on new invested capital** (RONIC). Next year's NOPAT rises by $b \times \text{NOPAT} \times \text{RONIC}$, so

$$g = b \times \text{RONIC}, \quad \text{equivalently} \quad b = \frac{g}{\text{RONIC}}.$$

Free cash flow is what is left after reinvestment, $\text{FCFF} = \text{NOPAT}(1 - b) = \text{NOPAT}(1 - g/\text{RONIC})$. Substituting into the growing perpetuity gives the **value-driver formula**.

Proposition 37.3. Value driver formula

A firm that grows its NOPAT at rate g forever by reinvesting at return RONIC has enterprise value

$$EV = \frac{\text{NOPAT}_1 \left(1 - \frac{g}{\text{RONIC}}\right)}{\text{WACC} - g}.$$

If $\text{RONIC} = \text{WACC}$, then $EV = \text{NOPAT}_1 / \text{WACC}$, whatever the growth rate.

Proof. The formula follows by substituting $\text{FCFF}_1 = \text{NOPAT}_1(1 - g/\text{RONIC})$ into $EV = \text{FCFF}_1 / (\text{WACC} - g)$. If $\text{RONIC} = \text{WACC}$, the numerator is $\text{NOPAT}_1(\text{WACC} - g)/\text{WACC}$, and the factor $\text{WACC} - g$ cancels. $\square$

Worked Example 37.4. Growth that creates, and destroys, value

A firm has NOPAT next year of 100, a WACC of 9 per cent and expected growth of 4 per cent a year.

- If RONIC is 16 per cent, it must reinvest $0.04/0.16 = 25$ per cent of NOPAT: free cash flow is 75 and $EV = 75/0.05 = 1,500$.
- If RONIC equals the WACC, 9 per cent, it must reinvest 44.4 per cent: free cash flow is 55.6 and $EV = 1,111 = 100/0.09$, the value it would have with no growth at all.
- If RONIC is 6 per cent, it must reinvest two-thirds: free cash flow is 33.3 and $EV = 667$. Growth *destroys* value.

Theorem 37.3 is the discounted-cash-flow twin of the residual-income result of Chapter 31: growth creates value only when new investment earns more than its cost of capital. It disciplines terminal values. Competition tends to push returns on new investment toward the cost of capital (Chapter 22), so a terminal value that combines high growth with high returns forever should be justified by an identifiable durable advantage—a patent, a network, a brand, a cost position. A common conservative assumption is that RONIC equals the WACC in the terminal period, which makes the terminal value $\text{NOPAT}_{T+1}/\text{WACC}$ and removes the growth assumption altogether.

► Inconsistent terminal assumptions

A frequent error is to grow the final year's cash flow at g without adjusting reinvestment: the final forecast year, which may reflect heavy investment or none, is capitalized as if it were a steady state. Another is to use an inflation-free (real) growth rate with a nominal discount rate, or the reverse (the Fisher relation of Chapter 9 must hold between them). A third is to let the terminal year's margins, working capital or capital expenditure differ sharply from the steady state the growth rate implies. Checking the implied reinvestment rate and RONIC in the terminal year catches all three.

37.3 Multiples

37.3.1 Trading and transaction comparables

Most valuations in practice are made, or at least checked, by **multiples**: the ratio of a value to a measure of performance, computed for comparable firms and applied to the firm being valued.

The most common are:

- **EV/EBITDA** and **EV/EBIT**, comparing enterprise value with operating profit, insensitive to capital structure;
- **EV/Sales**, for firms without meaningful profits;
- **price/earnings** (P/E), comparing equity value with net income;
- **price/book** (P/B), common for banks and insurers, whose assets are largely financial and carried near market value.

Trading comparables use the current market values of listed peers. *Precedent transactions* use prices paid in past acquisitions of comparable firms; these include a control premium (Chapter 38) and reflect market conditions at the time of each deal.

Worked Example 37.5. A comparables valuation

Three listed peers trade at EV/EBITDA multiples of 7, 8 and 12. The target's EBITDA is 20. The median multiple, 8, gives an enterprise value of 160; the mean, 9, would give 180, pulled up by the peer at 12. The median is less sensitive to an outlier (Chapter 11). Before relying on either, the analyst should ask why the third peer trades at 12: faster growth, higher margins or lower risk would justify a higher multiple, and if the target resembles it in those respects, 8 may be too low.

37.3.2 What multiples assume

A multiple is a compressed DCF. Dividing the value-driver formula by NOPAT gives the *justified* multiple

$$\frac{EV}{NOPAT_1} = \frac{1 - g/ROIC}{WACC - g},$$

which rises with growth (when ROIC exceeds WACC), with returns on new investment and with lower risk. Similarly, the Gordon model with dividends $D_1 = ROE \times B_0 \times (1 - g/ROE)$ gives a justified price-to-book ratio

$$\frac{P_0}{B_0} = \frac{ROE - g}{r_E - g},$$

above one if and only if ROE exceeds the cost of equity. A bank trading below book value is one the market expects to earn less than its cost of equity (Chapter 51).

Applying a peer's multiple therefore assumes that the target has the peer's growth, returns and risk. Multiples also inherit accounting differences: EBITDA excludes lease costs under IFRS 16 but includes them for many US operating leases; one-off items, acquisitions and differences in capitalization of development costs (Chapter 29) distort comparisons; cyclical firms' multiples look cheap at the peak of earnings and expensive at the trough. And multiples price the target relative to its peers, not absolutely: if the whole sector is overvalued, so will be the target.

Key idea. DCF and multiples are not rivals. A DCF makes assumptions explicit; multiples reveal how the market prices comparable assumptions. A good valuation uses both, reconciles them, and explains the differences—for example, by computing the growth or return that a peer's multiple implies.

37.4 Special Situations

37.4.1 Sum of the parts

A group with businesses of different risk and growth is valued by the **sum of the parts**: each division at a multiple or DCF appropriate to it, less the capitalized cost of central functions, less net debt. Comparing the result with the market value often reveals a *conglomerate discount*, one motive for break-ups and spin-offs (Chapter 38).

Worked Example 37.6. Sum of the parts

A group has an industrial division with EBITDA of 50, valued at 8 times, and a software division with EBITDA of 30, valued at 12 times; central costs are 10 a year, capitalized at 8 times. Enterprise value = $400 + 360 - 80 = 680$; after net debt of 200, equity is worth 480. If the group's shares are worth 400 in total, they trade at a discount of $1 - 400/480 = 17$ per cent to the sum of the parts.

37.4.2 Young and loss-making firms

For a firm with losses today and uncertain prospects, a single forecast is misleading. It is better to build explicit scenarios—failure, modest success, great success—value each, and weight them by probabilities, checking that the implied probability of success is plausible. Early-stage investors often use the **venture-capital method**: estimate the firm's value at a future exit, discount it at a high target rate that reflects both risk and the chance of failure, and compute the ownership share required for the investment.

Worked Example 37.7. The venture-capital method

A start-up could be worth 500 when sold in five years. A venture fund targets a 40 per cent annual return. The present value of the exit is $500/1.4^5 = 93.0$ (the *post-money* valuation). To invest 20, the fund needs $20/93.0 = 21.5$ per cent of the company, implying a *pre-money* valuation of 73.0. If later funding rounds will dilute its stake by a fifth, it must ask for $21.5/0.8 = 26.9$ per cent now.

The headline valuations of venture-backed firms deserve caution. They are usually computed by multiplying the price paid for the latest shares by all shares outstanding, but the latest investors typically receive preferred shares with protections—priority in a sale, guaranteed minimum returns in an initial public offering—that ordinary shares lack. Will Gornall and Ilya Strebulaev (2020) estimated that such terms made the reported valuations of US “unicorns” substantially higher, on average, than the fair value of their ordinary shares.

37.5 Common Errors

The errors that most often make valuations wrong are:

1. **Mismatching claims**: equity cash flows discounted at the WACC, or EV multiples applied to net income, or leases counted twice.
2. **Mismatching inflation**: real cash flows with nominal discount rates, or cash flows in one currency with a discount rate in another (Chapter 42).
3. **Growth without investment**: terminal growth that is not supported by reinvestment, or combined with returns above the cost of capital forever.

4. **Double counting risk:** discount rates padded for risks that are also reflected in pessimistic cash flows, or for diversifiable risks.
5. **Adjusted earnings:** multiples applied to “adjusted” profit measures that exclude recurring costs.
6. **Anchoring:** working backward from a desired answer, or from the price already agreed, rather than forward from evidence.

Case: WeWork, 2019

The valuation. WeWork leased office space on long contracts, refurbished it and sublet desks and offices to members on short contracts. It presented itself as a technology and community company, and investors priced it accordingly: SoftBank’s investment of January 2019 valued the parent company at about 47 billion dollars, many times the market value of IWG (formerly Regus), a listed firm with a broadly similar serviced-office business and profits.

The prospectus. WeWork’s filing in August 2019 disclosed revenue of about 1.8 billion dollars for 2018 and a net loss of about 1.9 billion; in the first half of 2019, revenue of about 1.5 billion and a net loss of about 0.9 billion. It had signed lease commitments of about 47 billion dollars—obligations that the equity valuation had to be set against. It had earlier promoted a measure called “community-adjusted EBITDA”, which excluded not only interest, tax and depreciation but also marketing, general and administrative expenses and the costs of developing and opening new locations. The filing also revealed governance arrangements that alarmed investors: shares giving the founder, Adam Neumann, voting control, and transactions between him and the company, including its payment of about 5.9 million dollars for the rights to the “We” trademark, which he later returned.

What followed. As investors pushed back, the valuation discussed for the listing fell toward 10 to 15 billion dollars. Neumann stepped down as chief executive on 24 September; the listing was withdrawn on 30 September. In October SoftBank agreed a rescue package that valued the company at about 8 billion dollars. WeWork eventually listed in 2021 by merging with a special-purpose acquisition company, and filed for Chapter 11 bankruptcy protection in November 2023 (Chapter 39).

In terms of this chapter. Almost every error on the list above was present. The 47 billion dollar valuation relied on multiples of technology firms rather than of property businesses, whose returns on capital are lower (Theorem 37.3); on adjusted earnings that excluded recurring costs; on a private-market price for preferred shares extrapolated to the whole company; and on growth that required enormous reinvestment and lease commitments—claims that belong in the EV bridge. WeWork’s fundamental problem, borrowing long through leases and lending short through memberships, is the maturity mismatch of banking (Chapter 48) without a lender of last resort. Once the prospectus forced the assumptions into view, the valuation fell to a level consistent with them.

Summary

- Enterprise value is the value of the operating business to all claimants; equity value is found by subtracting debt and other claims and adding cash and non-operating assets.
- Cash flows, discount rates and multiples must refer to the same claims: FCFF with WACC and EV; FCFE with the cost of equity and equity value.

- A DCF combines an explicit forecast with a terminal value, which usually dominates. As $g \rightarrow \text{WACC}$, the terminal value explodes; long-run growth must be modest and consistent.
- Growth requires reinvestment, $g = b \times \text{RONIC}$, and creates value only when RONIC exceeds the WACC; with RONIC equal to the WACC, value equals NOPAT/WACC.
- Multiples are compressed DCFs; justified multiples depend on growth, returns and risk. Use medians, check outliers, and adjust for accounting differences.
- Sum-of-the-parts values groups by division; scenario analysis and the venture-capital method value young firms, whose headline valuations often overstate the value of ordinary shares.
- Most valuation errors come from inconsistency, growth without investment, double-counted risk, adjusted earnings and anchoring.

Key terms

enterprise value; equity value; EV bridge; net debt; non-controlling interest; consistency principle; discounted cash flow; explicit forecast period; terminal value; exit multiple; sensitivity table; NOPAT; reinvestment rate; return on new invested capital; value-driver formula; multiple; trading comparables; precedent transactions; EV/EBITDA; price/earnings; price/book; justified multiple; sum of the parts; conglomerate discount; scenario analysis; venture-capital method; pre-money and post-money valuation; adjusted earnings; anchoring.

Exercises

Exercise 37.1 (EV bridge). A company has 200 million shares at 12, debt of 900 million, cash of 250 million, preferred shares of 100 million, non-controlling interests of 60 million and a stake in an associate worth 90 million. Find its enterprise value. If its EBITDA is 450 million, what is its EV/EBITDA multiple?

Exercise 37.2 (DCF). Free cash flow to the firm is forecast at 20, 22 and 24 in years 1 to 3, growing at 3 per cent a year thereafter. The WACC is 9 per cent, net debt 100 and there are 50 shares. Find the terminal value, enterprise value, equity value per share, and the share of EV contributed by the terminal value.

Exercise 37.3 (Terminal multiples). With a WACC of 8 per cent, compute the terminal-value multiple $(1 + g)/(WACC - g)$ for $g = 1, 2, 3, 4$ and 5 per cent. By what percentage does the terminal value rise when g moves from 3 to 4 per cent?

Exercise 37.4 (Value drivers). A firm has next-year NOPAT of 50, a WACC of 10 per cent and grows at 4 per cent. Find its enterprise value if RONIC is 20, 10 and 5 per cent. Explain the pattern.

Exercise 37.5 (Justified price-to-book). A bank has ROE of 15 per cent, a cost of equity of 10 per cent and long-run growth of 5 per cent. Find its justified P/B. What if ROE falls to 8 per cent? Interpret.

Exercise 37.6 (Comparables). Five peers trade at EV/EBITDA of 6.5, 7.5, 8, 9 and 15. The target has EBITDA of 40 and net debt of 90. Value its equity using the median and the mean. Which would you use, and what would you investigate?

Exercise 37.7 (Leases and multiples). Before adopting IFRS 16, a retailer had EBITDA of 80 and enterprise value of 640, excluding leases. It then recognizes lease liabilities of 150, and its EBITDA rises by its annual lease payments of 20. Compute the EV/EBITDA multiple before and after. Why does comparing it with a peer that has not made the same adjustment mislead?

Exercise 37.8 (Sum of the parts). A group has a consumer division (EBIT 60, peer EV/EBIT 14), an industrial division (EBIT 40, peer EV/EBIT 9) and central costs of 12 a year (capitalize at 10). Net debt is 400 and the market value of equity 800. Find the sum-of-the-parts equity value and the discount or premium.

Exercise 37.9 (Venture capital). A start-up expects to be worth 400 at exit in six years. An investor targets a 35 per cent return, will invest 15 now, and expects its stake to be diluted by 20 per cent in later rounds. Find the post-money value today and the ownership share required now.

Exercise 37.10 (Sustainable growth). A firm has NOPAT of 80 next year, reinvests 40 per cent of it, and earns 12 per cent on new investment; its WACC is 9 per cent. Find its growth rate and enterprise value, and compare with its value if it stopped growing and paid out all NOPAT.

Solutions

Solution 37.1. Equity 2,400; $EV = 2,400 + 900 - 250 + 100 + 60 - 90 = 3,120$ million. $EV/EBITDA = 3,120/450 = 6.9$.

Solution 37.2. $TV_3 = 24 \times 1.03/0.06 = 412$. $EV = 20/1.09 + 22/1.09^2 + (24 + 412)/1.09^3 = 18.35 + 18.52 + 336.67 = 373.54$. Equity = 273.54, or 5.47 a share. The terminal value contributes $412/1.09^3 = 318.14$, 85 per cent of EV.

Solution 37.3. 14.4, 17.0, 20.6, 26.0 and 35.0. From 3 to 4 per cent the multiple rises from 20.6 to 26.0, by 26 per cent.

Solution 37.4. Reinvestment $g/ROIC$: 20, 40 and 80 per cent; free cash flow 40, 30 and 10; $EV = FCF/0.06$: 667, 500 and 167. At $ROIC = WACC$, $EV = 50/0.10 = 500$, the no-growth value; growth adds value when $ROIC$ exceeds $WACC$ and destroys it when $ROIC$ falls short.

Solution 37.5. $P/B = (0.15 - 0.05)/(0.10 - 0.05) = 2.0$. With ROE of 8 per cent, $(0.08 - 0.05)/0.05 = 0.6$: the bank is expected to earn less than its cost of equity, so each unit of book equity is worth less than one, and retaining earnings to grow destroys value.

Solution 37.6. Median 8: $EV = 320$, equity 230. Mean 9.2: $EV = 368$, equity 278. Use the median, but investigate the peer at 15—a different business mix, a takeover bid in its price, or depressed EBITDA—and whether the target resembles it.

Solution 37.7. Before: $640/80 = 8.0$. After: $(640 + 150)/(80 + 20) = 7.9$. Leases enter both numerator and denominator, so the multiple changes little; but a peer that reports EBITDA before lease costs while excluding lease liabilities from EV (or the reverse) would have an inconsistent multiple, making the retailer look cheap or dear for purely accounting reasons.

Solution 37.8. $EV = 60 \times 14 + 40 \times 9 - 12 \times 10 = 840 + 360 - 120 = 1,080$; equity = 680. Market value 800 is a premium of $800/680 - 1 = 18$ per cent to the sum of the parts: the market either sees value the analysis misses or overprices the group.

Solution 37.9. Post-money = $400/1.35^6 = 66.1$. Ownership needed after dilution: $15/66.1 = 22.7$ per cent; now: $22.7/0.8 = 28.4$ per cent.

Solution 37.10. $g = 0.4 \times 12\% = 4.8$ per cent; $FCFF = 48$; $EV = 48/(0.09 - 0.048) = 1,143$. Without growth, $EV = 80/0.09 = 889$. Growth adds value because new investment earns 12 per cent, above the 9 per cent WACC.

Mergers, Buyouts and Private Equity

After studying this chapter you should be able to

1. Identify the sources of value in a merger and distinguish them from bad reasons to merge.
2. Compute the gain from an acquisition, the maximum price an acquirer should pay, and how the gain is divided.
3. Compare payment in cash and in shares, and explain why earnings-per-share accretion is not value creation.
4. Infer market-implied deal probabilities from merger-arbitrage spreads.
5. Explain the mechanics of a leveraged buyout, decompose its returns, and show how leverage magnifies gains and losses.
6. Describe the economics of private-equity funds: fees, carried interest, the J-curve, and performance measures including the public market equivalent.

A motivating question

In May 1998, Daimler-Benz, maker of Mercedes cars, and Chrysler, the third-largest American carmaker, announced a “merger of equals” worth about 36 billion dollars. The combination was expected to produce billions in savings and a global champion. Nine years later, Daimler sold most of Chrysler to a private-equity firm for a fraction of what it had paid; two years after that, Chrysler was in bankruptcy.

Mergers and acquisitions are among the largest investment decisions firms make, and the evidence on them is sobering: target shareholders usually gain handsomely, while acquirers’ shareholders gain little on average and often lose. Private-equity firms, which buy companies with borrowed money and sell them a few years later, have built an industry managing trillions of dollars on the claim that they can do better. This chapter applies the tools of valuation (Chapter 37) and the cost of capital (Chapter 35) to both.

38.1 Why Merge?

38.1.1 Good reasons

A merger creates value if the combined firm is worth more than the two firms apart. The difference is the **synergy**:

$$\text{Synergy} = V_{AB} - (V_A + V_B).$$

Its sources include:

- **Cost synergies:** eliminating duplicated head offices, plants and systems; economies of scale in purchasing and production (Chapter 21).

- **Revenue synergies:** selling each firm's products through the other's channels, or combining complementary technologies. These are harder to achieve and are usually forecast less reliably than cost savings.
- **Better management:** replacing a poorly run target's management, or using its assets more productively.
- **Tax and financing effects:** using the target's tax losses, or increasing debt capacity (Chapter 35).
- **Market power:** reducing competition, raising prices. This transfers value from customers rather than creating it, and competition authorities screen mergers to prevent it (Chapter 24).

38.1.2 Bad reasons

Several popular arguments do not survive analysis.

- **Diversification.** A conglomerate spanning unrelated industries is less risky than its parts, but shareholders can diversify on their own at almost no cost (Chapter 14). They will not pay a premium for diversification they can obtain for themselves.
- **Earnings-per-share growth.** Acquiring a firm with a lower price–earnings ratio in exchange for shares raises EPS mechanically, without creating value (Example 38.3).
- **Growth for its own sake.** Managers may value size—which brings pay, status and security—more than shareholders do (the agency problem of Chapter 25). Richard Roll (1986) added the *hubris hypothesis*: acquirers overestimate synergies and their own ability, and so overpay.

38.2 Value and Price

38.2.1 How the gain is divided

The acquirer's net gain is the synergy less the **premium** it pays over the target's standalone value, and less the costs of the deal:

$$\text{NPV to acquirer} = \text{Synergy} - \text{Premium} - \text{Transaction and integration costs.}$$

The maximum price the acquirer should pay is the target's standalone value plus the synergy, less costs; any lower price shares the gain with the acquirer's shareholders.

Worked Example 38.1. The walk-away price

A target is worth 100 on its own. Combining it with the acquirer is expected to produce synergies with a present value of 25, and integration will cost 8. The most the acquirer can pay without destroying value is $100 + 25 - 8 = 117$. If it pays 110, the target's shareholders receive a premium of 10 and the acquirer's shareholders gain $25 - 10 - 8 = 7$.

The evidence suggests that competition for targets and the hubris of bidders push prices toward, and sometimes beyond, the walk-away price. Premia in US public acquisitions have averaged of the order of 30 to 40 per cent above pre-announcement prices. Targets' shareholders earn large abnormal returns around the announcement; acquirers' shareholders earn close to zero on average, with wide variation and poor results for the largest deals. Sara Moeller,

Frederik Schlingemann and René Stulz (2005) estimated that acquirers' shareholders lost about 240 billion dollars in announcement-period returns on deals between 1998 and 2001, concentrated in a small number of very large acquisitions. Combined announcement returns of acquirer and target are modestly positive on average: mergers create some value, but most of it goes to the sellers.

38.2.2 Cash or shares?

An acquirer can pay in cash or in its own shares. With cash, the acquirer's shareholders bear all the risk that synergies disappoint or that the target was overvalued. With shares, the target's shareholders become shareholders of the combined firm and share that risk—and share in the synergies.

Worked Example 38.2. Paying with cash or shares

An acquirer has 100 shares at 50 (value 5,000); the target is worth 1,000 on its own; synergies are worth 300. *Cash offer of 1,200*: the acquirer's shareholders own a firm worth $5,000 + 1,000 + 300 - 1,200 = 5,100$, a gain of 100. *Share offer of 24 new shares* (worth 1,200 at the pre-deal price): the combined firm is worth 6,300 and has 124 shares, each worth 50.81. The target's shareholders receive $24 \times 50.81 = 1,219$, not 1,200, and the acquirer's shareholders gain only 81. With shares, the true cost depends on the post-deal price, which includes the synergies the target's holders now share.

The choice also conveys information. Managers who believe their shares are overvalued prefer to pay with them, and markets know it (the adverse-selection logic of Chapter 25): acquirers' announcement returns are, on average, lower for share offers than for cash offers.

38.2.3 Accretion and dilution

Analysts routinely report whether a deal is **accretive** (raises the acquirer's EPS) or **dilutive**. The measure is widely watched and largely uninformative about value.

Worked Example 38.3. EPS bootstrapping

In Example 38.2, suppose the acquirer earns 250 (EPS 2.50, P/E 20) and the target earns 100 (P/E 10), and there are no synergies. After the share offer, combined earnings are 350 on 124 shares: EPS rises to 2.82, an accretion of 13 per cent. Yet the acquirer has paid 1,200 for a firm worth 1,000: its shareholders own 100/124 of a combined firm worth 6,000, that is 4,839, less than the 5,000 they had. EPS rose while value fell.

Whenever an acquirer with a high P/E buys a target with a lower one using shares, the deal is accretive—regardless of whether it creates value. The combined earnings deserve a P/E between the two, not the acquirer's higher one; if the market applies the acquirer's P/E to them anyway, it is making a mistake. In the 1960s, conglomerates such as Ling-Temco-Vought grew rapidly by exploiting this arithmetic until their P/E ratios collapsed.

38.2.4 Accounting for the deal

In the consolidated accounts, the acquisition method records the target's identifiable assets and liabilities at fair value, and the excess of the price over them as goodwill (Chapter 29). Goodwill is not amortized but tested for impairment; a large impairment a few years after a deal is often the accounting acknowledgement that the acquirer overpaid. The largest example

remains AOL Time Warner, which recorded a loss of about 99 billion dollars in 2002, mostly from writing down goodwill created in the 2001 merger of AOL and Time Warner.

38.2.5 Merger arbitrage

Once a deal is announced, the target's shares usually trade below the offer price, because the deal might fail—through a competition ruling, a financing problem or a shareholder vote. Specialist investors buy targets and, in share deals, sell acquirers short to earn the spread, a strategy called **merger arbitrage**. The spread reveals the market's estimate of the probability of completion.

Worked Example 38.4. An implied probability

A cash offer of 50 a share is expected to close shortly. The target's shares trade at 47; if the deal failed they would be expected to fall to 35. Ignoring the time value of money and any risk premium, the market price is a probability-weighted average (Chapter 13): $47 = 50p + 35(1 - p)$, so $p = 12/15 = 0.8$. The arbitrageur earns 3 (6 per cent) with probability 0.8 and loses 12 with probability 0.2.

The payoff—small gains usually, large losses occasionally—resembles selling insurance, and merger-arbitrage returns tend to fall sharply when markets as a whole fall and deals are abandoned.

38.3 Leveraged Buyouts

38.3.1 Mechanics

In a **leveraged buyout** (LBO), a private-equity firm buys a company using a large amount of debt—often half to two-thirds of the price—secured on the company's assets and repaid from its cash flows. After several years it sells the company, lists it, or sells it to another fund. The buyer's return comes from three sources: growth in operating profit, a change in the valuation multiple, and the repayment of debt, which transfers value from lenders to the equity.

Worked Example 38.5. An LBO

A company with EBITDA of 12.5 is bought for 8 times EBITDA, an enterprise value of 100, financed with 60 of debt and 40 of equity. Over three years, EBITDA grows to 15 and debt is paid down to 45 from free cash flow. At exit, the company is sold at the same multiple of 8, for 120. The equity is worth $120 - 45 = 75$: a **multiple of invested capital** (MOIC) of $75/40 = 1.875$ and an internal rate of return of $1.875^{1/3} - 1 = 23.3$ per cent a year.

The gain of 35 decomposes into EBITDA growth ($2.5 \times 8 = 20$), multiple expansion (zero) and debt paydown (15). Without leverage, the same deal would turn 100 into 120: a MOIC of 1.2 and an IRR of 6.3 per cent.

Leverage magnifies losses as well as gains. If the company in the example were sold for 80, the equity would be worth 35—a loss of 12.5 per cent, against 20 per cent without leverage—but at 70 the equity would lose 37.5 per cent against 30 per cent unlevered, and below 45 it would be wiped out. This is the Modigliani–Miller logic of Chapter 30: leverage raises the expected return on equity by raising its risk. An LBO's high IRR is not in itself evidence of skill; the question is whether it exceeds what a similarly levered investment in public shares would have earned.

38.3.2 Sources of value

Michael Jensen (1989) argued that buyouts solve an agency problem of public companies with plentiful cash and few good investments. Heavy debt commits cash flow to lenders rather than to managers' pet projects; managers are given large equity stakes; and a concentrated owner monitors them closely. Studies of buyouts find operating improvements in many cases, especially in the buyouts of the 1980s. Critics point to other sources of returns: tax shields from the debt (Chapter 35); buying cheaply from sellers with less information; and transfers from employees, pre-existing creditors whose bonds lose value as leverage rises, and taxpayers. The relative importance of creation and transfer varies across deals and remains debated.

The risks are those of any highly levered firm (Chapter 39): a downturn, a rise in interest rates on floating-rate debt, or an operating setback can leave the company unable to service its debt. The largest buyout of its era, KKR's acquisition of RJR Nabisco for about 25 billion dollars, completed in 1989 after a bidding contest, produced disappointing returns for its buyers.

38.4 Private-Equity Funds

38.4.1 Structure and fees

Buyouts are made by **private-equity funds**: limited partnerships in which institutional investors (the *limited partners*, LPs, such as pension funds and endowments) commit capital for about ten years to a manager (the *general partner*, GP). The GP draws down the commitments as it finds deals and returns cash as it sells them. Its compensation traditionally has two parts:

- a **management fee**, often around 2 per cent a year of committed capital during the investment period, falling afterwards;
- **carried interest**, typically 20 per cent of the fund's profits, usually payable only after LPs have received their capital back plus a *hurdle* return, often 8 per cent a year.

Worked Example 38.6. Gross and net returns

A fund raises commitments of 1,000. Over its life, management fees total 200, so 800 is invested. The investments are sold for 2,000: a gross MOIC of 2.5. Profit over committed capital is $2,000 - 1,000 = 1,000$, of which the GP's carried interest is 20 per cent, 200. LPs receive 1,800 on the 1,000 they paid in: a net MOIC of 1.8. Fees and carry take 0.7 of the 2.5 gross multiple.

38.4.2 Measuring performance

Private-equity performance is hard to measure. Investments are not traded, so interim values are appraisals, which tend to be smoothed and slow to reflect market falls. The fund's cash flows follow a **J-curve**: negative in the early years, as capital is drawn and fees paid, turning positive only as investments are sold. Three measures are common:

- the **MOIC** (or TVPI, total value to paid-in capital), which ignores timing;
- the **IRR** of the LPs' cash flows, which accounts for timing but can be flattered by early exits and by borrowing against LP commitments to delay drawing their capital (*subscription lines*);
- the **public market equivalent** (PME) of Steven Kaplan and Antoinette Schoar (2005), which discounts the fund's cash flows at the return on a public index: the ratio of the

index-discounted distributions to the index-discounted contributions. A PME above one means the fund beat the index.

Worked Example 38.7. IRR, MOIC and PME

An LP invests 100 and receives 200 five years later: a MOIC of 2 and an IRR of 14.9 per cent. If a public share index rose from 100 to 180 over the same five years, the PME is $(200/1.8)/100 = 1.11$: the fund beat the index by 11 per cent over the period, not by the 100 per cent that the MOIC might suggest. A different fund that returned 1.5 times its money in one year has an IRR of 50 per cent but a lower MOIC; which is better depends on what the LP can do with the money in the remaining years.

Studies using PMEs, such as Robert Harris, Tim Jenkinson and Steven Kaplan (2014), found that US buyout funds on average outperformed public equity indices, net of fees, for most vintages before the mid-2000s, with a wide spread between the best and worst funds. Performance in more recent vintages has been closer to public markets. The comparison is sensitive to the benchmark: buyout funds hold smaller, more levered firms than the average listed company, so a fair benchmark should also be smaller and more levered, which narrows the measured outperformance. Private equity's returns, adjusted for leverage and illiquidity, are thus a matter of continuing research.

Case: DaimlerChrysler, 1998–2007

The deal. In May 1998 Daimler-Benz and Chrysler announced their combination as a “merger of equals”, with Daimler-Benz shareholders owning a little over half of the combined company, DaimlerChrysler, which was completed in November. The strategic case was complementary geography and products—Daimler's premium cars and European base, Chrysler's American trucks, minivans and Jeeps—and planned synergies of about 1.4 billion dollars in the first year, rising to about 3 billion.

What happened. Control quickly passed to Stuttgart; in 2000 Daimler's chief executive, Jürgen Schrempp, was quoted as saying that the “merger of equals” description had been adopted largely for psychological reasons, prompting a lawsuit by Chrysler's former largest shareholder, the investor Kirk Kerkorian, which was later dismissed. Many of Chrysler's senior managers left. The two companies' cultures, pay structures and engineering processes proved hard to integrate, and the promised synergies in shared components were slow to arrive. Chrysler suffered large losses in 2001 and again in 2006, when high petrol prices hit demand for its trucks and sport-utility vehicles. In 2007 Daimler sold 80.1 per cent of Chrysler to Cerberus Capital Management, a private-equity firm, in a transaction valued at about 7.4 billion dollars, most of which was invested in Chrysler itself rather than paid to Daimler. Under Cerberus, Chrysler entered the financial crisis highly exposed; it filed for Chapter 11 in April 2009 and emerged in an alliance with Fiat, backed by the US and Canadian governments.

In terms of this chapter. The deal illustrates almost every theme: synergies estimated confidently and realized slowly, especially in revenue and product development; a price that assumed success; integration costs—in people and culture as well as money—that were hard to foresee; and a sale that valued the business at a fraction of the purchase price, crystallizing a loss that Daimler's shareholders had long borne. The Cerberus sequel illustrates the other side: leverage and a highly cyclical business in a recession leave little room for error (Chapter 39).

Summary

- Mergers create value through synergies: cost savings, revenue gains, better management and tax effects. Market power transfers value from customers. Diversification and EPS growth are not good reasons to merge.
- The acquirer's NPV is synergy minus premium minus costs; the walk-away price is standalone value plus synergy minus costs. Target shareholders capture most of the gains on average.
- Paying with shares shares the synergies and the risk of overpayment with the target's holders, and may signal that the acquirer's shares are overvalued.
- EPS accretion is mechanical when a high-P/E acquirer buys a lower-P/E target with shares; it is not evidence of value creation.
- Merger-arbitrage spreads imply a probability of completion; the strategy resembles selling insurance.
- LBO returns come from EBITDA growth, multiple changes and debt paydown; leverage magnifies both gains and losses.
- Private-equity funds charge management fees and carried interest; performance should be measured net of fees and against public markets (PME), with attention to leverage and smoothing.

Key terms

synergy; cost synergy; revenue synergy; hubris hypothesis; acquisition premium; walk-away price; cash offer; share offer; exchange ratio; accretion; dilution; EPS bootstrapping; goodwill impairment; merger arbitrage; implied probability of completion; leveraged buyout; multiple of invested capital; value creation bridge; free cash flow hypothesis; private-equity fund; limited partner; general partner; management fee; carried interest; hurdle rate; J-curve; TVPI; subscription line; public market equivalent.

Exercises

Exercise 38.1 (Walk-away price). A target has a standalone value of 400. Synergies have a present value of 90; integration will cost 20 and advisers 10. Find the walk-away price. If the acquirer pays 430, how is the gain divided?

Exercise 38.2 (Cash or shares). An acquirer has 200 shares at 30; the target is worth 1,500 standalone; synergies are worth 500. Compare the acquirer's gain from a cash offer of 1,800 and from an offer of 60 new shares.

Exercise 38.3 (Bootstrapping). An acquirer earns 400 and has 100 shares at 80. It buys, with new shares, a target earning 100 and valued at 1,200, paying a 25 per cent premium. With no synergies, find the new EPS and the change in the value of the acquirer's original shareholders' stake.

Exercise 38.4 (Merger arbitrage). A target trades at 46 after a cash offer of 52. If the deal fails, the shares are expected to trade at 38. Find the implied probability of completion. If you believed the probability were 0.9, what would the shares be worth to you?

Exercise 38.5 (LBO returns). A company with EBITDA of 50 is bought at 9 times EBITDA with debt of 300. After five years, EBITDA is 65 and debt 180. Find the MOIC and IRR if the exit multiple is 9, and if it is 7. Decompose the equity gain in the first case.

Exercise 38.6 (Leverage). For the company in Exercise 38.5, compute the MOIC and IRR of an all-equity purchase with exit at 9 times. What explains the difference?

Exercise 38.7 (Coverage). In Exercise 38.5, the debt pays 8 per cent. Compute the initial ratio of EBITDA to interest. What would it be if EBITDA fell by 30 per cent in the first year? What might happen then?

Exercise 38.8 (Fund economics). A fund has commitments of 500. Fees total 100 and 400 is invested, returning 1,100. Carried interest is 20 per cent of profits over committed capital. Find the gross MOIC, the carry and the LPs' net MOIC.

Exercise 38.9 (IRR versus MOIC). Fund *A* returns 1.6 times its money in two years; fund *B* returns 2.5 times in seven years. Compute each IRR. Which would you prefer, and on what does the answer depend?

Exercise 38.10 (PME). An LP contributes 100 at time 0 and 50 at time 1, and receives 250 at time 5. A public index stands at 100 at time 0, 110 at time 1 and 170 at time 5. Compute the PME and interpret it.

Solutions

Solution 38.1. Walk-away price = $400 + 90 - 20 - 10 = 460$. At 430, the target's holders receive a premium of 30 and the acquirer's gain is $90 - 30 - 30 = 30$.

Solution 38.2. Cash: acquirer's gain = $500 - 300 = 200$. Shares: combined value $6,000 + 1,500 + 500 = 8,000$ on 260 shares, 30.77 each; the target receives $60 \times 30.77 = 1,846$; the acquirer's holders own $200 \times 30.77 = 6,154$, a gain of 154.

Solution 38.3. Price $1,500 = 18.75$ new shares at 80. EPS rises from 4.00 to $500/118.75 = 4.21$. Combined value $8,000 + 1,200 = 9,200$; original holders own $100/118.75$ of it, 7,747, a loss of 253 (roughly the 300 premium, less the part borne by the new shareholders).

Solution 38.4. $46 = 52p + 38(1 - p)$: $p = 8/14 = 0.57$. At $p = 0.9$, value = $0.9 \times 52 + 0.1 \times 38 = 50.6$ (before time value and risk adjustment): the shares would look cheap to you.

Solution 38.5. Entry $EV = 450$, equity 150. Exit at 9: $EV = 585$, equity 405, MOIC 2.70, $IRR\ 2.7^{0.2} - 1 = 22.0$ per cent. Exit at 7: $EV = 455$, equity 275, MOIC 1.83, IRR 12.9 per cent. Decomposition at 9: EBITDA growth $15 \times 9 = 135$; multiple change 0; debt paydown 120; total $255 = 405 - 150$.

Solution 38.6. $MOIC\ 585/450 = 1.30$; IRR 5.4 per cent. Leverage turns a 30 per cent gain in enterprise value, plus the cash flow used to repay debt, into a 170 per cent gain on a much smaller equity stake—at the cost of much greater risk.

Solution 38.7. Interest $0.08 \times 300 = 24$; $EBITDA/interest = 50/24 = 2.1$. After a 30 per cent fall, $35/24 = 1.46$; after capital expenditure and taxes, cash flow may not cover interest and scheduled repayments, covenants may be breached, and the owners may have to inject equity or negotiate with lenders (Chapter 39).

Solution 38.8. Gross MOIC $1,100/400 = 2.75$. Profit over commitments $1,100 - 500 = 600$; carry 120. LPs receive 980: net MOIC $980/500 = 1.96$.

Solution 38.9. *A*: $1.6^{1/2} - 1 = 26.5$ per cent. *B*: $2.5^{1/7} - 1 = 14.0$ per cent. If the LP can reinvest *A*'s proceeds at more than about 9.3 per cent a year for the remaining five years (since $1.6 \times 1.093^5 \approx 2.5$), *A* is better; otherwise *B* delivers more wealth. IRR alone does not decide.

Solution 38.10. Discounted distributions $250/1.70 = 147.1$; discounted contributions $100 + 50/1.10 = 145.5$; $PME = 1.01$. The fund roughly matched the index: the LP would have done about as well by investing the same cash flows in public shares.

Credit Analysis, Distress and Restructuring

After studying this chapter you should be able to

1. Explain why lenders focus on the downside, and assess a borrower with the “five Cs” and credit ratios.
2. Design and monitor covenants, and compute covenant headroom.
3. Compute expected loss from the probability of default, loss given default and exposure, and price a loan.
4. Show how default correlation fattens the tail of portfolio losses.
5. Use rating transition matrices to compute multi-year default probabilities, and estimate default risk with structural, reduced-form and statistical models.
6. Explain the costs of financial distress, including debt overhang and risk shifting.
7. Describe out-of-court and court-supervised restructuring, and apply the absolute priority rule to allocate value among creditors.

A motivating question

In 2021 China Evergrande Group, then the most indebted property developer in the world, with liabilities of about 2 trillion yuan (some 300 billion dollars), stopped paying its creditors. It owed money to banks, bondholders at home and abroad, suppliers who had accepted its IOUs, and more than a million home buyers who had paid for flats that had not been built. Its shares, once worth tens of billions of dollars, became almost worthless; in January 2024 a Hong Kong court ordered it to be wound up.

How should a lender judge whether a borrower will pay, and how much to charge? And when a borrower cannot pay, who gets what? This chapter develops the analysis of credit, building on the credit spreads and ratings of Chapter 33, the capital structure theory of Chapters 30 and 35, and the probability of Part II.

39.1 The Lender's Problem

39.1.1 An asymmetric payoff

A shareholder gains without limit if a business thrives. A lender gains at most the promised interest and loses up to the whole loan if the borrower fails. Credit analysis is therefore about the downside: not the most likely outcome but the bad ones, and what the lender recovers in them. The asymmetry also creates the information problems of Chapter 25. Raising the interest rate may attract riskier borrowers (adverse selection) and encourage borrowers to take more

risk (moral hazard), so that lenders may prefer to ration credit rather than raise its price, as Joseph Stiglitz and Andrew Weiss (1981) showed.

39.1.2 The five Cs

Bankers traditionally organize their judgement around five questions:

- **Character:** the borrower's integrity, record and governance—will it pay if it can?
- **Capacity:** its cash flow relative to its debt service—can it pay?
- **Capital:** its equity cushion—how much can it lose before creditors do?
- **Collateral:** assets pledged to the lender and their value in a forced sale.
- **Conditions:** the economy, the industry and the purpose of the loan.

39.1.3 Credit ratios

Capacity and capital are measured by ratios computed from the accounts (Chapter 31):

- **Leverage:** debt or net debt (debt minus cash) divided by EBITDA, the number of years of operating cash flow needed to repay the debt.
- **Coverage:** EBITDA or EBIT divided by interest expense, the number of times interest is earned.
- **Cash-flow ratios:** funds from operations or free cash flow divided by debt.
- **Liquidity:** cash and undrawn committed facilities relative to debt falling due within a year.

Rating agencies publish the ranges typical of each rating in each industry. Roughly, net debt below twice EBITDA and coverage above eight times are associated with strong investment-grade credits; net debt above five or six times EBITDA, with high-yield credits typical of leveraged buyouts (Chapter 38). Stable businesses, such as regulated utilities, can support more debt than cyclical ones.

39.1.4 Covenants

A loan agreement contains promises by the borrower, or **covenants**. *Affirmative* covenants require actions (providing accounts, maintaining insurance). *Negative* covenants restrict them (no further secured borrowing, no sale of key assets, limits on dividends). *Financial* covenants set limits on ratios. *Maintenance* covenants must be met at every test date, usually quarterly; *incurrence* covenants are tested only when the borrower takes a specified action, such as borrowing more or paying a dividend. A breach gives lenders the right to demand repayment—in practice, the right to renegotiate, often for a fee, a higher rate or tighter terms. Covenants are thus an early-warning system, which is why loans with few maintenance covenants, known as *covenant-lite*, give lenders less protection.

Worked Example 39.1. Covenant headroom

A company has debt of 540, cash of 60 and EBITDA of 120. Its loan requires net debt to EBITDA of at most 4.5 times. Currently, net debt is 480 and the ratio is $480/120 = 4.0$. Holding net debt constant, the covenant is breached if EBITDA falls below $480/4.5 = 106.7$, a decline of 11 per cent. Headroom measured in EBITDA is the useful figure: a fall of 11 per cent is well within the range of an ordinary recession for a cyclical firm.

39.2 Measuring Credit Risk

39.2.1 Expected loss

The **expected loss** on a loan over a period is

$$EL = PD \times LGD \times EAD,$$

where PD is the **probability of default**, LGD the **loss given default** (one minus the recovery rate, as a share of the exposure) and EAD the **exposure at default**—the amount owed at default, which for a credit line includes the part the borrower is likely to draw as it approaches trouble. A loan of 100 with PD of 4 per cent and LGD of 60 per cent has an expected loss of $0.04 \times 0.6 \times 100 = 2.4$.

Expected loss is a cost of doing business, to be covered by the interest rate. A bank prices a loan to cover, in addition, its funding cost, its operating costs and a return on the capital it must hold against *unexpected* losses:

$$\text{Loan rate} \approx \text{funding cost} + \frac{EL}{EAD} + \text{operating cost} + \text{capital} \times (\text{cost of equity} - \text{funding cost}).$$

Under accounting standards (IFRS 9 and the US CECL model), banks must also provide for expected credit losses on their loans from the start, rather than waiting for losses to occur (Chapter 28).

39.2.2 Portfolios and correlation

A bank holds many loans, and the risk of the portfolio depends on how defaults are correlated (Chapter 14). If defaults were independent, the law of large numbers would make portfolio losses highly predictable. They are not: defaults rise together in recessions, in particular industries and regions, and when property prices fall.

Worked Example 39.2. Correlation fattens the tail

A bank has 100 loans, each with a 2 per cent probability of default. *Independent defaults:* the number of defaults is binomial (Chapter 13), with mean 2; the probability of six or more is 1.5 per cent, and of eleven or more about 0.0006 per cent. *Correlated defaults:* suppose instead the economy is good with probability 0.8, when each loan defaults with probability 1 per cent, and bad with probability 0.2, when each defaults with probability 6 per cent. The average default probability is still $0.8 \times 1 + 0.2 \times 6 = 2$ per cent, and expected defaults are still 2. But the probability of six or more defaults is now 11 per cent, and of eleven or more 0.75 per cent—more than a thousand times the independent figure.

Expected losses are the same; the tail is not. Capital exists for the tail, which is why regulators' capital formulas (the Basel framework) depend on assumed default correlations, and why mistaken assumptions about correlation lay at the heart of the losses on mortgage securities in 2007–09 (Chapter 51).

39.2.3 Rating migration

Default is the end of a process. Chapter 33 noted that ratings migrate; a **transition matrix** records the probabilities of moving from each rating to each other over a year. Treating the ratings as the states of a Markov chain (Chapter 8), the probabilities over two years are given by the square of the matrix, over n years by its n th power.

Worked Example 39.3. Two-year default probabilities

With three states—investment grade (IG), high yield (HY) and default (D), which is absorbing—suppose the annual transition matrix is

$$P = \begin{pmatrix} 0.93 & 0.065 & 0.005 \\ 0.05 & 0.90 & 0.05 \\ 0 & 0 & 1 \end{pmatrix},$$

with rows giving today's state and columns next year's. The probability that an IG borrower has defaulted within two years is the (IG, D) entry of P^2 : default in the first year (0.005), or stay IG then default (0.93×0.005), or fall to HY then default (0.065×0.05):

$$0.005 + 0.00465 + 0.00325 = 0.0129.$$

The two-year probability is more than twice the one-year probability of 0.005, because a borrower can first be downgraded and then default from a riskier state. Over five years, the matrix gives cumulative default probabilities of 4.9 per cent for IG and 20.8 per cent for HY borrowers.

The same logic explains why cumulative default rates published by rating agencies rise more than proportionally with the horizon for high-grade issuers. Real transition probabilities are not constant: downgrades cluster in recessions, violating the Markov assumption of fixed probabilities.

39.2.4 Models of default

Structural models. Robert Merton (1974) observed that shareholders of a firm with debt have a call option on its assets (Chapter 41): at the debt's maturity, they pay the face value D and keep the assets if the assets are worth more; otherwise they walk away, and the lenders take the assets. Default occurs when the value of the assets V falls below D . If $\ln V$ is normally distributed (Chapter 13), with expected growth μ and volatility σ , the probability of default by time T is $\Phi(-DD)$, where the **distance to default** is

$$DD = \frac{\ln(V/D) + (\mu - \sigma^2/2)T}{\sigma\sqrt{T}},$$

the number of standard deviations by which the assets' expected log value exceeds the default point.

Worked Example 39.4. Distance to default

A firm's assets are worth 100, with volatility 25 per cent and expected return 6 per cent; debt of 70 is due in a year. Then $DD = [\ln(100/70) + 0.06 - 0.03125]/0.25 = (0.3567 + 0.0288)/0.25 = 1.54$, and the probability of default is $\Phi(-1.54) = 6.2$ per cent. If asset volatility rose to 35 per cent, the probability would rise to about 15 per cent.

The model links credit risk to equity: a falling share price or rising equity volatility signals rising default risk. Commercial versions, such as Moody's KMV, infer asset values and volatilities from share prices and map distances to default into probabilities using historical default data rather than the normal distribution, whose tails are too thin (Chapter 34).

Reduced-form models. Instead of modelling why default happens, these treat it as a random event arriving at a **hazard rate** λ per year. The probability of surviving t years is $e^{-\lambda t}$ (Chapter 4).

Since the spread compensates for expected loss, $\lambda \approx \text{spread}/\text{LGD}$: a spread of 3 per cent with an LGD of 60 per cent implies a risk-neutral hazard rate of 5 per cent, and a five-year survival probability of $e^{-0.25} = 78$ per cent. Such market-implied probabilities include risk premia and exceed actual default frequencies (Chapter 33).

Statistical models. Credit scoring predicts default from borrower characteristics by logit regression or machine learning (Chapter 19). The most famous early model is Edward Altman's (1968) *Z-score* for US listed manufacturers,

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5,$$

where X_1 is working capital, X_2 retained earnings, X_3 EBIT and X_5 sales, each divided by total assets, and X_4 is the market value of equity divided by total liabilities. Firms with Z below 1.81 were classed as likely to fail, above 2.99 as safe. A firm with ratios 0.10, 0.20, 0.08, 0.9 and 1.1 has $Z = 0.12 + 0.28 + 0.264 + 0.54 + 1.1 = 2.30$: in the grey zone. The weights were estimated on a small sample of firms in the 1940s to 1960s and should not be applied mechanically to other industries, countries or periods.

39.3 Financial Distress

39.3.1 The costs of distress

A firm is in **financial distress** when it has difficulty meeting its obligations. Distress is costly even before any default. *Direct* costs—lawyers, advisers, courts—are typically a few per cent of firm value for large firms. *Indirect* costs are larger and harder to measure: customers who fear for warranties and spare parts go elsewhere, suppliers demand cash on delivery, the best employees leave, and managers spend their time on creditors rather than the business. These costs are the counterweight to the tax benefits of debt in the trade-off theory (Chapter 35).

39.3.2 Conflicts between shareholders and creditors

Distress also distorts decisions, because shareholders of a firm near default are, in effect, gambling with creditors' money.

Worked Example 39.5. Debt overhang

A firm's assets are worth 90 with certainty, and it owes 100, due in a year. Its equity is worthless today. A project costs 10, to be provided by shareholders, and returns 15 in a year for certain: its NPV is positive. With the project, the firm will have 105; creditors take 100 and shareholders 5. Shareholders would invest 10 to receive 5, so they refuse: the gain goes to creditors. This is **debt overhang** (Stewart Myers, 1977): debt can prevent profitable investment.

Risk shifting is the mirror image. Suppose instead the firm can use its assets to make a gamble that pays 140 or 20 with equal probability (expected value 80, less than 90). Shareholders receive 40 or 0, worth 20 in expectation, rather than nothing: they prefer the gamble, though it destroys value, because creditors bear the downside.

Both problems make renegotiation valuable. In the overhang example, if creditors reduced their claim to 90, shareholders would invest (receiving 15 for 10) and creditors would still receive 90, what they would get anyway; the gain from the project would be shared. Much of the law and practice of restructuring is about making such bargains possible.

39.4 Restructuring

39.4.1 Out of court

A distressed firm first tries to renegotiate privately: extending maturities, reducing interest, exchanging debt for equity. Private workouts are cheaper and faster than court proceedings, and work well when debt is held by a few banks. With widely held bonds, they face a **holdout problem**: each bondholder prefers that others accept a reduction while it is paid in full, so the exchange may fail even if all creditors would gain from its success. Bond contracts increasingly include *collective action clauses* allowing a majority to bind a minority, a device also written into government bonds, whose crises are discussed in Chapter 52.

39.4.2 In court

When private negotiation fails, the law provides a collective procedure. In the United States, **Chapter 11** of the Bankruptcy Code allows a firm to reorganize while continuing to operate, usually under existing management:

- an **automatic stay** stops creditors from seizing assets or pursuing claims, preventing a destructive race to the courthouse;
- **debtor-in-possession** (DIP) financing, given priority over existing claims, lets the firm keep operating;
- creditors vote, class by class, on a **plan of reorganization**; the court can confirm a plan over the objection of a dissenting class (a **cram-down**) if the class receives at least what it would in liquidation and the plan respects priority.

Chapter 7 provides instead for liquidation. Other jurisdictions have moved toward similar tools: the United Kingdom's restructuring plan (introduced in 2020) allows cross-class cram-down, and Germany's StaRUG (2021) provides a court-supported restructuring framework outside insolvency.

39.4.3 The waterfall

The value of a reorganized or liquidated firm is distributed by priority. Under the **absolute priority rule**, each class is paid in full before the next receives anything: administrative and DIP claims first, then secured creditors (up to the value of their collateral), then senior unsecured, then subordinated creditors, and shareholders last.

Worked Example 39.6. Allocating value

A firm emerges from Chapter 11 with an enterprise value of 420. Its claims are DIP financing of 50, secured loans of 250, senior unsecured bonds of 200 and subordinated notes of 100. The DIP lenders and secured lenders are paid in full (300); the remaining 120 goes to the senior unsecured bonds, a recovery of $120/200 = 60$ per cent. The subordinated notes and the shares receive nothing. The senior unsecured bonds are the **fulcrum security**—the class in which value runs out—and typically receive most of the reorganized firm's equity. Investors who buy distressed debt aim to own the fulcrum security.

In practice, deviations from absolute priority are common: junior classes and even shareholders sometimes receive small amounts to secure their agreement and avoid delay. Historical recovery rates reflect priority: studies by rating agencies have found average recoveries of the

order of 60 to 70 per cent for secured bank loans, around 35 to 40 per cent for senior unsecured bonds, and less for subordinated debt, with recoveries lower in recessions, when many firms default at once and asset prices are depressed.

39.4.4 Private credit

Since the global financial crisis, a growing share of lending to mid-sized, often private-equity-owned companies has moved from banks to **private credit** funds, which lend directly, typically at floating rates, and hold the loans to maturity. The market grew to well over a trillion dollars. Direct lenders offer speed, certainty and flexibility, and a single lender or small group makes restructuring easier than with dispersed bonds. The risks are less visible: the loans are not traded, their valuations are appraisals, and the funds' own borrowing and links to banks and insurers make the system's exposure harder to see—questions that concern central banks (Chapter 51).

Case: China Evergrande

The business model. Evergrande grew from its founding in 1996 into one of China's largest developers, riding the country's urbanization and housing boom. Like many Chinese developers, it financed construction largely with buyers' money: flats were sold before completion, and the pre-sale receipts—contract liabilities in the accounts (Chapter 28)—funded the purchase of land for further projects. It also borrowed from banks and trust companies, issued bonds in China and in US dollars offshore, and paid suppliers with commercial paper. Growth required ever more land and pre-sales; the model worked as long as sales and prices rose.

The trigger. In August 2020 Chinese regulators, concerned about developers' debts, introduced the “three red lines”: limits on liabilities relative to assets (excluding advance receipts), net debt relative to equity, and cash relative to short-term debt. Developers breaching all three could not increase their borrowing. Evergrande breached them. Cut off from new credit, it cut prices to raise cash, which depressed sales across the market; suppliers went unpaid, construction stopped on many projects, and in September 2021 it missed interest payments on offshore bonds. In December 2021 Fitch declared it in restricted default.

What followed. Evergrande's accounts for 2021 and 2022, published only in 2023, showed combined losses of more than 580 billion yuan. A plan to restructure its offshore debt collapsed in 2023, and its founder, Hui Ka Yan, was detained by the authorities that September. On 29 January 2024 the High Court in Hong Kong, where the company was listed, ordered it to be wound up. In March 2024 China's securities regulator found that its main onshore subsidiary had overstated revenue by about 564 billion yuan in 2019 and 2020, and fined the company and banned its founder from the securities markets for life. Recoveries for offshore creditors were expected to be very low, and the court-appointed liquidators faced the difficulty of enforcing a Hong Kong order over assets mostly in mainland China.

In terms of this chapter. Every credit tool in this chapter points to warnings. Leverage and liquidity ratios were weak once pre-sale liabilities and off-balance-sheet obligations were counted. The lenders' claims were exposed to a correlated shock—a fall in the whole property market, which affected every project at once (Example 39.2). Offshore bondholders were structurally subordinated: their claims were on a Cayman Islands holding company, while the assets and the more senior local creditors, including home buyers, were in China, so the waterfall put them near the bottom. And the overstatement of revenue shows why the character of a borrower, the first of the five Cs, is not a

formality.

Summary

- Lenders' payoffs are capped, so credit analysis focuses on the downside and on recoveries. Information problems can lead to credit rationing.
- The five Cs organize judgement; leverage, coverage, cash-flow and liquidity ratios measure capacity and capital. Covenants provide early warning; headroom is best measured in EBITDA.
- $EL = PD \times LGD \times EAD$. Loan pricing covers funding, expected loss, operating costs and a return on capital held against unexpected loss.
- Default correlation leaves expected losses unchanged but fattens the tail of portfolio losses dramatically.
- Rating transition matrices are Markov chains: multi-year default probabilities are powers of the matrix.
- Structural models treat equity as a call option and measure distance to default; reduced-form models use hazard rates; statistical models score borrowers.
- Distress is costly and distorts incentives: debt overhang deters good investment, and risk shifting encourages bad gambles.
- Restructuring may be private or court-supervised; Chapter 11 provides a stay, DIP financing and cram-down. Value is allocated by priority, and the fulcrum security becomes the new equity.

Key terms

credit rationing; five Cs; net debt to EBITDA; interest coverage; liquidity; covenant; maintenance and incurrence covenants; covenant-lite; headroom; expected loss; probability of default; loss given default; exposure at default; unexpected loss; default correlation; transition matrix; cumulative default probability; structural model; distance to default; hazard rate; credit scoring; Z-score; financial distress; direct and indirect distress costs; debt overhang; risk shifting; holdout problem; collective action clause; Chapter 11; automatic stay; debtor-in-possession financing; cram-down; absolute priority rule; fulcrum security; recovery rate; private credit; structural subordination.

Exercises

Exercise 39.1 (Ratios). A company has EBITDA of 150, EBIT of 110, interest expense of 30, debt of 600 and cash of 75. Compute net debt to EBITDA, EBITDA coverage and EBIT coverage. What would you want to know before judging whether the leverage is high?

Exercise 39.2 (Headroom). In Exercise 39.1, the maintenance covenant sets net debt to EBITDA at no more than 4.25 times. By how much can EBITDA fall before a breach? If the company also spends 45 of cash on an acquisition that adds no EBITDA, what is the new headroom?

Exercise 39.3 (Loan pricing). A bank lends 10 million to a borrower with PD 1.5 per cent and LGD 45 per cent. Its funding cost is 4 per cent, operating costs are 0.5 per cent of the loan, it

holds capital of 8 per cent of the loan, and its cost of equity is 12 per cent. Estimate the loan rate.

Exercise 39.4 (Transition matrix). The annual transition matrix among IG, HY and D is $\begin{pmatrix} 0.90 & 0.08 & 0.02 \\ 0.10 & 0.80 & 0.10 \\ 0 & 0 & 1 \end{pmatrix}$. Find the two-year default probabilities for an IG and an HY borrower.

Exercise 39.5 (Distance to default). A firm's assets are worth 200 with volatility 30 per cent and expected return 7 per cent; debt of 150 is due in one year. Compute the distance to default and the probability of default. What happens if the share price falls sharply?

Exercise 39.6 (Hazard rates). A five-year bond trades at a spread of 300 basis points; the expected loss given default is 60 per cent. Estimate the risk-neutral hazard rate and the probability of default within five years. Why might the actual probability be lower?

Exercise 39.7 (Z-score). A manufacturer's ratios are: working capital/assets 0.05, retained earnings/assets 0.02, EBIT/assets -0.03 , market value of equity/liabilities 0.4, sales/assets 0.9. Compute its Z-score and interpret it.

Exercise 39.8 (Debt overhang). A firm's assets are worth 70 with certainty; it owes 100 in a year. A project costing 20, funded by shareholders, returns 30 in a year for certain. Will shareholders fund it? Find a reduction in the debt that makes both shareholders and creditors better off than without the project.

Exercise 39.9 (Waterfall). A firm's enterprise value in reorganization is 300. Claims: DIP 30, secured 150, senior unsecured 180, subordinated 60. Find each class's recovery under absolute priority, and identify the fulcrum security. How would the answer change if the enterprise value were 380?

Exercise 39.10 (Correlated defaults). A lender has 20 loans with default probability 5 per cent each. Compare the probability of four or more defaults if defaults are independent with the probability if the economy is good (probability 0.75, loan default probability 2 per cent) or bad (probability 0.25, default probability 14 per cent), with defaults independent within each state.

Solutions

Solution 39.1. Net debt 525; net debt/EBITDA = 3.5; EBITDA coverage $150/30 = 5.0$; EBIT coverage $110/30 = 3.7$. Judgement requires the industry and its cyclicity, the stability of EBITDA, capital-expenditure needs, the maturity profile of the debt and the liquidity available.

Solution 39.2. Breach when $\text{EBITDA} < 525/4.25 = 123.5$: a fall of 17.6 per cent. After the acquisition, net debt is 570; breach below $570/4.25 = 134.1$: headroom falls to 10.6 per cent.

Solution 39.3. Expected loss $0.015 \times 0.45 = 0.675$ per cent; capital charge $0.08 \times (12 - 4) = 0.64$ per cent; rate $\approx 4 + 0.675 + 0.5 + 0.64 = 5.8$ per cent.

Solution 39.4. IG: $0.02 + 0.90 \times 0.02 + 0.08 \times 0.10 = 0.046$. HY: $0.10 + 0.10 \times 0.02 + 0.80 \times 0.10 = 0.182$.

Solution 39.5. $DD = [\ln(200/150) + 0.07 - 0.045]/0.30 = (0.2877 + 0.025)/0.30 = 1.04$; $PD = \Phi(-1.04) = 14.9$ per cent. A sharp fall in the share price signals a fall in asset value (and usually a rise in volatility), reducing the distance to default and raising the probability.

Solution 39.6. $\lambda \approx 0.03/0.6 = 5$ per cent; five-year survival $e^{-0.25} = 0.779$, so a default probability of 22.1 per cent. Spreads also compensate for systematic risk, illiquidity and taxes, so the actual probability is likely lower.

Solution 39.7. $Z = 0.06 + 0.028 - 0.099 + 0.24 + 0.9 = 1.13$, below 1.81: in Altman's classification, a firm at high risk of failure—losses, thin equity relative to liabilities, and little working capital.

Solution 39.8. With the project, assets are 100; creditors take all of it, shareholders nothing, so they will not invest 20. Without the project, creditors receive 70. If the debt is reduced to F with $70 < F < 80$ —say 75—shareholders receive $100 - 75 = 25$ for an investment of 20 and creditors receive 75 rather than 70: both gain.

Solution 39.9. EV 300: DIP 30 and secured 150 in full; senior unsecured receive 120 of 180 (67 per cent) and are the fulcrum; subordinated and equity nothing. EV 380: senior unsecured are paid in full (180); subordinated receive 20 of 60 (33 per cent) and become the fulcrum.

Solution 39.10. Independent: $P(\geq 4) = 1.6$ per cent. Two states: $0.75 \times 0.06\% + 0.25 \times 30.4\% = 7.6$ per cent—nearly five times as likely, though expected defaults are 1 in both cases.

Project Finance, Real Estate and Real Assets

After studying this chapter you should be able to

1. Explain the structure of project finance—the special-purpose vehicle, non-recourse debt and the web of contracts—and the principle of allocating each risk to the party best able to bear it.
2. Trace a project's cash-flow waterfall, and compute the debt service coverage ratio and the loan life coverage ratio.
3. Size debt by sculpting repayments to a target coverage ratio.
4. Value income-producing property from net operating income and capitalization rates, and show how leverage magnifies changes in cap rates.
5. Analyse hotels and other operating real estate with occupancy, ADR and RevPAR, and explain their operating leverage.
6. Compute the levelized cost of energy, and explain how lenders size debt against uncertain production.

A motivating question

Between 2022 and 2024, the value of many office buildings in American and European cities fell by a third or more. Some owners—including large, sophisticated investment firms—handed buildings back to their lenders rather than put in more money. The buildings had not changed; many were still largely let. What had changed were interest rates and expectations about how much office space companies would need.

Real assets—power stations, toll roads, pipelines, office towers, hotels, apartment blocks—are among the largest investments in any economy. They share features that shape how they are financed and valued: long lives, large upfront costs, cash flows that can be forecast from contracts or from a few operating drivers, and the ability to support substantial debt. This chapter applies the tools of capital budgeting (Chapter 32), the cost of capital (Chapter 35) and credit analysis (Chapter 39) to them.

40.1 Project Finance

40.1.1 Structure

In **project finance**, a single asset is built and operated by a **special-purpose vehicle** (SPV), a company created for the purpose, owned by sponsors who provide equity. The SPV borrows on a **non-recourse** or limited-recourse basis: lenders can be repaid only from the project's cash flows and assets, not from the sponsors' other businesses. Debt typically finances 60 to 90 per

cent of the cost, depending on how predictable the cash flows are.

Because lenders rely on the project alone, its risks are allocated through contracts:

- a **construction contract** (often engineering, procurement and construction, EPC) at a fixed price and date, with penalties for delay;
- an **offtake agreement**, such as a power purchase agreement, under which a buyer commits to purchase output at agreed prices, or a **concession** granting the right to charge users or to receive availability payments from a government;
- **supply and operation and maintenance** contracts;
- **insurance**, and sometimes guarantees from governments or development banks.

The guiding principle is that each risk should be borne by the party best able to control or absorb it: construction risk by the contractor, demand risk by whoever can best influence demand (or by the government, if it cannot be controlled at all), and interest-rate and currency risks hedged in financial markets (Chapter 41). A project whose risks are left with parties who cannot bear them—such as the demand risk of the Channel Tunnel (Chapter 32)—is fragile however carefully it is modelled.

Why use this elaborate structure rather than financing the asset on the sponsors' balance sheets? Separating the project limits the sponsors' exposure to a single large asset; allows the project to carry more debt than the sponsors could; and, as Michael Jensen's free-cash-flow argument suggests (Chapter 38), commits the project's cash to lenders and investors rather than leaving it to managers' discretion. The costs are high transaction costs and inflexibility.

40.1.2 The cash-flow waterfall

A project's cash is applied in a strict order set by its loan agreement:

1. revenue, less operating costs and taxes, gives **cash flow available for debt service** (CFADS);
2. senior debt service (interest and scheduled principal);
3. top-up of a *debt service reserve account*, typically holding six months of debt service;
4. junior debt service;
5. distributions to equity, permitted only if coverage tests are met.

40.1.3 Coverage ratios

The central metric is the **debt service coverage ratio**, computed for each period:

$$\text{DSCR}_t = \frac{\text{CFADS}_t}{\text{Debt service}_t}.$$

A project with CFADS of 12 and debt service of 10 has a DSCR of 1.2: cash flow could fall by a sixth before it failed to cover its debt. Loan agreements set a *lock-up* level (commonly around 1.1 to 1.15), below which distributions to equity are blocked, and a lower default level. Minimum DSCRs required at the outset depend on the risk of the cash flows: perhaps 1.2 to 1.3 for a contracted power plant or availability-based infrastructure, and 1.5 or more for projects exposed to market prices or traffic.

The **loan life coverage ratio** looks at the whole remaining life of the loan:

$$\text{LLCR} = \frac{\text{PV of CFADS over the remaining loan life, at the loan's interest rate}}{\text{Debt outstanding}}.$$

It measures whether the project can repay its debt over the loan's life even if individual periods are weak.

40.1.4 Sizing and sculpting debt

Lenders size a project's debt so that the DSCR meets a target in every period. If debt were repaid as a level annuity, the weakest year would determine the size. Instead, repayments are commonly **sculpted**: debt service in each period is set equal to CFADS divided by the target DSCR, and the debt is the present value of that debt service.

Worked Example 40.1. Sculpting debt

A project's CFADS is forecast at 30, 32, 34, 36 and 38 over the five-year life of its loan. Lenders require a DSCR of 1.25; the interest rate is 6 per cent.

Sculpted debt service is $30/1.25 = 24$, then 25.6, 27.2, 28.8 and 30.4. The debt it can support is the present value of these payments at 6 per cent: 113.8. In the first year, interest is $0.06 \times 113.8 = 6.8$ and principal $24 - 6.8 = 17.2$; the debt is fully repaid after five years.

Level debt service constrained by the first year's CFADS would be 24 a year, supporting debt of only $24 \times 4.212 = 101.1$.

Sculpting raises the debt capacity by 12.7 while keeping the DSCR at 1.25 in every year. By construction, the LLCR is also 1.25: the present value of CFADS, 142.2, is 1.25 times the debt.

40.1.5 Public–private partnerships

Governments use project finance to procure infrastructure—roads, hospitals, schools, prisons—through *public–private partnerships* (PPPs). In an *availability* model, the government pays the SPV for keeping an asset available to a specified standard, and bears the demand risk; in a *concession* model, the SPV collects tolls or fares and bears it. PPPs can transfer construction and operating risks to parties able to manage them. But because private finance generally costs more than government borrowing, PPPs create value only if the risk transfer and efficiency gains exceed the higher financing cost. The United Kingdom's Private Finance Initiative, which financed several hundred projects from the 1990s, was criticized for inflexible long-term contracts and costly finance, and the government announced in 2018 that it would use no new PFI contracts.

40.2 Real Estate

40.2.1 Net operating income and cap rates

An income-producing property is valued from its **net operating income** (NOI): rental income, less allowances for vacancy and bad debts, less operating costs not recovered from tenants. NOI excludes financing costs and depreciation. The **capitalization rate** (cap rate) is the ratio of NOI to value:

$$\text{Value} = \frac{\text{NOI}}{\text{cap rate}}.$$

A property with NOI of 8 in a market where comparable properties sell at a cap rate of 5 per cent is worth 160.

The cap rate is the property market's version of an earnings yield, and the growing-perpetuity formula (Chapter 9) explains it. If NOI is expected to grow at g forever and

investors require a return r ,

$$\text{Value} = \frac{\text{NOI}_1}{r - g} \implies \text{cap rate} \approx r - g.$$

Cap rates rise when required returns rise—because interest rates or risk premia rise—or when expected rental growth falls. Because the cap rate is the difference between two rates, small changes in either can change values a great deal.

40.2.2 Leverage

Property is usually bought with substantial debt. Lenders limit the *loan-to-value* ratio (LTV) and require minimum *interest coverage* and *debt yield* (NOI divided by the loan). As in a leveraged buyout (Chapter 38), debt magnifies the equity's exposure to changes in value.

Worked Example 40.2. Cap rates and leverage

The property above, worth 160, is financed with a loan of 96 (60 per cent LTV) and equity of 64. If the market cap rate rises from 5 to 6 per cent, with NOI unchanged at 8, the value falls to $8/0.06 = 133.3$, a fall of 17 per cent. The equity falls to $133.3 - 96 = 37.3$, a fall of 42 per cent, and the LTV rises to 72 per cent. If the loan agreement caps LTV at 70 per cent, the owner is in breach and must inject equity or sell.

This mechanism explains why rising interest rates can cause property crises even when rents hold up: the cap rate rises, values fall, and leveraged owners and their lenders absorb losses. When loans come due and must be refinanced at lower values and higher rates, the owner must provide the difference—or hand the keys to the lender.

40.2.3 Leases and property types

The quality of NOI depends on the leases behind it: their remaining length (the *weighted average lease term*), the tenants' credit, break options, and whether rents are fixed, indexed to inflation or reviewed to market. Offices, retail, logistics warehouses, apartments and data centres differ in their lease lengths, cyclicity and exposure to structural change—online shopping for shopping centres, remote work for offices, e-commerce demand for warehouses. *Real estate investment trusts* (REITs), listed companies that own property and distribute most of their income, give investors liquid access to property, and their share prices often signal changes in value well before appraisal-based indices.

40.3 Operating Real Assets

40.3.1 Hotels

A hotel's income depends on daily sales rather than long leases. Its key drivers are the **occupancy** rate (the share of available rooms sold), the **average daily rate** (ADR) charged for occupied rooms, and their product, **revenue per available room**:

$$\text{RevPAR} = \text{ADR} \times \text{occupancy}.$$

Hotels have high fixed costs—staff, maintenance, property costs—so profit is highly sensitive to RevPAR: they have high operating leverage (Chapter 31).

Worked Example 40.3. Hotel operating leverage

A 200-room hotel charges an ADR of 150 at 75 per cent occupancy: RevPAR is 112.5, and annual room revenue $200 \times 112.5 \times 365 = 8.21$ million. Fixed costs are 4.0 million a year and variable costs 30 per occupied room-night (54,750 room-nights, 1.64 million), leaving profit of 2.57 million. If occupancy falls to 60 per cent at the same rate, room revenue falls 20 per cent to 6.57 million, variable costs to 1.31 million, and profit to 1.26 million: a fall of 51 per cent.

Hotel owners choose how much of this risk to keep. An owner who leases the hotel to an operator for fixed rent bears little operating risk; one who hires a manager under a management contract, paying fees based on revenue and profit, keeps most of it; franchising provides a brand and booking system for a fee.

40.3.2 Energy and commodities

The economics of a power project are summarized by its **levelized cost of energy** (LCOE): the constant price per unit of output at which the project would just break even in present-value terms,

$$\text{LCOE} = \frac{\text{PV of lifetime costs}}{\text{PV of lifetime output}}.$$

Output is discounted as well as costs because a unit of energy sold in the future brings revenue in the future.

Worked Example 40.4. The levelized cost of a solar plant

A solar plant costs 100 million to build, 2 million a year to operate, and produces 180 gigawatt-hours (GWh) a year for 25 years. At a discount rate of 7 per cent, the annuity factor for 25 years is 11.654, so the present value of costs is $100 + 2 \times 11.654 = 123.3$ million and the present value of output $180 \times 11.654 = 2,098$ GWh. The LCOE is 123.3 million divided by 2.098 million megawatt-hours: about 59 per megawatt-hour. Because most of the cost is upfront, the LCOE is highly sensitive to the discount rate: at 10 per cent it would be about 72.

For renewable projects, the main uncertainties are output—how much the sun shines or the wind blows—and price. Lenders size debt not on the expected, or “P50”, production estimate, but on a conservative one, such as the “P90” level that production is expected to exceed in nine years out of ten, and they prefer revenue fixed by a power purchase agreement to revenue at uncertain market prices (*merchant* risk). For mining and oil projects, lenders base their forecasts on conservative “price decks”, and sponsors often hedge part of production with forward contracts (Chapter 41).

Case: Office property after 2022

The shock. Two changes hit office markets at once. First, the pandemic made remote and hybrid work common, and in many cities physical office occupancy remained well below pre-pandemic levels years later; as leases expired, tenants took less space. Vacancy rates in US office markets rose to record levels of around a fifth of space. Second, central banks raised interest rates sharply from 2022 (Chapter 50): the Federal Reserve lifted its policy rate from near zero in March 2022 to over 5 per cent by mid-2023, and long-term yields rose with it.

The effect on values. Higher required returns and lower expected rental growth both raised

cap rates, and falling occupancy reduced NOI. Indices of US office values compiled from transactions showed falls of the order of a third or more from their 2022 peak, with much larger falls for older buildings in weaker locations; transaction volumes collapsed, making prices hard to observe. Owners of heavily leveraged buildings found that their equity had been wiped out. In February 2023, funds managed by Brookfield, one of the world's largest property investors, defaulted on loans secured on two office towers in Los Angeles rather than inject more equity. Banks with large exposures to commercial property came under scrutiny: in early 2024 the shares of New York Community Bancorp fell sharply after it raised provisions for losses on property loans.

In terms of this chapter. Example 40.2 is the mechanism in miniature: a rise of one point in the cap rate, with NOI unchanged, cut the equity of a 60 per cent levered owner by more than 40 per cent; falling NOI makes it worse. Non-recourse loans let owners walk away from a single building, limiting their losses and transferring the rest to lenders—the put option inherent in limited liability (Chapter 39). The lenders' losses depend on their LTVs at origination and on how long values stay low, which is why regulators monitored banks' property exposures closely. And the split between prime and older buildings shows that the drivers of NOI—lease terms, tenant demand, the quality of the asset—matter as much as the discount rate.

Summary

- Project finance places a single asset in an SPV financed with non-recourse debt; contracts allocate each risk to the party best able to bear it.
- Cash flows follow a waterfall: CFADS, senior debt service, reserves, junior debt, then distributions subject to coverage tests.
- $DSCR = CFADS / \text{debt service period by period}$; the LLCR compares the present value of CFADS over the loan life with the debt. Sculpting sets debt service at CFADS divided by the target DSCR, maximizing debt capacity.
- Property value equals NOI divided by the cap rate, and the cap rate is roughly the required return minus expected growth. Leverage magnifies changes in cap rates into large changes in equity value and LTV.
- Hotels depend on occupancy and ADR (RevPAR) and have high operating leverage; ownership, lease, management and franchise structures allocate the risk.
- The LCOE is the break-even price of energy in present-value terms; lenders size debt on conservative production and price estimates.

Key terms

project finance; special-purpose vehicle; sponsor; non-recourse debt; EPC contract; offtake agreement; power purchase agreement; concession; availability payment; risk allocation; cash-flow waterfall; cash flow available for debt service; debt service reserve account; debt service coverage ratio; lock-up; loan life coverage ratio; sculpting; public-private partnership; net operating income; capitalization rate; loan-to-value; debt yield; weighted average lease term; REIT; occupancy; average daily rate; RevPAR; management contract; franchise; levelized cost of energy; P50 and P90; merchant risk; price deck.

Exercises

Exercise 40.1 (DSCR). A project's CFADS is 18 and its debt service 15. Find the DSCR. The lock-up level is 1.15 and the default level 1.05. What happens if CFADS falls by 10 per cent?

Exercise 40.2 (Sculpting). CFADS over a three-year loan is forecast at 40, 42 and 44. The target DSCR is 1.3 and the interest rate 5 per cent. Find the sculpted debt service and the maximum debt.

Exercise 40.3 (LLCR). A project has debt outstanding of 100, and CFADS of 30 a year for the remaining four years of the loan, which bears interest at 6 per cent. Compute the LLCR and comment.

Exercise 40.4 (Cap rates). An office building has NOI of 5.4 million and is valued at a 6 per cent cap rate. It carries a loan of 60 million. Find the value, the LTV and the equity. If the cap rate rises to 7 per cent, recompute them.

Exercise 40.5 (Debt yield). A lender requires a debt yield of at least 10 per cent and an LTV of at most 65 per cent. For the building in Exercise 40.4 at the 6 per cent cap rate, what is the maximum loan? Which constraint binds?

Exercise 40.6 (Cap rates and growth). Investors require 8 per cent on a property whose NOI is expected to grow at 2.5 per cent a year. Find the cap rate. If the required return rises by one percentage point, by how much does the value fall?

Exercise 40.7 (RevPAR). A 150-room hotel has occupancy of 70 per cent and an ADR of 180. Find RevPAR and annual room revenue. The hotel can either cut its rate by 10 per cent, which raises occupancy to 78 per cent, or keep the rate. Which gives higher room revenue? What else would you need to know?

Exercise 40.8 (LCOE). A wind farm costs 150 million, has operating costs of 4 million a year and produces 300 GWh a year for 20 years. Compute its LCOE at 6 per cent (the 20-year annuity factor is 11.470).

Exercise 40.9 (P90 sizing). A solar project's annual output is normally distributed with mean 200 GWh and standard deviation 15 GWh. Power is sold at 50 per megawatt-hour under a contract; operating costs are 2 million a year. Find the P90 output. What maximum annual debt service can lenders accept on a P90 basis with a target DSCR of 1.3, and how does it compare with P50 sizing?

Exercise 40.10 (Risk allocation). For a new toll road financed as a PPP, suggest who should bear (a) construction cost overruns, (b) traffic volumes, (c) changes in law specific to the project, (d) interest-rate risk on the debt, and explain why.

Solutions

Solution 40.1. $DSCR = 18/15 = 1.20$. After a 10 per cent fall, $16.2/15 = 1.08$: below lock-up, so distributions to equity stop and cash is retained, but above the default level.

Solution 40.2. Debt service $40/1.3 = 30.77$, 32.31, 33.85. The maximum debt is their present value, $30.77/1.05 + 32.31/1.05^2 + 33.85/1.05^3 = 87.8$.

Solution 40.3. $PV \text{ of CFADS} = 30 \times 3.465 = 103.95$; $LLCR = 1.04$. The project can only just repay its debt over the loan life if CFADS holds; any shortfall would require a longer tenor (if a "tail" of cash flows remains after the loan's maturity) or a restructuring.

Solution 40.4. At 6 per cent: value 90 million, LTV $60/90 = 67$ per cent, equity 30. At 7 per cent: value 77.1 million, LTV 78 per cent, equity 17.1—a fall of 43 per cent for a 14 per cent fall in value.

Solution 40.5. Debt yield: $5.4/0.10 = 54$ million; LTV: $0.65 \times 90 = 58.5$ million. The debt-yield test binds: maximum loan 54 million.

Solution 40.6. Cap rate $\approx 8 - 2.5 = 5.5$ per cent. At 9 per cent required return, 6.5 per cent: value falls by $1 - 5.5/6.5 = 15.4$ per cent.

Solution 40.7. RevPAR = $180 \times 0.70 = 126$; room revenue $150 \times 126 \times 365 = 6.90$ million. With the rate cut: $162 \times 0.78 = 126.36$, marginally higher (6.92 million). But the extra 8 points of occupancy bring extra variable costs (cleaning, laundry, commissions), which probably outweigh the tiny revenue gain; the answer depends on variable cost per room-night, on ancillary spending by extra guests, and on whether the cut damages the hotel's pricing power.

Solution 40.8. PV of costs = $150 + 4 \times 11.470 = 195.9$ million; PV of output = $300 \times 11.470 = 3,441$ GWh; LCOE = $195.9/3.441 = 56.9$ per MWh.

Solution 40.9. $P_{90} = 200 - 1.2816 \times 15 = 180.8$ GWh; revenue $180.8 \times 50 = 9.04$ million; CFADS 7.04 million; maximum debt service $7.04/1.3 = 5.41$ million. At P_{50} , CFADS would be 8 million and debt service 6.15 million: P_{90} sizing reduces debt capacity by about 12 per cent.

Solution 40.10. (a) The construction contractor, under a fixed-price contract, since it controls construction. (b) Traffic is largely outside anyone's control and hard to forecast; unless the SPV can influence it, it may be better borne by the government through availability payments, or shared through revenue bands. (c) The government, which controls the law. (d) The SPV should hedge it in financial markets with interest-rate swaps, since neither party can control rates but markets can absorb the risk at low cost.

Derivatives and Risk Management

After studying this chapter you should be able to

1. Describe forwards, futures, swaps and options, and how they are used to hedge, to speculate and to arbitrage.
2. Price forwards and futures by the cost of carry, and explain marking to market and margin.
3. Compute the minimum-variance hedge ratio and the effectiveness of a hedge.
4. Value an interest-rate swap from discount factors.
5. State and prove put–call parity, and value options by replication and risk-neutral pricing in the binomial model.
6. Use the Black–Scholes–Merton formula and the Greeks, and interpret implied volatility and the volatility smile.
7. Explain hedge accounting, central clearing, and why a hedge that is sound in value can fail for lack of cash.

A motivating question

In 1993 the American subsidiary of Metallgesellschaft, one of Germany's largest industrial groups, had promised customers deliveries of oil products at fixed prices for up to ten years. To protect itself against rising oil prices, it bought short-dated oil futures. Then oil prices fell. On paper, its gains on the delivery contracts offset its losses on the futures. But the futures losses had to be paid in cash, every day, while the gains on the contracts would arrive over a decade. By the end of the year the group needed a rescue by its banks.

Derivatives—contracts whose value derives from the price of something else—allow firms to transfer risks they do not want to bear to others willing to bear them. They also allow large bets with little money down. This chapter explains what the main instruments are, how they are priced by the principle of no arbitrage, and how they are used and misused. It draws on the time value of money (Chapter 9), the lognormal distribution (Chapter 13), regression (Chapter 16) and the yield curve (Chapter 33).

41.1 Instruments and Uses

41.1.1 The main contracts

- A **forward contract** obliges one party to buy, and the other to sell, an asset at a future date for a price agreed today, the *forward price*. Forwards are negotiated privately (“over the counter”).
- A **futures contract** is a standardized forward traded on an exchange and settled daily through a clearing house.

- A **swap** exchanges one stream of payments for another—for example, fixed interest for floating interest—over a period of years.
- An **option** gives its holder the right, but not the obligation, to buy (a **call**) or sell (a **put**) an asset at a fixed **strike price** K , on a set date (a *European* option) or at any time up to it (an *American* option). The holder pays a price, the *premium*, for this right.

Forwards, futures and swaps are *symmetric*: gains on one side equal losses on the other, and they cost nothing to enter at market prices. Options are *asymmetric*: the holder's loss is limited to the premium. Before costs, derivatives are zero-sum between the two parties. Measured by notional amount, the global derivatives market is enormous—the Bank for International Settlements has reported notional amounts of over-the-counter derivatives of several hundred trillion dollars, mostly interest-rate contracts—but notional amounts overstate the value at risk; the market value of the contracts is a small fraction of the notional.

41.1.2 Hedging, speculation and arbitrage

A *hedger* uses derivatives to reduce an existing exposure: an airline buys fuel forward, an exporter sells foreign currency forward (Chapter 42), a borrower swaps floating-rate debt into fixed. A *speculator* takes a position to profit from a view, with leverage: a futures position requires only a margin deposit of a few per cent of the contract's value. An *arbitrageur* exploits price discrepancies between related instruments, and in doing so enforces the pricing relations of this chapter.

Why should a firm hedge? In the Modigliani–Miller world (Theorem 30.7), it should not bother: shareholders can hedge or diversify for themselves, so hedging cannot create value. Hedging creates value through the same frictions that make capital structure matter (Chapter 35): it reduces the probability and costs of financial distress, allows more debt and its tax shields, smooths taxable income when tax schedules are convex, and—as Kenneth Froot, David Scharfstein and Jeremy Stein (1993) argued—ensures that a firm has cash to fund good investments when external finance is costly.

41.2 Forwards and Futures

41.2.1 The cost of carry

What should the forward price be? Consider an asset that pays no income and costs nothing to store, such as gold held in a vault at negligible cost. An investor can obtain the gold at date T in two ways: buy a forward at price F , or borrow S today, buy the gold, and repay $S(1+r)^T$ at T . The two strategies deliver the same thing at the same date, so they must cost the same.

Proposition 41.1. Forward price by no arbitrage

For an asset with spot price S , no income and no storage costs, the forward price for delivery at T is $F = S(1+r)^T$. If the asset pays a continuous income yield q , and interest is compounded continuously at rate r , then $F = Se^{(r-q)T}$.

Proof. If $F > S(1+r)^T$, borrow S , buy the asset and sell it forward at F : at T , deliver the asset, receive F and repay $S(1+r)^T$, a riskless profit (*cash and carry*). If $F < S(1+r)^T$, sell the asset short, lend the proceeds and buy forward: at T , collect $S(1+r)^T$, pay F and return the asset. Either way, the arbitrage would be exploited until the prices align. With income, holding the asset earns q , which reduces the cost of carrying it. $\square$

Worked Example 41.2. Forward prices

Gold trades at 2,000 an ounce and the one-year interest rate is 5 per cent: the one-year forward price is 2,100. If a dealer quoted 2,150, an arbitrageur would borrow 2,000, buy gold, and sell it forward at 2,150, locking in 50 per ounce. For a share index at 5,000 with a dividend yield of 2 per cent and an interest rate of 5 per cent (both continuous), the one-year futures price is $5,000e^{0.03} = 5,152$.

The forward price is not a forecast of the future spot price; it is the spot price plus the net cost of carrying the asset. For commodities, storage costs raise the forward price, while the *convenience yield* of holding physical stock—the ability to keep a refinery running during a shortage—lowers it. When forward prices exceed spot prices the market is in **contango**; when they are below, in **backwardation**.

41.2.2 Marking to market

A futures position is settled every day. Each party posts *initial margin* with the clearing house; at the end of each day, gains and losses are credited and debited (**marking to market**), and a party whose margin falls below the *maintenance* level must restore it at once. Daily settlement nearly eliminates counterparty risk, but converts price changes into immediate cash flows—the heart of the Metallgesellschaft case.

41.2.3 Hedge ratios

A hedger rarely finds a futures contract on exactly the asset it is exposed to: an airline hedging jet fuel may use heating-oil futures. The residual risk from the imperfect match is **basis risk**. How many futures should it hold? Let the exposure change by ΔS and the futures price by ΔF , and let the hedger sell h futures per unit of exposure. The hedged position changes by $\Delta S - h\Delta F$, with variance (Theorem 14.2)

$$\sigma_S^2 - 2h\rho\sigma_S\sigma_F + h^2\sigma_F^2.$$

Minimizing over h gives the **minimum-variance hedge ratio**

$$h^* = \rho \frac{\sigma_S}{\sigma_F},$$

the slope of a regression of ΔS on ΔF (Chapter 16). The hedge removes the fraction ρ^2 of the variance: the regression's R^2 .

Worked Example 41.3. Hedging jet fuel

Monthly changes in jet-fuel prices have a standard deviation of 30 per cent a year; heating-oil futures, 25 per cent; the correlation is 0.9. The hedge ratio is $0.9 \times 0.30/0.25 = 1.08$: to hedge fuel worth 100 million, the airline sells heating-oil futures worth 108 million. The hedge removes $0.9^2 = 81$ per cent of the variance; the standard deviation of the hedged position is $\sqrt{0.19} = 44$ per cent of the unhedged one.

41.3 Swaps

In a plain **interest-rate swap**, one party pays a fixed rate on a notional principal and the other pays a floating rate, such as the overnight rate compounded over each period (Chapter 49).

The notional is never exchanged. A company with floating-rate debt can pay fixed and receive floating in a swap, converting its debt to fixed rate—a standard step in project finance (Chapter 40).

A swap can be valued as the difference between two bonds. The party receiving fixed is, in effect, long a fixed-rate bond and short a floating-rate note of the same notional. A floating-rate note is worth its face value at each reset date, because its coupon always equals the market rate (Chapter 33). At inception, the swap is worth zero, so the fixed bond must also be worth its face value: the fixed rate on a new swap, the **swap rate**, is the par yield. With discount factors d_t ,

$$c_n = \frac{1 - d_n}{\sum_{t=1}^n d_t}.$$

With the discount factors of Example 33.1, the three-year annual swap rate is 4.57 per cent, the par yield computed there. As rates move after inception, the swap acquires positive value for one party and negative value for the other.

41.4 Options

41.4.1 Payoffs and parity

At expiry, a call pays $\max(S_T - K, 0)$ and a put $\max(K - S_T, 0)$. An option is *in the money* if exercising it now would pay, *at the money* if $S = K$, and *out of the money* otherwise. Before expiry, its price exceeds its payoff from immediate exercise (its *intrinsic value*) by its *time value*, which reflects the chance that it will move further into the money.

Calls and puts on the same asset with the same strike and expiry are linked by a relation that requires no model.

Proposition 41.4. Put–call parity

For European options on a share paying no dividends before expiry,

$$C - P = S - \frac{K}{(1 + r)^T}.$$

Proof. Hold a call and sell a put. At expiry, if $S_T > K$, the call pays $S_T - K$ and the put is worthless; if $S_T < K$, the call is worthless and the put costs $K - S_T$. Either way the position is worth $S_T - K$ —the same as holding the share and owing K . Portfolios with identical payoffs must have identical prices. $\square$

Put–call parity shows that options are building blocks: a share plus a put (a *protective put*) has the same payoff as a call plus a bond. It is also the relation Merton used to value corporate debt (Chapter 39): the shareholders' position is a call on the firm's assets, and the creditors hold safe debt minus a put.

41.4.2 The binomial model

To value an option, suppose a share worth 100 today will be worth either 120 or 90 in a year, and the interest rate is 5 per cent. A call with strike 100 pays 20 or 0. Consider a portfolio of Δ shares financed partly by borrowing B . Choose them so that the portfolio replicates the call:

$$120\Delta - 1.05B = 20, \quad 90\Delta - 1.05B = 0.$$

Subtracting, $\Delta = 20/30 = 0.667$, and $B = 90 \times 0.667/1.05 = 57.14$. The portfolio costs $100 \times 0.667 - 57.14 = 9.52$. Since it pays exactly what the call pays in every state, the call must be worth 9.52; any other price would permit arbitrage.

The probability that the share goes up does not appear. Whatever investors believe, and however risk-averse they are, the call is worth 9.52—because its risk can be eliminated by holding shares. This suggests a shortcut. Define the **risk-neutral probability** q as the probability of an up-move that would make the share earn the riskless rate:

$$100 = \frac{120q + 90(1 - q)}{1.05} \implies q = \frac{1.05 - 0.90}{1.20 - 0.90} = 0.5.$$

Then the call's value is its expected payoff under q , discounted at the riskless rate: $0.5 \times 20/1.05 = 9.52$. **Risk-neutral valuation** is not an assumption that investors are indifferent to risk; it is a computational device that works because, in a world where the option can be replicated, risk preferences are already embodied in the share price. The put with the same strike pays 0 or 10 and is worth $0.5 \times 10/1.05 = 4.76$, as put–call parity confirms: $9.52 - 4.76 = 100 - 100/1.05$.

With many short steps, the binomial tree can approximate any distribution of the share price at expiry, and can value American options by checking at each node whether early exercise beats holding. The same method values the real options of Chapter 32.

41.4.3 Black–Scholes–Merton

As the steps of a binomial tree become shorter and more numerous, with up and down moves scaled appropriately, the share price at expiry becomes lognormal (Chapter 13). In continuous time, the price follows a *geometric Brownian motion*, whose proportional changes over any short interval are normal with mean μdt and variance $\sigma^2 dt$. Fischer Black and Myron Scholes (1973) and Robert Merton (1973) showed that an option can then be replicated by continuously adjusting a position in the share, and derived the value of a European call on a share paying no dividends:

$$C = S \Phi(d_1) - Ke^{-rT} \Phi(d_2), \quad d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T},$$

where Φ is the standard normal distribution function, r the continuously compounded riskless rate, and σ the volatility of the share's returns. The put follows from parity. The expected return μ does not appear, for the reason seen in the binomial model. Scholes and Merton received the Nobel prize in 1997; Black had died in 1995. The mathematics of continuous-time finance—stochastic calculus and Itô's lemma—lies beyond this book, but its conclusions can be used with the formula.

Worked Example 41.5. A Black–Scholes value

A share is at 100; a one-year call has strike 100; $r = 5$ per cent; $\sigma = 20$ per cent. Then $d_1 = [0 + (0.05 + 0.02) \times 1]/0.20 = 0.35$ and $d_2 = 0.15$; $\Phi(0.35) = 0.637$ and $\Phi(0.15) = 0.560$. The call is worth $100 \times 0.637 - 100e^{-0.05} \times 0.560 = 10.45$, and the put $10.45 - 100 + 95.12 = 5.57$.

The Greeks. The sensitivities of an option's value to its inputs guide hedging:

- **Delta** $\frac{\partial C}{\partial S} = \Phi(d_1)$ for a call, 0.637 in the example: the number of shares that replicate the option at the margin. A dealer who has sold the call hedges by holding delta shares and adjusts as delta changes (*delta hedging*).

- **Gamma**, the rate of change of delta with the share price: large for at-the-money options near expiry, whose hedges must be adjusted often.
- **Vega**, the sensitivity to volatility: about 0.38 per volatility point in the example; raising σ from 20 to 25 per cent raises the call's value to 12.34. Options are bets on volatility as much as on direction.
- **Theta**, the loss of value as time passes, all else equal.

41.4.4 Implied volatility and the smile

Every input to the formula except volatility can be observed. Given an option's market price, the volatility that equates the formula to the price is the **implied volatility**. Traders quote options in implied volatilities, and indices such as the VIX, computed from options on the S&P 500, measure the market's expected volatility over the following month.

If the Black–Scholes assumptions held, all options on the same share and expiry would have the same implied volatility. They do not. Since the crash of October 1987 (Chapter 13), implied volatilities of equity-index options have been higher for low strikes than for high ones, a pattern called the **volatility skew** or, more generally, the **volatility smile**. Out-of-the-money puts, which pay off in a crash, are priced as if volatility were higher. The smile is the market's correction for the model's thin normal tails and constant volatility (Chapter 34): investors pay more for crash protection than a lognormal distribution would justify. Practitioners use the formula as a quoting convention and interpolation device, and more elaborate models—with jumps or stochastic volatility—to fit the smile.

41.5 Risk Management in Practice

41.5.1 Hedge accounting

Derivatives are measured at fair value, with changes in value normally recognized in profit or loss (Chapter 30). The item being hedged, however, is often measured differently—a forecast purchase is not recognized at all until it happens—so a sound hedge can make reported profit *more* volatile. **Hedge accounting** under IFRS 9 and US GAAP (ASC 815) removes the mismatch for qualifying hedges. In a *fair-value hedge*, the hedged item is also remeasured for the hedged risk, so the changes offset in profit or loss. In a *cash-flow hedge* of a forecast transaction, the effective part of the derivative's gains and losses is held in other comprehensive income and released to profit or loss when the hedged transaction affects it. Firms must document the hedging relationship and show that it is expected to be effective.

41.5.2 Counterparty risk and clearing

In an over-the-counter contract, each party is exposed to the other's default. The failure of Lehman Brothers in 2008 and the rescue of AIG, which had sold credit protection on a vast scale, showed how opaque bilateral derivatives exposures could spread distress (Chapter 51). In 2009 the G20 leaders committed to requiring standardized over-the-counter derivatives to be cleared through *central counterparties*, to reporting all contracts to trade repositories, and to higher capital and margin requirements for uncleared ones. Central clearing replaces a web of bilateral exposures with exposures to a clearing house protected by margin and a default fund. It also concentrates risk in the clearing houses and makes margin calls—large and sudden in volatile markets—a source of liquidity strain, as the British pension funds of Chapter 33 discovered in 2022.

Key idea. A hedge offsets *values*; it does not necessarily offset *cash flows* at the same time. A derivative that is marked to market or margined daily requires cash when prices move against it, while the offsetting gain on the hedged exposure may arrive much later. Every hedging programme needs a funding plan for its worst-case margin calls.

Case: Metallgesellschaft, 1993

The strategy. In the early 1990s MG Refining and Marketing (MGRM), the US subsidiary of the German conglomerate Metallgesellschaft, offered independent petrol stations and other customers contracts to buy heating oil and petrol at fixed prices, for up to ten years. By late 1993 it had committed to supply about 160 million barrels. To hedge the risk that oil prices would rise, it bought oil futures and entered into swaps—mostly contracts expiring within months—in quantities roughly matching its total commitments, rolling them forward as they expired (a *stack-and-roll* hedge).

What happened. During 1993 oil prices fell sharply. The fixed-price supply contracts became more valuable to MGRM, but those gains would be realized only as oil was delivered over the coming decade. The futures lost value immediately, and the losses had to be paid in daily margin. At the same time, the oil market moved into contango, so that each roll required selling expiring futures and buying later ones at higher prices, adding further costs. MGRM's cash needs ran into the hundreds of millions of dollars. In December 1993 Metallgesellschaft's supervisory board replaced the management, and the new team closed out much of the hedge and the supply contracts. The group recorded losses of more than a billion dollars on the positions and survived only through a rescue organized by its bank creditors.

The debate. Christopher Culp and Merton Miller argued that the strategy was a sound hedge of value and that liquidating it had turned temporary cash outflows into permanent losses. Others, including Antonio Mello and John Parsons, replied that a one-for-one stack of short-dated futures against long-dated commitments was not the minimum-variance hedge, carried large basis and rollover risks, and exposed the firm to funding risk its parent was not prepared to bear. Both sides agreed on one point: whatever the merits of the hedge in value, it required a financing plan that MGRM did not have, and whose absence its bankers and board discovered only when the margin calls arrived.

In terms of this chapter. The case combines the cost of carry (contango raised the cost of rolling), the hedge ratio (a stack-and-roll hedge is not the minimum-variance hedge when spot and long-dated prices do not move one-for-one), marking to market (gains and losses in different periods), and the key idea above. It is also a lesson for accounting: under the rules then applying in Germany, the losses on the futures were recognized while the gains on the supply contracts were not—the mismatch that hedge accounting is designed to address.

Summary

- Forwards, futures and swaps are symmetric obligations; options give rights. Derivatives are used to hedge, speculate and arbitrage, and create value for firms only through frictions such as distress costs and costly external finance.
- No arbitrage sets the forward price at the spot price plus the net cost of carry: $F = Se^{(r-q)T}$. Futures are marked to market daily through margin.
- The minimum-variance hedge ratio is $\rho\sigma_S/\sigma_F$, the regression slope; it removes the fraction

ρ^2 of variance.

- A new swap's fixed rate is the par yield computed from discount factors.
- Put–call parity, $C - P = S - K/(1 + r)^T$, holds without a model. In the binomial model, options are valued by replication, equivalently by risk-neutral expectation discounted at the riskless rate.
- The Black–Scholes–Merton formula values European options when prices are lognormal; delta, gamma, vega and theta guide hedging. Implied volatilities form a smile or skew, revealing fat tails and crash fear.
- Hedge accounting aligns the recognition of hedges and hedged items; central clearing reduces counterparty risk but creates liquidity demands. A hedge of value can fail for lack of cash.

Key terms

derivative; forward contract; futures contract; swap; option; call; put; strike price; European and American options; premium; notional amount; hedger; speculator; arbitrageur; cost of carry; cash and carry; convenience yield; contango; backwardation; initial and maintenance margin; marking to market; clearing house; basis risk; minimum-variance hedge ratio; interest-rate swap; swap rate; intrinsic value; time value; put–call parity; protective put; binomial model; replicating portfolio; risk-neutral probability; risk-neutral valuation; geometric Brownian motion; Black–Scholes–Merton formula; delta; gamma; vega; theta; delta hedging; implied volatility; VIX; volatility smile and skew; hedge accounting; fair-value hedge; cash-flow hedge; central counterparty; stack-and-roll hedge.

Exercises

Exercise 41.1 (Forward price). A commodity trades at 50; the one-year interest rate is 4 per cent; storage costs 1 per unit, paid at the end of the year. Find the one-year forward price. If the market forward price is 55, describe the arbitrage and its profit.

Exercise 41.2 (Index futures). A share index stands at 4,000. The continuously compounded interest rate is 5 per cent and the dividend yield 1.5 per cent. Find the six-month futures price. Is the market in contango or backwardation?

Exercise 41.3 (Margin). A trader buys 10 gold futures, each for 100 ounces, at 2,000. Initial margin is 10,000 per contract and maintenance margin 7,500. The price falls to 1,970. Find the loss, the margin balance per contract and any margin call.

Exercise 41.4 (Hedge ratio). A firm has an exposure worth 10 million to an asset whose price changes have a standard deviation of 20 per cent; the futures available have a standard deviation of 25 per cent and a correlation of 0.8 with the asset. Find the minimum-variance hedge, the futures position, and the proportion of variance removed.

Exercise 41.5 (Swap rate). Discount factors for one, two and three years are 0.970, 0.935 and 0.900. Find the three-year swap rate. A company with a three-year floating-rate loan pays fixed in the swap: what rate does it effectively pay if its loan costs the floating rate plus 1.5 per cent?

Exercise 41.6 (Put–call parity). A share is at 50; a one-year European call with strike 50 costs 6.00; the one-year interest rate is 5 per cent. Find the price of the put implied by parity. If the put trades at 4.50, construct an arbitrage.

Exercise 41.7 (Binomial put). A share at 80 will be worth 100 or 64 in a year; the interest rate is 4 per cent. Value a one-year put with strike 80 by risk-neutral valuation, and find the replicating portfolio.

Exercise 41.8 (Black–Scholes). Value a six-month European call and put on a share at 50 with strike 55, $r = 4$ per cent (continuous) and $\sigma = 30$ per cent. ($\Phi(-0.249) = 0.4017$, $\Phi(-0.461) = 0.3224$.)

Exercise 41.9 (Delta hedging). A dealer sells 100 call contracts, each on 100 shares, with delta 0.60. How many shares must it hold to be delta-neutral? The next day the share price rises and delta becomes 0.65. What must it do? Why does frequent rehedge cost money when the share is volatile?

Exercise 41.10 (Hedge accounting). In year 1 an airline buys fuel futures to hedge fuel it will buy in year 2. Fuel prices rise: the futures gain 3 million in year 1, and the fuel costs 3 million more than budgeted in year 2. Show reported profit effects in each year with and without cash-flow hedge accounting.

Solutions

Solution 41.1. $F = 50 \times 1.04 + 1 = 53$. At 55: borrow 50, buy the commodity, sell forward at 55; at year-end deliver, receive 55, repay 52 and pay storage 1: profit 2 per unit.

Solution 41.2. $F = 4,000e^{(0.05-0.015) \times 0.5} = 4,070.6$. The futures price exceeds the spot: contango, because the interest rate exceeds the dividend yield.

Solution 41.3. Loss = $30 \times 100 \times 10 = 30,000$, 3,000 per contract; margin balance 7,000, below maintenance: a call of 3,000 per contract (30,000 in total) to restore 10,000.

Solution 41.4. $h^* = 0.8 \times 0.20/0.25 = 0.64$; sell futures worth 6.4 million; variance removed $0.8^2 = 64$ per cent, leaving a standard deviation 60 per cent of the unhedged one.

Solution 41.5. $c = (1 - 0.900)/(0.970 + 0.935 + 0.900) = 0.100/2.805 = 3.565$ per cent. The company pays floating plus 1.5 on the loan, receives floating and pays 3.565 in the swap: in effect a fixed 5.07 per cent.

Solution 41.6. $P = C - S + K/(1 + r) = 6.00 - 50 + 47.62 = 3.62$. The put at 4.50 is overpriced: sell the put, buy the call, short the share and lend 47.62. Initial cash $4.50 - 6.00 + 50 - 47.62 = 0.88$; at expiry the position nets to zero in every state (the call and put combine to $S_T - 50$, offset by returning the share and collecting 50), so 0.88 is riskless profit.

Solution 41.7. $q = (1.04 - 0.80)/(1.25 - 0.80) = 0.533$. Put payoffs 0 and 16: $P = (1 - 0.533) \times 16/1.04 = 7.18$. Replication: $\Delta = (0 - 16)/(100 - 64) = -0.444$ shares (a short position) and lending $(0 + 0.444 \times 100)/1.04 = 42.74$; cost $-35.56 + 42.74 = 7.18$.

Solution 41.8. $d_1 = [\ln(50/55) + (0.04 + 0.045) \times 0.5]/(0.30\sqrt{0.5}) = (-0.0953 + 0.0425)/0.2121 = -0.249$; $d_2 = -0.461$. $C = 50 \times 0.4017 - 55e^{-0.02} \times 0.3224 = 20.09 - 17.38 = 2.71$. $P = 2.71 - 50 + 55e^{-0.02} = 2.71 - 50 + 53.91 = 6.62$.

Solution 41.9. Delta exposure = $100 \times 100 \times 0.60 = 6,000$ shares; hold 6,000. When delta rises to 0.65, buy 500 more. The dealer buys after rises and sells after falls—buying high and selling low—so a volatile share generates rehedge losses; these are what the option premium pays for (the dealer is short gamma).

Solution 41.10. Without hedge accounting: year 1 profit +3 million (futures gain); year 2 –3 million (higher fuel cost). With cash-flow hedge accounting: the 3 million gain is held in other comprehensive income in year 1 and released in year 2, offsetting the higher fuel cost: profit effect zero in both years, reflecting the economics of the hedge.

International and Islamic Finance

After studying this chapter you should be able to

1. Read foreign-exchange quotations, compute cross rates, and detect triangular arbitrage.
2. Derive covered interest parity and use it to price forward exchange rates.
3. Distinguish transaction, translation and economic exposure, and account for foreign operations under IAS 21.
4. Hedge a foreign-currency exposure with forwards, money-market transactions and options, and compare the outcomes.
5. Organize a corporate treasury: netting, natural hedges and hedging policy.
6. Evaluate a foreign investment consistently in either currency, and treat country risk in the cash flows.
7. Explain the principles and main instruments of Islamic finance, and analyse their economic substance.

A motivating question

In June 2002, the government of Malaysia sold 600 million dollars of five-year certificates to investors around the world. The certificates paid a regular return and were repaid at the end of five years, like a bond. But they paid no interest. Investors instead owned, through a special-purpose company, a share in government land and buildings, which the government leased back; they received their share of the rent, and the government undertook to buy the property back at maturity. The issue was the first global sovereign *sukuk*, and it helped create a market in which governments and companies now raise large sums each year.

Firms and investors that operate across borders face prices in several currencies and, often, several legal and ethical frameworks for finance. This chapter covers the finance of foreign exchange from the perspective of a company—how exchange rates are quoted, how forward rates are set, what exposures a firm has and how to manage them, and how to value a foreign investment—and then introduces Islamic finance, a system of contracts designed to comply with Islamic law. Chapter 52 takes up exchange rates again from the macroeconomic side.

42.1 The Foreign-Exchange Market

42.1.1 Quotations

An exchange rate is the price of one currency in terms of another. A quotation names a *base* currency and a *quote* currency: $\text{EUR/USD} = 1.10$ means that one euro costs 1.10 dollars. Seen from the euro area, this is an *indirect* quote (units of foreign currency per unit of home currency); seen from the United States, a *direct* quote (home currency per unit of foreign currency). Because

either convention is common, every calculation should begin by stating which currency is being priced in which. Dealers quote a *bid* (the price at which they buy the base currency) and a higher *ask*; the spread is their compensation.

The foreign-exchange market is the largest financial market in the world: the Bank for International Settlements' triennial survey found average daily turnover of about 7.5 trillion dollars in April 2022, with the dollar on one side of close to nine-tenths of all trades. Most turnover is not spot trading but *FX swaps*, which combine a spot purchase with a forward sale, and are used by banks and companies to borrow one currency against another.

42.1.2 Cross rates and triangular arbitrage

Because most currencies trade actively against the dollar, the rate between two other currencies, a *cross rate*, is implied by their dollar rates. If EUR/USD is 1.10 and USD/JPY is 150, the implied rate of yen per euro (EUR/JPY) is $1.10 \times 150 = 165$ yen per euro. A quoted cross rate different from the implied one would permit **triangular arbitrage**: converting around the triangle of three currencies would return more than one started with. Arbitrage by dealers keeps cross rates within transaction costs of their implied values.

42.1.3 Forward rates and covered interest parity

A **forward exchange rate** is a rate agreed today for exchange at a future date. Like the forward prices of Chapter 41, it is set by no arbitrage. Write S for the spot rate and F for the forward rate, both in units of currency A per unit of currency B , and let i_A and i_B be the interest rates in the two currencies for the period of the forward.

Proposition 42.1. Covered interest parity

In the absence of arbitrage and transaction costs,

$$F = S \frac{1 + i_A}{1 + i_B}.$$

The currency with the higher interest rate trades at a forward discount that exactly offsets its interest advantage.

Proof. An investor with one unit of currency A can invest it at i_A , receiving $1 + i_A$. Alternatively, she can convert it into $1/S$ units of B , invest them at i_B , and sell the proceeds $(1 + i_B)/S$ forward today at F , receiving $F(1 + i_B)/S$ units of A . Both strategies are riskless and deliver currency A on the same date, so they must deliver the same amount: $1 + i_A = F(1 + i_B)/S$. If not, an arbitrageur would borrow in one currency and invest, covered, in the other. $\square$

Worked Example 42.2. A forward rate

EUR/USD spot is 1.10 dollars per euro. Six-month interest rates are 5 per cent a year in dollars and 3 per cent in euros, so 2.5 and 1.5 per cent for the half-year. Taking A as the dollar and B as the euro, the six-month forward rate is $1.10 \times 1.025/1.015 = 1.1108$ dollars per euro. The euro, with the lower interest rate, trades at a forward premium: dealers quote it as “forward points” of +108, the difference in the fourth decimal place.

Covered interest parity held very closely in major currencies before 2008; since then small, persistent deviations have appeared, because post-crisis regulation made arbitrage more costly for banks' balance sheets. The forward rate is not a forecast of the future spot rate: whether the

expected spot rate equals the forward rate—*uncovered* interest parity—is a separate question, taken up in Chapter 52, where the evidence is that it fails in the short run.

42.2 Foreign-Exchange Exposure

A company is exposed to exchange rates in three ways.

Transaction exposure arises from contracted cash flows in foreign currency: a receivable from a foreign customer, a payable to a foreign supplier, a foreign-currency loan. The amount in foreign currency is fixed; its value in home currency is not.

Translation exposure arises when a company consolidates foreign subsidiaries. Under IAS 21 (and ASC 830 in the United States), each entity measures its results in its *functional currency*, the currency of its main economic environment. Transactions in other currencies are recorded at the rate on the transaction date, and monetary items (cash, receivables, payables, loans) are retranslated at the closing rate, with differences recognized in profit or loss. To consolidate a foreign operation whose functional currency differs from the group's presentation currency, its assets and liabilities are translated at the closing rate and its income and expenses at (approximately) average rates; the resulting differences go to other comprehensive income, accumulating in a translation reserve within equity (Chapter 27), and are recycled to profit or loss if the operation is sold.

Worked Example 42.3. Translation exposure

A euro-area group owns a US subsidiary with net assets of 100 million dollars. At 1.10 dollars per euro they are translated as 90.9 million euros. If the dollar weakens to 1.25 dollars per euro, the same net assets translate to 80.0 million euros: a translation loss of 10.9 million euros in other comprehensive income. No cash has moved; the loss matters economically only to the extent that the subsidiary's future cash flows, measured in euros, have fallen.

Economic exposure is the sensitivity of the present value of a firm's future cash flows to exchange rates, including effects on its competitive position. A domestic manufacturer with no foreign sales or costs is exposed if it competes with importers: when its currency appreciates, imports become cheaper and its sales and margins fall. Economic exposure is the most important and hardest to measure; one estimate is the slope of a regression of the firm's share returns on exchange-rate changes (Chapter 16). It cannot be hedged fully with financial contracts, because it depends on future volumes and prices; firms manage it partly through *operational* choices, such as locating production in the markets where they sell.

42.3 Hedging and Treasury

42.3.1 Hedging a transaction

A company expecting a foreign-currency payment can fix its home-currency value with a forward contract; replicate the forward with borrowing and lending (a *money-market hedge*); or buy an option that protects against adverse moves while keeping the benefit of favourable ones.

Worked Example 42.4. Three hedges of a receivable

A German exporter will receive 10 million dollars in six months. Rates are as in Example 42.2.

- *Forward*: it sells 10 million dollars forward at 1.1108 and will receive $10/1.1108 = 9.002$ million euros for certain.
- *Money-market hedge*: it borrows $10/1.025 = 9.756$ million dollars today, to be repaid with the receivable; converts them at 1.10 into 8.869 million euros; and invests those at 1.5 per cent, receiving 9.002 million euros in six months. By covered interest parity, the result is identical to the forward.
- *Option*: it buys, for a premium of 0.1 million euros, the right to sell 10 million dollars at 1.12 dollars per euro. If the dollar weakens to 1.20, it exercises, receiving $10/1.12 = 8.929$ million euros, 8.829 million after the premium (unhedged: 8.333 million). If the dollar strengthens to 1.00, it lets the option lapse and sells at the market for 10.0 million euros, 9.9 million after the premium.

The forward eliminates uncertainty; the option sets a floor and preserves upside, at a cost.

42.3.2 Organizing the treasury

A multinational's treasury manages cash, funding and financial risk for the whole group. Several techniques reduce the need for external hedging:

- **Netting**. Subsidiaries that owe each other money in various currencies settle only their net positions through a central netting centre, reducing the number and size of transactions.
- **Natural hedges**. A subsidiary's foreign-currency assets can be financed with borrowing in the same currency, so that changes in the value of assets and debt offset; costs can be incurred in the currencies in which revenue is earned.
- **Cash pooling**. Surplus cash in one subsidiary funds deficits in others, reducing external borrowing.

The remaining exposures are hedged according to a written *hedging policy*: which exposures are hedged (contracted transactions fully, forecast ones partly, often in layers that decline with the horizon), with which instruments, with which counterparties and within which limits. Most companies treat their treasuries as cost centres whose job is to reduce risk, not as profit centres—a lesson reinforced by treasuries that speculated and lost, such as that of Metallgesellschaft (Chapter 41). Hedge accounting (Chapter 41) determines whether the hedges' effects are reported in the same periods as the exposures they offset.

42.4 International Capital Budgeting

A company evaluating a foreign project faces cash flows in one currency and investors who measure value in another. There are two consistent methods.

1. **Foreign-currency method**. Forecast the cash flows in the foreign currency, discount them at a foreign-currency discount rate, and convert the NPV at today's spot rate.
2. **Home-currency method**. Convert each year's expected cash flow into home currency at the forward rate for that year (from covered interest parity), and discount at a home-currency rate.

The two discount rates are linked by the same logic as covered parity: $1 + r_{\text{home}} = (1 + r_{\text{foreign}})(1 + i_{\text{home}})/(1 + i_{\text{foreign}})$. Applied consistently, the methods give the same answer.

Worked Example 42.5. A foreign project in two currencies

A euro-area company considers a US project costing 50 million dollars and returning 22 million dollars a year for three years. The dollar discount rate is 10 per cent; spot is 1.10 dollars per euro; one-year interest rates are 5 per cent in dollars and 3 per cent in euros. *Foreign-currency method:* $NPV = -50 + 22 \times 2.487 = 4.711$ million dollars, or $4.711/1.10 = 4.28$ million euros.

Home-currency method: the forward rates are $1.10 \times (1.05/1.03)^t$ dollars per euro, and the euro discount rate is $1.10 \times 1.03/1.05 - 1 = 7.90$ per cent. Converting each cash flow at its forward rate and discounting at 7.90 per cent gives an NPV of 4.28 million euros—the same.

Mixing the methods—converting at today’s spot rate and discounting at a home rate, or forecasting that a high-interest currency will hold its value while using its high discount rate—produces errors that can be larger than the project’s NPV (Chapter 37).

Country risk. Foreign projects face risks beyond the exchange rate: expropriation, restrictions on converting and remitting profits, war and changes in tax. It is common to add a “country risk premium” to the discount rate. But these risks usually concern specific events that reduce cash flows; they are better modelled explicitly, with scenarios and probabilities, and mitigated by structuring—local partners, financing from development banks, political-risk insurance from agencies such as the World Bank’s Multilateral Investment Guarantee Agency. A premium added to the discount rate penalizes distant cash flows more than near ones, which may or may not match the risk. What should be included in the discount rate is only the part of country risk that diversified investors cannot avoid (Chapter 35).

42.5 Islamic Finance

42.5.1 Principles

Islamic finance is finance conducted in accordance with the *Shariah*, Islamic law as interpreted by scholars. Its main prohibitions are:

- **riba**: interest, or any predetermined return on a loan of money. Money is regarded as a medium of exchange, not a commodity that earns a return by itself;
- **gharar**: excessive uncertainty or ambiguity in the terms of a contract, such as selling what one does not own or cannot deliver;
- *maysir*: gambling and speculation;
- investment in prohibited (*haram*) activities, such as alcohol, pork and conventional interest-based finance.

Positively, Islamic finance emphasizes that returns should be linked to the ownership of real assets or to sharing in the risks of an enterprise: a financier may earn a profit from trade, from renting out an asset, or from a share of business profits, but not from lending money at interest.

42.5.2 Instruments

The main contracts replicate the economic functions of conventional finance within these constraints:

- **Murabaha** (cost-plus sale): the bank buys an asset that the customer needs and sells it to the customer at a disclosed markup, payable in instalments. It is the most common form of Islamic financing.
- **Ijara** (lease): the bank buys an asset and leases it to the customer for rent; in *ijara wa iqtina*, ownership passes to the customer at the end.
- **Musharaka** (partnership): bank and customer contribute capital to a venture and share profits in agreed proportions and losses in proportion to capital. In *diminishing musharaka*, used for home finance, the customer buys the bank's share of the property gradually while paying rent on the part the bank still owns.
- **Mudaraba** (investment partnership): one party provides capital and the other management; profits are shared as agreed, and financial losses fall on the capital provider. Islamic banks' investment accounts are often structured this way.
- **Sukuk**: certificates giving investors an undivided ownership interest in assets, or in the cash flows of a venture, from which their returns are paid. They are the Islamic counterpart of bonds.
- **Takaful**: cooperative insurance, in which participants contribute to a fund from which claims are paid, with an operator managing it for a fee or a share of profits.

Worked Example 42.6. The economics of a murabaha

A customer needs a machine costing 100,000. The bank buys it and sells it to the customer for 112,000, payable in two instalments of 56,000 at the end of each year. The bank's return is a trading profit, not interest. But its economic substance is a loan: the implied rate of return r solves

$$100,000 = \frac{56,000}{1+r} + \frac{56,000}{(1+r)^2},$$

giving $r = 7.9$ per cent (an internal rate of return, found as in Chapter 5). Banks commonly set the markup with reference to conventional benchmark interest rates. Under IFRS, such a receivable is typically measured at amortized cost using this effective rate (Chapter 30): accounting follows substance.

The example raises the central debate within Islamic finance. Critics, including some Islamic scholars and economists, argue that many instruments replicate interest-bearing debt in form while retaining its substance, and that the industry has moved away from the profit-and-loss-sharing contracts (*musharaka* and *mudaraba*) its founders envisaged. Defenders reply that the requirement to deal in real assets ties finance to the real economy, restricts speculation and excessive leverage, and gives customers ethical choices; and that differences of legal form matter, for example in who bears the risks of an asset's ownership. Both sides agree that the distinction between *asset-based* sukuk, in which investors rely on the issuer's promise to repurchase the assets, and *asset-backed* sukuk, in which investors have genuine recourse to the assets, is economically significant: most sukuk are asset-based, and their credit risk is essentially that of the issuer.

42.5.3 Governance and markets

Each Islamic financial institution has a *Shariah supervisory board* of scholars who approve its products. Because interpretations differ across schools and countries, standard-setting bodies have developed: the Accounting and Auditing Organization for Islamic Financial Institutions (AAOIFI, based in Bahrain) issues Shariah, accounting and governance standards, and the Islamic Financial Services Board (IFSB, based in Kuala Lumpur) issues prudential standards for regulators. Malaysia has built one of the most comprehensive legal frameworks: its Islamic Financial Services Act 2013 regulates Islamic banks and takaful operators, and a Shariah Advisory Council at Bank Negara Malaysia issues rulings that bind courts and institutions on Shariah questions in Islamic finance. Islamic finance assets are estimated at several trillion dollars, concentrated in Iran, Saudi Arabia, Malaysia, the other Gulf states and a few others.

The legal robustness of the contracts has been tested. In 2017 Dana Gas, a company in the United Arab Emirates, argued that its sukuk had become non-compliant with Shariah and therefore unenforceable under UAE law. Part of the documentation was governed by English law, and the English High Court held the obligations enforceable, one of several episodes that led investors to focus on the legal structure of sukuk as much as their religious approval.

Case: Malaysia's 2002 global sovereign sukuk

The background. Malaysia had developed a domestic Islamic capital market from the 1980s: it enacted an Islamic Banking Act in 1983, and in 1990 Shell MDS issued what is often cited as the first corporate sukuk, in ringgit. By 2002 the government sought to show that Islamic instruments could raise funds from international investors on terms comparable to conventional bonds.

The structure. In June 2002 the government raised 600 million dollars through a five-year *ijara* sukuk—the first global sovereign sukuk. A special-purpose company, Malaysia Global Sukuk Inc., bought land and buildings from the government with the investors' money and leased them back to the government. The government's rental payments, set at a floating rate linked to dollar interbank rates, were passed through to the certificate holders twice a year. At maturity the government was committed to buy the properties back for the original amount, which was paid out to investors. The certificates were rated, and listed on the Labuan and Luxembourg exchanges, and were bought by conventional as well as Islamic investors.

What followed. The issue became a template. Sovereigns and companies in the Gulf, Asia, Africa and Europe—including the United Kingdom in 2014—issued sukuk, and Malaysia has remained one of the largest issuers and the leading centre of sukuk issuance, especially in local currency.

In terms of this chapter. For investors, the certificates' cash flows—regular payments and repayment of principal at maturity, backed by the government's undertakings—were economically equivalent to a floating-rate government bond; their credit risk was Malaysia's, not that of the buildings (an asset-based structure). Their Shariah compliance rested on the ownership and lease of real assets, which also created real legal steps—property transfers, a special-purpose vehicle, a purchase undertaking—that conventional bonds do not need. Pricing followed the logic of Chapter 33: a spread over a benchmark rate reflecting Malaysia's credit. And the case shows how financial innovation can serve investors with particular preferences—here religious ones—while fitting the structures of global markets.

Summary

- Exchange rates are prices of one currency in another; always state the convention. Cross rates are implied by dollar rates, and triangular arbitrage enforces them.
- Covered interest parity, $F = S(1 + i_A)/(1 + i_B)$, sets forward rates by no arbitrage; the forward rate is not a forecast.
- Transaction exposure concerns contracted foreign-currency flows; translation exposure, the consolidation of foreign operations under IAS 21 (differences in OCI); economic exposure, the effect of exchange rates on future cash flows and competitiveness.
- Forwards and money-market hedges fix a home-currency amount and are equivalent under CIP; options set a floor for a premium.
- Treasuries reduce exposures by netting, natural hedges and pooling, and hedge the rest under a written policy, as cost centres rather than profit centres.
- Foreign projects can be valued in foreign currency at a foreign discount rate or in home currency at forward rates and a home discount rate; consistent use gives the same NPV. Country risk belongs mainly in cash-flow scenarios.
- Islamic finance prohibits *riba*, *gharar* and *maysir* and ties returns to assets or shared risk. *Murabaha*, *ijara*, *musharaka*, *mudaraba*, *sukuk* and *takaful* replicate conventional functions; debates concern whether form or substance differs. Standards come from AAOIFI and IFSB, and Malaysia has a comprehensive legal framework.

Key terms

base and quote currency; direct and indirect quotation; bid–ask spread; FX swap; cross rate; triangular arbitrage; forward exchange rate; forward points; covered interest parity; transaction exposure; translation exposure; functional currency; presentation currency; translation reserve; economic exposure; money-market hedge; currency option; netting; natural hedge; cash pooling; hedging policy; foreign-currency and home-currency methods; country risk; political-risk insurance; Shariah; *riba*; *gharar*; *maysir*; *murabaha*; *ijara*; *musharaka*; diminishing *musharaka*; *mudaraba*; *sukuk*; asset-based and asset-backed *sukuk*; *takaful*; Shariah supervisory board; AAOIFI; IFSB.

Exercises

Exercise 42.1 (Triangular arbitrage). EUR/USD is 1.10 and USD/JPY is 150. A bank quotes EUR/JPY at 167. Starting with 1 million euros, describe the arbitrage and compute the profit.

Exercise 42.2 (Forward rate). GBP/USD spot is 1.25 dollars per pound. One-year interest rates are 4.5 per cent in dollars and 5 per cent in pounds. Find the one-year forward rate and the forward points. Which currency trades at a forward premium?

Exercise 42.3 (Forward and money-market hedges). A US company will receive 5 million pounds in a year. Using the rates in Exercise 42.2, find its dollar receipts with a forward hedge and with a money-market hedge.

Exercise 42.4 (Option hedge). A US importer must pay 2 million euros in three months. The three-month forward rate is 1.103 dollars per euro. A call option on euros with strike 1.10 costs 0.03 dollars per euro. Compare the dollar cost of the payment, unhedged, with a forward and with the option, if the spot rate in three months is 1.00, 1.10 or 1.20.

Exercise 42.5 (Translation). A UK group's euro-area subsidiary has net assets of 50 million euros. The rate moves from 0.85 to 0.80 pounds per euro during the year. Find the translation difference and where it is reported. Does it affect cash?

Exercise 42.6 (Netting). Within a group, subsidiary *A* owes *B* 10 million, *B* owes *C* 7 million, *C* owes *A* 4 million and *B* owes *A* 3 million (all in dollar equivalents). Find each subsidiary's net position and the payments required after multilateral netting.

Exercise 42.7 (Foreign project). A UK company considers a US project costing 30 million dollars and returning 12 million dollars a year for three years. The dollar discount rate is 9 per cent; GBP/USD spot is 1.25; one-year interest rates are 4.5 per cent in dollars and 5 per cent in pounds. Find the NPV in pounds by both methods.

Exercise 42.8 (Murabaha). A bank buys equipment for 200,000 and sells it to a customer for 230,000, payable in two equal annual instalments. Find the implied rate of return. How would you describe the transaction's economic substance?

Exercise 42.9 (Sukuk). A special-purpose company buys buildings from a government for 500 million, leases them back at an annual rent of 22.5 million for five years, and will sell them back for 500 million at the end. Describe the investors' cash flows and yield. In what circumstances would investors' position differ from that of holders of a 4.5 per cent government bond?

Exercise 42.10 (Economic exposure). An Italian furniture maker sells only in the euro area but competes with imports priced in dollars. A regression of its monthly share returns on the monthly percentage change in the dollar price of a euro has a slope of -0.8 . Interpret the coefficient. How might the company reduce its exposure?

Solutions

Solution 42.1. The quoted 167 exceeds the implied 165, so euros are dear in yen. Sell 1 million euros for 167 million yen; buy $167/150 = 1.11333$ million dollars; buy $1.11333/1.10 = 1.01212$ million euros. Profit: about 12,100 euros, before costs.

Solution 42.2. $F = 1.25 \times 1.045/1.05 = 1.2440$ dollars per pound; forward points -60 (-59.5). The dollar, with the lower interest rate, trades at a forward premium (the pound at a discount).

Solution 42.3. Forward: $5 \times 1.2440 = 6.220$ million dollars. Money market: borrow $5/1.05 = 4.762$ million pounds, convert at 1.25 to 5.952 million dollars, invest at 4.5 per cent: 6.220 million dollars. Identical, by CIP.

Solution 42.4. Premium 60,000 dollars. At 1.00: unhedged 2.00 million; forward 2.206; option 2.06 (lapse). At 1.10: unhedged 2.20; forward 2.206; option 2.26. At 1.20: unhedged 2.40; forward 2.206; option 2.26 (exercise at 1.10). The option caps the cost at 2.26 million while allowing the importer to benefit if the euro falls.

Solution 42.5. Translated net assets fall from 42.5 to 40.0 million pounds: a translation loss of 2.5 million pounds, reported in other comprehensive income and accumulated in the translation reserve (recycled to profit or loss only if the subsidiary is sold). No cash moves.

Solution 42.6. *A*: receives $4 + 3 = 7$, pays 10: net -3 . *B*: receives 10, pays $7 + 3 = 10$: net 0. *C*: receives 7, pays 4: net $+3$. One payment of 3 million from *A* to *C* replaces four payments totalling 24 million.

Solution 42.7. Dollar NPV $= -30 + 12 \times 2.531 = 0.376$ million, or 0.30 million pounds at 1.25. Home method: forwards $1.25 \times (1.045/1.05)^t$; sterling discount rate $1.09 \times 1.05/1.045 - 1 = 9.52$ per cent; the NPV is again 0.30 million pounds.

Solution 42.8. $200,000 = 115,000/(1 + r) + 115,000/(1 + r)^2$ gives $r = 9.8$ per cent. In substance, a two-year amortizing loan at 9.8 per cent, secured on the equipment; in form, a sale at a markup, in which the bank briefly owns the asset.

Solution 42.9. Investors pay 500 million, receive 22.5 million a year for five years and 500 million at the end: a yield of 4.5 per cent, the same cash flows as a 4.5 per cent bond. They differ if the structure is tested: if the government failed to pay rent or to repurchase the buildings, asset-backed sukuk holders might have recourse to the buildings, whereas holders of asset-based sukuk (the usual case) rely on the government's undertaking, like bondholders; if the contracts' Shariah compliance or enforceability were challenged; and in tax and accounting treatment in some jurisdictions.

Solution 42.10. A 1 per cent rise in the dollar price of a euro (a euro appreciation) is associated with a 0.8 per cent fall in the share price: a stronger euro makes dollar-priced imports cheaper in Europe, squeezing the company's sales and margins. It could reduce the exposure by sourcing materials or production in dollars, by differentiating its products to reduce competition with imports, or partially by hedging expected future cash flows with forwards—though financial hedges cover only a limited horizon.

Part VI

Macroeconomics

National Accounts and the Open-Economy Accounting Framework

After studying this chapter you should be able to

1. Describe the institutional sectors and the sequence of accounts in the System of National Accounts.
2. Decompose GDP by expenditure, by income and by industry, and compute contributions to growth.
3. Distinguish GDP, GNI, national income and disposable income, and compute saving rates.
4. Explain how labour-market statistics are defined and measured.
5. Derive the national saving–investment identity for an open economy and relate it to the current account.
6. Construct and interpret the balance of payments and the international investment position, including valuation effects.
7. Use sectoral financial accounts to trace who lends to whom.
8. Interpret quarterly data correctly, including annualization and seasonal adjustment.

A motivating question

In July 2020 the U.S. Bureau of Economic Analysis reported that real GDP had fallen at an annual rate of about 33 per cent in the second quarter of the year. Headlines announced the largest collapse in American history. Yet the economy had not shrunk by a third: output in the second quarter was roughly 9 per cent lower than in the first. Both statements were accurate; the difference lay in a convention. At the same time, the U.S. household saving rate jumped to over 30 per cent of disposable income, the federal deficit exploded, and the current-account deficit barely changed. Were these developments connected? The accounting identities of this chapter say that they had to be.

Chapter 10 introduced GDP, price indexes and the sector-balance identity. This chapter develops the full national-accounting framework on which the rest of Part VI is built. It parallels Part IV: just as a firm's balance sheet, income statement and cash-flow statement articulate, a nation's production, income, saving, investment, borrowing and wealth are linked by identities. Understanding those identities—and remembering that they are identities, not theories—is the first step in macroeconomics.

43.1 The System of National Accounts

43.1.1 An international standard

Almost every country compiles its national accounts according to the **System of National Accounts (SNA)**, an international standard published by the United Nations, the IMF, the OECD, the World Bank and the European Commission. The current version is the 2008 SNA; an update (the 2025 SNA) has been agreed, with countries implementing it over the following years. The European Union applies a closely aligned regulation, ESA 2010. The United States publishes its National Income and Product Accounts (NIPAs), prepared by the Bureau of Economic Analysis (BEA), which are broadly consistent with the SNA though differently organized; the Federal Reserve publishes the U.S. *Financial Accounts* (the Z.1 release), which track financial flows and balance sheets by sector. China's National Bureau of Statistics, Germany's Federal Statistical Office (Destatis) and Malaysia's Department of Statistics (DOSM) all follow the SNA framework.

43.1.2 Institutional sectors

The SNA divides resident units into **institutional sectors** according to their economic behaviour:

- **Non-financial corporations:** firms producing goods and non-financial services.
- **Financial corporations:** banks, the central bank, insurance companies, pension funds, investment funds.
- **General government:** central, state and local governments and social security funds.
- **Households:** including unincorporated businesses owned by households (sole proprietors).
- **Non-profit institutions serving households (NPISH):** charities, churches, unions.

Together these make up the total economy. Transactions with non-residents are recorded in the accounts of the **rest of the world**.

43.1.3 The sequence of accounts

For each sector and for the economy as a whole, the SNA records a *sequence of accounts*, each of which ends with a *balancing item* that is carried to the next:

Account	Records	Balancing item
Production	Output minus intermediate consumption	Value added (GDP)
Generation of income	Value added paid to labour, capital and government	Operating surplus, mixed income
Allocation of primary income	Plus property income received, minus paid	Primary income (GNI)
Secondary distribution	Taxes on income, social contributions and benefits, other transfers	Disposable income
Use of income	Final consumption	Saving
Capital account	Saving plus capital transfers, minus investment	Net lending (+) / borrowing (–)
Financial account	Net acquisition of financial assets minus net incurrence of liabilities	Net lending (+) / borrowing (–)
Balance sheet	Stocks of assets and liabilities	Net worth

The structure mirrors business accounting closely. Value added corresponds to a firm's gross profit before paying its workers; saving corresponds to retained earnings; net lending corresponds to the difference between internally generated funds and investment, which must be matched by net financing. And the capital account and the financial account yield the *same* balancing item, net lending, computed in two ways—from real flows and from financial flows. This is the macroeconomic counterpart of a firm's cash-flow statement, in which the sum of operating and investing cash flows must be matched by financing cash flows.

43.2 GDP in Detail

43.2.1 Expenditure components

The expenditure identity of Chapter 10, $Y \equiv C + I + G + X - M$, can be refined using the categories in the U.S. NIPAs:

- **Personal consumption expenditures (C):** durable goods (cars, appliances), non-durable goods (food, fuel, clothing) and services (housing, health care, financial services). Services dominate spending in rich economies.
- **Gross private domestic investment (I):** non-residential fixed investment (structures, equipment and *intellectual property products*—software, research and development, and entertainment originals), residential investment (new housing and improvements), and the change in private inventories.
- **Government consumption expenditures and gross investment (G):** purchases of goods and services, compensation of government employees (government output is valued at cost), and government investment (roads, buildings, military equipment, government R&D). Transfers such as pensions and unemployment benefits are excluded.
- **Net exports ($X - M$).**

The composition of spending differs greatly across economies. In the United States, personal consumption is roughly two-thirds of GDP and net exports are negative. In China, household consumption has been under 40 per cent of GDP for much of the past two decades, while gross capital formation has exceeded 40 per cent—among the highest investment shares ever recorded in a large economy. Germany combines a household consumption share of roughly one-half with persistent positive net exports. Malaysia is an exceptionally open economy: its exports of goods and services have exceeded two-thirds of GDP in most recent years, and its imports are nearly as large, reflecting its role in global electronics supply chains. These differences, discussed in Chapters 44 and 52, shape how each economy responds to shocks.

43.2.2 Contributions to growth

Because GDP is a sum, its growth rate is a weighted sum of the growth rates of its components. If s_j is component j 's share of GDP in the previous period and g_j its growth rate, the **contribution to growth** of component j is $s_j g_j$, and

$$g_Y \approx \sum_j s_j g_j, \quad \text{with imports entering negatively.}$$

The BEA publishes these contributions every quarter; they are more informative than component growth rates alone, because a small component growing fast may contribute less than a large one growing slowly.

Worked Example 43.1. Contributions to GDP growth

In a year, the shares of GDP are C 68%, I 18%, G 17%, exports 11%, imports 14% (so net exports are -3%). Real growth rates: C 2.5%, I 4%, G 1%, exports 3%, imports 5%. Contributions:

Consumption	$0.68 \times 2.5 = 1.70$
Investment	$0.18 \times 4 = 0.72$
Government	$0.17 \times 1 = 0.17$
Exports	$0.11 \times 3 = 0.33$
Imports	$-0.14 \times 5 = -0.70$
GDP growth	2.22

Net exports subtract 0.37 percentage points, because imports grew faster than exports and are a larger share. (With chain-weighted real data, contributions are computed with a formula that makes them add up exactly; the principle is the same.)

► Common error: “imports reduce GDP”

Imports are subtracted in the expenditure identity only because they are already included in C , I and G (and in exports, when imported components are re-exported). Subtracting them removes foreign production from a measure of domestic production. A rise in imports that replaces domestic production does reduce GDP; a rise in imports that meets additional spending—a household buying an imported car it would not otherwise have bought—leaves GDP unchanged, since C and M rise together.

43.2.3 The income side and the statistical discrepancy

Measured by income, GDP is the sum of compensation of employees, gross operating surplus (including depreciation), gross mixed income of the self-employed, and taxes on production and imports less subsidies. The BEA reports **gross domestic income** (GDI) from the income side. In principle GDI equals GDP; in practice they are estimated from different source data, and the difference is recorded as the **statistical discrepancy**. It can be large enough to change the story: in some quarters GDP and GDI have moved in different directions. Many economists average the two.

The share of compensation in value added is the **labour share**. In the United States and several other advanced economies it declined from the early 2000s, a trend that economists have attributed variously to technology, globalization, the rise of highly profitable “superstar” firms, and weaker worker bargaining power (Chapter 45).

43.2.4 GDP by industry

The production approach also yields value added by industry. In rich economies, services account for around three-quarters of value added and manufacturing for well under a fifth, though Germany’s manufacturing share (around a fifth of gross value added) is high by advanced-economy standards. In China, industry’s share rose above 40 per cent during the 2000s before declining as services grew. Structural change—the shift of output and employment from agriculture to industry to services as incomes rise—is one of the most robust facts of economic development (Chapter 44).

43.3 From GDP to Disposable Income

43.3.1 National income concepts

Starting from GDP:

$$\begin{aligned}\text{GNI} &= \text{GDP} + \text{primary income from abroad} - \text{primary income paid abroad}, \\ \text{NNI} &= \text{GNI} - \text{consumption of fixed capital (depreciation)}, \\ \text{GNDI} &= \text{GNI} + \text{net current transfers from abroad}.\end{aligned}$$

Gross national disposable income (GNDI) is the income available to residents for consumption and saving. For a country receiving large remittances from citizens working abroad—the Philippines, for instance—GNDI substantially exceeds GNI.

43.3.2 Household disposable income and saving

Household **disposable income** is primary income (wages, mixed income, net property income) plus social benefits and other transfers received, minus taxes on income and wealth and social contributions paid. Household saving is disposable income minus consumption, and the **household saving rate** is saving divided by disposable income. Saving rates differ widely: China's household saving rate has been well above 30 per cent of disposable income in many recent years, Germany's around 10 per cent or higher, and the U.S. rate typically in single digits outside of the pandemic.

43.4 Labour-Market Measurement

43.4.1 Definitions

Macroeconomics cares as much about employment as about output. Labour-market statistics follow definitions set by the International Labour Organization (ILO):

- A person is **employed** if she did any work for pay or profit during the reference week (even one hour), or had a job from which she was temporarily absent.
- A person is **unemployed** if she was not employed, was available for work, and had actively looked for work in the previous four weeks (or was waiting to start a job).
- The **labour force** is the employed plus the unemployed; everyone else of working age is *not in the labour force*.

The **unemployment rate** is unemployed divided by the labour force; the **participation rate** is the labour force divided by the working-age population; the **employment–population ratio** is employed divided by working-age population.

43.4.2 Surveys and alternative measures

In the United States, the unemployment rate comes from the monthly Current Population Survey of about 60,000 households, conducted for the Bureau of Labor Statistics; payroll employment comes from the Current Employment Statistics survey of businesses (the “establishment survey”). The two sometimes diverge, and each has characteristic errors. The headline unemployment rate (U-3) excludes discouraged workers, who want work but have stopped looking, and part-time workers who want full-time work; the broader U-6 measure includes them. Germany's Federal Employment Agency publishes a registered unemployment

count that differs from the ILO measure published by Destatis. China's urban surveyed unemployment rate, published since 2018, covers only urban areas and has faced questions about coverage of rural migrant workers.

► **Common error: a falling unemployment rate always means more people are working**

The unemployment rate can fall because unemployed people find jobs or because they stop looking and leave the labour force. In the latter case, the employment–population ratio does not rise. After the 2008 recession, the U.S. unemployment rate fell steadily while participation also fell; analysts debated how much of the participation decline reflected discouragement (cyclical) and how much population ageing (structural).

43.5 Saving, Investment and the Current Account

43.5.1 The open-economy identity

National saving is gross national disposable income minus consumption by households and government:

$$S_N \equiv \text{GNDI} - C - G_C,$$

where G_C is government consumption. Treating government investment as part of I , and using $\text{GNDI} \equiv Y + \text{NPI} + \text{NCT}$ (net primary income and net current transfers from abroad), together with the expenditure identity $Y \equiv C + G_C + I + X - M$:

$$S_N - I \equiv X - M + \text{NPI} + \text{NCT} \equiv CA,$$

where CA is the **current-account balance**.

Saving, investment and the current account. A country's current-account surplus equals its national saving minus its domestic investment. A country that invests more than it saves must borrow the difference from abroad, running a current-account deficit; one that saves more than it invests lends abroad.

Splitting national saving into private saving S_P and government saving $S_G = T - G_C$ (taxes net of transfers minus government consumption):

$$CA \equiv (S_P - I_P) + (S_G - I_G),$$

the sector-balance identity of Chapter 10 with government investment separated.

43.5.2 Examples

The identity organizes discussions of global imbalances. Germany's large current-account surpluses in the 2010s reflected national saving well above domestic investment, with contributions from households, corporations and, for several years, a government surplus. China's surplus peaked in the mid-2000s, when national saving approached half of GDP, and then fell as investment and consumption rose relative to saving. The U.S. deficit reflects national saving below investment, with large government deficits in most years since 2001. Malaysia ran persistent surpluses after the Asian financial crisis of 1997–98, when its investment rate fell sharply from very high pre-crisis levels while saving remained high, and it rebuilt its foreign-exchange reserves (Chapter 52).

43.6 The Balance of Payments

43.6.1 Structure

The **balance of payments** (BoP) records all economic transactions between residents and non-residents over a period. It follows the IMF's *Balance of Payments and International Investment Position Manual*, sixth edition (BPM6). It has three accounts:

1. The **current account**:

- goods (exports minus imports of merchandise);
- services (travel, transport, financial and business services, intellectual-property charges);
- *primary income*: compensation of employees and investment income (dividends, reinvested earnings, interest);
- *secondary income*: current transfers such as workers' remittances, foreign aid and contributions to international organizations.

2. The **capital account**: capital transfers (debt forgiveness, migrants' transfers of assets) and acquisitions and disposals of non-produced, non-financial assets such as patents and land rights. Usually small.

3. The **financial account**: net acquisition of foreign financial assets by residents minus net incurrence of liabilities to non-residents, classified as *direct investment* (a lasting interest, conventionally 10% or more of voting power), *portfolio investment* (shares and debt securities), *financial derivatives*, *other investment* (loans, deposits, trade credit) and *reserve assets* (foreign assets controlled by the central bank).

43.6.2 The balance-of-payments identity

Every international transaction is recorded twice, as in double-entry bookkeeping: an export of goods is matched by an acquisition of a financial asset (a foreign bank deposit) or a reduction in a liability. Under BPM6's sign conventions, the identity is

$$CA + KA \equiv FA,$$

where *FA* is the net financial account (net acquisition of assets minus net incurrence of liabilities). A country with a current-account surplus must, on net, acquire foreign assets or reduce its foreign liabilities. In practice, because the entries are compiled from different sources, there is a residual, *net errors and omissions*.

Worked Example 43.2. Recording transactions

A German firm exports machinery worth 10 million euros to a U.S. customer, who pays into the German firm's account at a U.S. bank. Germany's current account records exports of 10 million (credit); its financial account records a 10 million increase in German residents' foreign assets (other investment, deposits). Both sides of the identity rise by 10 million.

If the German firm then uses the deposit to buy U.S. Treasury bonds, the financial account records a decrease in deposit assets and an increase in portfolio assets of the same amount: no net change. If the Bundesbank instead acquired the dollars (in a currency intervention, Chapter 52), the entry would be in reserve assets.

► **Common error: “a current-account deficit means a country is losing money”**

A current-account deficit means that residents are spending more than their income and financing the difference by selling assets to, or borrowing from, non-residents. Whether that is a problem depends on why. A country borrowing to finance productive investment that will generate future exports may be making a sensible intertemporal choice (the two-period model of Chapter 9 applied to a nation). One borrowing to finance consumption, or relying on short-term capital inflows that can reverse suddenly, may be heading for a crisis (Chapter 52).

43.7 The International Investment Position

43.7.1 Stocks and flows again

The **international investment position** (IIP) is a balance sheet: the stock of a country's external financial assets and liabilities at a date. The **net international investment position** (NIIP) is assets minus liabilities. By the stock-flow logic of Chapter 1,

$$\text{NIIP}_t \equiv \text{NIIP}_{t-1} + FA_t + \text{valuation changes}_t + \text{other volume changes}_t,$$

where FA_t (approximately the current account) is the net financial flow, and *valuation changes* reflect changes in asset prices and exchange rates.

43.7.2 Valuation effects

Valuation effects can be as large as years of current-account flows. Consider the United States, whose foreign assets are largely equity and direct investment denominated in foreign currencies, while its liabilities are largely dollar-denominated debt (Treasury bonds and deposits). When the dollar depreciates, the dollar value of U.S. foreign assets rises while liabilities are unchanged: the NIIP improves without any current-account surplus. When foreign stock markets underperform U.S. markets, the reverse happens. The U.S. NIIP deteriorated from a position close to balance in the late 1980s to a net liability exceeding 20 trillion dollars by 2024, driven not only by persistent current-account deficits but also, in the 2010s and 2020s, by the strong performance of U.S. equity markets, which raised the value of foreigners' holdings of U.S. shares.

Many emerging economies have the opposite currency structure: they borrow in dollars but hold assets in their own currency. A depreciation then *increases* the domestic-currency value of their liabilities, worsening their net position just when the economy is under stress—a *currency mismatch* that was central to the Asian financial crisis (Chapter 52).

43.7.3 Reserves

Official reserve assets are foreign-currency assets held by monetary authorities and available for intervention or emergency needs. China's reserves have exceeded 3 trillion dollars since 2011, the world's largest. Malaysia, after the 1997–98 crisis, built up reserves as insurance against sudden capital outflows. Holding reserves is costly (they typically earn low returns) but provides self-insurance against sudden stops, a trade-off discussed in Chapter 52.

43.8 Sectoral Financial Accounts

The financial account of each sector records how it lends and borrows. In the U.S. Financial Accounts (Z.1), for example, one can trace how, in a given year, households acquired deposits and shares and incurred mortgage debt, how non-financial corporations issued bonds and bought back equity, how the federal government issued Treasury securities, and how the rest of the world bought them. Because every financial asset is some other sector's liability, the flows form a closed matrix: the sum of all sectors' net lending, including the rest of the world, is zero.

Worked Example 43.3. Sector balances in 2020

In 2020 the U.S. federal government ran a deficit of around 15 per cent of GDP, as it paid stimulus cheques, enhanced unemployment benefits and loans to firms. The sector-balance identity required that the other sectors' surpluses rise by the same amount (to the extent that the current account did not move). They did: households, whose incomes were supported by transfers while their spending on services was restricted by the pandemic, ran record surpluses, and firms reduced investment and borrowed heavily at low rates to hold cash. The household saving rate rose above 30 per cent of disposable income in April 2020. The identity does not tell us that the deficit "caused" the saving or vice versa; it tells us that the two were two sides of the same set of transactions. Chapter 46 analyses the behaviour behind them.

43.9 Reading Quarterly Data

43.9.1 Annualization

Quarterly GDP growth in the United States is reported at a **seasonally adjusted annual rate**: the quarterly growth rate compounded over four quarters. If real GDP falls by 9 per cent from one quarter to the next, the annualized rate is $0.91^4 - 1 = -31.4\%$. A quarterly rise of 0.5 per cent is reported as $1.005^4 - 1 = 2.0\%$. This explains the headline in the motivating question. Most other countries, including Germany and Malaysia, headline the quarter-on-quarter rate without annualizing, and China headlines the year-on-year rate—the change from the same quarter a year earlier. Comparing growth across countries requires converting to a common basis.

43.9.2 Seasonal adjustment

Retail sales rise every December; construction falls every winter in cold countries; Chinese output dips around the Lunar New Year, whose date moves between January and February. **Seasonal adjustment** removes these regular calendar patterns so that underlying movements can be seen. It is a statistical estimate, revised as more data arrive, and it can be distorted by unusual events: the pandemic's sharp fall and rebound in 2020 confused seasonal-adjustment models for several years afterward.

43.9.3 Revisions

As Chapter 10 noted, estimates are revised. The BEA's advance estimate of 2020Q2 was later revised, as were most quarterly estimates. When evaluating a past policy decision, use the data available at the time.

43.10 Identities and Theories

This chapter has presented identities: $Y \equiv C + I + G + NX$; $S_N - I \equiv CA$; $CA + KA \equiv FA$; the sector balances sum to zero. They hold in every economy at every date, whatever anyone does. They are the scaffolding of macroeconomics, not its content. Each constrains outcomes—a government deficit must be matched somewhere; a current-account deficit must be financed—but none says what causes what. To explain why saving rose, why a current account moved, or how a policy will affect output, we need behavioural relationships: how households decide to consume and save, how firms decide to invest, how prices and wages adjust, and how monetary policy works. The rest of Part VI supplies them.

Case: China's high saving and investment

From the early 2000s to the early 2010s, China's gross national saving rose to around half of GDP—an unusually high rate for any economy—and its investment rate to over 45 per cent. Because saving exceeded investment, China ran current-account surpluses that peaked close to 10 per cent of GDP in 2007, and its central bank accumulated foreign reserves on an enormous scale.

The national accounts identify who did the saving. Household saving rates were high, which economists have linked to rapid income growth (consumption habits adjust slowly), the one-child policy's effects on old-age support and on competition in marriage markets, weak social insurance that encouraged precautionary saving, and limited access to consumer credit. Corporate saving was also high, as firms—including state-owned enterprises that paid little in dividends—retained profits; and the government saved too, through budgetary surpluses of its operating accounts and investment through state entities. On the other side, investment was driven by rapid urbanization, infrastructure and manufacturing capacity, financed largely by state-influenced banks.

After the global financial crisis, as export demand fell, China's surplus shrank. The identity required that saving fall relative to investment; it did so partly because investment rose still further, in a credit-financed stimulus. The resulting build-up of corporate and local-government debt is a central theme of discussions of China's growth model (Chapter 44). The case shows how the national accounts frame a question—why was the surplus so large?—whose answer requires the behavioural theories of later chapters.

Summary

- The System of National Accounts records production, income, consumption, saving, investment, lending and wealth by institutional sector in a sequence of linked accounts.
- GDP by expenditure comprises consumption, investment (including intellectual property products and inventories), government purchases and net exports; contributions to growth weight component growth rates by shares. GDI and GDP differ by a statistical discrepancy.
- GNI adds net primary income from abroad; disposable income adds transfers. Saving is disposable income minus consumption.
- Labour statistics follow ILO definitions; the unemployment rate can fall because people stop looking for work.
- National saving minus domestic investment equals the current-account balance.

- The balance of payments records current, capital and financial accounts; $CA + KA \equiv FA$. The international investment position changes with financial flows and valuation effects.
- Sectoral financial accounts show who lends to whom; sector balances sum to zero.
- U.S. quarterly growth is annualized; China headlines year-on-year growth; seasonal adjustment and revisions matter.
- Identities constrain outcomes but do not explain them.

Key terms

System of National Accounts; institutional sectors; sequence of accounts; net lending/borrowing; personal consumption expenditures; gross private domestic investment; intellectual property products; contribution to growth; gross domestic income; statistical discrepancy; labour share; GNI; NNI; GNDI; disposable income; household saving rate; employed; unemployed; labour force; unemployment rate; participation rate; employment–population ratio; U-6; national saving; current-account balance; balance of payments; capital account; financial account; direct investment; portfolio investment; reserve assets; international investment position; net international investment position; valuation effects; currency mismatch; official reserve assets; sectoral financial accounts; seasonally adjusted annual rate; seasonal adjustment.

Exercises

Exercise 43.1 (Components). Classify each item in GDP by expenditure, or explain why it is excluded: (a) a firm's purchase of accounting software; (b) a government's payment of old-age pensions; (c) a family's purchase of a newly built flat; (d) an increase in a car maker's stock of unsold cars; (e) a German tourist's hotel bill in Kuala Lumpur (from Malaysia's and Germany's perspectives); (f) a university's spending on research, if it is a government institution.

Exercise 43.2 (Contributions). Real GDP components (in constant prices) change as follows from last year to this year: C from 700 to 714, I from 200 to 190, G from 150 to 156, exports from 120 to 126, imports from 170 to 172. Compute GDP in each year, GDP growth, and each component's contribution. Which component was the main drag on growth?

Exercise 43.3 (Income concepts). A country has GDP of 1,000, depreciation of 150, primary income received from abroad of 40 and paid abroad of 90, and net current transfers from abroad of 25. Compute GNI, NNI and GNDI.

Exercise 43.4 (Labour statistics). A country has a working-age population of 50 million, of whom 30 million are employed and 2 million unemployed. Compute the unemployment rate, the participation rate and the employment–population ratio. If 0.5 million unemployed people stop searching for work, what happens to each measure?

Exercise 43.5 (Saving and the current account). A country has GNDI of 2,000, household consumption of 1,200, government consumption of 400, private investment of 350 and government investment of 100. Compute national saving and the current account. If government consumption rises by 50 with no change in other items, what must happen to the current account?

Exercise 43.6 (Balance of payments). Record in Malaysia's balance of payments: (a) a Malaysian factory exports chips worth 100 to a U.S. buyer, who pays into the factory's dollar account in Kuala Lumpur; (b) a Japanese firm builds a new plant in Penang for 200, financed by

transferring funds to a Malaysian bank; (c) Malaysian workers in Singapore remit 30 to their families; (d) Bank Negara Malaysia buys 50 of U.S. Treasury bills with dollars from its deposits at a foreign bank.

Exercise 43.7 (Valuation effects). A country has foreign assets worth 1,000 (in its own currency), all denominated in euros, and foreign liabilities of 1,200, all in its own currency. During the year it runs a current-account surplus of 30, used to acquire more euro assets, and at the end of the year the euro becomes 10% more expensive in terms of the country's currency. What is its NIIP at the start and at the end of the year? How large is the valuation effect?

Exercise 43.8 (Annualization). Convert: (a) quarterly growth of 1.2% to an annualized rate; (b) an annualized rate of -4% to a quarterly rate; (c) quarterly growth rates of 0.5%, 0.8%, -0.3% and 0.6% into a year-on-year growth rate for the fourth quarter.

Exercise 43.9 (Identities and causation). A commentator argues: "Germany's current-account surplus proves that German firms are more competitive than their neighbours." Use the saving–investment identity to identify at least two other explanations consistent with the same surplus.

Solutions

Solution 43.1. (a) Investment: intellectual property products (software). (b) Excluded: a transfer payment, not a purchase of output. (c) Residential investment. (d) Investment: change in inventories. (e) For Malaysia: export of services (travel); for Germany: import of services, and included in German consumption, netting to zero in German GDP. (f) Government gross investment (government R&D is treated as investment in the current SNA and NIPAs).

Solution 43.2. GDP last year = $700 + 200 + 150 + 120 - 170 = 1,000$; this year = $714 + 190 + 156 + 126 - 172 = 1,014$. Growth 1.4%. Contributions (change divided by last year's GDP): $C + 1.4$; $I - 1.0$; $G + 0.6$; exports $+0.6$; imports -0.2 . Sum = 1.4. Investment was the main drag.

Solution 43.3. GNI = $1,000 + 40 - 90 = 950$. NNI = $950 - 150 = 800$. GNDI = $950 + 25 = 975$.

Solution 43.4. Labour force 32 million. Unemployment rate $2/32 = 6.25\%$; participation $32/50 = 64\%$; employment–population ratio $30/50 = 60\%$. After 0.5 million stop searching: labour force 31.5, unemployment 1.5, unemployment rate 4.76%; participation 63%; employment–population ratio unchanged at 60%.

Solution 43.5. $S_N = 2,000 - 1,200 - 400 = 400$. $I = 450$. $CA = 400 - 450 = -50$. If government consumption rises by 50 with GNDI, household consumption and investment unchanged, national saving falls to 350 and the current account to -100 . (In reality GNDI, consumption and investment would respond; the identity holds whatever happens, but "no change in other items" is a behavioural assumption.)

Solution 43.6. (a) Current account: goods exports $+100$; financial account: increase in assets (other investment, deposits) $+100$. (b) Financial account: direct investment liabilities $+200$ (a foreign firm's lasting interest in Malaysia); and an increase in Malaysian banks' foreign assets or decrease in liabilities of 200 as the funds are transferred—net financial account unchanged. (The construction spending itself is Malaysian output and investment.) (c) Current account: secondary income $+30$; financial account: increase in deposits abroad or reduction of liabilities $+30$. (d) Financial account: reserve assets shift from deposits to Treasury bills; no change in the total.

Solution 43.7. Start: $1,000 - 1,200 = -200$. The surplus raises euro assets to 1,030 at the old exchange rate. When the euro becomes 10% dearer, their value in domestic currency rises to $1,030 \times 1.10 = 1,133$. Liabilities, in domestic currency, are unchanged at 1,200. End NIIP = -67 . The improvement of 133 consists of the current-account surplus (30) and a valuation gain (103) more than three times as large.

Solution 43.8. (a) $1.012^4 - 1 = 4.89\%$. (b) $0.96^{1/4} - 1 = -1.02\%$. (c) $1.005 \times 1.008 \times 0.997 \times 1.006 - 1 = 1.61\%$.

Solution 43.9. Since $CA \equiv S_N - I$, a surplus may reflect high saving (households saving for retirement in an ageing society; firms retaining profits; government budget surpluses under the debt brake) or low domestic investment (weak investment demand, perhaps because of demographics or regulation), not only competitiveness. The identity cannot distinguish these explanations; evidence on saving and investment behaviour is required.

Economic Growth

After studying this chapter you should be able to

1. Describe the main facts of long-run growth and the wide differences in income across countries.
2. Set up and solve the Solow growth model, find its steady state, and analyse transitional dynamics.
3. Explain why saving affects the level but not the long-run growth rate of income in the Solow model, and derive the golden rule.
4. Derive the speed of convergence and explain conditional convergence.
5. Carry out growth accounting and interpret the Solow residual.
6. Explain the Ramsey–Cass–Koopmans model in outline and the modified golden rule.
7. Explain endogenous growth through ideas and creative destruction.
8. Assess the role of institutions, geography and structural change, and discuss the middle-income trap.

A motivating question

In 1978 China's real GDP per person was lower than that of most countries in sub-Saharan Africa. Over the next four decades it grew, on average, by close to 9 per cent a year. At that rate, income doubles roughly every eight years; over forty years it rises about thirty-fold. Hundreds of millions of people left extreme poverty in what was probably the fastest large-scale improvement in living standards in history. Over the same period, many other countries grew slowly or not at all, and some—Malaysia among them—grew rapidly for decades and then found it hard to make the final step to high income. Germany, already rich in 1978, grew more slowly, as rich countries tend to.

Why are some countries so much richer than others, and why do growth rates differ so much? Robert Lucas wrote that once one starts to think about these questions, “it is hard to think about anything else.” Small differences in growth rates compound into enormous differences in living standards: by the rule of 70 (Chapter 4), an economy growing at 2 per cent doubles its income every 35 years, while one growing at 1 per cent takes 70. This chapter develops the theory of long-run growth, beginning with the Solow model and moving to theories in which growth is driven by ideas and institutions.

44.1 Facts about Growth

Several facts, some first summarized by Nicholas Kaldor in 1961, frame the theory.

1. **Sustained growth is recent.** For most of human history, output per person grew very slowly. Sustained growth of 1–2 per cent a year in output per person began with the

Industrial Revolution in Britain around 1800 and spread unevenly thereafter.

2. **In rich countries, output per worker has grown at a roughly steady rate** over long periods—about 2 per cent a year in the United States since the late nineteenth century—and the capital–output ratio, the real interest rate and the shares of labour and capital in income have been roughly stable (*Kaldor’s stylized facts*), though the labour share has fallen somewhat in recent decades (Chapter 43).
3. **Income differences across countries are enormous.** GDP per person, adjusted for differences in price levels (at purchasing-power parity), is more than 50 times higher in the richest countries than in the poorest.
4. **Growth rates vary widely across countries and over time.** Some economies have grown at 7–10 per cent for decades (Japan in the 1950s and 1960s; South Korea, Taiwan, Singapore and Hong Kong from the 1960s; China from 1978); others have stagnated or declined.
5. **There is no unconditional convergence.** Poor countries as a group have not grown faster than rich ones. But among countries with similar characteristics—the OECD members, or the U.S. states—poorer ones have tended to grow faster: *conditional convergence*.

44.2 The Solow Model

Robert Solow’s 1956 model is the starting point of modern growth theory. It asks how far capital accumulation can explain growth.

44.2.1 Assumptions

Production. Output is produced from capital K , labour L and the level of technology A (labour-augmenting, or “effective labour”) according to

$$Y = F(K, AL),$$

with constant returns to scale in K and AL , positive and diminishing marginal products, and the Inada conditions (the marginal product of capital tends to infinity as $K \rightarrow 0$ and to zero as $K \rightarrow \infty$). The Cobb–Douglas case is $Y = K^\alpha (AL)^{1-\alpha}$, $0 < \alpha < 1$.

Growth of labour and technology. Labour grows at rate n and technology at rate g , both exogenous:

$$\frac{\dot{L}}{L} = n, \quad \frac{\dot{A}}{A} = g.$$

Saving and investment. A constant fraction s of output is saved and invested (the model is for a closed economy, so saving equals investment by the identity of Chapter 43). Capital depreciates at rate δ :

$$\dot{K} = sY - \delta K.$$

44.2.2 Per effective worker

Because of constant returns, we can divide by effective labour AL . Define capital per effective worker $k = K/(AL)$ and output per effective worker $y = Y/(AL)$. Then

$$y = F\left(\frac{K}{AL}, 1\right) = f(k), \quad f(k) = k^\alpha \text{ in the Cobb–Douglas case.}$$

To find how k evolves, use the growth-rate rules of Chapter 4: $\dot{k}/k = \dot{K}/K - \dot{A}/A - \dot{L}/L$. Substituting $\dot{K}/K = sY/K - \delta$ and $Y/K = f(k)/k$:

Proposition 44.1. The fundamental equation of the Solow model

$$\dot{k} = sf(k) - (n + g + \delta)k.$$

The first term is actual investment per effective worker. The second is **break-even investment**: the investment needed just to keep k constant, replacing depreciated capital (δk) and equipping new workers (nk) and new effective labour from technical progress (gk).

44.2.3 The steady state

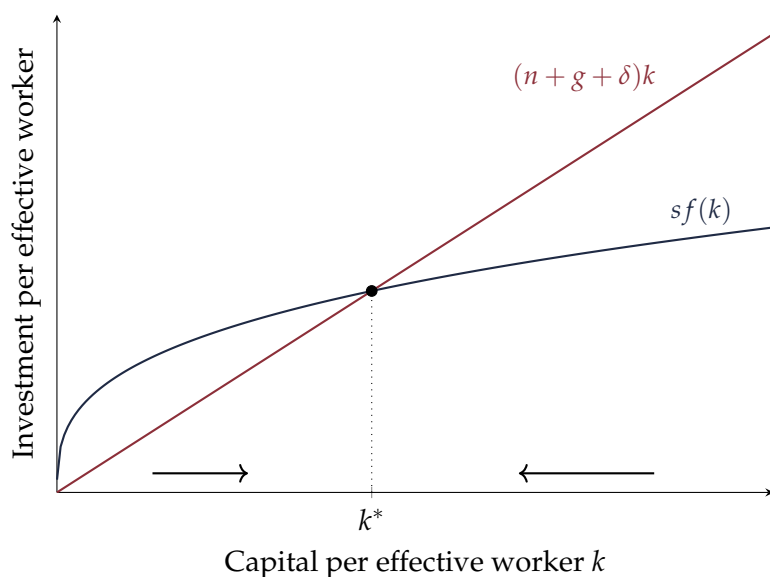


Figure 44.1: The Solow diagram for $f(k) = k^{1/3}$, $s = 0.2$, $n + g + \delta = 0.08$. Below k^* , actual investment exceeds break-even investment and k rises; above k^* , it falls. The arrows show the direction of movement.

Because f is concave and satisfies the Inada conditions, the curve $sf(k)$ starts steeper than the line $(n + g + \delta)k$ and ends flatter, so they cross exactly once at a positive k^* (Figure 44.1). At $k < k^*$, $\dot{k} > 0$; at $k > k^*$, $\dot{k} < 0$. Whatever the starting point, the economy converges to the **steady state** k^* , defined by

$$sf(k^*) = (n + g + \delta)k^*.$$

Worked Example 44.2. Cobb–Douglas steady state

With $f(k) = k^\alpha$: $sk^{*\alpha} = (n + g + \delta)k^*$, so

$$k^* = \left(\frac{s}{n + g + \delta} \right)^{1/(1-\alpha)}, \quad y^* = \left(\frac{s}{n + g + \delta} \right)^{\alpha/(1-\alpha)}.$$

With $\alpha = 1/3$, $s = 0.2$, $n = 0.01$, $g = 0.02$, $\delta = 0.05$: $n + g + \delta = 0.08$, $k^* = 2.5^{1.5} = 3.95$, $y^* = 2.5^{0.5} = 1.58$, and consumption per effective worker $c^* = (1 - s)y^* = 1.26$.

44.2.4 Balanced growth

In the steady state, k and y are constant. Since $Y = y \cdot AL$, total output grows at $n + g$, and output per worker $Y/L = yA$ grows at g . Capital per worker grows at g too, so the capital–output ratio is constant; with competitive factor markets, the return to capital $f'(k^*)$ is constant and the wage per effective worker is constant, so wages per worker grow at g . The model reproduces Kaldor’s facts. This path is called **balanced growth**.

Key idea. In the Solow model, sustained growth in output per worker comes only from technological progress. Capital accumulation alone cannot sustain growth, because diminishing returns to capital eventually make each additional unit of capital add too little output to cover its own depreciation and dilution.

44.2.5 Comparative statics: the saving rate

Suppose the saving rate rises permanently from s to s' . The $sf(k)$ curve shifts up, and the steady state rises to $k^{*'} > k^*$. During the transition, k and y grow faster than in the old steady state; once the new steady state is reached, the growth rate of output per worker returns to g . The saving rate therefore has a **level effect** but no long-run **growth effect**.

How large is the level effect? From the Cobb–Douglas solution, the elasticity of y^* with respect to s is $\alpha/(1 - \alpha)$. With $\alpha = 1/3$, a 10% rise in the saving rate (from 20% to 22%, say) raises steady-state output by about 5%. Differences in saving can therefore explain only modest differences in income. To explain a fifty-fold gap by saving alone would require implausibly large differences in saving rates—a first indication that capital accumulation is not the whole story.

44.2.6 The golden rule

Higher saving raises output, but it also means consuming a smaller share of it. What saving rate maximizes steady-state consumption? Steady-state consumption per effective worker is

$$c^* = f(k^*) - (n + g + \delta)k^*.$$

Maximizing over k^* :

Proposition 44.3. The golden rule

Steady-state consumption is maximized at the capital stock k_G where

$$f'(k_G) = n + g + \delta :$$

the marginal product of capital equals the break-even investment rate. With Cobb–Douglas technology, the golden-rule saving rate is $s_G = \alpha$.

Proof. Differentiating c^* with respect to k^* gives $f'(k^*) - (n + g + \delta) = 0$. With $f(k) = k^\alpha$, $\alpha k_G^{\alpha-1} = n + g + \delta$, and the saving rate that sustains k_G is $s_G = (n + g + \delta)k_G / f(k_G) = (n + g + \delta)k_G^{1-\alpha} = \alpha$. $\square$

In Example 44.2, $s_G = 1/3$: the economy saves too little to reach the golden rule. At the golden rule, $k_G = (\alpha/0.08)^{1.5} = 8.51$, $y_G = 2.04$ and $c_G = 2.04 - 0.08 \times 8.51 = 1.36$, above 1.26. But moving from $s = 0.2$ to $s = 1/3$ requires lower consumption during the transition; whether the move is worth it depends on how the future is weighted against the present,

which the Solow model, with its fixed saving rate, cannot say. An economy that saves *more* than the golden rule is *dynamically inefficient*: by saving less, it could raise consumption both immediately and in every future period. Empirical tests suggest that most advanced economies are not dynamically inefficient.

44.2.7 Transitional dynamics and convergence

How fast does the economy approach its steady state? Linearizing the fundamental equation around k^* gives

$$\dot{k} \approx -\lambda(k - k^*), \quad \lambda = (1 - \alpha)(n + g + \delta)$$

in the Cobb–Douglas case. The gap between k and k^* (and between y and y^*) shrinks at rate λ per year. This is the continuous-time counterpart of the difference-equation stability result of Chapter 4. With $\alpha = 1/3$ and $n + g + \delta = 0.08$, $\lambda = 5.3\%$, implying a half-life of $\ln 2 / \lambda \approx 13$ years.

Empirical estimates of the speed of convergence, by contrast, are typically around 2 per cent a year, implying a half-life of about 35 years. The model predicts too-rapid convergence with $\alpha = 1/3$. Mankiw, Romer and Weil (1992) showed that if the model is augmented with human capital, so that the combined share of broad capital (physical plus human) is around two-thirds, both the predicted speed of convergence and the explanatory power for cross-country income differences improve considerably.

Proposition 44.4. Conditional convergence

In the Solow model, an economy grows faster the further it is below its own steady state. Economies with the same saving rates, population growth, depreciation and technology converge to the same path of output per worker; economies with different parameters converge to different steady states.

This is exactly the pattern in the data: poor countries do not, in general, grow faster than rich ones (no unconditional convergence), but controlling for determinants of the steady state, they do (conditional convergence).

44.3 Growth Accounting

44.3.1 The Solow residual

How much of observed growth reflects more capital, more labour, and better technology? Write the production function as $Y = A_T K^\alpha L^{1-\alpha}$, where A_T is **total factor productivity** (TFP). Taking growth rates (Chapter 4):

$$g_Y = \alpha g_K + (1 - \alpha) g_L + g_{A_T}.$$

Given data on output, capital and labour growth, and an estimate of α (with competitive factor markets, α is capital's share of income, by Euler's theorem of Chapter 5), TFP growth is the residual:

$$g_{A_T} = g_Y - \alpha g_K - (1 - \alpha) g_L.$$

This is the **Solow residual**—Moses Abramovitz called it “a measure of our ignorance.”

Worked Example 44.5. Growth accounting

An economy's output grows at 4% a year, its capital stock at 5% and its labour input at 1%. Capital's share of income is 0.35. Then

$$g_{A_T} = 4 - 0.35 \times 5 - 0.65 \times 1 = 4 - 1.75 - 0.65 = 1.6\%.$$

Capital accounts for 1.75 points of growth, labour for 0.65, and TFP for 1.6—40 per cent of growth. Output per worker grows at 3%, of which capital deepening ($\alpha(g_K - g_L) = 1.4$) accounts for 1.4 points and TFP for 1.6.

44.3.2 Findings

Solow's own 1957 study attributed most of the growth of U.S. output per worker in the first half of the twentieth century to the residual. Later work refined the inputs: measuring labour input by hours and by educational attainment ("labour quality"), and capital by type, shrinks the residual somewhat. Alwyn Young's (1995) study of the East Asian "tigers" found that their extraordinary growth from the 1960s to the 1990s was largely explained by rapid accumulation of physical capital, rising participation and education, with TFP growth unremarkable in some cases—Singapore most notably. The finding implied that their growth would slow as the scope for further accumulation narrowed, as the Solow model predicts. Growth-accounting studies of China since 1978 similarly find a large role for capital accumulation, with estimates of TFP growth that vary widely depending on the data used and that suggest TFP growth slowed after the late 2000s.

The residual is not a direct measure of technical change. It captures everything that raises output from given measured inputs: new technology, better management, reallocation of resources from low- to high-productivity uses (for example from agriculture to industry), changes in the utilization of capital over the business cycle, and measurement error.

44.4 Optimal Saving: The Ramsey–Cass–Koopmans Model

The Solow model takes the saving rate as given. The Ramsey–Cass–Koopmans model, building on Frank Ramsey (1928) and developed by David Cass and Tjalling Koopmans (1965), makes it the outcome of household optimization, extending the two-period Euler equation of Chapter 9 to an infinite horizon.

Advanced Analysis: The Ramsey model

A representative household maximizes $\int_0^\infty e^{-\rho t} u(c(t)) dt$, with $u(c) = c^{1-\theta}/(1-\theta)$ (constant relative risk aversion θ , Chapter 25), subject to the economy's resource constraint. For simplicity set $n = g = 0$, so $\dot{k} = f(k) - c - \delta k$. The first-order conditions of the optimal-control problem yield the continuous-time Euler equation

$$\frac{\dot{c}}{c} = \frac{1}{\theta} [f'(k) - \delta - \rho].$$

Consumption grows when the net return to capital exceeds the rate of time preference, and $1/\theta$ is the elasticity of intertemporal substitution. In the steady state $\dot{c} = 0$, so

$$f'(k^*) = \rho + \delta :$$

the **modified golden rule**. Because $\rho > 0$, $f'(k^*) > \delta$ (the golden rule with $n = g = 0$): the optimal capital stock is below the golden-rule level. Impatient households do not find

it worth sacrificing current consumption to reach the capital stock that would maximize steady-state consumption. The economy converges to the steady state along a unique *saddle path*; any other path either violates the resource constraint or leaves capital unused. The model rules out dynamic inefficiency, and it provides the foundation for the real business cycle and New Keynesian models of Chapter 47.

44.5 Endogenous Growth

The Solow model explains growth in output per worker by technological progress, but it does not explain where technology comes from. **Endogenous growth** theories, developed from the mid-1980s, make technological progress the outcome of economic decisions.

44.5.1 Why diminishing returns must be avoided

Sustained growth requires that some input can be accumulated without running into diminishing returns. In the simplest *AK model*, output is $Y = AK$, where K is a broad notion of capital including human capital and knowledge. Then $\dot{K}/K = sA - \delta$: the growth rate depends on the saving rate. The model shows what is needed, but offers no account of why returns to broad capital should be constant.

44.5.2 Ideas

Paul Romer (1990) identified the key economic property of ideas: they are *non-rival*. A design, a formula or a piece of software can be used by any number of people at once without being used up, unlike a machine or a worker's time. This has a profound implication. If producing output requires ideas and rival inputs (capital and labour), then doubling the rival inputs, with the same ideas, doubles output: constant returns to rival inputs. But if ideas are also doubled, output more than doubles. There are *increasing returns* to ideas and rival inputs together.

Increasing returns cannot be sustained with price-taking firms that pay each input its marginal product, since the payments would exceed output (Euler's theorem, Chapter 5). Romer's resolution is that the producers of ideas earn temporary monopoly profits—through patents, trade secrets and first-mover advantages—which reward the costly research that creates them. This links growth theory to the industrial organization of Chapter 24.

In Romer's model, the growth rate of ideas depends on the number of researchers. Growth is *endogenous*: it depends on how many resources an economy devotes to research, and therefore on policies (patent protection, research subsidies, education) that affect that choice. The model also implies that the social return to research exceeds the private return, because each new idea makes future research easier (a positive externality, Chapter 26), so markets under-invest in research.

44.5.3 Creative destruction

Philippe Aghion and Peter Howitt (1992), building on Joseph Schumpeter, modelled growth as a sequence of quality improvements, each of which makes the previous technology obsolete. Innovators are motivated by the monopoly rents they earn until they are displaced by the next innovator: growth involves **creative destruction**. The framework explains why growth is associated with high rates of firm entry and exit and of job reallocation, why competition can either encourage innovation (firms innovate to escape competition) or discourage it (by reducing the rewards to innovation), and why policies protecting incumbents can slow growth.

Joel Mokyr, Philippe Aghion and Peter Howitt were awarded the 2025 Nobel prize in economics for their work on innovation-driven growth.

44.5.4 Scale effects and research productivity

Early endogenous growth models predicted that larger economies, with more researchers, would grow faster. But the number of researchers in the United States has increased many-fold since the 1950s without a rise in the growth rate. Charles Jones (1995) and later Bloom, Jones, Van Reenen and Webb (2020) documented that *research productivity*—the growth of ideas per researcher—has fallen substantially: ideas are getting harder to find. In “semi-endogenous” growth models consistent with this evidence, long-run growth depends on the growth of the research workforce, which in turn depends on population growth and on the share of people who become researchers.

44.6 Beyond Accumulation: Institutions, Geography and Structure

44.6.1 Proximate and fundamental causes

Capital accumulation, education and technology are the *proximate* causes of growth. Why do some societies accumulate and innovate while others do not? The *fundamental* causes are the subject of an active literature.

Institutions. Daron Acemoglu, Simon Johnson and James Robinson argued that **inclusive institutions**—secure property rights, rule of law, constraints on rulers, broad access to economic opportunity—encourage investment and innovation, while **extractive institutions** that concentrate power and wealth in an elite discourage them (Chapter 1). In a famous study (2001), they used differences in the mortality of European settlers in former colonies as an instrument for institutions (Chapter 17): where settlers faced high mortality, colonizers established extractive institutions rather than settling, and those institutions persisted. They were awarded the 2024 Nobel prize for this work. The empirical strategy has been debated, but the importance of institutions is widely accepted.

Geography. Access to the sea, disease environments, climate and natural resources affect productivity directly and shape institutions. The *resource curse*—the tendency of some resource-rich countries to grow slowly—suggests that geography works partly through institutions.

Culture, trust and social capital have been linked to growth, with the difficulty that they are hard to measure and may themselves be shaped by institutions and income.

44.6.2 Structural transformation

As economies grow, labour moves from agriculture to manufacturing and then to services. W. Arthur Lewis (1954) modelled development as the transfer of labour from a traditional sector with low productivity (and surplus labour) to a modern sector with higher productivity. Each worker who moves raises aggregate productivity, even if productivity within each sector is unchanged. Much of China’s growth since 1978 reflects this reallocation: hundreds of millions of workers moved from farms to factories and cities. Reallocation is a finite source of growth: once agriculture’s share of employment has fallen, it contributes little more.

44.6.3 The middle-income trap

Several economies have grown rapidly to middle-income levels and then slowed before reaching high income. Growth based on moving labour out of agriculture, on accumulating capital and on adopting foreign technology eventually runs into diminishing returns and rising wages; continued growth requires domestic innovation, better institutions, higher-quality education and deeper financial markets. Malaysia grew by around 7 per cent a year for much of the period from the 1970s to the Asian financial crisis, and has grown more slowly since; it set out to reach high-income status by 2020, and although it came close, it had not been classified as high-income by the World Bank by that date. Whether a distinct “trap” exists is debated—the evidence for a statistically special slowdown at middle income is mixed—but the policy challenge of moving from growth by accumulation to growth by innovation is real, and it is the challenge China now faces.

Case: China: growth by accumulation, reallocation and catch-up

China’s growth since the reforms that began in 1978 illustrates each part of this chapter. *Catch-up and convergence.* China started far below the technological frontier. By opening to foreign investment and trade, it could adopt existing technologies rather than invent them, which allows much faster growth than frontier innovation—a form of conditional convergence once institutions were reformed to permit it.

Accumulation. Investment rose to over 40 per cent of GDP (Chapter 43). In the Solow model, a rising saving rate raises growth during a long transition. Growth-accounting studies attribute a large share of China’s growth to capital accumulation.

Reallocation. Hundreds of millions of workers moved from agriculture to manufacturing and services, raising aggregate productivity. Agriculture’s share of employment fell from around 70 per cent in 1978 to around a quarter by the early 2020s.

Institutions. Reforms introduced market incentives—the household responsibility system in agriculture, the growth of township and village enterprises, the opening of special economic zones, and later the restructuring of state-owned enterprises and accession to the WTO in 2001—without the full set of Western-style institutions. Economists debate how far China’s experience supports or challenges the institutional view.

Slowdown. Growth slowed from around 10 per cent in the 2000s to around 5 per cent by the mid-2020s. Each of the chapter’s mechanisms predicts deceleration: diminishing returns to capital as the capital stock grows (with evidence of falling returns on investment and rising debt); the working-age population peaking in the 2010s and then declining; fewer workers left to move from farms; and a shrinking distance to the technological frontier in many industries. Continued growth depends increasingly on TFP: innovation, the reallocation of capital from less to more productive firms, and the reform of state-owned enterprises. China’s large investments in research, and its leading position in industries such as batteries, solar panels and electric vehicles (Chapter 21), suggest a capacity for frontier innovation in some sectors; whether that can sustain economy-wide productivity growth is one of the central questions for the world economy.

Summary

- Sustained growth is recent; income differences are enormous; there is conditional but not unconditional convergence.
- In the Solow model, $\dot{k} = sf(k) - (n + g + \delta)k$. The economy converges to a steady state in which output per worker grows at the rate of technical progress g .

- Saving affects the level of income, not its long-run growth rate; the golden rule $f'(k) = n + g + \delta$ maximizes steady-state consumption.
- Economies converge to their own steady states at rate $(1 - \alpha)(n + g + \delta)$ in the Cobb–Douglas case; empirical convergence is slower, consistent with a larger share of broad capital.
- Growth accounting decomposes growth into contributions of capital, labour and TFP (the Solow residual).
- The Ramsey model derives saving from optimization; its steady state satisfies the modified golden rule.
- Endogenous growth models explain technology: non-rival ideas create increasing returns, rewarded by temporary monopoly; creative destruction replaces old technologies. Ideas appear to be getting harder to find.
- Institutions, geography and structural transformation are fundamental determinants of growth; the transition from accumulation to innovation is the challenge of middle-income economies.

Key terms

Kaldor's stylized facts; conditional convergence; Solow model; effective labour; capital per effective worker; break-even investment; steady state; balanced growth; level effect; growth effect; golden rule; dynamic inefficiency; speed of convergence; total factor productivity; growth accounting; Solow residual; Ramsey–Cass–Koopmans model; modified golden rule; saddle path; endogenous growth; AK model; non-rivalry; creative destruction; research productivity; inclusive and extractive institutions; resource curse; structural transformation; middle-income trap.

Exercises

Exercise 44.1 (Steady state). With $f(k) = k^{0.4}$, $s = 0.25$, $n = 0.01$, $g = 0.015$, $\delta = 0.05$, find k^* , y^* and c^* . What is the growth rate of output per worker and of total output in the steady state?

Exercise 44.2 (Saving rate). In Exercise 44.1, the saving rate rises to 0.30. Find the new steady-state y^* and the percentage increase. Compare with the elasticity $\alpha/(1 - \alpha)$.

Exercise 44.3 (Golden rule). For the economy of Exercise 44.1, find the golden-rule capital stock, output and consumption. Is the economy saving too much or too little relative to the golden rule?

Exercise 44.4 (Population growth). Using the Solow model, explain the effect of a permanent fall in population growth on the steady-state level of output per worker and on its long-run growth rate. Relate your answer to China's and Germany's demographic prospects.

Exercise 44.5 (Convergence speed). For the economy of Exercise 44.1, compute the speed of convergence and the half-life. If broad capital's share is 0.7 instead of 0.4, what are they?

Exercise 44.6 (Growth accounting). From 1990 to 2010, an economy's output grew at 9% a year, its capital stock at 11%, and employment at 1%; average years of schooling rose, raising labour quality by 1% a year. With capital's share 0.45, compute TFP growth (a) ignoring and (b) including labour quality. What share of growth does each factor explain in (b)?

Exercise 44.7 (Ramsey). In the Ramsey model with $f(k) = k^{0.3}$, $\delta = 0.06$, $\rho = 0.03$ and $n = g = 0$, find the modified-golden-rule capital stock and compare it with the golden-rule capital stock.

Exercise 44.8 (AK model). In an AK economy with $A = 0.4$, $\delta = 0.05$ and $s = 0.2$, what is the growth rate? What if s rises to 0.25? Contrast with the Solow model.

Exercise 44.9 (Ideas). Explain why the non-rivalry of ideas implies increasing returns to scale, and why this means that a perfectly competitive economy cannot fully reward the creators of ideas.

Exercise 44.10 (Institutions). Explain the logic of using settler mortality as an instrument for institutions. What assumption must hold for the instrument to be valid? Suggest one reason it might fail.

Solutions

Solution 44.1. $n + g + \delta = 0.075$. $k^* = (0.25/0.075)^{1/0.6} = 3.333^{1.667} = 7.44$. $y^* = 3.333^{0.4/0.6} = 3.333^{0.667} = 2.23$. $c^* = 0.75 \times 2.23 = 1.67$. Output per worker grows at $g = 1.5\%$; total output at $n + g = 2.5\%$.

Solution 44.2. $y^* = (0.30/0.075)^{0.667} = 4^{0.667} = 2.52$, an increase of about 12.9% (from 2.23). The elasticity $0.4/0.6 = 0.667$ times the 20% increase in s gives about 13% for a small change, consistent with the exact calculation ($1.2^{0.667} = 1.129$).

Solution 44.3. $0.4k_G^{-0.6} = 0.075$, so $k_G = (0.4/0.075)^{1/0.6} = 5.333^{1.667} = 16.3$. $y_G = 16.3^{0.4} = 3.05$; $c_G = 3.05 - 0.075 \times 16.3 = 1.83$. The golden-rule saving rate is $\alpha = 0.4 > 0.25$, so the economy saves too little relative to the golden rule ($c^* = 1.67 < 1.83$).

Solution 44.4. A fall in n lowers break-even investment, rotating the $(n + g + \delta)k$ line down: k^* and y^* rise, so the level of output per worker rises. Long-run growth of output per worker is unchanged at g ; total output grows more slowly. For China and Germany, whose working-age populations are shrinking, the Solow model implies higher capital per worker but slower growth of total output, and a shift in the composition of growth toward productivity. (Models with endogenous or semi-endogenous growth suggest that slower population growth may also slow the growth of ideas.)

Solution 44.5. $\lambda = 0.6 \times 0.075 = 4.5\%$; half-life $\ln 2/0.045 = 15.4$ years. With $\alpha = 0.7$: $\lambda = 0.3 \times 0.075 = 2.25\%$; half-life 30.8 years, closer to empirical estimates.

Solution 44.6. (a) $g_A = 9 - 0.45 \times 11 - 0.55 \times 1 = 9 - 4.95 - 0.55 = 3.5\%$. (b) Labour input grows at 2%: $g_A = 9 - 4.95 - 1.1 = 2.95\%$. Shares of growth in (b): capital $4.95/9 = 55\%$; labour (including quality) $1.1/9 = 12\%$; TFP $2.95/9 = 33\%$.

Solution 44.7. Modified golden rule: $0.3k^{-0.7} = \rho + \delta = 0.09$, so $k^* = (0.3/0.09)^{1/0.7} = 3.333^{1.4286} = 5.58$. Golden rule (with $n = g = 0$): $0.3k^{-0.7} = 0.06$, $k_G = 5^{1.4286} = 9.97$. The optimal capital stock is well below the golden rule because of impatience.

Solution 44.8. Growth = $sA - \delta = 0.08 - 0.05 = 3\%$. With $s = 0.25$: $0.10 - 0.05 = 5\%$. In the AK model a higher saving rate raises the growth rate permanently; in the Solow model it raises growth only temporarily.

Solution 44.9. Output depends on ideas A and rival inputs X : $Y = F(A, X)$. Replicating an activity requires doubling rival inputs only, since the same ideas can be used again: $F(A, 2X) = 2F(A, X)$ (constant returns to rival inputs). Doubling ideas as well must then give more than double output: $F(2A, 2X) > 2F(A, X)$. If rival inputs are paid their marginal products, Euler's theorem says the payments to them exhaust output, leaving nothing to pay for ideas. Ideas must therefore be rewarded by something other than competitive marginal-product pricing—temporary monopoly profits, public funding or prizes.

Solution 44.10. Settler mortality affected where Europeans settled and hence what institutions they set up (the relevance condition); those institutions persisted and affect income today. For validity, settler mortality must affect current income *only* through institutions (the exclusion restriction). It might fail if the disease environment that killed settlers also affects health and productivity today directly, or if settlers brought human capital and technology as well as institutions.

Consumption, Saving, Investment and Labour Markets

After studying this chapter you should be able to

1. Explain the Keynesian consumption function and why it fails to fit long-run data.
2. Derive consumption from the intertemporal budget constraint and interpret the permanent-income and life-cycle hypotheses.
3. Distinguish the marginal propensity to consume out of transitory and permanent income, and explain the evidence on actual spending responses.
4. Explain the random-walk hypothesis, precautionary saving and liquidity constraints.
5. State Ricardian equivalence and the conditions under which it fails.
6. Derive the desired capital stock from the user cost of capital and explain the accelerator, Tobin's q and irreversibility.
7. Analyse labour supply and demand, classify unemployment, and derive steady-state unemployment from worker flows.
8. Explain the matching model and the Beveridge curve, and use them to interpret short-time work schemes.

A motivating question

In the spring of 2020 governments in many advanced economies sent money directly to households. In the United States, most adults received cheques of \$1,200 in April 2020, followed by further payments in December 2020 and March 2021. In Malaysia the government allowed workers to withdraw savings from the Employees Provident Fund, the national retirement scheme, and eventually well over RM100 billion was withdrawn. In Germany, by contrast, the main instrument was not a transfer to households at all but a subsidy to firms that kept workers on their payrolls at reduced hours—*Kurzarbeit*.

How much of a transfer will a household spend? Will a firm facing a sudden collapse in sales lay workers off or keep them? When the central bank raises interest rates, how much does business investment fall? These are questions about the behaviour of households and firms, and they are the building blocks of every macroeconomic model. The national accounts of Chapter 43 tell us that $Y \equiv C + I + G + NX$; they do not tell us what determines C and I , or why some of the people who want to work cannot find jobs. This chapter builds those behavioural relationships from the microeconomics of Parts I and III.

45.1 Consumption: From Keynes to the Budget Constraint

45.1.1 The Keynesian consumption function

Consumption is the largest component of expenditure: about two-thirds of GDP in the United States, somewhat over half in Germany and Malaysia, and under 40 per cent in China (Chapter 43). John Maynard Keynes proposed in 1936 that consumption depends mainly on current disposable income:

$$C = c_0 + c_1(Y - T), \quad 0 < c_1 < 1,$$

where $c_0 > 0$ is autonomous consumption and c_1 is the **marginal propensity to consume** (MPC). Because $c_0 > 0$, the *average* propensity to consume $C/(Y - T)$ falls as income rises: richer households save a larger share of their income.

Cross-sectional data support this: at any date, high-income households save a larger fraction of their income than low-income households. Short time series also fit reasonably well. But Simon Kuznets found in the 1940s that over long periods the U.S. saving rate had not risen as incomes rose several-fold: the long-run average propensity to consume was roughly constant. A consumption function that fits cross sections and short samples but not long samples is telling us that current income is not really what drives consumption. The resolution came from putting the household's intertemporal budget constraint, developed in Chapter 9 and interpreted in Chapter 20, at the centre.

45.1.2 The intertemporal budget constraint

Consider a household that lives for T periods, starts with financial wealth A_0 , earns labour income y_t and can borrow or lend at the real interest rate r . Its wealth evolves as

$$A_{t+1} = (1 + r)(A_t + y_t - c_t).$$

This is a stock–flow relation of exactly the kind introduced in Chapter 1 and used in the balance sheets of Chapter 10: the stock of wealth at the start of next period equals the stock this period plus the flow of saving, with interest. If the household cannot die in debt ($A_T \geq 0$) and gains nothing from dying with wealth, iterating forward gives

$$\sum_{t=0}^{T-1} \frac{c_t}{(1+r)^t} = A_0 + \sum_{t=0}^{T-1} \frac{y_t}{(1+r)^t} \equiv A_0 + H_0,$$

where H_0 is **human wealth**, the present value of expected labour income. The present value of lifetime consumption equals total wealth, financial plus human. This is the many-period version of the two-period constraint in Chapter 9.

45.2 Permanent Income and the Life Cycle

45.2.1 Consumption smoothing

Suppose, as in Chapter 9, that lifetime utility is $\sum_t \beta^t u(c_t)$ with u concave. Maximizing it subject to the budget constraint gives, period by period, the Euler equation

$$u'(c_t) = \beta(1+r) u'(c_{t+1}).$$

In the benchmark case $\beta(1+r) = 1$ —the household's impatience exactly balances the return to saving—the Euler equation says $u'(c_t) = u'(c_{t+1})$, so consumption is constant. The household consumes the *annuity value* of its total wealth, whatever the timing of its income.

Proposition 45.1. The permanent-income consumption rule

If $\beta(1+r) = 1$ and the horizon is infinite, consumption in every period equals

$$c = \frac{r}{1+r} (A_0 + H_0),$$

the annuity value of total wealth. If $r = 0$ and the horizon is T periods, $c = (A_0 + H_0)/T$.

Proof. With constant consumption c forever, the present value of consumption is $c \sum_{t=0}^{\infty} (1+r)^{-t} = c(1+r)/r$. Setting this equal to $A_0 + H_0$ gives the result. With $r = 0$ and T periods, the present value of consumption is simply cT . $\square$

Milton Friedman (1957) called this annuity value **permanent income**: consumption depends on permanent income, not on current income. Franco Modigliani and Richard Brumberg (1954) developed the closely related **life-cycle hypothesis**, emphasizing that income is low in youth, high in middle age and low in retirement, so households save in their working years and dissave in retirement.

Worked Example 45.2. Life-cycle saving and the MPC

A worker aged 25 expects to live to 75 (50 more years), to work until 65 (40 years) at a real income of 60,000 per year, and has no wealth. Let $r = 0$ and $\beta = 1$. Lifetime resources are $40 \times 60,000 = 2,400,000$, so consumption is $2,400,000/50 = 48,000$ per year. The worker saves 12,000 per year while working; at 65 she has accumulated $40 \times 12,000 = 480,000$, which finances ten years of dissaving at 48,000.

A *one-off windfall* of 10,000 raises lifetime resources by 10,000 and consumption by $10,000/50 = 200$ per year. The MPC out of the windfall in the year it arrives is $200/10,000 = 0.02$.

A *permanent raise* of 1,000 per year for the rest of her working life raises lifetime resources by 40,000 and consumption by $40,000/50 = 800$ per year: an MPC of 0.8.

The example resolves Kuznets's puzzle. In a cross section, many high-income households are having an unusually good year (their income is temporarily above its permanent level) and many low-income households an unusually bad one, so the rich appear to save a larger share of their income. Over long periods, income growth is mostly permanent, so consumption rises in proportion. The Keynesian consumption function is a statistical artefact of mixing transitory and permanent income.

Key idea. The effect of a change in income on consumption depends on how much it changes *lifetime* resources. Transitory changes have small effects; permanent changes have large ones. Policy-makers who want to predict the effect of a tax cut or a transfer must ask whether households expect it to last.

45.2.2 The life cycle and aggregate saving

Because the young save and the old dissave, aggregate saving depends on the age structure of the population and on the rate of income growth. If income is growing, each cohort of young savers is richer than the cohort of old dissavers, so aggregate saving is positive even though each household's lifetime saving is zero. Modigliani used this logic to explain why fast-growing economies such as Japan in the 1960s had high saving rates. It also predicts that ageing populations—Germany, Japan, and increasingly China—will save less as the share

of retirees rises, a prediction that is broadly, though not precisely, borne out; saving by the elderly has fallen by less than the simple model predicts, partly because of bequest motives and uncertainty about longevity and health costs.

45.3 Uncertainty, Precaution and Liquidity Constraints

45.3.1 The random walk

Robert Hall (1978) drew a striking implication of the permanent-income hypothesis under uncertainty. With quadratic utility and $\beta(1+r) = 1$, the Euler equation becomes $c_t = \mathbb{E}_t[c_{t+1}]$: today's consumption is the best forecast of tomorrow's. Consumption therefore follows a **random walk**: it changes only when news arrives that changes permanent income, and changes in consumption should not be predictable from any information available in advance, including predictable changes in income.

This prediction can be tested. If a household knows that its income will rise next quarter—because of a scheduled pay rise or a tax rebate announced months ahead—its consumption should rise *when the news arrives*, not when the money arrives. Studies of predictable income changes tend to find **excess sensitivity**: consumption rises when the income arrives. The evidence on U.S. tax rebates is especially clean because the timing of payments was randomized by the last digits of taxpayers' identification numbers. Johnson, Parker and Souleles (2006) found that households spent 20–40 per cent of the 2001 rebates on non-durable goods within three months of receiving them; Parker, Souleles, Johnson and McClelland (2013) found that households spent 12–30 per cent of the 2008 stimulus payments on non-durables and 50–90 per cent including durables such as cars. These responses are far larger than the permanent-income benchmark predicts for a one-off payment.

45.3.2 Liquidity constraints and hand-to-mouth households

The simplest explanation is that many households cannot borrow against future income. A household whose current income is below its permanent income would like to borrow; if lenders will not lend to it—perhaps because its future income cannot be used as collateral, an information problem of the kind analysed in Chapter 25—it is **liquidity constrained** and consumes its current income. For such a household, the MPC out of a transfer is close to one.

Kaplan and Violante (2014) documented that many households are “wealthy hand-to-mouth”: they own substantial illiquid wealth (a house, a retirement account) but hold little in liquid form, so they behave as if liquidity constrained. This explains why even middle-income households respond strongly to transfers, and why allowing access to illiquid retirement savings, as Malaysia did in 2020–22, can raise spending.

45.3.3 Precautionary saving

If future income is uncertain, a prudent household saves more than it would under certainty. Formally, if marginal utility is convex ($u''' > 0$, a property called *prudence*, satisfied by the constant-relative-risk-aversion utility of Chapter 25), then by Jensen's inequality $\mathbb{E}[u'(c_{t+1})] > u'(\mathbb{E}[c_{t+1}])$: uncertainty raises the expected marginal value of future consumption, so the household shifts resources toward the future. **Precautionary saving** explains why households hold buffer stocks of liquid wealth, why saving rises when unemployment risk rises—as in 2008–09 and 2020—and why high saving in China is sometimes linked to limited social insurance for health, education and old age (Chapter 43).

Advanced Analysis: The log-linear Euler equation

With constant relative risk aversion $u(c) = c^{1-\theta}/(1-\theta)$, the Euler equation under certainty is $c_{t+1}/c_t = [\beta(1+r)]^{1/\theta}$. Writing $\beta = e^{-\rho}$ and taking logs,

$$\Delta \ln c_{t+1} \approx \frac{1}{\theta} (r - \rho).$$

Consumption growth rises with the real interest rate, with slope $1/\theta$, the *elasticity of intertemporal substitution* (EIS). With $\theta = 2$, $\beta = 0.96$ and $r = 0.05$: $\beta(1+r) = 1.008$ and consumption grows at $1.008^{0.5} - 1 = 0.40\%$ a year. If r rises to 0.07, $\beta(1+r) = 1.0272$ and growth rises to 1.35%: a 2-point rise in r raises consumption growth by about 1 point, as $1/\theta = 0.5$ implies. Under uncertainty with lognormal consumption, a precautionary term $\frac{\theta}{2} \text{Var}(\Delta \ln c_{t+1})$ is added to the right-hand side. This equation, written for the aggregate economy and combined with goods-market equilibrium, becomes the New Keynesian IS curve of Chapter 47. Estimates of the EIS from aggregate data are often small (below 0.5), which is one reason the interest-rate channel of monetary policy works partly through other channels (Chapter 49).

45.4 Saving, Debt and Ricardian Equivalence

What happens to private saving if the government cuts taxes today and finances the cut by borrowing? The government's intertemporal budget constraint, like the household's, says that the present value of its spending must equal the present value of its taxes (plus its initial net wealth). If spending is unchanged, a tax cut today must be matched by tax increases later with the same present value.

Proposition 45.3. Ricardian equivalence

Suppose households have the same horizon as the government, can borrow and lend at the same interest rate as the government, and correctly anticipate future taxes, and suppose taxes are lump-sum. Then, for a given path of government spending, the timing of taxes does not affect household consumption: a debt-financed tax cut raises private saving by exactly the amount of the tax cut, leaving national saving unchanged.

Proof. Human wealth net of taxes is the present value of $y_t - T_t$. The tax cut lowers T_0 by Δ and raises some future T_s by $\Delta(1+r)^s$, whose present value is Δ . Net human wealth, and therefore consumption, is unchanged. Disposable income rises by Δ today, so saving rises by Δ . $\square$

Robert Barro (1974) revived the argument, which David Ricardo had discussed and rejected in the early nineteenth century. Ricardian equivalence fails when its assumptions fail: when households are liquidity constrained (they spend the tax cut), when the future taxes will fall on people not yet born and current households do not care about them, when taxes distort behaviour, and when households do not look far ahead. The evidence suggests partial offset: private saving rises by some fraction of a debt-financed tax cut, but not all of it. Chapter 47 returns to the implications for public debt.

► “Saving equals investment, so higher household saving raises investment”

The identity $S_{\text{private}} + (T - G) \equiv I + NX$ (Chapter 43) must hold. But it does not say that a rise in desired saving *causes* a rise in investment. If households try to save more

in a recession, lower spending can reduce output and income until actual saving is no higher than before—Keynes’s paradox of thrift, analysed in Chapter 46. Which variables adjust to keep the identity satisfied is a question about behaviour, not accounting.

45.5 Investment

Investment—spending on new structures, equipment, intellectual property and inventories—is about a fifth of GDP in the United States and Germany and over two-fifths in China, but it is far more volatile than consumption. In a typical U.S. recession, investment falls by several times as much as GDP in percentage terms. Understanding investment is therefore central to understanding fluctuations.

45.5.1 The desired capital stock

Chapter 21 showed that a firm uses capital up to the point at which its marginal revenue product equals its cost. For a firm that owns its capital, that cost is the user cost

$$uc = P_K(i + \delta - g_{P_K}),$$

or, in real terms, $P_K(r + \delta)$ when the relative price of capital goods is constant and r is the real interest rate. With a Cobb–Douglas technology $Y = AK^\alpha L^{1-\alpha}$, the marginal product of capital is $\alpha Y/K$. Setting its value equal to the user cost (with output price normalized to 1):

$$K^* = \frac{\alpha Y}{uc}.$$

This is the neoclassical theory of investment developed by Dale Jorgenson in the 1960s. The desired capital stock rises with expected output and falls with the user cost; in the Cobb–Douglas case its elasticity with respect to the user cost is -1 .

Worked Example 45.4. Interest rates and desired capital

A firm expects to produce output of 1,000 (real units per year) with $\alpha = 0.3$. Capital goods cost $P_K = 1$, depreciate at $\delta = 0.10$, and their price is stable. At an interest rate of 5 per cent, $uc = 0.15$ and

$$K^* = \frac{0.3 \times 1,000}{0.15} = 2,000.$$

If the central bank raises rates so that the firm’s borrowing cost rises to 7 per cent, $uc = 0.17$ and $K^* = 300/0.17 = 1,765$: a fall of 11.8 per cent. A tax change that allows faster depreciation for tax purposes (Chapter 29) lowers the effective user cost and works in the opposite direction.

45.5.2 From capital to investment: the accelerator

Investment is the flow that adjusts the stock:

$$I_t = K_{t+1} - (1 - \delta)K_t = (K_{t+1} - K_t) + \delta K_t.$$

Gross investment consists of net investment plus replacement investment. If firms keep the capital stock close to $K^* = \alpha Y/uc$, then net investment depends on the *change* in expected output. This is the **accelerator**: because the capital stock is several times annual output, a small change in the growth of expected output produces a large proportional change in investment.

Suppose the capital–output ratio is 3, depreciation is 10 per cent, and output has been growing at 3 per cent, so that $K = 3Y$ and investment is $3 \times 0.03Y + 0.1 \times 3Y = 0.39Y$. If expected output growth falls to zero, investment falls to $0.30Y$: a fall of 23 per cent, from a 3-point change in output growth. This is why investment and profits are so much more volatile than GDP, as Chapter 31 noted from firms’ cash-flow statements and as Chapter 46 develops.

45.5.3 Adjustment costs and Tobin’s q

Firms do not jump to K^* . Installing capital takes time and disrupts production; ordering, building and bringing a factory into operation can take years. These *adjustment costs* make it optimal to adjust gradually. James Tobin (1969) proposed a measure that summarizes the incentive to invest:

$$q = \frac{\text{market value of the firm's capital}}{\text{replacement cost of that capital}}.$$

If $q > 1$, the stock market values an extra unit of installed capital at more than it costs to buy, so firms should invest; if $q < 1$, they should let capital depreciate. With convex (quadratic) adjustment costs, optimal investment satisfies

$$\frac{I}{K} = \delta + \frac{q - 1}{\phi},$$

where ϕ measures the size of adjustment costs. With $\phi = 10$, $\delta = 0.1$ and $q = 1.2$, the firm invests 12 per cent of its capital stock a year: 10 per cent to replace depreciation and 2 per cent to expand.

Accounting bridge. Tobin’s q is closely related to the market-to-book ratio of Chapter 31, but they differ in an important way: book value records assets at historical cost less depreciation and omits most internally generated intangibles such as brands and software (Chapter 29), whereas q ’s denominator is the replacement cost of the capital. Firms rich in unrecorded intangibles—Apple, Microsoft—have market-to-book ratios far above their true q . Measured investment in intangibles has grown steadily in advanced economies; the national accounts capitalize software and research and development, but company accounts under IFRS and U.S. GAAP generally expense research.

Fumio Hayashi (1982) showed that under constant returns and perfect competition, the *marginal* q that matters for investment equals the *average* q observable in stock prices. Empirically, q explains investment less well than the theory predicts, and cash flow adds explanatory power, especially for small firms. This is consistent with the credit-market frictions of Chapter 25: firms that cannot borrow freely invest out of internal funds, so investment depends on cash flow as well as on the return to capital. The same frictions are central to the financial accelerator of Chapter 51.

45.5.4 Irreversibility and uncertainty

Much investment is irreversible: a specialized factory cannot be resold for its cost. When the return to investment is uncertain, waiting has value, because it allows the firm to learn more before committing. Avinash Dixit and Robert Pindyck (1994) showed that this *option value of waiting* raises the return a firm requires before it invests, sometimes well above the user cost. The theory predicts that increases in uncertainty—about demand, policy or trade rules—cause firms to postpone investment, which is one channel through which uncertainty shocks may cause recessions.

45.6 The Labour Market

45.6.1 Labour demand and supply

A competitive firm hires labour up to the point at which the marginal product of labour equals the real wage (Chapter 21): $MPL = W/P$. With Cobb–Douglas technology, $MPL = (1 - \alpha)Y/L$, so labour’s share of income is $(1 - \alpha)$. Aggregate labour demand slopes down in the real wage and shifts out when productivity or the capital stock rises.

Labour supply comes from the household’s choice between consumption and leisure (Chapter 20). A higher real wage has a substitution effect that raises hours and an income effect that lowers them. Over long periods the income effect has been strong: average annual hours per worker in rich countries have fallen substantially over the past century while real wages have risen many-fold. Over short periods, the relevant elasticity is the *intertemporal* elasticity of labour supply—the response of hours to a temporary change in the wage—which microeconomic studies usually find to be small for prime-age men and larger for women and for decisions about whether to work at all (the extensive margin). Chapter 47 shows that the size of this elasticity is a central point of disagreement between theories of the business cycle.

In the frictionless model, the real wage adjusts until everyone who wants to work at the going wage has a job. Any unemployment would be voluntary. Actual labour markets are different.

45.6.2 Types of unemployment

Chapter 43 defined the unemployment rate. Economists distinguish:

- **Frictional unemployment:** the time it takes workers and employers to find each other, even when jobs are available.
- **Structural unemployment:** a mismatch between the skills or locations of workers and the jobs available, or a real wage held above the market-clearing level by institutions.
- **Cyclical unemployment:** unemployment above the normal level because aggregate demand is weak (Chapter 46).

The **natural rate of unemployment** is the rate that prevails when cyclical unemployment is zero: frictional plus structural. It is not constant and it is not “natural” in any normative sense; it depends on institutions and on the efficiency with which workers and jobs are matched. Chapter 46 relates it to the non-accelerating-inflation rate of unemployment.

45.6.3 Why might wages not clear the market?

Several mechanisms can keep the real wage above the level at which labour supply equals demand.

Minimum wages and collective bargaining. A legal minimum above the equilibrium wage for low-skilled workers creates a surplus of labour in that market (Chapter 22), although if employers have monopsony power the effect on employment can be zero or positive (Chapter 24). Unions that bargain for wages above the competitive level raise the wages of insiders at the expense of outsiders.

Efficiency wages. Firms may choose to pay more than the market-clearing wage because higher wages raise productivity. In the Shapiro–Stiglitz (1984) model, effort is not observable

(the moral-hazard problem of Chapter 25), and the threat of dismissal disciplines shirking only if losing the job is costly. If wages cleared the market, a dismissed worker could immediately find another job at the same wage, and the threat would be empty. Equilibrium therefore requires some unemployment, which acts as a worker-discipline device. Other efficiency-wage mechanisms include lower turnover and the ability to attract better applicants (adverse selection, Chapter 25).

45.6.4 Worker flows and steady-state unemployment

Unemployment is not a pool of the same people who stay idle. Every month in the United States, millions of workers lose or leave jobs and millions find them. Let s be the monthly **separation rate** (the fraction of employed workers who lose their jobs) and f the **job-finding rate** (the fraction of unemployed workers who find jobs). With a labour force of fixed size and unemployment rate u ,

$$\Delta u = s(1 - u) - fu.$$

Unemployment is constant when inflows equal outflows:

Proposition 45.5. Steady-state unemployment

With separation rate s and job-finding rate f , the steady-state unemployment rate is

$$u^* = \frac{s}{s + f}.$$

Worked Example 45.6. Unemployment from flows

With $s = 0.02$ and $f = 0.30$ per month, $u^* = 0.02/0.32 = 6.25\%$. Suppose a recession lowers the job-finding rate to 0.20 while separations are unchanged. The new steady state is $0.02/0.22 = 9.09\%$. The adjustment is fast: starting from 6.25 per cent, next month's rate is $0.0625 + 0.02 \times 0.9375 - 0.20 \times 0.0625 = 0.0625 + 0.01875 - 0.0125 = 6.875\%$. The gap to the new steady state ($9.09 - 6.25 = 2.84$ points) shrinks by the factor $1 - s - f = 0.78$ each month, so about half of it closes within three months.

The formula clarifies policy debates. High unemployment in some European countries in the 1990s and 2000s reflected mainly low job-finding rates—long spells of unemployment—rather than high separation rates; U.S. unemployment is characterized by high flows in both directions. In U.S. recessions, both a rise in separations and a fall in job finding contribute, with the fall in job finding accounting for the larger share of the rise in unemployment in most studies.

45.6.5 Search and matching

Why is the job-finding rate what it is? Peter Diamond, Dale Mortensen and Christopher Pissarides, who shared the 2010 Nobel prize, developed the **search and matching** model. The number of new hires per period is given by a *matching function* $M(U, V)$ of the numbers of unemployed workers U and vacancies V , increasing in both and usually assumed to have constant returns, for example $M = \mu U^{1/2} V^{1/2}$. Define labour-market **tightness** as $\theta = V/U$. Then

$$f = \frac{M}{U} = \mu \theta^{1/2}, \quad \frac{M}{V} = \mu \theta^{-1/2}.$$

A tighter market makes it easier for workers to find jobs and harder for firms to fill vacancies. Firms post vacancies until the expected profit from a filled job, discounted by the time it takes

to fill it, just covers the cost of the vacancy; wages are set by bargaining over the surplus that a match creates, since both sides would lose by separating and searching again.

Combining $u^* = s/(s + f)$ with $f = \mu(v/u)^{1/2}$, where v is the vacancy rate, traces out a downward-sloping relation between unemployment and vacancies: the **Beveridge curve**. Movements *along* the curve reflect changes in labour demand—in a recession, vacancies fall and unemployment rises. *Shifts* of the curve reflect changes in matching efficiency μ or the separation rate: if the curve moves out, the economy has more unemployment at every level of vacancies, a sign of worse matching or more structural mismatch.

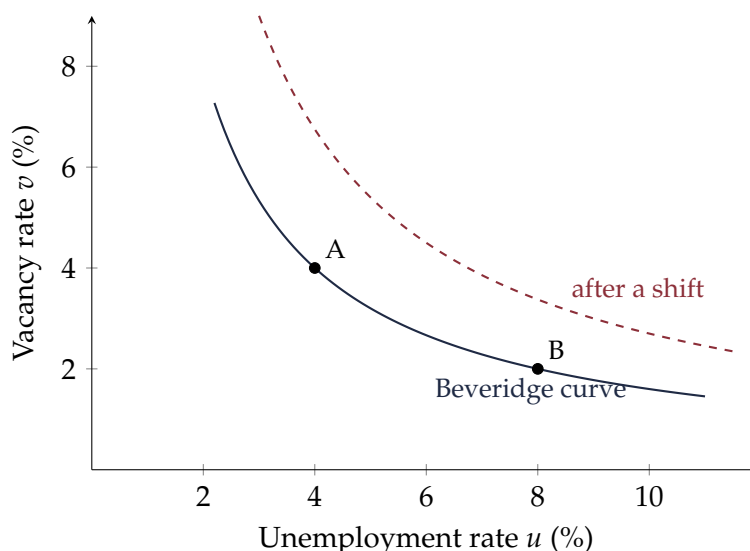


Figure 45.1: The Beveridge curve. A fall in labour demand moves the economy along the curve from A to B: fewer vacancies, more unemployment. A fall in matching efficiency or a rise in separations shifts the whole curve outward (dashed), raising unemployment at every vacancy rate.

In 2021–22 many advanced economies experienced unusually high vacancy rates for their unemployment rates, a shift that reflected reallocation after the pandemic, changes in labour-force participation and possibly lower matching efficiency. The ratio of vacancies to unemployed workers in the United States reached about two in 2022, far above its pre-pandemic level; central banks watched this measure of tightness closely as a signal of wage pressure (Chapter 50).

Case: Germany's short-time work: labour hoarding by design

In the recession of 2008–09, German real GDP fell by about 5.7 per cent, more than in the United States. Yet German unemployment barely rose, while U.S. unemployment doubled to 10 per cent. In the COVID-19 shock of 2020, the contrast was even sharper: U.S. unemployment jumped from 3.5 per cent in February to almost 15 per cent in April, while German unemployment rose by less than one percentage point.

The mechanism. Under *Kurzarbeit*, a firm facing a temporary fall in demand can cut its employees' hours; the Federal Employment Agency replaces a large part (typically 60–67 per cent, and more during the pandemic) of the net earnings lost on the hours not worked. Workers keep their jobs; the firm keeps its trained workforce. The number of short-time workers peaked at around 1.4–1.5 million in 2009 and at around 6 million in April 2020, nearly a fifth of employees subject to social insurance contributions.

In terms of this chapter. Short-time work holds the separation rate s down in a recession, so that the adjustment happens through hours per worker rather than through the number

of workers. In the matching model, a match has value—firm-specific skills, the cost of recruiting and training—that is destroyed when the worker is laid off and must be rebuilt by costly search afterwards. If the shock is temporary, subsidizing the preservation of matches can be efficient. The data are consistent with this: German employment and output recovered quickly after both recessions without the long spells of unemployment that followed the U.S. layoffs of 2008–09.

In accounting terms. For the firm, the subsidy reduces personnel expenses in the income statement and conserves cash (Chapter 31); for the government, it is a transfer that raises the fiscal deficit (Chapter 43) and acts as an automatic stabilizer (Chapter 46).

The limits. If the shock is not temporary—if demand shifts permanently away from a sector—short-time work preserves matches that should be dissolved, slowing reallocation of workers to growing firms. The United States relied instead on unemployment insurance and direct transfers, and its labour market reallocated faster, with many workers returning to different employers. Which approach is better depends on whether the shock is temporary or permanent, which is exactly what is hard to know in real time.

45.7 Putting the Pieces Together

This chapter has built three behavioural relationships:

1. *Consumption* depends on total wealth—financial and human—and on the real interest rate, with a larger response to current income for liquidity-constrained households. In compact form, $C = C(Y - T, \text{wealth}, r)$, increasing in the first two arguments and, on balance, decreasing in r .
2. *Investment* depends on expected output and on the user cost of capital, and therefore on the real interest rate: $I = I(r, Y^e, q)$, decreasing in r .
3. *Unemployment* depends on the rates at which workers separate from and find jobs, which in turn depend on labour demand, matching efficiency and labour-market institutions.

Chapter 46 combines the first two with the goods-market equilibrium condition to build the IS curve, and relates the third to output through Okun's law. Chapter 47 derives all three from explicit optimization in dynamic general equilibrium.

Summary

- The Keynesian consumption function fits cross sections and short samples but not long-run data, because it mixes transitory and permanent income.
- The intertemporal budget constraint says the present value of consumption equals financial plus human wealth. With $\beta(1 + r) = 1$, consumption is the annuity value of total wealth (permanent income).
- The MPC out of a transitory change in income is small; out of a permanent change, close to one. Life-cycle saving makes aggregate saving depend on growth and demography.
- Consumption should follow a random walk under the permanent-income hypothesis, but actual spending responds strongly to predictable and transitory income because of liquidity constraints and hand-to-mouth behaviour. Prudent households save for precaution.

- Ricardian equivalence holds under strong assumptions; in practice private saving offsets debt-financed tax cuts only partly.
- The desired capital stock is $K^* = \alpha Y / uc$; investment responds to the user cost, to changes in expected output (the accelerator), to Tobin's q , to cash flow and to uncertainty.
- Steady-state unemployment is $u^* = s / (s + f)$. Matching models explain job finding by market tightness and generate the Beveridge curve. Efficiency wages and institutions can keep wages above market-clearing levels.

Key terms

marginal propensity to consume; average propensity to consume; human wealth; intertemporal budget constraint; consumption smoothing; permanent income; life-cycle hypothesis; random walk; excess sensitivity; liquidity constraint; hand-to-mouth household; precautionary saving; prudence; elasticity of intertemporal substitution; Ricardian equivalence; user cost of capital; desired capital stock; accelerator; adjustment costs; Tobin's q ; option value of waiting; intertemporal elasticity of labour supply; frictional, structural and cyclical unemployment; natural rate of unemployment; efficiency wage; separation rate; job-finding rate; search and matching; matching function; tightness; Beveridge curve; short-time work.

Exercises

Exercise 45.1 (Life-cycle consumption). A household aged 30 expects to live to 80 and to retire at 65. It has financial wealth of 50,000 and expects real labour income of 45,000 a year until retirement. With $r = 0$ and $\beta = 1$, find annual consumption, annual saving while working, and wealth at retirement.

Exercise 45.2 (Transitory versus permanent). A household with an infinite horizon faces $r = 0.04$ and $\beta(1 + r) = 1$. By how much does its annual consumption rise if (a) it receives a one-off transfer of 1,000 today; (b) its income rises by 1,000 in every year, starting today? Interpret the results as MPCs.

Exercise 45.3 (Euler equation with CRRA). With $u(c) = c^{1-\theta} / (1 - \theta)$, $\theta = 3$, $\beta = 0.97$: find the growth rate of consumption when $r = 0.02$ and when $r = 0.05$. What does the difference tell you about the strength of the intertemporal-substitution channel?

Exercise 45.4 (Liquidity constraints). In Example 9.7 of Chapter 9, the household with $y_1 = 100$, $y_2 = 110$, $r = 0.1$, $\beta = 0.9$ and log utility wished to borrow 5.26. Suppose it cannot borrow at all. What does it consume in each period? Now it receives a transfer of 10 in period 1. How much of the transfer does it spend in period 1, (a) if it can borrow freely, (b) if it cannot borrow?

Exercise 45.5 (Ricardian equivalence). The government cuts lump-sum taxes by 100 this year and will raise them next year by $100(1 + r)$ to repay the debt. Explain what happens to private saving, national saving and consumption if Ricardian equivalence holds. Give two reasons it might not hold, and say for each whether consumption would rise by more or less.

Exercise 45.6 (Desired capital). A firm with $\alpha = 0.35$ expects output of 5,000. Capital costs $P_K = 1$ and depreciates at 8 per cent; the real interest rate is 3 per cent. (a) Find K^* . (b) If the real interest rate rises to 4.5 per cent, find the new K^* and the percentage change. (c) If the firm adjusts halfway to its new desired stock in the first year, what is net investment in that year?

Exercise 45.7 (Accelerator). In an economy with capital–output ratio 2.5 and depreciation 8 per cent, output has been growing at 4 per cent. Compute gross investment as a share of output. If

expected growth falls to 1 per cent, compute the new investment share and the percentage fall in investment.

Exercise 45.8 (Tobin's q). Explain why a firm with $q < 1$ might still invest. Explain why Apple's market-to-book ratio is a poor measure of its q .

Exercise 45.9 (Flows). In a labour market the monthly separation rate is 1.5 per cent and the job-finding rate is 25 per cent. (a) Find steady-state unemployment. (b) Short-time work lowers the separation rate in a recession to 1.2 per cent while the finding rate falls to 18 per cent; without short-time work the separation rate would have risen to 2 per cent. Compare the steady-state unemployment rates with and without the scheme.

Exercise 45.10 (Matching). With $M = 0.5 U^{1/2} V^{1/2}$ (per month) and separation rate 2 per cent, find the job-finding rate and steady-state unemployment when $v/u = 1$ and when $v/u = 0.5$. Explain how an outward shift of the Beveridge curve would appear in this model.

Exercise 45.11 (Diagnose the error). "Unemployment benefits cause unemployment, so the country with the highest unemployment rate must have the most generous benefits." Identify which parts of this statement are behavioural claims, which are empirical claims, and what else could explain cross-country differences in unemployment.

Solutions

Solution 45.1. Remaining life 50 years; working life 35 years. Lifetime resources = $50,000 + 35 \times 45,000 = 50,000 + 1,575,000 = 1,625,000$. Consumption = $1,625,000/50 = 32,500$. Saving while working = $45,000 - 32,500 = 12,500$. Wealth at 65 = $50,000 + 35 \times 12,500 = 487,500$, which finances $15 \times 32,500 = 487,500$ of retirement consumption. Check: the accounts balance exactly.

Solution 45.2. (a) Total wealth rises by 1,000; consumption rises by $1,000 \times 0.04/1.04 = 38.46$. MPC ≈ 0.04 . (b) The present value of 1,000 a year starting today is $1,000 \times 1.04/0.04 = 26,000$; consumption rises by $26,000 \times 0.04/1.04 = 1,000$. MPC = 1. The household spends a permanent increase in full and a transitory one only in proportion to its annuity value.

Solution 45.3. $c_{t+1}/c_t = [\beta(1+r)]^{1/3}$. At $r = 0.02$: $0.97 \times 1.02 = 0.9894$, $0.9894^{1/3} = 0.99645$: consumption falls by 0.36 per cent a year, because the household is more impatient than the interest rate rewards. At $r = 0.05$: $0.97 \times 1.05 = 1.0185$, $1.0185^{1/3} = 1.00613$: growth 0.61 per cent. A 3-point rise in r raises consumption growth by about 1 point, consistent with an EIS of $1/3$. With a low EIS, a change in interest rates changes the timing of consumption only modestly.

Solution 45.4. Without borrowing, the household would like $c_1 = 105.26 > y_1$, so the constraint binds: $c_1 = 100$, $c_2 = 110$. (a) With free borrowing, W rises by 10 to 210; $c_1 = 210/1.9 = 110.53$, up by 5.26: it spends 53 per cent of the transfer in period 1 and saves the rest for period 2. (b) Without borrowing, it would now like $c_1 = 110.53$, but its period-1 resources are 110. The constraint still binds, so $c_1 = 110$ and $c_2 = 110$: it spends the whole transfer in period 1. The constrained household's MPC is 1.

Solution 45.5. Disposable income rises by 100 this year; households anticipate taxes of $100(1+r)$ next year, whose present value is 100, so human wealth net of taxes is unchanged and consumption does not change. Private saving rises by 100; public saving falls by 100; national saving is unchanged. It might fail (i) because some households are liquidity constrained and spend the tax cut: consumption rises; (ii) because households are myopic or expect the taxes to fall on future generations they do not care about: consumption rises. (If households expected the debt to be repaid with a distortionary tax that they feared more than the lump-sum tax, consumption could even fall.)

Solution 45.6. (a) $uc = 0.03 + 0.08 = 0.11$; $K^* = 0.35 \times 5,000/0.11 = 1,750/0.11 = 15,909$. (b) $uc = 0.125$; $K^* = 14,000$; change -12.0% . (c) Net investment = $0.5 \times (14,000 - 15,909) = -955$: gross investment is 955 less than replacement investment $0.08 \times 15,909 = 1,273$, i.e. gross investment of about 318.

Solution 45.7. Initially $I/Y = 2.5 \times 0.04 + 0.08 \times 2.5 = 0.10 + 0.20 = 0.30$. With 1 per cent growth, $I/Y = 0.025 + 0.20 = 0.225$. Investment falls by $0.075/0.30 = 25\%$ when output growth slows by 3 points.

Solution 45.8. Average q may differ from marginal q : a firm with some unprofitable old assets can have average $q < 1$ while a new project has a high marginal return. Market values may also be depressed temporarily, or the firm may need to replace worn-out capital to stay in business. Apple's book value omits internally generated intangibles—brand, software, design capability, its ecosystem—and its equity is reduced by large buybacks (Chapter 30), so market-to-book greatly exceeds the ratio of market value to the replacement cost of all its capital.

Solution 45.9. (a) $u^* = 0.015/0.265 = 5.66\%$. (b) With the scheme: $0.012/(0.012 + 0.18) = 6.25\%$. Without: $0.02/(0.02 + 0.18) = 10.0\%$. The scheme keeps steady-state unemployment 3.75 points lower by preventing separations, at the fiscal cost of the subsidy.

Solution 45.10. $f = M/U = 0.5(v/u)^{1/2}$. With $v/u = 1$: $f = 0.5$, $u^* = 0.02/0.52 = 3.85\%$. With $v/u = 0.5$: $f = 0.5 \times 0.707 = 0.354$, $u^* = 0.02/0.374 = 5.35\%$. An outward shift of the Beveridge curve corresponds to a fall in matching efficiency (0.5 falling to, say, 0.4) or a rise in the separation rate: at each vacancy rate, the job-finding rate is lower or inflows are higher, so unemployment is higher.

Solution 45.11. “Benefits cause unemployment” is a behavioural claim (more generous benefits may lower search effort or raise reservation wages, reducing f); its size is an empirical question. The inference that the highest-unemployment country has the most generous benefits is a logical error: unemployment also depends on separation rates, matching efficiency, labour-market regulation, the minimum wage, the age and skill structure of the labour force, the sectoral structure of the economy, and—crucially—the business cycle: a country in recession has high cyclical unemployment whatever its benefits. Establishing causation requires the methods of Chapter 17, such as comparing workers just eligible and just ineligible for longer benefits.

Short-Run Fluctuations and Stabilization

After studying this chapter you should be able to

1. Describe the main facts of business cycles and the co-movement of output, employment, investment and inflation.
2. Solve the Keynesian cross, derive the expenditure multipliers, and explain the paradox of thrift.
3. Derive the IS curve and combine it with a monetary-policy (MP) curve in which the central bank sets the real interest rate.
4. Explain Okun's law and the expectations-augmented Phillips curve, and the idea of a natural rate or NAIRU.
5. Build an aggregate demand–aggregate supply model in output–inflation space and use it to analyse demand and supply shocks.
6. Evaluate fiscal policy: multipliers, automatic stabilizers, crowding out and policy lags.
7. Interpret the COVID-19 recession as a combination of supply and demand shocks.

A motivating question

Between the fourth quarter of 2007 and the second quarter of 2009, U.S. real GDP fell by about 4 per cent and unemployment rose from under 5 per cent to 10 per cent. Nothing had happened to the country's capital stock, its technology or the skills of its workers. Factories stood idle not because they could not produce but because nobody was buying what they could produce. In the spring of 2020, output fell even faster, but for a different reason at first: governments told people to stay at home, and many businesses could not operate at all.

The growth theory of Chapter 44 explains the path of *potential* output—what the economy can produce with its resources fully employed. It does not explain why actual output sometimes falls far below potential for years at a time. In the long run, prices and wages adjust and the economy returns to potential; in the short run, they do not adjust quickly enough, and spending determines output. This chapter develops the short-run framework used by central banks and finance ministries: an IS curve for spending, a monetary-policy rule for interest rates, and a Phillips curve for inflation.

46.1 Facts about Fluctuations

1. **Fluctuations are irregular.** Recessions—periods of falling output—occur at irregular intervals and vary greatly in depth and length. In the United States the dating of recessions is done by a committee of the National Bureau of Economic Research (NBER), which looks at a range of indicators rather than the popular rule of thumb of “two consecutive quarters of falling GDP.”

2. **Output components co-move but with different amplitudes.** Consumption of non-durables and services is smoother than GDP; spending on durable goods, residential construction and business investment is several times as volatile. Inventory investment swings sharply and accounts for a large share of the fall in GDP in many recessions.
3. **Employment and unemployment move with output.** Unemployment rises in recessions and falls in expansions, typically lagging output by a few months. Hours and employment fall less, in percentage terms, than output: labour productivity tends to be procyclical.
4. **Inflation tends to fall in and after recessions**, and to rise when the economy runs hot, but the relationship is loose and has changed over time.
5. **Asset prices, credit and interest rates are procyclical**; financial crises are followed by unusually deep and long recessions (Chapter 51).

We measure the cyclical position of the economy by the **output gap**: the percentage deviation of actual real GDP from potential output Y^* ,

$$\tilde{Y} = \frac{Y - Y^*}{Y^*} \times 100.$$

Potential output is not observed; it is estimated by statistical filters or production-function methods (Chapter 44), and estimates are revised, sometimes substantially, as new data arrive. Central banks set policy using gap estimates they know to be imprecise (Chapter 50).

46.2 The Keynesian Cross

46.2.1 Planned expenditure

Suppose that, in the short run, firms produce whatever is demanded at existing prices. Then output is determined by planned expenditure. Using the consumption function of Chapter 45 in its simplest Keynesian form, planned expenditure is

$$PE = c_0 + c_1(Y - T) + I + G + NX,$$

with investment I , government purchases G , taxes T and net exports NX taken as given for now. Equilibrium requires that output equal planned expenditure: $Y = PE$. If output exceeded planned expenditure, unsold goods would pile up as unplanned inventory investment (which the national accounts record as investment, so the *identity* $Y \equiv C + I + G + NX$ always holds—Chapter 43); firms would cut production. Solving $Y = PE$:

$$Y = \frac{1}{1 - c_1} [c_0 - c_1 T + I + G + NX].$$

Worked Example 46.1. Multipliers in the Keynesian cross

Let $c_0 = 200$, $c_1 = 0.75$, $T = 400$, $I = 500$, $G = 400$ and $NX = 0$. Then

$$Y = \frac{200 - 0.75 \times 400 + 500 + 400}{0.25} = \frac{800}{0.25} = 3,200.$$

Check: disposable income is 2,800, consumption $200 + 0.75 \times 2,800 = 2,300$, and $2,300 + 500 + 400 = 3,200$.

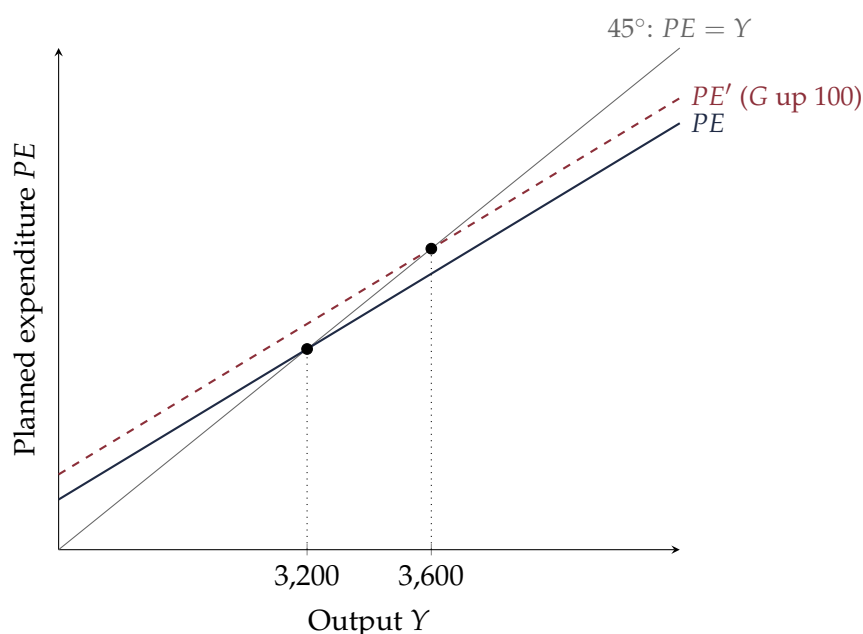


Figure 46.1: The Keynesian cross for Example 46.1. Equilibrium is where planned expenditure crosses the 45-degree line. A rise of 100 in government purchases shifts PE up by 100 and raises equilibrium output by 400.

Government purchases. If G rises by 100, Y rises by $100/0.25 = 400$ to 3,600. The **government-purchases multiplier** is $1/(1 - c_1) = 4$.

Taxes. If T rises by 100, Y falls by $0.75 \times 100/0.25 = 300$. The **tax multiplier** is $-c_1/(1 - c_1) = -3$: smaller in absolute value, because part of a tax change comes out of saving.

Balanced budget. If G and T both rise by 100, Y rises by $400 - 300 = 100$: the **balanced-budget multiplier** is 1.

The multiplier arises because spending is someone else's income. An extra 100 of government purchases raises the income of the firms and workers who produce the goods by 100; they spend 75 of it, which raises others' income by 75; those people spend 56.25, and so on. The total increase is $100(1 + 0.75 + 0.75^2 + \dots) = 100/(1 - 0.75) = 400$, a geometric series (Chapter 4).

46.2.2 Leakages and automatic stabilizers

The multiplier of 4 is too large to be realistic, because the simple model omits several *leakages* from the spending chain. If taxes rise with income, $T = tY$, and imports rise with income, $M = mY$, the multiplier becomes

$$\frac{1}{1 - c_1(1 - t) + m}.$$

With $c_1 = 0.75$, $t = 0.25$ and $m = 0.2$, it is $1/(1 - 0.5625 + 0.2) = 1.57$. Small, very open economies—Malaysia, whose imports exceed 60 per cent of GDP, or Ireland—have much smaller domestic multipliers than large, relatively closed economies such as the United States, because much of any extra spending leaks abroad.

Income-related taxes and transfers are **automatic stabilizers**. When output falls, tax receipts fall and unemployment benefits rise automatically, cushioning disposable income without

any new decision by government. The same feature that lowers the multiplier on fiscal policy lowers the multiplier on private shocks: in the numbers above, a fall of 100 in investment reduces output by 157 rather than 400. Countries with large welfare states, such as Germany, have strong automatic stabilizers; the short-time work scheme of Chapter 45 is a particularly powerful one.

46.2.3 The paradox of thrift

Suppose households decide to save more at every level of income: c_0 falls by 100. In Example 46.1, output falls by 400 to 2,800. What happens to saving? Before the change, private saving was $Y - T - C = 2,800 - 2,300 = 500$. After it, disposable income is 2,400 and consumption is $100 + 0.75 \times 2,400 = 1,900$, so saving is again 500. The attempt to save more has reduced income until saving is unchanged. This is Keynes's **paradox of thrift**. It follows from the identity $S_{\text{private}} + (T - G) \equiv I + NX$: with investment, the budget and net exports unchanged, private saving cannot change, so income must fall. If investment falls when income falls, saving actually declines. What is prudent for one household can be self-defeating for all households together—a fallacy of composition that applies only when output is demand-determined.

46.3 The IS Curve

Investment is not fixed. Chapter 45 showed that it depends on the real interest rate through the user cost of capital; consumption of durables and housing does too. Write

$$I = I_0 - b r, \quad b > 0.$$

Substituting into the equilibrium condition,

$$Y = \frac{1}{1 - c_1} [c_0 - c_1 T + I_0 - b r + G + NX].$$

This relation between the real interest rate and the level of output at which the goods market is in equilibrium is the **IS curve** (for “investment equals saving”). It slopes down: a higher real interest rate lowers investment and, through the multiplier, output. Anything that raises spending at a given interest rate—higher government purchases, lower taxes, more optimistic firms, stronger foreign demand, higher household wealth—shifts it to the right.

It is often convenient to write the IS curve in gap form:

$$\tilde{Y} = -a (r - \bar{r}) + \epsilon_d,$$

where $\bar{r}$ is the **natural** (or neutral) **real interest rate**—the rate at which output equals potential when there are no demand shocks— $a > 0$ measures the sensitivity of spending to interest rates (it includes the multiplier), and ϵ_d is a *demand shock*. When the central bank sets r above $\bar{r}$, policy is *contractionary*; below $\bar{r}$, *expansionary*. Chapter 50 discusses how the natural rate is estimated and why many estimates suggest it fell substantially between the 1990s and the 2010s.

46.4 Monetary Policy and the MP Curve

46.4.1 The central bank sets the interest rate

How is the interest rate determined? Older textbooks answered with the *LM curve*: the central bank fixes the quantity of money, and the interest rate adjusts to equate money demand

to money supply. That is not how modern central banks operate. The Federal Reserve, the European Central Bank, the Bank of England, Bank Negara Malaysia and most others announce a target for a short-term nominal interest rate and supply whatever quantity of central-bank money is needed to achieve it. Chapter 49 describes exactly how they do so. The People's Bank of China has historically placed more weight on quantities, but it too has moved toward steering short-term interest rates.

Following David Romer (2000), we therefore replace the LM curve with an **MP curve**: the central bank sets the real interest rate. Because prices adjust slowly, a change in the nominal policy rate changes the real rate $r = i - \pi^e$ in the short run. In the simplest case, the MP curve is horizontal at the rate the central bank chooses. More usefully, the central bank follows a rule in which it raises the real rate when inflation is above its target π^* :

$$r = \bar{r} + \phi (\pi - \pi^*), \quad \phi > 0.$$

Chapter 50 develops such rules—including the famous rule proposed by John Taylor—in detail and shows why $\phi > 0$ is essential for stability.

Advanced Analysis: IS–LM and the money supply

In the IS–LM model developed by John Hicks (1937) from Keynes's *General Theory*, money market equilibrium requires $M/P = L(i, Y)$, where money demand L falls with the nominal interest rate i (the opportunity cost of holding money) and rises with income. For a fixed money supply M and price level P , this defines an upward-sloping LM curve in (Y, i) space: higher income raises money demand, which requires a higher interest rate to restore equilibrium. IS–LM and IS–MP give the same answers when the central bank targets the money supply and the IS–MP answer when it targets the interest rate. The practical case against money targeting is that money demand has proved unstable—financial innovation shifts $L(\cdot)$ unpredictably—so a money target transmits these shifts into interest rates and output. Most central banks that tried monetary targeting in the late 1970s and 1980s abandoned it; the Bundesbank retained money-growth targets until 1998 but in practice often missed them, using them more as a communication device (Chapter 50).

46.5 Unemployment, Output and Inflation

46.5.1 Okun's law

Arthur Okun observed in 1962 that U.S. unemployment and output moved together in a stable way. In gap form, **Okun's law** states

$$u - u^* = -\omega \tilde{Y},$$

with ω around 1/2 in U.S. data: each percentage point by which output is below potential raises unemployment by about half a point above its natural rate. The coefficient is less than one because in a downturn firms hoard labour, hours per worker fall, and some workers leave the labour force rather than becoming unemployed. It is smaller still in Germany and Japan, where employment protection and schemes such as short-time work make firms adjust hours rather than headcount (Chapter 45), and it is not a law of nature: Okun's relation shifted during the COVID-19 shock, when unemployment first rose far more than the output gap would predict and then fell faster.

46.5.2 The Phillips curve

In 1958 A. W. Phillips documented a negative relationship between wage inflation and unemployment in the United Kingdom from 1861 to 1957. Paul Samuelson and Robert Solow (1960) found a similar relation between price inflation and unemployment in the United States and suggested that policy-makers faced a menu: lower unemployment at the cost of higher inflation.

Milton Friedman (1968) and Edmund Phelps (1967) argued that the menu was an illusion. Workers and firms care about *real* wages. If everyone expects 5 per cent inflation, wages and prices will rise at 5 per cent a year when the labour market is at its natural rate; only when unemployment is below the natural rate will they rise faster than expected. The **expectations-augmented Phillips curve** is

$$\pi = \pi^e + \kappa \tilde{Y} + \epsilon_s,$$

where $\kappa > 0$ measures how strongly inflation responds to the output gap, and ϵ_s is a *supply* (or *cost-push*) *shock*, such as a rise in the price of oil. Using Okun's law, it can equivalently be written in terms of unemployment: $\pi = \pi^e - (\kappa/\omega)(u - u^*) + \epsilon_s$.

Proposition 46.2. The natural rate hypothesis

If expected inflation eventually equals actual inflation, then in the long run output is at potential and unemployment at its natural rate, whatever the rate of inflation. There is no long-run trade-off between inflation and unemployment.

With *adaptive expectations*, $\pi_t^e = \pi_{t-1}$, the Phillips curve becomes $\pi_t - \pi_{t-1} = \kappa \tilde{Y}_t + \epsilon_s$: holding output above potential causes inflation to *accelerate*, year after year. The unemployment rate at which inflation neither rises nor falls is called the **non-accelerating-inflation rate of unemployment** (NAIRU); in this model it is the natural rate. Friedman and Phelps made their argument when inflation was still low; the stagflation of the 1970s, in which inflation and unemployment rose together, was widely seen as its vindication (Chapter 50).

► “Low unemployment causes inflation”

The Phillips curve relates inflation to unemployment *relative to* expected inflation and the natural rate. Low unemployment is inflationary only if it is below the natural rate, which is not directly observable and changes over time. U.S. unemployment fell below 4 per cent in 2018–19 without a rise in inflation, leading many to argue the natural rate was lower than previously thought. Conversely, when supply shocks hit, inflation can rise with unemployment flat or rising.

In the decades before 2020, the Phillips curve in many advanced economies appeared to have become very flat: large swings in unemployment were associated with small changes in inflation. Explanations include better-anchored inflation expectations (Chapter 50), globalization, and the possibility that the curve is non-linear—flat when the economy is slack and steep when it is very tight. The inflation surge of 2021–23 supported the non-linear view: with very tight labour markets and large supply shocks, inflation rose rapidly.

46.6 Aggregate Demand and Aggregate Supply

46.6.1 The model

We can now combine the three relationships into a model of output and inflation. Substituting the monetary-policy rule into the IS curve gives the **aggregate demand** (AD) curve:

$$\tilde{Y} = -a\phi(\pi - \pi^*) + \epsilon_d.$$

It slopes down in $(\tilde{Y}, \pi)$ space, but for a different reason from the demand curve of a single good: when inflation rises, the central bank raises the real interest rate, which reduces spending. The Phillips curve is the short-run **aggregate supply** (AS) curve:

$$\pi = \pi^e + \kappa \tilde{Y} + \epsilon_s.$$

It slopes up and shifts with expected inflation. The long-run aggregate supply curve is vertical at $\tilde{Y} = 0$.

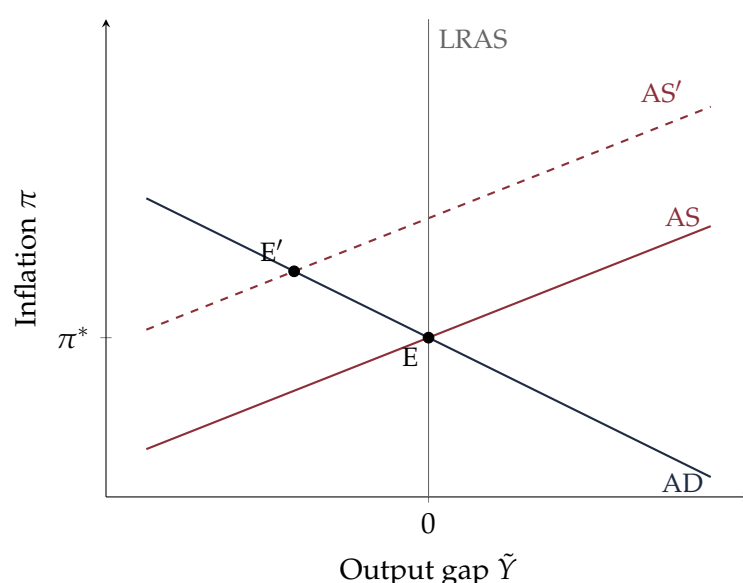


Figure 46.2: Aggregate demand and supply. Starting at E, an adverse supply shock shifts AS up to AS'; the new short-run equilibrium E' has higher inflation and a negative output gap. The central bank raises the real interest rate in response to higher inflation, which moves the economy along AD.

46.6.2 Demand shocks

A negative demand shock—a collapse in business confidence, a fall in house prices that reduces household wealth, a fall in exports—shifts AD to the left. Output falls below potential, and through the Phillips curve inflation falls. The central bank's rule automatically cuts the real interest rate, cushioning the fall. Because demand shocks move output and inflation in the *same* direction, stabilizing inflation and stabilizing output are complementary goals: there is no dilemma.

Worked Example 46.3. A demand shock with adaptive expectations

Let $a = 1$, $\phi = 0.5$, $\kappa = 0.25$, $\pi^* = 2$, with $\pi_t^e = \pi_{t-1}$ and an initial inflation rate of 2. A demand shock $\epsilon_d = -2$ lasts for three years. In each year, the AD and AS equations must

hold simultaneously. In year 1, writing $x = \pi_1 - 2$:

$$\tilde{Y}_1 = -0.5x - 2, \quad x = 0.25 \tilde{Y}_1.$$

So $\tilde{Y}_1 = -0.125\tilde{Y}_1 - 2$, giving $\tilde{Y}_1 = -1.78$ and $\pi_1 = 2 - 0.44 = 1.56$. The central bank cuts the real rate by $\phi x = 0.22$ points below $\bar{r}$. Continuing the same calculation:

Year	Output gap	Inflation	$r - \bar{r}$
1	-1.78	1.56	-0.22
2	-1.58	1.16	-0.42
3	-1.40	0.81	-0.60
4 (shock over)	+0.53	0.94	-0.53
5	+0.47	1.06	-0.47

Because expectations adapt, low inflation becomes embedded: after the shock ends, inflation is still below target, so the central bank keeps rates low and output runs *above* potential for several years until inflation climbs back. A more aggressive response (larger ϕ) would cushion output more in the first years. A central bank whose target is credible, so that π^e stays at π^* , avoids the drift in expectations altogether—the theme of Chapter 50.

46.6.3 Supply shocks and the policy dilemma

An adverse supply shock—a rise in oil prices, a harvest failure, a disruption of supply chains—shifts AS up (Figure 46.2). Inflation rises *and* output falls: **stagflation**. Now the central bank faces a genuine dilemma. Raising interest rates to fight inflation deepens the recession; holding them down to support output lets inflation rise further and risks lifting expected inflation, which would shift AS up again. How much weight to put on each goal, and how to judge whether a price shock will pass quickly or feed into expectations, is the central problem of Chapter 50. The oil shocks of 1973–74 and 1979–80, and the energy shock that followed Russia’s full-scale invasion of Ukraine in 2022, are the classic cases.

Micro–macro bridge. A supply shock in the aggregate model is, at the micro level, a shift in firms’ marginal cost curves (Chapter 21). How much of a rise in the cost of oil is passed on to consumers depends on the elasticities of supply and demand and on market structure (Chapters 22 and 24), and it appears in firms’ income statements as higher cost of goods sold and lower margins (Chapter 31). The aggregate Phillips curve compresses all of this into the single term ϵ_s .

46.7 Fiscal Policy

46.7.1 How large are fiscal multipliers?

The Keynesian cross suggests multipliers well above one. In practice, several forces make them smaller. *Leakages* into taxes, saving and imports reduce the chain of spending. *Monetary offset*: if the central bank follows a rule like the one above, it raises interest rates when fiscal expansion raises output and inflation, **crowding out** private investment. *Forward-looking households* may save part of a debt-financed tax cut in anticipation of future taxes (Ricardian equivalence, Chapter 45). *Supply constraints*: when the economy is at full capacity, extra spending raises

prices rather than output.

Empirical estimates vary widely with method and circumstances. Valerie Ramey's (2019) survey concluded that most estimates of the multiplier on government purchases in the United States lie between about 0.6 and 1. Estimates based on regional variation within a monetary union, where the central bank cannot offset spending in one region, are higher: Nakamura and Steinsson (2014) estimated a relative multiplier of about 1.5 for U.S. military procurement across states. There is evidence—contested—that multipliers are larger in deep recessions and, in particular, when interest rates are at their lower bound, because the central bank then does not offset fiscal expansion by raising rates (Chapter 51). Multipliers on transfers to liquidity-constrained households can be substantial, for the reasons given in Chapter 45.

46.7.2 Policy lags

Stabilization policy acts with lags. The *recognition lag* is the time it takes to realize that the economy has turned: GDP data arrive with a delay of a month or more and are revised. The *decision lag* is the time it takes to agree on a response: short for a central-bank committee that meets every six weeks, potentially long for a legislature. The *implementation lag* is the time it takes for the measure to be put into effect—for infrastructure projects, often years. The *effect lag* is the time before the measure affects spending: Friedman described the effects of monetary policy as operating with “long and variable lags,” and central banks generally assume that the peak effect of an interest-rate change on inflation comes one to two years later.

Lags make activist policy risky: a stimulus designed for a recession may arrive during the recovery and overheat the economy. This is one argument for relying on automatic stabilizers, and for delegating short-run stabilization mainly to an independent central bank (Chapter 50), with discretionary fiscal policy reserved for large shocks. Chapter 47 adds a deeper objection: if people's behaviour depends on their expectations of policy, the relationships estimated from past data may change when policy changes—the Lucas critique.

Case: COVID-19: a shock to both supply and demand

In March and April 2020, governments closed large parts of their economies to slow the spread of the virus. U.S. real GDP fell in the second quarter of 2020 at an annualized rate of around 30 per cent; euro-area GDP fell by more than 11 per cent in the quarter; Malaysia's economy contracted by about 17 per cent year on year in the second quarter during its Movement Control Order. China, hit first, contracted in the first quarter and recovered from the second.

Supply or demand? Both. Lockdowns stopped firms from producing—a supply shock that, taken alone, would raise inflation. But they also stopped people from spending on restaurants, travel and entertainment, and fear of infection and job loss depressed spending further—a demand shock that, taken alone, would lower inflation. In the AD–AS diagram both curves shifted left; output collapsed, and the effect on inflation was ambiguous. In the event, inflation initially fell (U.S. consumer-price inflation dipped close to zero in May 2020), suggesting that the demand effect dominated at first.

The fiscal response. Governments responded on an unprecedented scale. The U.S. CARES Act of March 2020 authorized about \$2.2 trillion, and further packages followed in December 2020 and March 2021; total U.S. fiscal support amounted to roughly a quarter of a year's GDP. Germany used short-time work (Chapter 45), loans and guarantees for firms and a temporary cut in value-added tax. Malaysia announced its PRIHATIN package in March 2020, combining cash transfers, wage subsidies, loan moratoria and access to retirement savings. Automatic stabilizers worked at full strength: tax revenues

fell and benefit payments rose.

The monetary response. Central banks cut rates to their lower bounds—the Federal Reserve to a range of 0–0.25 per cent on 15 March 2020; Bank Negara Malaysia lowered its overnight policy rate in several steps to 1.75 per cent by July 2020—and launched large asset purchases and lending programmes (Chapter 51).

In the accounts. Household disposable income in the United States *rose* in 2020, because transfers more than offset lost wages, while consumption fell; the personal saving rate spiked above 30 per cent in April 2020. Households accumulated “excess savings” of the order of \$2 trillion, which were spent down in 2021–23. Government deficits widened sharply, the mirror image of the rise in private saving (Chapter 43’s sectoral balances).

The aftermath. As economies reopened in 2021, demand—supported by accumulated savings and continued stimulus—recovered faster than supply, which was held back by bottlenecks in shipping and semiconductors, shifts in spending from services to goods, and, from 2022, the energy shock. Inflation rose to its highest levels in four decades. Whether the fiscal response was too large, whether central banks reacted too slowly, and how much of the inflation came from demand rather than supply are the questions of Chapter 50.

Summary

- Fluctuations are irregular deviations of output from potential. Investment and durables are much more volatile than consumption; unemployment moves inversely with output.
- In the Keynesian cross, output is determined by planned expenditure; the government-purchases multiplier is $1/(1 - c_1)$ and the tax multiplier $-c_1/(1 - c_1)$. Leakages to taxes and imports lower the multiplier and make automatic stabilizers work. The paradox of thrift shows that an attempt to save more can lower income without raising saving.
- The IS curve relates output to the real interest rate. Modern central banks set short-term interest rates, which the MP curve (or a policy rule) represents.
- Okun’s law links unemployment to the output gap. The expectations-augmented Phillips curve $\pi = \pi^e + \kappa\tilde{Y} + \epsilon_s$ implies no long-run trade-off between inflation and unemployment.
- Combining IS, the policy rule and the Phillips curve gives AD and AS in output–inflation space. Demand shocks move output and inflation together; supply shocks move them in opposite directions and create a policy dilemma.
- Fiscal multipliers depend on leakages, monetary offset, expectations and slack; estimates are often around 0.6–1 for government purchases, higher when monetary policy is constrained. Policy acts with lags.

Key terms

recession; output gap; potential output; planned expenditure; Keynesian cross; multiplier; government-purchases multiplier; tax multiplier; balanced-budget multiplier; leakages; automatic stabilizers; paradox of thrift; IS curve; natural real interest rate; demand shock; LM curve; MP curve; policy rule; Okun’s law; Phillips curve; expectations-augmented Phillips curve; natural rate hypothesis; adaptive expectations; NAIRU; supply shock; aggregate demand; aggregate supply; stagflation; crowding out; policy lags.

Exercises

Exercise 46.1 (Keynesian cross). With $c_0 = 150$, $c_1 = 0.8$, $T = 250$, $I = 300$, $G = 250$, $NX = 0$: find equilibrium output, consumption and private saving. Find the government-purchases and tax multipliers.

Exercise 46.2 (Leakages). In Exercise 46.1, suppose instead that taxes are $T = 0.2Y$ and imports $M = 0.1Y$, with exports fixed at 200. Find the multiplier. How much does output fall if investment falls by 50?

Exercise 46.3 (Paradox of thrift). In the economy of Example 46.1, suppose investment depends on income: $I = 100 + 0.125Y$. Find the new equilibrium output. Autonomous consumption then falls by 100. Find the new output and private saving, and explain why saving falls.

Exercise 46.4 (IS curve). With $c_0 = 200$, $c_1 = 0.75$, $T = G = 400$, $NX = 0$ and $I = 700 - 4,000r$, derive the IS curve. If the central bank sets $r = 0.05$, what is output? If potential output is 3,400, what real interest rate closes the output gap?

Exercise 46.5 (Okun's law). Unemployment is 6.5 per cent and the natural rate is 4.5 per cent. With Okun's coefficient $\omega = 0.5$, what is the output gap? If potential output is 20 trillion, how much output is being lost per year?

Exercise 46.6 (Phillips curve). With $\pi_t = \pi_{t-1} + 0.3\tilde{Y}_t$, inflation is 2 per cent and the government holds output 2 per cent above potential for four years. Find inflation in each year. What must happen to bring inflation back to 2 per cent?

Exercise 46.7 (AD–AS). In the model of Example 46.3, suppose instead an adverse supply shock $\epsilon_s = 1$ hits for one year, starting from $\pi = 2$ and $\tilde{Y} = 0$. Find output and inflation in year 1. Compare the outcome with $\phi = 1.5$. Explain the trade-off.

Exercise 46.8 (Multipliers). Explain why the fiscal multiplier is likely to be larger (a) in a recession with interest rates at their lower bound than in a boom; (b) for transfers to low-income households than for tax cuts for high-income households; (c) in the United States than in Malaysia.

Exercise 46.9 (Lags). A government legislates an infrastructure programme during a recession. Identify the four lags and explain how they could make the programme destabilizing.

Exercise 46.10 (COVID-19). Using the AD–AS diagram, show the effects of a lockdown that shifts both AD and AS left. Under what conditions does inflation fall? Using the sectoral balances identity, explain why the government deficit and private saving both rose in 2020.

Solutions

Solution 46.1. $Y = (150 - 0.8 \times 250 + 300 + 250)/0.2 = 500/0.2 = 2,500$. Disposable income 2,250; $C = 150 + 0.8 \times 2,250 = 1,950$; private saving $= 2,250 - 1,950 = 300 = I$ (the budget is balanced and $NX = 0$). Multipliers: $1/0.2 = 5$ for G , $-0.8/0.2 = -4$ for T .

Solution 46.2. Multiplier $= 1/(1 - 0.8 \times 0.8 + 0.1) = 1/0.46 = 2.17$. A fall of 50 in investment lowers output by $50 \times 2.17 = 109$.

Solution 46.3. $Y = 200 + 0.75(Y - 400) + 100 + 0.125Y + 400 = 400 + 0.875Y$, so $Y = 400/0.125 = 3,200$ (unchanged, by construction, since $I = 500$ at that output). With $c_0 = 100$: $Y = 300/0.125 = 2,400$: the multiplier is now 8. Investment falls to $100 + 0.125 \times 2,400 = 400$, and since $G = T$ and $NX = 0$, private

saving equals investment: 400, down from 500. Households tried to save more, but the fall in income reduced investment and hence saving.

Solution 46.4. $Y = [200 - 300 + 700 - 4,000r + 400]/0.25 = 4,000 - 16,000r$. At $r = 0.05$, $Y = 3,200$. To reach 3,400: $16,000r = 600$, $r = 3.75\%$. The natural real rate in this economy is 3.75 per cent.

Solution 46.5. $u - u^* = 2 = -0.5\tilde{Y}$, so $\tilde{Y} = -4\%$. Lost output $= 0.04 \times 20 = 0.8$ trillion a year.

Solution 46.6. Each year inflation rises by $0.3 \times 2 = 0.6$: 2.6, 3.2, 3.8, 4.4 per cent. Output must be held *below* potential until inflation falls back: for example, a gap of -2 per cent for four years would bring it back to 2 per cent. Under adaptive expectations, bringing inflation down requires a recession—the logic behind the Volcker disinflation (Chapter 50).

Solution 46.7. With $x = \pi_1 - 2$: $\tilde{Y}_1 = -a\phi x$ and $x = \kappa\tilde{Y}_1 + 1$. With $\phi = 0.5$: $x = -0.125x + 1$, $x = 0.889$, $\tilde{Y}_1 = -0.444$. With $\phi = 1.5$: $x = -0.375x + 1$, $x = 0.727$, $\tilde{Y}_1 = -1.091$. A more aggressive central bank holds inflation lower at the cost of a deeper recession: with supply shocks, stabilizing inflation and output conflict.

Solution 46.8. (a) At the lower bound the central bank does not raise rates in response, so there is no monetary offset, and with slack the extra demand raises output rather than prices. (b) Low-income households are more likely to be liquidity constrained and have higher MPCs (Chapter 45). (c) Malaysia is far more open, so a larger share of extra spending leaks into imports.

Solution 46.9. Recognition (realizing the recession has begun), decision (legislating the programme), implementation (planning, tendering and starting construction) and effect (spending feeding through to income). If the total exceeds the length of the recession, the spending arrives during the recovery, adding demand when output is returning to potential and raising inflation and interest rates—destabilizing rather than stabilizing.

Solution 46.10. Both curves shift left; output falls unambiguously. Inflation falls if the AD shift is larger (relative to the slopes) than the AS shift. The identity $(S - I) + (T - G) \equiv NX$: with net exports roughly unchanged, a large rise in the government deficit $(G - T)$ —through transfers, stabilizers and lower tax receipts—must be matched by a rise in private saving net of investment. Households could not spend on many services and received transfers, so their saving rose.

Modern Macroeconomic Frameworks

After studying this chapter you should be able to

1. Explain rational expectations and the Lucas critique, and why they led macroeconomics toward explicit microfoundations.
2. Describe the real business cycle model: its mechanism, its method of calibration and the main criticisms of it.
3. Explain why nominal rigidities give monetary policy real effects, and interpret Calvo pricing.
4. State the three equations of the New Keynesian model, explain the Taylor principle and derive impulse responses to a monetary-policy shock.
5. Describe what a DSGE model is, how central banks use such models, and how financial frictions and heterogeneous households have been added.
6. Derive the government debt dynamics equation, the debt-stabilizing primary balance and the role of $r - g$.
7. Explain fiscal–monetary interactions, fiscal dominance and “unpleasant monetarist arithmetic.”

A motivating question

In the late 1960s many economists believed that governments could keep unemployment permanently low by accepting a moderate, stable rate of inflation, as the Phillips curve seemed to promise. A decade later, the United States and much of Europe had both high inflation and high unemployment. The models that had guided policy had not merely been wrong about a number; they had failed to anticipate that people would change their behaviour once policy changed.

Modern macroeconomics was built from the wreckage. Its central methodological commitment is that aggregate relationships—the IS curve, the Phillips curve, the consumption function—should be derived from the decisions of households and firms who look ahead, rather than assumed. Chapter 45 already derived consumption from the intertemporal budget constraint. This chapter shows how that approach, combined with nominal rigidities, produces the New Keynesian model that underlies the forecasting and policy analysis of almost every central bank, and how the same forward-looking logic governs the sustainability of public debt.

47.1 Expectations and the Lucas Critique

47.1.1 Rational expectations

The Phillips curve of Chapter 46 contained expected inflation. How are expectations formed? *Adaptive expectations* assume that people extrapolate from the past: $\pi_t^e = \pi_{t-1}$. Under adaptive expectations, a government that is willing to let inflation rise steadily can hold unemployment below its natural rate indefinitely, because expectations always lag behind. That implies people make the same mistake year after year.

John Muth (1961) proposed instead that people's expectations are the forecasts implied by the relevant economic model, given the information they have: **rational expectations**. Formally, $\pi_t^e = \mathbb{E}_{t-1}[\pi_t]$, the mathematical expectation conditional on information available when the expectation is formed (Chapter 13). Rational expectations do not require that people be right; they require that their errors be unpredictable—not systematically biased in a way they could learn to correct.

47.1.2 The Lucas critique

Robert Lucas (1976) drew the implication for policy analysis. A macroeconometric model estimated on past data contains coefficients—the marginal propensity to consume, the slope of the Phillips curve—that reflect how people behaved *under the policy that was in place*. If policy changes, people's expectations of future policy change, and so do their decisions, so the estimated coefficients change. Using a model estimated under one policy regime to predict the effects of a different regime is therefore unreliable.

Worked Example 47.1. The Lucas critique and the Phillips curve

Suppose that in the 1960s inflation had always averaged about 2 per cent, so people expected 2 per cent. An economist estimates $\pi_t = 2 + 0.5\tilde{Y}_t$ from the data and concludes that running output 4 per cent above potential would permanently raise inflation to 4 per cent—a price worth paying. But the true relation is $\pi_t = \pi_t^e + 0.5\tilde{Y}_t$, and the “2” in the estimated equation is expected inflation, which was stable only because policy kept inflation stable. Once policy aims for 4 per cent inflation, expectations rise to 4, and holding the gap at 4 per cent requires inflation of 6, then 8, and so on. The estimated coefficient was not a structural feature of the economy; it was a by-product of the old policy.

The remedy Lucas proposed was to build models from parameters that are plausibly **structural**—invariant to policy—such as preferences, technology and the rules governing price setting, and to derive behaviour from optimization under rational expectations. Under the same banner, Thomas Sargent and Neil Wallace (1975) showed that in a model with flexible prices and rational expectations, systematic monetary policy has no effect on output (the *policy-ineffectiveness proposition*); only surprises matter. That strong result depends on flexible prices; the New Keynesian model below keeps rational expectations but restores a role for systematic policy by adding nominal rigidities.

47.2 Real Business Cycles

47.2.1 The model

Finn Kydland and Edward Prescott (1982) took Lucas's programme literally. They asked whether the Ramsey model of Chapter 44, extended to include a labour–leisure choice and random shocks to total factor productivity, could explain business cycles with no role for

money, rigid prices or market failures. In their **real business cycle** (RBC) model:

- a representative household maximizes expected lifetime utility from consumption and leisure, choosing how much to work and save;
- competitive firms produce output with a Cobb–Douglas technology $Y_t = A_t K_t^\alpha L_t^{1-\alpha}$;
- productivity A_t fluctuates randomly around its trend, with persistent shocks: $\ln A_t = \rho \ln A_{t-1} + \varepsilon_t$.

A positive productivity shock raises the marginal product of labour and hence the real wage, and raises the marginal product of capital and hence the return to saving. Because the shock is temporary, households take advantage of it: they work more while the wage is high (intertemporal substitution in labour supply) and save much of the extra income to smooth consumption (Chapter 45). Output rises by more than productivity, investment rises by much more than consumption, and hours rise—reproducing the co-movements of Chapter 46. Business cycles are the optimal response of the economy to changes in its productive opportunities, and stabilization policy can only make things worse.

47.2.2 Calibration

Kydland and Prescott introduced a new method. Rather than estimating the model statistically, they *calibrated* it: they chose parameters from long-run averages and microeconomic evidence (capital's share of income from the national accounts, the discount factor from average real interest rates, depreciation from the investment–capital ratio), measured productivity shocks by the Solow residual (Chapter 44), fed the shocks into the model, and compared the simulated fluctuations with the data. The model could account for a large share of the variance of U.S. output and for the relative volatility of consumption and investment.

47.2.3 Criticisms

1. **What are the shocks?** Large falls in TFP would mean technological regress. The Solow residual falls in recessions partly because firms use capital and labour less intensively—they hoard labour (Chapter 45)—so measured TFP is endogenous to the cycle. Utilization-adjusted TFP series are far less cyclical.
2. **Labour supply.** The model needs a large intertemporal elasticity of labour supply to generate the observed swings in hours; microeconomic estimates for individuals are small (Chapter 45). Models with an extensive margin (the decision whether to work) and indivisible labour can reconcile the two partly, but the issue remains a point of disagreement.
3. **Money matters.** Episodes such as the Volcker disinflation (Chapter 50), in which a monetary tightening was followed by a deep recession, and a large body of evidence on the effects of monetary-policy shocks, are hard to reconcile with a model in which money is neutral.
4. **Unemployment.** In the basic model all variation in hours is voluntary.

Despite these criticisms, the RBC programme permanently changed the discipline. Its method—dynamic general equilibrium with optimizing agents, rational expectations and random shocks, solved numerically and compared with data—became the common language of macroeconomics. Kydland and Prescott were awarded the Nobel prize in 2004.

47.3 Nominal Rigidities

47.3.1 Why prices are sticky

For money to affect real variables, some nominal prices or wages must adjust slowly. But why would an optimizing firm fail to adjust its price? Three strands of theory give answers.

Market power. Only firms that set prices can have sticky prices. The New Keynesian model therefore assumes *monopolistic competition*: many firms, each selling a differentiated product and facing a downward-sloping demand curve (Chapter 24). Each firm sets a price as a markup over marginal cost.

Menu costs and near-rationality. Changing prices is costly—reprinting catalogues, updating systems, the risk of antagonizing customers. For a firm with market power, the profit lost by leaving its price slightly away from the optimum is *second order* (by the envelope theorem, Chapter 6: at the optimum, the first derivative of profit with respect to price is zero), whereas the menu cost is first order. N. Gregory Mankiw (1985) and George Akerlof and Janet Yellen (1985) showed that small menu costs can therefore cause large nominal rigidity—and that the resulting fluctuations in output can have large welfare costs, because each firm ignores the effect of its price on aggregate demand (an aggregate-demand externality, Chapter 26).

Evidence. Studies of the micro data underlying consumer price indexes find that prices change frequently on average but that the median price, excluding temporary sales, lasts several months to about a year. Bils and Klenow (2004) found a median duration of about 4 to 5 months for U.S. consumer prices including sales; Nakamura and Steinsson (2008) found about 8 to 11 months excluding sales. Wages change less often than most prices, typically once a year. During the 2021–22 inflation surge, the frequency of price changes rose sharply—a sign that the degree of stickiness itself responds to the size of shocks.

47.3.2 Calvo pricing

Guillermo Calvo (1983) proposed a tractable device. In each period, each firm has a fixed probability $1 - \theta$ of being able to reset its price, independently of how long it has kept its current price. A fraction θ of firms therefore keep last period's price. The expected duration of a price is

$$\sum_{k=1}^{\infty} k (1 - \theta) \theta^{k-1} = \frac{1}{1 - \theta}.$$

With $\theta = 0.75$ in a quarterly model, prices last on average four quarters. A firm that can reset its price knows it may be stuck with it for some time, so it sets the price as a markup over a weighted average of current and *expected future* marginal costs. This forward-looking behaviour is what makes expectations central to inflation.

47.4 The New Keynesian Model

47.4.1 The three equations

Log-linearizing the optimality conditions of households and firms around a zero-inflation steady state yields three equations. Let x_t be the output gap, π_t inflation and i_t the nominal

interest rate (all per quarter, measured as deviations from the steady state), and let r_t^n be the natural real interest rate.

The dynamic IS curve. From the household's Euler equation (Chapter 45's advanced box) and the goods-market condition $Y = C$:

$$x_t = \mathbb{E}_t x_{t+1} - \frac{1}{\sigma} (i_t - \mathbb{E}_t \pi_{t+1} - r_t^n),$$

where $1/\sigma$ is the elasticity of intertemporal substitution. The current output gap depends on the expected future gap and on the gap between the real interest rate and the natural rate. Iterating forward, today's output depends on the whole expected path of future real interest rates—which is why central-bank communication about future policy matters (Chapter 50).

The New Keynesian Phillips curve. From Calvo pricing:

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t + u_t,$$

where κ is larger when prices are more flexible (θ smaller) and u_t is a cost-push shock. Iterating forward, inflation today depends on the whole expected path of future output gaps: firms raise prices now if they expect costs to be high in the future.

A monetary-policy rule. The central bank sets the nominal rate by a Taylor-type rule (Chapter 50):

$$i_t = \phi_\pi \pi_t + \phi_x x_t + v_t,$$

where v_t is a monetary-policy shock—a deviation from the rule.

Bridge. Compare Chapter 46. The dynamic IS curve replaces $\tilde{Y} = -a(r - \bar{r}) + \epsilon_d$; the New Keynesian Phillips curve replaces $\pi = \pi^e + \kappa \tilde{Y} + \epsilon_s$, with *rational* expectations of *next* period's inflation instead of adaptive expectations based on last period's; and the interest-rate rule replaces the MP curve. The structure is the same; the difference is that every coefficient is derived from preferences and technology, and expectations are forward-looking.

47.4.2 The Taylor principle

A forward-looking model raises a question the backward-looking model of Chapter 46 did not: is the equilibrium unique? Suppose people came to expect higher inflation for no fundamental reason. If the central bank responded weakly, the real interest rate would fall, stimulating demand, raising output and validating the higher inflation: expectations would be self-fulfilling. To rule this out, the central bank must raise the nominal rate by *more* than one-for-one with inflation, so that the real rate rises.

Proposition 47.2. The Taylor principle

In the three-equation model with $\phi_\pi, \phi_x \geq 0$, the rational-expectations equilibrium is unique (*determinate*) if and only if

$$\kappa(\phi_\pi - 1) + (1 - \beta)\phi_x > 0.$$

Since $1 - \beta$ is small, this essentially requires $\phi_\pi > 1$.

The proof uses the eigenvalues of the system's transition matrix and is given in Galí (2015). The **Taylor principle** has a historical interpretation: Clarida, Galí and Gertler (2000) estimated that the Federal Reserve's response to inflation before 1979 had a coefficient below one, and after 1979 well above one. The high and volatile inflation of the 1970s may partly reflect self-fulfilling expectations permitted by an insufficiently aggressive central bank.

47.4.3 The effects of a monetary-policy shock

Suppose the shock follows $v_t = \rho v_{t-1} + \varepsilon_t$ and there are no other shocks. Guess that the solution takes the form $x_t = \psi_x v_t$, $\pi_t = \psi_\pi v_t$ (the method of undetermined coefficients). Then $\mathbb{E}_t x_{t+1} = \rho \psi_x v_t$ and $\mathbb{E}_t \pi_{t+1} = \rho \psi_\pi v_t$. Substituting into the three equations and matching coefficients on v_t :

Proposition 47.3. Responses to a monetary shock

$$x_t = -(1 - \beta\rho)\Lambda v_t, \quad \pi_t = -\kappa\Lambda v_t, \quad \Lambda = \frac{1}{(1 - \beta\rho)[\sigma(1 - \rho) + \phi_x] + \kappa(\phi_\pi - \rho)}.$$

Proof. The Phillips curve gives $\psi_\pi = \beta\rho\psi_\pi + \kappa\psi_x$, so $\psi_\pi = \kappa\psi_x/(1 - \beta\rho)$. The IS curve with the rule substituted in gives $\sigma\psi_x = \sigma\rho\psi_x - (\phi_\pi\psi_\pi + \phi_x\psi_x + 1 - \rho\psi_\pi)$, i.e. $\psi_x[\sigma(1 - \rho) + \phi_x] + \psi_\pi(\phi_\pi - \rho) = -1$. Substituting for ψ_π and solving gives $\psi_x = -(1 - \beta\rho)\Lambda$ and hence $\psi_\pi = -\kappa\Lambda$. $\square$

Worked Example 47.4. A 1-point tightening

Take a quarterly calibration close to Galí (2015): $\sigma = 1$, $\beta = 0.99$, $\kappa = 0.1$, $\phi_\pi = 1.5$, $\phi_x = 0.125$, $\rho = 0.5$. A shock $\varepsilon_0 = 0.25$ is an unexpected tightening of 1 percentage point at an annual rate. Then

$$\Lambda = \frac{1}{0.505 \times (0.5 + 0.125) + 0.1 \times 1.0} = \frac{1}{0.4156} = 2.406.$$

On impact, the output gap falls by $0.505 \times 2.406 \times 0.25 = 0.30$ per cent and quarterly inflation by $0.1 \times 2.406 \times 0.25 = 0.06$ points (0.24 points at an annual rate). The nominal rate rises by $1.5 \times (-0.060) + 0.125 \times (-0.304) + 0.25 = 0.122$ (0.49 points annualized): less than the shock, because the rule responds to the fall in inflation and output. The real rate $i_0 - \mathbb{E}_0 \pi_1 = 0.122 + 0.5 \times 0.060 = 0.152$ rises by more than the nominal rate. All variables then decay at rate $\rho = 0.5$ per quarter (Figure 47.1).

The model's responses decay monotonically because the shock does; estimated responses in the data are hump-shaped and delayed, with the peak effect on output after about a year and on inflation after one to two years. Adding habit formation in consumption, adjustment costs in investment, indexation of prices to past inflation and sticky wages produces those humps—which brings us to medium-scale DSGE models.

47.4.4 The divine coincidence

If there are no cost-push shocks ($u_t = 0$), a policy that keeps inflation at zero in every period also keeps the output gap at zero: set $\pi_t = \mathbb{E}_t \pi_{t+1} = 0$ in the Phillips curve and $x_t = 0$ follows. Blanchard and Galí (2007) called this the **divine coincidence**: there is no trade-off between stabilizing inflation and stabilizing the welfare-relevant output gap. It is the formal counterpart

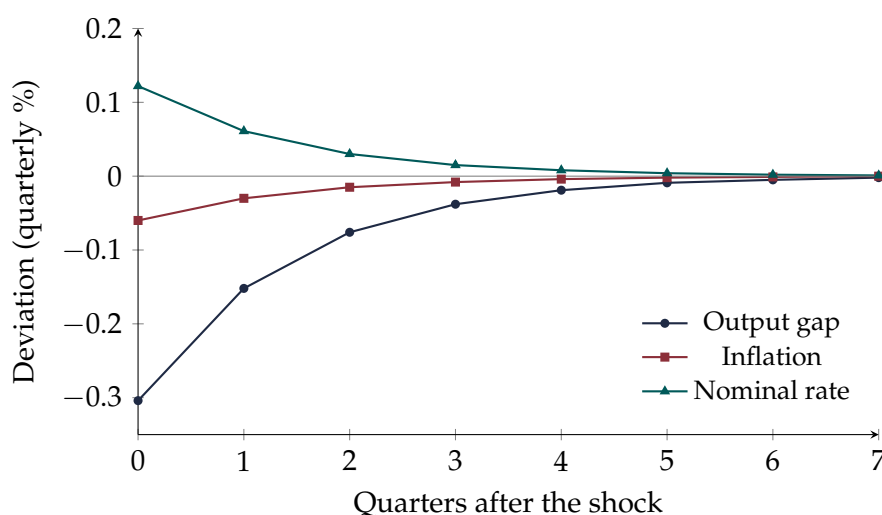


Figure 47.1: Impulse responses to a monetary-policy tightening in the three-equation New Keynesian model (Example 47.4).

of Chapter 46's observation that demand shocks create no dilemma. Cost-push shocks—sticky wages, markup shocks, energy prices—break the coincidence and force a trade-off, which is the subject of optimal-policy analysis in Chapter 50.

47.5 DSGE Models in Practice

47.5.1 From three equations to policy models

A **dynamic stochastic general equilibrium** (DSGE) model is a model in which aggregate outcomes are derived from the optimizing decisions of households, firms and policy-makers who form rational expectations, and in which random shocks drive fluctuations. The RBC model and the three-equation New Keynesian model are small DSGE models. Frank Smets and Raf Wouters (2003, 2007) estimated a medium-scale version—with sticky prices and wages, habit formation, investment adjustment costs, variable capital utilization and seven shocks—using Bayesian methods on euro-area and U.S. data, and showed that it could forecast about as well as purely statistical models. Versions of such models are used at the ECB, the Federal Reserve, the Bank of England, the Bank of Canada and many other central banks, alongside simpler models, statistical forecasting tools and judgement.

47.5.2 What DSGE models missed

Before 2008, most DSGE models used by central banks had no meaningful financial sector: households and firms borrowed at the policy rate, banks did not appear, and defaults could not occur. They were therefore silent about the forces that produced the Global Financial Crisis. The response has been to incorporate:

- **Financial frictions.** In the *financial accelerator* of Bernanke, Gertler and Gilchrist (1999), firms' borrowing costs depend on their net worth because lenders face costly verification of borrowers' outcomes (an information problem of the kind analysed in Chapter 25). A shock that lowers asset prices lowers net worth, raises the external finance premium, reduces investment, and lowers asset prices further. In the Kiyotaki–Moore (1997) model, borrowing is limited by the value of collateral, producing a similar amplification. Chap-

ter 51 develops these mechanisms and their role in banking crises.

- **Banks.** Models in which banks' own capital constrains their lending, following Gertler and Kiyotaki (2010), link the bank balance sheets of Chapter 48 to the macroeconomy.
- **Heterogeneous households.** Heterogeneous-agent New Keynesian (HANK) models, such as Kaplan, Moll and Violante (2018), replace the representative household with a distribution of households, many of them hand-to-mouth (Chapter 45). In these models monetary policy works less through intertemporal substitution and more through its effects on income, employment and asset values, and the distribution of wealth matters for aggregate outcomes.
- **The lower bound on interest rates.** When the nominal rate cannot fall below a floor, the linear model's policy rule breaks down; Chapter 51 discusses what central banks do instead.

► “DSGE models assume markets always work perfectly”

The RBC model does; the New Keynesian model does not. Nominal rigidities, monopolistic competition, financial frictions, search frictions in labour markets (Chapter 45) and incomplete markets for household risk are all standard ingredients. The more serious criticisms concern whether the particular frictions chosen are the important ones, whether rational expectations are a good approximation in unusual times, and whether log-linear approximations around a steady state can capture crises.

47.6 Government Debt Dynamics

47.6.1 The debt accumulation equation

The government's budget is a flow; its debt is a stock. Let B_t be nominal government debt at the end of period t , i the average nominal interest rate on it, and D_t the **primary deficit**—spending other than interest minus revenue. Then

$$B_t = (1 + i)B_{t-1} + D_t.$$

Divide by nominal GDP, $P_t Y_t$, which grows at rate $(1 + \pi)(1 + g) - 1$, where g is real growth. Writing $b_t = B_t / (P_t Y_t)$, $d_t = D_t / (P_t Y_t)$ and r for the real interest rate, so that $(1 + i) = (1 + r)(1 + \pi)$,

$$b_t = \frac{1 + r}{1 + g} b_{t-1} + d_t \approx b_{t-1} + (r - g) b_{t-1} + d_t.$$

Proposition 47.5. The debt-stabilizing primary balance

The debt-to-GDP ratio is constant when the primary *surplus* $s = -d$ equals

$$s^* = \frac{r - g}{1 + g} b.$$

If $r > g$, stabilizing the ratio requires a primary surplus; if $r < g$, the ratio is stable, or falling, even with a primary deficit.

Worked Example 47.6. The snowball effect

Debt is 100 per cent of GDP, the real interest rate is 3 per cent and real growth 2 per cent. The debt-stabilizing primary surplus is $0.01/1.02 \times 100 = 0.98$ per cent of GDP. With a balanced primary budget, the debt ratio grows by the factor $1.03/1.02$ each year, reaching 110 per cent after ten years—the “snowball.” If instead $r = 3\%$ and $g = 4\%$, a balanced primary budget lets the ratio *fall* to about 91 per cent in ten years.

The equation shows why the difference between interest rates and growth rates, $r - g$, dominates discussions of debt sustainability. For much of the period after 2008, interest rates in advanced economies were below growth rates, and Olivier Blanchard (2019) argued that the fiscal cost of debt was correspondingly lower than generally assumed. The rise in interest rates from 2022 narrowed or reversed that gap for many countries, raising the primary surpluses needed to stabilize debt.

47.6.2 The intertemporal budget constraint

Iterating the debt equation forward, and requiring that debt not grow forever faster than the interest rate (the government cannot run a Ponzi scheme), gives the government’s intertemporal budget constraint:

$$b_{t-1} \frac{1+r}{1+g} = \sum_{k=0}^{\infty} \left(\frac{1+g}{1+r} \right)^k s_{t+k} \quad (r > g) :$$

current debt must be matched by the present value of future primary surpluses. This is the public-sector counterpart of the household constraint in Chapter 45 and of the valuation of a firm as the present value of its future free cash flows (Chapter 31). Debt is sustainable when there is a credible path of future surpluses that satisfies it. Markets judge credibility; when they doubt it, interest rates rise, which raises $r - g$ and makes the doubt self-fulfilling—a coordination problem with multiple equilibria (Chapter 23) central to the euro-area crisis (Chapter 52).

47.7 Fiscal–Monetary Interactions**47.7.1 Seigniorage and unpleasant monetarist arithmetic**

The government’s consolidated budget, including the central bank, has one more source of finance: issuing money. The revenue from creating central-bank money is **seigniorage**. If the central bank buys government bonds with newly created reserves and remits its profits to the treasury, the private sector ends up holding money instead of bonds, and interest costs are lower.

Thomas Sargent and Neil Wallace (1981) showed that monetary and fiscal policy cannot be chosen independently. Suppose fiscal policy sets a path of primary deficits that does not satisfy the intertemporal budget constraint with bond finance alone. Then eventually the central bank must supply the missing seigniorage, whatever it wishes. If it tightens today, holding down money growth, the government issues more bonds, the debt grows, and even more seigniorage is needed later. Under this **fiscal dominance**, tight money today can mean higher inflation tomorrow—“unpleasant monetarist arithmetic.” Under *monetary dominance*, by contrast, the central bank sets its policy independently and fiscal policy must adjust its surpluses to satisfy the budget constraint.

Advanced Analysis: The fiscal theory of the price level

Write the intertemporal constraint in nominal terms: $B_{t-1}/P_t =$ present value of real primary surpluses. If the government commits to a path of surpluses that does not adjust to the level of debt (an “active” fiscal policy), then the equation can hold only if the price level P_t adjusts: the price level is determined by fiscal policy. This is the *fiscal theory of the price level* (Leeper 1991; Sims 1994; Woodford 1995; Cochrane 2023). In Leeper’s terminology, a determinate equilibrium requires one active and one passive authority: an active central bank obeying the Taylor principle with a passive fiscal authority that adjusts surpluses, or a passive central bank with an active fiscal authority. Some economists interpret the 2021–22 inflation partly as the fiscal theory predicts: large transfers financed by debt that people did not expect to be repaid by future surpluses. The theory is contested, but it sharpens a real point: inflation control requires fiscal backing.

Case: Japan: high debt, low interest rates and the central bank’s balance sheet

Japan’s general-government gross debt rose from around 60 per cent of GDP in 1990 to roughly 250 per cent by the 2020s, the highest among advanced economies. On the arithmetic of this chapter, this looks alarming. Yet for three decades Japanese government bond yields were among the lowest in the world, and there was no fiscal crisis. Why? $r - g$. Interest rates fell below nominal growth for long periods, and with deflation or very low inflation, the Bank of Japan held short rates near zero from the late 1990s and slightly below zero from 2016. With r close to g , even high debt is cheap to carry.

Who holds the debt. Most Japanese government bonds are held domestically—by banks, insurers, pension funds and, increasingly, the Bank of Japan itself, which through quantitative easing (Chapter 51) came to hold around half of all outstanding government bonds. In the consolidated public-sector balance sheet, bonds held by the central bank are replaced by bank reserves, which pay the policy rate. The effective maturity of consolidated public debt is therefore much shorter than that of the bonds, and its cost is sensitive to the policy rate.

Yield-curve control. From 2016 the Bank of Japan targeted the 10-year government bond yield near zero, committing to buy whatever quantity of bonds was needed. This is a clear example of monetary policy that interacts with debt management: the central bank capped the government’s long-term borrowing cost.

The exit. As inflation rose above 2 per cent from 2022, the Bank of Japan loosened and then, in March 2024, ended both negative interest rates and yield-curve control, raising its policy rate further in 2024 and 2025. Each increase in the policy rate raises interest paid on the reserves created by QE, reducing the Bank’s remittances to the treasury, and gradually raises the government’s borrowing costs as debt is rolled over. Japan illustrates each element of this chapter: high debt is sustainable when $r - g$ is low and holders are confident; the central bank’s balance sheet is part of the government’s financing; and the transition to higher rates tests the fiscal–monetary framework.

Summary

- Rational expectations assume that people’s forecasts are model-consistent and their errors unpredictable. The Lucas critique warns that relationships estimated under one policy may change when policy changes.
- Real business cycle models explain fluctuations as optimal responses to productivity

shocks; they introduced calibrated dynamic general equilibrium methods but struggle with the nature of the shocks, labour supply and monetary non-neutrality.

- Nominal rigidities with monopolistic competition give monetary policy real effects; small menu costs can cause large rigidities. Calvo pricing implies average price duration $1/(1 - \theta)$ and forward-looking price setting.
- The New Keynesian model consists of a dynamic IS curve, the New Keynesian Phillips curve and an interest-rate rule. Determinacy requires the Taylor principle. A monetary tightening lowers output and inflation and raises the real rate.
- Without cost-push shocks, stabilizing inflation stabilizes the output gap (the divine coincidence).
- Medium-scale DSGE models are used by central banks; post-2008 models add financial frictions, banks and heterogeneous households.
- Debt dynamics: $\Delta b \approx (r - g)b - s$. The debt-stabilizing primary surplus is $(r - g)b/(1 + g)$. Debt must equal the present value of future primary surpluses.
- Monetary and fiscal policy interact through the consolidated budget constraint; under fiscal dominance, the central bank may be forced to accommodate deficits.

Key terms

adaptive expectations; rational expectations; Lucas critique; structural parameters; policy-ineffectiveness proposition; real business cycle model; calibration; intertemporal substitution in labour supply; monopolistic competition; menu costs; near-rationality; aggregate-demand externality; Calvo pricing; dynamic IS curve; New Keynesian Phillips curve; Taylor principle; determinacy; impulse response; divine coincidence; DSGE model; financial accelerator; HANK model; primary deficit; debt dynamics; snowball effect; $r - g$; debt-stabilizing primary balance; intertemporal budget constraint of the government; seigniorage; fiscal dominance; monetary dominance; unpleasant monetarist arithmetic; fiscal theory of the price level; yield-curve control.

Exercises

Exercise 47.1 (Lucas critique). A government observes that in past recessions, temporary tax rebates raised consumption substantially. It announces that from now on it will pay a rebate in every recession. Using the Lucas critique and Chapter 45, explain why the effect of future rebates might differ from the effect of past ones.

Exercise 47.2 (RBC intuition). In an RBC model, explain why a temporary productivity improvement raises hours worked by more than a permanent one of the same size. What does this imply about the importance of the intertemporal elasticity of labour supply?

Exercise 47.3 (Calvo). In a quarterly model, what is the expected duration of prices if $\theta = 0.6$? If $\theta = 0.8$? Using the evidence cited in the text, which value seems closer to the data for prices excluding sales?

Exercise 47.4 (Menu costs). Explain, using the envelope theorem, why a monopolistically competitive firm loses little profit by leaving its price unchanged after a small shock, while a perfectly competitive firm could not leave its price unchanged at all.

Exercise 47.5 (Taylor principle). With $\beta = 0.99$, $\kappa = 0.1$, check whether the equilibrium is determinate for (a) $\phi_\pi = 1.5$, $\phi_x = 0.5$; (b) $\phi_\pi = 0.9$, $\phi_x = 0$; (c) $\phi_\pi = 0.9$, $\phi_x = 2$. Interpret (c).

Exercise 47.6 (Impulse responses). Repeat Example 47.4 with $\phi_\pi = 3$, keeping the other parameters. How do the responses of output and inflation change? Explain.

Exercise 47.7 (Divine coincidence). Explain why a cost-push shock $u_t > 0$ breaks the divine coincidence. If the central bank kept inflation at zero in the face of such a shock, what would happen to the output gap?

Exercise 47.8 (Debt dynamics). A country has debt of 130 per cent of GDP, an average real interest rate on its debt of 2.5 per cent and real growth of 0.8 per cent. (a) Find the debt-stabilizing primary surplus. (b) If it runs a primary deficit of 1 per cent of GDP, what is the debt ratio after one year? (c) What happens to your answer to (a) if the real interest rate rises to 4 per cent?

Exercise 47.9 (Intertemporal constraint). Debt is 60 per cent of GDP, $r = 3\%$, $g = 1\%$. What constant primary surplus satisfies the intertemporal budget constraint? Compare with the debt-stabilizing surplus and explain why they are the same.

Exercise 47.10 (Fiscal dominance). Explain the Sargent–Wallace argument that tight monetary policy today can lead to higher inflation in the future. Under what institutional arrangement is the argument irrelevant?

Solutions

Solution 47.1. Past rebates were unexpected and temporary; many recipients were liquidity constrained and spent them. Once rebates become a systematic rule, households anticipate them: those who can borrow may raise spending as soon as a recession starts (or before the payment arrives), and forward-looking households treat the rebates as part of their expected income over the cycle, so the response when the cheque arrives may be smaller. The coefficient estimated under the old policy is not structural.

Solution 47.2. A temporary rise in productivity raises the wage now relative to the future, so households substitute work into the present (a strong intertemporal substitution effect) with a small income effect, because lifetime wealth rises only a little. A permanent rise raises wages in every period, so there is no reason to shift work across time and the income effect offsets the substitution effect. The RBC model's ability to generate large fluctuations in hours from temporary shocks therefore depends on a large intertemporal elasticity of labour supply.

Solution 47.3. $1/(1 - 0.6) = 2.5$ quarters; $1/(1 - 0.8) = 5$ quarters. Nakamura–Steinsson's 8–11 months excluding sales is about 3–4 quarters, between the two; values around $\theta = 0.7$ – 0.75 are common.

Solution 47.4. Profit $\Pi(p)$ is maximized at p^* , where $\Pi'(p^*) = 0$. A shock moves the optimum slightly to p^{**} ; keeping p^* loses approximately $\frac{1}{2}|\Pi''|(p^{**} - p^*)^2$, which is second order in the size of the shock. A perfectly competitive firm faces a horizontal demand curve: if the market price rises and it keeps its price below, it faces unlimited demand at a loss-making price; if the market price falls and it keeps its price above, it sells nothing. Its loss is first order, so it cannot have a sticky price.

Solution 47.5. (a) $0.1 \times 0.5 + 0.01 \times 0.5 = 0.055 > 0$: determinate. (b) $0.1 \times (-0.1) + 0 = -0.01 < 0$: indeterminate. (c) $-0.01 + 0.01 \times 2 = 0.01 > 0$: determinate. A strong response to the output gap can compensate for a weak response to inflation because, in the long run, a permanent rise in inflation of 1 point raises the output gap by $(1 - \beta)/\kappa$ points through the Phillips curve, which the rule then responds to.

Solution 47.6. $\Lambda = 1/[0.505 \times 0.625 + 0.1 \times 2.5] = 1/(0.3156 + 0.25) = 1/0.5656 = 1.768$. Output gap on impact $-0.505 \times 1.768 \times 0.25 = -0.223$; inflation $-0.1 \times 1.768 \times 0.25 = -0.044$. Both fall less than before, because the rule responds more strongly to the fall in inflation by cutting the rate, offsetting more of the shock (the nominal rate now rises by $3 \times (-0.044) + 0.125 \times (-0.223) + 0.25 = 0.089$).

Solution 47.7. With $u_t > 0$, $\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t + u_t$. Setting $\pi_t = \mathbb{E}_t \pi_{t+1} = 0$ requires $x_t = -u_t/\kappa$: a negative output gap. Zero inflation is achievable only by a recession; the central bank must trade inflation stability against output stability.

Solution 47.8. (a) $s^* = (0.025 - 0.008)/1.008 \times 130 = 2.19\%$ of GDP. (b) $b = (1.025/1.008) \times 130 + 1 = 132.19 + 1 = 133.2\%$. (c) $s^* = (0.04 - 0.008)/1.008 \times 130 = 4.13\%$ of GDP: the required surplus nearly doubles.

Solution 47.9. A constant surplus s forever has present value $s \sum_{k=0}^{\infty} \left(\frac{1.01}{1.03}\right)^k = s \times 1.03/0.02 = 51.5s$ (in the timing of the formula in the text). The constraint requires $0.6 \times 1.03/1.01 = 51.5s$, so $s = 0.6 \times 0.02/1.01 = 1.19\%$ of GDP. The debt-stabilizing surplus is $(0.02/1.01) \times 60 = 1.19\%$: the same, because a constant debt ratio is one path satisfying the constraint, and with a constant surplus it is the only one.

Solution 47.10. If fiscal policy runs deficits that bond finance cannot sustain forever, the central bank will eventually have to monetize. Tightening now means more bond finance now, a higher future debt and interest bill, and therefore more future money creation—higher future inflation, and, if money demand depends on expected inflation, possibly higher inflation today. The argument is irrelevant under monetary dominance: when the fiscal authority is committed (by rules, institutions or politics) to adjusting surpluses to keep debt sustainable, and the central bank is independent and prohibited from financing the government.

Part VII

Money, Banking and Integration

Money, Commercial Banking and Payments

After studying this chapter you should be able to

1. Explain the functions of money and distinguish central-bank money, bank deposits and other money-like claims.
2. Define the main monetary aggregates and explain why definitions differ across countries.
3. Read a commercial bank's balance sheet and income statement, and compute leverage, return on equity and the effect of loan losses on capital.
4. Record a bank loan, a payment between banks and a loan repayment with journal entries, and explain how banks create and destroy money.
5. Explain why the textbook money multiplier is misleading as a causal account, and what actually constrains bank lending.
6. Derive the Baumol–Tobin demand for money and state the determinants of money demand.
7. State the quantity theory of money, assess its long-run and short-run evidence, and relate it to hyperinflation.
8. Describe how payment systems settle transactions in central-bank money, and the role of non-bank finance and stablecoins.

A motivating question

Where does money come from? Most people would answer that it is printed by the government or the central bank. Yet banknotes and coins make up only a small fraction of the money in a modern economy. In the United Kingdom, the Bank of England estimated in 2014 that about 97 per cent of the money held by households and firms consisted of bank deposits; the proportion is similar in the euro area and the United States. And most of those deposits were not created when someone deposited cash in a bank. They were created when a bank made a loan.

This chapter explains how. It builds directly on the accounting of Part IV. A bank is a firm whose balance sheet is unusual: its assets are mostly loans and securities, and its liabilities are mostly deposits that other people use as money. Once you can record a loan in a bank's journal, the mechanics of money creation, the meaning of bank capital and the fragility of banks all become concrete. Chapter 25 has already shown why banks face information problems; this chapter shows the balance sheets on which those problems play out, and Chapters 49 and 51 build on them to explain central banking and financial crises.

48.1 What Is Money?

48.1.1 Functions

Money is whatever is generally accepted in payment for goods and services and in settlement of debts. It performs three functions:

1. **Medium of exchange.** Without money, trade requires a *double coincidence of wants*: a baker who wants shoes must find a shoemaker who wants bread. Money breaks the exchange into two transactions, lowering transaction costs (Chapter 1).
2. **Unit of account.** Prices, wages, debts and financial statements are expressed in money. A common unit makes comparison possible; accounting (Part IV) presupposes it.
3. **Store of value.** Money carries purchasing power from the present into the future. It is not the only store of value—bonds, shares and houses also store value, usually with higher returns—but it is the most *liquid*: it can be spent immediately at no cost.

Inflation erodes money's function as a store of value and, when severe, as a unit of account: in hyperinflations, prices are quoted in foreign currency, and firms' financial statements lose meaning (IAS 29 requires restatement of financial statements in hyperinflationary economies).

48.1.2 Kinds of money

Commodity money has value in its own right: gold and silver coins, or cigarettes in prisoner-of-war camps.

Fiat money has no intrinsic value; it is accepted because others accept it and because the state accepts it in payment of taxes and declares it legal tender. Since the end of the Bretton Woods system in 1971, all major currencies have been fiat currencies.

A hierarchy of promises. It is useful to think of modern money as a hierarchy of IOUs:

- **Central-bank money:** banknotes, and the **reserves** (deposits) that commercial banks hold at the central bank. These are liabilities of the central bank. Together they form the **monetary base**.
- **Commercial-bank money:** deposits at commercial banks. These are liabilities of commercial banks—promises to pay central-bank money on demand. Most payments are made by transferring them.
- *Near-money:* money-market fund shares, repurchase agreements, and stablecoins—promises by non-banks, usually to pay bank deposits.

Each layer promises to pay in the layer above. A bank deposit is valuable because the bank can convert it into central-bank money on demand; a bank run is a doubt about that promise (Chapter 23, Chapter 51).

48.1.3 Monetary aggregates

Central banks publish several measures of the money stock, ranked by how liquid the included assets are:

- The *monetary base* (M0): currency in circulation plus bank reserves.
- *Narrow money* (M1): currency held by the public plus overnight (demand) deposits.

- *Broad money* (M2 in the United States; M3 in the euro area; M3 in Malaysia): M1 plus savings and time deposits, and in wider measures money-market fund shares, repos and short-term debt securities.

The exact definitions differ from country to country and change over time: in 2020 the Federal Reserve reclassified savings deposits into M1 when it removed limits on withdrawals from them, causing M1 to jump. Any analysis of money data must check the definition and its history, as Chapter 43 advised for national-accounts data.

48.2 The Commercial Bank's Accounts

48.2.1 The balance sheet

Table 48.1 shows a stylized bank. As Chapter 27 noted, banks present their balance sheets broadly in order of liquidity rather than dividing items into current and non-current.

Table 48.1: Balance sheet of a stylized commercial bank (currency units, millions)

Assets		Liabilities and equity	
Cash and reserves at central bank	50	Customer deposits	800
Government and other securities	150	Wholesale funding (interbank, repo)	100
Loans to households and firms (net)	750	Bonds issued	40
Other assets	50	<i>Total liabilities</i>	940
		Equity	60
Total assets	1,000	Total liabilities and equity	1,000

Three features distinguish it from the balance sheet of a manufacturer (Chapter 27):

1. **High leverage.** Equity is only 6 per cent of assets: assets are $1,000/60 = 16.7$ times equity. Non-financial firms typically finance a third to a half of their assets with equity.
2. **Maturity and liquidity transformation.** Most deposits can be withdrawn on demand or at short notice; most loans are long-term and cannot be sold quickly at full value. The bank borrows short and lends long.
3. **Its liabilities are money.** Customer deposits are the main means of payment in the economy.

48.2.2 The income statement and profitability

A bank earns *net interest income*—interest received on loans and securities minus interest paid on deposits and wholesale funding—plus fees and commissions, and pays operating expenses and taxes. A key expense is the *provision for credit losses*: under IFRS 9 and U.S. GAAP's current expected credit loss model, banks recognize expected losses on loans before defaults occur (Chapter 28).

Because leverage is high, a small return on assets becomes a large return on equity:

$$\text{ROE} = \text{ROA} \times \frac{\text{Assets}}{\text{Equity}}.$$

This is the DuPont identity of Chapter 31 applied to a bank. If the bank in Table 48.1 earns a return on assets of 1 per cent (10 million), its return on equity is $10/60 = 16.7$ per cent. Leverage cuts both ways.

Worked Example 48.1. Loan losses and bank capital

Suppose 4 per cent of the bank's loans (30 million) turn out to be uncollectable. The bank writes them down: *Dr Credit-loss expense 30 / Cr Loans (allowance) 30*. Assets fall to 970; liabilities are unchanged at 940; equity falls from 60 to 30—half of it is gone. The bank's depositors are still protected: assets exceed liabilities. But a further loss of 30 million would wipe out equity entirely. Equity is the buffer that absorbs losses before creditors bear them; this is why regulators impose *capital requirements* (Section 48.4 and Chapter 51).

48.2.3 Why do banks exist?

Chapter 25 provided the answer. Lenders cannot easily judge the quality of borrowers (adverse selection) or observe what borrowers do with the money (moral hazard). Douglas Diamond (1984) showed that a bank can economize on these costs by acting as a *delegated monitor*: it screens and monitors many borrowers on behalf of many depositors, and by diversifying across borrowers it can offer depositors a safe claim. Banks also provide liquidity: depositors can withdraw when they need to, while the bank funds illiquid long-term projects (Diamond and Dybvig 1983). Both functions create fragility, because the safe, liquid claims are backed by risky, illiquid assets. Diamond and Dybvig shared the 2022 Nobel prize with Ben Bernanke, whose work on the Great Depression features in the case at the end of this chapter.

48.3 How Banks Create Money**48.3.1 A loan creates a deposit**

A household borrows 20,000 from Bank A to buy a car. Bank A does not hand over banknotes, and it does not transfer someone else's deposit. It credits the household's deposit account:

Bank A			
DR	Loans to customers	20,000	
	CR Customer deposits		20,000

Both sides of Bank A's balance sheet grow by 20,000. The household's balance sheet also grows: it has a new asset (the deposit) and a new liability (the loan). Nobody's deposit has fallen. Broad money has risen by 20,000. *Bank lending creates deposits*.

48.3.2 Spending the deposit: interbank settlement

The household pays the car dealer, who banks with Bank B. The household's deposit at Bank A falls; the dealer's deposit at Bank B rises. Bank A must pay Bank B, and it does so by transferring central-bank reserves:

Bank A			
DR	Customer deposits (household)	20,000	
	CR Reserves at central bank		20,000
Bank B			
DR	Reserves at central bank	20,000	
	CR Customer deposits (dealer)		20,000

On the central bank's own books, the reserves simply move from Bank A's account to Bank B's. The deposit created by Bank A now sits at Bank B; money in the economy is still 20,000 higher than before the loan. But Bank A has lost reserves. In practice, payments flow in both directions all day, and much of the outflow is offset by inflows from customers of other banks paying Bank A's customers; only the net amount must be settled. Still, a bank that lends much faster than its competitors will tend to lose reserves, and must borrow them in the interbank market or from the central bank, or attract deposits by offering higher interest rates.

48.3.3 Repayment destroys money

When the household repays the loan out of its deposit, the entries reverse: *Dr Customer deposits / Cr Loans*. Deposits and loans both fall; money is destroyed. The money stock therefore depends on the net flow of new lending: when households and firms borrow more than they repay, money grows; when they pay down debt, as many did after 2008, money growth slows even if the central bank supplies abundant reserves.

Key idea. Commercial banks create money by lending and destroy it when loans are repaid. They are not intermediaries that wait for deposits and lend them out; lending comes first and creates the deposit. What limits lending is not the quantity of deposits but the bank's profitability, its capital and liquidity, the demand for loans, regulation, and the interest rate set by the central bank.

48.3.4 The money multiplier and why it misleads

Many textbooks present a different account. Suppose banks hold reserves equal to a fraction rr of deposits D and the public holds currency C equal to a fraction c of deposits. Then the monetary base is $B = C + R = (c + rr)D$ and money is $M = C + D = (1 + c)D$, so

$$M = \frac{1 + c}{c + rr} B \equiv m B.$$

With $c = 0.1$ and $rr = 0.1$, the **money multiplier** is $1.1/0.2 = 5.5$; with $rr = 0.02$, it is 9.2. On this account, the central bank controls the base, and the base determines the money stock through a stable multiplier: each unit of reserves supports $1/rr$ units of deposits, created by a chain of lending and redepositing.

The formula is an *identity* once c and rr are defined as observed ratios. Read as a causal theory, it is misleading for three reasons:

1. **Central banks set interest rates, not reserves.** As Chapters 46 and 49 explain, a central bank that targets a short-term interest rate must supply whatever reserves banks demand at that rate. Causation runs largely from lending and deposits to reserves, not the reverse.
2. **Reserve requirements often do not bind.** Many central banks (the Bank of England, the Bank of Canada) impose no reserve requirements; the Federal Reserve reduced its requirement to zero in March 2020. Since 2008, banks in the United States, the euro area, the United Kingdom and Japan have held reserves far in excess of any requirement.
3. **The ratios are not stable.** When the Federal Reserve roughly quintupled the monetary base between 2008 and 2014 through quantitative easing (Chapter 51), broad money grew only modestly: the multiplier collapsed. The multiplier moved because banks chose to hold more reserves and lend less, not because of a mechanical rule.

► “A bank with 100 million of reserves can lend only a fixed multiple of 100 million”

This confuses accounting, behaviour and institutions. A bank’s lending is limited by its capital (a regulatory and economic constraint), by its expected ability to meet outflows (a liquidity constraint, which it can manage by borrowing reserves at the market rate), and by the profitability of loans at the prevailing interest rate. The central bank supplies reserves to the banking system as a whole at its policy rate. The quantity of reserves a bank happens to hold is therefore not the binding constraint in most modern systems.

The multiplier framework remains useful for one purpose: as an *ex post* decomposition of changes in money into changes in the base and in the public’s and banks’ behaviour. The case at the end of the chapter uses it this way.

48.4 Bank Capital and Liquidity

48.4.1 Capital

A bank’s **capital** in the regulatory sense is, roughly, its equity plus certain loss-absorbing instruments. Capital is *not* a pile of cash held in reserve; it is a source of funding, on the right-hand side of the balance sheet, like deposits. The difference is that equity holders absorb losses first and cannot demand repayment. A bank with more capital can suffer larger losses before becoming insolvent.

Regulators require banks to hold capital in proportion to their *risk-weighted assets*: each asset is multiplied by a weight that reflects its riskiness (zero for most domestic government bonds, higher for corporate loans), and capital must exceed a percentage of the total. Under the international Basel III framework (Chapter 51), banks must hold common equity of at least 4.5 per cent of risk-weighted assets plus a conservation buffer of 2.5 per cent, with further buffers for systemically important banks, and a minimum leverage ratio of equity to unweighted exposures. Capital requirements are a response to the moral hazard created by deposit insurance and implicit guarantees (Chapter 25): if depositors do not monitor a bank because they are insured, someone else must limit the bank’s risk-taking.

48.4.2 Liquidity

A bank can be solvent—assets worth more than liabilities—and still fail if it cannot meet withdrawals because its assets cannot be sold quickly without large losses. This is a *liquidity* problem. Banks manage liquidity by holding reserves and securities that can be sold or pledged as collateral, by matching the maturity of some assets and liabilities, and by relying on the central bank as lender of last resort (Chapter 51). Basel III added liquidity requirements: a *liquidity coverage ratio* requiring enough high-quality liquid assets to survive a 30-day stress, and a *net stable funding ratio* requiring stable funding for illiquid assets.

Accounting bridge. Chapter 30 showed how Silicon Valley Bank held long-term bonds classified as held to maturity, carried at amortized cost, while their market value fell as interest rates rose. That was a liquidity and solvency problem hidden by a measurement choice: when depositors withdrew, the bank had to sell the bonds and realize the losses, revealing that its equity was much smaller than reported. The classification choices of Part IV are not technicalities for a bank; they determine how quickly losses become visible.

48.5 The Demand for Money

Why hold money, which pays little or no interest, rather than bonds? Because money is needed for transactions, as a precaution against unexpected payments, and as a safe, liquid asset in a portfolio.

48.5.1 The Baumol–Tobin model

William Baumol (1952) and James Tobin (1956) modelled the transactions demand as an inventory problem. A household spends Y per year at an even rate. It keeps its wealth in bonds paying interest i and makes n trips a year to convert bonds into money, each trip costing b (time or a fee). After each trip it holds Y/n in money, which it spends down to zero, so its average money holding is $Y/(2n)$. Its total cost is the interest forgone plus the trip costs:

$$\text{Cost}(n) = i \frac{Y}{2n} + b n.$$

Minimizing: $-iY/(2n^2) + b = 0$, so

$$n^* = \sqrt{\frac{iY}{2b}}, \quad M^d = \frac{Y}{2n^*} = \sqrt{\frac{bY}{2i}}.$$

Worked Example 48.2. The square-root rule

A household spends 60,000 a year, bonds pay 4 per cent, and each conversion costs 2. Then $n^* = \sqrt{0.04 \times 60,000/4} = \sqrt{600} = 24.5$ trips a year and average money holdings are $60,000/(2 \times 24.5) = 1,225$. If the interest rate doubles to 8 per cent, money holdings fall by a factor of $\sqrt{2}$ to 866. The income elasticity of money demand is $1/2$ and the interest elasticity $-1/2$.

The model predicts *economies of scale* in money holding and a negative response to interest rates. Its specific numbers are less important than its logic: money demand depends positively on transactions (income) and negatively on the opportunity cost of holding money—the gap between the interest rate on bonds and the rate paid on money. Financial innovation that lowers b —ATMs, online banking, card and instant payments—lowers money demand. In general form, real money demand is $M^d/P = L(i, Y)$, the function used in the LM curve of Chapter 46.

48.6 The Quantity Theory of Money

48.6.1 From identity to theory

Chapter 1 introduced the quantity equation $MV \equiv PY$, where V is velocity, defined as PY/M . In growth rates (Chapter 4),

$$g_M + g_V \equiv \pi + g_Y.$$

The **quantity theory of money** adds two behavioural assumptions: velocity is stable ($g_V \approx 0$), and real output is determined by real factors—technology, capital, labour—independently of money (*long-run neutrality*). Then

$$\pi = g_M - g_Y:$$

inflation equals money growth minus real output growth. If money grows at 7 per cent and real output at 3 per cent, inflation is 4 per cent. As Milton Friedman put it, “inflation is always and everywhere a monetary phenomenon.”

48.6.2 Evidence

The long run and high inflation. Across countries and over long periods, the quantity theory fits well. Countries with money growth of 50 per cent a year have inflation near 50 per cent; countries with money growth of 5 per cent have low inflation. The relationship is especially clear when inflation is high, because variations in money growth then swamp variations in velocity and real growth.

The short run and low inflation. At low inflation and over short periods, the relationship is weak. Velocity fluctuates with interest rates and financial innovation; U.S. M2 velocity fell sharply after 2008 and again in 2020 as money holdings surged while spending did not. The money stock in modern systems is largely determined by bank lending, which responds to the economy, so money is as much a consequence of economic activity as a cause of it. This is why central banks no longer target money growth (Chapter 46's advanced box and Chapter 50), although many, including the ECB, still monitor monetary and credit developments as a cross-check on inflation risks.

48.6.3 Hyperinflation

A **hyperinflation**, conventionally defined as inflation above 50 per cent a month, is almost always the result of a government financing large deficits by creating money when it cannot borrow or tax—the fiscal dominance of Chapter 47 in its extreme form. Germany in 1922–23, Hungary in 1945–46 (the most extreme recorded case), Zimbabwe in 2007–08 and Venezuela from 2016 are the classic cases. As inflation accelerates, people reduce their money holdings as quickly as possible, so velocity rises and prices rise even faster than money. Seigniorage revenue can then *fall* as money growth rises, because the tax base—real money holdings—shrinks. Hyperinflations end when the fiscal problem is solved and the central bank is freed from financing the government, often with a new currency and a credible commitment, as in Germany's currency reform of late 1923.

48.7 Payments

48.7.1 Settlement in central-bank money

When a customer of Bank A pays a customer of Bank B, the two banks must settle. In most systems, final settlement takes place in central-bank money: reserves move between the banks' accounts at the central bank, as in the journal entries above. Large-value payments are usually settled in **real-time gross settlement** (RTGS) systems, which settle each payment individually and immediately: Fedwire in the United States, T2 (which replaced TARGET2 in 2023) in the euro area, CHAPS in the United Kingdom, RENTAS in Malaysia. Retail payments—card payments, direct debits—are often cleared and *netted* over a period, with only net positions settled in central-bank money, which saves reserves but creates credit exposures between banks until settlement.

Central banks supply intraday credit to banks, usually against collateral, so that payments do not stall because a bank is waiting for incoming funds. Payment systems are therefore closely linked to central-bank operations (Chapter 49).

48.7.2 Fast payments and new forms of money

Many countries have introduced instant retail payment systems that settle payments between accounts at different banks within seconds, around the clock: India's UPI, Brazil's Pix (launched 2020), Malaysia's DuitNow, the euro area's instant payments, and the Federal Reserve's Fed-Now (launched 2023). These systems compete with cards and cash and have greatly reduced the use of cash in some countries.

Alongside them, privately issued *stablecoins*—crypto-assets that promise to maintain a fixed value against a currency, usually by holding short-term government securities and bank deposits—grew rapidly in the 2020s. A stablecoin is a promise by a non-bank to pay money; if its backing assets lose value or cannot be sold quickly, holders may run, as happened with the algorithmic stablecoin TerraUSD in 2022. The European Union's Markets in Crypto-Assets regulation (MiCA) and, in 2025, U.S. federal legislation established regulatory frameworks for stablecoin issuers, requiring high-quality reserves. Many central banks are also studying or piloting central bank digital currencies (Chapter 51).

48.7.3 Non-bank finance

A large share of credit and liquidity transformation takes place outside banks, in what is often called *non-bank financial intermediation* (or “shadow banking”): money-market funds, investment funds, securities dealers, finance companies, and the repo market in which securities are sold with an agreement to repurchase them—in effect, collateralized short-term loans. These institutions perform bank-like functions without bank-like regulation or direct access to the central bank, and several of the most severe episodes of the 2007–09 crisis and of March 2020 began in them (Chapter 51).

Case: The Great Depression: when the money supply collapsed

Between 1929 and 1933, U.S. real GDP fell by about a quarter, the price level by about a quarter, and unemployment rose to about 25 per cent. In *A Monetary History of the United States* (1963), Milton Friedman and Anna Schwartz argued that the depth of the Depression was due largely to a collapse of the money stock by about a third, which the Federal Reserve failed to prevent.

Bank failures. Waves of bank runs began in late 1930. About 9,000 banks suspended operations between 1930 and 1933. There was no federal deposit insurance: depositors in a failed bank could lose their savings, so the rational response to rumours was to withdraw (the coordination game of Chapter 23).

The multiplier as accounting. The monetary base actually *rose* over the period, by roughly a fifth. What collapsed was the multiplier. Frightened depositors converted deposits into currency, so the currency–deposit ratio c roughly doubled, from about 0.09 to about 0.20; banks, fearing runs, held more reserves, raising the reserve–deposit ratio from about 0.08 to about 0.13. With these approximate values,

$$m_{1929} = \frac{1.09}{0.09 + 0.077} = 6.5, \quad m_{1933} = \frac{1.20}{0.20 + 0.125} = 3.7.$$

The multiplier fell by about 43 per cent; with the base up about 18 per cent, money fell by about a third ($0.57 \times 1.18 \approx 0.67$). The identity does not say *why* the ratios changed; that required the behavioural story of runs and precautionary reserve holding.

The credit channel. Ben Bernanke (1983) added that bank failures destroyed the information capital that banks had accumulated about their borrowers (Chapter 25). Small firms and farmers who depended on their banks could not obtain credit elsewhere, so the

real economy suffered beyond what the fall in money alone would imply. Falling prices raised the real value of debts—Irving Fisher’s *debt deflation*—pushing more borrowers into default and more banks into insolvency.

In the accounts. Every step was a balance-sheet event: loan defaults reduced bank equity; withdrawals reduced deposits and reserves; forced sales of securities lowered their prices, imposing losses on other banks that held them. A downward spiral of the kind the financial accelerator formalizes (Chapter 47 and Chapter 51).

The response. The banking crisis ended after the bank holiday of March 1933, when banks were closed, inspected and reopened with government backing; federal deposit insurance began in 1934; and the United States left the gold standard in 1933, allowing monetary expansion. Money and output then grew rapidly. The lesson that central banks must prevent collapses of the banking system and the money stock shaped the response to 2008 and 2020 (Chapter 51)—Bernanke himself was chairman of the Federal Reserve in 2008.

Summary

- Money is a medium of exchange, unit of account and store of value. Modern money is a hierarchy of promises: central-bank money (notes and reserves), commercial-bank deposits and near-money.
- Monetary aggregates range from the monetary base to broad money; definitions differ and change.
- Banks are highly leveraged, transform maturities and liquidity, and issue liabilities that serve as money. ROE equals ROA times leverage; loan losses reduce equity directly.
- A bank loan creates a deposit; payments move deposits between banks and are settled in reserves; repayment destroys money.
- The money multiplier is an identity, not a causal theory; central banks supply reserves at their policy rate, and lending is constrained by capital, liquidity, profitability and loan demand.
- Capital absorbs losses; liquidity allows a bank to meet outflows. Basel III imposes minimum capital and liquidity ratios.
- Money demand rises with income and falls with the opportunity cost of holding money (Baumol–Tobin: $M^d = \sqrt{bY/2i}$).
- The quantity theory $\pi = g_M - g_Y$ fits long-run and high-inflation data well but not short-run low-inflation data. Hyperinflations result from monetary financing of deficits.
- Payments settle in central-bank money through RTGS and netting systems; fast payments, stablecoins and non-bank finance are changing the payments landscape.

Key terms

medium of exchange; unit of account; store of value; double coincidence of wants; liquidity; commodity money; fiat money; central-bank money; reserves; monetary base; commercial-bank money; M1; broad money; leverage; maturity transformation; net interest income; provision for credit losses; delegated monitoring; money creation; interbank settlement; money multiplier; currency–deposit ratio; reserve–deposit ratio; capital; risk-weighted assets; liquidity coverage ratio; net stable funding ratio;

Baumol–Tobin model; transactions demand; velocity; quantity theory of money; long-run neutrality; hyperinflation; real-time gross settlement; netting; fast payments; stablecoin; non-bank financial intermediation; repo; debt deflation.

Exercises

Exercise 48.1 (Leverage and ROE). A bank has assets of 500, deposits of 440, other liabilities of 25 and equity of 35. Its return on assets is 0.9 per cent. (a) Compute leverage and ROE. (b) If it raises equity to 50 by issuing shares (using the proceeds to repay 15 of its other liabilities), what is ROE if ROA is unchanged? (c) What loss on its assets would make the bank insolvent in each case?

Exercise 48.2 (Journal entries). Record the following in the books of the banks involved: (a) Bank X lends 5,000 to a firm; (b) the firm pays 5,000 to a supplier who banks with Bank Y; (c) the supplier withdraws 1,000 in cash from Bank Y; (d) the firm later repays 2,000 of its loan (having received a payment of 2,000 from a customer of Bank Y). What is the net change in broad money after all four transactions?

Exercise 48.3 (Money multiplier). The currency–deposit ratio is 0.15 and the reserve–deposit ratio 0.05. (a) Compute the multiplier. (b) If the central bank raises the base by 100 and the ratios are unchanged, what happens to money? (c) Why might the ratios not be unchanged? Refer to the experience after 2008.

Exercise 48.4 (Diagnose the error). “When the central bank buys bonds from banks, it gives them more reserves, which they will lend out, so the money supply will rise by a multiple of the purchase.” Identify the accounting, behavioural and institutional components of this claim and correct it.

Exercise 48.5 (Baumol–Tobin). A firm makes payments of 2,400,000 a year at an even rate. Each transfer from its investment account costs 50, and the investment account pays 5 per cent. (a) Find the optimal number of transfers and average money holdings. (b) By what percentage do money holdings change if the transfer cost falls to 12.5?

Exercise 48.6 (Quantity theory). Money grows at 12 per cent a year, real output at 3 per cent and velocity is constant. (a) What is inflation? (b) If financial innovation raises velocity by 2 per cent a year, what is inflation? (c) Why does the quantity theory predict inflation poorly in the short run in low-inflation economies?

Exercise 48.7 (Hyperinflation). Explain why seigniorage revenue can fall as money growth rises during a hyperinflation. Relate your answer to the concept of a tax base.

Exercise 48.8 (Capital). A bank has risk-weighted assets of 600 and total assets of 1,000. Its common equity is 45. (a) Compute its common equity ratio (to RWA) and its leverage ratio (equity to total assets). (b) Does it meet a requirement of 4.5 per cent plus a 2.5 per cent buffer? (c) If it makes a new corporate loan of 50 with a risk weight of 100 per cent, what are the new ratios?

Exercise 48.9 (Payments). Explain why a payment between two customers of the same bank requires no settlement in central-bank money, while a payment between customers of different banks does. What problem does intraday central-bank credit solve?

Exercise 48.10 (Great Depression). Using the multiplier identity, explain why the U.S. money stock could fall by a third between 1929 and 1933 while the monetary base rose. Which parts of the story are accounting and which are behaviour?

Solutions

Solution 48.1. (a) $\text{Leverage} = 500/35 = 14.3$; $\text{ROE} = 0.9\% \times 14.3 = 12.9\%$ (profit 4.5, ROE 4.5/35). (b) Assets unchanged at 500; leverage $500/50 = 10$; $\text{ROE} = 9\%$. (c) A loss of 35 (7 per cent of assets) wipes out equity in the first case; 50 (10 per cent) in the second. More capital lowers ROE but makes the bank safer.

Solution 48.2. (a) Bank X: Dr Loans 5,000 / Cr Deposits (firm) 5,000. Money +5,000. (b) Bank X: Dr Deposits (firm) 5,000 / Cr Reserves 5,000; Bank Y: Dr Reserves 5,000 / Cr Deposits (supplier) 5,000. Money unchanged. (c) Bank Y: Dr Deposits 1,000 / Cr Cash 1,000. Broad money unchanged (a deposit becomes currency held by the public; both are money), though the bank's reserves fall. (d) Receipt: Bank Y: Dr Deposits (customer) 2,000 / Cr Reserves 2,000; Bank X: Dr Reserves 2,000 / Cr Deposits (firm) 2,000. Repayment at Bank X: Dr Deposits (firm) 2,000 / Cr Loans 2,000. Money –2,000. Net change in broad money: +3,000, equal to the net new lending.

Solution 48.3. (a) $m = 1.15/0.20 = 5.75$. (b) Money rises by 575. (c) After 2008, banks held reserves far above requirements because reserves became safe, liquid and (from October 2008 in the United States) interest-bearing, loan demand was weak, and banks were repairing their capital. The reserve ratio rose enormously, so the multiplier fell and money rose far less than the base.

Solution 48.4. Accounting: when the central bank buys bonds from banks, banks' reserves rise and their bond holdings fall; deposits are unchanged (if it buys from non-banks, deposits and reserves both rise). Behavioural: whether banks then lend more depends on loan demand, profitability and capital, not on the reserves themselves. Institutional: banks cannot "lend out" reserves to households, who do not hold accounts at the central bank; reserves circulate only among banks, and their total is set by the central bank. The corrected statement: asset purchases raise reserves; they may raise broad money directly (if bought from non-banks) and may encourage lending by lowering long-term interest rates (Chapter 51), but there is no mechanical multiple.

Solution 48.5. (a) $n^* = \sqrt{0.05 \times 2,400,000 / (2 \times 50)} = \sqrt{1,200} = 34.6$ transfers; $M = 2,400,000 / (2 \times 34.6) = 34,641$. (b) M is proportional to $\sqrt{b}$; a fall in b to a quarter halves money holdings, to 17,321: a 50 per cent fall.

Solution 48.6. (a) $12 - 3 = 9\%$. (b) $12 + 2 - 3 = 11\%$. (c) Velocity fluctuates with interest rates and innovation; the money stock is largely endogenous to lending and activity; and at low inflation these variations are as large as the variations in money growth, so money growth has little predictive power.

Solution 48.7. Seigniorage in real terms is approximately (money growth) $\times$ (real money balances). Money growth is the tax rate; real balances are the tax base. As inflation rises, people economize on money holdings, so real balances fall. Beyond some rate, the base shrinks faster than the rate rises and revenue falls, as with a Laffer curve for any tax.

Solution 48.8. (a) $45/600 = 7.5\%$ of RWA; leverage ratio $45/1,000 = 4.5\%$. (b) Yes: 7.5 per cent exceeds 7.0 per cent. (c) RWA = 650, assets = 1,050: ratios $45/650 = 6.9\%$ and $45/1,050 = 4.3\%$. The bank now falls just short of the 7 per cent combined requirement and must raise capital or reduce risk-weighted assets.

Solution 48.9. Within one bank, the payment simply debits one customer's deposit and credits another's; the bank's total liabilities are unchanged. Between banks, the paying bank's liabilities fall and the receiving bank's rise, so the paying bank must transfer an asset the receiving bank accepts—central-bank reserves. Intraday credit lets banks make payments before receiving incoming ones, preventing gridlock in which every bank waits for others to pay first.

Solution 48.10. $M = mB$ with $m = (1 + c)/(c + rr)$. The rise in c (depositors converting deposits into currency) and in rr (banks hoarding reserves) reduced m by more than the rise in B , so M fell. The identity is accounting; the reasons for the changes in c and rr —fear of bank failures without deposit insurance, precautionary liquidity—are behaviour; the absence of deposit insurance and of adequate lender-of-last-resort action are institutional.

Central-Bank Operations and Monetary Transmission

After studying this chapter you should be able to

1. Read a central bank's balance sheet and record open-market operations, currency issuance and government payments.
2. Explain how a central bank controls an overnight interest rate in a corridor system and in a floor system.
3. Describe open-market operations, repos, standing facilities and collateral frameworks.
4. Explain how the policy rate passes through to money-market rates and along the yield curve, using the expectations hypothesis and the term premium.
5. Trace a policy action through the interest-rate, credit, asset-price, exchange-rate and expectations channels to spending and inflation.
6. Quantify the effect of a rate increase on the financial statements of households, firms and banks.
7. Analyse the 2022–23 monetary tightening in the United States and compare it with other central banks.

A motivating question

On 16 March 2022 the Federal Open Market Committee announced that it was raising its target range for the federal funds rate from 0–0.25 per cent to 0.25–0.50 per cent. Over the next sixteen months it raised the range ten more times, to 5.25–5.50 per cent. What exactly did the Federal Reserve *do* on each of those days? Nobody at the Fed telephoned banks and told them to charge more. The Fed did not sell mountains of bonds. Yet within minutes, the interest rates at which banks lend to each other overnight moved up by almost exactly the announced amount, and within weeks, mortgage rates, corporate bond yields and the dollar had all moved.

This chapter explains the plumbing and the transmission. The plumbing is accounting: a central bank is a bank, with a balance sheet, and its policy is implemented through transactions recorded on that balance sheet and on the balance sheets of commercial banks (Chapter 48). The transmission is economics: how a change in one overnight rate alters the decisions of households, firms and banks that we modelled in Chapters 45–20.

49.1 The Central Bank's Balance Sheet

49.1.1 A stylized balance sheet

The composition of the balance sheet reveals a great deal about a country's monetary framework:

Table 49.1: Stylized central-bank balance sheet

Assets	Liabilities and equity
Gold and foreign-exchange reserves	Banknotes (currency in circulation)
Domestic securities (mainly government bonds)	Reserves (deposits of commercial banks)
Loans to banks (repos, refinancing operations)	Government deposits
Other assets	Reverse repos and other liabilities
	Capital and revaluation accounts

- The **Federal Reserve's** assets are dominated by U.S. Treasury securities and agency mortgage-backed securities acquired through quantitative easing (Chapter 51); its largest liabilities are currency and bank reserves. Before 2008 its balance sheet was under \$1 trillion; it peaked near \$9 trillion in 2022.
- The **Eurosystem** (the ECB and the national central banks of the euro area, including the Bundesbank) historically provided most reserves by lending to banks against collateral; from 2015 its asset purchases made securities the largest item. Large loans to banks in 2020–21 (targeted longer-term refinancing operations) were repaid in 2022–24.
- **Bank Negara Malaysia's** and the **People's Bank of China's** assets are dominated by foreign-exchange reserves, accumulated through intervention in the currency market (Chapter 52). The PBoC has long used *reserve requirement ratios*—the fraction of deposits banks must hold as reserves—as an active policy tool, cutting them repeatedly in the 2010s and 2020s to release liquidity, alongside interest-rate instruments.

49.1.2 Transactions on the balance sheet

Open-market purchase. When the central bank buys a government bond worth 100 from a commercial bank, it pays by crediting the bank's reserve account:

Central bank			
DR	Government securities	100	
	CR Reserves (bank's deposit)		100
Commercial bank			
DR	Reserves at central bank	100	
	CR Government securities		100

The central bank creates reserves “out of nothing”—by crediting an account—just as a commercial bank creates a deposit when it lends (Chapter 48). A sale reverses the entries and drains reserves.

Currency demand. When households withdraw banknotes, commercial banks obtain them from the central bank by giving up reserves: *central bank: Dr Reserves / Cr Banknotes*. Total central-bank liabilities are unchanged, but reserves fall. Before holidays, currency demand rises and reserves fall; the central bank offsets this if it wants to keep reserves stable.

Government payments. Most governments bank with their central bank (in the United States, the Treasury General Account). When taxpayers pay taxes, their bank deposits fall, their banks' reserves fall and the government's deposit at the central bank rises: *central bank: Dr Reserves / Cr Government deposits*. When the government spends, the reverse happens. Such *autonomous factors*—movements in currency and government deposits that the central bank

does not control—change the quantity of reserves every day, and the central bank must forecast and offset them when reserves are scarce.

49.2 Implementing the Policy Rate

49.2.1 The market for reserves

Commercial banks need reserves to settle payments with each other (Chapter 48) and, where they apply, to meet reserve requirements. Banks with more reserves than they need lend them overnight to banks with fewer, in the *interbank* or *money market*. The overnight rate on these loans—the federal funds rate in the United States, the euro short-term rate (€STR) in the euro area, SONIA in the United Kingdom, the overnight interbank rate in Malaysia—is the rate the central bank aims to control.

The central bank sets two **administered rates**:

- a **deposit rate** (or interest on reserves), paid on reserves banks hold at the central bank. No bank will lend overnight in the market at less than this, since it can earn it risk-free by leaving the money at the central bank;
- a **lending rate**, at which banks can borrow overnight from a **standing facility** against collateral (the Fed's discount window, the ECB's marginal lending facility). No bank will pay more than this in the market.

These two rates form a **corridor**. The demand curve for reserves slopes down between them: at high market rates, banks economize on reserves; at low rates, they hold more. Above the lending rate, demand is capped because banks can borrow unlimited amounts from the facility; at the deposit rate, the curve becomes flat, because banks will hold any quantity of reserves that earns the deposit rate.

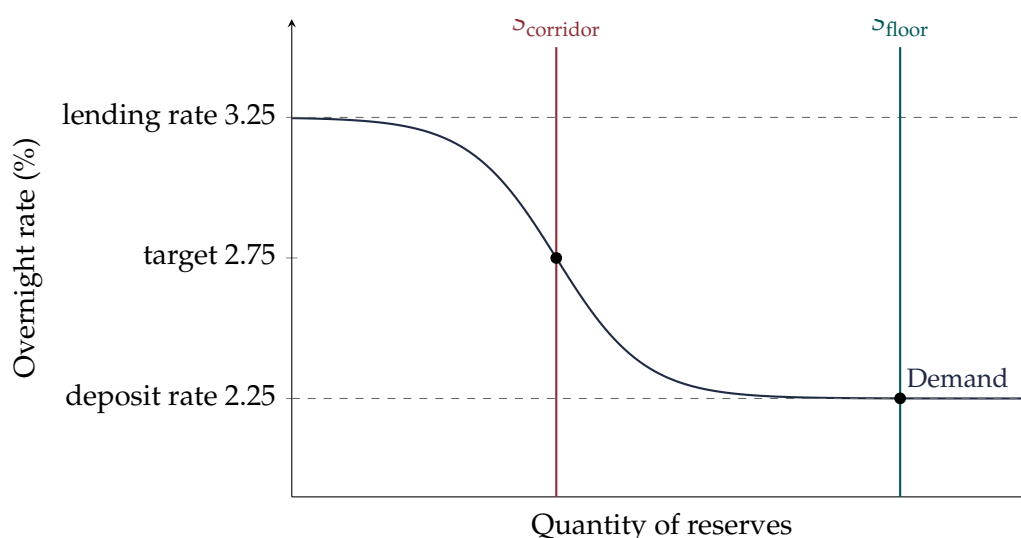


Figure 49.1: The market for reserves. In a corridor system (S_{corridor}) the central bank keeps reserves scarce and supplies just the quantity at which demand crosses its target. In a floor system (S_{floor}) reserves are abundant, demand is flat, and the market rate settles at the deposit rate.

49.2.2 The corridor system

Before 2008, most central banks operated a **corridor system** with scarce reserves. The central bank estimated the demand for reserves and, through daily or weekly open-market operations, supplied just the quantity at which the demand curve crossed its target rate, usually in the middle of the corridor (Figure 49.1). To raise the rate, it drained reserves; to lower it, it added them. The Fed's desk in New York conducted repo operations almost every morning to offset autonomous factors; the ECB supplied reserves mainly through weekly main refinancing operations, lending to banks at its main refinancing rate.

A corridor system requires precise forecasting of reserve demand and autonomous factors, and the market rate fluctuates when forecasts are wrong. Its virtue is a small central-bank balance sheet.

49.2.3 The floor system

When central banks bought large quantities of securities after 2008 (Chapter 51), reserves became abundant and the market rate fell to the deposit rate: the supply curve moved to the flat part of the demand curve. The central bank now controls the market rate by setting the deposit rate itself—a **floor system**. To raise rates, it simply raises the interest it pays on reserves; no change in the quantity of reserves is needed. This separates the *price* decision (the policy rate) from the *quantity* decision (the size of the balance sheet), allowing the central bank to buy or sell assets for other purposes—for example, to lower long-term rates or to stabilize markets—without losing control of short-term rates.

Worked Example 49.1. Raising rates in a floor system

The Federal Reserve announces a rise of 0.25 points in its target range, from 4.75–5.00 to 5.00–5.25 per cent. It implements the decision by setting, effective the next day, the interest rate on reserve balances (IORB) at 5.15 per cent (up from 4.90), the overnight reverse repo (ON RRP) offering rate at 5.05 per cent (up from 4.80), and the minimum bid rate at its standing repo facility at 5.25 per cent. No bank will lend reserves for less than it earns at the Fed, and money-market funds, which cannot earn IORB, will not lend for less than the ON RRP rate. The effective federal funds rate moves up by about 0.25 points, to around 5.08 per cent. The balance sheet need not change at all.

The Federal Reserve formally adopted an “ample reserves” framework in 2019, and the Bank of England, the Bank of Japan and, from 2015 to 2024, the ECB have operated floors. Two practical complications matter:

1. **Leakage.** Some institutions that lend in money markets cannot hold reserves or earn the deposit rate (in the United States, government-sponsored enterprises and money-market funds). They lend below the deposit rate, pulling market rates below the floor. The Fed's ON RRP facility, open to money-market funds, sets a sub-floor.
2. **How ample is ample?** As a central bank shrinks its balance sheet (quantitative tightening, Chapter 51), reserves may become scarce again, and money-market rates can spike—as U.S. repo rates did in September 2019. The Fed created a standing repo facility in 2021 as a ceiling tool, and announced in late 2025 that it would stop reducing its securities holdings to keep reserves ample.

In March 2024 the ECB announced a new operational framework in which reserves would be supplied in response to banks' demand through its lending operations, with the spread between its main refinancing rate and its deposit facility rate narrowed to 0.15 points from

September 2024—a “demand-driven floor” intended to keep rates near the deposit rate with a smaller balance sheet than a pure floor.

49.3 Operations, Repos and Collateral

49.3.1 Repos

A **repurchase agreement** (repo) is a sale of securities with an agreement to buy them back at a slightly higher price on a future date. Economically it is a collateralized loan: the seller receives cash and pays interest (the difference between the prices); the buyer holds the securities as collateral. A central-bank repo *adds* reserves temporarily; a *reverse* repo drains them. Because repos are short and reversible, they are the main instrument for fine-tuning reserves in a corridor system; outright purchases change reserves permanently.

49.3.2 Collateral and haircuts

Central banks lend only against collateral, and they apply a *haircut*: a bank pledging a bond worth 100 might receive 95. Haircuts protect the central bank against a fall in the collateral’s value (Chapter 25’s logic of collateral as a response to information problems). The range of collateral the central bank accepts—its *collateral framework*—is a powerful policy lever. The Eurosystem has always accepted a wide range of collateral, including bank loans; in crises central banks widen their frameworks, and deciding how far to go is a central question of lender-of-last-resort policy (Chapter 51).

49.4 From the Policy Rate to Market Rates

49.4.1 Money markets

Changes in the overnight rate pass almost completely and immediately to other overnight rates: unsecured interbank rates, repo rates (the Secured Overnight Financing Rate, SOFR, in the United States) and the rates paid by money-market funds. These overnight benchmarks, rather than the discredited LIBOR (whose manipulation scandal led to its replacement between 2021 and 2023), are now the reference rates for trillions of dollars of loans and derivatives. A floating-rate corporate loan priced at “SOFR plus 2 per cent” reprices directly when the Fed moves.

49.4.2 The yield curve

Longer-term interest rates depend on the *expected path* of short rates. Chapter 33 introduced the two components: under the **expectations hypothesis**, an n -year yield equals (approximately) the average of the short rates expected over the next n years; in addition, investors typically demand a **term premium** TP_n for the risk of holding long bonds, whose prices fall when rates rise unexpectedly. For monetary policy the relation to remember is

$$i_{n,t} \approx \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{E}_t i_{1,t+k} + TP_{n,t},$$

because the central bank sets the first term of the average and influences the expected later ones.

Worked Example 49.2. Reading the yield curve

The one-year rate is 4 per cent, and markets expect one-year rates of 3.5 per cent next year and 3 per cent the year after. Under the pure expectations hypothesis, the three-year yield is $(4 + 3.5 + 3)/3 = 3.5$ per cent (exactly, $(1.04 \times 1.035 \times 1.03)^{1/3} - 1 = 3.50$ per cent). With a term premium of 0.3 points, it is 3.8 per cent. Conversely, if the one-year yield is 4 per cent and the two-year yield 3.8 per cent, the *implied forward rate* for the second year is $1.038^2/1.04 - 1 = 3.60$ per cent: markets expect rates to fall (unless the term premium is negative).

The expectations component explains why central banks care so much about communication (Chapter 50): by shaping expectations of future policy rates, they can move long-term rates—which matter most for mortgages and investment—without changing today’s rate. An *inverted* yield curve, with long rates below short rates, signals that markets expect rates to fall, often because they expect a recession; in the United States, inversions have preceded most recessions since the 1960s, although the long inversion of 2022–24 was not followed by a recession within the usual interval.

49.4.3 Pass-through to bank rates

Banks’ lending and deposit rates respond to the policy rate, but not one-for-one and not immediately. Pass-through depends on the structure of loans (floating or fixed rates), on banks’ funding costs, and on competition. Banks with market power (Chapter 24) raise deposit rates slowly when the policy rate rises—widening their net interest margins—which is exactly what happened in many countries in 2022–23. Deposit rates lagging the policy rate also encouraged depositors to move money into money-market funds and government bills, an outflow that contributed to the pressures on U.S. regional banks in 2023 (Chapter 51).

49.5 The Transmission Mechanism

How does a change in the policy rate affect spending, output and inflation? Economists distinguish several channels, each built on earlier chapters.

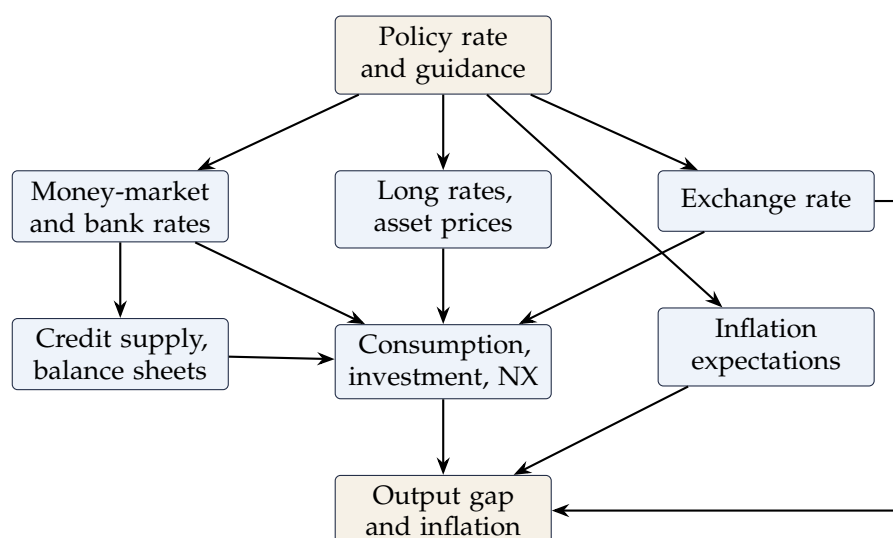


Figure 49.2: Channels of monetary transmission. The exchange rate also affects inflation directly through import prices.

49.5.1 The interest-rate channel

Because prices are sticky, a rise in the nominal policy rate raises real interest rates (Chapter 47). Higher real rates raise the user cost of capital and reduce desired investment (Chapter 45); they reward saving and tilt consumption toward the future through the Euler equation; and they raise the cost of financing durable goods and houses. This is the channel embedded in the IS curve of Chapters 46 and 20.

49.5.2 The credit channel

Bernanke and Gertler (1995) argued that the interest-rate channel alone cannot explain the size and timing of monetary effects, and emphasized two credit channels:

- **The balance-sheet channel.** Higher rates lower the value of borrowers' collateral and their cash flow (by raising interest payments), weakening their net worth and raising the external finance premium they pay (Chapter 25). Small firms and households with little collateral are hit hardest.
- **The bank-lending channel.** Higher rates may reduce banks' capacity or willingness to lend—by raising funding costs, lowering the value of their securities, or tightening their capital positions—so that bank-dependent borrowers face credit rationing (Chapter 25), a reduction in the quantity rather than just the price of credit.

Chapter 28 noted that higher rates raise the cost of financing working capital: a firm with a long cash conversion cycle must fund more inventory and receivables at the higher rate.

49.5.3 The cash-flow channel

The effect of a rate rise on household spending depends heavily on how mortgages are structured. In the United States, most mortgages carry a fixed rate for 30 years, so existing borrowers are protected and the effect comes mainly through new borrowing and house prices. In the United Kingdom, Australia and Malaysia, a large share of mortgages have variable rates or rates fixed only for a few years, so a rate rise quickly raises monthly payments and cuts disposable income—a direct cash-flow effect that makes monetary policy act faster.

49.5.4 The asset-price and wealth channel

Higher rates lower the present value of future cash flows (Chapter 9), reducing the prices of bonds, shares and property. Lower wealth reduces consumption (Chapter 45); lower share prices reduce Tobin's q and investment. For example, a 10-year bond with a 3 per cent coupon priced at par falls from 100 to about 88 when its yield rises to 4.5 per cent—a loss that appears, depending on its accounting classification, in profit, in other comprehensive income, or not at all (Chapter 30).

49.5.5 The exchange-rate channel

Higher domestic rates, relative to foreign rates, attract capital inflows and tend to appreciate the currency (the uncovered-interest-parity logic developed in Chapter 52). An appreciation lowers net exports and reduces inflation directly by lowering the domestic price of imports. In small open economies such as Malaysia, this channel is often the fastest and strongest. It also transmits policy *across* borders: when the Federal Reserve raises rates, the dollar tends to appreciate, and other central banks face pressure on their currencies and inflation.

49.5.6 The expectations channel

A credible central bank that raises rates to fight inflation can lower inflation expectations, which directly lowers inflation through the Phillips curve (Chapters 46 and 47), reducing the output cost of disinflation. Communication about the future path of rates shifts the yield curve. Chapter 50 develops this channel.

49.5.7 Lags and magnitudes

Estimating the effects of monetary policy is hard, because central banks respond to the economy: rates tend to rise when the economy is strong, so a naive correlation might suggest that higher rates cause booms. Researchers identify *monetary-policy shocks*—changes in rates not explained by the central bank’s own forecasts (Romer and Romer 2004) or surprises measured in narrow windows around announcements (the high-frequency methods of Gürkaynak, Sack and Swanson 2005 and others)—and trace their effects (the causal inference of Chapter 17). A common finding for the United States is that an unexpected tightening of 1 percentage point reduces industrial production and output by somewhere between about 0.5 and 2 per cent at the peak, one to two years later, and lowers the price level more slowly, with the largest effects after two years or more. The ranges are wide, and the effects may vary with the state of the economy and the structure of the financial system.

49.6 A Rate Increase Through the Accounts

The integrated flowchart of this book asks, for any shock: *whose constraint changes?* For a rate increase, the answer is everyone’s, but differently.

Worked Example 49.3. A firm’s interest expense and coverage

A firm has operating profit (EBIT) of 30 million and debt of 200 million: 120 million of floating-rate loans at SOFR plus 2 points, and 80 million of fixed-rate bonds at 4 per cent. With SOFR at 0.1 per cent, interest expense is $120 \times 0.021 + 80 \times 0.04 = 2.52 + 3.20 = 5.72$ million, and interest coverage (EBIT/interest, Chapter 31) is 5.2. After the central bank raises rates by 4.5 points, SOFR is 4.6 per cent and interest expense is $120 \times 0.066 + 3.20 = 11.12$ million: coverage falls to 2.7. Pre-tax profit falls from 24.3 to 18.9 million, operating cash flow falls by the same 5.4 million, and if the firm’s loan covenants require coverage above 3, it is in breach and must renegotiate. The fixed-rate bonds protect the firm until they mature and must be refinanced at higher rates.

- **Households** with variable-rate mortgages pay more interest; savers earn more on deposits (with a lag); homeowners see house prices fall. Net debtors cut spending more than net savers raise it, so the distribution of debt matters for the aggregate effect (the HANK insight of Chapter 47).
- **Firms** face higher interest expense on floating-rate debt, higher discount rates on investment projects, and tighter credit. Highly leveraged firms and those with short-term debt are most affected.
- **Banks** earn more on floating-rate assets and on reserves, and pay more on deposits (with a lag), so net interest margins often widen at first. But the market value of their fixed-rate bonds and loans falls, credit losses rise as borrowers struggle, and depositors may move to higher-yielding alternatives—risks that crystallized in 2023.

- **The government** pays more interest as debt is rolled over, raising $r - g$ (Chapter 47); the central bank pays more interest on reserves, reducing or eliminating its profits and remittances to the treasury.
- **Measured GDP** responds as consumption, investment and net exports adjust; company revenue and earnings respond in turn (Chapter 43's link between firm accounts and national accounts).

Case: The 2022–23 tightening: the fastest in four decades

By early 2022 U.S. consumer-price inflation had reached 7.5 per cent and was still rising; it peaked at 9.1 per cent in June 2022. Between March 2022 and July 2023 the Federal Reserve raised its target range eleven times, from 0–0.25 to 5.25–5.50 per cent, including four consecutive increases of 0.75 points in 2022—the fastest tightening since the early 1980s.

Implementation. Because reserves were abundant, each increase was implemented through administered rates, as in Example 49.1. Money-market funds placed record amounts in the Fed's ON RRP facility—around \$2.5 trillion at its peak at the end of 2022—because it paid more than private repo. From June 2022 the Fed also began quantitative tightening, allowing securities to mature without replacement. The balance sheet and the policy rate were adjusted separately, as the floor system allows.

Transmission. The average rate on new 30-year fixed mortgages roughly doubled, from around 3 per cent at the end of 2021 to nearly 8 per cent in October 2023; home sales fell sharply, though house prices fell only modestly because existing owners, locked into low fixed rates, stopped selling. The dollar appreciated strongly in 2022, transmitting the tightening abroad. Bank deposit rates lagged, and deposits flowed into money-market funds. The market value of banks' securities portfolios fell by several hundred billion dollars, contributing to the failure of Silicon Valley Bank in March 2023 (Chapters 30 and 51).

Outcomes. Inflation fell to about 3 per cent by mid-2023 and continued to decline more slowly, while unemployment remained below 4 per cent through 2023—a better outcome than many forecasts, which had predicted a recession. Economists debate how much of the disinflation came from monetary tightening, through demand and expectations, and how much from the unwinding of supply shocks (Chapter 50). The Fed began cutting rates in September 2024.

A comparison. Other central banks tightened too, from different starting points. The ECB raised its deposit facility rate from –0.5 per cent in July 2022 to 4 per cent by September 2023, ending eight years of negative rates; euro-area inflation peaked above 10 per cent in October 2022, driven by energy prices after Russia's invasion of Ukraine. Bank Negara Malaysia raised its overnight policy rate more modestly, from 1.75 to 3 per cent between May 2022 and May 2023, as Malaysian inflation, helped by fuel subsidies and price controls, peaked below 5 per cent. The People's Bank of China moved in the opposite direction, lowering rates and reserve requirements to support an economy facing a property downturn and very low inflation. The divergence illustrates that monetary policy responds to domestic conditions—and that the dollar's appreciation forced many central banks to weigh the exchange-rate channel heavily (Chapter 52).

Summary

- A central bank's liabilities are currency, bank reserves and government deposits; its assets are securities, loans to banks and foreign-exchange reserves. Open-market operations,

currency withdrawals and government payments change reserves.

- The central bank controls the overnight rate using administered rates. In a corridor system, it keeps reserves scarce and supplies the quantity that hits its target; in a floor system, reserves are abundant and the market rate follows the deposit rate, separating rate policy from balance-sheet policy.
- Repos provide temporary reserves against collateral with haircuts; standing facilities cap market rates; collateral frameworks are a key policy lever.
- Long rates reflect expected future short rates plus a term premium; forward rates reveal market expectations.
- Policy is transmitted through interest-rate, credit (balance-sheet and bank-lending), cash-flow, asset-price, exchange-rate and expectations channels, with long and variable lags.
- A rate increase changes the financial statements of households, firms, banks, the government and the central bank, each differently.

Key terms

central-bank balance sheet; open-market operation; autonomous factors; reserve requirement ratio; money market; overnight rate; administered rates; deposit rate; interest on reserve balances; lending rate; standing facility; corridor system; floor system; ample reserves; overnight reverse repo facility; standing repo facility; repurchase agreement; reverse repo; haircut; collateral framework; SOFR; €STR; expectations hypothesis; term premium; forward rate; inverted yield curve; pass-through; interest-rate channel; balance-sheet channel; bank-lending channel; cash-flow channel; wealth channel; exchange-rate channel; expectations channel; monetary-policy shock; quantitative tightening.

Exercises

Exercise 49.1 (Balance-sheet entries). Record the following on the central bank's balance sheet and explain the effect on reserves: (a) it buys 50 of government bonds from a pension fund (which banks with Bank Z); (b) the Treasury receives 30 of tax payments; (c) households withdraw 10 of banknotes before a holiday; (d) the central bank lends 20 to a bank for one week in a repo.

Exercise 49.2 (Corridor). A central bank's corridor is 1.50–2.50 per cent with a target of 2 per cent. Demand for reserves is $R = 100 - 40(i - 2)$ within the corridor (billions). (a) How many reserves must it supply to hit the target? (b) An autonomous factor unexpectedly drains 10 billion. What happens to the market rate if the central bank does not respond? (c) How would your answer differ under a floor system with reserves of 400 billion?

Exercise 49.3 (Floor). Explain why, in a floor system, a central bank can raise its policy rate without selling any assets. Why might the effective market rate lie slightly below the interest rate on reserves?

Exercise 49.4 (Yield curve). One-year rates are expected to be 5, 4, 3 and 3 per cent over the next four years; the four-year term premium is 0.5 points. (a) What is the four-year yield? (b) If the central bank convinces markets that rates will stay at 5 per cent for all four years, what is the four-year yield? (c) What does this imply about the power of communication?

Exercise 49.5 (Forward rates). The two-year yield is 3 per cent and the three-year yield 3.4 per cent. Compute the implied one-year forward rate for year three. What might explain a rising forward rate?

Exercise 49.6 (Bond losses). A bank holds 10 billion (face value) of 10-year bonds with a 2 per cent coupon, bought at par. Market yields rise to 4.5 per cent. (a) What is the market value of the bonds? (b) How is the loss reported if the bonds are classified at amortized cost, and if at fair value through other comprehensive income? (c) Why does the classification matter if depositors withdraw?

Exercise 49.7 (Firm finances). A firm has EBIT of 50, 300 of floating-rate debt at the policy rate plus 2.5 points, and no fixed-rate debt. The policy rate rises from 1 to 5 per cent. (a) Compute interest expense and coverage before and after. (b) The firm's investment projects are valued by discounting at its cost of capital; explain how the rate rise affects which projects it undertakes.

Exercise 49.8 (Mortgages). Explain why a 1-point rise in the policy rate is likely to reduce consumption faster in Malaysia or the United Kingdom than in the United States. What other channel matters relatively more in the United States?

Exercise 49.9 (Identification). Why is the simple correlation between policy rates and output a poor guide to the effect of monetary policy? Describe one method of identifying monetary-policy shocks.

Exercise 49.10 (Integration). Trace a 1-point rise in the ECB's deposit rate through (a) the balance sheet of a German bank holding large reserves and fixed-rate mortgages; (b) a German exporter; (c) euro-area GDP and inflation. Identify one channel in each through which the effect operates.

Solutions

Solution 49.1. (a) Dr Securities 50 / Cr Reserves 50 (Bank Z's reserves rise; the pension fund's deposit at Bank Z rises): reserves +50, broad money +50. (b) Dr Reserves 30 / Cr Government deposits 30: reserves −30. (c) Dr Reserves 10 / Cr Banknotes 10: reserves −10. (d) Dr Loans to banks (repo) 20 / Cr Reserves 20: reserves +20 for one week. Net: reserves +30.

Solution 49.2. (a) At $i = 2$, $R = 100$ billion. (b) Supply 90: $90 = 100 - 40(i - 2)$, so $i = 2.25$ per cent. (c) Under a floor, demand is flat at the deposit rate over a wide range; a drain of 10 from 400 leaves reserves abundant and the rate at the floor, unchanged. This insensitivity to autonomous factors is an advantage of floor systems.

Solution 49.3. With abundant reserves, the market rate is determined by the rate banks can earn by holding reserves, since no bank will lend for less. Raising that administered rate raises the market rate directly. Institutions without access to interest on reserves (such as GSEs in the United States) lend reserves to banks below IORB, and banks, which face balance-sheet costs, arbitrage only partly, so the market rate sits slightly below IORB, with the ON RRP rate as a sub-floor.

Solution 49.4. (a) Average expected rate = 3.75%; yield $\approx 4.25\%$. (b) $5 + 0.5 = 5.5\%$. (c) Without changing today's rate, credible communication about future rates can raise the four-year yield by 1.25 points—a powerful lever on the rates that matter for investment and mortgages (Chapter 50 on forward guidance).

Solution 49.5. $f = 1.034^3/1.03^2 - 1 = 1.10549/1.0609 - 1 = 4.20\%$. A rising forward rate may reflect expected future policy tightening, expected higher inflation, or a rising term premium for longer maturities.

Solution 49.6. (a) Price per 100 face value $= \sum_{t=1}^{10} 2/1.045^t + 100/1.045^{10} = 15.83 + 64.39 = 80.22$: market value about 8.02 billion, a loss of about 1.98 billion. (b) At amortized cost, the loss is not recognized (only disclosed in the notes); at FVOCI it reduces equity through OCI but not profit. For regulatory capital, treatments differ by jurisdiction and bank size. (c) If depositors withdraw and the bank must sell the bonds, the loss is realized in profit and reduces equity immediately; a bank whose

equity is smaller than its unrecognized losses is economically insolvent, which depositors who notice will run from (Chapter 51).

Solution 49.7. (a) Before: rate 3.5%, interest 10.5, coverage 4.8. After: rate 7.5%, interest 22.5, coverage 2.2. (b) A higher cost of capital lowers the net present value of every project, especially long-lived ones whose cash flows arrive far in the future; projects that were marginally positive become negative and are dropped, reducing investment (the user-cost logic of Chapter 45).

Solution 49.8. In Malaysia and the United Kingdom many mortgages have variable or short-fixed rates, so payments rise quickly and disposable income falls (the cash-flow channel). In the United States most mortgages are fixed for 30 years, so the effect on existing borrowers is small; the effects work more through new borrowing, housing activity, asset prices, credit conditions and the dollar.

Solution 49.9. Central banks raise rates when they expect the economy to be strong and inflation to rise, so high rates coincide with strong output even if they reduce output relative to what would otherwise happen—reverse causality and omitted variables (Chapter 17). One method: the Romer–Romer approach regresses rate changes on the central bank’s own forecasts and treats the residual as a shock; another uses changes in futures prices in a 30-minute window around announcements, during which nothing else should systematically move rates.

Solution 49.10. (a) The bank earns 1 point more on its reserves at once; it pays more on deposits with a lag; its fixed-rate mortgages lose market value and earn the old rate, so its margin on the mortgage book is squeezed (interest-rate risk; bank-lending channel if it tightens credit). (b) The euro tends to appreciate, making the exporter’s goods dearer abroad (exchange-rate channel); its borrowing costs rise (interest-rate and balance-sheet channels). (c) Investment and consumption fall through higher real rates and lower asset prices; net exports fall; the output gap widens and inflation falls with a lag, helped by lower import prices and, if the ECB is credible, lower expected inflation.

Inflation, Expectations, Credibility and Policy Rules

After studying this chapter you should be able to

1. Explain the costs of expected and unexpected inflation and of deflation, and why most central banks target inflation of about 2 per cent.
2. Use the Fisher relation to distinguish nominal and real interest rates, ex ante and ex post, and read inflation expectations from indexed bonds and surveys.
3. Explain the concept of the natural (neutral) real interest rate and why estimates of it fell.
4. State the Taylor rule, compute its prescriptions, and explain the Taylor principle.
5. Derive the inflation bias of discretionary policy in the Barro–Gordon model and explain how commitment, independence and conservative central bankers address it.
6. Derive optimal policy under a cost-push shock and explain flexible inflation targeting.
7. Describe inflation targeting and central-bank communication in practice.
8. Analyse the Volcker disinflation and the post-pandemic inflation of 2021–24.

A motivating question

In March 1980 U.S. consumer prices were rising at almost 15 per cent a year. Inflation had been drifting upward for fifteen years, and each attempt to bring it down had been abandoned when unemployment rose. People had come to expect inflation, built it into wage contracts and interest rates, and behaved in ways that made it self-perpetuating. Three years later, inflation was below 4 per cent. The cost was the deepest recession since the 1930s, with unemployment reaching almost 11 per cent.

Forty years later, inflation in the United States, the euro area and the United Kingdom rose to around 10 per cent, the highest in four decades. This time it fell back to around 3 per cent within about two years, with far smaller increases in unemployment. Why was the second disinflation so much cheaper than the first? A large part of the answer is *expectations*: in 2022, unlike 1980, most people believed that central banks would bring inflation back to target, and they behaved accordingly. This chapter explains why expectations matter, how central banks earn the credibility that anchors them, and the rules and frameworks they use.

50.1 Why Inflation Matters

50.1.1 Costs of expected inflation

Even fully anticipated inflation imposes costs:

- **Shoe-leather costs.** Inflation raises the opportunity cost of holding money, so people hold less and make more trips to the bank (the Baumol–Tobin model of Chapter 48). The resources spent economizing on money are wasted.
- **Menu costs.** Firms must change prices more often (Chapter 47).
- **Relative-price distortions.** With staggered price setting, inflation causes the prices of firms that recently reset to differ from those that have not, misallocating demand among goods. In the New Keynesian model this is the main welfare cost of inflation, and it is the reason optimal policy in that model aims at price stability.
- **Tax distortions.** Tax systems are largely defined in nominal terms. Taxes on nominal interest and nominal capital gains tax part of the inflation compensation, raising the effective tax on real returns; historical-cost depreciation (Chapter 29) is worth less in real terms when prices rise, raising the effective tax on investment.
- **Accounting distortions.** Historical-cost accounting mixes currency units of different purchasing power; in high inflation, reported profits overstate real profits, and inventory methods such as FIFO inflate margins (Chapter 28).

50.1.2 Costs of unexpected inflation

Unexpected inflation redistributes wealth. It reduces the real value of nominal debts, transferring wealth from creditors to debtors: from savers holding bonds and deposits to borrowers with fixed-rate mortgages, and from bondholders to governments. Unexpected disinflation does the reverse. Because inflation that is high tends also to be variable, high inflation makes long-term nominal contracts risky, shortening the horizons over which people are willing to lend and plan.

50.1.3 Deflation and the choice of target

Deflation—a falling price level—has costs of its own. It raises the real value of debts (Fisher’s debt deflation, Chapter 48), and with a lower bound on nominal interest rates it can raise real rates when the economy needs them to fall (Chapter 51). Most central banks in advanced economies aim for inflation of about 2 per cent rather than zero, for three reasons: measured inflation tends to overstate true inflation because of quality improvements and new goods (Chapter 10); a positive inflation rate keeps nominal interest rates higher on average, leaving more room to cut them in a recession; and because nominal wages rarely fall, some inflation allows real wages to adjust downward in declining sectors.

50.2 Nominal and Real Interest Rates

50.2.1 The Fisher relation

Irving Fisher (1930) distinguished nominal and real interest rates. If a lender receives nominal interest i and prices rise by π , the real return is $(1 + i)/(1 + \pi) - 1$. The *ex ante* real rate—the one that matters for decisions—uses expected inflation:

$$1 + r = \frac{1 + i}{1 + \pi^e}, \quad \text{so} \quad i \approx r + \pi^e.$$

This is the **Fisher relation**. The *ex post* real rate uses actual inflation and differs from the *ex ante* rate by the inflation surprise. The *Fisher effect* is the hypothesis that, in the long run, a

permanent rise in expected inflation raises nominal interest rates one-for-one, leaving real rates unchanged. Across countries and over long periods, high-inflation countries do have high nominal interest rates, as the Fisher effect predicts.

Worked Example 50.1. Real rates and inflation surprises

A bank lends at 5 per cent for one year when expected inflation is 3 per cent. The ex ante real rate is $1.05/1.03 - 1 = 1.94$ per cent (approximately 2). Inflation turns out to be 8 per cent. The ex post real rate is $1.05/1.08 - 1 = -2.78$ per cent: the borrower gained and the lender lost about 4.7 per cent of the principal in real terms relative to expectations.

50.2.2 Measuring expected inflation

Expectations cannot be observed directly. Central banks use:

- **Surveys** of households (the University of Michigan survey in the United States; the European Commission's consumer survey), firms, and professional forecasters (the Survey of Professional Forecasters, the ECB's SPF).
- **Market prices.** Governments issue inflation-indexed bonds (Treasury Inflation-Protected Securities in the United States, index-linked gilts in the United Kingdom) whose payments rise with a price index, so they pay a real yield. The difference between the yield on a nominal bond and an indexed bond of the same maturity is the **breakeven inflation rate**. If the 10-year nominal yield is 4.2 per cent and the indexed yield 1.9 per cent, breakeven inflation is 2.3 per cent—although it also contains risk and liquidity premia, so it is not a pure measure of expectations. Forward measures, such as expected inflation from five to ten years ahead ("5y5y"), are used to judge whether long-term expectations are anchored.

Household expectations are typically higher and more dispersed than professional forecasts, and more influenced by the prices people see often, such as food and petrol. They matter because households set wages and make spending decisions.

Key idea. Expectations are *anchored* when long-term expected inflation stays near the central bank's target even when current inflation moves away from it. Anchored expectations make the Phillips curve behave as if $\pi^e = \pi^*$: shocks to inflation are temporary, and bringing inflation back to target requires only a modest output cost.

50.3 The Neutral Real Interest Rate

Chapter 46 introduced the natural real interest rate $\bar{r}$, the real rate consistent with output at potential and stable inflation; it is often denoted r^* and called the **neutral rate**. Policy is tight when the real policy rate is above r^* and loose when it is below. The concept goes back to Knut Wicksell (1898). The difficulty is that r^* is not observable and must be inferred from the behaviour of output and inflation.

The most widely cited estimates, from the model of Thomas Laubach and John Williams (2003) and its successors, suggest that r^* in the United States and other advanced economies fell from around 2–3 per cent in the 1990s to around zero or below in the 2010s, and may have risen somewhat since 2022, although the uncertainty around any estimate is several percentage points. Explanations for the decline include slower productivity growth (which lowers the return to capital, Chapter 44), population ageing (which raises desired saving, Chapter 45), high saving in China and other emerging economies (a "global saving glut"), and strong demand

for safe assets. A lower r^* means that interest rates are more likely to hit their lower bound in recessions—one motivation for the unconventional tools of Chapter 51.

50.4 Policy Rules

50.4.1 The Taylor rule

John Taylor (1993) observed that U.S. monetary policy from 1987 to 1992 was well described by a simple rule:

$$i = r^* + \pi + 0.5(\pi - \pi^*) + 0.5\tilde{Y},$$

with $r^* = 2$ and $\pi^* = 2$. The nominal rate equals the neutral real rate plus current inflation, plus half the gap between inflation and target, plus half the output gap. Rearranging, the response of the nominal rate to inflation is 1.5: the rule satisfies the Taylor principle of Chapter 47. When inflation rises by 1 point, the nominal rate rises by 1.5 points and the real rate by 0.5 point, so policy tightens.

Worked Example 50.2. Taylor-rule prescriptions

Suppose inflation is 6 per cent, the target 2 per cent and the output gap +1 per cent. With Taylor's original $r^* = 2$:

$$i = 2 + 6 + 0.5 \times 4 + 0.5 \times 1 = 10.5\%.$$

If the neutral rate is 0.5 per cent instead, the prescription is 9 per cent. If the output gap is -1 per cent, the prescriptions fall by 1 point. In mid-2022, when U.S. inflation was near these values, the actual federal funds rate was still below 2 per cent: by the Taylor rule's standards, policy was far behind the curve. Defenders of the Fed argued that inflation was driven by supply shocks that policy should partly look through, and that the Fed was tightening quickly by other means, including forward guidance, which raised longer-term rates well before the policy rate itself.

The Taylor rule became a benchmark rather than a mechanical guide. Central banks publish comparisons of rule prescriptions with actual policy; variations use forecasts of inflation rather than current inflation, a larger coefficient on the output gap, or *inertia* (partial adjustment of the rate toward the rule's prescription each quarter, $i_t = \rho_i i_{t-1} + (1 - \rho_i) i_t^{\text{rule}}$), which fits the data better and, by making future rates predictable, strengthens the effect of policy on long-term rates. The choice of r^* and of the measure of the output gap can change the prescription by several percentage points, which is why rules inform rather than dictate policy.

50.5 Rules Versus Discretion

50.5.1 Time inconsistency

Finn Kydland and Edward Prescott (1977) identified a problem that arises whenever expectations depend on anticipated policy. A policy that is optimal when announced may not be optimal to carry out once people have acted on the announcement. Such a policy is **time inconsistent**. A government may promise not to tax existing capital heavily, to encourage investment; once the capital is installed, taxing it is tempting. People anticipate the temptation, and invest less. Game theory (Chapter 23) calls this a problem of *commitment*: the outcome under commitment is better than the subgame-perfect outcome without it.

50.5.2 The Barro–Gordon model

Robert Barro and David Gordon (1983) applied the idea to monetary policy. Let the output gap be determined by an expectations-augmented Phillips curve, $x = (\pi - \pi^e)/\kappa$. The central bank dislikes inflation, and it would like output to be *above* its natural level by $k > 0$ —for example because taxes or market power make the natural level inefficiently low (Chapters 22 and 24). Its loss function is

$$L = \frac{1}{2}\pi^2 + \frac{1}{2}\lambda(x - k)^2,$$

where $\lambda > 0$ is the weight it places on output. The public forms rational expectations. The timing is: the public sets π^e (in wage contracts, say); then the central bank chooses π .

Proposition 50.3. The inflation bias of discretion

Under discretion, equilibrium inflation is $\pi = \lambda k/\kappa > 0$ and output is at its natural level ($x = 0$). Under commitment to $\pi = 0$, output is also at its natural level, and the loss is lower.

Proof. Taking π^e as given, the central bank minimizes L with $x = (\pi - \pi^e)/\kappa$. The first-order condition is $\pi + \lambda(x - k)/\kappa = 0$. With rational expectations, $\pi^e = \pi$ in equilibrium, so $x = 0$ and $\pi = \lambda k/\kappa$. The loss is $\frac{1}{2}(\lambda k/\kappa)^2 + \frac{1}{2}\lambda k^2$. Under commitment to $\pi = 0$, expectations are $\pi^e = 0$, $x = 0$, and the loss is $\frac{1}{2}\lambda k^2$, lower by $\frac{1}{2}(\lambda k/\kappa)^2$. $\square$

Worked Example 50.4. Inflation bias

With $\lambda = 0.5$, $k = 2$ and $\kappa = 0.5$, discretionary inflation is $0.5 \times 2/0.5 = 2$ per cent, and the loss is $\frac{1}{2} \times 4 + \frac{1}{2} \times 0.5 \times 4 = 3$. Under commitment, inflation is zero and the loss is 1. Output is the same in both cases. The public, knowing that the central bank would be tempted to create surprise inflation if expected inflation were zero, expects exactly enough inflation that the temptation disappears—and the economy gets inflation with no output gain.

The model captures an important feature of the 1970s: policy-makers did not intend to produce high inflation, but the absence of a credible commitment to low inflation produced it.

50.5.3 Solutions

Reputation. In a repeated game (Chapter 23), a central bank that values its reputation may resist the temptation, because surprise inflation today would raise expected inflation in the future. Reputation can sustain low inflation if the central bank is patient enough.

A conservative central banker. Kenneth Rogoff (1985) showed that society can gain by appointing a central banker who places *less* weight on output than society does (a smaller λ), because the inflation bias $\lambda k/\kappa$ falls. The cost is a weaker response to supply shocks. The Bundesbank, with its famous aversion to inflation after the hyperinflation of 1923 (Chapter 48), was often cited as the model.

Independence. Delegating monetary policy to an independent central bank, with a clear mandate and protection from short-term political pressure, removes the government's temptation from the decision. Cross-country studies from the early 1990s (Alesina and Summers 1993) found that countries with more independent central banks had lower average inflation without lower growth. Since then most countries have granted their central banks operational

independence: the Bank of England in 1997, the ECB (whose independence is written into the EU Treaties) from its creation in 1998, Bank Negara Malaysia under its Central Bank of Malaysia Act 2009. Independence is *instrument* independence—freedom to set rates—with goals usually set by law or by government.

Targets and transparency. An explicit numerical inflation target, with public accountability for meeting it, makes deviations costly and visible. This is inflation targeting.

50.6 Optimal Policy and Inflation Targeting

50.6.1 Responding to cost-push shocks

Chapter 47 showed that cost-push shocks break the divine coincidence. Suppose the Phillips curve is $\pi = \kappa x + u$ (with expected future inflation anchored at zero), and the central bank minimizes $\frac{1}{2}(\pi^2 + \lambda x^2)$, now with $k = 0$. The first-order condition gives the **targeting rule**

$$x = -\frac{\kappa}{\lambda} \pi :$$

the central bank “leans against the wind,” accepting a negative output gap when inflation is above target. Substituting into the Phillips curve,

$$\pi = \frac{\lambda}{\lambda + \kappa^2} u, \quad x = -\frac{\kappa}{\lambda + \kappa^2} u.$$

With $\kappa = 0.5$, $\lambda = 0.25$ and a shock $u = 1$, inflation rises by 0.5 point and the output gap falls by 1 point: the shock is split between inflation and output. A central bank with a smaller λ lets less of the shock through to inflation and more to output.

Advanced Analysis: Commitment and history dependence

In the forward-looking New Keynesian model, $\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t + u_t$, the central bank can do better than discretion even without an inflation bias, by committing to a *history-dependent* policy. Under optimal commitment, the targeting rule becomes $\pi_t = -(\lambda/\kappa)(x_t - x_{t-1})$: the central bank promises to keep output below potential for a while even after the shock has passed. Because firms are forward-looking, the promise of future restraint lowers expected inflation and hence current inflation, improving the trade-off today. This *stabilization bias* of discretion, analysed by Clarida, Galí and Gertler (1999) and Woodford (2003), is the formal foundation of forward guidance: credible promises about future policy are themselves a policy instrument.

50.6.2 Inflation targeting in practice

New Zealand adopted the first formal inflation target in 1990; Canada followed in 1991 and the United Kingdom in 1992, and dozens of central banks in advanced and emerging economies have since done so. **Inflation targeting** has five elements: a public numerical target for inflation; an institutional commitment to price stability as the primary long-run goal; the use of all available information, not a single intermediate target such as money; transparency about strategy and decisions; and accountability for meeting the target.

In practice inflation targeting is *flexible*: central banks aim to return inflation to target over a medium-term horizon of perhaps two years, allowing temporary deviations to avoid excessive output volatility—which is the targeting rule above in words. Arrangements differ:

- The **ECB** has a mandate of price stability; since its 2021 strategy review it has aimed for 2 per cent inflation over the medium term, with the target *symmetric*—deviations above and below are equally undesirable. Its 2025 strategy assessment confirmed the target and treated the post-pandemic surge as a test of the framework, emphasizing forceful responses to large, persistent deviations in either direction.
- The **Federal Reserve** has a dual mandate from Congress: maximum employment and stable prices. It adopted an explicit 2 per cent target for PCE inflation in 2012. In 2020 it adopted *flexible average inflation targeting*, promising to let inflation run moderately above 2 per cent after periods below it; its 2025 review returned to a flexible inflation-targeting framework without that make-up element.
- **Bank Negara Malaysia** does not have a formal inflation target; its mandate is to promote monetary and financial stability conducive to sustainable growth, and it sets its overnight policy rate with regard to both inflation and growth. Many emerging-market central banks combine inflation objectives with concern for the exchange rate (Chapter 52).

50.6.3 Communication

Since the 1990s central banks have moved from secrecy toward transparency: publishing their decisions immediately, holding press conferences, releasing minutes and forecasts, and in some cases publishing projections of the policy-rate path (the Fed's "dot plot"). **Forward guidance**—statements about the likely future path of policy—works through the expectations component of long-term rates (Chapter 49) and, in the advanced box above, through expected inflation. It can be *Delphic* (a forecast of what the central bank expects to do) or *Odyssean* (a commitment that binds, like Odysseus tied to the mast). Communication becomes especially important when the policy rate is at its lower bound (Chapter 51).

Case: The Volcker disinflation, 1979–83

The inheritance. U.S. inflation rose from under 2 per cent in the early 1960s to over 5 per cent by 1970 and, after the oil shocks of 1973–74 and 1979, to double digits. The Fed repeatedly tightened and then eased when unemployment rose. Estimates of the Fed's reaction function for this period suggest that the Taylor principle was violated (Chapter 47): real rates were often negative. Expectations became unanchored.

The shock. Paul Volcker became chairman in August 1979. On 6 October 1979 the Fed announced that it would target the growth of bank reserves rather than the federal funds rate, letting interest rates rise as far as necessary—partly a device to deflect political responsibility for the rates themselves. The federal funds rate rose to a peak of around 19–20 per cent in 1981. After a short recession in 1980 and a premature easing, the Fed held rates high through a second, deeper recession in 1981–82. Consumer-price inflation, which peaked near 15 per cent in 1980, fell to below 4 per cent in 1983. Unemployment peaked at 10.8 per cent at the end of 1982.

The cost. The **sacrifice ratio**—the cumulative loss of output (in per cent of a year's GDP) per percentage point of reduction in inflation—was roughly 2 to 3 in most estimates. Expectations adjusted only slowly: survey measures of expected inflation remained well above actual inflation for several years, consistent with a central bank whose commitment was not yet believed. With adaptive expectations, as in Example 46.3 of Chapter 46, disinflation requires a period of output below potential; with rational expectations and full credibility, it could in principle be nearly costless. The Volcker episode lay between the two: the Fed's credibility had to be *earned* by accepting a recession.

In the accounts. High rates had direct balance-sheet effects: savings and loan associations, which held long-term fixed-rate mortgages funded by short-term deposits, became insolvent as deposit rates rose above mortgage yields—a maturity mismatch whose resolution in the late 1980s cost U.S. taxpayers over \$100 billion. Heavily indebted developing countries, which had borrowed in dollars at floating rates, faced the Latin American debt crisis from 1982 (Chapter 52).

The legacy. The disinflation established the principle that central banks can and should control inflation, and the credibility of the Fed under Volcker and his successors underpinned the long period of low and stable inflation—the “Great Moderation”—that followed.

50.7 The Post-Pandemic Inflation

50.7.1 What happened

Inflation rose from 2021 across most economies. U.S. consumer-price inflation peaked at 9.1 per cent in June 2022; euro-area inflation at 10.6 per cent in October 2022; U.K. inflation at 11.1 per cent in October 2022. Inflation was lower but still elevated in much of Asia, including Malaysia, where fuel subsidies and price controls damped it, and remained very low in China. Central banks initially judged the rise to be *transitory*, the result of reopening bottlenecks; the Federal Reserve dropped the word in November 2021 and, as Chapter 49 described, then tightened rapidly.

50.7.2 Why

Using the tools of this book, the causes fall into three groups:

1. **Supply shocks** (ϵ_s in Chapter 46; u_t in Chapter 47): shipping and semiconductor bottlenecks, the shift of spending from services to goods, and from 2022 the surge in energy and food prices following Russia’s invasion of Ukraine—especially severe in Europe, which depended on Russian gas.
2. **Demand:** fiscal support and accumulated savings (Chapter 46), low interest rates and quantitative easing, and the release of pent-up demand on reopening. In the United States, where fiscal support was largest, demand pressure played a larger role than in the euro area.
3. **Labour-market tightness:** very high vacancies relative to unemployment (the Beveridge curve of Chapter 45) raised wage growth, which became the main driver of persistent services inflation as supply shocks faded.

Ben Bernanke and Olivier Blanchard (2023) decomposed the U.S. surge and concluded that most of the initial rise came from shocks to prices in particular sectors—energy, food, and shortages—while the effect of labour-market tightness was smaller but more persistent.

50.7.3 Why the disinflation was cheap

Inflation fell from its peak to around 3 per cent in about two years in the United States and the euro area, with unemployment rising little. Three features explain the contrast with 1979–83:

- **Anchored expectations.** Long-term market-based and professional expectations remained close to 2 per cent throughout, so the rise in current inflation did not become

embedded; the Phillips curve behaved as if expected inflation were at target.

- **Supply shocks reversed.** Energy prices fell from their 2022 peaks and bottlenecks eased, shifting the AS curve back down.
- **A steep Phillips curve at high tightness.** If the Phillips curve is non-linear, steep when labour markets are very tight (Chapter 46), then a moderate cooling of demand—visible in falling vacancies with little rise in unemployment, a movement down a steep part of the Beveridge curve—can lower inflation substantially.

The episode vindicated the value of credibility built over decades. It also raised hard questions: whether central banks were too slow to respond in 2021, whether frameworks designed for a world of too-low inflation (such as the Fed's average inflation targeting) delayed the response, and how to distinguish in real time a supply shock to look through from one that will unanchor expectations.

Summary

- Expected inflation imposes shoe-leather, menu, relative-price, tax and accounting costs; unexpected inflation redistributes wealth. Deflation raises real debts and can trap policy at the lower bound. Most central banks target about 2 per cent.
- The Fisher relation $i \approx r + \pi^e$ links nominal and real rates. Expected inflation is measured by surveys and by breakeven rates from indexed bonds; anchored long-term expectations are central to inflation control.
- The neutral real rate r^* is unobservable; estimates fell substantially between the 1990s and 2010s.
- The Taylor rule $i = r^* + \pi + 0.5(\pi - \pi^*) + 0.5\tilde{Y}$ satisfies the Taylor principle and is a benchmark for policy.
- Time inconsistency creates an inflation bias under discretion ($\pi = \lambda k / \kappa$ in Barro–Gordon). Reputation, conservative central bankers, independence and explicit targets address it.
- Under cost-push shocks, optimal policy leans against the wind: $x = -(\kappa / \lambda)\pi$. Commitment to history-dependent policy improves the trade-off and underpins forward guidance.
- Flexible inflation targeting with transparency and communication is the dominant framework.
- The Volcker disinflation was costly because credibility had to be earned; the post-pandemic disinflation was cheaper because expectations stayed anchored and supply shocks reversed.

Key terms

shoe-leather costs; menu costs; relative-price distortion; redistribution; deflation; Fisher relation; ex ante and ex post real interest rate; Fisher effect; inflation-indexed bond; breakeven inflation rate; anchored expectations; neutral real interest rate; Taylor rule; Taylor principle; inertia; time inconsistency; commitment; discretion; Barro–Gordon model; inflation bias; reputation; conservative central banker; central-bank independence; targeting rule; stabilization bias; inflation targeting; flexible inflation targeting; symmetric target; dual mandate; average inflation targeting; forward guidance; Delphic and Odyssean guidance; sacrifice ratio; Great Moderation.

Exercises

Exercise 50.1 (Fisher). A 2-year loan has a nominal rate of 6 per cent a year. Expected inflation is 3.5 per cent a year. (a) What is the ex ante real rate (exactly)? (b) Actual inflation is 2 per cent in year 1 and 9 per cent in year 2. What is the ex post real return over the two years? Who gained?

Exercise 50.2 (Breakeven). The 5-year nominal government yield is 3.8 per cent, the 5-year indexed yield 1.4 per cent, the 10-year nominal yield 4.1 per cent and the 10-year indexed yield 1.6 per cent. (a) Compute 5- and 10-year breakeven inflation. (b) Compute approximate average breakeven inflation for years 6–10. (c) Why might breakevens not equal expected inflation?

Exercise 50.3 (Taylor rule). Compute the Taylor-rule rate ($r^* = 1$, $\pi^* = 2$) for: (a) $\pi = 2$, $\tilde{Y} = 0$; (b) $\pi = 4$, $\tilde{Y} = -2$; (c) $\pi = 1$, $\tilde{Y} = -4$. Comment on (c).

Exercise 50.4 (Inertia). The policy rate is 1 per cent and the Taylor-rule prescription jumps to 5 per cent and stays there. With inertia $\rho_i = 0.8$ per quarter, find the policy rate for the next four quarters. Why might a central bank prefer gradual adjustment?

Exercise 50.5 (Barro–Gordon). With $\kappa = 1$, $k = 1$ and $\lambda = 1$, find discretionary inflation and the loss under discretion and commitment. Society's λ is 1; it appoints a central banker with $\lambda = 0.25$. What is inflation? Why might society not want $\lambda = 0$?

Exercise 50.6 (Cost-push shock). With $\pi = 0.4x + u$, loss $\frac{1}{2}(\pi^2 + \lambda x^2)$ and $u = 2$: find π and x for $\lambda = 1$ and $\lambda = 0.16$. Interpret.

Exercise 50.7 (Sacrifice ratio). A disinflation reduces inflation from 10 to 4 per cent. Output is below potential by 3, 4, 3 and 2 per cent in four successive years. Compute the sacrifice ratio. How would credibility affect it?

Exercise 50.8 (Why 2 per cent?). Give three reasons for targeting 2 per cent inflation rather than zero, and one argument for a higher target such as 3 or 4 per cent. What are the risks of raising the target?

Exercise 50.9 (Diagnose the error). “An increase in the central bank’s balance sheet mechanically creates an equal increase in consumer-price inflation.” Identify which part of the statement is accounting, which is behavioural and which is causal, and correct it.

Exercise 50.10 (Post-pandemic inflation). Using the AD–AS model of Chapter 46 with anchored expectations, show how a supply shock and a demand boom could together produce high inflation. Explain why the subsequent disinflation required a smaller recession than the Volcker disinflation.

Solutions

Solution 50.1. (a) $1.06/1.035 - 1 = 2.42\%$ a year. (b) Nominal gross return $1.06^2 = 1.1236$; price level $1.02 \times 1.09 = 1.1118$; real gross return $1.1236/1.1118 = 1.0106$: about 0.53 per cent a year, below the expected 2.42 per cent. The borrower gained from unexpectedly high inflation.

Solution 50.2. (a) 5-year: 2.4 per cent; 10-year: 2.5 per cent. (b) Approximately $2 \times 2.5 - 2.4 = 2.6$ per cent a year for years 6–10. (c) Breakevens include an inflation risk premium (nominal bondholders want compensation for inflation uncertainty, raising breakevens) and a liquidity premium (indexed bonds are less liquid, raising their yields and lowering breakevens), and can be distorted by supply and demand in either market.

Solution 50.3. (a) $1 + 2 + 0 + 0 = 3\%$. (b) $1 + 4 + 1 - 1 = 5\%$. (c) $1 + 1 - 0.5 - 2 = -0.5\%$: the rule prescribes a negative nominal rate, which may be infeasible at the lower bound—the situation that led central banks to unconventional policy (Chapter 51).

Solution 50.4. $i_1 = 0.8 \times 1 + 0.2 \times 5 = 1.8$; $i_2 = 0.8 \times 1.8 + 1 = 2.44$; $i_3 = 2.952$; $i_4 = 3.36$. Gradualism reduces the risk of overreacting to noisy data, avoids disrupting financial markets, and—because markets expect the rate to keep rising—moves long-term rates more than a one-off change of the same size, so it can achieve much of the tightening immediately.

Solution 50.5. Discretion: $\pi = 1$, loss $\frac{1}{2} + \frac{1}{2} = 1$. Commitment: $\pi = 0$, loss $\frac{1}{2}$. With $\lambda = 0.25$: $\pi = 0.25$. With $\lambda = 0$ the central banker would ignore output entirely and, when supply shocks hit, would impose large output losses to hold inflation at zero; the Rogoff trade-off is between a lower average bias and worse stabilization of shocks.

Solution 50.6. $\pi = \lambda u / (\lambda + 0.16)$, $x = -0.4u / (\lambda + 0.16)$. $\lambda = 1$: $\pi = 2/1.16 = 1.72$, $x = -0.8/1.16 = -0.69$. $\lambda = 0.16$: $\pi = 0.32/0.32 = 1.0$, $x = -0.8/0.32 = -2.5$. A central bank that cares less about output lets less of the shock into inflation but imposes a much deeper recession.

Solution 50.7. Cumulative loss = $3 + 4 + 3 + 2 = 12$ per cent of a year's GDP; inflation fell 6 points; sacrifice ratio = 2. With full credibility and forward-looking price setting, expected inflation would fall with the announcement, shifting the Phillips curve down and lowering the ratio; with adaptive expectations it would be higher.

Solution 50.8. Reasons for 2 rather than 0: measurement bias in price indexes; room to cut nominal rates in recessions; downward nominal wage rigidity. For a higher target: with a low neutral rate, 2 per cent leaves too little room above the lower bound, so recessions may be deeper. Risks: higher costs of inflation (Section 50.1), possible loss of credibility if the target is seen as movable, and less anchored expectations.

Solution 50.9. Accounting: when the central bank buys assets, its balance sheet expands and reserves (its liabilities) rise equally. Behavioural: whether banks lend more and households and firms spend more depends on their decisions, interest rates and confidence; reserves themselves are not spent by the public (Chapter 48). Causal: inflation depends on aggregate demand relative to supply and on expectations; large balance-sheet expansions after 2008 were followed by below-target inflation in the United States, the euro area and Japan. Corrected: asset purchases can raise inflation if they raise demand (by lowering long-term rates and raising asset prices) or expectations, but there is no mechanical one-for-one link.

Solution 50.10. The supply shock shifts AS up; the demand boom shifts AD right: inflation rises sharply, with output near or above potential. With anchored expectations, π^e stays near target, so as the supply shock unwinds AS shifts back down, and a moderate tightening that shifts AD back left is enough; the economy does not need a prolonged negative output gap to wring expectations down, as it did in 1979–83 when π^e itself had risen to double digits.

Unconventional Policy and Financial Stability

After studying this chapter you should be able to

1. Explain why there is an effective lower bound on nominal interest rates and why it matters for stabilization policy.
2. Describe forward guidance, quantitative easing, negative rates, yield-curve control and targeted lending, record their balance-sheet effects and assess their channels and evidence.
3. Analyse bank runs with the Diamond–Dybvig model and show how deposit insurance and a lender of last resort eliminate the run equilibrium.
4. State Bagehot’s rule and explain the moral hazard created by rescues.
5. Explain how leverage, fire sales and the financial accelerator amplify shocks, and why credit booms predict crises.
6. Describe macroprudential policy and the Basel III capital and liquidity standards.
7. Discuss central bank digital currencies and stablecoins.
8. Analyse the Global Financial Crisis of 2007–09 and the banking stress of 2023 using accounting, micro and macro tools together.

A motivating question

On Monday 15 September 2008, Lehman Brothers, the fourth-largest U.S. investment bank, filed for bankruptcy. On Tuesday a large money-market fund that held Lehman’s short-term debt announced that it could no longer repay its investors a dollar for every dollar invested; investors withdrew hundreds of billions of dollars from similar funds within days. By the end of the week, the U.S. Treasury had guaranteed money-market funds, the Federal Reserve had lent tens of billions of dollars to the insurer AIG, and the Treasury Secretary and the Fed Chairman were asking Congress for \$700 billion to buy troubled assets. Within months the Fed had cut its policy rate to zero—and had to find other ways to support an economy in free fall.

This chapter covers two linked subjects. The first is what central banks do when they cannot lower interest rates any further: the *unconventional* tools used after 2008, in 2020, and in Japan for much longer. The second is *financial stability*: why financial systems are prone to crises, and what central banks and regulators do to prevent and manage them. Both subjects draw on every earlier part of the book: the information economics of Chapter 25, the coordination games of Chapter 23, the bank accounting of Chapters 30 and 48, and the macro models of Chapters 46 and 20.

51.1 The Effective Lower Bound

51.1.1 Why rates cannot fall without limit

Banknotes pay a nominal interest rate of zero. If deposit and bond rates fell far below zero, people could withdraw cash and hold it instead. Because storing and insuring large amounts of cash is costly, rates can go somewhat below zero—as several central banks demonstrated after 2012—but not far. There is therefore an **effective lower bound** (ELB) on nominal interest rates, close to or slightly below zero.

51.1.2 Why the bound matters

The Taylor rule of Chapter 50 can prescribe a negative nominal rate: in a deep recession with low inflation, Exercise 50.3 gave -0.5 per cent. At the bound, the central bank cannot deliver the prescribed stimulus. Worse, if inflation falls, the real rate $r = i - \pi^e$ rises with the nominal rate stuck at zero, tightening policy just when the economy needs the opposite, and risking a deflationary spiral: falling prices raise real rates and real debts, reducing spending and pushing prices down further (Chapter 48's debt deflation). In the New Keynesian model (Chapter 47), a large negative shock to the natural rate r^n with the nominal rate at zero produces a deep recession and deflation, and makes promises about the future especially powerful. The lower the neutral rate (Chapter 50), the more often the bound binds—one reason it became a central concern after 2008. Fiscal multipliers are plausibly larger at the bound, because the central bank does not offset fiscal expansion by raising rates (Chapter 46).

51.2 Unconventional Monetary Tools

51.2.1 Forward guidance

If today's rate cannot fall, the central bank can lower long-term rates by promising to keep short rates low for longer than markets expected (the expectations hypothesis, Chapter 49). The Fed's guidance evolved from open-ended language (rates would stay low "for an extended period," 2009) to calendar dates ("at least through mid-2013," 2011) to thresholds (no increase at least as long as unemployment remained above 6.5 per cent and projected inflation no more than half a point above 2 per cent, December 2012). The ECB and the Bank of England adopted guidance in 2013.

Guidance is effective only if credible. The advanced box of Chapter 50 showed that under commitment the central bank would like to promise future restraint—or, at the bound, future stimulus, keeping rates low even after the economy recovers and allowing inflation to overshoot—but that such promises are time inconsistent: when the time comes, the central bank is tempted to renege. In standard New Keynesian models, promises about the distant future have implausibly large effects (the "forward guidance puzzle" of Del Negro, Giannoni and Patterson), which has led to models with less forward-looking households and firms.

51.2.2 Quantitative easing

Mechanics. In **quantitative easing** (QE), the central bank buys large quantities of longer-term securities—government bonds, and in some programmes mortgage-backed securities, corporate bonds or equities—financed by creating reserves. If it buys 100 of government bonds from a pension fund:

Central bank			
DR	Government bonds	100	
	CR Reserves (of the pension fund's bank)		100
Pension fund's bank			
DR	Reserves at central bank	100	
	CR Deposits (pension fund)		100
Pension fund			
DR	Bank deposits	100	
	CR Government bonds		100

The central bank's balance sheet expands; bank reserves rise; the pension fund holds a deposit instead of a bond, so broad money rises by 100. The private sector as a whole now holds fewer long-term bonds and more short-term, floating-rate claims (reserves and deposits).

Channels. QE is thought to work through several channels:

- **Portfolio balance and duration.** By removing long-duration bonds from private portfolios, the central bank reduces the interest-rate risk investors must bear, lowering the term premium (Chapter 49). Investors who sold bonds rebalance into other assets—corporate bonds, equities, foreign assets—raising their prices and lowering their yields.
- **Signalling.** Buying long-term bonds signals that the central bank intends to keep policy rates low for a long time, reinforcing forward guidance (it would lose money on its bonds if it raised rates soon).
- **Market functioning.** In a panic, when dealers cannot absorb sales, central-bank purchases restore liquidity and prevent fire sales—the main purpose of purchases in March 2020, when even the U.S. Treasury market became dysfunctional.

Evidence. Event studies of announcements find large effects of the first rounds: Gagnon, Raskin, Remache and Sack (2011) estimated that the Fed's purchases announced in 2008–09 reduced 10-year yields by somewhere between about 30 and 100 basis points, mainly through lower term premia. Later rounds, anticipated by markets, had smaller measured announcement effects. Effects on output and inflation are harder to identify. The ECB's asset purchase programme from 2015 and its pandemic emergency purchase programme (PEPP, with an envelope of €1.85 trillion) also compressed spreads between euro-area sovereign bonds, an important consideration in a monetary union (Chapter 52).

The fiscal side. QE swaps long-term government debt for reserves that pay the policy rate. While rates are low, the central bank earns more on its bonds than it pays on reserves and remits profits to the treasury. When rates rise, the position reverses. From late 2022 the Federal Reserve paid more interest on reserves and reverse repos than it earned, suspended remittances and recorded a “deferred asset” that exceeded \$200 billion; the Bank of England's losses on its asset purchases are indemnified by the Treasury; the Bundesbank used up its provisions and reported losses. These losses do not impair a central bank's ability to operate—it cannot run out of its own currency—but they have fiscal consequences and can attract political criticism. *Quantitative tightening* (QT)—letting bonds mature without reinvestment, or selling them—reverses QE, draining reserves toward the level at which the floor system of Chapter 49 still works.

51.2.3 Negative interest rates

Several central banks set their deposit rates below zero: Danmarks Nationalbank from 2012, the ECB from June 2014 (reaching -0.5 per cent in 2019), the Swiss National Bank and Sweden’s Riksbank from 2015, and the Bank of Japan from 2016. Negative rates pass through to money-market rates and, via expectations, to longer yields, and they encourage depreciation of the currency. Their limits lie in the banks: if banks cannot pass negative rates on to retail depositors (few did), their margins are squeezed, which may reduce lending—and at some “reversal rate” further cuts could become contractionary. Central banks mitigated this with *tiering*, exempting part of banks’ reserves from the negative rate. All had returned to positive rates by 2024, the Bank of Japan last in March 2024.

51.2.4 Yield-curve control and targeted lending

Yield-curve control targets a longer-term yield directly, promising unlimited purchases to hold it. The Bank of Japan targeted its 10-year yield from 2016 to 2024 (Chapter 47’s case); the Reserve Bank of Australia targeted its 3-year yield in 2020–21, abandoning the target abruptly in November 2021 when inflation rose and market pressure mounted—a lesson in the credibility costs of exiting a commitment early.

Targeted lending programmes lend to banks cheaply on condition that they lend to firms and households: the Bank of England’s Funding for Lending Scheme (2012) and the ECB’s targeted longer-term refinancing operations (TLTROs, from 2014), whose most generous terms in 2020–21 allowed banks to borrow at rates as low as -1 per cent. Central banks also lent directly to non-banks in emergencies, as the Fed did through its commercial-paper facilities in 2008 and 2020 and its facilities for corporate bonds and mid-sized firms in 2020, often with fiscal backing from the Treasury to absorb credit losses.

► “QE is printing money to finance the government”

QE creates reserves to buy bonds *from the private sector*, in secondary markets, at a time chosen by the central bank to meet its inflation objective; the government must still issue and service the debt, and the central bank can reverse the purchases. Direct financing of the government by the central bank—*monetary financing*, prohibited for the ECB by Article 123 of the EU Treaty—is different: it is determined by the government’s needs, which is the fiscal dominance of Chapter 47. The distinction lies in who decides and why, not in the accounting entries, which look similar. That is also why the credibility of the institutional framework matters so much.

51.3 Bank Runs and the Lender of Last Resort

51.3.1 The Diamond–Dybvig model revisited

Chapter 23 presented a simple bank-run game. Two depositors each placed 100 in a bank that invested the 200 in a project yielding 240 at maturity but only 160 if liquidated early. The payoffs were:

		Depositor 2	
		Wait	Withdraw
Depositor 1	Wait	120, 120	60, 100
	Withdraw	100, 60	80, 80

Both (Wait, Wait) and (Withdraw, Withdraw) are equilibria. Douglas Diamond and Philip Dybvig (1983) showed that this fragility is inherent in the bank's useful function of turning illiquid assets into liquid deposits.

Proposition 51.1. Eliminating the run equilibrium

(a) *Lender of last resort.* If the central bank will lend the bank 100 against the project, repayable from the project's proceeds at maturity, then "Wait" is a strictly dominant strategy and the only equilibrium is (Wait, Wait). (b) *Deposit insurance.* If a credible insurer guarantees each depositor 100, "Wait" is weakly dominant and a depositor who expects the other to wait strictly prefers to wait; the run equilibrium disappears if depositors break ties by waiting. In both cases the guarantee is never used in equilibrium.

Proof. (a) If the other depositor withdraws, the bank borrows 100 from the central bank to pay him, keeps the project, and at maturity repays 100 from the 240, leaving 140 for the depositor who waited (ignoring interest). Waiting gives 140 against 100 for withdrawing; if the other waits, waiting gives 120 against 100. Wait dominates. (b) With insurance, a depositor who waits when the other withdraws receives at least 100, the same as withdrawing; if the other waits, waiting gives 120 against 100. No depositor gains by running. $\square$

The key insight is that a run is a coordination failure; a credible commitment to pay removes the reason to run, and therefore costs nothing if the bank is solvent. The United States introduced federal deposit insurance in 1934, after the Great Depression (Chapter 48); most countries now insure deposits up to a limit (\$250,000 per depositor per bank in the United States; €100,000 in the European Union; RM250,000 in Malaysia).

51.3.2 Bagehot's rule

Walter Bagehot's *Lombard Street* (1873) set out the classic doctrine: in a panic, the central bank should *lend freely*, to *solvent* institutions, against *good collateral*, at a *penalty rate*. Lending freely stops the run; restricting it to solvent institutions with good collateral protects the central bank and avoids bailing out the insolvent; the penalty rate discourages banks from relying on the central bank in normal times. The rule separates liquidity problems, which the central bank should solve, from solvency problems, which require capital from shareholders, creditors or, as a last resort, taxpayers.

In practice the distinction is hard to draw in a crisis, because asset values depend on whether the panic is stopped: assets that would be worth enough if held to maturity may be worth too little in a fire sale. And guarantees create **moral hazard** (Chapter 25): if banks, their creditors and uninsured depositors expect to be rescued, they take more risk and monitor less. Institutions whose failure would threaten the system are *too big to fail*—or too interconnected—and their creditors enjoy an implicit subsidy. Much post-2008 regulation, including capital surcharges for systemically important banks and resolution regimes that impose losses on bondholders (*bail-in*), aims to make failure possible without catastrophe.

Case: March 2023: SVB, the Bank Term Funding Program and Credit Suisse

Chapter 30 described how Silicon Valley Bank's unrealized losses on held-to-maturity securities roughly equalled its equity, and how uninsured depositors tried to withdraw about \$42 billion in a single day. The bank was closed on Friday 10 March 2023; Signature Bank followed on Sunday.

The response. On Sunday 12 March, U.S. authorities invoked a "systemic risk exception"

allowing the FDIC to protect *all* depositors of the two banks, including uninsured ones. The Federal Reserve created the Bank Term Funding Program, lending to banks for up to a year against Treasury and agency securities valued at *par*—their face value—rather than at their depressed market value. In Bagehot’s terms, the Fed lent freely but not against collateral valued at market prices: it lent against the value the securities would have at maturity, which is exactly the amount a bank holding them to maturity would receive. This neutralized the specific run incentive created by unrealized losses, without buying the bonds or recognizing the losses. Borrowing under the programme rose above \$160 billion before it stopped making new loans in March 2024. A few weeks later, First Republic Bank failed and was sold.

Across the Atlantic. Credit Suisse, a global systemically important bank weakened by years of losses and scandals, suffered large deposit outflows. On 19 March 2023 the Swiss authorities arranged its takeover by UBS, backed by liquidity assistance from the Swiss National Bank and a government guarantee; about CHF16 billion of Credit Suisse’s additional tier 1 bonds—a form of loss-absorbing capital—were written off entirely, while shareholders received UBS shares.

Lessons. Uninsured, concentrated deposits can run far faster than in the past, coordinated through social media and moved by mobile banking. Interest-rate risk, not only credit risk, can make a bank insolvent. The accounting classification of securities affects when losses become visible (Chapter 30). And the resolution frameworks designed after 2008 were bypassed in the United States by the systemic risk exception and in Switzerland by an emergency merger, rekindling debates about moral hazard and too-big-to-fail.

51.4 Why Financial Systems Are Fragile

51.4.1 Leverage and fire sales

Leverage amplifies losses (Chapter 48), and when many institutions manage their leverage actively, it amplifies price movements too. Tobias Adrian and Hyun Song Shin (2010) documented that investment banks’ leverage was *procyclical*: they expanded their balance sheets when asset prices rose and contracted them when prices fell.

Worked Example 51.2. A fire-sale spiral

A bank has assets of 100, funded by debt of 95 and equity of 5: leverage of 20. Asset prices fall by 1 per cent. Assets are now worth 99, debt is still 95, and equity has fallen to 4—a 20 per cent loss. Leverage has risen to $99/4 = 24.75$. If the bank wants to restore leverage of 20 (because its lenders or regulators require it), it must shrink its assets to $20 \times 4 = 80$, selling 19 of assets and repaying debt. If many banks hold similar assets and sell at the same time, prices fall further, reducing equity again and forcing more sales. A 1 per cent fall in prices has triggered sales of almost a fifth of the balance sheet.

Fire sales are an externality (Chapter 26): each bank’s sales lower the prices at which others must value their assets, but no bank takes that into account. They explain why the correlations among assets that looked diversified in normal times rise sharply in a crisis (Chapter 14’s point that pooling fails when risks are correlated), and why concentrated exposures to a common factor—one housing market, one sovereign—are so dangerous.

51.4.2 The financial accelerator and credit cycles

The *financial accelerator* of Bernanke, Gertler and Gilchrist (Chapter 47) operates through borrowers' balance sheets: a fall in asset prices lowers net worth, raising the external finance premium and reducing investment and consumption, which lowers asset prices further. In the Kiyotaki–Moore model, borrowing is limited by collateral values, so a fall in land or house prices reduces borrowing, spending and prices in a loop. The same mechanisms operate in reverse in booms: rising collateral values permit more borrowing, which raises prices further.

Moritz Schularick and Alan Taylor (2012), using data for 14 advanced economies since 1870, found that rapid growth of bank credit relative to GDP is the best single predictor of financial crises; Òscar Jordà, Schularick and Taylor showed that recessions preceded by credit booms are deeper and recoveries slower. Carmen Reinhart and Kenneth Rogoff's (2009) survey of eight centuries of crises found a pattern—rising leverage, asset-price booms, then busts followed by long recoveries and rising public debt—that recurs across countries and eras. Hyman Minsky's "financial instability hypothesis" holds that stability itself breeds instability, as borrowers and lenders become more confident and take on more risk.

51.4.3 Provisioning and procyclicality

Accounting rules can themselves amplify cycles. Under the expected-credit-loss models of IFRS 9 and U.S. GAAP's CECL (Chapter 28), banks recognize losses when they expect them, not when defaults occur. This is more timely than the old incurred-loss model, which was criticized for recognizing losses "too little, too late" in 2008. But it means that provisions—and hence reductions in equity—rise sharply at the *start* of a downturn, when forecasts deteriorate, which could lead banks to cut lending just when the economy weakens. In 2020, regulators allowed banks to phase in the capital effect of new provisions to limit this.

51.5 Macroprudential Policy

51.5.1 From micro to macro

Traditional *microprudential* regulation asks whether each bank, considered alone, is safe. The crisis of 2007–09 showed that the system can be fragile even when each bank appears sound, because of the fire-sale externalities, common exposures and interconnections described above. **Macroprudential policy** aims at the stability of the financial system as a whole. It has two dimensions:

- the *time dimension*: limiting the build-up of risk in booms (procyclicality);
- the *cross-sectional dimension*: limiting risks arising from interconnections and from institutions that are systemically important.

51.5.2 Basel III

The Basel Committee on Banking Supervision, hosted by the Bank for International Settlements, sets international standards that national authorities implement. The **Basel III** reforms, agreed after the crisis and finalized in 2017, include:

- **More and better capital.** Common equity tier 1 of at least 4.5 per cent of risk-weighted assets, tier 1 capital of at least 6 per cent and total capital of at least 8 per cent, plus a *capital conservation buffer* of 2.5 per cent of common equity, which banks can draw down in stress at the cost of restrictions on dividends and bonuses.

- **A countercyclical capital buffer** of up to 2.5 per cent (higher at national discretion), raised when credit grows rapidly and released in downturns—a direct time-dimension tool. Several authorities released their buffers in March 2020.
- **Surcharges for global systemically important banks** of 1 to 3.5 per cent, and domestic surcharges for domestically important banks.
- **A leverage ratio** of at least 3 per cent of tier 1 capital to total (unweighted) exposures, as a backstop against errors in risk weights.
- **Liquidity standards:** the liquidity coverage ratio and net stable funding ratio (Chapter 48), each at least 100 per cent.

Implementation of the final elements (sometimes called “Basel III endgame” or “Basel IV”) has proceeded unevenly across jurisdictions in the mid-2020s, with delays and modifications in the United States, the United Kingdom and the European Union.

Bridge to Chapter 30. Chapter 30 showed that, in the Modigliani–Miller benchmark, the required return on a firm’s equity falls as it becomes less levered, so the overall cost of funding is unchanged. Anat Admati and Martin Hellwig (2013) argued that this logic applies to banks: much higher equity requirements—20 to 30 per cent of assets—would make banks far safer at little social cost, because the apparent private cost of equity reflects mainly the tax advantage of debt and implicit guarantees, which are transfers rather than social costs. Bankers and some economists reply that deposits have special value as money and that more equity would push activity into less regulated non-banks. The debate shapes the level of the requirements above.

51.5.3 Other tools

- **Borrower-based measures:** limits on loan-to-value (LTV) and debt-to-income (DTI) ratios for mortgages, which constrain leverage directly at the source. Bank Negara Malaysia has capped LTV ratios on borrowers’ third and subsequent housing loans since 2010; many European and Asian authorities use DTI or debt-service limits.
- **Sectoral tools:** higher risk weights on particular exposures, such as commercial real estate. China’s “three red lines,” introduced in 2020, capped the borrowing of property developers according to their leverage, liquidity and debt ratios—a sectoral macroprudential measure that exposed the fragility of heavily indebted developers such as Evergrande.
- **Stress tests:** supervisors project banks’ capital under severe hypothetical scenarios and require those that fall short to raise capital. The U.S. Supervisory Capital Assessment Program of 2009, whose credible results helped restore confidence in U.S. banks, established the model.
- **Resolution regimes** that allow a failing bank to be restructured without taxpayer support, with loss-absorbing debt that can be converted into equity (bail-in), and central clearing of derivatives to reduce opaque bilateral exposures.

51.5.4 Monetary policy and financial stability

Should monetary policy “lean against the wind” of credit booms, raising rates beyond what inflation and output alone would warrant? The case for leaning is that macroprudential tools are imperfect and “rates get in all the cracks” (Jeremy Stein). The case against, made by Lars

Svensson among others, is that the costs of leaning—weaker output and inflation below target—are certain and immediate, whereas the benefits—a lower probability of a crisis—are small and uncertain. The prevailing view is a version of the Tinbergen principle that each objective needs its own instrument: interest rates for price stability, macroprudential tools for financial stability, with coordination between them.

51.6 Central Bank Digital Currencies and Stablecoins

Only banks can hold central-bank money in electronic form; the public holds central-bank money only as cash. A **central bank digital currency** (CBDC) would give the public direct access to electronic central-bank money. A *wholesale* CBDC, limited to financial institutions, would modernize settlement; a *retail* CBDC would be a digital form of cash.

The Bahamas launched the first retail CBDC in 2020; China's e-CNY has been piloted widely since 2020; the ECB began a preparation phase for a digital euro in 2023, and EU legislation to establish it has been under discussion. In January 2025, by contrast, the United States prohibited federal agencies from developing a CBDC. The main concern is *disintermediation*: if households moved deposits into a riskless CBDC, banks would lose cheap funding, and in a crisis the CBDC could make runs faster—the Diamond–Dybvig problem in digital form. Proposed designs therefore limit holdings (a limit of around €3,000 per person has been discussed for the digital euro) and pay no interest.

Privately issued stablecoins (Chapter 48) raise similar questions from the other direction: a large stablecoin holding short-term government securities is economically a narrow bank or money-market fund, subject to runs if its backing is doubted, and could draw deposits from banks. Regulation requiring high-quality reserves, redemption at par and supervision—as in the EU's MiCA and the 2025 U.S. legislation—treats them accordingly.

Case: The Global Financial Crisis, 2007–09

The build-up. From the late 1990s to 2006, U.S. house prices roughly doubled, and mortgage lending to riskier borrowers expanded rapidly. Mortgages were pooled and sliced into securities (Chapter 25's case on securitization), many rated AAA on the assumption that defaults across regions would be only weakly correlated. The securities were held, often with high leverage and short-term funding, by investment banks, by off-balance-sheet vehicles of commercial banks funded with asset-backed commercial paper, and by banks in Europe. Leverage of the large U.S. investment banks reached about 30.

The trigger. House prices peaked in 2006 and began to fall nationwide—the correlated shock that the ratings had treated as very unlikely. Defaults rose, and the value of mortgage securities became uncertain. On 9 August 2007 BNP Paribas suspended redemptions from three funds because it could not value their assets; the market for asset-backed commercial paper froze. Central banks injected liquidity. In March 2008 Bear Stearns, facing a run by its repo lenders, was sold to JPMorgan with Federal Reserve support.

The panic. In September 2008 the government-sponsored mortgage agencies Fannie Mae and Freddie Mac were taken into conservatorship; Lehman Brothers failed on 15 September; the Reserve Primary Fund “broke the buck” on 16 September, triggering a run on money-market funds; AIG, which had sold insurance against the default of mortgage securities, was rescued. Interbank lending seized up as every bank doubted the solvency of every other—the adverse selection of Chapter 25 in its most acute form.

Through the four lenses. In *microeconomic* terms, the crisis combined information asym-

metries (lenders who did not hold the loans they originated had weak incentives to screen, Chapter 25), moral hazard (implicit guarantees for large institutions), and a coordination failure (runs on repo, commercial paper and money-market funds, Chapter 23). In *accounting* terms, losses flowed through fair-value measurement of securities and through provisions on loans (Chapters 28 and 30), reducing equity at highly levered institutions; off-balance-sheet vehicles came back onto balance sheets when sponsors stood behind them. In *macroeconomic* terms, falling house prices and tighter credit cut consumption and investment—the financial accelerator—and U.S. unemployment rose to 10 per cent in October 2009. In *monetary* terms, the Fed cut its policy rate to 0–0.25 per cent in December 2008 and then turned to unconventional tools.

The response. The Federal Reserve lent on a vast scale, through new facilities for banks, dealers, commercial-paper issuers and money-market funds, and through currency swap lines with foreign central banks to meet the dollar funding needs of European banks. Its balance sheet grew from under \$1 trillion to over \$2 trillion by the end of 2008. It began QE with purchases of agency debt and mortgage-backed securities announced in November 2008, expanded in March 2009 to Treasury securities. Congress passed the \$700 billion Troubled Asset Relief Program in October 2008, used mainly to inject capital into banks. The 2009 stress tests, with a government commitment to provide capital to any bank that could not raise it privately, restored confidence. The transcripts of the Federal Open Market Committee’s meetings in 2008, published with a five-year lag, allow students to follow policy-makers’ real-time reasoning as the crisis unfolded.

The aftermath. Output losses were large and persistent, as Reinhart and Rogoff’s evidence predicted. The crisis produced the Basel III reforms, stress testing, resolution regimes, central clearing of derivatives, and macroprudential policy as a new field; it also set the stage for the euro-area sovereign crisis (Chapter 52).

Summary

- Because cash pays zero interest, nominal rates face an effective lower bound near zero. At the bound, falling inflation raises real rates, and conventional policy cannot provide stimulus.
- Unconventional tools include forward guidance, QE (which swaps long-term bonds for reserves and works through portfolio balance, signalling and market functioning), negative rates, yield-curve control and targeted lending. QE has fiscal consequences when rates rise.
- Bank runs are coordination failures (Diamond–Dybvig). A lender of last resort or credible deposit insurance eliminates the run equilibrium at no cost if banks are solvent. Bagehot’s rule: lend freely to solvent institutions against good collateral at a penalty rate. Guarantees create moral hazard.
- Leverage, fire sales, the financial accelerator and provisioning rules amplify shocks. Credit booms are the best predictor of crises.
- Macroprudential policy targets systemic risk in its time and cross-sectional dimensions, through Basel III capital and liquidity standards, countercyclical buffers, surcharges, borrower-based limits, stress tests and resolution regimes.
- CBDCs and stablecoins raise questions of bank disintermediation and run risk.
- The Global Financial Crisis and the 2023 banking stress illustrate the interaction of information problems, accounting, leverage, runs and central-bank responses.

Key terms

effective lower bound; deflationary spiral; forward guidance; forward guidance puzzle; quantitative easing; portfolio-balance channel; signalling channel; market functioning; quantitative tightening; deferred asset; negative interest rates; tiering; reversal rate; yield-curve control; targeted lending; monetary financing; bank run; Diamond–Dybvig model; lender of last resort; deposit insurance; Bagehot’s rule; moral hazard; too big to fail; bail-in; procyclical leverage; fire sale; financial accelerator; credit boom; financial instability hypothesis; expected-credit-loss provisioning; microprudential and macroprudential policy; Basel III; capital conservation buffer; countercyclical capital buffer; G-SIB surcharge; leverage ratio; loan-to-value limit; stress test; lean against the wind; central bank digital currency; disintermediation.

Exercises

Exercise 51.1 (The lower bound). The neutral real rate is 0.5 per cent, the inflation target 2 per cent, inflation 0.5 per cent and the output gap -5 per cent. (a) Compute the Taylor-rule rate. (b) If the policy rate is stuck at zero, what is the real rate, and how far is it above the rate the rule implies? (c) If inflation falls to -1 per cent, what happens to the real rate?

Exercise 51.2 (QE entries). The central bank buys 200 of government bonds: 120 from commercial banks and 80 from an insurance company. Record the entries for the central bank, the banks and the insurer. By how much do reserves and broad money change?

Exercise 51.3 (Central-bank income). A central bank holds 3,000 of bonds yielding an average of 2 per cent, financed by 3,000 of reserves. (a) Compute its net interest income when the interest on reserves is 0.25 per cent and when it is 4.5 per cent. (b) Explain why losses do not prevent the central bank from operating, and what their fiscal implications are.

Exercise 51.4 (Bank run). In the Diamond–Dybvig example, suppose the lender of last resort charges a penalty rate so that the bank must repay 110 at maturity for a loan of 100. Is “Wait” still dominant? What is the highest repayment consistent with “Wait” being dominant?

Exercise 51.5 (Insurance limits). Explain why SVB’s depositors ran even though the United States has deposit insurance. What does the case imply about the design of deposit insurance and about the value of the lender of last resort?

Exercise 51.6 (Fire sales). A bank has assets of 200, debt of 188 and equity of 12, and targets leverage of about 16.7. (a) Asset prices fall by 2 per cent. What are equity and leverage? (b) How many assets must it sell to restore its target? (c) Explain the externality involved when many banks do this.

Exercise 51.7 (Capital requirements). A bank has risk-weighted assets of 800, total exposures of 1,500 and common equity of 50. Its authorities require CET1 of 4.5 per cent plus a 2.5 per cent conservation buffer and a 1 per cent countercyclical buffer, and a leverage ratio of 3 per cent. (a) Does it comply? (b) Which constraint binds? (c) If the countercyclical buffer is released in a downturn, how much more risk-weighted lending could it support with the same equity?

Exercise 51.8 (Provisioning). Explain how expected-credit-loss provisioning could make bank lending more procyclical than incurred-loss provisioning, and why regulators might nevertheless prefer it.

Exercise 51.9 (Macroprudential). House prices and mortgage credit are rising rapidly, but inflation is at target. Compare raising the policy rate, raising the countercyclical buffer, and imposing a loan-to-value limit. Which would you recommend, and why?

Exercise 51.10 (GFC through four lenses). Explain in one paragraph each how the collapse of Lehman Brothers can be analysed as (a) a problem of asymmetric information, (b) an accounting event, (c) a macroeconomic shock and (d) a monetary-policy problem.

Solutions

Solution 51.1. (a) $i = 0.5 + 0.5 + 0.5(0.5 - 2) + 0.5(-5) = 1 - 0.75 - 2.5 = -2.25\%$. (b) Real rate $\approx 0 - 0.5 = -0.5\%$; the rule's real rate is $-2.25 - 0.5 = -2.75\%$, so policy is about 2.25 points too tight. (c) The real rate rises to $+1\%$ with the nominal rate stuck at zero: deflation tightens policy automatically.

Solution 51.2. Central bank: Dr Bonds 200 / Cr Reserves 200. Banks (selling 120): Dr Reserves 120 / Cr Bonds 120. Insurer's bank: Dr Reserves 80 / Cr Deposits (insurer) 80. Insurer: Dr Deposits 80 / Cr Bonds 80. Reserves rise by 200; broad money rises by 80 (only the purchase from the non-bank creates a deposit).

Solution 51.3. (a) Income = 60. Expense = 7.5 at 0.25 per cent: net income 52.5. At 4.5 per cent, expense = 135: net loss 75. (b) The central bank's liabilities are its own currency and reserves; it can always pay interest by creating reserves, and nobody can force it into insolvency. But its losses reduce or eliminate remittances to the treasury, which must be made up by other revenue or borrowing—they are a fiscal cost of the earlier QE, offsetting the profits earned while rates were low.

Solution 51.4. If the other withdraws, the waiting depositor receives $240 - 110 = 130 > 100$: Wait still dominates. Wait remains dominant as long as $240 - R > 100$, i.e. repayment $R < 140$. A penalty rate up to 40 per cent over the loan's term would still eliminate the run—so Bagehot's penalty rate is compatible with stopping runs.

Solution 51.5. Over 90 per cent of SVB's deposits exceeded the \$250,000 insurance limit, so most depositors were uninsured and had every reason to run once they doubted solvency. Insurance protects against runs only for the deposits it covers. The case suggests either broader coverage for operating accounts (with associated moral hazard and pricing), much tighter supervision of banks with concentrated uninsured deposits, or greater reliance on the lender of last resort—which requires that banks have pre-positioned collateral and that the central bank values it in a way that meets the need (as the BTFP did).

Solution 51.6. (a) Assets 196, equity $196 - 188 = 8$, leverage $196/8 = 24.5$. (b) Target assets = $16.7 \times 8 \approx 133$ (with target leverage $200/12 = 16.67$): sell about 63, almost a third of the balance sheet. (c) Each bank's sales depress prices, lowering the equity of all banks holding the same assets and forcing further sales; no bank accounts for the losses it imposes on others (a pecuniary externality), which justifies macroprudential limits on leverage.

Solution 51.7. (a) CET1 ratio = $50/800 = 6.25\%$, below the 8 per cent combined requirement; leverage ratio = $50/1,500 = 3.33\%$, above 3 per cent. It does not comply with the risk-weighted requirement. (b) The risk-weighted requirement binds (needs 64 of CET1). (c) Without the 1 per cent buffer, the requirement is 7 per cent: equity of 50 supports RWA of $50/0.07 = 714$ versus $50/0.08 = 625$ with it—about 89 more. (With current RWA of 800, it would still fall short and need to raise about 6 of equity.)

Solution 51.8. Under ECL, provisions rise as soon as forecasts deteriorate, reducing equity at the start of a downturn and possibly causing banks to cut lending then; under incurred loss, provisions rose later, when defaults occurred. Regulators prefer ECL because it recognizes losses earlier, giving a truer picture of banks' solvency and preventing the build-up of hidden losses that delayed the response in 2008; procyclicality can be managed with capital buffers that are released in downturns.

Solution 51.9. Raising the policy rate would slow credit but also tighten conditions for the whole economy, pushing inflation below target—a blunt instrument. A higher countercyclical buffer makes banks more resilient and modestly raises lending costs across all loans. An LTV limit targets mortgage leverage directly, reducing the risk of defaults and fire sales in the housing market. A combination of the LTV limit (and/or DTI limit) with the buffer is the usual recommendation, reserving the policy rate for inflation (the Tinbergen principle).

Solution 51.10. (a) Nobody could judge which institutions held how many losses on mortgage securities; lenders withdrew from all of them (adverse selection), and repo lenders ran. (b) Losses on fair-valued securities and provisions reduced Lehman's equity; with leverage around 30, a small fall in asset values wiped out equity; off-balance-sheet exposures and repo accounting obscured its position. (c) The failure froze credit markets and destroyed wealth and confidence, cutting investment and consumption worldwide—a large negative demand shock amplified by the financial accelerator. (d) The central bank cut rates to the lower bound, acted as lender of last resort to the system, provided dollar swap lines abroad, and then turned to QE—while facing the moral-hazard question of whom to rescue.

Open-Economy Monetary Economics, Crises and the Integrated Capstone

After studying this chapter you should be able to

1. Define nominal, real and effective exchange rates, and explain purchasing-power parity and its limits.
2. Apply covered interest parity, derive uncovered interest parity, and explain the evidence against uncovered parity.
3. Use Mundell–Fleming intuition to compare monetary and fiscal policy under fixed and floating exchange rates, and explain exchange-rate overshooting.
4. State the monetary-policy trilemma and classify exchange-rate regimes; explain the theory of optimal currency areas.
5. Record foreign-exchange intervention on the central bank's balance sheet and distinguish sterilized from unsterilized intervention.
6. Explain exchange-rate pass-through, sudden stops, currency mismatch and the main models of currency, banking and sovereign crises.
7. Analyse the Asian financial crisis, the euro-area sovereign crisis and Ukraine's wartime monetary policy by integrating microeconomics, accounting, macroeconomics and monetary economics.
8. Explain a single policy action—a central-bank rate increase—consistently at all four levels.

A motivating question

On 2 July 1997 Thailand stopped defending the peg of the baht to the U.S. dollar. Within months the currencies of Indonesia, Malaysia, the Philippines and South Korea had lost a third or more of their value; stock markets collapsed; banks and firms that had borrowed in dollars found their debts ballooning in local currency. In September 1998 Malaysia, rejecting the standard prescription of the International Monetary Fund, imposed controls on capital outflows and fixed the ringgit at 3.80 to the dollar—a peg it kept for almost seven years.

Why did economies that had grown rapidly for decades suffer such abrupt crises? Why did Malaysia's choice—fixed rate, controlled capital, independent interest rates—provoke such controversy? And what do these episodes have in common with the euro-area crisis of 2010–12 and with Ukraine's efforts to keep its currency, banks and prices stable under full-scale invasion? Answering requires every lens of this book at once: the incentives of borrowers, lenders and speculators; the balance sheets of firms, banks, governments and central banks; the macroeconomics of saving, investment and the current account; and the monetary economics of interest rates, reserves and credibility. This final chapter first adds the open-economy tools and then uses them, with everything else, to analyse crises and to complete the book's central

exercise.

52.1 Exchange Rates

52.1.1 Definitions

The **nominal exchange rate** E is the price of foreign currency in terms of domestic currency: for Malaysia, ringgit per U.S. dollar. A *rise* in E is a **depreciation** of the domestic currency (more ringgit are needed to buy a dollar); a fall is an **appreciation**. (Some countries, including the euro area and the United Kingdom, conventionally quote the inverse; always check the convention.)

The **real exchange rate** measures the relative price of foreign and domestic goods:

$$Q = \frac{E P^*}{P},$$

where P is the domestic price level and P^* the foreign price level. A rise in Q is a *real depreciation*: foreign goods become more expensive relative to domestic goods, which tends to raise exports and reduce imports. In growth rates,

$$g_Q = g_E + \pi^* - \pi.$$

Because countries trade with many partners, central banks also publish *effective* exchange rates, weighted averages against trading partners' currencies, in nominal and real terms.

52.1.2 Purchasing-power parity

Absolute purchasing-power parity (PPP) holds that the same basket of goods costs the same in every country when converted at the market exchange rate, so $Q = 1$ and $E = P/P^*$. It follows from the law of one price applied to all goods—an arbitrage argument (Chapter 22). *Relative* PPP holds that the real exchange rate is constant, so

$$g_E = \pi - \pi^* :$$

a country with inflation 3 points higher than its trading partners' sees its currency depreciate by about 3 per cent a year.

PPP fails badly in the short run: real exchange rates swing widely with nominal exchange rates, because goods prices are sticky (Chapter 47) while exchange rates move like asset prices. In the long run, deviations shrink, but slowly—Kenneth Rogoff's (1996) "PPP puzzle" is that estimated half-lives of real-exchange-rate deviations are three to five years, too long to be explained by price stickiness alone. PPP also fails systematically in levels: price levels are higher in rich countries. The *Balassa–Samuelson effect* explains why: productivity growth in rich countries is concentrated in traded goods (manufacturing), whose prices are equalized internationally, so wages are high everywhere in the economy, raising the prices of non-traded services such as haircuts and restaurant meals (Chapter 44's productivity differences reappear in price levels). This is why GDP comparisons across countries use PPP conversion rather than market exchange rates (Chapter 43).

52.2 Interest Parity

52.2.1 Covered interest parity

Chapter 42 derived **covered interest parity** (CIP) from an arbitrage argument (Theorem 42.1). In the notation of this chapter, with E the spot rate in ringgit per dollar, F the one-year forward rate, and i and i^* the Malaysian and U.S. interest rates, it reads

$$1 + i = \frac{F}{E} (1 + i^*) :$$

investing a ringgit at home must yield the same as converting it into dollars, investing at the U.S. rate and selling the proceeds forward. If the spot rate is 4.50 ringgit per dollar and one-year rates are 3 per cent in Malaysia and 5 per cent in the United States, the forward rate must be $4.50 \times 1.03/1.05 = 4.414$: the dollar trades at a forward discount that exactly offsets its higher interest rate. CIP held very closely before 2008; since then small but persistent deviations (the “cross-currency basis”) have appeared, reflecting the balance-sheet costs to banks of arbitrage under post-crisis regulation (Chapter 51) and strong demand for dollar funding.

52.2.2 Uncovered interest parity

If investors do not cover their positions but are risk-neutral, they will equalize *expected* returns:

$$1 + i = \frac{\mathbb{E}[E_{+1}]}{E} (1 + i^*), \quad \text{approximately} \quad \frac{\mathbb{E}[E_{+1}] - E}{E} = i - i^*.$$

This is **uncovered interest parity** (UIP): the currency with the higher interest rate is expected to depreciate by the interest differential. In the example, the ringgit is expected to *appreciate* against the dollar by about 1.9 per cent, compensating investors for its lower interest rate.

► **“High interest rates mean a currency will depreciate, so they are bad for the currency”**

UIP concerns the *expected change* in the exchange rate *given* the interest differential. When a central bank *unexpectedly* raises its rate, the currency typically *appreciates immediately*—so that it can then be expected to depreciate at the new, larger differential. The level jumps up; the expected path afterwards slopes down. Confusing the two is a common error in reading exchange-rate news.

Empirically, UIP fails over short and medium horizons: high-interest currencies have tended, on average, to depreciate by less than the differential and often to appreciate—the *forward premium puzzle*. This makes the *carry trade* (borrowing in low-interest currencies such as the yen and investing in high-interest ones) profitable on average, though it is exposed to sudden crashes when risk appetite collapses and investors unwind positions together. The usual interpretation is a time-varying *risk premium*: investors demand compensation for holding currencies that tend to fall in bad times. Adding a risk premium ρ , UIP becomes $\mathbb{E}[\Delta E]/E = i - i^* - \rho$, and shifts in ρ —capital flight in a crisis, or a sudden preference for dollars—move exchange rates independently of interest rates.

52.3 Monetary and Fiscal Policy in the Open Economy

52.3.1 Mundell–Fleming intuition

Robert Mundell and Marcus Fleming extended the IS–LM model to a small open economy with perfect capital mobility in the early 1960s. The essential logic can be expressed with the

IS–MP framework of Chapter 46 and UIP. Net exports now depend on the real exchange rate, so the IS curve becomes

$$\tilde{Y} = -a(r - \bar{r}) + b(q - \bar{q}) + \epsilon_d,$$

where q is the (log) real exchange rate. With perfect capital mobility, the domestic interest rate cannot differ from the world rate except by expected depreciation.

Floating exchange rate. *Monetary policy is powerful.* A cut in the domestic interest rate lowers investment and consumption costs *and* depreciates the currency, raising net exports: two channels reinforce each other. *Fiscal policy is weakened.* A fiscal expansion raises demand and pushes up domestic interest rates; capital inflows appreciate the currency, and falling net exports crowd out much of the stimulus.

Fixed exchange rate. *Monetary policy is lost.* To hold the exchange rate, the central bank must keep its interest rate equal to the anchor country's: any attempt to cut rates triggers capital outflows and pressure for depreciation, which the central bank must resist by selling reserves and draining domestic money until rates return to the world level. *Fiscal policy is powerful.* A fiscal expansion that pushes up interest rates attracts inflows; to prevent appreciation, the central bank must buy foreign currency and expand the money supply, accommodating the expansion.

The model's lessons are qualitative but durable: under a peg, a country imports its monetary policy from the anchor and must rely on fiscal policy for stabilization—the predicament of individual euro-area members (below).

Advanced Analysis: Exchange-rate overshooting

Rudiger Dornbusch (1976) explained why floating exchange rates are so volatile. Suppose a permanent monetary expansion will, in the long run, raise the price level and hence (by PPP) the exchange rate by 5 per cent. Prices are sticky, so in the short run the expansion lowers the domestic interest rate—say by 1 point below the world rate for two years. By UIP, the domestic currency must then be expected to *appreciate* by 1 per cent a year during those two years. The only way the currency can be expected to appreciate toward a long-run level 5 per cent weaker than before is if it first depreciates by *more* than 5 per cent: it jumps by about 7 per cent on impact and then appreciates by 1 per cent a year to its new long-run level. The exchange rate *overshoots*. Rational expectations plus sticky prices generate exchange-rate volatility larger than the volatility of the fundamentals, with no irrationality or speculation required.

52.3.2 The trilemma

The fixed-rate analysis generalizes into one of the most useful propositions in international economics.

Proposition 52.1. The monetary-policy trilemma

A country cannot simultaneously have (1) a fixed exchange rate, (2) free capital mobility, and (3) an independent monetary policy. It can choose at most two.

Proof. With free capital mobility and a credibly fixed exchange rate, expected depreciation is zero, so UIP forces $i = i^*$ (up to a risk premium the central bank does not control). Monetary

policy cannot then be set independently. Giving up any one of the three removes the contradiction: capital controls allow $i \neq i^*$ with a fixed rate; a floating rate allows $i \neq i^*$ with expected depreciation absorbing the difference. $\square$

Choice	Gives up	Examples
Float + capital mobility	Fixed rate	United States, euro area as a whole, United Kingdom, Japan; Malaysia since 2005 (managed float)
Fixed + capital mobility	Monetary independence	Individual euro-area members (Germany, Greece); Hong Kong's currency board; Denmark's peg to the euro
Fixed + monetary independence	Capital mobility	Bretton Woods system (1944–71); Malaysia 1998–2005; China for much of its history (managed rate with capital controls)

Hélène Rey (2013) argued that for many countries the choice is harsher: a *global financial cycle* driven largely by U.S. monetary policy and global risk appetite moves capital flows, credit and asset prices worldwide, so even floating does not fully insulate a country with open capital markets—a “dilemma” rather than a trilemma, unless capital flows are managed through macroprudential tools or controls. The IMF's position has shifted accordingly: its 2012 institutional view, updated in 2022, accepts capital-flow management measures in some circumstances, a notable change since it criticized Malaysia's controls in 1998.

52.4 Exchange-Rate Regimes

52.4.1 The spectrum

Regimes range from *hard pegs*—currency unions (the euro), *currency boards* (Hong Kong since 1983, which backs every Hong Kong dollar issued with U.S. dollar reserves at a fixed rate), and official dollarization (Ecuador, El Salvador)—through *soft pegs* (fixed but adjustable rates, crawling pegs, bands), *managed floats* (the exchange rate is market-determined but the central bank intervenes to smooth movements), to *free floats*. The IMF classifies regimes by what countries do, not what they say: many countries that claim to float intervene heavily—a “fear of floating” (Calvo and Reinhart 2002) driven by currency mismatches and pass-through to inflation.

52.4.2 Choosing a regime

A peg offers low transaction costs and exchange-rate risk in trade and investment with the anchor, and can import the anchor's monetary credibility—a way for a high-inflation country to tie its hands (Chapter 50's commitment problem). Its costs are the loss of monetary policy to respond to domestic shocks and vulnerability to speculative attack if the peg is not fully credible. A float allows monetary independence and lets the exchange rate absorb shocks, at the cost of volatility.

Mundell's (1961) theory of **optimal currency areas** asks when a group of regions benefits from sharing a currency. Sharing is more attractive when trade among members is intensive; when shocks are symmetric, so a single monetary policy suits all; when labour moves freely between regions; when wages and prices are flexible; and when a common fiscal system transfers resources to regions in recession. The United States scores well on labour mobility

and fiscal transfers; the euro area, at its creation in 1999, scored less well on both, which became painfully evident after 2010.

52.5 Foreign-Exchange Intervention

52.5.1 On the balance sheet

When a central bank buys foreign currency to resist appreciation, it pays with newly created reserves. Suppose Bank Negara Malaysia buys 1 billion U.S. dollars from a commercial bank at 4.50 ringgit per dollar:

Central bank (RM billions)			
DR	Foreign-exchange reserves	4.5	
	CR Reserves of commercial banks		4.5

This is *unsterilized* intervention: domestic reserves rise, pushing down domestic interest rates, which itself tends to weaken the currency. To keep its interest-rate target, the central bank can **sterilize** the intervention by draining the reserves—selling government securities or, as many emerging-market central banks do, issuing its own bills:

DR	Reserves of commercial banks	4.5	
	CR Central-bank bills issued		4.5

After sterilization, the monetary base is unchanged, but the central bank holds more foreign assets and the private sector holds more domestic-currency bills and fewer dollars. In the balance of payments, the purchase is recorded as an increase in reserve assets (Chapter 43).

52.5.2 Does sterilized intervention work?

Under perfect capital mobility and perfect substitutability of domestic and foreign bonds, sterilized intervention has no effect: it merely swaps one asset for another that investors regard as equivalent. It can work if assets are imperfect substitutes (a *portfolio-balance* effect: investors must be induced by a change in the exchange rate to hold fewer dollars), if it signals future monetary policy, or if it provides liquidity in a disorderly market. Evidence suggests sizeable effects in emerging markets with less liquid currency markets and smaller effects for major currencies.

There is a fundamental asymmetry. A central bank resisting *appreciation* creates domestic currency, of which it has unlimited supply; it can accumulate reserves indefinitely, as China did in the 2000s, though at the cost of sterilization (paying interest on its bills while earning low returns on reserves) and possibly of inflation. A central bank resisting *depreciation* must sell foreign currency, of which it has a limited stock; once markets believe reserves will run out, they attack.

52.5.3 Reserves as insurance

This asymmetry explains why many emerging economies accumulated large reserves after the crises of the late 1990s. A common benchmark, the *Greenspan–Guidotti rule*, holds that reserves should at least cover external debt falling due within a year; the IMF uses broader metrics. Malaysia's reserves rose from about \$20 billion at the trough in 1998 to over \$100 billion; China's have exceeded \$3 trillion since 2011. Reserves are costly—they earn less than the country pays on its external liabilities—but they insure against sudden stops.

52.6 Pass-Through, Sudden Stops and Balance Sheets

52.6.1 Exchange-rate pass-through

A depreciation raises the domestic-currency price of imports. The proportion of an exchange-rate change that shows up in import prices and consumer prices is **pass-through**. If imports are 30 per cent of the consumer basket and half of a 10 per cent depreciation is passed through to their prices, consumer prices rise by about $0.3 \times 0.5 \times 10 = 1.5$ per cent. Pass-through to consumer prices is typically lower in advanced economies with credible inflation targets than in emerging economies, and has declined as inflation has become better anchored (Chapter 50). Because most world trade is invoiced in dollars—the “dominant currency paradigm” of Gita Gopinath and co-authors—a dollar appreciation raises import prices in many countries at once, regardless of their bilateral trade with the United States.

52.6.2 Sudden stops

A country with a current-account deficit is borrowing from abroad (Chapter 43): $CA \equiv S - I$. The deficit can be sustained only as long as foreigners are willing to lend. A **sudden stop**—a sharp reversal of capital inflows, identified by Guillermo Calvo—forces the current account toward balance almost overnight. If a country with a deficit of 6 per cent of GDP loses its financing and has few reserves, domestic absorption ($C + I + G$) must fall by around 6 per cent of GDP relative to income, through some combination of a large real depreciation, which switches spending to domestic goods, and a fall in output, which reduces imports. Sudden stops have preceded most emerging-market crises.

52.6.3 Currency mismatch and original sin

Chapter 43 introduced currency mismatch: when a country’s residents owe debts in foreign currency but earn income in domestic currency, a depreciation raises the domestic-currency value of their debts. For a firm with ringgit revenue and dollar debt, a 40 per cent depreciation raises its debt by 40 per cent in ringgit, possibly wiping out its equity overnight—a balance-sheet effect that can make depreciation *contractionary*, overwhelming the expansionary effect on exports. Barry Eichengreen and Ricardo Hausmann called the inability of many emerging economies to borrow abroad in their own currency **original sin**. Its partial redemption—the growth of local-currency government bond markets held by foreigners in the 2000s and 2010s—shifted exchange-rate risk to foreign investors, but made capital flows sensitive to global risk appetite.

52.6.4 Models of crises

- **First-generation models** (Krugman 1979): a government runs deficits financed by money creation while pegging the exchange rate. Reserves fall steadily; speculators, anticipating the inevitable collapse, attack when reserves reach a threshold, exhausting them at once. The crisis is caused by inconsistent *fundamentals*—fiscal dominance (Chapter 47) meeting a peg.
- **Second-generation models** (Obstfeld 1994): a government weighs the benefits of keeping a peg (credibility) against its costs (high interest rates and unemployment). If speculators attack, defending the peg becomes costlier, which may make abandoning it optimal: an attack can be self-fulfilling, with multiple equilibria (Chapter 23). The 1992–93 crisis of the European Exchange Rate Mechanism, in which the United Kingdom left the mechanism on “Black Wednesday,” fits this pattern.

- **Third-generation models** emphasize balance sheets: currency mismatches, bank fragility and implicit guarantees. Kaminsky and Reinhart (1999) documented *twin crises*, in which banking and currency crises occur together and reinforce each other: depreciation damages borrowers with dollar debts, which damages banks, which prompts capital flight, which deepens the depreciation.
- **Sovereign-debt crises:** a government that borrows in a currency it does not control—foreign currency, or a shared currency such as the euro—can be forced to default, and fear of default can be self-fulfilling through higher interest rates (Chapter 47). When domestic banks hold large amounts of their own government's debt and the government stands behind the banks, a *sovereign–bank doom loop* links their solvency.

Case: The Asian financial crisis and Malaysia's capital controls

Build-up. In the early and mid-1990s, Thailand, Malaysia, Indonesia and South Korea grew rapidly with high investment (Chapter 44), much of it financed by capital inflows. Currencies were pegged or tightly managed against the dollar, so borrowers treated dollar loans as cheaper than local-currency loans and ignored the exchange-rate risk—a moral hazard encouraged by the apparent official commitment to the pegs (Chapter 25). Thailand's short-term foreign debt came to exceed its usable reserves, and banks and finance companies funded long-term property and corporate loans with short-term foreign borrowing: maturity *and* currency mismatch on the same balance sheets. Malaysia's external debt was less short-term, but its corporate sector was highly leveraged and its stock market had boomed on inflows.

Crisis. After exhausting much of its reserves defending the baht, Thailand floated on 2 July 1997. Contagion spread through the region as investors reassessed similar vulnerabilities. The ringgit fell from about 2.5 per dollar before the crisis to a low near 4.9 in January 1998. Firms with dollar debts suffered huge losses; banks' non-performing loans soared (the twin-crisis dynamic). Output fell by about 7 per cent in Malaysia in 1998 and by around 13 per cent in Indonesia.

Policy choices. Thailand, Indonesia and Korea accepted IMF programmes, which initially required high interest rates to stabilize currencies and fiscal tightening, and closure of weak financial institutions. Malaysia refused. On 1 September 1998 it imposed controls on capital outflows (restricting the repatriation of portfolio investment for a year and ending offshore ringgit trading), pegged the ringgit at 3.80 per dollar, and cut interest rates. In trilemma terms, Malaysia gave up capital mobility to combine a fixed rate with monetary independence. It also set up agencies to buy banks' bad loans (Danaharta) and recapitalize banks (Danamodal)—restoring bank equity so that lending could resume (Chapters 48 and 51).

Assessment. Malaysia's recovery was at least as fast as its neighbours', and Kaplan and Rodrik (2002) argued that the controls allowed a quicker recovery with lower interest rates; critics noted that by September 1998 regional markets had begun to stabilize, and that the controls may have raised Malaysia's cost of capital later. The controls on outflows were relaxed from 1999, and the peg was replaced by a managed float in July 2005. Malaysia subsequently ran persistent current-account surpluses and rebuilt its reserves (Chapter 43), developed a deep local-currency bond market, and strengthened bank supervision and capital—lessons that most emerging economies drew in similar ways.

Case: The euro-area sovereign crisis, 2010–12

A currency union without fiscal union. When the euro was created in 1999, national interest rates converged: Greek, Portuguese, Spanish and Irish governments and banks could borrow almost as cheaply as Germany's. Capital flowed from the core to the periphery, financing construction booms in Ireland and Spain and government deficits in Greece. Wages and prices in the periphery rose faster than in Germany, which restrained its own wages in the 2000s; the periphery's real exchange rates appreciated and its current-account deficits widened, mirrored by Germany's surpluses (Chapter 43). Within the union, no exchange rate could adjust.

The crisis. In October 2009 a new Greek government revealed that its deficit was far larger than reported—a failure of fiscal accounting, later revised to over 15 per cent of GDP. Greek yields soared; in May 2010 Greece received its first EU–IMF programme, followed by Ireland (whose government had guaranteed its banks' liabilities, turning a banking crisis into a sovereign one), Portugal, Spain (for its banks) and Cyprus. In 2012 Greece restructured its privately held debt, imposing losses of over half the face value on bondholders. Italian and Spanish ten-year spreads over Germany rose above 5 percentage points in 2011–12.

Four lenses. *Micro:* moral hazard (markets had priced periphery debt as if the union would stand behind it) and a coordination game among bondholders—if others sell, the risk of default rises, so sell (Chapter 23). *Accounting:* banks held large amounts of their own sovereign's bonds, often at low risk weights; falling bond prices eroded bank capital, while governments' guarantees of banks added to sovereign debt—the doom loop. Under the Eurosystem's payment system, deposit flight from periphery banks to German banks showed up as rising TARGET2 claims of the Bundesbank on the ECB (and liabilities of periphery central banks), which exceeded €700 billion in 2012—balance-sheet entries recording that the Eurosystem was refinancing the periphery's banks as private funding fled. *Macro:* with no exchange rate to adjust, periphery economies had to regain competitiveness by *internal devaluation*—falling wages and prices relative to the core—in the middle of fiscal austerity, a combination that produced deep recessions; Greek GDP fell by about a quarter. *Monetary:* the ECB could not lower the policy rate for one country, and its mandate prohibited monetary financing (Chapter 51).

“Whatever it takes.” On 26 July 2012 ECB President Mario Draghi said that “within our mandate, the ECB is ready to do whatever it takes to preserve the euro. And believe me, it will be enough.” In September the ECB announced Outright Monetary Transactions: potentially unlimited purchases of the short-term bonds of countries that agreed to an EU programme. Spreads fell sharply, although OMT was never used. A lender of last resort for sovereigns, if credible, eliminates the bad equilibrium of a self-fulfilling debt run—exactly as for banks in Chapter 51. The crisis also produced a banking union: from 2014 the ECB became the supervisor of the euro area's significant banks, with a single resolution mechanism.

Case: Ukraine: monetary policy under full-scale war

The National Bank of Ukraine (NBU) offers a rare case of monetary and financial management under an extreme real shock, documented in its own Inflation Reports and Financial Stability Reports.

Before the invasion. Ukraine had adopted inflation targeting with a floating hryvnia in 2015–16, after a crisis in 2014–15 in which the currency lost two-thirds of its value and inflation exceeded 40 per cent. The NBU had cleaned up the banking system, closing about a hundred insolvent banks and nationalizing PrivatBank, the largest, in 2016. Its

target was 5 per cent inflation.

The shock (micro and macro). After 24 February 2022, a large part of the capital stock was destroyed, millions of people were displaced, ports were blockaded and energy infrastructure was attacked. Real GDP fell by about 29 per cent in 2022. The government's revenue collapsed while defence spending soared; the budget deficit, excluding grants, reached around a quarter of GDP. In AD–AS terms (Chapter 46), both curves shifted violently left.

The monetary response. On the first day of the invasion, the NBU fixed the exchange rate at 29.25 hryvnias per dollar and imposed capital controls and restrictions on foreign-currency withdrawals. In trilemma terms, it gave up capital mobility to stabilize the currency: in wartime, a fixed rate was the only credible nominal anchor, and a panic-driven depreciation would have fed inflation and damaged balance sheets. The NBU financed the government directly in the first months—buying about 400 billion hryvnias of government bonds in 2022—accepting temporary monetary financing (Chapter 47's fiscal dominance) because there was no alternative before foreign aid arrived. It raised its policy rate from 10 to 25 per cent in June 2022, and in July devalued the fixed rate by about 25 per cent, to 36.57.

The accounts. Monetary financing appeared on the NBU's balance sheet as government bonds on the asset side and reserves or government deposits on the liability side (Chapter 49). To prevent the resulting liquidity from flowing into the foreign-exchange market, the NBU made hryvnia assets more attractive: it paid its policy rate on banks' holdings of its certificates of deposit and, from 2023, allowed banks to meet part of their required reserves with designated government bonds—tools that absorbed liquidity and supported the government-bond market. The banking system, recapitalized before the war, remained liquid and profitable, with deposits growing as households saved.

Inflation. Inflation peaked at 26.6 per cent at the end of 2022. As foreign financing from the European Union, the United States, the IMF and others replaced monetary financing from 2023, and as reserves rose to over \$40 billion by the end of 2023—a record—inflation fell to 5.1 per cent in December 2023. The NBU began cutting its policy rate in July 2023.

The return toward inflation targeting. In October 2023 the NBU moved from the fixed rate to “managed flexibility,” letting the hryvnia float with intervention to smooth movements, and gradually eased capital controls—its announced strategy of greater exchange-rate flexibility and a return to inflation targeting. Inflation rose again in 2024, driven by poor harvests, electricity shortages from attacks on power plants, and labour shortages, and the NBU raised its policy rate from 13 to 15.5 per cent between December 2024 and March 2025.

Four lenses. *Micro:* households' incentives to hold dollars (dollarization), exporters' incentives under surrender requirements, and banks' incentives shaped by the returns the NBU offered on hryvnia instruments. *Accounting:* the balance sheets of the NBU, the government and banks recorded the financing of the war and the absorption of liquidity. *Macro:* a collapse of potential output, a huge fiscal deficit financed by foreign aid (a transfer in the current account and a liability-free financing of $G - T$, Chapter 43), and supply-driven inflation. *Monetary:* the choice of nominal anchor under extreme uncertainty, the temporary acceptance of fiscal dominance, the use of capital controls as the trilemma dictates, and the gradual rebuilding of credibility. That inflation fell to single digits within two years of the invasion's inflation peak, with a functioning banking system, is a striking demonstration of the value of institutional preparation and credible policy frameworks.

52.7 The Integrated Capstone: One Event at Four Levels

The book began by promising that a reader who had mastered it could take a single event and explain it at four levels without contradiction. This section carries out that exercise for the event with which Chapter 49 began: a central-bank rate increase. The recurring question is the one in Figure 52.1: *whose constraint changes?*

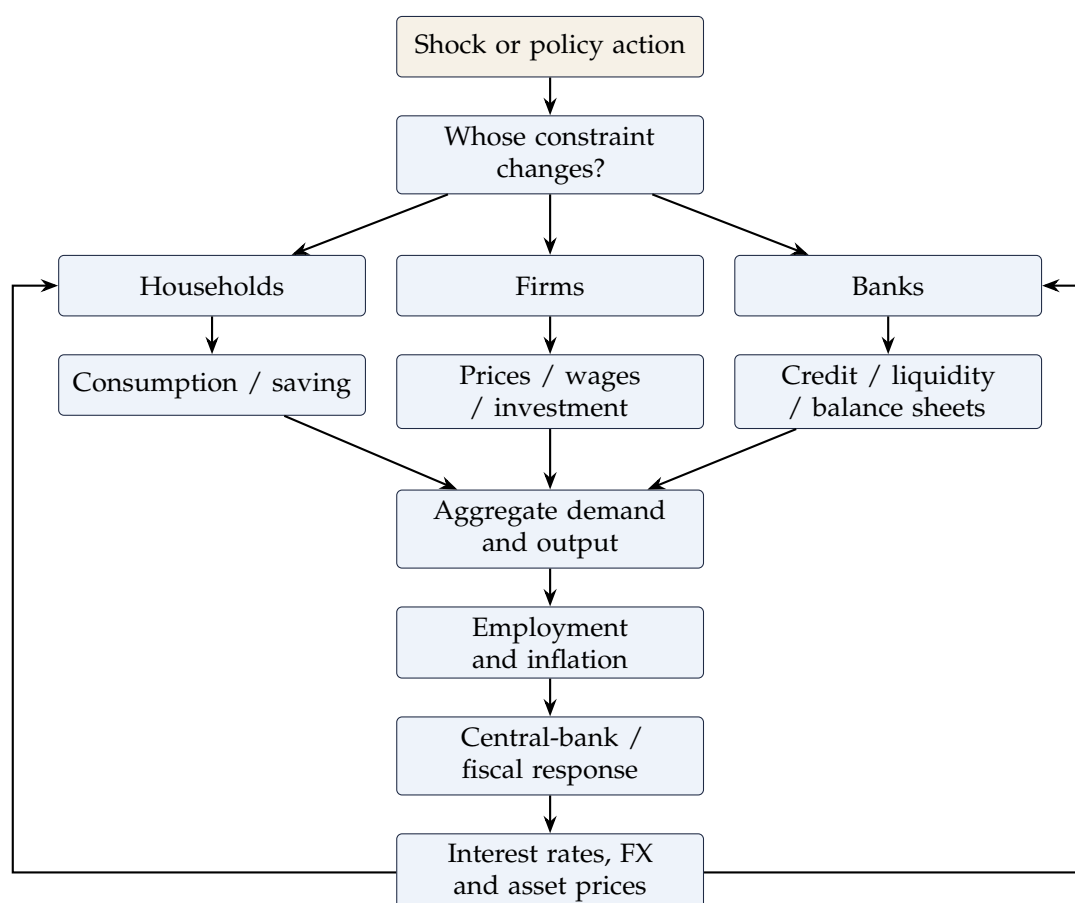


Figure 52.1: The integrated flowchart. Every policy case in this book can be traced through it: a shock changes some agents' constraints; their decisions aggregate into demand, output, employment and inflation; policy responds; financial prices change; and the new prices feed back into everyone's constraints.

52.7.1 The event

A small open economy with an inflation target of 2 per cent has inflation of 4 per cent and a positive output gap. Its central bank, operating a floor system, raises its policy rate by 1 percentage point, from 3 to 4 per cent, and signals that further increases are possible.

52.7.2 Level 1: implementation (monetary)

The central bank raises the interest rate it pays on reserves and the rate on its standing facilities by 1 point (Chapter 49). No quantity of reserves needs to change. Overnight interbank and repo rates rise by about 1 point within a day. Longer-term yields rise by less or more than 1 point, depending on how the move and the accompanying guidance change expectations of

future policy rates and the term premium (the expectations hypothesis, Chapter 49): if markets had already expected the increase, long rates barely move; if the guidance signals more to come, they rise further. By uncovered interest parity, the currency appreciates on impact (the pitfall in Section 52.2). If the central bank is credible, measures of expected inflation edge down (Chapter 50).

52.7.3 Level 2: decisions (microeconomic)

- **Households** face a higher real interest rate, because inflation expectations move less than the nominal rate: the Euler equation tilts consumption toward the future (Chapter 45), modestly if the elasticity of intertemporal substitution is small. More importantly, a household with a variable-rate mortgage of 200,000 pays about 2,000 a year more in interest—a cash-flow effect that, for a liquidity-constrained household, reduces spending almost one-for-one (Chapters 45 and 49). Savers earn more, but with lower marginal propensities to consume.
- **Firms** face a higher user cost of capital: with depreciation of 10 per cent and a borrowing rate rising from 5 to 6 per cent, the user cost rises from 15 to 16 per cent, and the desired capital stock falls by about 6 per cent (Chapter 45's $K^* = \alpha Y / uc$). How much of their higher financing costs they pass on in prices depends on market structure (Chapter 24): firms with market power and inelastic demand pass on more. Exporters lose competitiveness from the appreciation.
- **Banks** earn more on reserves and floating-rate loans at once and pay more on deposits with a lag—how long a lag depends on competition for deposits (Chapter 24). Riskier borrowers face higher premia; credit standards may tighten, especially for borrowers with little collateral (the credit channel, Chapter 25).

52.7.4 Level 3: the accounts

- **Firm income statements:** interest expense rises on floating-rate debt; interest coverage falls; for a firm with 120 of floating-rate debt, a 1-point rise adds 1.2 to annual interest expense (Chapter 49's example scaled down). Cash flow from operations falls by the same amount (interest paid is an operating cash flow under U.S. GAAP; under IFRS it has been a choice between operating and financing until IFRS 18 takes effect; Chapters 27 and 31). Discount rates used in impairment tests rise, so the recoverable amounts of assets and goodwill may fall, triggering impairments (Chapter 29).
- **Firm and bank balance sheets:** the market value of fixed-rate bonds and loans falls. For a bank holding a 10-year bond, a 1-point rise in yields lowers its value by roughly 8 per cent (its duration). Where the bond is at fair value through profit or loss or through other comprehensive income, the loss reduces equity at once; where it is at amortized cost, the loss is disclosed but not recognized (Chapter 30)—unless the bank must sell (Chapter 51's SVB case). Expected-credit-loss provisions rise as forecasts of defaults deteriorate (Chapter 28).
- **Liabilities:** the present value of long-term obligations such as pensions and provisions falls, because they are discounted at higher rates—the same interest-rate change can therefore *raise* reported equity for a company with large pension obligations (Chapter 30).
- **The central bank and government:** the central bank pays more interest on reserves, reducing its profits and remittances to the treasury (Chapter 51); the government pays more on new and floating-rate debt, raising $r - g$ in the debt-dynamics equation (Chapter 47).

52.7.5 Level 4: the aggregates (macroeconomic)

In the IS curve (Chapters 46 and 47), higher real rates reduce consumption and investment; the appreciation reduces net exports. With the multiplier and the accelerator (Chapter 45), output falls relative to potential over the following quarters; the output gap narrows. Okun's law translates the smaller gap into higher unemployment—by about half a point for each point of output gap (Chapter 46). The Phillips curve translates a smaller gap into lower inflation, with a lag of one to two years; the appreciation lowers import prices sooner, through pass-through: if the currency appreciates by 3 per cent, imports are 30 per cent of the basket and half the change passes through, consumer prices are about $0.3 \times 0.5 \times 3 = 0.45$ per cent lower than otherwise. If expectations are anchored, inflation returns toward target with a modest rise in unemployment; if not, a larger output loss is needed (the sacrifice ratio, Chapter 50).

In the national accounts (Chapter 43), the effects appear as lower growth of consumption and investment, lower net exports, and—on the income side—lower corporate profits (higher interest paid, lower sales) and eventually slower growth of compensation. *In the sectoral balances*, if firms cut investment more than they cut saving, the corporate sector's net borrowing falls; if the government's interest bill rises, its deficit widens unless it adjusts. The identity $(S - I) + (T - G) \equiv CA$ holds throughout; what the model tells us is which components move.

52.7.6 Consistency checks

The four levels must agree. Three checks guard against contradiction:

1. **Identities are not causes.** The rise in bank interest income is matched by higher interest expense of borrowers; the fall in a bond's market value is a loss to its holder and not a gain to anyone else until the bond matures—but the higher future yield is a gain to anyone buying it now. Every entry has a counterpart somewhere (Chapter 1).
2. **Stocks and flows reconcile.** Lower investment flows mean slower growth of the capital stock and, in the long run, lower potential output only if the policy persists (Chapter 44). Losses on bonds change stocks (equity) without directly changing current flows (GDP): GDP does not fall because a bond's price falls, although spending may fall because wealth has.
3. **Uncertainty is part of the answer.** The size of each effect depends on parameters—the elasticity of intertemporal substitution, the share of variable-rate debt, the degree of market power, the credibility of the central bank, the degree of pass-through—that differ across countries and change over time. A complete answer states the mechanisms, the accounting, the likely signs and the plausible magnitudes, and says what data would tell us which mechanisms are strongest.

Key idea. Microeconomics explains *why* households, firms and banks respond; accounting records *what* happens to their resources and obligations; macroeconomics *adds up* the responses subject to identities that must hold; and monetary economics explains *how* the central bank's action reaches the economy and why its effects are uncertain and lagged. None of the four is sufficient alone. Together they allow one economic event to be explained without contradiction—the ability this book set out to teach.

Summary

- The nominal exchange rate E is the domestic price of foreign currency; the real exchange rate is $Q = EP^*/P$. PPP holds only in the long run and only in relative terms; Balassa–Samuelson explains why price levels are higher in rich countries.
- Covered interest parity holds closely; uncovered interest parity fails at short horizons because of risk premia, making carry trades profitable but crash-prone. An unexpected rate rise appreciates the currency on impact.
- Under floating rates, monetary policy is powerful and fiscal policy weakened; under fixed rates, the reverse. Exchange rates overshoot when prices are sticky.
- The trilemma: fixed exchange rate, capital mobility, monetary independence—choose two. A global financial cycle limits insulation even under floating.
- Regimes range from hard pegs to free floats; optimal currency areas require symmetric shocks, mobility, flexibility and fiscal transfers.
- Intervention appears on the central bank's balance sheet; sterilization neutralizes its effect on the monetary base. Defending against depreciation is limited by reserves; reserves are self-insurance.
- Pass-through, sudden stops, currency mismatch and sovereign–bank loops explain why emerging-market and currency-union crises are so severe. Crisis models range from inconsistent fundamentals to self-fulfilling attacks and balance-sheet crises.
- The Asian crisis, the euro-area crisis and Ukraine's wartime policy each require all four lenses.
- A central-bank rate increase can be explained consistently at four levels: implementation, decisions, accounts and aggregates.

Key terms

nominal exchange rate; depreciation; appreciation; real exchange rate; effective exchange rate; purchasing-power parity; law of one price; PPP puzzle; Balassa–Samuelson effect; forward rate; covered interest parity; cross-currency basis; uncovered interest parity; forward premium puzzle; carry trade; currency risk premium; Mundell–Fleming model; overshooting; trilemma; global financial cycle; capital controls; hard peg; currency board; dollarization; managed float; fear of floating; optimal currency area; foreign-exchange intervention; sterilization; reserve adequacy; Greenspan–Guidotti rule; pass-through; dominant currency paradigm; sudden stop; currency mismatch; original sin; first-, second- and third-generation crisis models; twin crises; sovereign–bank doom loop; internal devaluation; TARGET2 balances; Outright Monetary Transactions; monetary financing; managed flexibility.

Exercises

Exercise 52.1 (Real exchange rate). Over a year, the ringgit depreciates from 4.40 to 4.62 per dollar; Malaysian inflation is 2 per cent and U.S. inflation 3 per cent. (a) Compute the percentage changes in the nominal and real exchange rates. (b) What does relative PPP predict for the nominal rate?

Exercise 52.2 (Interest parity). The spot rate is 1.10 dollars per euro. One-year interest rates are 4 per cent in the United States and 2.5 per cent in the euro area. (a) Using CIP, find the forward

rate in dollars per euro. (b) Under UIP, what is the expected dollar-per-euro rate in a year? (c) If the Fed unexpectedly raises its rate by 1 point, what happens to the spot rate, holding the expected future rate fixed?

Exercise 52.3 (Overshooting). A permanent monetary expansion will raise the long-run exchange rate (domestic per foreign) by 8 per cent. The domestic interest rate falls 2 points below the foreign rate for 1.5 years and then returns. Using UIP, find the impact change in the exchange rate and describe its path.

Exercise 52.4 (Trilemma). Classify each regime and state which element of the trilemma it gives up: (a) Hong Kong; (b) China in 2010; (c) Germany since 1999; (d) Malaysia in 1999; (e) the United States; (f) Ukraine in 2022.

Exercise 52.5 (Intervention entries). A central bank sells 2 billion dollars of reserves at 40 domestic units per dollar to banks to support its currency. (a) Record the entry. (b) What happens to domestic reserves and short-term interest rates if it does not sterilize? (c) Record a sterilization using a purchase of government bonds from banks. (d) Why can this operation not continue indefinitely?

Exercise 52.6 (Currency mismatch). A firm has assets worth 1,000 in domestic currency, domestic debt of 300 and dollar debt of 500 (in domestic currency at the current rate). (a) Compute its equity. (b) The currency depreciates by 40 per cent. Compute its equity. (c) What does this imply for the effect of depreciation on investment in economies with large currency mismatches?

Exercise 52.7 (Pass-through). Imports are 40 per cent of the consumer basket and short-run pass-through is 60 per cent. The currency depreciates by 15 per cent. (a) Estimate the effect on consumer prices. (b) Why might pass-through be lower in a country with a credible inflation target?

Exercise 52.8 (Sudden stop). A country has a current-account deficit of 5 per cent of GDP, financed by short-term inflows, and reserves equal to 3 per cent of GDP. Inflows stop. (a) How long can reserves finance the deficit? (b) What must happen afterwards to absorption? (c) Explain two ways the adjustment can occur and their consequences.

Exercise 52.9 (Optimal currency areas). Using Mundell's criteria, assess whether the euro area was an optimal currency area in 1999, and explain which of the criteria the crisis of 2010–12 revealed to be weakest. What institutional changes since then address them?

Exercise 52.10 (Doom loop). Explain, with balance-sheet reasoning, the sovereign–bank doom loop. How did the ECB's OMT announcement and the banking union aim to break it?

Exercise 52.11 (Ukraine). Explain why the NBU fixed the exchange rate and imposed capital controls in February 2022, and why it later moved to managed flexibility. Use the trilemma, the concept of a nominal anchor, and the role of reserves.

Exercise 52.12 (Capstone). Choose a real central-bank decision from the past two years in any country. Using official sources (the central bank's statement and balance sheet, the national statistical office, and at least one listed company's annual report), write a two-page memo explaining the decision and its likely effects at the four levels of Section 52.7. Separate facts, assumptions, mechanisms and uncertainty.

Solutions

Solution 52.1. (a) Nominal: $4.62/4.40 - 1 = 5.0\%$ depreciation. Real: $g_Q = 5.0 + 3 - 2 = 6.0\%$ real depreciation. (b) Relative PPP predicts $g_E = \pi - \pi^* = 2 - 3 = -1\%$: a 1 per cent appreciation. The actual depreciation deviates from PPP by 6 points, typical of short-run movements.

Solution 52.2. (a) In dollars per euro, the domestic (dollar) rate is 4 per cent: $F = 1.10 \times 1.04/1.025 = 1.1161$. (b) Under UIP, the expected rate equals the forward rate: 1.116 (the euro is expected to appreciate by about 1.5 per cent). (c) With $i = 5\%$ and the expected future rate fixed at 1.116, UIP requires $E = 1.116 \times 1.025/1.05 = 1.0895$: the dollar appreciates on impact (fewer dollars per euro), by about 1 per cent.

Solution 52.3. During the 1.5 years, UIP requires the domestic currency to be expected to appreciate by 2 per cent a year, i.e. by 3 per cent in total. To end at +8 per cent, the rate must jump by about $8 + 3 = 11$ per cent on impact, then appreciate steadily by 2 per cent a year for 1.5 years to its long-run level.

Solution 52.4. (a) Currency board: fixed rate and capital mobility; gives up monetary independence. (b) Managed rate with capital controls; gives up (much) capital mobility. (c) Member of a currency union with capital mobility; gives up national monetary independence. (d) Peg with controls on outflows; gives up capital mobility. (e) Floating with capital mobility; gives up a fixed rate. (f) Fixed rate with capital controls; gives up capital mobility.

Solution 52.5. (a) Central bank: Dr Reserves of banks 80 billion / Cr FX reserves 80 billion. (b) Domestic reserves fall by 80 billion, tightening money-market conditions and raising short-term rates, which also supports the currency. (c) Dr Government bonds 80 / Cr Reserves of banks 80, restoring reserves. (d) FX reserves are finite; once markets expect them to run out, speculators attack (first-generation logic), and sterilization prevents the interest-rate rise that might otherwise stem the outflow.

Solution 52.6. (a) $1,000 - 300 - 500 = 200$. (b) Dollar debt becomes $500 \times 1.4 = 700$; equity = $1,000 - 300 - 700 = 0$ (assuming domestic assets unchanged in value). (c) Depreciation wipes out the net worth of firms with dollar debts, raising their external finance premium and cutting investment (the financial accelerator, Chapter 51); this balance-sheet effect can outweigh the boost to exports, making depreciation contractionary.

Solution 52.7. (a) $0.4 \times 0.6 \times 15 = 3.6$ per cent. (b) Firms expect the central bank to keep overall inflation at target and may expect the depreciation to be partly reversed, so they are less willing to raise prices; lower and more stable inflation also reduces the frequency of price changes (Taylor's argument).

Solution 52.8. (a) About $3/5$ of a year, roughly seven months. (b) Absorption must fall by about 5 per cent of GDP relative to income, so that the current account moves to balance. (c) Through real depreciation (expenditure switching toward domestic goods), which raises the burden of foreign-currency debts; and through a fall in output and imports (expenditure reduction), which raises unemployment. Most sudden stops involve both.

Solution 52.9. Trade integration was high, but shocks were not symmetric (booms in the periphery, stagnation in Germany in the early 2000s), labour mobility across countries was low, wages and prices adjusted slowly, and there was no common fiscal capacity to transfer resources. The crisis revealed the absence of fiscal and banking integration and of a lender of last resort for sovereigns as the weakest points. Responses: the European Stability Mechanism (a crisis-lending fund), OMT, the banking union (single supervision and resolution), and in 2020 the NextGenerationEU programme, financed by common EU borrowing.

Solution 52.10. Banks hold their government's bonds; when the sovereign's creditworthiness falls, bond prices fall, reducing bank equity. Weaker banks may need government support, and the government guarantees or recapitalizes them, raising its debt and further lowering its creditworthiness. Credit tightens and the economy weakens, lowering tax revenue. OMT removed the self-fulfilling component of sovereign risk by promising potentially unlimited purchases; the banking union moved supervision and resolution to the euro-area level, so that bank failures need not fall on a single national budget.

Solution 52.11. In the first days of war, a floating rate would likely have collapsed in a panic, feeding inflation and damaging balance sheets; a fixed rate provided a clear nominal anchor. With a fixed rate, the trilemma required giving up capital mobility (controls) if the NBU wanted to set interest rates for domestic purposes and preserve reserves. As foreign aid rebuilt reserves to record levels and inflation fell, the NBU could afford more exchange-rate flexibility, which lets the rate absorb shocks and restores the role of the policy rate within an inflation-targeting framework; the easing of controls proceeded gradually to avoid destabilizing outflows.

Solution 52.12. There is no single answer. A strong memo states the decision and the central bank's stated reasons; describes implementation (administered rates, balance-sheet operations); traces effects on at least one household type, one firm (using its reported debt structure, interest expense and coverage) and the banking system; estimates macro effects with explicit assumptions (for example, Okun's coefficient and pass-through); checks consistency with identities; and ends with the main uncertainties and the data that would resolve them.

Difference and Differential Equations

A.1 First-Order Equations: A Reminder

Chapter 4 solved the first-order linear difference equation $x_{t+1} = ax_t + b$: if $a \neq 1$, it has the steady state $x^* = b/(1 - a)$ and solution $x_t = x^* + a^t(x_0 - x^*)$, and it converges if and only if $|a| < 1$ (Theorem 4.4). Examples in the book include capital accumulation (Chapter 44), the evolution of unemployment from worker flows (Chapter 45, where $a = 1 - s - f$), public debt dynamics (Chapter 47, where $a = (1 + r)/(1 + g)$, so that debt explodes without primary surpluses when $r > g$), and adaptive expectations (Chapter 46). This appendix extends the analysis in four directions.

Forward solutions. Some equations are naturally solved *forward*. The New Keynesian Phillips curve $\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t$ with $0 < \beta < 1$ has the unique bounded solution

$$\pi_t = \kappa \sum_{k=0}^{\infty} \beta^k \mathbb{E}_t x_{t+k} :$$

inflation is the discounted sum of expected future output gaps. Asset prices are solved the same way: a share price is the discounted sum of expected future dividends (Chapters 31 and 37).

A.2 Second-Order Equations and Cycles

The equation $x_{t+2} = a_1 x_{t+1} + a_2 x_t$ has solutions of the form λ^t , where λ solves the *characteristic equation* $\lambda^2 - a_1 \lambda - a_2 = 0$. If the roots are real, the solution is a combination of λ_1^t and λ_2^t . If they are complex, the solution oscillates, with the modulus of the roots governing whether oscillations grow or die out. For $x_{t+2} = 1.2x_{t+1} - 0.5x_t$, the roots are $0.6 \pm 0.374i$, with modulus $\sqrt{0.5} = 0.707$: damped cycles. Paul Samuelson's (1939) multiplier–accelerator model, combining the consumption function (Chapter 46) with an accelerator for investment (Chapter 45), produced exactly this kind of equation and showed how cycles could arise from the interaction of the two.

A.3 Stochastic Difference Equations

Adding a random shock gives the first-order **autoregressive process** AR(1):

$$x_t = \rho x_{t-1} + \varepsilon_t, \quad \mathbb{E}[\varepsilon_t] = 0, \quad \text{Var}(\varepsilon_t) = \sigma^2.$$

If $|\rho| < 1$, x is stationary with mean zero and variance $\sigma^2/(1 - \rho^2)$, and the effect of a shock decays geometrically: the impulse response is ρ^k after k periods. Productivity shocks in real business cycle models and policy shocks in the New Keynesian model (Chapter 47) are typically AR(1); Chapter 18 discusses their estimation. If $\rho = 1$, x is a *random walk*, as consumption is under Hall's hypothesis (Chapter 45): shocks are permanent and the variance grows without limit.

A.4 Differential Equations

Chapter 7 solved the continuous-time equation $\dot{x} = -\lambda(x - x^*)$: the gap to the steady state shrinks at the proportional rate λ , with half-life $\ln 2/\lambda$. Chapter 44 used this result, with $\lambda = (1 - \alpha)(n + g + \delta)$, to derive the speed of convergence in the Solow model. Non-linear equations $\dot{x} = h(x)$ can be *linearized* around a steady state x^* where $h(x^*) = 0$, using the first-order Taylor approximation $h(x) \approx h'(x^*)(x - x^*)$ (Chapter 5): the steady state is locally stable if $h'(x^*) < 0$, with $\lambda = -h'(x^*)$.

Phase diagrams. Two-variable systems, such as the Ramsey model's (k, c) , are analysed with phase diagrams: plot the curves on which $\dot{k} = 0$ and $\dot{c} = 0$, determine the direction of motion in each region, and identify the saddle path along which the system converges to the steady state. Saddle-path stability—one stable and one unstable eigenvalue, with one predetermined variable (k) and one jump variable (c)—is the continuous-time counterpart of the Blanchard–Kahn condition of Chapter 8.

Dynamic Optimization

B.1 Dynamic Programming

Many problems in macroeconomics involve choosing a sequence of actions over time, where today's choice affects tomorrow's opportunities. Richard Bellman's principle of optimality says that an optimal plan has the property that, whatever the initial state and first decision, the remaining decisions form an optimal plan for the state that results. This allows an infinite-horizon problem to be written recursively.

Let the *state* W summarize everything relevant about the past (for example, wealth), and let $V(W)$, the *value function*, be the maximized lifetime utility from state W . Then V satisfies the **Bellman equation**

$$V(W) = \max_c \{u(c) + \beta V(W')\}, \quad W' = (1+r)(W-c).$$

Worked Example B.1. Eating a cake

Let $u(c) = \ln c$ and $r = 0$, so $W' = W - c$: a consumer must divide a cake of size W over an infinite future. Guess $V(W) = A + B \ln W$. The first-order condition of the Bellman equation is $1/c = \beta B/(W - c)$, so $c = W/(1 + \beta B)$. Substituting back and matching the coefficients on $\ln W$ gives $B = 1 + \beta B$, so $B = 1/(1 - \beta)$, and

$$c = (1 - \beta) W :$$

the consumer eats a constant fraction $1 - \beta$ of the remaining cake each period. With $\beta = 0.95$, she eats 5 per cent of what remains.

The Euler equation from the Bellman equation. Differentiating the Bellman equation with respect to c gives $u'(c) = \beta(1+r)V'(W')$; the envelope theorem (Theorem 6.5) gives $V'(W) = u'(c)$. Combining them yields the Euler equation of Chapters 9 and 45: $u'(c_t) = \beta(1+r)u'(c_{t+1})$. Under uncertainty, the same argument gives $u'(c_t) = \beta(1+r)\mathbb{E}_t[u'(c_{t+1})]$.

Dynamic programming is the workhorse of modern macroeconomics: RBC and DSGE models (Chapter 47), heterogeneous-agent models with borrowing constraints (Chapter 45) and models of optimal policy (Chapter 50) are all solved by computing value functions, usually numerically.

B.2 Optimal Control

In continuous time, a problem such as the Ramsey model (Chapter 44),

$$\max \int_0^\infty e^{-\rho t} u(c) dt \quad \text{subject to} \quad \dot{k} = f(k) - c - \delta k,$$

is solved with the *current-value Hamiltonian*

$$\mathcal{H} = u(c) + \mu [f(k) - c - \delta k],$$

where μ is the shadow value of capital. The conditions for an optimum are:

$$\frac{\partial \mathcal{H}}{\partial c} = 0 \Rightarrow u'(c) = \mu, \quad \dot{\mu} = \rho\mu - \frac{\partial \mathcal{H}}{\partial k} = \mu [\rho + \delta - f'(k)],$$

together with the resource constraint and the *transversality condition* $\lim_{t \rightarrow \infty} e^{-\rho t} \mu k = 0$, which rules out accumulating capital that is never consumed. Differentiating $u'(c) = \mu$ with respect to time and using CRRA utility, for which $-u''(c)c/u'(c) = \theta$, gives

$$\frac{\dot{c}}{c} = \frac{1}{\theta} [f'(k) - \delta - \rho],$$

the continuous-time Euler equation of Chapter 44. The Hamiltonian's conditions are the continuous-time counterparts of the Lagrangian's: μ plays the role of λ , and $\partial \mathcal{H} / \partial c = 0$ is the familiar condition that the marginal utility of consumption equal the marginal value of the resource.

Sources and Further Reading

This list follows the book's source hierarchy. For institutions, standards and data, the primary sources are the issuing bodies themselves; they revise their documents, and readers should consult current versions. For models, the canonical original papers are listed, followed by textbook treatments. For cases, the official documents of central banks, regulators, courts and companies come first. Works are listed once, under the part where they are used most.

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